

1     **An entanglement perspective on**  
2     **generalized symmetries, dualities**  
3     **and anomalies**

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## 84 Summary

85 The *quantum many-body problem* has taken centre stage in theoretical physics  
86 for the better part of the last century. Its central challenge is to predict  
87 and understand the *phase diagram* of a microscopic *quantum Hamiltonian*  
88 describing a macroscopic number of interacting degrees of freedom. Due to  
89 the *exponential scaling* of the dimension of the many-body *Hilbert space*, it is  
90 typically unfeasible to obtain analytical solutions for more than a handful  
91 of particles. Progress on this front therefore calls for *efficient* and *scalable*  
92 *approximate* parametrisations of the wave functions of interest.

93 Having emerged from the field of *quantum information theory*, and in  
94 particular *entanglement theory*, *tensor networks* have gained significant at-  
95 tention over the last twenty years in the study of local quantum many-body  
96 Hamiltonians in low spatial dimensions. Tensor networks parametrise the  
97 class of wave functions satisfying an *area law* scaling of their *entanglement*  
98 *entropy*, which is observed to hold for local *gapped* Hamiltonians. Their  
99 central idea is that *global* properties of the wave function can be efficiently  
100 compressed into *local* tensors that contain the variational parameters of the  
101 wave function. One of the main merits of this local parametrisation is that  
102 the presence of *symmetries* translates into local invariance properties of the  
103 constituent tensors.

104 *Symmetry*, and in particular *symmetry breaking*, has long been appreciated  
105 as an organising principle for classical phase transitions, before becoming  
106 equally central to the study of quantum many-body phase diagrams. Tra-  
107 ditionally, symmetries act on the underlying quantum degrees of freedom  
108 by unitary or anti-unitary representations of *groups*. Within the *Landau*  
109 *paradigm* of phase transitions, a quantum phase of matter is then characterised  
110 by the *unbroken* symmetry of the ground state subspace.

111 In recent years, however, it has been recognised that there exist phases that  
112 go beyond the traditional Landau paradigm. These phases are characterised  
113 by the existence of *hidden* symmetry operators that act on the entanglement  
114 patterns of the ground state. Since tensor networks provide direct access to  
115 these entanglement degrees of freedom, they reveal these hidden symmetries  
116 and form a natural tool for their study. Remarkably, these hidden symmetry  
117 operators need not be invertible and may fall outside the scope of group  
118 theory. Rather, they are naturally described by *(higher-)fusion categories*. In  
119 a sense, tensor networks provide concrete *representations* of these generalised  
120 symmetry structures.

121 These *categorical* symmetries have also been recognised to be inherited by  
122 the *edge* theories of these beyond-Landau phases. The edge theories therefore

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live in one lower dimension. This *holographic* way of describing generalised notions of symmetry has gained significant attention in recent years and forms one of the central topics of this thesis. Concretely, this modern perspective on global symmetries allows us to use established tools from *topological quantum field theory* in their study. This point of view facilitates in particular the study of *gauging* and *dualities* of these symmetries, as well as the classification of phases protected by them, thereby fitting these phases into a *generalised Landau paradigm*.

This thesis is about various aspects of (generalised) symmetry, in particular their associated dualities and anomalies, within the context of tensor networks. It is divided into two parts, *Foundations* and *Applications*, that respectively contain introductory chapters to the topics of this thesis and the novel research results in the form of five publications, as well as a set of lecture notes.

In the introductory chapter 1, we begin by describing in detail the challenge posed by the quantum many-body problem. We recapitulate some key facts about the traditional Landau paradigm and the role that symmetry plays in it.

Chapter 2 is devoted to the introduction of tensor network states and some of their properties. Both the one-dimensional matrix product states (MPSs) and the two-dimensional projected entangled-pair states are discussed. By making use of the fundamental theorem of matrix product states — which stipulates in particular that an MPS is invariant under a certain symmetry if and only if its local tensors are left invariant by the symmetry — we review in detail the classification of one-dimensional gapped phases of matter. These phases are fully described by the pattern of symmetry breaking in their ground state subspace, together with a topological invariant for their residual symmetry, representing the symmetry-protected topological order.

In chapter 3, we first provide an introduction to the necessary notions of fusion category theory. Next, we revisit the construction of Levin and Wen’s string-net models. These provide a framework for lattice realisations of non-chiral (2+1)-dimensional topological orders with gapped boundary conditions. We discuss in particular the classification of those gapped boundary conditions, as well as their ground states and anyonic excitation spectrum. Crucially, the ground states of these string-net models admit exact tensor network representations that are characterised by topological symmetries that act only on the entanglement degrees of freedom. These form precisely the hidden symmetries mentioned above.

Finally, in chapter 4 we construct general boundary Hamiltonians for the string-net models. These Hamiltonians can be either gapped or gapless, but



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inherit the (generically non-invertible) symmetries from the bulk. We then sketch how these symmetries can be gauged or dualised, and illustrate this via the Kramers-Wannier duality of the transverse field Ising model. When defined on periodic chains, duality transformations act as non-invertible linear operators. They can be promoted to genuine isometries by enlarging the Hilbert space with additional degrees of freedom that implement symmetry-twisted boundary conditions, on which the duality operators can act non-trivially. Subsequently, we study the action of the two-dimensional Kramers-Wannier duality and argue that it leads to dual models with a symmetry structure described naturally by a fusion 2-category. After outlining the relevant algebraic framework, we conclude with an outlook towards a general theory of dualities in 2+1 dimensions.

The first chapter of the second part consists of lecture notes on tensor networks. This is the only application chapter that does not contain fundamentally new results. Moreover, parts of the introductory chapters are based on relevant material from these lecture notes.

The first publication, presented in chapter 6, classifies one-dimensional bosonic phases of matter protected by quasi-one-dimensional spatial symmetries. These symmetries correspond to the so-called frieze symmetry groups. The chapter uses techniques very similar to those discussed in chapter 2 and provides, among other results, canonical forms for the phases that arise in this classification. We also give a practical algorithm to compute group cohomology and representative cocycles.

The publication presented in chapter 7 establishes the result that different Morita equivalent string-net models realise the same non-chiral topological phase. Although these string-net models are microscopically very distinct, we construct an explicit constant-depth quantum circuit that maps one model to the other. This circuit acts not only on the ground state subspace, but also on the excited states when supplemented with suitable auxiliary degrees of freedom.

Chapter 8 then demonstrates that the one-dimensional duality transformations introduced in chapter 4 can be interpreted as gauging a particular collection of symmetries encoded in the symmetry fusion category. In particular, we demonstrate that the MPO duality intertwiners are retrieved from a gauging map, acted upon by a unitary constant-depth quantum circuit that disentangles the gauge degrees of freedom. This generalises an earlier construction due to Haegeman et al. for group symmetries. We also discuss the role of anomalies in this framework, and determine under which conditions this gauging procedure can be viewed as implementing a local version of the symmetry.

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202        In chapter 9, we build on an earlier construction by José Garre-Rubio.  
203        By starting from an abelian symmetry in arbitrary dimension and iteratively  
204        gauging both the symmetry and its dual symmetry, we obtain states in one  
205        dimension higher. We show that the states obtained in this way are naturally  
206        stabiliser codes, and provide numerous examples fitting into this framework.  
207        Finally, in chapter 10, we turn to  $(2+1)$ -dimensional dualities. We discuss  
208        a duality transformation in both one and two spatial dimensions, that amounts  
209        to the composition of a gauging transformation followed by a twisted gauging  
210        transformation. This project solves the open problem of constructing  $(2+1)$ -  
211        dimensional self-dual models, meaning models that are left invariant under a  
212        particular duality transformation. We also lift these duality transformations  
213        to unitary transformations by taking into account the appropriate boundary  
214        conditions.

## Samenvatting

Het *kwantumveeldeeltjesprobleem* bevindt zich in het middelpunt van de theoretische fysica sinds het begin van de vorige eeuw. Het centrale probleem is het voorspellen en begrijpen van het *fasediagram* van microscopische *kwantumhamiltonianen* die een macroscopisch aantal interagerende vrijheidsgraden beschrijven. Omwille van de *exponentiële* schaling van de dimensie van de veeldeeltjes *Hilbertruimte* is het typisch onmogelijk om analytische oplossingen te construeren voor meer dan een handvol deeltjes. Vooruitgang hierin vraagt daarom om *efficiënte* en *schaalbare benaderende* parameterisaties van de relevante golffuncties.

*Tensor netwerken* zijn voortgekomen uit het domein van de *kwantuminformatietheorie*, en in het bijzonder uit de *kwantum verstrengelingstheorie*. Ze hebben de afgelopen twintig jaar veel aandacht gekregen in de studie van lokale kwantumhamiltonianen in lage spatiale dimensies. Tensor netwerken parametriseren de klasse van golffuncties die voldoen aan een *oppervlak-tewetschaling* van de *verstrengelingsentropie*, die wordt waargenomen dat ze geldt voor lokale Hamiltonianen met een *energie kloof*. Het centrale idee is dat *globale* eigenschappen van de golffunctie efficiënt kunnen worden gecomprimeerd in *lokale* tensoren die de variationele parameters van de golffunctie bevatten. Een van de belangrijkste voordelen van deze lokale parametrisatie is dat de aanwezigheid van *symmetrieën* zich vertaalt in lokale invariantie-eigenschappen van de onderliggende tensoren.

*Symmetrie*, en in het bijzonder *symmetriebreking*, wordt sinds lang geapprecieerd als een organiserend principe voor klassieke fasetransities, vooraleer zij even centraal kwam te staan in de studie van kwantumveeldeeltjesfasediagrammen. Traditioneel werken symmetrieën in op de onderliggende kwantumvrijheidsgraden via unitaire of anti-unitaire representaties van *groepen*. Binnen *Landau's paradigma* van fasetransities wordt een kwantum fase van materie gekarakteriseerd door de *ongebroken* symmetrie van de grondtoestandsdeelruimte.

In recente jaren werd echter erkend dat er fasen bestaan die voorbij het traditionele Landau paradigma gaan. Deze fasen worden gekarakteriseerd door het bestaan van *verborgen* symmetrieoperatoren die inwerken op de verstrengelingspatronen van de grondtoestand. Omdat tensor netwerken directe toegang verschaffen tot deze verstrengelingsvrijheidsgraden, onthullen ze deze verborgen symmetrieën en vormen ze een natuurlijke taal voor hun studie. Opmerkelijk genoeg hoeven deze verborgen symmetrieoperatoren niet inverteerbaar te zijn en kunnen ze daarom buiten het kader van groepentheorie vallen. Ze worden daarentegen op natuurlijke wijze beschreven door

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254 (*hogere*) *fusie*categorieën. In zekere zin vormen tensor netwerken concrete  
255 *representaties* van deze gegeneraliseerde symmetriestructuren.

256 Deze categorische symmetrieën blijken ook te worden overgeërfd door de  
257 *randtheorieën* van deze fasen voorbij het Landau paradigma. Deze randthe-  
258 orieën leven in een dimensie lager. Deze *holografische* manier om gegener-  
259 aliseerde noties van symmetrie te beschrijven, heeft de voorbije jaren aanzien-  
260 lijke aandacht gekregen en vormt een van de centrale onderwerpen van deze  
261 thesis. Concreet laat dit moderne perspectief op globale symmetrieën ons  
262 toe om gevestigde technieken uit de *topologische kwantumveldentheorie* te  
263 gebruiken in hun studie. Dit gezichtspunt faciliteert in het bijzonder de studie  
264 van het *ijken* en *dualiseren* van deze symmetrieën, maar ook de classificatie  
265 van de fasen die erdoor worden beschermd, en past deze fasen zo in een  
266 *gegeneraliseerd Landau paradigma*.

267 Deze thesis behandelt verschillende aspecten van gegeneraliseerde symmetrie,  
268 in het bijzonder de bijbehorende dualiteiten en anomalieën, binnen de context  
269 van tensor netwerken. Ze is verdeeld in twee delen, *Foundations* and *Applica-*  
270 *tions*, die respectievelijk inleidende hoofdstukken over de onderwerpen van  
271 deze thesis en nieuwe onderzoeksresultaten in de vorm van vijf publicaties en  
272 een verzameling cursusnota's bevatten.

273 In het inleidende hoofdstuk 1 beginnen we met een gedetailleerde beschri-  
274 jving van de uitdaging die het kwantumveeldeeltjesprobleem stelt. We re-  
275 capituleren enkele van de belangrijkste eigenschappen van het traditionele  
276 Landau paradigma en de rol die symmetrie daarin speelt.

277 Hoofdstuk 2 is gewijd aan de introductie van tensor netwerktoestanden en  
278 enkele van hun eigenschappen. Zowel de eendimensionale matrixproducttoes-  
279 tanden als de tweedimensionale geprojecteerde verstrengelde-paartoestanden  
280 worden besproken. Door gebruik te maken van de fundamentele stelling van  
281 matrixproducttoestanden, die in het bijzonder beschrijft hoe een symmetrie  
282 van een MPS zich vertaalt in een lokale invariantie van de onderliggende ten-  
283 soren, herhalen we in detail de classificatie van eendimensionale kwantumfasen  
284 van materie. Deze fasen worden volledig beschreven door het patroon van  
285 symmetriebreking in de grondtoestandsruimte, samen met een topologische  
286 invariant voor de overblijvende symmetrie, die de symmetrie-beschermd  
287 topologische orde karakteriseert.

288 In hoofdstuk 3 geven we eerst een introductie tot de nodige begrippen uit  
289 fusiecategorietheorie. Vervolgens herhalen we de constructie van de string-net  
290 modellen van Levin en Wen. Deze vormen een kader voor roosterrealisaties  
291 van niet-chirale topologische ordes in  $(2+1)$  dimensies met randvoorwaarden  
292 die de energie kloof behouden. We bespreken in het bijzonder de classificatie

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van deze randvoorwaarden, evenals hun grondtoestanden en het anyonische excitatiespectrum. Cruciaal is dat deze grondtoestanden exacte tensor netwerkrepresentaties hebben die worden gekarakteriseerd door topologische symmetrieën die uitsluitend inwerken op de verstrengelingsvrijheidsgraden. Deze vormen precies de eerder vermelde verborgen symmetrieën.

Tot slot construeren we in hoofdstuk 4 algemene randhamiltonianen voor de string-net modellen. Deze Hamiltonianen erven de (in het algemeen niet-inverteerbare) symmetrieën van de bulk over. Vervolgens schetsen we hoe deze symmetrieën kunnen worden geïjkt of gedualiseerd, en illustreren we dit aan de hand van de Kramers-Wannier-dualiteit van het Ising model in een transversaal veld. Op periodieke randvoorwaarden gedragen dualiteits-transformaties zich als niet-inverteerbare lineaire operatoren. Ze kunnen worden gepromoveerd tot isometrieën door de Hilbertruimte uit te breiden met bijkomende vrijheidsgraden die symmetrie-compatibele randvoorwaarden implementeren, waarop de dualiteitsoperatoren niet-triviaal kunnen inwerken. Vervolgens bestuderen we de werking van de tweedimensionale Kramers-Wannier-dualiteit en beargumenteren we dat deze leidt tot duale modellen met een symmetriestructuur die op natuurlijke wijze wordt beschreven door een fusie-2-categorie. Na een schets van het relevante algebraïsche kader besluiten we dit hoofdstuk met een vooruitblik op een algemene theorie van dualiteiten in  $(2+1)$  dimensies.

Het eerste hoofdstuk van het tweede deel bestaat uit cursusnota's over tensor netwerken. Dit is het enige toepassingshoofdstuk dat geen fundamenteel nieuwe onderzoeksresultaten bevat. Bovendien zijn delen van de inleidende hoofdstukken gebaseerd op relevant materiaal uit deze cursusnota's.

De eerste publicatie, gepresenteerd in hoofdstuk 6, classificeert eendimensionale bosonische fasen van materie die worden beschermd door quasi-eendimensionale ruimtelijke symmetrieën. Deze symmetrieën corresponderen met de zogenaamde frieze symmetriegroepen. Het hoofdstuk gebruikt technieken die sterk gelijken op die uit hoofdstuk 2 en levert, naast andere resultaten, canonieke vormen voor de fasen die uit deze classificatie voortkomen. We geven ook een praktisch algoritme voor het berekenen van groepencohomologie en representatieve cocycles.

In de publicatie die in hoofdstuk 7 wordt gepresenteerd, tonen we aan dat verschillende Morita-equivalente string-net modellen dezelfde niet-chirale topologische fase realiseren. Hoewel deze string-net modellen microscopisch sterk verschillen, construeren we een expliciet kwantumcircuit van constante diepte dat het ene model afbeeldt op het andere. Dit circuit werkt niet alleen in op de grondtoestandsdeelruimte, maar ook op geëxciteerde toestanden wanneer bijkomende hulpvrijheidsgraden in rekening worden gebracht.

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333       Hoofdstuk 8 toont aan dat de eendimensionale dualiteitstransformaties  
334       die in hoofdstuk 4 worden geïntroduceerd, kunnen worden geïnterpreteerd als  
335       het ijen van een specifieke collectie symmetrieën uit een symmetriefusiecat-  
336       egorie. In het bijzonder tonen we aan dat de MPO dualiteitsoperatoren  
337       worden verkregen uit een ijkingsafbeelding waarop vervolgens een unitair  
338       kwantumcircuit van constante diepte wordt toegepast dat de ijkvrijheids-  
339       graden ontstrengelt. Dit veralgemeent een eerdere constructie van Haegeman  
340       et al. voor groepssymmetrieën. We bespreken ook de rol van anomalieën  
341       binnen dit kader en bepalen onder welke voorwaarden deze ijkingsprocedure  
342       kan worden beschouwd als de implementatie van een lokale versie van de  
343       symmetrie.

344       In hoofdstuk 9 bouwen we voort op een eerdere constructie van José Garre-  
345       Rubio. We vertrekken van een abelse symmetrie in een arbitraire dimensie  
346       en ijen op iteratieve wijze zowel de symmetrie als haar duale symmetrie. Zo  
347       bekomen we toestanden in een dimensie hoger. We tonen aan dat de aldus  
348       verkregen toestanden op natuurlijke wijze stabilisator codes definiëren en  
349       geven verschillende voorbeelden die binnen dit kader passen.

350       Tot slot gaan we in hoofdstuk 10 over tot dualiteiten in  $(2+1)$  dimensies.  
351       We bespreken een dualiteitstransformatie in zowel een als twee ruimtelijke  
352       dimensies die neerkomt op de compositie van een ijkingstransformatie en  
353       een daaropvolgende getwiste ijkingstransformatie. Dit project lost het open  
354       probleem op van de constructie van zelfduale modellen in  $(2+1)$  dimensies,  
355       waarmee we modellen bedoelen die invariant zijn onder een specifieke du-  
356       aliteitstransformatie. We promoveren deze dualiteitstransformaties bovendien  
357       tot unitaire transformaties door de gepaste randvoorwaarden in rekening te  
358       brengen.

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## 1

## Prologue

The *quantum many-body problem* stands as one of the central problems of physics of the last century. It concerns the description of the *collective behaviour* of a *macroscopic* number of quantum constituents. Despite the relative simplicity of the one-particle Schrödinger equation, the *exponential growth* of the many-body Hilbert space poses a fundamental challenge to analytical and computational approaches, rendering exact solutions essentially intractable for all but the smallest system sizes.

The significance of the many-body problem can hardly be overstated as it lies at the heart of quantum field theory, condensed matter physics, quantum chemistry and quantum computation. Any progress on this front therefore carries implications across a wide spectrum of fields.

A crucial observation, however, is that the typical states of physical interest in the many-body Hilbert space occupy only a *low-dimensional manifold*. A physically motivated approach to the many-body problem should therefore aim to efficiently parametrise this tiny low-complexity manifold and develop methods to project and diagonalise the Hamiltonian within it. This philosophy exactly underlies the *tensor network* approach, which forms the central technical toolbox of this thesis. In essence, tensor network states provide efficient parametrisations of states satisfying an *area law* for their *entanglement entropy*.

## 1.1 The exponential wall

The hallmark of a quantum many-body system is the *tensor product structure* of its Hilbert space. By this, it is meant that the full many-body Hilbert space  $\mathcal{H}$  can be written as a tensor product of constituent spaces hosting

462 localised *quantum modes* or *particles*:

$$\mathcal{H} = \bigotimes_{i=1}^N \mathcal{H}_i. \quad (1.1)$$

463 Throughout, we will assume the local Hilbert space dimension  $d_i := \dim_{\mathbb{C}} \mathcal{H}_i$   
 464 to be finite, which is naturally true in many cases of interest, for example  
 465 when  $\mathcal{H}_i$  is the Hilbert space of a qubit  $\mathcal{H}_i \simeq \mathbb{C}^2$ . Even when  $\mathcal{H}_i$  is inherently  
 466 infinite-dimensional such as in the case of bosonic modes, a truncation of the  
 467 local basis is often permitted for many physically relevant systems.

468 Provided an (orthonormal) basis for the local modes,  $\mathcal{H}_i := \text{Span}_{\mathbb{C}}\{|i\rangle_i \mid i =$   
 469  $1, 2, \dots, d_i\}$ , let us consider the (orthonormal) tensor product basis for  $\mathcal{H}$   
 470 denoted by

$$|i_1, i_2, \dots, i_N\rangle := |i_1\rangle \otimes |i_2\rangle \otimes \dots \otimes |i_N\rangle. \quad (1.2)$$

471 In this basis, a many-body wave function  $|\Psi\rangle$  can be expressed as

$$|\Psi\rangle = \sum_{\{i_j\}} \Psi_{i_1, i_2, \dots, i_N} |i_1, i_2, \dots, i_N\rangle, \quad (1.3)$$

472 with amplitudes  $\Psi_{i_1, i_2, \dots, i_N} := \langle i_1, i_2, \dots, i_N | \Psi \rangle$ . Notably, a generic wave  
 473 function is specified by a number of complex coefficients  $\Psi_{i_1, i_2, \dots, i_N}$  that scales  
 474 *exponentially* in the number of constituent particles. Consequently, an exact  
 475 treatment of even a moderate number of particles very quickly becomes  
 476 computationally intractable. However, as argued below, most many-body  
 477 states cannot be created by any reasonable physical mechanism due to the  
 478 *superposition principle* of quantum mechanics, which leads to the existence  
 479 of highly *entangled* many-body states.

480 A precise mathematical characterisation of entanglement is as follows. Con-  
 481 sider a *bipartition* of the Hilbert space in a subsystem  $A$  and its complement  
 482  $\bar{A}$ , i.e.  $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}$ , and write  $d_A = \dim(\mathcal{H}_A)$ ,  $d_{\bar{A}} = \dim(\mathcal{H}_{\bar{A}})$ . By the  
 483 *singular value decomposition* (SVD), any state in  $\mathcal{H}$  can be written in the  
 484 form

$$|\Psi\rangle = \sum_{k=1}^r s_k |\psi_k^A\rangle \otimes |\psi_k^{\bar{A}}\rangle, \quad (1.4)$$

485 where  $|\psi_k^A\rangle$  and  $|\psi_k^{\bar{A}}\rangle$ ,  $r \leq \min(d_A, d_{\bar{A}})$ , are orthonormal bases for  $\mathcal{H}_A$  and  
 486  $\mathcal{H}_{\bar{A}}$  respectively. The non-negative numbers  $s_k$  for  $k = 1, 2, \dots, r$  are called  
 487 *singular values* or *Schmidt coefficients*. The state  $|\Psi\rangle$  is said to be *entangled*  
 488 between the parties  $A$  and  $\bar{A}$  when the Schmidt rank  $r$  is strictly larger than  
 489 1, and to be *maximally entangled* when all singular values are equal (and

non-zero). The state  $|\Psi\rangle$  is thus *unentangled* if and only if it is a (tensor) product state, viz.  $|\Psi\rangle = |\psi^A\rangle \otimes |\psi^{\bar{A}}\rangle$ .

Information about the entanglement can be compressed into one number, the *entanglement entropy*  $S_A$ , which is defined as the *von Neumann entropy* of the reduced density matrix  $\rho_A$  (or equivalently  $\rho_{\bar{A}}$ ), and which can simply be expressed as:

$$\rho_A = \text{Tr}_{\bar{A}}(|\Psi\rangle\langle\Psi|) = \sum_k s_k^2 |\psi_k^A\rangle\langle\psi_k^A|, \quad (1.5)$$

such that:

$$S_A = -\text{Tr}(\rho_A \log(\rho_A)) = -\sum_k s_k^2 \log(s_k^2). \quad (1.6)$$

More refined information about the entanglement structure of the state is encoded into the *entanglement spectrum*, defined as the eigenvalues  $\{s_k^2\}_k$  of the reduced density matrix  $\rho_A$ . A central result in the field of quantum information theory due to Nielsen stipulates that by making use of *local operations supplemented with classical communication*, so-called LOCC transformations, a bipartite state  $|\Psi_1\rangle$  can be converted *deterministically* into a state  $|\Psi_2\rangle$  if and only if the entanglement spectrum of  $|\Psi_1\rangle$ ,  $\{s_{k,1}^2\}_k$ , *majorises* that of  $|\Psi_2\rangle$ ,  $\{s_{k,2}^2\}_k$ , meaning that  $\sum_{k=1}^m s_{k,1}^2 \geq \sum_{k=1}^m s_{k,2}^2$ , for all  $m = 1, \dots, r$ , and where the Schmidt coefficients are assumed to be *decreasingly* ordered [1].

For many cases of interest — notably the low-energy states of *local gapped* Hamiltonians — the rapid decay of the entanglement spectrum is precisely what permits an efficient *tensor network* parametrisation of the state  $|\Psi\rangle$ . As they form the central technical tool of this thesis, chapter 2 is devoted to introduce and discuss them.

From the expression eq. (1.5) we can also conclude that  $\rho_A$  only describes a pure state, i.e.  $\text{rank}(\rho_A) = 1$ , when  $|\Psi\rangle$  is a product state; or, in other words, that the subsystem  $A$  behaves as a classical mixture if the state  $|\Psi\rangle$  is entangled. This neatly illustrates Schrödinger’s observation that [...] *the best possible knowledge of a whole does not necessarily include the best possible knowledge of all its part* [...] [2].

Having defined the notion of entanglement, we can make precise the above claim that most states in the quantum many-body Hilbert space are physically inaccessible. Focusing on the Hilbert space of  $N$  qubits, let us consider to this end the following question put forward in ref. [3]: what portion of the many-body Hilbert space can be prepared and explored by means of a physical mechanism, specifically by time-evolution under a local time-dependent Hamiltonian for a time that scales at most *polynomially* in the

524 system size? Following ref. [3], any such Hamiltonian is approximated to  
525 arbitrary precision by a polynomial-depth *quantum circuit* consisting of few-  
526 qubit gates drawn from a *universal gate set*, and  $\varepsilon$ -balls around the outputs  
527 of these circuits are considered. In this context, an  $\varepsilon$ -ball around a certain  
528 state  $|\Psi_0\rangle$  is defined as the collection of all states  $|\Psi\rangle$  such that

$$|\langle\Psi|\Psi_0\rangle| \geq 1 - \varepsilon. \quad (1.7)$$

529 A straightforward counting argument demonstrates that such balls occupy  
530 a fraction of the many-body Hilbert space that scales as  $\mathcal{O}(\varepsilon^{2^N})$ , i.e. as a  
531 double exponential!

532 From these considerations we must, perhaps quite shockingly, conclude  
533 that the vast majority of states in the Hilbert space should be deemed *unphys-*  
534 *ical* in any operationally meaningful sense. Indeed, there is an overwhelming  
535 time barrier to prepare a generic state consisting of any more than a handful  
536 of constituent particles.

537 Any physically motivated approach to the many-body problem should there-  
538 fore consist of identifying the relevant low-complexity *manifold of states*  
539 and developing the techniques needed to deal with them. Perhaps the most  
540 well-known such manifold is that of *Slater determinants* of fermions. This  
541 parametrisation and its associated self-consistent Hartree-Fock method has  
542 been extremely successful for almost-free fermion systems and has elucidated  
543 the structure of the periodic table of elements [4–6]. However, when correla-  
544 tions are strong, perturbative approaches such as Hartree-Fock fail to capture  
545 the relevant physics. It turns out that for systems satisfying an *area law* of  
546 entanglement entropy, the natural low-complexity manifold is precisely that  
547 of tensor network states. In particular, such an area law can be proven to  
548 hold for one-dimensional *local gapped lattice models*, and is observed in higher  
549 dimensions.

## 550 1.2 Quantum lattice models

551 *Quantum lattice models* appear ubiquitously as an effective, compressed  
552 description of the low-energy physics of a condensed matter system or as  
553 spatial discretisation of a quantum field theory. In the context of this thesis,  
554 we will roughly think of a *lattice*  $\Lambda$  as a discretisation of the spatial manifold  
555 on which the many-body Hilbert space is supported. It consists of a collection  
556 of vertices, edges, plaquettes, ... that we will denote by respectively  $V(\Lambda)$ ,  
557  $E(\Lambda)$ ,  $P(\Lambda)$ , ...

Provided a lattice, the many-body Hilbert space eq. (1.1) is constructed by assigning a local Hilbert space to the vertices, edges, plaquettes or combinations thereof. The model itself is then defined through a microscopic Hamiltonian  $H$  consisting of *geometrically local* terms so that it can be written as  $H = \sum_i h_{i,K}$  where  $h_{i,K}$  denotes a term acting non-trivially on at most  $K$  sites in the neighbourhood of site  $i$ . Moreover, such Hamiltonians typically depend on a small number of tunable parameters that are meant to mimic among other things external magnetic fields and chemical potentials.

Throughout this thesis we are primarily interested in the *ground state subspace* and *low-lying excited states* of local lattice models in *one* and *two spatial dimensions*. Our focus on the low-energy spectrum is physically motivated by the observation that the Fermi temperature of valence electrons in materials of interest is of the order  $10^4 K$ , well above the temperature scale of experiments. Also in the perturbative computation of cross sections in quantum field theory, the ground state is usually the reference state on top of which excitations are created.

Finding the (approximate) ground state of a generic quantum lattice model is notoriously difficult due to the impossibility to minimise the energy of all Hamiltonian terms simultaneously [7]. This phenomenon is referred to as *quantum frustration* and is intimately related to the concept of *monogamy of entanglement*. As a paradigmatic example, consider the antiferromagnetic spin 1/2 Heisenberg chain:

$$H = \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1}, \quad (1.8)$$

with  $\mathbf{S} = (S^x, S^y, S^z)$  the  $\mathfrak{su}(2)$  generators in the spin 1/2 representation. On two sites, the unique ground state is the (maximally entangled) spin singlet  $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ . One might therefore hope that the ground state on any chain length is one for which all reduced density matrices  $\rho_{i,i+1}$  are equal to a spin singlet. Such a state would indeed clearly minimise all exchange terms  $\mathbf{S}_i \cdot \mathbf{S}_{i+1}$ . However, such a state does not exist in virtue of *monogamy of entanglement* which stipulates that a maximally entangled pair of particles cannot be entangled with a third one [8, 9]. Instead, each spin must optimally distribute its entanglement among its neighbours, and it is precisely this frustration that gives rise to a strongly correlated ground state characterised by intricate *entanglement patterns*.

The nature of these entanglement structures depends strongly on whether the model is gapped. A model is said to possess an (*energy*) *gap* if the

excitation energy of the lowest-lying excited state(s) is non-vanishing in the thermodynamic limit  $N \rightarrow \infty$ .<sup>1</sup> Such a system is characterised by exponentially decaying *connected correlation functions* to which a finite *correlation length*  $\xi$  can be associated, viz.:

$$\langle O(\mathbf{r})O(\mathbf{0}) \rangle_{\text{conn.}} \sim \exp(-|\mathbf{r}|/\xi), \quad (1.9)$$

for a local operator  $O$  [10, 11]. The entanglement structure of gapped systems differs vastly from that of arbitrary states in the many-body Hilbert space. By virtue of the *locality* of physical interactions, gapped ground states obey an *area law* of their entanglement entropy. In generic spatial dimensions, the area law of entanglement entropy stipulates that the von Neumann entropy between any region  $A$  and its complement does not scale as the volume of its interior (as could be expected for generic quantum states) but rather by the surface area of its boundary  $\partial A$ . Specialising to the case of one-dimensional chains, the area of a contiguous region is simply a constant and for this case the area law was formally proven by Hastings by a clever application of Lieb-Robinson bounds [12, 13]. In fact, it can be shown that the entanglement entropy scales with the correlation length according to  $\log(\xi)$  [14].<sup>2</sup> In higher dimensions, the area law is not formally proven, but observed to hold in important cases of interest.

The low-lying spectrum on top of gapped ground states likewise exhibit a characteristic structure. Elementary excitations are typically organised in a few well-defined isolated branches of *quasiparticle* excitations with well-defined energy and momentum [15–19]. At higher energies, quasiparticle combine into *scattering states* whose energy is bounded from below by the two-particle excitation energy.

### 1.3 Symmetries and quantum phases of matter

The central challenge in the many-body problem is to determine and characterise the *phase diagram* of a given microscopic model, provided a *symmetry* of the model.

A symmetry is, roughly speaking, a collection of transformations that leave invariant the dynamics of the model. A standard distinction is made between *internal* and *spacetime* symmetries. Spacetime symmetries act on

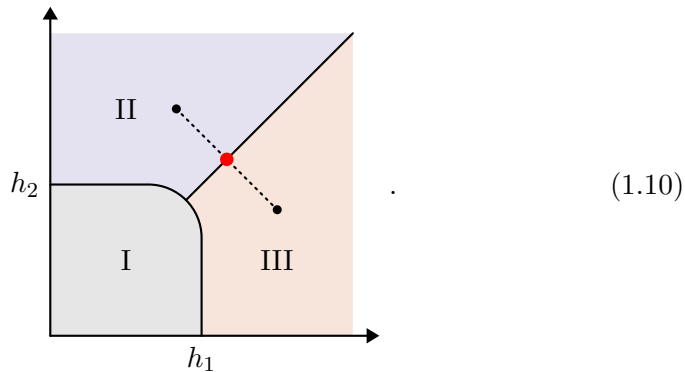
<sup>1</sup>A state which has a finite energy gap above the true ground state for any finite system size, but whose energy splitting vanishes exponentially in the thermodynamic limit  $N \mapsto \infty$  is thus also a ground state.

<sup>2</sup>The entanglement entropy of *critical* systems mildly violates the area law, since in that case the entanglement entropy of a contiguous region of  $L$  spins scales as  $\log(L)$ .



the underlying lattice itself — for instance via spatial translations, rotations or reflections — or as time translation or time-reversal. The one-dimensional spacetime symmetries are the topic of chapter 6. In what follows, we restrict attention to internal symmetries, which act directly on the quantum degrees of freedom rather than on the underlying spacetime. Concretely, we here assume the internal symmetries to be implemented *globally* via a unitary *on-site* (tensor product) representation of a group  $G$ . Crucially, the ground state projector commutes with the symmetry. The *ground state subspace*, or *ground space*, however, need not be trivial: even though the ground space may be one-dimensional, in which case it is *symmetric*, it could also transform in a higher-dimensional representation of  $G$ , which signals *spontaneous symmetry breaking* of  $G$ . In that case, a reference ground state is stabilised under the  $G$ -action by a maximal subgroup  $H \subseteq G$ , and one speaks of the *symmetry-breaking pattern*  $G \mapsto H$ .

A gapped quantum phase of matter can roughly be understood as an *equivalence class* of gapped Hamiltonians whose properties vary smoothly with the Hamiltonian parameters. More precisely, two models are said to belong to the same phase when they can be interpolated by a smooth path of gapped, bounded-strength, local Hamiltonians that commute with the symmetry group  $G$ . From the point of view of quantum information theory, this is equivalent to the existence of a symmetric quantum circuit of sub-linear depth that maps their ground states into each other [20–22]. Distinct gapped phases are separated by *continuous phase transitions* characterised by a diverging correlation length and *algebraic* decay of correlation functions whose universal scaling behaviour is captured by a *conformal field theory* [23]. For a Hamiltonian with two controllable parameters, a fictitious phase diagram with three gapped phases is sketched below. The dotted line represents a Hamiltonian path interpolating between phases II and III. Along this path the gap closes at the phase boundary.



Built upon intuition about classical many-body systems, *Landau's paradigm* has since long provided the unifying framework of phases and phase transition [24]. This framework states that phases of matter are in one-to-one correspondence with symmetry-breaking patterns of the symmetry. Moreover, the paradigm posits that phase transitions can be diagnosed by local *order parameters* that are charged under the symmetry group that acquire a non-vanishing vacuum expectation value in the symmetry-broken phase. Since Kadanoff and Ceva [25] it has also been understood that *symmetric* phases are analogously recognised via *disorder parameters*.

The canonical example is the one-dimensional transverse field *Ising model*, governed by the Hamiltonian [26]

$$H = -J \sum_i Z_i Z_{i+1} - \lambda \sum_i X_i, \quad (1.11)$$

which possesses a global  $\mathbb{Z}_2$  *spin-flip symmetry* generated by  $\prod_i X_i$ . In the *ordered* or *ferromagnetic* phase ( $J > \lambda$ ), the spontaneous breaking of this symmetry is characterised by a non-vanishing expectation value of the local magnetisation  $\langle Z_i \rangle \neq 0$ , which serves as the local order parameter.<sup>3</sup> In the *disordered* or *paramagnetic* phase ( $\lambda > J$ ), the symmetry is restored and  $\langle Z_i \rangle = 0$ , but now the disorder parameter  $\langle \prod_{i=k}^l X_i \rangle$  attains a non-zero vacuum expectation value.

Since its conception, quantum phases of matter that defy Landau's paradigm have emerged. Perhaps unsurprisingly, these *new* phases are characterised by the distinct ways in which the symmetry can manifest itself in the entanglement patterns of the ground state [21, 27–32]. Given the prominent role of some of these phases throughout this thesis, let us take some time here to introduce them and discuss some of their properties.

In the presence of an on-site group symmetry, gapped *symmetry-protected topological* or *trivial* (SPT) phases are characterised by the uniqueness of the ground state on periodic boundary conditions and are adiabatically connected to product states, when the symmetry is explicitly broken [28, 33]. On open boundary conditions they exhibit gapless modes that are exponentially localised near the boundaries, and which transform *projectively* under the symmetry group. In one dimension, they can be completely described within

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<sup>3</sup>One typically defines this order parameter by introducing a small longitudinal field  $h \sum_i Z_i$  into the Hamiltonian, whose role is to select one of the two symmetry-broken ground states depending on the sign of  $h$ , which is then taken to zero,  $h \mapsto 0$ , after going to the thermodynamic limit  $N \mapsto +\infty$ .

the framework of one-dimensional tensor network states or *matrix product states*, and we will revisit their classification and physical properties within this framework in section 2.4.

Unlike SPT phases, (*intrinsic*) *topological phases* exist only in two or more spatial dimensions, and do not require the presence of a (physical) symmetry. They are characterised by a ground state degeneracy that depends solely on the topology of the spatial manifold and their gapped excitations exhibit *anyonic* braiding statistics. A particular class of topological phases, which admit a microscopic lattice realisation in terms of *string-net models*, form the topic of chapter 3, and are also naturally described in terms of tensor networks. As shown below, these tensor networks host *freely deformable* symmetry operators that act only on the *entanglement degrees of freedom*. Such symmetries furnish representations of *fusion categories*, thereby extending familiar notions from finite group and representation theory to algebraic structures whose multiplication law does not generally have an inverse.

## 1.4 Outline of this thesis

This thesis is divided into two parts: *Foundations* and *Applications*. The latter entails the publications around which the thesis is centred and contains the research results obtained over the past five years. All of this work was carried out in collaboration with others. Each chapter begins with a brief summary of the main results, references the corresponding publication or preprint, and details my own contributions to the project. The research article is then reproduced in its published form or in its most recent version. One of these chapters is a set of lecture notes on tensor networks and as such does not contain new research results.

The first part on the other hand, consists of three chapters that serve both as an extended introduction to the topics underlying the research presented in the second part and as a guide to the relevant literature. Chapter 2 introduces the framework of tensor networks, which constitute the primary technical tool employed throughout this thesis. Chapter 3 provides an overview of two-dimensional non-chiral topological orders realised by string-net models and discusses their tensor network representations. Finally, chapter 4 is devoted to the study of the corresponding one-lower-dimensional, viz. (1+1)-dimensional, *edge theories* and their inherited *generalised symmetry structures*, with an outlook towards the (2+1)-dimensional setting.

The thesis concludes with a summary of the main results, together with some general remarks and perspectives for future research.



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# Foundations



# Tensor networks

Having established the importance of the *area law* for gapped quantum many-body systems, we now introduce *tensor networks* as the main tool used throughout this thesis. In brief, tensor networks are a class of many-body trial wave functions that, by construction, satisfy an area law for the entanglement entropy with respect to any bipartition. Parametrised by a number of variational parameters that only scales as a *polynomial* in the system size, they also provide an exponentially more compressed description of the many-body wave functions of interest.

At their core, tensor networks bring a holographic approach to the many-body problem in which the information of the wave function is encapsulated in local constituent tensors encoding both the *physical* and *entanglement* degrees of freedom. All the information — local quantities such as the energy density, as well as non-local ones such as the correlation functions — is stored locally into these tensors. Moreover, the parametrisation of states and operators in terms of tensor networks also allows one to formulate them directly in the *thermodynamic limit*.

Crucial for our purposes, all symmetries of a given model, both *spacetime* symmetries and *internal* symmetries, can typically be easily incorporated in tensor network ansätze. Enforcing symmetry significantly improves both the stability and implementation of numerical algorithms. Moreover, it facilitates the classification of gapped phases of matter. Indeed, in the language of tensor networks, phase classification often effectively boils down to classifying the inequivalent symmetry actions on the constituent tensors. But there is more: in recent years it has been understood that even in the absence of any physical symmetry, distinct *intrinsically topologically ordered* phases may arise that can be characterised by the symmetry structure of the *entanglement*

*patterns* of the ground states. In this way, tensor networks reveal the *hidden topological* symmetries that characterise the long-range entanglement present in these gapped phases. The algebraic structure of these virtual symmetries is naturally described by a *(higher-)fusion category*, and is inherited by their one-lower-dimensional *edge theories*, which themselves may be gapped, gapless or topologically ordered. The description of the *virtual MPO symmetries* of *two-dimensional* non-chiral topological orders and their edge Hamiltonians is the content of chapter 3 and chapter 4 respectively.

The goal of this chapter is to provide a concise introduction to tensor network states and operators, and review the classification of gapped one-dimensional bosonic of phases of matter. After sketching the *projected entangled-pair construction* of quantum many-body states with an area law for their entanglement entropy, we introduce the widely used graphical tensor network calculus. Next, we zoom in on the one-dimensional tensor network states, called *matrix product states*, and formulate the *fundamental theorem of MPS* which provides necessary and sufficient conditions for two MPS tensors to generate the same quantum state. We also briefly introduce *matrix product operators*, as tensor network representations of linear operators acting on one-dimensional lattices. We comment on the two-dimensional *projected entangled-pair states* and *operators*. In conclusion we review in detail the classification of gapped one-dimensional phases. It will be shown that in the presence of an on-site symmetry group  $G$ , phases are classified by pairs  $(H \subseteq G, \psi)$ , where the subgroup  $H$  defines the pattern of symmetry breaking  $G \mapsto H \subseteq G$ , and  $\psi$  is a *2-cocycle* of  $G$  that is associated to the *projective* symmetry action on the virtual degrees of freedom of the ground space matrix product state.

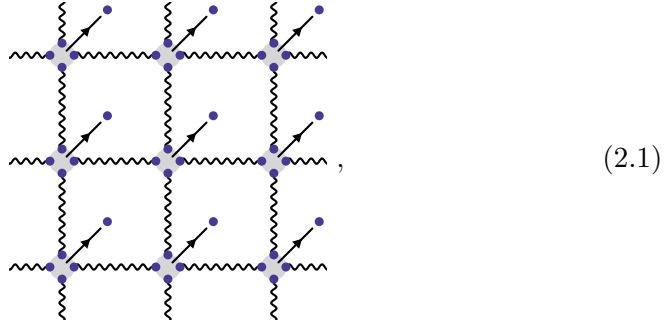
## 2.1 Tensor network states

The main lesson of the area law is that distribution of entanglement happens via the boundary of spatial regions. Trial wave functions whose entanglement entropy exhibits such a pattern, and at the same time respect all lattice symmetries can be constructed via a *projected entangled-pair construction*, as advocated in ref. [34]. In this framework, maximally entangled pairs of *virtual* spins are first distributed between all pairs of neighbouring spins according to the connectivity of the lattice. Subsequently, one ‘*projects*’<sup>1</sup> onto the physical degrees of freedom of the model by means of a linear *tensor map* on each site. These tensors contain a number of degrees of freedom which play the role of

<sup>1</sup>The local tensors are usually not idempotent linear maps.



787 *variational parameters* of the quantum state they generate. Schematically,  
 788 this construction can be visualised on the two-dimensional square lattice as:



789 where the nodes connected by wiggly lines represent the virtual pairs, and the  
 790 arrows denote the local linear projection maps. The *virtual* or *entanglement*  
 791 degrees of freedom are associated to a Hilbert space  $\mathbb{C}^\chi$ , where  $\chi$  is called  
 792 the *bond dimension*. Though illustrated here for the two-dimensional square  
 793 lattice, this construction generalises straightforwardly to any dimension,  
 794 lattice geometry, choice of boundary conditions, or assignment of different  
 795 tensors on each site. By choosing the tensor identical on every site, this  
 796 construction provides an explicit way to construct generic quantum states  
 797 with area law entanglement that enjoy all lattice symmetries.

798 The area law is manifest in this construction, since the entanglement  
 799 entropy across the boundary of any contiguous region of spins is upper  
 800 bounded by  $\log \chi$  times the number of virtual pairs that cross the boundary,  
 801 or thus the surface area of the boundary of that region. This will be made  
 802 more precise in what follows for the one-dimensional case.

803 Throughout this thesis, we use a graphical calculus in which tensors are repre-  
 804 sented by nodes with a number of legs representing their indices. Concretely,  
 805 a rank-5 tensor  $A$  can be expressed as:

$$\begin{aligned}
 \text{Diagram of tensor } A \text{ with 5 legs} &= \sum_{\substack{i,j,k \\ l,m}} j \text{ --- } \text{Diagram of tensor } A \text{ with 5 legs} \text{ --- } l \, |i, j, k\rangle\langle l, m| \\
 &= \sum_{\substack{i,j,k \\ l,m}} A_{jklm}^i |i, j, k\rangle\langle l, m|
 \end{aligned}
 \tag{2.2}$$

806 We adopt the graphical convention that connected legs between tensors stand  
 807 for a sum over the corresponding indices, in which case it is said that the

legs are *contracted*. Using this calculus, the state schematically represented  
in eq. (2.1) can be written more explicitly as



810 In this figure, and similar figures below, it is understood that the tensor  
811 networks extend to infinity in all lattice directions, unless specified otherwise.

## 812 2.2 Matrix product states

In one dimension, tensor network states boil down to the class of *matrix product states* (MPS) [35]. In the translation invariant case, they can be parametrised directly in the thermodynamic limit by a single rank-3 tensor  $A \in \mathbb{C}^{d \times \chi^2}$  as:

$$|\Psi[A]\rangle = \text{---} \bigcirc_A \text{---} \bigcirc_A \text{---} \dots \text{---} \bigcirc_A \text{---} \bigcirc_A \text{---} . \quad (2.4)$$

A matrix product state on a finite chain with boundary conditions specified by vectors  $|v_L\rangle, |v_R\rangle$  is then defined as

$$|\Psi[A], v_L, v_R\rangle = \langle v_L| \begin{array}{c} | \\ \textcircled{A} \\ | \end{array} \begin{array}{c} | \\ \textcircled{A} \\ | \end{array} \dots \begin{array}{c} | \\ \textcircled{A} \\ | \end{array} \begin{array}{c} | \\ \textcircled{A} \\ | \end{array} |v_R\rangle . \quad (2.5)$$

As discussed below, when the MPS describes the ground space of a *symmetry-protected topological phase*, the virtual spaces near the boundaries provide a natural basis for the symmetry-protected *edge modes*.

Continuing along these lines, we can define MPSs with larger unit cells or with distinct tensors  $A[i]$  on every site. Notably, every state in the many-body Hilbert space of a finite chain can be represented by an MPS with a bond dimension that scales exponentially in the system size. To every matrix product state one can also associate a *gapped* and *frustration-free parent Hamiltonian* for which the MPS is a ground state. When the MPS has a certain symmetry described by a group  $G$ , the parent Hamiltonian can be symmetrised so as to commute with  $G$ . We refer to this as the *symmetric parent Hamiltonian*. Moreover, for a generic MPS it is the unique ground state of its parent Hamiltonian, as detailed below.



863 *matrix* defined as

$$\mathbb{T}_A := \sum_i A^i \otimes \bar{A}^i = \begin{array}{c} \textcircled{\bar{A}} \\ | \\ \textcircled{A} \end{array}, \quad (2.9)$$

864 and which is typically interpreted as a matrix from the right to the left (virtual)  
 865 indices. A generic MPS has the property that  $\mathbb{T}_A$  has a unique dominant  
 866 eigenvalue  $\lambda_1$  by virtue of the *quantum Frobenius-Perron theorem* [37]. By  
 867 rescaling  $A^i \mapsto A^i/\sqrt{\lambda_1}$ , this eigenvalue can be made 1. Its corresponding  
 868 left (right) eigenvectors are denoted by  $\Lambda$  ( $\rho$ ), both are full rank and can be  
 869 made positive. Furthermore, they can be normalised to  $\text{Tr}(\Lambda\rho) = 1$ :

$$\begin{array}{c} \textcircled{\Lambda} \quad \textcircled{\rho} \\ | \quad | \\ \textcircled{\Lambda} \quad \textcircled{\rho} \end{array} = 1, \quad (2.10)$$

870 which ensures the normalisation of the state in the thermodynamic limit, i.e.  
 871  $\langle \Psi[A] | \Psi[A] \rangle = 1$ .

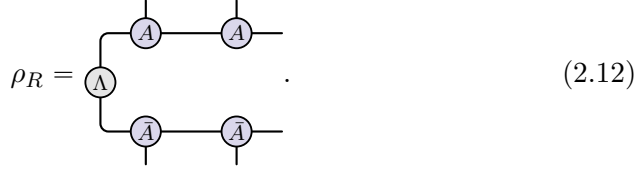
872 Similarly, we can express the two-point correlation function of local  
 873 operators  $O_1$  and  $O_2$  separated by a distance  $L$  as:

$$\langle O_1[1] O_2[L+1] \rangle_{|\Psi[A]\rangle} = \begin{array}{c} \textcircled{\bar{A}} \quad \textcircled{\bar{A}} \\ | \quad | \\ \textcircled{\Lambda} \quad \boxed{O_1} \quad \textcircled{A} \\ | \quad | \\ \textcircled{A} \quad \textcircled{A} \end{array} \quad \dots \quad \begin{array}{c} \textcircled{\bar{A}} \quad \textcircled{\bar{A}} \\ | \quad | \\ \textcircled{A} \quad \boxed{O_2} \quad \textcircled{\rho} \\ | \quad | \\ \textcircled{A} \quad \textcircled{A} \end{array}. \quad (2.11)$$

874 By exploiting the tensor network structure of the transfer matrix, this cor-  
 875 relation function can be computed with a computational cost that scales as  
 876  $\mathcal{O}(L\chi^3)$ . Subtracting from eq. (2.11) the disconnected part  $\langle O_1 \rangle_{\Psi[A]} \langle O_2 \rangle_{\Psi[A]}$   
 877 and invoking the eigendecomposition of  $\mathbb{T}_A$  reveals that the dominant decay  
 878 is exponential, i.e.  $\sim \exp(-L/\xi)$ , where the *correlation length*  $\xi$  is related to  
 879 the second largest eigenvalue  $\lambda_2$  of  $\mathbb{T}_A$  via  $\xi = -1/\log(|\lambda_2|)$ .

880 The uniqueness of the dominant eigenvalue of the transfer matrix is a  
 881 property called *injectivity*. As mentioned above this is a generic property  
 882 and a useful equivalent definition is the following. An MPS is injective if  
 883 there exists a finite *injectivity length*  $L$  such that the collection of matrices  
 884  $\{A^{i_1} A^{i_2} \dots A^{i_L}\}$  constitutes a basis for the  $\chi \times \chi$  matrix algebra. Injective  
 885 MPSs have the properties that the implicit choice of boundary conditions  
 886 in eq. (2.4) in the thermodynamic limit is immaterial and that they are the  
 887 *unique* ground state of their corresponding parent Hamiltonian. Below, we  
 888 will interpret non-injective matrix product states as exhibiting spontaneous  
 889 symmetry-breaking. As such they naturally show up in the classification of  
 890 gapped phases of matter.

As argued above, MPSs satisfy an area law for the entanglement entropy across any bipartition [7, 12, 38]. To make this precise in the case of a uniform MPS, notice that across any entanglement cut, the reduced density matrix with respect to the right-half chain can be depicted as:



$$\rho_R = \begin{array}{c} \text{---} \text{A} \text{---} \text{A} \text{---} \\ | \quad | \\ \text{A} \\ | \quad | \\ \text{---} \text{A} \text{---} \text{A} \text{---} \end{array} . \quad (2.12)$$

This demonstrates that the reduced density matrix  $\rho_R$  can be interpreted as the product of a  $d^L \times \chi$ -dimensional matrix with a  $\chi \times d^L$ -dimensional one, where  $L$  is formally the number of spins to the right of the cut. Hence,  $\text{rank}(\rho_R) \leq \chi$ . This means that the entanglement entropy is upper bounded independent of the size of the right subsystem by  $S_L \leq \log \chi$ , which is the maximal amount of entanglement entropy for a state with Schmidt rank  $\chi$ .

Central to the classification of symmetric gapped phases is the following observation. Two MPSs  $|\Psi[A]\rangle$  and  $|\Psi[B]\rangle$  whose MPS matrices are related via a gauge transformation eq. (2.6) obviously generate the same state. In virtue of the *fundamental theorem of MPS* the converse also holds generically. More precisely, this theorem states that two uniform *injective* matrix product states with periodic boundary conditions represent the same state independent of the system size, if and only if the MPS tensors are related by a non-singular gauge transformation:

$$|\Psi[A]\rangle \sim |\Psi[B]\rangle \iff A^i = e^{i\chi} X B^i X^{-1}, \quad \forall i. \quad (2.13)$$

Despite its relevance, the proof of the fundamental theorem is surprisingly simple and can be formulated as a straightforward application of the Cauchy-Schwarz inequality for which we refer to e.g. ref. [39].

Importantly, two non-injective periodic MPSs can generate the same state while not simply being related by a gauge transformation eq. (2.6). Consider an MPS tensor which in a certain basis reads

$$A^i = \begin{pmatrix} A_1^i & A_o^i \\ 0 & A_2^i \end{pmatrix}, \quad \forall i. \quad (2.14)$$

It follows directly from the block structure of this tensor that for any choice of the off-diagonal block  $A_o^i$ ,  $A^i$  generates the same state on periodic boundary conditions as the trace of the matrices  $A^i$  is independent of  $A_o^i$ . The upper

triangular block structure of the tensor signals the presence of an invariant subspace  $W_1$ , i.e.  $A^i W_1 \subseteq W_1, \forall i$ . Writing  $\mathcal{P}_{W_1}$  ( $\mathcal{P}_{W_1^\perp}$ ) for the projector onto  $W_1$  (and its orthogonal complement  $W_1^\perp$ ), we can thus replace

$$A^i \mapsto A_1^i \oplus A_2^i = \mathcal{P}_{W_1} A^i \mathcal{P}_{W_1} + \mathcal{P}_{W_1^\perp} A^i \mathcal{P}_{W_1^\perp} \quad (2.15)$$

so as to bring  $A_i$  in block diagonal form. By repeating this procedure a finite number of times we can thus bring any MPS tensor in the form

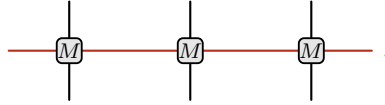
$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i, \quad (2.16)$$

where all  $\{A_k^i\}_k$  are injective, and are referred to as the *injective blocks*. Such an MPS is said to be in *canonical form* [35].

Linear operators in turn can naturally be expressed in terms of *matrix product operators* (MPOs) [40, 41]. In full, these are written as

$$\sum_{\{i_k, j_k\}} (\dots M^{i_k, j_k} M^{i_{k+1}, j_{k+1}} \dots) |\dots, i_k, i_{k+1}, \dots \rangle \langle \dots, j_k, j_{k+1}, \dots|, \quad (2.17)$$

and can graphically be depicted as



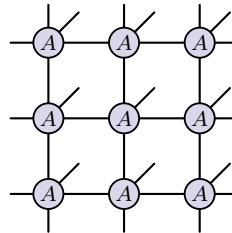
$$\quad (2.18)$$

A simple example is the pure density matrix  $\rho = |\Psi[A]\rangle \langle \Psi[A]|$  with MPO tensor  $M^{ij} = A^i \otimes \bar{A}^j$ .

Similar to the case of MPS, an MPO is called *injective* when there exists a finite length for which the MPO matrices span the full matrix algebra.

## 2.3 Projected entangled-pair states

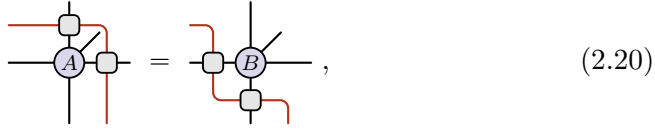
The natural generalisation of matrix product states to two dimensions is in terms of *projected entangled-pair states* (PEPSs) [42]. In the uniform case, we can parametrise a PEPS on a square lattice in terms of a single rank-5 tensor as follows:



$$|\Psi[A]\rangle = \quad (2.19)$$

Unlike the case of MPS, the contraction of a generic PEPS is computationally significantly more costly [43]. To tackle this issue, approximate contraction schemes have been and are being developed and used with success in variational ground state optimisation [44–46].

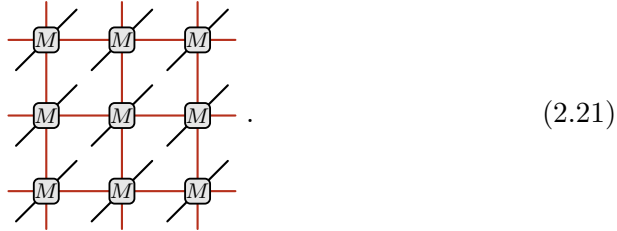
A further distinction from the one-dimensional case is the absence of a completely general fundamental theorem of PEPS that provides necessary and sufficient conditions for two PEPS tensors to generate the same wave function. For many systems of interest — and in particular the *string-net* ground states that are the topic of chapter 3 — PEPS tensors generating the same state  $|\Psi[A]\rangle \sim |\Psi[B]\rangle$  are intertwined via an MPO living purely on the virtual level obeying the *pulling-through* equation:



$$(2.20)$$

together with the analogous relations in all other orientations.

Finally, analogous to matrix product operators, *projected entangled-pair operators* are tensor network representations of linear operators in two dimensions:



$$(2.21)$$

## 2.4 Classification of gapped one-dimensional phases

Recall that in the classical Landau paradigm phases of matter are classified by their symmetry-breaking pattern  $G \rightarrow H \subseteq G$  [24, 32]. Additional phases arise in the quantum mechanical setting since the entanglement degrees of freedom can transform in discrete distinct ways under the residual symmetry group  $H$ . Concretely, for one-dimensional *bosonic* systems, these classes are organised in the *second cohomology group* of  $H$  over  $U(1)$ , denoted by  $H^2(H, U(1))$  and defined below. Notably, for finite  $H$  the second cohomology group  $H^2(H, U(1))$  is itself finite<sup>2</sup>, so that the number of gapped phases with

<sup>2</sup>Even though the second cohomology group is also finite for some Lie groups such as  $H^2(SO(3), U(1)) \simeq \mathbb{Z}_2$ , this is not generically true, not even for compact groups.

residual  $H$  symmetry is also finite. One of the great achievements of tensor networks and in particular MPS, is that this one-dimensional classification can be completely carried out and understood in that framework, also in the fermionic case [27, 28, 47]. Let us also note at this point that for one-dimensional bosonic systems, there exist no non-trivial phases without imposing a symmetry. This is different in the fermionic case, due to the existence of a fermionic superselection rule.

This classification rests on two pillars. The first is the fact that MPS provide a *faithful* representation of gapped ground spaces [7]; the second is the (symmetric) parent Hamiltonian construction of MPS [48]. Indeed, as argued above, two Hamiltonians  $H_1, H_2$  with the same symmetry belong to the same phase if and only if there exists a smooth interpolating Hamiltonian path starting in  $H_1$  and ending in  $H_2$ . Assuming an exact MPS representation of the ground space of both Hamiltonians, one can restate this as the search for an interpolating path of the *symmetric parent* Hamiltonians. Equivalently, one can rephrase this problem as the search for a smooth interpolating MPS path.

Let us for a moment assume that the on-site symmetry  $G$  is the full symmetry group of the model; we return to the case where  $G$  is a subsymmetry below. In that case, the ground state corresponds to a non-injective MPS,  $A^i = \bigoplus_{k=1}^r A_k^i$ , with each of the blocks generating an injective MPS. In a certain basis, the symmetry acts on the virtual level of the MPS via the *induced* representation

$$P_g \bigoplus_{k=1}^r (U_k(h) \otimes \bar{U}_k(h)), \quad \forall g \in G, \quad (2.22)$$

where  $h \equiv h_{g,r_k}$  depends on  $g$  and  $k$  as specified below. The  $\{U_k(h)\}_h$  are unitary matrices acting on the block  $k$ . The representation  $P_g$  permutes the injective blocks, and it is assumed that this permutation acts *transitively*. If we denote by  $\pi_g$  the action of  $P_g$  on the labels  $k$ , transitivity means that for every two blocks  $k_1, k_2$  there exists some  $g \in G$  such that  $\pi_g(k_1) = k_2$ .

We subsequently choose an arbitrary but fixed *reference* block denoted by  $k_0$ . This thereby specifies the *stabiliser* subgroup  $H \subseteq G$  defined as:

$$H := \{g \in G \mid \pi_g(k_0) = k_0\}. \quad (2.23)$$

Note that choosing a different reference block  $k'_0$  would result in a stabiliser group *conjugate*, and thus *isomorphic*, to  $H$ .

From the *orbit-stabiliser theorem*, combined with transitivity of  $P_g$ , it follows that the number of blocks is given by  $|G/H|$ . Let  $r_k$ , with  $k =$



994  $1, 2, \dots, r = |G/H|$ , denote a set of representatives of the cosets  $G/H$ , with  
 995  $r_{k_0} = \mathbb{1}$ . The group element  $h_{g,r_k} \in H$  is then defined via:

$$gr_k = r_{k'} h_{g,r_k}. \quad (2.24)$$

996 In fact, it can be shown that the  $U_k(h)$ 's furnish a *projective* representation  
 997 of  $H$ , and the corresponding *topological index* is precisely what encodes the  
 998 SPT phase of the surviving symmetry. We relegate the discussion of this to  
 999 the next paragraph.

1000 As anticipated above,  $G$  could be a subsymmetry of the full symmetry of  
 1001 the model. In that case, the MPS could exhibit an additional block structure,  
 1002 viz.  $A^i = \bigoplus_K \bigoplus_k A_k^i$ , signalling the breaking of the larger symmetry. This  
 1003 degeneracy is however not protected by the symmetry  $G$ , meaning it can  
 1004 be lifted by perturbing the Hamiltonian with symmetric terms. Altogether,  
 1005 this makes precise the above statement that non-injectivity of an MPS is the  
 1006 tensor network manifestation of symmetry-breaking, and clarifies the role of  
 1007 the subgroup  $H \subseteq G$ .

1008 We are left to explain the distinct ways an injective MPS can transform under  
 1009 a symmetry group, viz.:

$$U(g)^{\otimes N} |\Psi[A]\rangle \simeq |\Psi[A]\rangle, \quad \forall g \in G. \quad (2.25)$$

1010 We use here  $G$  to denote the symmetry group of the MPS  $|\Psi[A]\rangle$ , but this  
 1011 could be the residual symmetry in the ground state subspace. A direct  
 1012 application of the fundamental theorem eq. (2.13) reveals that the MPS  
 1013 tensor itself has to transform, for all  $g \in G$ , according to:

$$\sum_j U(g)_{ij} A^j = e^{i\chi_g} X_g A^i X_g^\dagger \quad (2.26)$$

1014 for unitary gauge matrices  $\{X_g | g \in G\}$  and complex phases  $\{e^{i\chi_g} | g \in G\}$   
 1015 that may depend on the group element  $g$ . We can now act with two distinct  
 1016 group elements  $g, h$  on the MPS in two different ways: by acting with  $U(gh)$   
 1017 directly and invoking the fundamental theorem once as in eq. (2.26) or by  
 1018 sequentially acting with  $g$  and  $h$  and making use of eq. (2.26) twice. Equating  
 1019 both yields:

$$e^{i\chi_{gh}} X_{gh} A^i X_{gh}^\dagger = e^{i(\chi_g + \chi_h)} X_g X_h A^i X_h^\dagger X_g^\dagger, \quad \forall i, g, h. \quad (2.27)$$

1020 Injectivity of the MPS subsequently implies that  $\{e^{i\chi_g} | g \in G\}$  constitutes  
 1021 a one-dimensional representation of  $G$ , whereas the  $X_g$ 's form a *projective*

representation of  $G$ , i.e. they multiply according to  $X_g X_h = \psi(g, h) X_{gh}$  for  $U(1)$  phases  $\psi(g, h)$ . Imposing associativity of the matrix multiplication constrains these phases and shows that they adhere to a consistency condition known as the *2-cocycle condition*:

$$\psi(g, h)\psi(gh, k) = \psi(g, hk)\psi(h, k), \quad \forall g, h, k \in G. \quad (2.28)$$

Solutions of eq. (2.28) form an abelian group  $Z^2(G, U(1))$  under pointwise multiplication. Among these, the trivial solutions, or *coboundaries*, take the form  $\psi(g, h) = \frac{\phi_{gh}}{\phi_g \phi_h}$ , for any function  $\phi : G \rightarrow U(1)$ . In the jargon of group cohomology such a trivial solution is referred to as a *coboundary* and they constitute a normal subgroup  $B^2(G, U(1)) \trianglelefteq Z^2(G, U(1))$ . Notably, rescaling of the gauge matrices with arbitrary phases  $X_g \mapsto \phi_g X_g$  maps the corresponding 2-cocycle to one differing by the coboundary  $\phi_g$ , leading to a physical redundancy. Therefore, one *quotients by* such solutions in solving eq. (2.28), leading to equivalence classes of 2-cocycles. Two 2-cocycles  $\psi \sim \psi'$  are thus considered cohomologically equivalent if they are related via a coboundary and these equivalence classes form the quotient group  $H^2(G, U(1)) := Z^2(G, U(1))/B^2(G, U(1))$  which is the anticipated *second cohomology group* of  $G$  over  $U(1)$ .

An MPS that transforms projectively on the virtual level as per eq. (2.26) is characterised by *symmetry-protected degeneracies* in its entanglement spectrum, as well as the existence of localised *edge modes* on open chains [49, 50]. The existence of both follows from the fact that, contrary to linear representations, there exist *no* cohomologically non-trivial *one-dimensional* projective representations. As an illustrative example, consider the  $SO(3)$ -symmetric ground state of the antiferromagnetic spin 1 Heisenberg model. Since the *projective* representations of  $SO(3)$  can be identified with the *half-integer linear* representations of its universal cover,  $SU(2)$ , the entanglement spectrum of the ground state is organised in multiplets of *even* dimension. This is demonstrated in fig. 2.1 where the first 48 Schmidt values of the ground state are plotted.

As anticipated above, a natural basis for the edge modes is given by the virtual degrees of freedom of the MPS ground state near the boundaries. Indeed, a uniform MPS on open boundary conditions,

$$|\Psi[A]\rangle = \sum_{\{i_j\}} (A^{i_1} A^{i_2} \dots A^{i_N})_{\alpha\beta} |i_1, i_2, \dots, i_N\rangle, \quad (2.29)$$

corresponds to  $\chi^2$  states  $|\Psi[A], \alpha, \beta\rangle$  that have the same energy for the parent Hamiltonian. Naively, we would thus count  $\chi^2$  ground states on the open

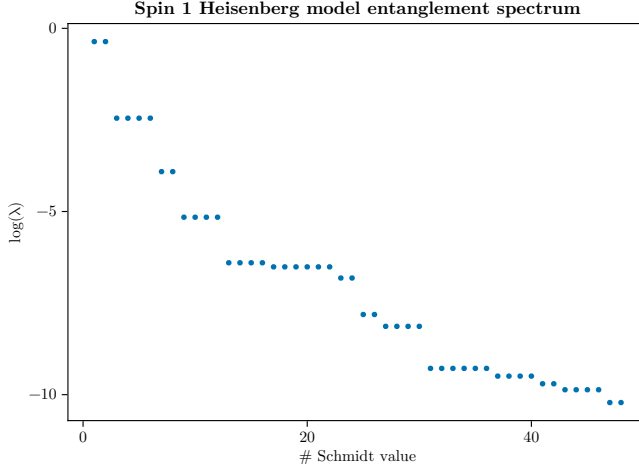


Figure 2.1: The entanglement spectrum of the spin 1 Heisenberg model ground state in the thermodynamic limit with bond dimension 48. Created with MPSKit.jl [51].

chain, but some of these modes can be gapped out by turning on additional terms in the Hamiltonian. However, as long as these additional terms respect the symmetry, a certain amount of degeneracy is persistent; these edge modes transform under the symmetry action according to

$$U(g)^{\otimes N} |\Psi[A], \alpha, \beta\rangle = \sum_{\alpha', \beta'} (X_g)_{\alpha\alpha'} (\bar{X}_g)_{\beta\beta'} |\Psi[A], \alpha', \beta'\rangle, \quad (2.30)$$

so that at least a two-fold degeneracy persists.

This concludes phase classification in the one-dimensional setting where the symmetry acts via a unitary (tensor product) group representation. To summarise, phases are classified by pairs  $(H, [\psi])$  where  $H \subseteq G$  is a subgroup up to conjugacy and  $[\psi]$  is a class taking values in  $H^2(H, \text{U}(1))$  [27, 28].

Remarkably, these data also classify (indecomposable) *module categories*  $\mathcal{R}(H, \psi)$  over the *fusion category*  $\text{Vec}_G$  [52, 53]. *Simple objects* of the corresponding module category are labelled by cosets  $\{r_1 H, r_2 H, \dots\}$  and  $\mathcal{R}(H, \psi)$  carries a natural  $\text{Vec}_G$  action induced by  $g \triangleright rH := (gr)H$ . This observation suggests the physical interpretation where the gapped ground states are identified with the simple objects, and the action  $\triangleright$  with the action of the on-site  $G$  symmetry on the ground state subspace.

This observation is more than a formal analogy and is the first indication of a correspondence between module categories and gapped phases of matter. The *generalised Landau paradigm*, the subject of section 4.3, stipulates

1075 that this correspondence persists to one-dimensional (lattice) models with a  
1076 symmetry described by a generic fusion category  $\mathcal{C}$  [54–57]. More precisely,  
1077 it states that the different gapped phases are given by the different module  
1078 categories  $\mathcal{P}$  over  $\mathcal{C}$ , with the simple objects labelling the ground states and  
1079 the *module action* describing the symmetry action on them.

# Non-chiral topological order: string-nets and categories

In the previous chapter we outlined the classification of one-dimensional bosonic gapped phases of matter, which required the presence of a symmetry to protect a non-trivial phase. This radically changes in one dimension higher since even in the absence of any symmetries, gapped *intrinsic topological phases* arise. These phases are characterised by a *robust ground state degeneracy* that depends only on the topology of the underlying spatial manifold, *anyonic* quasiparticle excitations and in some cases *edge modes*.

In this chapter we focus on *non-chiral* topological phases that admit *gapped boundary conditions*. It is believed that these phases are fully characterised by their anyon content, corresponding fusion rules and braiding statistics. This brings us into the realm of *category theory* since these data are naturally encapsulated in a *modular fusion category*, specifically the *Drinfeld* or *monoidal centre*  $\mathcal{Z}(\mathcal{D})$  of a generically non-braided *input* fusion category  $\mathcal{D}$ .

Importantly, every such topological order admits a lattice realisation directly in terms of this input  $\mathcal{D}$ . This is precisely the *Levin-Wen string-net condensation model* that provides a frustration-free Hamiltonian whose torus ground state sectors and excitation spectrum exactly reproduce  $\mathcal{Z}(\mathcal{D})$ . From the point-of-view of tensor networks, these string-net ground states admit particularly elegant PEPS representations characterised by the existence of MPO symmetries that live purely on the *virtual* level, form a representation of a *Morita equivalent* fusion category  $\mathcal{C}$  and that can be continuously deformed through the lattice. Consequently, these MPO symmetries survive small physical perturbations and characterise the long-range entanglement structure.

In this chapter, we review the string-net models and their PEPS representations. Significant attention is devoted to the classification and construction of their gapped boundaries. Beyond being interesting in their own right, the data characterising the gapped boundaries also classify *gapped phases* of the string-net boundary Hamiltonians that inherit the (non-invertible) symmetries of the bulk topological order. This is the content of the *generalised Landau paradigm* and is treated in the next chapter.

## 3.1 Fusion categories for topological orders

The description of non-chiral topological orders and their realisation in terms of string-net models requires the language of *fusion categories* and *modular fusion categories*. Whereas the former is used to define the microscopic Hilbert space and Hamiltonian of the string-net models, the exchange statistics of the anyonic excitations are described by the *non-degenerate, modular, braiding* encapsulated in the latter.

In what follows we will provide the necessary aspects of (braided) fusion categories, and thereby introduce their *graphical calculus*. We invite the reader to consult the literature for a more precise and more complete technical treatment of these topics [53, 58–60].

It should be kept in mind that even though fusion categories are strictly richer structures than finite groups and their representation theory, they may in fact be regarded as a natural generalisation of both. Indeed, much of the structure of a fusion category can be understood as the categorical counterpart of familiar concepts from group and representation theory, a perspective that we illustrate in detail below. As a result, fusion categories provide a unified framework in which conventional symmetries encoded in groups and *topological, generalised* symmetries can be treated on equal footing.

### 3.1.1 Algebraic structure of fusion categories

#### Simple objects and fusion

Succinctly, a *unitary fusion (1-)category* (UFC)  $\mathcal{C}$  consists of a collection of objects  $\text{ob}(\mathcal{C})$  to which we will refer as *topological defects, topological lines, sectors* or *charges*, depending on the context. Every  $X \in \text{ob}(\mathcal{C})$  admits a decomposition as a direct sum of finitely many *simple objects*  $\mathcal{I}_{\mathcal{C}} = \{X_i\}_i$ , i.e.:

$$X = \bigoplus_{X' \in \mathcal{I}_{\mathcal{C}}} N_X^{X'} X', \quad (3.1)$$

for non-negative integers  $N_X^{X'}$  counting the multiplicity of each direct summand. The decomposition is unique for every object  $X$ , up to a permutation of the factors. The set  $\text{ob}(\mathcal{C})$  is endowed with a *monoidal* or *tensor product*  $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$  encoding the *fusion* of defects. We will express this product in terms of an integer rank-3 tensor  $N$  via:

$$X_1 \otimes X_2 = \bigoplus_{X_3 \in \mathcal{I}_{\mathcal{C}}} N_{X_1 X_2}^{X_3} X_3, \quad (3.2)$$

for simple objects  $X_1, X_2$ . These fusion rules are *associative* so that for any four simple objects we have that  $\sum_X N_{X_1 X_2}^X N_{X X_3}^{X_4} = \sum_X N_{X_1 X}^{X_4} N_{X_2 X_3}^X$ . Generically  $N_{X_1 X_2}^{X_3} \neq N_{X_2 X_1}^{X_3}$ . Moreover, since the tensor product  $\otimes$  distributes over the direct sum  $\bigoplus$ , the product of all objects is defined fully by eq. (3.2). Among the simple objects, there exists a unique *monoidal unit* notated as  $\mathbb{1}_{\mathcal{C}}$  or simply  $\mathbb{1}$  that satisfies  $\mathbb{1} \otimes X = X \otimes \mathbb{1} = X$ ,  $\forall X \in \text{ob}(\mathcal{C})$ . To every object  $X$  there corresponds a unique *dual*  $X^*$  characterised by  $N_{X X^*}^{\mathbb{1}} = 1$ .

Below, we also make use of the *quantum dimensions*  $d_X$  of the simple objects which are completely specified by

$$d_X > 0, \quad d_{X_1} d_{X_2} = \sum_{X_3 \in \mathcal{I}_{\mathcal{C}}} N_{X_1 X_2}^{X_3} d_{X_3}. \quad (3.3)$$

In particular we have  $d_{\mathbb{1}} = 1$  and  $d_X = d_{X^*}$ , for all  $X \in \text{ob}(\mathcal{C})$ . Also, the quantum dimensions are additive over the direct sum, so that  $d_X = \sum_{X' \in \mathcal{I}_{\mathcal{C}}} N_X^{X'} d_{X'}$  for any  $X \in \text{ob}(\mathcal{C})$ .

### Graphical calculus

The different ways in which simple objects  $X_1$  and  $X_2$  can fuse to  $X_3$  is captured by the *fusion space*  $\mathcal{C}(X_1 \otimes X_2, X_3) \simeq \mathbb{C}^{N_{X_1 X_2}^{X_3}}$  for which we use the shorthand  $\mathcal{C}_{X_3}^{X_1 X_2}$ . In category theoretic parlance, this Hilbert space for every triple of objects is referred to as *Hom*, *fusion* or *junction space*. Notably, for simple objects  $X_1, X_2$  we have  $\mathcal{C}(X_1, X_2) = \delta_{X_1, X_2} \mathbb{C}$ . Basis vectors are notated as  $|X_1 X_2 X_3; i\rangle \in \mathcal{C}_{X_3}^{X_1 X_2}$  with  $i = 1, 2, \dots, N_{X_1 X_2}^{X_3}$ . These vectors and their duals can be depicted in terms of strings diagrams as follows:

$$|X_1 X_2 X_3; i\rangle \equiv \begin{array}{c} X_1 \quad X_2 \\ \diagdown \quad \diagup \\ \textcircled{i} \\ \diagup \\ X_3 \end{array}, \quad \langle X_1 X_2 X_3; i| \equiv \begin{array}{c} X_3 \\ \diagup \\ \textcircled{i} \\ \diagdown \quad \diagup \\ X_1 \quad X_2 \end{array}. \quad (3.4)$$

Here and below, all string diagrams are implicitly oriented downwards. We adopt a normalisation of these basis vectors such that the following graphical identity holds:

$$\left| \begin{array}{c} X_3 \\ \text{---} i \\ \text{---} j \\ X'_3 \end{array} \right\rangle = \delta_{X_3, X'_3} \delta_{i,j} \sqrt{\frac{d_{X_1} d_{X_2}}{d_{X_3}}} \left| \begin{array}{c} X_3 \\ \text{---} \\ \text{---} \\ X_3 \end{array} \right\rangle. \quad (3.5)$$

Note that the basis vectors are not normalised to one. Indeed the inner product of basis vectors can be depicted as:

$$\langle X_1 X_2 X_3; i | X_1 X_2 X_3; j \rangle = \left| \begin{array}{c} X_3 \\ \text{---} i \\ \text{---} j \\ X_3 \end{array} \right\rangle = \delta_{i,j} \sqrt{d_{X_1} d_{X_2} d_{X_3}}, \quad (3.6)$$

which for the particular case of  $X_1 = \mathbb{1}$ ,  $X_2 = X_3 \equiv X$  specialises to

$$\left| \begin{array}{c} X \\ \text{---} \\ \text{---} \end{array} \right\rangle = d_X. \quad (3.7)$$

This normalisation is referred to as the *isotopy invariant* normalisation convention, but other conventions are used throughout the literature [59]. In this normalisation, the completeness of the basis then boils down to:

$$\left| \begin{array}{c} X_1 \\ \text{---} \\ \text{---} \end{array} \right\rangle \left| \begin{array}{c} X_2 \\ \text{---} \\ \text{---} \end{array} \right\rangle = \sum_{X_3, i} \sqrt{\frac{d_{X_3}}{d_{X_1} d_{X_2}}} \left| \begin{array}{c} X_1 \quad X_2 \\ \text{---} i \\ \text{---} i \\ X_1 \quad X_2 \end{array} \right\rangle. \quad (3.8)$$

Consider now the fusion of three simple objects. In virtue of the associativity of the fusion rules, the associated Hom space capturing the fusion of these three objects decomposes in two distinct but isomorphic ways as follows:

$$\mathcal{C}_{X_4}^{X_1 X_2 X_3} \simeq \bigoplus_{X_5} \mathcal{C}_{X_5}^{X_1 X_2} \otimes \mathcal{C}_{X_4}^{X_5 X_3} \simeq \bigoplus_{X_6} \mathcal{C}_{X_4}^{X_1 X_6} \otimes \mathcal{C}_{X_6}^{X_2 X_3}, \quad (3.9)$$

for all simples  $X_4$  appearing in the decomposition of  $X_1 \otimes X_2 \otimes X_3$ . The unitary isomorphism between both decompositions is given by the *F-symbol*



or *associator*<sup>1</sup>  $F^{X_1 X_2 X_3}$  and its matrix components are conveniently depicted in terms of the *F-move*

$$\begin{array}{c} X_1 \quad X_2 \quad X_3 \\ \diagdown \quad \diagup \\ i \\ \diagup \quad \diagdown \\ X_5 \quad j \\ | \\ X_4 \end{array} = \sum_{X_6, k, l} (F_{X_4}^{X_1 X_2 X_3})_{X_5, ij}^{X_6, kl} \begin{array}{c} X_1 \quad X_2 \quad X_3 \\ \diagdown \quad \diagup \\ k \\ \diagup \quad \diagdown \\ X_6 \quad l \\ | \\ X_4 \end{array} . \quad (3.10)$$

An *F*-symbol is set to zero when not all of the involved fusion rules are fulfilled. Crucially, they also satisfy a *consistency equation* or *coherence axiom* colloquially known as the *pentagon equation* which in full reads:

$$\begin{aligned} & \sum_o \left( F_{X_5}^{X_6 X_3 X_4} \right)_{X_7, lm}^{X_8, no} \left( F_{X_5}^{X_1 X_2 X_8} \right)_{X_6, ko}^{X_9, pq} \\ &= \sum_{X_{10}, rst} \left( F_{X_7}^{X_1 X_2 X_3} \right)_{X_6, kl}^{j, rs} \left( F_{X_5}^{X_1 X_{10} X_4} \right)_{X_7, sm}^{X_9, tq} \left( F_{X_9}^{X_2 X_3 X_4} \right)_{X_{10}, rt}^{X_8, np} . \end{aligned} \quad (3.11)$$

In virtue of *Ocneanu rigidity*, the number of solutions to the pentagon equation, up to basis transformations of the Hom spaces, is finite for any valid set of fusion rules.

### 3.1.2 Examples

Before proceeding, let us provide a few paradigmatic examples of these rather abstract definitions.

#### Finite group symmetry: $\text{Vec}_G^\omega$

Let  $G$  be a finite group. The fusion category  $\text{Vec}_G$  is the category of (complex finite-dimensional)  $G$ -graded vector spaces. Its fusion rules are completely defined by

$$(V \otimes W)_g = \bigoplus_{hk=g} V_h \otimes W_k \quad (3.12)$$

$\forall g, h, k \in G$ , where  $V_g$  stands for the  $g$ -graded component of  $V$ . Representative simple objects of  $\text{Vec}_G$  are the one-dimensional vector spaces  $\{\mathbb{C}_g\}_{g \in G}$  and

<sup>1</sup>The *F*-symbol has an interpretation as the complex number that  $\mathcal{C}$  assigns to the planar projection of a tetrahedron whose edges are decorated by objects in  $\text{ob}(\mathcal{C})$  and vertices decorated by junction spaces. In what follows we will assume the *F*-symbol to be independent of the chosen projection. This property is known as *tetrahedral symmetry*, and not every UFC satisfies this. We refer to ref. [61] for the generalisation of the string-net construction to fusion categories that do not enjoy this symmetry.

are thus in bijection with the group elements of  $G$ . Abusing notation, they are sometimes simply denoted by  $g \equiv \mathbb{C}_g$ . The non-vanishing  $F$ -symbols are all equal to one when all fusion rules are fulfilled. Whenever  $G$  possesses a non-trivial *third cohomology group*  $H^3(G, \mathbb{U}(1))$ , a non-trivial 3-cocycle  $\omega$  can be used to construct  $\text{Vec}_G^\omega$ . This fusion category is constructed analogously to  $\text{Vec}_G$  where now the non-zero components of the  $F$ -symbols evaluate to  $\omega$ . In that case, the pentagon equation is equivalent to the 3-cocycle condition satisfied by  $\omega$ .

### Finite group representations: $\text{Rep}(G)$

Intimately connected to  $\text{Vec}_G$  is the category  $\text{Rep}(G)$ , again for a finite group  $G$ .<sup>2</sup> This category encodes as objects the (unitary finite-dimensional) representations of  $G$ . Representative *irreducible* representations then constitute the simple objects and their fusion is given by the usual tensor product of  $G$ -representations. It can then be deduced that the 6-j symbols are in this case the matrix coefficients of the monoidal associator.

### The Fibonacci fusion category: $\text{Fib}$

The *Fibonacci* UFC  $\text{Fib}$  consists of two objects labelled by  $\{\mathbb{1}, \tau\}$  whose only non-trivial fusion rule reads  $\tau \otimes \tau = \mathbb{1} \oplus \tau$ . The quantum dimension of  $\tau$  is easily found to be the golden ratio  $d_\tau = \varphi = \frac{1+\sqrt{5}}{2} = 1.618\dots$ , indicative of the fact that  $\text{Fib}$  is not realised as the representation category of any finite group as this would require  $d_\tau$  to be a non-negative integer. The non-trivial  $F$ -symbols read:

$$F_\tau^{\tau\tau\tau} = \frac{1}{\varphi} \begin{pmatrix} 1 & \sqrt{\varphi} \\ \sqrt{\varphi} & -1 \end{pmatrix}. \quad (3.13)$$

As anticipated above, we come to the intermediate conclusion that fusion categories provide a framework that not only unifies finite group and representation theory — as exemplified by the fusion categories  $\text{Vec}_G$  and  $\text{Rep}(G)$  — but also extends it — e.g. in the form of the Fibonacci category. As a guide to the reader, we compactly summarise this in table 3.1:

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<sup>2</sup>For all  $G$ ,  $\text{Vec}_G$  and  $\text{Rep}(G)$  are *Morita equivalent*, as explained below. This means that their Drinfeld centres  $\mathcal{Z}(\text{Vec}_G)$  and  $\mathcal{Z}(\text{Rep}(G))$  are equivalent as braided fusion categories.

| $\text{Vec}_G$                                                                                   | $\text{Rep}(G)$                                                                                                                                    | Categorical notion                                                       |
|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| $G$ -graded vector spaces<br>$V, W, U, \dots$                                                    | $G$ -representations<br>$(V_i, \rho_i)$                                                                                                            | Objects $\text{ob}(\mathcal{C})$                                         |
| Grading-preserving<br>linear maps $\phi : V \rightarrow W$                                       | Intertwiners<br>$f \circ \rho_1 = \rho_2 \circ f$                                                                                                  | Hom spaces $\mathcal{C}(X, X')$                                          |
| $\text{Vec}_G(\mathbb{C}_{g_1}, \mathbb{C}_{g_2}) =$<br>$\delta_{g_1, g_2} \mathbb{C}$           | $\text{Rep}(G)(\alpha_i, \alpha_j) = \delta_{i,j} \mathbb{C},$<br>w/ $\alpha_i, \alpha_j$ irreps (Schur)                                           | Simple objects<br>$\mathcal{C}(X_1, X_2) = \delta_{X_1, X_2} \mathbb{C}$ |
| $(V \otimes W)_g =$<br>$\bigoplus_{hk=g} V_h \otimes W_k$                                        | $\alpha_i \otimes \alpha_j \simeq \bigoplus_k N_{ij}^k \alpha_k$                                                                                   | Monoidal product &<br>fusion rules                                       |
| $\mathbb{C}_1 \otimes V \simeq V \simeq V \otimes \mathbb{C}_1$                                  | $\underline{1} \otimes \rho \simeq \rho \simeq \rho \otimes \underline{1},$ w/ $\underline{1}$<br>the trivial irrep                                | Monoidal unit $\underline{1}$                                            |
| $\mathbb{C}_g \otimes \mathbb{C}_{g^{-1}} \simeq \mathbb{C}_1$                                   | $\alpha \otimes \alpha^* \ni \underline{1}$                                                                                                        | Dual object                                                              |
| $F : (V \otimes W) \otimes U \xrightarrow{\sim}$<br>$V \otimes (W \otimes U),$ w/ $F$<br>trivial | $F : (\alpha_1 \otimes \alpha_2) \otimes \alpha_3 \xrightarrow{\sim}$<br>$\alpha_1 \otimes (\alpha_2 \otimes \alpha_3),$ w/ $F$<br>the 6-j symbols | $F$ -symbol (Monoidal<br>associator)                                     |

Table 3.1: Comparison of  $\text{Vec}_G$ ,  $\text{Rep}(G)$ , and their categorical generalisations.

### 3.1.3 Braiding

Finally, we introduce the *braiding* of two defects diagrammatically as:

$$R^{X_1 X_2} := \begin{array}{c} X_1 \quad X_2 \\ \diagdown \quad \diagup \\ X_2 \quad X_1 \end{array}, \quad (R^{X_1 X_2})^{-1} := \begin{array}{c} X_2 \quad X_1 \\ \diagup \quad \diagdown \\ X_1 \quad X_2 \end{array}. \quad (3.14)$$

Typically,  $R^{X_1 X_2} \circ R^{X_2 X_1} \neq \text{id}_{X_1 \otimes X_2}$  so that in the graphical calculus one should keep track of the distinction between *over* and *underbraiding* of the strands. Categories for which  $R^{X_1 X_2} \circ R^{X_2 X_1}$  is the identity map on  $X_1 \otimes X_2$ , are called *symmetric*.

The components of the braiding eq. (3.14) in our choice of basis then gives rise to the unitary *R-matrix* defined via:

$$\begin{array}{c} X_1 \quad X_2 \\ \diagdown \quad \diagup \\ \text{---} i \text{---} \\ | \\ X_3 \end{array} = \sum_j (R_{X_3}^{X_1 X_2})_{ij} \begin{array}{c} X_1 \quad X_2 \\ \diagup \quad \diagdown \\ j \text{---} \\ | \\ X_3 \end{array}. \quad (3.15)$$

The braiding is required to satisfy a consistency equation called the *hexagon equation* that also involves the *F-symbols*. A fusion category is called *braided* or *pre-modular* when it is endowed with a braiding that satisfies the hexagon axiom. For any braided fusion category we then define the *topological S-matrix*  $S_{X_1, X_2}$  via the amplitudes

$$\begin{aligned} S_{X_1, X_2} &:= \frac{1}{\text{FPdim } \mathcal{C}} \begin{array}{c} \bigcirc_{X_1} \quad \bigcirc_{X_2} \end{array} \\ &= \frac{1}{\text{FPdim } \mathcal{C}} \sum_{X_3} N_{X_1 X_2}^{X_3} d_{X_3} \text{Tr}(R_{X_3}^{X_1 X_2} R_{X_3}^{X_2 X_1}), \end{aligned} \quad (3.16)$$

where we have defined the *total quantum dimension* or *Frobenius-Perron dimension of  $\mathcal{C}$*  as  $\text{FPdim } \mathcal{C} := \sum_{X \in \mathcal{I}_{\mathcal{C}}} d_X^2$ . Physically, the amplitudes eq. (3.16) correspond to an interferometry process in which two defects  $X_1$  and  $X_2$  are created from the vacuum, exchanged and annihilated again. Notably, if for every  $X_1 \neq \mathbb{1}$  there exists at least one  $X_2 \neq \mathbb{1}$  such that their double-braiding  $R^{X_1 X_2} \circ R^{X_2 X_1}$  is non-trivial, the S-matrix is non-degenerate [62, Cor. 2.16]. Physically, this means that every non-trivial defect can be distinguished from

the vacuum by double-braiding it with a suitable probe defect. This is called the *principle of remote detectability* [63, 64]. This motivates the definition of a *modular* fusion category, which is a pre-modular fusion category with non-degenerate S-matrix.

### 3.1.4 Examples

#### Finite group symmetry: $\text{Vec}_G^\omega$

Evidently,  $\text{Vec}_G$  only admits a braiding if  $G$  is abelian. In that case the braiding is provided by a choice of *bilinear form*  $b$ , meaning a map  $b : G \times G \rightarrow \text{U}(1)$  satisfying

$$\begin{aligned} b(g_1, g_3) \cdot b(g_2, g_3) &= b(g_1 g_2, g_3), \\ b(g_1, g_2) \cdot b(g_1, g_3) &= b(g_1, g_2 g_3), \end{aligned} \quad (3.17)$$

for all  $g_1, g_2, g_3 \in G$ . When the associator belongs to a non-trivial class  $[\omega] \in H^3(G, \text{U}(1))$ , admissible associator-braiding pairs are in one-to-one correspondence with the elements of the *abelian* cohomology group  $H_{\text{ab}}^3(G, \text{U}(1))$ . We refer to the relevant literature for a precise definition and construction of the braiding [53].

#### Finite group representations: $\text{Rep}(G)$

The category  $\text{Rep}(G)$  admits a symmetric braiding that amounts to the transposition of representation spaces, i.e.  $R : V_1 \otimes V_2 \rightarrow V_2 \otimes V_1$  for  $(V_1, \rho_1), (V_2, \rho_2)$  in  $\text{ob}(\text{Rep}(G))$ .

#### The Fibonacci fusion category: $\text{Fib}$

The Fibonacci category admits a *modular* braiding. Indeed, its  $S$ -matrix

$$S = \frac{1}{\sqrt{2+\varphi}} \begin{pmatrix} 1 & \varphi \\ \varphi & -1 \end{pmatrix}, \quad (3.18)$$

is non-degenerate [65].

## 3.2 String-net models

### 3.2.1 Hilbert space and Hamiltonian

Let  $\mathcal{D}$  be a UFC whose objects we denote by  $\{Y\}$ , and  $\Lambda$  be an *oriented trivalent lattice*  $\Lambda$  without boundary. The microscopic string-net Hilbert space on  $\Lambda$  is constructed from the *input*  $\mathcal{D}$  as follows [66, 67]. A basis state

corresponds to an assignment, or  $\mathcal{D}$ -colouring [67], of simple objects  $Y \in \mathcal{I}_{\mathcal{D}}$  to all the edges  $E(\Lambda)$ , with the requirement that the fusion rules are satisfied on all vertices  $V(\Lambda)$ . Conventionally, reversing the orientation of an edge  $e \rightarrow -e$  amounts to taking the dual object on that edge. Then, to every vertex with adjacent edges labelled  $Y_1, Y_2, Y_3$  the junction space  $\mathcal{C}(Y_1 \otimes Y_2, Y_3)$  is attached. In what follows we specialise to the *hexagonal lattice*. On it, the Hilbert space  $\mathcal{H}_{\mathcal{D}}(\Lambda)$  is thus defined as:

$$\mathcal{H}_{\mathcal{D}}(\Lambda) := \text{Span}_{\mathbb{C}} \left\{ \begin{array}{c} \text{Diagram of a hexagonal plaquette with vertices } i_1, i_2, i_3, i_4, i_5, i_6 \text{ and edges } Y_1, Y_2, Y_3, Y_4, Y_5, Y_6, Y_7, Y_8, Y_9, Y_{10}, Y_{11}, Y_{12} \end{array} \mid \{Y\}, \{i\} \right\}, \quad (3.19)$$

where it is understood that the picture extends over the entire lattice. In correspondence with our graphical calculus, we also assume all edges to be oriented downwards.

On every plaquette  $p \in P(\Lambda)$  and for every  $Y \in \mathcal{I}_{\mathcal{D}}$  we subsequently define a local operator  $B_p^Y$ . This operator admits a particularly intuitive definition as inserting a *non-contractible* counter-clockwise loop labelled by the simple object  $Y$  in the plaquette and fusing it into the physical lattice with local applications of the resolution of the identity  $F$ -moves, and normalisation of the basis vectors eqs. (3.5), (3.8) and (3.10). In order for this to be consistent with our graphical calculus, and in particular the quantum dimension relation eq. (3.7), we should therefore think of the underlying manifold as being *punctured* in the interior of every plaquette in  $P(\Lambda)$ . Schematically, the action can therefore be depicted as:

$$B_p^Y \cdot |\Psi\rangle = \begin{array}{c} \text{Diagram of a hexagonal plaquette with a central shaded circle and a counter-clockwise loop labeled } Y \end{array} \xrightarrow{(3.8)} \sum \{d\} \begin{array}{c} \text{Diagram of a hexagonal plaquette with a central shaded circle and a counter-clockwise loop with nodes and arrows} \end{array} \quad (3.20)$$

$$\begin{array}{c}
 \begin{array}{c}
 \text{(3.10)} \\
 \text{(3.5)} \\
 \mapsto
 \end{array}
 \sum \{d\} \{F\}
 \end{array}
 \cdot
 \begin{array}{c}
 \text{Diagram: A hexagon with a shaded circle in the center. Each of the six vertices of the hexagon has a line segment extending outwards. The shaded circle has diagonal hatching.}
 \end{array}
 \quad (3.21)$$

3

1289 Herein, we have for simplicity omitted all edge and vertex labels;  $\{d\}$  and  $\{F\}$   
 1290 are placeholders for the quantum dimensions and  $F$ -symbols obtained via  
 1291 repeated use of the aforementioned identities. We refer to [66] for an explicit  
 1292 computation of these matrix elements. We can also infer from application of  
 1293 our graphical calculus that these plaquette operators form a representation  
 1294 of the fusion rules, viz.:

$$B_{\mathbf{p}}^{Y_1} \cdot B_{\mathbf{p}}^{Y_2} = \sum_{Y_3} N_{Y_1 Y_2}^{Y_3} B_{\mathbf{p}}^{Y_3}. \quad (3.22)$$

1295 With these plaquette operators at our disposal, the local Hamiltonian terms  
 1296 are constructed as:

$$B_{\mathbf{p}}^{\mathcal{D}} = \sum_{Y \in \mathcal{I}_{\mathcal{D}}} \frac{d_Y}{\text{FPdim} \mathcal{D}} B_{\mathbf{p}}^Y =: \text{Diagram: A dashed circle.} \quad (3.23)$$

1297 From combining eq. (3.22) with the identity

$$d_Y \text{FPdim} \mathcal{D} = \sum_{Y_1, Y_2} N_{Y_1 Y_2}^Y d_{Y_1} d_{Y_2}, \quad (3.24)$$

1298 it directly follows that  $B_{\mathbf{p}}^{\mathcal{D}}$  is an idempotent for every  $\mathbf{p} \in \mathbf{P}(\Lambda)$ , i.e.  $B_{\mathbf{p}}^{\mathcal{D}} \cdot B_{\mathbf{p}}^{\mathcal{D}} =$   
 1299  $B_{\mathbf{p}}^{\mathcal{D}}, \forall \mathbf{p}$ . Commutation of plaquette operators  $B_{\mathbf{p}}^{\mathcal{D}}$  and  $B_{\mathbf{p}' \neq \mathbf{p}}^{\mathcal{D}}$  is immediate  
 1300 when  $\mathbf{p}$  and  $\mathbf{p}'$  are far separated. When  $\mathbf{p}$  and  $\mathbf{p}'$  share an edge this follows  
 1301 from comparing the matrix elements of both application orders and invoking  
 1302 the pentagon equation eq. (3.11) on their matrix elements. The string-net  
 1303 Hamiltonian is then finally given by

$$H_{\mathcal{D}}(\Lambda) = - \sum_{\mathbf{p} \in \mathbf{P}(\Lambda)} B_{\mathbf{p}}^{\mathcal{D}}. \quad (3.25)$$

1304 Remark that a ground state of  $H_{\mathcal{D}}(\Lambda)$  can be constructed by acting with the  
 1305 ground state projector  $\prod_{\mathbf{p}} B_{\mathbf{p}}^{\mathcal{D}}$  — which is well-defined since all  $B_{\mathbf{p}}^{\mathcal{D}}$ 's commute

— on a generic string-net state in  $\mathcal{H}_{\mathcal{D}}(\Lambda)$ . This is exactly the approach taken in section 3.4 to write down the ground state PEPS representations.

Importantly, the ground state subspace is one-dimensional when the underlying lattice has the topology of a two-sphere [67]. However, on the two-torus the ground space is degenerate, and the degeneracy does not depend on the system size. Due to the absence of any global symmetry, this degeneracy cannot be traced back to ordinary symmetry breaking as discussed in section 2.4. Rather, the ground state degeneracy can be understood as the breaking of a more general type of symmetry whose operators are supported on oriented *freely deformable* loops  $\ell$  on the lattice. Such operators constitute an example of a *global 1-form* symmetry [31, 68, 69].<sup>3</sup>

These loop operators, and their multiplication rules, are encoded in the *monoidal* or *Drinfeld centre*  $\mathcal{Z}(\mathcal{D})$  of  $\mathcal{D}$ .

### 3.2.2 Loop operators and ground states

Formally, the centre  $\mathcal{Z}(\mathcal{D})$  is a unitary fusion category whose simple objects are pairs  $(Z, \gamma_{Z, \bullet})$  where  $Z$  is a (generically non-simple) object in  $\mathcal{D}$  and  $\gamma_{Z, \bullet}$  is a natural isomorphism, called the *half-braiding*. Its matrix elements can be expressed graphically as

$$\begin{array}{c} Y_1 \quad Z \\ \diagdown \quad \diagup \\ \text{---} \quad \text{---} \\ \diagup \quad \diagdown \\ Z \quad Y_1 \end{array} = \sum_{Y_2} \sum_{i,j} (\Gamma_{Y_2}^{Y_1 Z})_{ij} \begin{array}{c} Y_1 \quad Z \\ \diagdown \quad \diagup \\ \text{---} \quad \text{---} \\ \diagup \quad \diagdown \\ Z \quad Y_1 \end{array} \quad , \quad (3.26)$$

$\forall Y_1 \in \mathcal{I}_{\mathcal{D}}$ . The half-braiding satisfies a hexagon consistency axiom involving the associator of  $\mathcal{D}$ . Consequently,  $\mathcal{Z}(\mathcal{D})$  can be endowed with a *modular* braiding, defined directly in terms of  $\gamma_{Z, \bullet}$  [70]. Naturality of  $\gamma_{Z, \bullet}$  implies the following graphical equality:

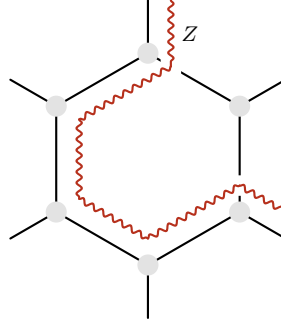
$$\begin{array}{c} Y_1 \quad Y_2 \\ \diagdown \quad \diagup \\ \text{---} \quad \text{---} \\ \diagup \quad \diagdown \\ Y_3 \end{array} = \begin{array}{c} Y_1 \quad Y_2 \\ \diagdown \quad \diagup \\ \text{---} \quad \text{---} \\ \diagup \quad \diagdown \\ Y_3 \end{array} \quad , \quad (3.27)$$

for all admissible junctions  $|Y_1 Y_2 Y_3, i\rangle$ . Below, we often abuse notation and write simply  $Z$  instead of  $(Z, \gamma_{Z, \bullet})$ .

<sup>3</sup>In general, a  $(D+1)$ -dimensional quantum theory is said to host a *global  $p$ -form symmetry* if there exist symmetry operators supported on freely-deformable closed submanifolds of *codimension  $p$*  with respect to the spatial dimension [31, 69].



In order to define the action of a loop labelled by  $(Z, \gamma_{Z, \bullet})$  on a state, one proceeds in a similar fashion as in the definition of the plaquette operators. One inserts the object  $Z$  along the path  $\ell$ , and locally fuses it into the physical lattice, thereby making use of the half-braiding eq. (3.26) at every crossing with the physical lattice and the usual local recoupling identities eqs. (3.5), (3.8) and (3.10). An illustration of such a loop is:



(3.28)

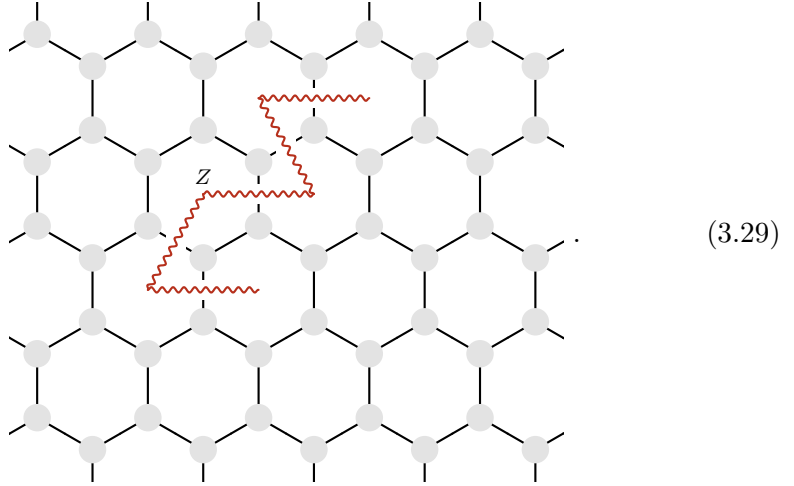
Although drawn as a line, it should be remarked that these operators act (diagonally) on edges adjacent to their path and are therefore more precisely interpreted as *ribbon operators* with a finite width. These coincide with those introduced by Kitaev in ref. [71] and Levin and Wen in ref. [66], see also refs. [72, 73].

The *topological* nature of the loop operator follows directly from the naturality condition eq. (3.27), which ensures that  $Z$  can slide freely across any junction of the physical lattice without altering the resulting operator. Note that this topological invariance also accounts for the commutativity with the Hamiltonian. When a loop and a plaquette operator have disjoint support, they commute trivially. When they have overlapping support, one can deform the loop away from the plaquette before acting with the latter, and subsequently restore the path of the loop, so that the combined action remains unchanged.

Provided a reference string-net ground state one can construct a complete basis of torus ground states by acting with all simple loop operators  $(Z, \gamma_{Z, \bullet}) \in \mathcal{I}_{\mathcal{Z}(\mathcal{D})}$  along a non-contractible cycle of the torus. We therefore conclude that the torus ground state degeneracy is given by  $|\mathcal{I}_{\mathcal{Z}(\mathcal{D})}|$ . This also implies that orthogonal ground states are locally indistinguishable, as the precise location of the  $Z$ -loop can be continuously deformed and is thus invisible for any local operator. Acting on the other hand with loop operators along the perpendicular non-contractible cycle provides a different set of ground states that are related to the former via the unitary basis transformation given by the  $S$ -matrix of  $\mathcal{Z}(\mathcal{D})$  defined in eq. (3.16) [74].

### 3.2.3 String operators and excited states

Similar to the construction of ground states via loop operators, excited states are constructed by acting on the ground states with *open* string operators, also labelled by objects  $(Z, \gamma_Z, \bullet) \in \mathcal{I}_{\mathcal{Z}(\mathcal{D})}$ . Such a state corresponds to an  $(Z, Z^*)$ -anyon-anti-anyon pair, and schematically can be depicted as



In this sense, the different ground states constructed in section 3.2.2 can be understood as the result of creating a  $(Z, Z^*)$ -pair from a reference ground state, dragging one of the anyons around a non-contractible cycle before annihilating the pair.

Importantly, note that in fusing the open string in eq. (3.29) to the physical lattice, one is in general left with dangling vertical edges on the plaquettes on which the string starts and terminates, if  $Z$  decomposes as a non-trivial object of  $\mathcal{D}$ . In that case the obtained state does not belong to the string-net Hilbert space, and the corresponding anyons are said to possess a non-trivial *charge*. Accommodating this type of excitations naturally leads us to consider the *extended* string-net models, first introduced in ref. [75]. In this framework, the string-net Hilbert space is extended with extra open strands or *tails*, one for every vertex of the original lattice. Consequently, all anyonic excitations, also those with a non-trivial charge, can be considered without the need to leave the Hilbert space or violate the  $\mathcal{D}$  fusion rules. Concretely, on the

hexagonal lattice, the extended Hilbert space is defined as:

$$\mathcal{H}_D^{\text{Ext.}}(\Lambda) := \text{Span}_{\mathbb{C}} \left\{ \begin{array}{c} \text{Diagram of a hexagonal lattice with vertices } i_1, i_2, i_3, i_4, i_5, i_6, i_7, i_8 \text{ and edges } Y_1, \dots, Y_{16} \end{array} \mid \{Y\}, \{i\} \right\}. \quad (3.30)$$

In this depiction, an identification between every tail and the closest vertex of the hexagonal lattice is assumed. The extended Hamiltonian consists of the plaquette terms which generalise those introduced in eq. (3.23), and are defined in detail in ref. [75], and additionally *vertex terms*  $Q_v^D$  for every  $v \in V$ . These act by projecting the tail corresponding to  $v$  to the identity object. Concretely, locally:

$$Q_v^D \left| \begin{array}{c} Y_4 \quad Y_5 \\ \quad \quad k \\ Y_3 \quad j \text{---} Y \\ Y_2 \quad i \text{---} Y' \\ \quad \quad Y_1 \end{array} \right\rangle = \delta_{Y, \mathbb{1}} \left| \begin{array}{c} Y_4 \quad Y_5 \\ \quad \quad k \\ Y_2 \quad 1 \text{---} \dots \text{---} 1 \\ Y_2 \quad i \text{---} Y' \\ \quad \quad Y_1 \end{array} \right\rangle. \quad (3.31)$$

Collecting everything, the extended string-net Hamiltonian is of the form

$$H_D^{\text{Ext.}}(\Lambda) = - \sum_{p \in P(\Lambda)} B_p^D - \sum_{v \in V(\Lambda)} Q_v^D. \quad (3.32)$$

Notably, in the subspace defined by  $Q_v^D = 1$ ,  $\forall v \in V$ , the extended string-net model coincides with the original Levin-Wen string-net.

At this point we will not attempt to describe in further detail the action of the loop operators, the explicit ground states or excited states. Rather, we demonstrate in section 3.4 the construction of distinct PEPS representations of string-net ground states. The upshot of the tensor network approach is that a complete basis of torus ground states can be constructed by inserting a network of *matrix product operators* living purely on the *virtual level* of the

PEPS. The network to insert corresponds concretely to *central idempotents* of the *tube algebra*. In fact, the representation theory of the tube algebra is often used to construct an explicit realisation of the Drinfeld centre. This description of the torus ground states is significantly simpler than working with the ribbon operators. Also, crucially, since the MPOs live on the virtual level, they survive physical perturbations and hence furnish a representation of the symmetries of the boundary Hamiltonians discussed in section 4.2. In a similar spirit, by introducing a new type of tensors whose support is specified by a central idempotent of the tube algebra, excited states also admit a PEPS representation.

### 3.3 Gapped boundaries and domain walls

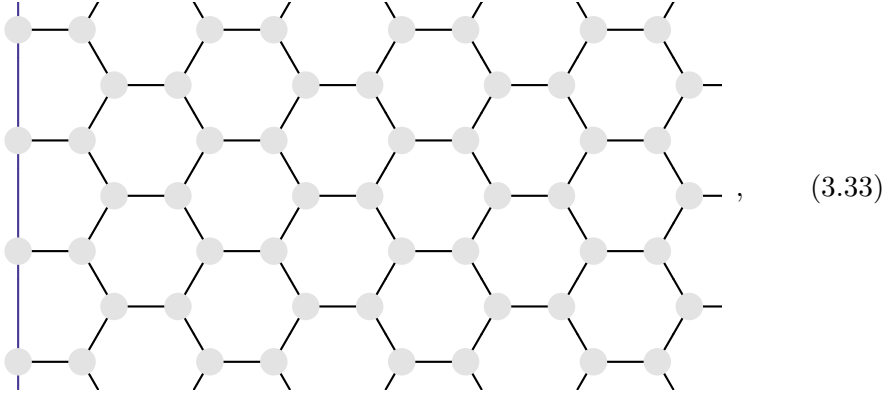
#### 3.3.1 Gapped boundaries

So far we have restricted our attention to the case of closed boundary conditions. As anticipated in the introduction, we can define string-net models with different types of open boundary conditions while preserving the gap. One choice of boundary is obtained by simply terminating the lattice, in which case the edge degrees of freedom are labelled by objects and junctions in  $\mathcal{D}$ . This demonstrates that every string-net model has at least one gapped boundary, to which we refer as the *regular* or *smooth* boundary condition, but depending on the input  $\mathcal{D}$ , other choices might be possible [76, 77].

The boundary conditions that we will study momentarily are constructed in such a way that the bulk plaquette operators can be extended to the boundary degrees of freedom while retaining their mutual commutativity. Commutation of the plaquette operators follows, similar as in the bulk, from local relations involving the  $F$ -symbols of  $\mathcal{D}$ .

To this end, let us introduce a new set of labels  $\mathcal{I}_{\mathcal{R}} = \{R_1, R_2, \dots\}$  for these boundary degrees of freedom, and an action of  $\mathcal{D}$  on those labels implemented via  $R_1 \triangleleft Y = \sum_{R_2 \in \mathcal{I}_{\mathcal{R}}} N_{R_1 Y}^{R_2} R_2$  so that vertices on the boundary host a new type of vector space with basis vectors  $|R_1 Y R_2; i\rangle$ ,  $i = 1, 2, \dots, N_{R_1 Y}^{R_2}$ . Schematically, the lattice of the string-net states in the presence of such a

1426 boundary looks like:



1427 where we draw edges that carry labels in  $\mathcal{I}_{\mathcal{R}}$  in blue. The type of Hom space  
 1428 can be inferred from the colour of the adjacent edges. Similar as in the bulk  
 1429 we define a unitary isomorphism  $\triangleleft F$ , generically distinct from the earlier  
 1430 defined  $F$ -symbol, whose matrix entries are defined graphically in terms of  
 1431 an  $\triangleleft F$ -move:

$$\begin{array}{c} R_1 \quad Y_1 \quad Y_2 \\ \diagup \quad \diagup \quad \diagup \\ \bullet \quad \bullet \quad \bullet \\ \diagdown \quad \diagdown \quad \diagdown \\ R_3 \quad R_2 \end{array} \quad \begin{array}{c} i \\ j \end{array} = \sum_{Y_3, ij} \left( \triangleleft F_{R_2}^{R_1 Y_1 Y_2} \right)^{Y_3, kl}_{R_3, ij} \quad \begin{array}{c} R_1 \quad Y_1 \quad Y_2 \\ \diagup \quad \diagup \quad \diagup \\ \bullet \quad \bullet \quad \bullet \\ \diagdown \quad \diagdown \quad \diagdown \\ R_3 \quad R_2 \end{array} \quad \begin{array}{c} k \\ l \end{array} . \quad (3.34)$$

1432 We refer to  $\triangleleft F$  as the (*right*) *module associator*, and explain this terminology  
 1433 below. We require the  $\triangleleft F$ -symbols to adhere to a consistency axiom involving  
 1434 the  $F$ -symbols of  $\mathcal{D}$  that is of the form

$$\triangleleft F \triangleleft F = \sum F \triangleleft F \triangleleft F. \quad (3.35)$$

1435 Its explicit form is straightforward to derive and spelled out in many places,  
 1436 e.g. ref. [78]. For later reference, let us also introduce quantum dimensions  
 1437 of objects in  $\mathcal{I}_{\mathcal{R}}$  in the sense of refs. [53, 79] via:

$$\begin{aligned} d_R &> 0 \\ d_R d_Y &= \sum_{R' \in \mathcal{I}_R} N_{RY}^{R'} d_{R'}, \end{aligned} \quad (3.36)$$

1438 for all  $Y \in \mathcal{I}_{\mathcal{D}}$ . The quantum dimensions  $d_R$  are defined up to a global  
 1439 rescaling  $d_R \mapsto \lambda d_R$ ,  $\forall R \in \mathcal{I}_R$  for any  $R$ -independent positive real scalar  $\lambda$ .

1440 To fix it, we will adopt the canonical normalisation

$$\text{FPdim}\mathcal{R} := \sum_{R \in \mathcal{I}_{\mathcal{R}}} d_R^2 = \sum_{Y \in \mathcal{I}_{\mathcal{D}}} d_Y^2 =: \text{FPdim}\mathcal{D}. \quad (3.37)$$

1441 In this normalisation, we have the resolution of the identity

$$R \left| \begin{array}{c} Y \end{array} \right. = \sum_{R', i} \sqrt{\frac{d_{R'}}{d_R d_Y}} \begin{array}{c} R \quad Y \\ \diagdown \quad \diagup \\ i \\ \diagup \quad \diagdown \\ R' \quad Y \\ R \quad Y \end{array}, \quad (3.38)$$

1442 and the inner product

$$\begin{array}{c} R_2 \\ | \\ i \\ \diagdown \quad \diagup \\ R_1 \quad Y \\ \diagup \quad \diagdown \\ j \\ | \\ R_2' \end{array} = \delta_{R_2, R_2'} \delta_{i, j} \sqrt{\frac{d_{R_1} d_Y}{d_{R_2}}} \begin{array}{c} R_2 \\ | \end{array}. \quad (3.39)$$

1443 Succinctly, provided a unitary fusion category  $\mathcal{D}$ , the set of boundary labels  
 1444  $\mathcal{I}_{\mathcal{R}}$ , together with the action  $\triangleleft$  and a solution to the pentagon equation  
 1445 eq. (3.35) form the defining data of a (unitary) *right module category*  $\mathcal{R}$  over  
 1446  $\mathcal{D}$  [52, 53]. Similarly, we can define a *left* module category to construct a  
 1447 gapped right boundary condition.

1448 Notably, the *opposite* of a *right* module category  $\mathcal{R}$  over  $\mathcal{D}$ , notated as  
 1449  $\mathcal{R}^{\text{op}}$ , is naturally a *left* module category whose module action is defined  
 1450 as [53, 79]:

$$Y \triangleright^{\text{op}} R := R \triangleleft Y^*. \quad (3.40)$$

1451 The objects of this opposite module category can be interpreted as *duals* of  
 1452 the objects in  $\mathcal{R}$ . This leads to the following graphical identity

$$R^* \left| \begin{array}{c} R \end{array} \right. = \sum_{Y, i} \sqrt{\frac{d_Y}{d_R d_{R^*}}} \begin{array}{c} R^* \quad R \\ \diagdown \quad \diagup \\ i \\ Y \\ \diagup \quad \diagdown \\ i \\ R^* \quad R \end{array}. \quad (3.41)$$

1453 For a generic choice of module category  $\mathcal{R}$ , the bulk plaquette operators  
 1454 are extended to plaquettes adjacent to the boundary by again inserting  
 1455 a punctured counter-clockwise loop labelled by  $\bigoplus_{X \in \mathcal{I}_{\mathcal{D}}} \frac{d_X}{\text{FPdim}\mathcal{D}} X$  in the  
 1456 boundary plaquettes and absorbing them into the physical lattice by making  
 1457 use of the different resolutions of the identity,  $F$ -moves and normalisations

eqs. (3.8), (3.38), (3.10), (3.34), (3.5) and (3.39). It can be verified that the boundary plaquette operators commute with all the other Hamiltonian terms by virtue of eq. (3.35), so that the Hamiltonian remains gapped near the boundary.

### 3.3.2 Examples

#### The regular module category

Every UFC  $\mathcal{D}$  trivially defines a unitary indecomposable module category over itself. We will refer to this as the *regular* module category. For this choice, the corresponding boundary condition is the smooth boundary condition.

#### $\text{Vec}_G$ -module categories

As anticipated in section 2.4, indecomposable (left) module categories over  $\text{Vec}_G$  are – up to conjugacy – specified by pairs  $(H \subseteq G, \psi)$  where  $[\psi] \in H^2(H, \text{U}(1))$ . In what follows we write the corresponding module category as  $\mathcal{R}(H, \psi)$ .

The explicit construction of  $\mathcal{R}(H, \psi)$  from the data  $(H, \psi)$  is reviewed in detail in chapter 8. Here we only recall a few key facts. For one, the object set  $\mathcal{I}_{\mathcal{R}}$  is a transitive left  $G$ -set so that it can be identified with the cosets  $G/H$  endowed with the canonical left  $G$ -action. The module associator  ${}^{\triangleright}F$  can then be constructed from the cocycle  $\psi$  by making use of Shapiro’s isomorphism between the cohomology groups  $H^2(H, \text{U}(1)) \simeq H^2(G, \text{Fun}(G/H, \text{U}(1)))$ , where  $\text{Fun}(G/H, \text{U}(1))$  is the  $G$ -module of functions  $G/H \rightarrow \text{U}(1)$ .

Notably, for the choice  $H = G$  and trivial 2-cocycle, the corresponding module category  $\mathcal{R}(G, 1)$  contains only one simple object, so that  $\mathcal{R}(G, 1)$  is equivalent to the category  $\text{Vec}$  of finite-dimensional complex vector spaces.

For  $H = \{1\}$ , the regular module category is obtained:  $\mathcal{R}(\{1\}, 1) = \text{Vec}_G$ .

#### $\text{Vec}_G^{\omega}$ -module categories

Analogously to the untwisted case, module categories over  $\text{Vec}_G^{\omega}$  are in one-to-one correspondence with pairs  $(H, \psi)$  where now  $\psi$  trivialises the 3-cocycle  $\omega$  on the subgroup  $H$ . This imposes constraints on the allowed subgroups  $H \subseteq G$ .

The object set  $\mathcal{I}_{\mathcal{R}}$  can again be identified with the left cosets  $G/H$ , and in this case the module associator can be expressed in terms of both  $\omega$  and  $\psi$ , see e.g. ref. [55] for an explicit formula.

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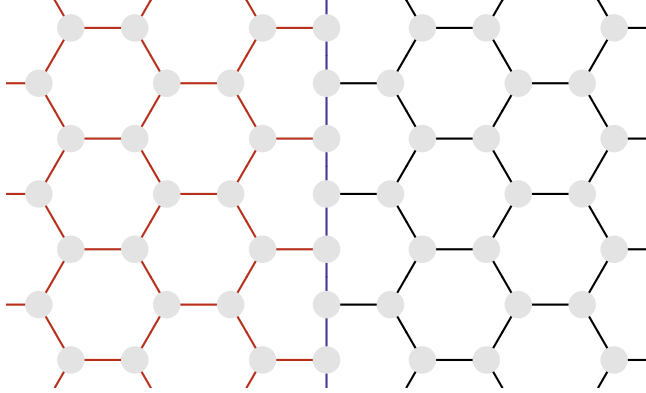
that are of the form

$$\bowtie F \triangleright F = \sum \triangleright F \bowtie F \bowtie F, \quad (3.44)$$

$$\triangleleft F \bowtie F = \sum \bowtie F \bowtie F \triangleleft F. \quad (3.45)$$

With this additional structure,  $\mathcal{R}$  is a  $(\mathcal{C}, \mathcal{D})$ -bimodule category.

Similar as to the string-net Hilbert space in the presence of an  $\mathcal{R}$  gapped boundary, the Hilbert space in the presence of an  $\mathcal{R}$  domain wall takes the form:



$$, \quad (3.46)$$

where the edges in red are valued in  $\text{ob}(\mathcal{C})$ .

Note that up to this point we have assumed  $\mathcal{C}$  and  $\mathcal{D}$  to be completely unrelated. In particular we have not assumed the topological orders  $\mathcal{Z}(\mathcal{C})$  and  $\mathcal{Z}(\mathcal{D})$  to be related in any way. Categories  $\mathcal{C}$  and  $\mathcal{D}$  for which  $\mathcal{Z}(\mathcal{C})$  and  $\mathcal{Z}(\mathcal{D})$  are equivalent as braided fusion categories are said to be *Morita equivalent*. Importantly, Morita equivalence admits an equivalent characterisation as the existence of an *invertible* bimodule category  $\mathcal{R}$  between  $\mathcal{C}$  and  $\mathcal{D}$ . Physically, such an invertible bimodule category describes a completely transparent gapped interface between the  $\mathcal{C}$ - and  $\mathcal{D}$ -string-nets for which all anyons in either model can tunnel through the domain wall without *condensing* on it. From the mathematical point of view, invertibility of a bimodule category can be deduced from the explicit matrix components of the bimodule associators  $\bowtie F$ , as proven in ref. [81].

Turning things around, for any fusion category  $\mathcal{C}$  and module category  $\mathcal{R}$ , there exists a *unique* fusion category  $\mathcal{D}$  for which  $\mathcal{R}$  can be endowed with the structure of an invertible  $(\mathcal{C}, \mathcal{D})$ -bimodule category. In category theoretic jargon, this fusion category  $\mathcal{D}$  is called the *Morita dual* category of  $\mathcal{C}$  with respect to  $\mathcal{R}$  and is often notated as  $\mathcal{D} = \mathcal{C}_{\mathcal{R}}^*$ . Vice versa also  $\mathcal{D}_{\mathcal{R}}^* = \mathcal{C}$ . By definition, the Morita dual  $\mathcal{C}_{\mathcal{R}}^*$  is the category of  $\mathcal{C}$ -module endofunctors of  $\mathcal{R}$ , denoted  $\text{Func}_{\mathcal{C}}(\mathcal{R}, \mathcal{R})$ . This is explained below in section 4.2 in more detail.

### Example

Consider the left  $\text{Vec}_G$ -module category  $\text{Vec}$ . The Morita dual  $(\text{Vec}_G)_{\text{Vec}}^*$  corresponds to  $\text{Rep}(G)$  [53]. Given a representation  $(V, \rho) \in \text{ob}(\text{Rep}(G))$ , and  $g \equiv \mathbb{C}_g \in \mathcal{I}_{\text{Vec}_G}$ , the  $\bowtie F$ -symbol evaluates to

$$(\bowtie F_{\mathbb{C}}^{g\mathbb{C}\rho})_{\mathbb{C},1j}^{\mathbb{C},i1} = \rho(g)_{ij}, \quad (3.47)$$

for basis vectors  $i, j$  in  $V$ .

## 3.4 Tensor network representations

### 3.4.1 Ground state representations

As anticipated in section 3.2, PEPS representations of string-net ground states are straightforward to construct since the ground state projector can be expressed as a product of local commuting projectors, namely  $\prod_{\mathbf{p}} B_{\mathbf{p}}^{\mathcal{D}}$ . To this end, one acts with the projector on an admissible reference state in the string-net Hilbert space, typically chosen as the one with the identity object  $\mathbb{1}_{\mathcal{D}}$  on every edge  $\mathbf{e} \in \mathbf{E}(\Lambda)$ . Explicitly, the resulting state is a superposition of all admissible string configurations, with amplitudes given by products of  $F$ -symbols, one for each vertex  $\mathbf{v} \in \mathbf{V}(\Lambda)$ . The local PEPS tensor therefore evaluates to an  $F$ -symbol multiplied by appropriate quantum dimension factors. These PEPS ground states are characterised by the existence of MPOs that live purely on the virtual level and can be freely moved through the lattice, without implementing a physical symmetry action. In fact, the (indecomposable) MPOs form a representation of the fusion rules of  $\mathcal{D}$  under multiplication, which is implemented locally via *fusion tensors* that intertwine them. Also the topological invariance of the MPOs in the ground states is characterised locally by a *pulling-through condition* satisfied by the PEPS and MPO tensors. Algebraically, this condition follows from the pentagon equation satisfied by the  $F$ -symbols. Since these MPOs are not related to any physical symmetry action, they survive small perturbations on the physical level. This construction of string-net ground states is covered in refs. [82–84], but turns out to not be the most general construction.

Indeed, provided the input category  $\mathcal{D}$ , distinct PEPS representations may exist [78]. In the tensor network framework, the defining feature of a string-net wave function is the existence of locally deformable MPO symmetries. As demonstrated in ref. [78], the collection of consistency equations that these PEPS and MPO tensors have to satisfy are precisely the six pentagon

equations associated to an invertible  $(\mathcal{C}, \mathcal{D})$ -bimodule category  $\mathcal{R}$ . For every such choice of  $\mathcal{R}$ , the corresponding PEPS tensors evaluate to the components of the  $\mathcal{R}$ -module associator  ${}^{\triangleleft}F$ , while the MPO symmetries now form a representation of the Morita dual category  $\mathcal{C} = \mathcal{D}_{\mathcal{R}}^*$ . Although the *physical* degrees of freedom are still specified by  $\mathcal{D}$ , the choice of  $\mathcal{R}$  together with  $\mathcal{D}$  defines the *virtual* level of the PEPS. The Morita equivalence between  $\mathcal{D}$  and  $\mathcal{C}$  is equivalent to *MPO injectivity*, a technical condition that guarantees a ground state degeneracy independent of system size [78, 81, 85, 86]. As in the canonical construction, these generalised string-net ground states can be obtained by acting on a reference product state with a ground state projector, now defined in terms of  $\mathcal{R}$  and denoted by  $\prod_{\mathbf{p}} B_{\mathbf{p}}^{\mathcal{R}}$ .

Let us unpack this construction in more detail. First, we consider for any (right)  $\mathcal{D}$ -module category  $\mathcal{R}$  the following *module* plaquette operators:

$$B_{\mathbf{p}}^{\mathcal{R}} = \sum_{R \in \mathcal{I}_{\mathcal{R}}} \frac{d_R}{\text{FPdim} \mathcal{R}} B_{\mathbf{p}}^R =: \text{[Diagram: a dashed blue circle with a central dot]} . \quad (3.48)$$

The action of  $B_{\mathbf{p}}^{\mathcal{R}}$  is now defined via the insertion of a loop labelled by  $R \in \mathcal{I}_{\mathcal{R}}$  in the punctured plaquette  $\mathbf{p}$  and fusing it into the physical lattice, thereby making use of eqs. (3.34), (3.39) and (3.41). From these local rules it also follows readily that these module plaquette operators together with the  $B_{\mathbf{p}}^{\mathcal{D}}$ 's and  $B_{\mathbf{p}}^{\mathcal{C}}$ 's as defined in eq. (3.23) obey

$$B_{\mathbf{p}}^{\mathcal{R}} \cdot B_{\mathbf{p}}^{\mathcal{C}} = B_{\mathbf{p}}^{\mathcal{R}}, \quad B_{\mathbf{p}}^{\mathcal{D}} \cdot B_{\mathbf{p}}^{\mathcal{R}} = B_{\mathbf{p}}^{\mathcal{R}}, \quad (3.49)$$

for every  $\mathbf{p}$ . Hence, acting with the projector  $\prod_{\mathbf{p}} B_{\mathbf{p}}^{\mathcal{R}}$  on any reference state yields a ground state of the  $\mathcal{D}$ -string-net model.<sup>5</sup> Note, however, that the reference state is now necessarily one in  $\mathcal{H}_{\mathcal{C}}(\Lambda)$ , for example one that corresponds to assigning  $\mathbb{1}_{\mathcal{C}}$  to all edges  $\mathbf{e} \in \mathbf{E}(\Lambda)$ .

Putting everything together, we can thus summarise the derivation of the ground state PEPS representation of the  $\mathcal{D}$ -string-net with a generic module

<sup>5</sup>By reversing the orientation of the loops in eq. (3.48) — or, equivalently, considering the module plaquette operators  $B_{\mathbf{p}}^{\mathcal{R}^{\text{op}}}$  for the *opposite module category*  $\mathcal{R}^{\text{op}}$  — one obtains a ground state projector for the  $\mathcal{C}$ -string-net model.

category as:

$$\begin{array}{c} \text{Diagram 1: A central shaded circle surrounded by a dashed blue hexagon, which is further enclosed by a dashed red hexagon.} \end{array} \xrightarrow{(3.41)} \sum \{d\} \begin{array}{c} \text{Diagram 2: A central shaded circle surrounded by a solid blue hexagon with vertices marked by grey dots. Each vertex has a small loop extending outwards.} \end{array} \quad (3.50)$$

$$\begin{array}{c} \text{Diagram 3: A central shaded circle surrounded by a solid black hexagon with vertices marked by grey dots. Each vertex has a line segment extending outwards.} \end{array} \xrightarrow[(3.39)]{(3.34)} \sum \{d\} \{F\} \quad (3.51)$$

Notably, all the wave function amplitudes derived via eq. (3.51) are written in terms of  $|\mathbf{V}(\Lambda)|$   $\triangleleft F$ -symbols. Their specific form invites us to consider tensors whose non-vanishing components are depicted by:

$$\begin{array}{c} \text{Diagram 4: A Y-junction with three legs labeled } R_1, R_2, R_3. \text{ The legs are colored blue, red, and blue respectively. The junction is labeled with } Y_1, Y_2, Y_3 \text{ and indices } i, j, k, l. \end{array} \equiv \begin{array}{c} \text{Diagram 5: A Y-junction with three legs labeled } (R_1 Y_1 R_3, i), (R_3 Y_2 R_2, j), \text{ and } (R_1 Y_3 R_2, l). \text{ The junction is labeled with } (Y_1 Y_2 Y_3, k). \end{array}, \quad (3.52)$$

$$= \frac{1}{\sqrt{d_{R_3}}} \left( \frac{d_{Y_1} d_{Y_2}}{d_{Y_3}} \right)^{1/4} (\triangleleft F_{R_2}^{R_1 Y_1 Y_2})_{R_3, ij}^{Y_3, kl}$$

as well as a mirrored version evaluating to the complex conjugate  $F$ -symbol. In what follows we typically suppress the orientation of the tensor legs. Note that every tensor index corresponds to a triple of objects and a multiplicity index. As per our convention that  $F$ -symbols for which not all fusion rules are fulfilled are zero, some of the objects need to be shared among different tensor legs for the component to be non-vanishing. Due to their internal structure, we thus refer to these kinds of tensors as *triple line tensors*. Contraction of these

tensors happens via identification of objects on legs of concatenated tensors, identification of basis vectors and summing over them. We also assume the convention that every closed loop in such a contraction is weighted with an additional factor of a quantum dimension [84].

It follows from the module pentagon equation eq. (3.35) that two distinct ways to contract these tensors are related via an  $F$ -symbol as follows:

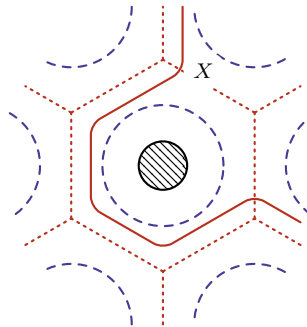
$$\begin{array}{c} Y_1 \quad Y_2 \quad Y_3 \\ \diagdown \quad \diagup \quad \diagup \\ \textcircled{i} \\ \diagup \quad \diagdown \quad \diagdown \\ Y_5 \quad \textcircled{j} \\ \diagup \quad \diagdown \quad \diagdown \\ Y_4 \end{array} = \sum_{Y_6, k, l} (F_{Y_4}^{Y_1 Y_2 Y_3})_{Y_5, i j}^{Y_6, k l} \begin{array}{c} Y_1 \quad Y_2 \quad Y_3 \\ \diagdown \quad \diagup \quad \diagup \\ \textcircled{k} \\ \diagup \quad \diagdown \quad \diagdown \\ Y_6 \quad \textcircled{l} \\ \diagup \quad \diagdown \quad \diagdown \\ Y_4 \end{array} . \quad (3.53)$$

The tensors eq. (3.52) can thus be viewed as an explicit representation of the graphical calculus of the fusion category  $\mathcal{D}$ .

Finally, it can be easily checked that contraction of tensors eq. (3.52) according to the connectivity of the underlying lattice  $\Lambda$  and weighted with the appropriate factors of quantum dimensions exactly reproduces the ground state eq. (3.51).

### 3.4.2 MPO symmetries

The PEPS ground state representation enjoys topological MPO symmetries acting only on the virtual level as foreshadowed above. Provided  $\mathcal{D}$  and  $\mathcal{R}$ , these topological symmetries are organised in the Morita dual category  $\mathcal{C} = \mathcal{D}_{\mathcal{R}}^*$ . In order to derive the MPO representation of these topological lines, we consider similarly as in eq. (3.51) the ground state projector acting on an empty string configuration with a loop labelled by an object  $X \in \text{ob}(\mathcal{C})$  inserted, for instance:



$$\text{Diagram (3.54)}$$

Note that the intersections of the  $X$ -labelled loop with the physical lattice are trivial. Topological invariance of these loops is guaranteed by the following

1631 *sliding property* of the ground state projectors:

$$(3.55)$$

1632 for every  $X \in \text{ob}(\mathcal{C})$ . This sliding property is immediately derived from the  
 1633 by now familiar graphical calculus. In a way the ground state projector can  
 1634 thus be interpreted as a defect that hides the punctures from the topological  
 1635 lines and is therefore sometimes referred to as a *cloaking* defect or boundary  
 1636 condition [87, 88], or *Kirby colour* [89].

1637 By carefully working out the recoupling of configurations with strings,  
 1638 one finds that in the presence of a loop the resulting PEPS ground state  
 1639 should be modified by inserting triple line tensors of the form

$$(3.56)$$

$$= \frac{1}{\sqrt{d_{R_1} d_{R_2}}} (\bowtie F_{R_2}^{X R_1 Y})_{R_3, i j}^{R_4, k l},$$

1640 along the path of the loop at every intersection with the physical lattice. The  
 1641 full MPO is then obtained by contracting these tensor along the ‘horizontal’  
 1642 direction. In fact, the simple objects  $X \in \mathcal{I}_{\mathcal{C}}$  and  $Y \in \mathcal{I}_{\mathcal{D}}$  label the *injective*  
 1643 *blocks* of these topological MPOs in either direction. It follows from the  
 1644 consistency equation eq. (3.45) that these tensors together with the ground  
 1645 state PEPS tensors satisfy the *pulling-through* or *zipper equation*:

$$(3.57)$$

1646 that does not modify the physical index  $i$ , so that these MPOs can move  
 1647 freely on the virtual level of the PEPS. In fact, by turning things around, one

can interpret the PEPS tensors as *intertwiners* for the MPO tensors in the ‘vertical’ direction.

Continuing along these lines, we can show that two MPOs labelled by  $X_1$  and  $X_2$  supported on homologous loops  $\ell_1 \sim \ell_2$  can be fused and decomposed according to  $X_1 \otimes X_2 = \sum_{X_3 \in \mathcal{I}_C} N_{X_1 X_2}^{X_3} X_3$ . This fusion product can be implemented *locally* in terms of *fusion* tensors defined in terms of the left module associator  $\triangleright F$ :

$$\begin{aligned}
 & \equiv \text{Fusion Tensor} \\
 & = \frac{1}{\sqrt{d_{R_3}}} \left( \frac{d_{X_1} d_{X_2}}{d_{X_3}} \right)^{1/4} (\triangleright F_{R_2}^{X_1 X_2 R_1})_{X_3, ij}^{R_3, kl},
 \end{aligned} \tag{3.58}$$

that together with the MPO tensors eq. (3.56) satisfy a pulling-through equation akin to a rotated version of eq. (3.57).

In summary, a  $\mathcal{D}$ -string-net PEPS representation fully exploits the structure provided by an invertible  $(\mathcal{C}, \mathcal{D})$ -bimodule category  $\mathcal{R}$ . The bimodule category encodes a collection of five unitary  $F$ -symbols in terms of which the PEPS tensors, MPO tensors and their fusion tensors are defined. These  $F$ -symbols simultaneously satisfy six coupled pentagon equations. Two of these pentagon equations involve only the associators of  $\mathcal{D}$  and  $\mathcal{C}$  respectively. Two others — eq. (3.35) and its left counterpart — involve also the left and right module associators and are interpreted as recoupling relations for the PEPS and fusion tensors akin to eq. (3.53). The remaining two involve the bimodule associator  $\bowtie F$  and are in turn interpreted as intertwining relations of the MPO tensors in either direction, as in eq. (3.57).

As an aside, let us note that techniques very similar to those presented in this chapter can be used to construct explicit PEPS representations of the gapped boundaries and domain walls discussed in section 3.3.3 [78]. In this way, one obtains a complete tensor network realisation of the Kitaev-Kong framework for gapped boundaries and interfaces in string-net models [76, 90].

The reader may also wonder how PEPS torus ground states of the same  $\mathcal{D}$ -string net model corresponding to different choices of module categories  $\mathcal{R}$  and  $\mathcal{R}'$  are related. As argued in ref. [78], one can construct MPO *intertwiners* that locally change the choice of module category by pulling this MPO through

the PEPS tensors. The existence of these intertwiners guarantees that the distinct PEPS representations are *locally* indistinguishable, but may differ *globally*. In the next chapter, these intertwiners are reviewed in detail and shown to implement *duality* transformations of the boundary theories.

### 3.4.3 Example: the toric code

Let us illustrate the above formalism for arguably the simplest non-chiral topological order: Kitaev's *toric code* [71]. Within the string-net formalism, the toric code model corresponds to the input category  $\text{Vec}_{\mathbb{Z}_2}$ . In virtue of the  $\text{Vec}_G$ -module category classification reviewed in section 3.3.2, there exist two module categories over  $\text{Vec}_{\mathbb{Z}_2}$ :  $\text{Vec}_{\mathbb{Z}_2}$  itself and  $\text{Vec}$ . For both cases the triple line tensors from which their PEPS ground states are constructed can be drastically simplified. In fact, they can be fully expressed in terms of *GHZ tensors* in the Pauli- $X$  and Pauli- $Z$  basis:

$$\begin{array}{c} \text{---} \cdot \cdot \cdot \text{---} \\ \diagup \quad \diagdown \\ \bigcirc \\ \diagdown \quad \diagup \\ \text{---} \cdot \cdot \cdot \text{---} \end{array} = |00 \cdots 0\rangle + |11 \cdots 1\rangle, \quad \begin{array}{c} \diagup \quad \diagdown \\ \bigoplus \\ \text{---} \end{array} = |+++\rangle + |--\rangle, \quad (3.59)$$

with  $\{|0\rangle, |1\rangle\}$  denoting the Pauli- $Z$  eigenbasis, and  $|\pm\rangle := 1/\sqrt{2}(|0\rangle \pm |1\rangle)$ . These tensors satisfy a number of invariance properties under conjugation by the Pauli matrices. Concretely, the first one is invariant under conjugation by two Pauli- $Z$  operators on any pair of legs, as well as simultaneous conjugation with Pauli- $X$  operators on all legs:

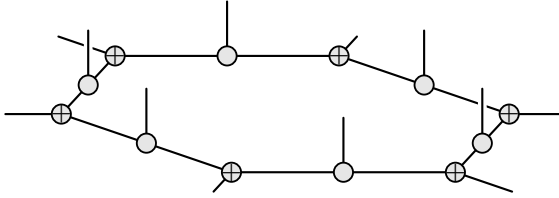
$$\begin{array}{c} Z \quad \cdot \cdot \cdot \\ \diagup \quad \diagdown \\ \bigcirc \\ \diagdown \quad \diagup \\ Z \quad \cdot \cdot \cdot \end{array} = \begin{array}{c} X \quad \cdot \cdot \cdot X \\ \diagup \quad \diagdown \\ \bigcirc \\ \diagdown \quad \diagup \\ X \quad \cdot \cdot \cdot X \end{array} = \begin{array}{c} \cdot \cdot \cdot \\ \diagup \quad \diagdown \\ \bigcirc \\ \diagdown \quad \diagup \\ \cdot \cdot \cdot \end{array}, \quad (3.60)$$

whereas the second tensor in eq. (3.59) has the same symmetries with  $X$  and  $Z$  interchanged. Using them, the toric code ground states for  $\mathcal{R} = \text{Vec}_{\mathbb{Z}_2}$  and  $\mathcal{R} = \text{Vec}$  are respectively written as:

$$\begin{array}{c} \text{---} \quad \text{---} \quad \text{---} \\ \diagup \quad \diagdown \quad \diagup \quad \diagdown \\ \bigoplus \quad \bigoplus \quad \bigoplus \\ \diagdown \quad \diagup \quad \diagdown \quad \diagup \\ \text{---} \quad \text{---} \quad \text{---} \end{array}, \quad (3.61)$$

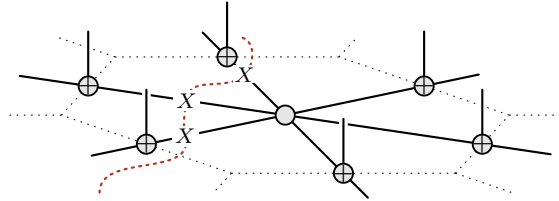


1698 and

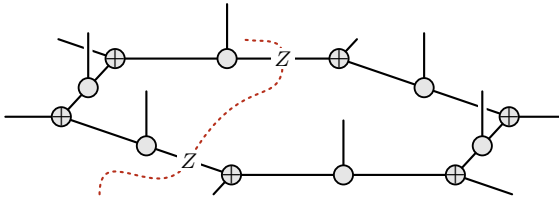

(3.62)

1699 The dotted lines in the first PEPS representation serve as a guide to the  
 1700 eye and represent the primal hexagonal lattice. It can be readily verified by  
 1701 making use of the symmetry properties eq. (3.60) that both states are indeed  
 1702 eigenstates of the toric code plaquette operators.

1703 Also the virtual MPO symmetries simplify drastically, as they have bond  
 1704 dimension 1 in both cases. In fact, they are implemented via loops of Pauli- $X$   
 1705 and  $-Z$  operators on the virtual level. For example:


(3.63)

1706 and


(3.64)

1707 The red dotted line in either figure traces out the path of the corresponding  
 1708 loop operator. The topological invariance of these loop operators follows  
 1709 directly from the symmetries of the tensors eq. (3.60).

### 1710 3.4.4 Anyonic excitations and the tube algebra

1711 We now proceed to discuss the construction of excited states and complete  
 1712 bases of torus ground states. As mentioned above, both excited states and  
 1713 distinct ground states can be obtained by studying the action of loop and  
 1714 string operators labelled by elements of  $\mathcal{Z}(\mathcal{D})$  on a reference ground state.  
 1715 From the perspective of tensor networks it is however more natural to describe  
 1716 these states in terms of the *tube algebra*. In fact, the tube algebra provides  
 1717 a natural framework for computing the Drinfeld centre of a fusion category

and the associated half-braidings that appear in the definition of string operators. Formally, this follows from the existence of a braided equivalence between the category of representations of the tube algebra and the Drinfeld centre [67, 70, 91–94].

Up to this point, we have been somewhat vague about the precise nature of the anyonic excitations appearing in the string-net models. Physically, anyons can be viewed as quasiparticle excitations whose local energy density is slightly higher than that of the ground state. They are always created in particle-antiparticle pairs on top of a reference torus ground state, a feature that also becomes manifest in the tensor network anyon ansatz discussed below. Since the (extended) string-net Hamiltonian is a local commuting-projector Hamiltonian, anyonic excitations correspond to violations of a small number of plaquette terms, vertex terms, or a mixture of both.

In essence, the idea underlying the tube algebra is to regard anyon excitations as different *ground state sectors* of the theory defined on a *punctured surface*, where the punctures coincide with the locations of the anyons, rather than viewing them as defects above a ground state on the unpunctured lattice. The ground space on such a punctured surface decomposes into sectors characterised by fixed boundary conditions around the punctures [84, 86, 94, 95]. These sectors form indecomposable *modules* under the action of the tube algebra, and the irreducible representations of the tube algebra precisely label the different anyon species. Since the tube algebra can be endowed with the structure of a (finite-dimensional)  $C^*$ -algebra, the existence of these irreducible modules is guaranteed by the *Artin-Wedderburn theorem*.

Let us now provide a qualitative definition of the tube algebra. As an algebra, it is spanned by basis elements that may be represented diagrammatically as cylinders decorated by  $\mathcal{C}$ -coloured graphs and containing a number of marked points on each boundary. Different choices for the number of marked boundary points give rise to different tube algebras. Although these algebras are not isomorphic, they are all *Morita equivalent*. Recall that Morita equivalence is a strictly weaker notion of equivalence than algebra isomorphism but guarantees that the corresponding categories of representations are equivalent, implying that all such tube algebras describe the same anyon theory. The multiplication of tubes is defined diagrammatically by stacking two decorated cylinders on top of one another and subsequently applying the graphical calculus to rewrite the resulting diagram as a linear combination of basis tubes. In this way, the structure factors are determined entirely by the associator of the underlying fusion category.

The attentive reader might have noticed that we discuss tube algebras based on  $\mathcal{C}$ , even though we argued that the ground states and excitations are labelled by  $\mathcal{Z}(\mathcal{D})$ . In virtue of  $\mathcal{Z}(\mathcal{C}) \simeq \mathcal{Z}(\mathcal{D})$ , the ground states and excitations that we construct momentarily thus physically correspond to the associated sectors of  $\mathcal{Z}(\mathcal{D})$  under this equivalence. In fact, this equivalence can be implemented via the aforementioned bimodule tubes.

Without loss of generality, we restrict ourselves to the tube algebra with a single boundary label on each side of the cylinder. Within the tensor network formalism, the basis elements of the tube algebra can be represented as matrix product operators on periodic boundary conditions of the following form [84–86]:

$$\mathfrak{T}^{A,A',X,X',i,j} := \text{Diagram} , \quad (3.65)$$

for all  $A, A', X, X' \in \mathcal{I}_{\mathcal{C}}$  such that  $X' \in X \otimes A$ ,  $X' \in A' \otimes X$  and where  $i \in \mathcal{C}(X', A' \otimes X)$  and  $j \in \mathcal{C}(X \otimes A, X')$ . In this figure, we explicitly show the arrows on the tensor legs, in preparation of the definition of the Hermitian conjugate below. Crucially, the multiplication rules of the tube algebra are independent of the number of MPO tensors appearing in eq. (3.65). This follows from the zipper equation eq. (3.57) that they satisfy. We then compute the product  $\mathfrak{T}^{A',A'',X_1,X'_1,i,j} \cdot \mathfrak{T}^{A,\tilde{A},X_2,X'_2,k,l}$ , which is non-vanishing only when  $A' = \tilde{A}$ , by recoupling the tensor network diagram

$$\delta_{A',\tilde{A}} \text{Diagram} , \quad (3.66)$$

1775 which yields:

$$\begin{aligned}
 & \mathfrak{T}^{A', A'', X_1, X'_1, i, j} \cdot \mathfrak{T}^{A, \bar{A}, X_2, X'_2, k, l} \\
 &= \sum_{\substack{X_3, X'_3 \\ m, n, o \\ p, q}} \sqrt{\frac{d_{X'_3} d_{A'} d_{X_1} d_{X_2}}{d_{X_3} d_{X'_1} d_{X'_2}}} (F_{X'_3}^{X_1 A' X_2})_{X'_1, jn}^{X'_2, lo} (\bar{F}_{X'_3}^{X_1 X_2 A})_{X_3, mq}^{X'_2, lo} \\
 & \quad \times (\bar{F}_{X'_3}^{A'' X_1 X_2})_{X'_1, in}^{X_2, mp} \cdot \mathfrak{T}^{A, A'', X_3, X'_3, p, q}.
 \end{aligned} \tag{3.67}$$

1776 Moreover, the algebra is closed under taking the *Hermitian conjugate* defined  
 1777 as:

$$(\mathfrak{T}^{A, A', X, X', i, j})^\dagger := \text{Diagram} \tag{3.68}$$

1778 We can express this conjugate in the basis of tubes eq. (3.65) by making  
 1779 use of the  $F$ -moves on the fusion tensors, as well as additional graphical  
 1780 identities for reinstating the orientation of the arrows on the tensor legs [86].  
 1781 Eventually, one finds

$$\begin{aligned}
 (\mathfrak{T}^{A, A', X, X', i, i'})^\dagger &= d_X^2 \sqrt{\frac{d_{X''}}{d_{X'}}} \sum_{\substack{j, j' \\ k, k'}} (F_{A'}^{A' X X})_{X', ik}^{\mathbb{1}, 11} (\bar{F}_{A'}^{X A \bar{X}})^{X'', jk'}_{X', i'k} \\
 & \quad \times (F_{\bar{X}}^{\bar{X} X \bar{X}})_{\mathbb{1}, 11}^{\mathbb{1}, 11} (\bar{F}_{X''}^{\bar{X} X X''})_{\mathbb{1}, 11}^{A', k'j'} \cdot \mathfrak{T}^{A', A, \bar{X}, X'', j, j'}
 \end{aligned} \tag{3.69}$$

1782 Altogether this endows the collection of tubes with the structure of a  $C^*$ -  
 1783 algebra.

1784 Invoking the Artin-Wedderburn theorem on this algebra, we are guaranteed  
 1785 the existence of an isomorphism between the tube algebra and a direct  
 1786 sum of simple matrix algebras, the number of which is equal to the simple  
 1787 objects in  $\mathcal{Z}(\mathcal{C})$ . Concretely, we can construct linear combinations of tubes  
 1788 that behave as *matrix units* within each block under the multiplication  
 1789 eq. (3.67) [86, 92, 96, 97], viz.:

$$e^{Z, A_i, B_j} \cdot e^{Z', C_k, D_l} = \delta_{Z, Z'} \delta_{B, C} \delta_{j, k} e^{Z, A_i, D_l}, \tag{3.70}$$

1790 where  $Z, Z'$  label simple objects of  $\mathcal{Z}(\mathcal{C})$ , and  $A, B$  ( $C, D$ ) denote the simple  
 1791 objects appearing in the decomposition of  $Z$  ( $Z'$ ) as object of  $\mathcal{C}$ , with  $i, j$

( $k, l$ ) denoting their multiplicity within this decomposition. Under taking the conjugate, they are transposed according to

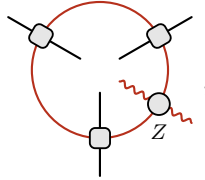
$$(e^{Z, A_i, B_j})^\dagger = e^{Z, B_j, A_i}. \quad (3.71)$$

Notably, matrix units of the form  $e^{Z, A_i, A_i}$  are (*simple*) *idempotents*, while off-diagonal elements  $e^{Z, A_i, B_j}$ ,  $A_i \neq B_j$  are *nilpotents*. Summing over all simple idempotents within the same block provides the *minimal central idempotents*

$$\mathcal{P}^Z := \sum_{A_i} e^{Z, A_i, A_i}. \quad (3.72)$$

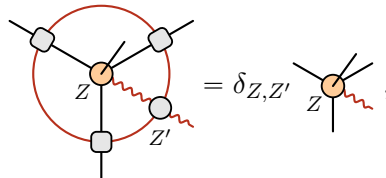
Connecting back to the string operators, it can be shown that the coefficients of the central idempotents when expressed in the basis eq. (3.65) exactly coincide with the coefficients of the half-braiding isomorphisms eq. (3.26)  $\mathcal{Z}(\mathcal{C})$  [92, 95, 97].

Let us denote the minimal central idempotents now as follows



$$, \quad (3.73)$$

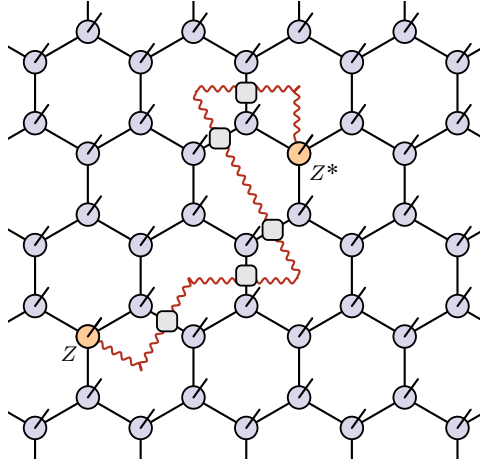
wherein compared to eq. (3.65) we have explicitly depicted the periodic boundary condition, and conflated the fusion tensors into one tensor labelled by  $Z$ . The anyon ansatz of ref. [86] now stipulates that in order to define a  $(Z, Z^*)$ -anyon-anti-anyon pair, one replaces one of the PEPS ground state tensors by a new type of tensor supported on the image of the corresponding idempotent, interpreted as a matrix from the ‘outer’ to the ‘inner’ indices, i.e.:



$$= \delta_{Z, Z'} \text{ (network with bottom vertex } Z \text{)}, \quad (3.74)$$

and analogously for  $Z^*$ , with the additional requirement that the pair  $(Z, Z^*)$  is in the vacuum sector, as spelled out in detail in ref. [86]. Putting everything

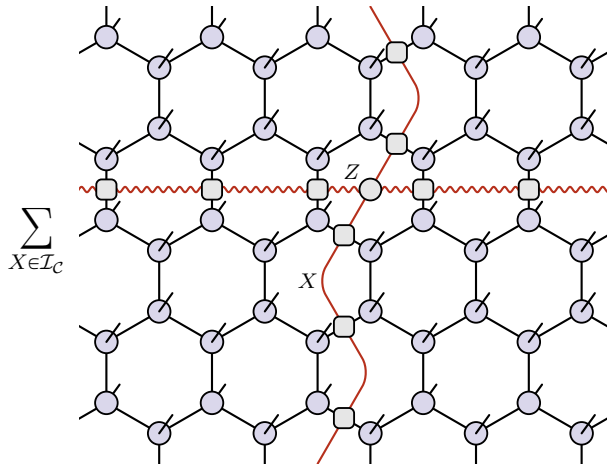
together, we could depict such a state as:



(3.75)

Let us stress once again that the location of the string connecting the anyons is not detectable as it can be freely moved through the lattice in virtue of the pulling-through condition eq. (3.57). With this we conclude our discussion of the anyon ansatz, but let us remark that from it, the anyonic braiding statistics and the fusion rules of the anyons, as encoded in  $\mathcal{Z}(\mathcal{C})$ , can be computed directly, as demonstrated in ref. [86].

Given these tubes, a complete basis for the  $\mathcal{D}$ -string-net torus ground space is also straightforward to construct. For each anyon  $Z \in \mathcal{I}_{\mathcal{Z}(\mathcal{C})}$  one constructs a ground state with an anyon flux  $Z$  threading the ‘vertical’ cycle of the torus by inserting the corresponding tube and closing it. Schematically:



(3.76)

1822 Note that we also could have used a simple idempotent to construct a ground  
1823 state. As argued in ref. [86], all simple idempotents contained in the same  
1824 central idempotent yield the same ground state.





# Generalised symmetries and dualities

In this final introductory chapter, we study the *edge Hamiltonians* of two-dimensional string-net models and three-dimensional (*twisted*) *Dijkgraaf–Witten theories* [98, 99]. As discussed in the previous chapter, the *virtual* symmetries of topological models can exhibit a non-invertible fusion structure. Consequently, these lower-dimensional edge theories inherit the corresponding symmetry structures as genuine *internal symmetries*.

In recent years, this holographic perspective on generalised symmetries has appeared in the literature under various names, including *SymTFT* [100–102], *topological holography* [103–105], the *strange correlator* construction [106–108], or simply *symmetric tensor networks* [109, 110]. The principal advantage of this approach is that it separates the *dynamical* and *topological aspects* of the boundary theory while making manifest the modern viewpoint that a global symmetry is defined by the existence of a collection of topological defects [69, 111, 112]. In particular, this framework enables us to employ the toolbox of *topological quantum field theory* to analyse these one-lower-dimensional edge theories. As argued in the previous chapter, tensor networks provide an especially powerful framework for realising such generalised symmetry structures, as the correlated action of these operators on neighbouring sites can be encoded *locally* within the tensor. Moreover, this formulation allows these symmetries to be defined directly in the thermodynamic limit and for arbitrary boundary conditions.

We begin by reviewing the representation theory of injective (non-invertible) MPO symmetries in terms of module categories, and the construction of generic Hamiltonians that commute with them. We then turn to the study of

*dualities* in these systems, and formulate a *generalised Landau paradigm* for gapped phases protected by categorical symmetries [32, 54–57]. As advocated in chapter 8, such dualities can always be interpreted as *gauging* an appropriate collection of MPO symmetries, followed by local *disentangling* unitary transformations. We subsequently incorporate *symmetry-twisted boundary conditions* into the framework in order to construct *isometric* duality maps between dual models.

Finally, we illustrate the two-dimensional Kramers–Wannier duality and argue that its dual symmetry is described by a genuine *fusion 2-category*, namely  $2\text{Rep}(\mathbb{Z}_2)$ . In the concluding section, we explore in more detail the general symmetry structures encoded in fusion 2-categories, with a particular emphasis on  $2\text{Vec}_G$  and  $2\text{Rep}(G)$ , as well as their duality transformations.

## 4.1 MPO representations of categorical symmetries

Reversing the logic of the previous chapter, we are now in a position to construct distinct injective MPO representations of a unitary fusion category  $\mathcal{C}$  via a choice of (left)  $\mathcal{C}$ -module category  $\mathcal{R}$ . Indeed, for any module category  $\mathcal{R}$  over  $\mathcal{C}$ , there exists a unique Morita dual  $\mathcal{D}$  such that  $\mathcal{R}$  is endowed with the structure of an invertible  $(\mathcal{C}, \mathcal{D})$ -bimodule category, as discussed in the previous chapter. From the bimodule associators  ${}^{\triangleright}F$  we can construct four-valent MPO tensors such that on periodic boundary conditions and independent of system size we have the graphical identities

$$\begin{array}{c} \text{Diagram 1: A vertical line with three horizontal red lines passing through it. Each intersection is marked with a small square. The top horizontal line is labeled  $X_1$  and the bottom horizontal line is labeled  $X_2$ . } \\ \text{Diagram 2: A sum over  $X_3$  of  $N_{X_1 X_2}^{X_3}$  times a diagram with three vertical lines and three horizontal red lines. The top horizontal line is labeled  $X_3$ . } \end{array} = \sum_{X_3} N_{X_1 X_2}^{X_3} \begin{array}{c} \text{Diagram 3: Three vertical lines with three horizontal red lines. The top horizontal line is labeled  $X_3$ . } \end{array}, \quad (4.1)$$

for structure constants  $N$  coinciding with the fusion rules of  $\mathcal{C}$ . In virtue of the fundamental theorem eq. (2.13), there exists  $N_{X_1 X_2}^{X_3}$  linearly independent *fusion tensors* that locally implement these fusion rules via

$$\begin{array}{c} \text{Diagram 1: A vertical line with two horizontal red lines passing through it. Each intersection is marked with a small square. The top horizontal line is labeled  $X_1$  and the bottom horizontal line is labeled  $X_2$ . A red line labeled  $X_3$  enters from the left, loops around the bottom square, and exits to the right. } \\ \text{Diagram 2: A vertical line with two horizontal red lines passing through it. Each intersection is marked with a small square. The top horizontal line is labeled  $X_1$  and the bottom horizontal line is labeled  $X_2$ . A red line labeled  $X_3$  enters from the left, passes through the bottom square, and exits to the right. } \end{array} = \begin{array}{c} \text{Diagram 3: A vertical line with two horizontal red lines passing through it. Each intersection is marked with a small square. The top horizontal line is labeled  $X_1$  and the bottom horizontal line is labeled  $X_2$ . A red line labeled  $X_3$  enters from the left, passes through the bottom square, and exits to the right. } \end{array}. \quad (4.2)$$

These fusion tensors are constructed from the  ${}^{\triangleright}F$ -symbols, and are themselves recoupled via the  $F$ -symbols of  $\mathcal{C}$ .

Note that in sharp contrast to the group case, generic categorical MPO symmetries are defined on *constrained* Hilbert spaces. These constraints are implemented via the module action  $\triangleleft : \mathcal{R} \times \mathcal{D} \rightarrow \mathcal{R}$ , as detailed in chapter 8. In what follows we therefore sometimes refer to  $\mathcal{R}$  as describing the *kinematical degrees of freedom* of the model. In fact, the  $\mathcal{C}$ -symmetric models that we consider below can be thought of as a tensor network representation and a generalisation of the *anyonic spin chain construction* first considered in ref. [113].

**Example: on-site group symmetries** Recall that  $\text{Vec}$  can be regarded as an invertible  $(\text{Vec}_G, \text{Rep}(G))$ -bimodule category. The corresponding MPOs eq. (4.1) in this case reduce to an on-site (bond dimension 1) representation of  $G$ :

$$\begin{array}{c} \text{---} \square \text{---} \square \text{---} \square \text{---} \\ | \quad | \quad | \\ u_{i-1} \quad u_i \quad u_{i+1} \end{array} \stackrel{g}{=} \bigotimes_i u_i(g). \quad (4.3)$$

## 4.2 Symmetric boundary Hamiltonians and dualities

Having these MPO representations of categorical symmetries at our disposal, we now proceed to constructing local  $(1+1)d$  Hamiltonians that commute with these symmetries.

Before proceeding to the construction of Hamiltonians with a genuine categorical symmetry  $\mathcal{C}$ , let us first specialise to the group case to attain some intuition and put things in a perhaps more familiar context. Considering an *irreducible* unitary representation  $(V, \alpha)$  of a group  $G$ , and a rank four tensor — representing for example a local nearest-neighbour Hamiltonian term — whose domain and codomain are the tensor product space  $V \otimes V$ . Invariance under the action of  $G$  translates graphically to

$$\begin{array}{c} \alpha(g) \quad \alpha(g) \\ | \quad | \\ \boxed{h} \\ | \quad | \\ \alpha(g)^\dagger \quad \alpha(g)^\dagger \end{array} = \begin{array}{c} | \quad | \\ \boxed{h} \\ | \quad | \end{array}, \quad \forall g \in G. \quad (4.4)$$

The *Wigner-Eckart* theorem then stipulates that the tensor  $h$  can be written as the contraction of two Clebsch-Gordan coefficients implementing the unitary basis transformation

$$\begin{bmatrix} \alpha & \alpha \\ - & - \end{bmatrix} \begin{bmatrix} \beta & k \\ - & - \end{bmatrix} : V \otimes V \rightarrow W, \quad k = 1, 2, \dots, N_{\alpha\alpha}^\beta, \quad (4.5)$$

for all irreps  $(W, \beta)$  appearing in the decomposition  $\alpha \otimes \alpha \simeq \bigoplus_{\beta} N_{\alpha\alpha}^{\beta} \beta$ , and *reduced matrix elements* depending only on the irrep labels [114, 115]. Concretely putting things together, we find that  $h$  can be expressed in its components as:

$$\begin{array}{c} i \quad i' \\ | \quad | \\ \boxed{h} \\ | \quad | \\ j \quad j' \end{array} = \sum_{\beta, k} h(\beta, k) \left( \sum_l \begin{bmatrix} \alpha & \alpha & \beta, k \\ i & i' & l \end{bmatrix} \begin{bmatrix} \alpha & \alpha & \beta, k \\ j & j' & l \end{bmatrix}^* \right) |i, i' \times j, j'|, \quad (4.6)$$

for complex coefficients  $h(\beta, k)$  and  $i, i', j, j' = 1, 2, \dots, \dim_{\mathbb{C}} \alpha$ . The explicit form of the tensor  $h$  in terms of Clebsch-Gordan coefficients invites us to equivalently express it in terms of the fusion tensors eq. (3.52) specialised to the case  $\mathcal{C} = \text{Vec}_G$ ,  $\mathcal{R} = \text{Vec}$ ,  $\mathcal{D} = (\text{Vec}_G)^*_{\text{Vec}} = \text{Rep}(G)$ :

$$\begin{array}{c} | \quad | \\ \boxed{h} \\ | \quad | \end{array} = \sum_{\beta, k} h(\beta, k) \begin{array}{c} \alpha \quad \alpha \\ | \quad | \\ \textcircled{k} \\ | \\ \beta \\ | \\ \textcircled{k} \\ | \quad | \\ \alpha \quad \alpha \end{array}, \quad (4.7)$$

where we have left the module labels implicit.<sup>1</sup> Notably, in this more abstract formulation the origin of the invariance of  $h$  under the  $G$ -action can be traced back to the pulling-through property eq. (3.57). Indeed, remember from section 4.1 that for the case at hand the symmetry MPOs exactly implement the on-site group symmetry, so that eq. (4.4) boils down to two applications of the pulling-through equation. We refer the reader to refs. [39, 57, 97, 116] for a number of concrete examples.

Continuing along these lines, we ought to consider the triple line tensors eq. (3.52) evaluating to  ${}^{\triangleleft}F$  as the elementary  $\mathcal{C}$ -symmetric tensors, or *generalised* Clebsch-Gordan coefficients [81]. For these, the symmetry is implemented via the pulling-through equation eq. (3.57) and, using them, local symmetric Hamiltonians can be constructed. Viewed from this perspective, the topological invariance of the MPO symmetries guaranteed by the pulling-through equation can be seen as the *defining* property of a *generalised, categorical symmetry*. This construction of symmetric Hamiltonians in terms of tensor networks thereby provides an explicit one-dimensional lattice realisation of the viewpoint advocated first in ref. [69] to identify the existence of a collection of topological operators as the definition of a *global* symmetry.

<sup>1</sup>Note that compared to eq. (3.52) we also dropped factors of quantum dimension for ease of notation.



even though the ground state entanglement may differ significantly. On the level of the local symmetric operators eq. (4.8) this boils down to changing the module associators to which the symmetric tensors evaluate, while preserving the reduced matrix elements  $h$ . Crucially, the dual symmetry can be a genuine categorical symmetry even when the initial symmetry structure is a group, as exemplified by  $(\text{Vec}_G)_{\text{Vec}}^* = \text{Rep}(G)$  for any non-abelian group  $G$ . Notwithstanding the abstract nature of the formalism, most well-known duality transformations in the condensed matter and high-energy physics literature — in particular Kramers–Wannier and Kennedy–Tasaki duality — can be rephrased in this language given an appropriate choice of symmetry of the model under scrutiny [97, 116, 122–126]. Moreover, the extension of this duality framework to the two-dimensional setting follows a completely analogous logic as presented below.

**Example: Kramers–Wannier duality – Part I** Let us consider the transverse field Ising model encountered above in eq. (1.11). As already mentioned, the model enjoys a  $\mathbb{Z}_2$  spin-flip symmetry acting on-site and generated by  $\prod_i X_i$ . Abstractly, the symmetry is encoded in the fusion category  $\mathcal{C} = \text{Vec}_{\mathbb{Z}_2}$ . It is then straightforward to verify that the Hamiltonian terms can be represented on the Hilbert space  $\mathbb{C}[\mathbb{Z}_2]^{\otimes L}$ ,  $\mathbb{C}[\mathbb{Z}_2] = \text{Span}_{\mathbb{C}}\{|0\rangle, |1\rangle\}$ , naturally associated to the choice  $\mathcal{R} = \text{Vec}_{\mathbb{Z}_2}$ , and explicitly read:

$$Z_i Z_{i+1} = \sum_{g_i, g_{i+1}} (-1)^{g_{i+1}} \begin{array}{c} g_i \quad g_{i+1} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ g_i \quad g_{i+1} \end{array}, \quad X_i = \sum_{g_i, g_{i+1}} \begin{array}{c} g_{i+1} \quad g_{i+1}+1 \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ g_i \quad g_{i+1} \end{array}. \quad (4.10)$$

Recall that  $\text{Vec}_{\mathbb{Z}_2}$  only admits  $\text{Vec}$  as non-trivial module category. Hence, the only non-trivial duality transformation amounts to changing  $\text{Vec}_{\mathbb{Z}_2} \mapsto \text{Vec}$ . It is straightforward to derive that upon this change, the Hamiltonian terms transmute according to the prescription

$$Z_i Z_{i+1} \mapsto Z_{i+\frac{1}{2}}, \quad X_i \mapsto X_{i-\frac{1}{2}} X_{i+\frac{1}{2}}. \quad (4.11)$$

Notably, the dual symmetry is generated by  $\prod_i Z_{i+1/2}$ , described by the symmetry category  $\text{Rep}(\mathbb{Z}_2) = (\text{Vec}_{\mathbb{Z}_2})_{\text{Vec}}^*$ , in accordance with the prediction of the framework. Also, the order and disorder operators are mapped onto each other

$$Z_i Z_j \mapsto \prod_{k=i}^{j-1} Z_{k+\frac{1}{2}}, \quad \prod_{k=i}^j X_k \mapsto X_{i-\frac{1}{2}} X_{j+\frac{1}{2}}, \quad (4.12)$$



the category of  $\mathcal{D}$ -module *endofunctors* of  $\mathcal{R}$ ,  $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R})$ , whose fusion structure is defined by the composition of module functors. Similarly, the category  $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$  is naturally endowed with the structure of an invertible  $(\mathcal{D}_{\mathcal{R}'}^{\star}, \mathcal{D}_{\mathcal{R}}^{\star})$ -bimodule category.

Provided a (simple) object  $X$  in  $\mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')}$ , we construct the MPO tensor of the duality intertwiner as

$$\begin{aligned}
 \begin{array}{c} \text{Diagram 1: A box labeled } X \text{ with four legs. Top-left: } R'_1 \text{ (blue), top-right: } R'_2 \text{ (blue), bottom-left: } R_1 \text{ (blue), bottom-right: } R_2 \text{ (blue). Left leg has index } i \text{ (blue), right leg has index } l \text{ (blue). Top leg has index } j \text{ (blue), bottom leg has index } k \text{ (blue).} \end{array} \equiv \begin{array}{c} \text{Diagram 2: A box labeled } X \text{ with four legs. Top: } (R'_1 Y R'_2, j) \text{ (blue), bottom: } (R_1 Y R_2, k) \text{ (blue), left: } (R_1 X R'_1, i) \text{ (red), right: } (R'_2 X R_2, l) \text{ (red).} \end{array} \quad (4.15) \\
 = \left( {}^X \omega_{R'_2}^{R_1 Y} \right)_{R'_1, ij}^{R_2, kl}.
 \end{aligned}$$

Note that we have indicated the change in module category by coupons in the second depiction of the tensor.

Finally, the duality transformation eq. (4.13) then follows from the intertwining property

$$\begin{array}{c} \text{Diagram 3: A box labeled } X \text{ with four legs. Top-left: } Y_1 \text{ (blue), top-right: } Y_2 \text{ (blue), bottom-left: } Y_3 \text{ (blue), bottom-right: } Y_4 \text{ (blue). Left leg has index } i \text{ (blue), right leg has index } j \text{ (blue).} \end{array} = \begin{array}{c} \text{Diagram 4: A box labeled } X \text{ with four legs. Top-left: } Y_1 \text{ (blue), top-right: } Y_2 \text{ (blue), bottom-left: } Y_3 \text{ (blue), bottom-right: } Y_4 \text{ (blue). Left leg has index } i \text{ (blue), right leg has index } j \text{ (blue).} \end{array}, \quad (4.16)$$

which is a graphical representation of the aforementioned coherence axiom satisfied by  $\omega$ .

This concludes our discussion on the transformation of local symmetric operators under duality. It remains to understand the fate of *charged* local operators. As is already apparent in the example of Kramers-Wannier duality, local charged operators are not typically mapped to *local* operators but rather to *string* operators. In fact, given a local order parameter in a given phase, a duality transformation can be used to obtain the disorder parameter in the dual phase [25, 32]. We will not pursue this direction further here, but we refer to chapter 10 for more concrete examples when the duality amounts to a *twisted gauging* procedure, and to ref. [127] for recent developments on order parameters in one-dimensional lattice models with categorical symmetries.

Collecting all these structures in a table:



| Physical notion                     | Categorical notion                               | Notation                                              |
|-------------------------------------|--------------------------------------------------|-------------------------------------------------------|
| Algebra local symmetric operators   | Fusion category                                  | $\mathcal{D}$                                         |
| Kinematical degrees of freedom      | $\mathcal{D}$ -module category                   | $\mathcal{R}$                                         |
| Global symmetry original model      | Dual fusion category                             | $\mathcal{C} = \mathcal{D}_{\mathcal{R}}^{\star}$     |
| Duality operators                   | $(\mathcal{C}', \mathcal{C})$ -bimodule category | $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$ |
| Dual kinematical degrees of freedom | $\mathcal{D}$ -module category                   | $\mathcal{R}'$                                        |
| Dual global symmetry                | Dual fusion category                             | $\mathcal{C}' = \mathcal{D}_{\mathcal{R}'}^{\star}$   |

Table 4.1: Dictionary of the key physical concepts, corresponding category-theoretical structures, and their notation used throughout the thesis.

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**Example: Kramers-Wannier duality – Part II** This leaves us to explain how the mapping of  $\mathbb{Z}_2$ -symmetric terms under the Kramers-Wannier duality eq. (4.11) fits into this framework. It follows from  $\text{Fun}_{\text{Vec}_{\mathbb{Z}_2}}(\text{Vec}_{\mathbb{Z}_2}, \text{Vec}) \simeq \text{Vec}$  that there exists a unique injective MPO intertwiner implementing the duality  $\text{Vec}_{\mathbb{Z}_2} \rightarrow \text{Vec}$ .<sup>2</sup> Removing the trivial lines and Hom spaces from the module structure eq. (4.15) for the case at hand and rewriting the MPO in more conventional tensor network notation, we obtain:

$$D_{\text{KW}} = \text{---} \begin{array}{c} \square \\ | \\ \square \\ | \\ \square \\ | \\ \square \\ | \\ \square \\ | \\ \square \end{array} \text{---} , \quad (4.17)$$

$\begin{array}{c} \boxed{\text{Vec}} \\ \boxed{\text{Vec}_{\mathbb{Z}_2}} \end{array}$

where the constituent tensors are GHZ tensors in the Pauli-Z and -X basis, as defined in eq. (3.59) earlier:

$$\text{---} \begin{array}{c} \square \\ | \end{array} \text{---} = |000\rangle + |111\rangle, \quad \text{---} \begin{array}{c} | \\ \square \end{array} \text{---} = |+++ \rangle + |-- \rangle. \quad (4.18)$$

Note in particular that the physical legs on either side of the intertwiner are shifted with half a lattice site with respect to each other, in accordance with the expected mapping of operators eq. (4.11). One subsequently finds that the mapping of symmetric operators eq. (4.11) follows directly from the local symmetry properties eq. (3.60).

<sup>2</sup>It holds more generally that  $\text{Fun}_{\mathcal{D}}(\mathcal{D}, \mathcal{R}) \simeq \mathcal{R}$ . Concretely, the simple module functors in  $\text{Fun}_{\mathcal{D}}(\mathcal{D}, \mathcal{R}) \simeq \mathcal{R}$  are in one-to-one correspondence with simple objects  $R$  in  $\mathcal{I}_{\mathcal{R}}$  such that  $R_{\omega} = {}^{\omega}F^{R--}$ .

### 4.3 The Landau paradigm for categorical symmetries

We are thus finally led to the problem of classifying gapped phases of matter protected by a categorical symmetry in one-dimensional lattice models. Returning to the example of the Ising model of the previous section, we observe that under Kramers-Wannier duality the symmetric and the symmetry-broken phase are interchanged. More generally, examples such as the Kennedy-Tasaki duality discussed at length in chapter 10 illustrate that for any gapped phase there exists a *unique* duality transformation that can map it to a dual model whose *dual symmetry* is *spontaneously broken*. This suggests a one-to-one correspondence between *module categories*, labelling duality transformations, and gapped phases of matter.

This observation is formalised by the *generalised Landau paradigm* for one-dimensional gapped phases protected by a symmetry encoded in a UFC  $\mathcal{C}$  [32, 54–57]. The paradigm asserts that gapped phases are in one-to-one correspondence with module categories  $\mathcal{P}$  over the symmetry category  $\mathcal{C}$ . The simple objects of the module category  $\mathcal{P}$  label the ground states on periodic boundary conditions, while the module action encodes the action of the symmetry on these ground states. Crucially, for genuinely non-invertible symmetries, the action of a symmetry operator on a given ground state does not in general produce another unique ground state. Instead, it yields a *superposition* of ground states, in sharp contrast with the ordinary invertible case.

The generalised Landau paradigm has been established rigorously in the context of matrix product states in ref. [55]. Their arguments make use of tensor network techniques similar to those employed in section 2.4 for the classification of gapped phases protected by a group symmetry. Notably, in the case of the symmetry category  $\mathcal{C} = \text{Vec}_G$ , the classification in terms of module categories exactly reproduces the conventional Landau classification in terms of subgroups  $H \subseteq G$  and 2-cocycles  $[\psi] \in H^2(H, \text{U}(1))$  as already anticipated in section 2.4.

Within this framework, a symmetry  $\mathcal{C}$  is sometimes said to admit an *SPT phase* if there exists a module category over it that is equivalent to  $\text{Vec}$  [54, 128–131]. Indeed, such a module category would describe a phase characterised by a unique ground state on periodic boundary conditions; a direct analogue of the hallmark of an SPT phase protected by an ordinary group symmetry. The category  $\mathcal{C}$  is then necessarily the representation category  $\text{Rep}(\mathcal{A})$  of a finite-dimensional semisimple *Hopf algebra*  $\mathcal{A}$  [53].<sup>3</sup>

<sup>3</sup>Notably, every fusion category can be realised as the representation category of a *weak* Hopf algebra.

Consequently, simple objects of such a fusion category have *integer* quantum dimensions. The converse, however, is not true: the fusion category  $\text{Vec}_G^\omega$  has integer quantum dimensions, yet it does not admit  $\text{Vec}$  as module category when the class  $[\omega]$  is non-trivial. An equivalent characterisation of such a symmetry category is then to say that it is (*t Hooft*) *anomaly-free*. By this we mean that the symmetry can be fully *gauged* in the sense of ref. [132] and chapter 8. As demonstrated in the latter, this implies the existence of an injective MPS, symmetric under the full symmetry category  $\mathcal{C}$ .

Two of the most important applications of the generalised Landau paradigm are as follows. First, it implies that there always exists a dual description of a given phase in which the entanglement is minimised. Indeed, in the spontaneously broken dual model, symmetry-protected degeneracies in the Schmidt spectrum are lifted, thereby maximally reducing the redundancy in the matrix product state description of both the ground states and the low-lying excitations. This has been demonstrated in ref. [57].

Second, in the case of a continuous  $SU(2)$  symmetry, where the *Mermin-Wagner theorem* forbids spontaneous symmetry breaking, a duality transformation can map the system to a dual model with a  $\text{Rep}(SU(2))$  symmetry. Since this dual symmetry is discrete rather than continuous, the assumptions underlying the Mermin–Wagner theorem no longer apply, allowing for symmetry-broken phases in the dual description. In fact, this exactly underlies standard numerical tensor network methods that exploit continuous symmetries [57, 109, 110].

## 4.4 Symmetry-twisted boundary conditions

Up until this point, we have considered only periodic boundary conditions. Notably, duality MPOs typically have a non-trivial kernel when defined on periodic boundary conditions and are therefore *non-invertible* linear operators. This was already observed in the example of the Kramers–Wannier duality of the Ising model. In that case, the duality projects out states with non-trivial charge under the spin-flip symmetry.

To access and relate the full spectrum of dual theories, it is necessary to consider *symmetry-twisted boundary conditions* [97, 133–136]. A symmetry twist can heuristically be understood as a spatially localised defect that extends along the time direction. Within our framework, symmetry twists are incorporated as additional degrees of freedom of the Hilbert space [97, 135]. They are conventionally inserted between sites  $L$  and  $L + 1 \sim 1$ , and are labelled by objects in the symmetry fusion category  $\mathcal{C}$ . In their presence,

with  $X \in \mathcal{I}_{\mathcal{C}}$ . The full Hamiltonian can hence be written as  $H^{\mathcal{R},X} = \sum_{i \neq L} h_{i,i+1}^{\mathcal{R}} + h_{L,1}^{\mathcal{R},X}$ . Notably, these symmetry twists are *topological* in the sense that their position can be shifted by one lattice site via a local unitary transformation. A duality still corresponds to a change of module category  $\mathcal{R} \mapsto \mathcal{R}'$ , and the corresponding intertwining MPOs are now promoted to *bimodule tubes* that can act on the symmetry twist [96, 137]. Focusing for concreteness on the case where one side of the duality corresponds to the regular module category, and using the equivalence  $\text{Fun}_{\mathcal{D}}(\mathcal{D}, \mathcal{R}) \simeq \mathcal{R}$ , these tubes reduce to

$$\mathfrak{Z}_{D|\mathcal{R}}^{Y,X,R,R',i,j} := \text{Diagram} \quad (4.20)$$

The collection of all tubes  $\{\mathfrak{T}_{\mathcal{R}|\mathcal{R}}, \mathfrak{T}_{\mathcal{R}|\mathcal{R}'}, \mathfrak{T}_{\mathcal{R}'|\mathcal{R}}, \mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}\}$  constitutes an enlarged tube algebra which can be shown to also form a finite-dimensional  $C^*$ -algebra. Consequently, it admits an Artin-Wedderburn decomposition. In particular, the collections of  $\{\mathfrak{T}_{\mathcal{R}|\mathcal{R}}\}$  and  $\{\mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}\}$  tubes furnish subalgebras that encode the symmetry of either model in the presence of twists. Their corresponding matrix units coincide with those discussed in section 3.4.4. The associated central idempotents are said to label the *topological sectors* of either model, and by virtue of the Morita equivalence between  $\mathcal{D}_{\mathcal{R}}^*$  and  $\mathcal{D}_{\mathcal{R}'}^*$ , characterised by  $\mathcal{Z}(\mathcal{D}_{\mathcal{R}}^*) \simeq \mathcal{Z}(\mathcal{D}_{\mathcal{R}'}^*)$ , these sectors are in bijection with one another.

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a (typically non-simple) boundary condition, as well as a charge of the sub-symmetry that preserves that boundary condition. For any  $Z$  in  $\mathcal{Z}(\mathcal{D}_{\mathcal{R}}^*)$  that contains a given  $X \in \mathcal{I}_{\mathcal{D}_{\mathcal{R}}^*}$  in its decomposition as object of  $\mathcal{D}_{\mathcal{R}}^*$ , the Hamiltonian in the topological sector  $Z$  with multiplicity  $i$  is defined via the simple idempotents as

$$H_{Z,i}^{\mathcal{R},X} := (e_{\mathcal{R}|\mathcal{R}}^{Z,X_i,X_i})^\dagger H^{\mathcal{R},X} e_{\mathcal{R}|\mathcal{R}}^{Z,X_i,X_i}. \quad (4.21)$$

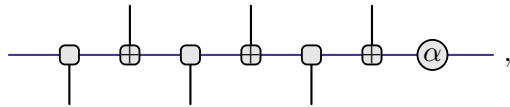
Note that all other twisted Hamiltonians in the same sector, i.e.  $H_{Z,j}^{\mathcal{R},X'}$ , have the same spectrum as  $H_{Z,i}^{\mathcal{R},X}$ , since they are related by the nilpotent  $e_{\mathcal{R}|\mathcal{R}}^{Z,X'_j,X_i}$  in the same block  $Z$ .

Additionally, the enlarged tube algebra admits matrix units of the type  $\mathfrak{T}_{\mathcal{R}|\mathcal{R}'}^{Z,X_i,Z',X'_j}$  and  $\mathfrak{T}_{\mathcal{R}'|\mathcal{R}}^{Z',X'_i,Z,X_j}$ , where  $Z' \in \mathcal{I}_{\mathcal{Z}(\mathcal{D}_{\mathcal{R}'}^*)}$  denotes the image of  $Z \in \mathcal{I}_{\mathcal{Z}(\mathcal{D}_{\mathcal{R}}^*)}$  under the equivalence  $\mathcal{Z}(\mathcal{D}_{\mathcal{R}}^*) \simeq \mathcal{Z}(\mathcal{D}_{\mathcal{R}'}^*)$ . Using them, isometries between the dual models are constructed via

$$H_{Z',j}^{\mathcal{R}',X'} = (e_{\mathcal{R}|\mathcal{R}'}^{Z,X_i,Z',X'_j})^\dagger H_{Z,i}^{\mathcal{R},X} e_{\mathcal{R}|\mathcal{R}'}^{Z,X_i,Z',X'_j}. \quad (4.22)$$

By identifying the anyonic sectors of the string-net models with the topological sectors of their edge Hamiltonians, we have completed the holographic dictionary relating the bulk and edge. We conclude by exemplifying this correspondence in the case of the Kramers–Wannier duality.

**Example: Kramers–Wannier duality – Part III** The Ising model admits periodic or antiperiodic boundary conditions, and for each boundary condition decomposes into even and odd charge sectors, resulting in a total of four topological sectors. At the Hamiltonian level, working with antiperiodic boundary conditions amounts to changing the sign of one of the ferromagnetic interactions.<sup>4</sup> The intertwining tubes eq. (4.20) can then be written as



$$, \quad (4.23)$$

where the boundary matrix  $\alpha$  in each sector is given by one of the Pauli matrices, as per the dictionary in table 4.2. Notably, it is easy to verify by making use of the symmetry properties eq. (3.60) that the  $(P, +1)$  and

<sup>4</sup>Because the symmetry twists are topological, the Ising Hamiltonian with any odd number of twists is unitarily equivalent to one with a single twist.

( $AP, -1$ ) sectors are mapped identically under the Kramers-Wannier duality, while the other two are transposed. It is precisely in this sense that one should interpret the aphorism that Kramers-Wannier duality *exchanges boundary conditions and charge sectors*.

| Boundary condition | Charge sector | Dual boundary condition | Dual charge sector | Boundary matrix $\alpha$ |
|--------------------|---------------|-------------------------|--------------------|--------------------------|
| P                  | +1            | P                       | +1                 | $\mathbb{1}$             |
| P                  | -1            | AP                      | +1                 | $Z$                      |
| AP                 | +1            | P                       | -1                 | $X$                      |
| AP                 | -1            | AP                      | -1                 | $Y$                      |

Table 4.2: Mapping of topological sectors under the Kramers-Wannier duality. Here,  $(A)P$  stands for *(anti)periodic*.

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## 4.5 Two-dimensional Kramers-Wannier duality and fusion 2-categorical symmetry

Having established the tensor network framework of categorical symmetries and dualities in one-dimensional lattice models, we now proceed to the two-dimensional Kramers-Wannier duality and the associated higher symmetry structures described by *fusion 2-categories* [138–142]. Since fusion 2-categories are considerably more intricate structures than their 1-categorical counterparts, we keep the following discussion as concrete as possible. We relegate a sketch of the general framework of fusion 2-categorical symmetries and their dualities to the following section. We begin by constructing the intertwining PEPO that implements the two-dimensional Kramers-Wannier duality.

### 4.5.1 Two-dimensional Kramers-Wannier duality

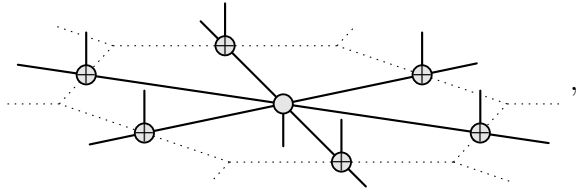
Let us consider for concreteness the two-dimensional *triangular lattice*, endowed with qubits on the vertices  $V$ . Similar to the one-dimensional case, we will study the fate of the two-dimensional transverse field Ising model under Kramers-Wannier duality. In two dimensions, the Hamiltonian reads

$$H = -J \sum_{\mathbf{e}} Z_{s(\mathbf{e})} Z_{t(\mathbf{e})} - \lambda \sum_{\mathbf{v}} X_{\mathbf{v}}. \quad (4.24)$$

Herein,  $s$  and  $t$  are functions that assign to every edge their *source* and *target*, vertices. Let us assume *periodic* boundary conditions, so that effectively the

underlying topology is that of the two-torus. Evidently, the Hamiltonian commutes with a global  $\mathbb{Z}_2$  spin-flip symmetry generated by  $\prod_v X_v$ . In the context of higher-form symmetries, such a symmetry operator is called a *0-form* symmetry operator or *surface operator* since the operator extends over all of space, or thus a submanifold of *codimension* 0 with respect to the spatial dimensionality. Notably, one could add to eq. (4.24) additional terms that commute with the 0-form  $\mathbb{Z}_2$  symmetry without altering much of the following discussion [141].

In analogy with the one-dimensional Kramers-Wannier duality operator eq. (4.17), its two-dimensional counterpart admits a PEPO representation of bond dimension two that can be depicted as [139]



where the constituent tensors were defined in eq. (3.59). By pulling the Hamiltonian terms through this intertwiner and thereby making use of the symmetry properties listed in eq. (3.60), one finds the following mapping:

$$Z_{s(e)}Z_{t(e)} \mapsto Z_e, \quad X_v \mapsto \prod_{e|v} X_e, \quad (4.26)$$

where  $e|v$  stands for the collection of all edges  $e \in E$  adjacent to  $v$ . Collecting everything, the dual Hamiltonian is thus found to be

$$H^\vee = -J \sum_e Z_e - \lambda \sum_v \prod_{e|v} X_e. \quad (4.27)$$

This is however not the complete story, since it can be easily verified that the dual model obeys the kinematical constraints

$$\prod_{e \in \ell} Z_e \stackrel{!}{=} \mathbb{1} \quad (4.28)$$

for every closed loop  $\ell$  on the triangular lattice, including the non-contractible ones. Similarly as in one dimension, this signals in particular that the dual model is in the even sector of a dual *1-form* symmetry that we discuss momentarily. In conclusion, the dual model can thus be thought of as a toric code in a transverse field, wherein the plaquette terms are enforced kinematically rather than dynamically, and where the terms in the second sum of eq. (4.27) are the vertex terms.

## 4.5.2 Dual symmetry structure

Let us proceed by unpacking the full symmetry structure of the dual model  $H^\vee$ . First, for every closed loop  $\ell$ , operators of the form

$$\mathbb{U}[\ell] := \prod_{e \in \ell} Z_e \quad (4.29)$$

are immediately verified to commute with  $H^\vee$ . Moreover, they are *topological* by virtue of the kinematical constraints eq. (4.28), and in the context of gauge theories referred to as *Wilson loops* [138, 143–145]. Taken together, this means that the collection of loop operators  $\mathbb{U}[\ell]$  for all closed loops  $\ell$  generate a 1-form  $\text{Rep}(\mathbb{Z}_2)$  symmetry.<sup>5</sup>

Recently, it has been appreciated that besides the Wilson loops, the dual model hosts *surface operators* that are obtained by *condensing* the loop operators on the spatial manifold [64, 146, 147]. Surface operators that admit this interpretation are commonly referred to in the literature as *condensation defects* and are often thought of as resulting from a *higher gauging* procedure [148–152]. More concretely, by this it is meant that the surface operator results from summing over a *sufficiently fine mesh* of Wilson lines [132]. By making use of the topological invariance and fusion rules of the loop operators, this network can be reduced to the following *minimal* network of Wilson loops:

$$\mathbb{U}^{\{1\}} := \frac{1}{2} \left( \sum_{s_x \in \{0,1\}} \mathbb{U}[\ell_x]^{s_x} \right) \cdot \left( \sum_{s_y \in \{0,1\}} \mathbb{U}[\ell_y]^{s_y} \right), \quad (4.30)$$

where  $\ell_x$  and  $\ell_y$  denote representative loops along the two non-contractible cycles of the two-torus. The notation  $\mathbb{U}^{\{1\}}$  for this condensation defect is motivated below.

From its definition eq. (4.30) we immediately infer the following key properties. First, in stark contrast with the 0-form  $\mathbb{Z}_2$  symmetry of the original model,  $\mathbb{U}^{\{1\}}$  acts trivially on any local operator. This follows directly from the topological invariance of the 1-form symmetry operators appearing in the definition eq. (4.30). An easy computation also shows that  $\mathbb{U}^{\{1\}}$  multiplies according to  $\mathbb{U}^{\{1\}} \cdot \mathbb{U}^{\{1\}} = 2 \cdot \mathbb{U}^{\{1\}}$ , so that in particular it does not admit a

<sup>5</sup>Let us note at this point that in general, the gauging of a  $(d+1)$ -dimensional theory with a  $q$ -form symmetry is expected to result in a dual model with a  $(d-q)$ -form symmetry [69], see also chapter 9.

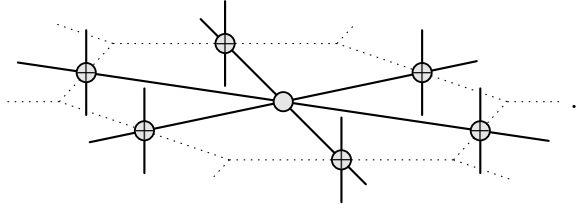


2261 multiplicative inverse.<sup>6</sup> Finally, note that all Wilson loops condense on this  
 2262 surface in the sense that  $\mathbb{U}[\ell] \cdot \mathbb{U}^{\{1\}} = \mathbb{U}^{\{1\}} = \mathbb{U}^{\{1\}} \cdot \mathbb{U}[\ell]$ .

2263 Using the tensor network calculus developed in refs. [141, 150] and chapter 10,  
 2264 a PEPO representation of  $\mathbb{U}^{\{1\}}$  can be constructed. Defining the tensor

$$\text{Diagram} := \text{Diagram}, \quad (4.31)$$

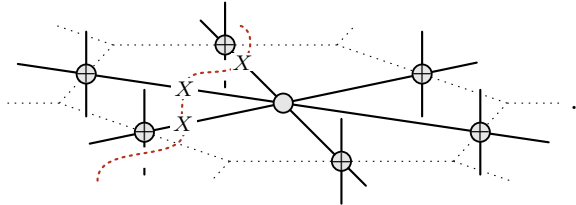
2265 this PEPO is depicted as:



(4.32)

2266 Remark the striking similarity with the  $\mathcal{R} = \text{Vec}_{\mathbb{Z}_2}$  tensor network represen-  
 2267 tation of the toric code ground state constructed in section 3.4.3. Indeed, in  
 2268 a very loose but not meaningless sense, one can think of the condensation  
 2269 operator as the toric code ground state in which the physical degrees of  
 2270 freedom are ‘doubled’ so as to turn it into a non-invertible surface operator.

2271 The similarity with the toric code ground state is more than an analogy  
 2272 in the sense that the condensation defect can be decorated with topological  
 2273 lines labelled by elements in  $\mathbb{Z}_2$ . In the language of gauge theories, such lines  
 2274 are referred to as ‘t Hooft lines. Very concretely, one constructs for any loop  
 2275  $\ell^\vee$  on the *dual* lattice, the non-trivial ‘t Hooft line by inserting a string of  
 2276 Pauli- $X$  operators on the virtual level along  $\ell^\vee$ , cf. eq. (3.63). For example:



(4.33)

2277 Topological invariance follows directly from the symmetry properties of the  
 2278 local tensors, as usual.

<sup>6</sup>In fact, a more careful computation reveals that on a generic closed and oriented spatial manifold  $\Sigma$  the fusion rule of  $\mathbb{U}^{\{1\}}$  reads  $\mathbb{U}^{\{1\}} \cdot \mathbb{U}^{\{1\}} = Z_{\mathbb{Z}_2}(\Sigma) \cdot \mathbb{U}^{\{1\}}$ , with  $Z_{\mathbb{Z}_2}(\Sigma) = 2^{b_1 - b_0}$  the partition function of pure  $\mathbb{Z}_2$  gauge theory on  $\Sigma$  and  $b_{0,1}$  the respective Betti numbers of  $\Sigma$ . Hence, the fusion rules of this symmetry operator are *topology dependent* [141, 152].

Let us recapitulate what we have found so far. We have argued that the dual model  $H^\vee$  enjoys 1-form  $\text{Rep}(\mathbb{Z}_2)$  symmetry. By condensing these Wilson loops, one obtains a condensation defect  $\mathbb{U}^{\{1\}}$  that itself hosts 't Hooft lines organised in  $\text{Vec}_{\mathbb{Z}_2}$ . By similar methods, one can construct a topological interface between  $\mathbb{U}^{\{1\}}$  and the trivial identity surface operator. Formally, these structures combine into a *fusion 2-category* denoted  $2\text{Rep}(\mathbb{Z}_2)$ . Notably,  $2\text{Rep}(\mathbb{Z}_2)$  can be thought of as a *categorification* of the category  $\text{Rep}(\mathbb{Z}_2)$ , which is the symmetry structure one expects after performing the Kramers-Wannier duality on a  $\text{Vec}_{\mathbb{Z}_2}$ -symmetric model. And indeed, as seen below, the 0-form  $\mathbb{Z}_2$  symmetry of the Ising model provides a concrete realisation of the symmetry 2-category  $2\text{Vec}_{\mathbb{Z}_2}$ . Remarkably, unlike in the one-dimensional setting, the original and dual symmetry are not monoidally equivalent. This prevents us from constructing a model *self-dual* under the Kramers-Wannier duality. In turn, this raises the question whether other duality transformations exist that give rise to (2+1)-dimensional self-dual models. The answer is affirmative. In chapter 10, we consider a more refined duality transformation that amounts to the composition of a *gauging* and *twisted gauging* procedure that admits self-dual models.

Before unpacking at a more abstract level fusion 2-categorical symmetry structures, let us discuss the *topological sectors* of the models from this section and study their permutation under the duality transformation.

### 4.5.3 Topological sectors

In complete analogy with the one-dimensional case, the two-dimensional Ising model admits periodic or antiperiodic boundary conditions along either non-contractible cycle of the torus. Furthermore, the Hilbert space for each choice of boundary conditions decomposes into two  $\mathbb{Z}_2$  charge sectors, so that the total number of topological sectors is  $2^3 = 8$ .

To identify these sectors in the dual model, we note first that the Wilson loop operators  $\mathbb{U}[\ell_x]$ ,  $\mathbb{U}[\ell_y]$  measure the  $\mathbb{Z}_2$  *flux* through the corresponding cycles of the torus. This labels four distinct charge sectors with respect to the dual 1-form symmetry. These can be accessed by choosing the boundary conditions of the Ising model. It can also be verified explicitly that the odd  $\mathbb{Z}_2$  charge sector of the *original* model can be accessed by changing the sign of one of the vertex terms in eq. (4.27). Heuristically, this can be seen by taking the product of all vertex terms and tracking that under duality this amounts to

$$\prod_v X_v = - \prod_v \prod_{e|v} X_e = -1. \quad (4.34)$$

Changing the sign of one of these vertex terms is sometimes referred to as choosing a *1-form twisted boundary condition* [119, 120].

In summary, we obtain a very similar mapping as in the one-dimensional case, whereby the boundary conditions and charge sectors are interchanged.

## 4.6 Algebraic structure of fusion 2-categories

Let us now unpack in some more detail what a fusion 2-categorical symmetry entails and develop its graphical calculus in parallel [140, 150].

### 4.6.1 Algebraic structure of fusion 2-categories

Loosely speaking, a *fusion 2-category*  $\mathfrak{C}$  describes a collection of *topological surfaces* which constitute the object set  $\text{ob}(\mathfrak{C})$ . In what follows we will use (*topological*) *surfaces* and *objects* interchangeably. The physically inclined reader might want to think about these surfaces as being implemented in a lattice model via PEPOs. Surfaces which cannot be written as the direct sum of other surfaces are said to be *simple*. We write this set as  $\mathcal{I}_{\mathfrak{C}}$ . Fusing two simple surfaces and decomposing the result can be depicted as

$$\begin{array}{c} \text{Diagram: A square labeled } X_1 \text{ containing a smaller square labeled } X_2 \text{ with a line connecting them.} \end{array} = \bigoplus_{X_3 \in \mathcal{I}_{\mathfrak{C}}} N_{X_1 X_2}^{X_3} \begin{array}{c} \text{Diagram: A square labeled } X_3. \end{array}, \quad (4.35)$$

for an integer rank-3 tensor  $N$ . The fusion rules are associative. Among the simple objects, there exists a unique *monoidal unit* denoted by  $\mathbb{1}_{\mathfrak{C}}$ , or simply  $\mathbb{1}$ . Sometimes we will refer to  $\mathbb{1}_{\mathfrak{C}}$  as the *trivial* or *transparent* surface. *Topological defect lines* on the interface between generic surfaces  $X$  and  $X'$  are organised in a *Hom (1-)category* denoted by  $\mathfrak{C}(X, X')$ , and we will use the shorthand  $\mathfrak{C}^{X, X'} \equiv \mathfrak{C}(X, X')$ . On the lattice, these might be represented via MPOs, living on the *virtual level* of the PEPOs, akin to those defined in eq. (4.33). Topological lines are said to encode the *1-morphisms* of the 2-category. Graphically:

$$\begin{array}{c} \text{Diagram: A square divided vertically by a red line labeled } f, \text{ with } X' \text{ on the left and } X \text{ on the right.} \end{array} = \begin{array}{c} \text{Diagram: A square divided by a curved red line labeled } f, \text{ with } X' \text{ on the left and } X \text{ on the right.} \end{array}, \quad (4.36)$$

for  $f \in \text{ob}(\mathfrak{C}(X, X'))$ . Lines between compatible surface operators can be fused ‘horizontally’:

$$\begin{array}{|c|c|c|} \hline f & & g \\ \hline X_3 & X_2 & X_1 \\ \hline \end{array} = \begin{array}{|c|c|} \hline f \otimes g & \\ \hline X_3 & X_1 \\ \hline \end{array}. \quad (4.37)$$

It can be shown that the Hom category  $\mathfrak{C}^{X,X}$  for any simple object  $X$  naturally forms a *fusion 1-category*. Specialising to the identity object  $X = \mathbb{1}$ , it is found that  $\mathfrak{C}^{\mathbb{1},\mathbb{1}}$  is in fact a *braided* fusion category, but the braiding is not usually modular. And remarkably,  $\mathfrak{C}^{X,X'}$  is an indecomposable  $(\mathfrak{C}^{X,X}, \mathfrak{C}^{X',X'})$ -*bimodule category* for any pair of simple objects  $X, X'$ .

Finally, the *morphisms* in  $\mathfrak{C}^{X,X'}$  encode topological *point operators* on the defect lines, and are said to constitute the *2-morphisms* of  $\mathfrak{C}$ . These can be composed ‘vertically’, which is pictorially represented as:

$$\begin{array}{|c|c|} \hline f & \\ \hline \varphi & \\ \hline g & \\ \hline \psi & \\ \hline X' & h & X \\ \hline \end{array} = \begin{array}{|c|c|} \hline f & \\ \hline \chi & \\ \hline X' & h & X \\ \hline \end{array}, \quad (4.38)$$

where  $f, g, h \in \text{ob}(\mathfrak{C}(X, X'))$ ,  $\varphi \in \mathfrak{C}^{X,X'}(f, g)$ ,  $\psi \in \mathfrak{C}^{X,X'}(g, h)$ , and  $\chi = \psi \circ \varphi$ .

Associativity of the tensor product is encoded by natural *associator* 1-isomorphisms  $\alpha^{X_1 X_2 X_3} : (X_1 \otimes X_2) \otimes X_3 \xrightarrow{\sim} X_1 \otimes (X_2 \otimes X_3)$ . The associator satisfies the pentagon equation up to natural 2-isomorphisms called the *pentagonator* [140, 150].

In stark contrast with the 1-categorical setting, fusion 2-categories thus allow for the existence of non-trivial topological defects on the interface between distinct simple objects, as well as *self-interfaces*. This naturally leads to the definition of a *connected component*, which is an equivalence class of simple objects with the equivalence relation  $X \sim X' \iff \mathfrak{C}(X, X') \neq 0$ . Physically, all surfaces that belong to the same connected component are related via *condensation* of an appropriate set of topological lines, as we will exemplify below [64, 146, 147, 152]. Moreover, the surfaces belonging to the identity component are exactly those that can be defined on *open* patches. As such, any fusion 2-category admits a finite direct sum decomposition:

$$\mathfrak{C} \simeq \bigoplus_{i=1}^r \mathfrak{C}_i. \quad (4.39)$$

## 4.6.2 Examples

### Finite group symmetry: $2\text{Vec}_G^\pi$

Arguably the simplest fusion 2-category is  $2\text{Vec}_G$ , for any finite group  $G$ . The simple objects coincide with  $\text{Vec}_G$ , often identified with the group elements, since the fusion rules simply follow the multiplication of  $G$ :  $\text{Vec}_{g_1} \otimes \text{Vec}_{g_2} = \text{Vec}_{g_3}$ . For the particular case of  $G = \mathbb{Z}_2$ , we obtain the symmetry structure of the two-dimensional Ising model introduced and discussed in the previous section. Notably, there are no non-trivial topological lines, i.e.  $2\text{Vec}_G(\text{Vec}_{g_1}, \text{Vec}_{g_2}) = \delta_{g_1, g_2} \text{Vec}$ .

Similarly as in the case of the fusion category  $\text{Vec}_G$ , the  $2\text{Vec}_G$  fusion rules can be twisted by a choice of non-trivial pentagonator that can be expressed in terms of a 4-cocycle  $[\pi] \in H^4(G, \text{U}(1))$ . As such, one obtains the 2-category denoted  $2\text{Vec}_G^\pi$ .

### Finite group 2-representation: $2\text{Rep}(G)$

The fusion 2-category  $2\text{Rep}(G)$  admits a few equivalent definitions. For one, it can be understood as the *Morita dual* of  $2\text{Vec}_G$  with respect to the *module 2-category*  $2\text{Vec}$  [150], so that it can be realised as the fusion 2-category of  $2\text{Vec}_G$ -*module 2-endofunctors* of  $2\text{Vec}$ , viz.  $2\text{Rep}(G) = 2\text{Fun}_{2\text{Vec}_G}(2\text{Vec}, 2\text{Vec})$ . More useful for our discussion is that it can also be constructed as the category of  $\text{Vec}_G$ -*module categories* and  $\text{Vec}_G$ -*module functors*. This is notated as  $2\text{Rep}(G) \equiv \text{Mod}(\text{Vec}_G)$ , in direct analogy with the fusion 1-categorical counterpart  $\text{Rep}(G) = \text{Mod}(\mathbb{C}[G])$ . Elements of  $2\text{Rep}(G)$  are therefore often said to encode  $G$ -*2-representations*.

Either way, simple objects are in one-to-one correspondence with the indecomposable  $\text{Vec}_G$ -module categories and thus indexed by pairs  $(H \subseteq G, \psi)$ ,  $[\psi] \in H^2(H, \text{U}(1))$  (section 3.3.2). We denote the associated defect by  $\mathbb{U}^{H, \psi}$ . Its fusion rules are then specified by

$$\mathbb{U}^{H, \psi} \otimes \mathbb{U}^{K, \varphi} = \bigoplus_{[r] \in H \backslash G / K} \mathbb{U}^{(H \cap rKr^{-1})_{\varrho_r}}. \quad (4.40)$$

Here,  $\varrho_r$  is a 2-cocycle constructed from  $\psi$  and  $\varphi$  on the group  $H \cap rKr^{-1}$ , derived in ref. [80]. From these fusion rules we infer that the monoidal unit is given by  $\mathbb{U}^G$ . The 1-morphisms between simple surfaces  $\mathbb{U}^{H, \psi}$ ,  $\mathbb{U}^{K, \varphi}$  are organised in the category of module functors

$$\mathfrak{C}(\mathbb{U}^{H, \psi}, \mathbb{U}^{K, \varphi}) = \text{Fun}_{\text{Vec}_G}(\mathcal{R}(H, \psi), \mathcal{R}(K, \varphi)), \quad (4.41)$$

which concretely is given by

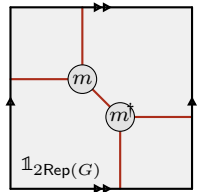
$$\mathrm{Fun}_{\mathrm{Vec}_G}(\mathcal{R}(H, \psi), \mathcal{R}(K, \varphi)) = \bigoplus_{[r] \in H \backslash G / K} \mathrm{Rep}^{\varpi_r}(H \cap rKr^{-1}), \quad (4.42)$$

where  $\varpi_r$  is a 2-cocycle on  $H \cap rKr^{-1}$  whose explicit expression in terms of  $\psi$  and  $\varphi$  can be found in ref. [52]. In particular, this illustrates the fact that self-interfaces on the identity surface are encoded in the symmetrically braided fusion category  $(\mathrm{Vec}_G)_{\mathrm{Vec}}^* = \mathrm{Rep}(G)$ . These are said to encode the *Wilson loop* operators, generalising those discussed above in the context of the toric code to general groups  $G$ . At the other extreme, one has  $(\mathrm{Vec}_G)_{\mathrm{Vec}_G}^* = \mathrm{Vec}_G$ , which encodes the *'t Hooft lines*.

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Despite their Morita equivalence, the structure of  $2\mathrm{Vec}_G$  and  $2\mathrm{Rep}(G)$  differs significantly. Clearly,  $2\mathrm{Vec}_G$  and  $2\mathrm{Rep}(G)$  are not monoidally equivalent, not even when  $G$  is abelian. This highlights once more a remarkable distinction with the 1-categorical setting where  $\mathrm{Vec}_G \simeq \mathrm{Rep}(G)$  as fusion categories, for abelian  $G$ .

Note that any two simple objects in  $2\mathrm{Rep}(G)$  belong to the same connected component, as we can conclude from inspecting eq. (4.42). In particular, all simple objects are connected to the transparent surface, which implies that all surfaces can be obtained via condensation of an appropriate set of Wilson loops encoded in  $\mathrm{Rep}(G)$ . More precisely, as argued in chapter 10, the appropriate set of loops to condense is encoded in an *algebra object* of  $\mathrm{Rep}(G)$ . In brief, an algebra object  $A$  is a collection of lines  $A = \bigoplus_{\rho} N_{\rho}^A \rho$  together with a *multiplication map*  $m : A \otimes A \mapsto A$ , whose main property is that its composition is associative on the nose. We refer to chapter 8 for a more precise definition of algebra objects in unitary fusion categories. Provided an algebra object in  $\mathrm{Rep}(G)$ , the condensation defect thus schematically boils down to the following network of defects:



$$(4.43)$$

that formally lives on the identity surface. We note that a representative algebra object associated with a surface  $\mathbb{U}^H$  corresponds, at the level of the objects, to the *induced representation*  $\mathrm{Ind}_H^G(\underline{1}_H)$  in  $G$  of the trivial representation  $\underline{1}_H$  of  $H$  [153]. The associated multiplication morphism admits a simple expression in terms of the  $\mathrm{Rep}(H)$  module associator over  $\mathrm{Rep}(G)$  [132, 154, 155].

For the case of  $G = \mathbb{Z}_2$ , the algebra object  $\mathbb{C}[\mathbb{Z}_2] \simeq \text{Ind}_{\{1\}}^{\mathbb{Z}_2}(\mathbb{1}_{\{1\}})$ , viewed as the *regular*  $\mathbb{Z}_2$ -representation, results in the condensation defect eq. (4.30) upon expressing eq. (4.43) in terms of the associated MPO symmetries.

Conversely, the surface  $\mathbb{U}^{H_\psi}$  can be constructed by condensing on  $\mathbb{U}^{\{1\}}$  the algebra object  $\mathbb{C}^\psi[H]$  in the category  $\text{Vec}_G$  of 't Hooft lines, viewed as the  $\psi$ -twisted group algebra of  $H$ . As an object,  $\mathbb{C}^\psi[H]$  decomposes as  $\bigoplus_{h \in H} h$ , with the multiplication map  $m$  implementing the  $H$  group multiplication twisted by the 2-cocycle  $\psi$ .

## 4.7 Tensor network representations of 2-categorical symmetries and dualities

Having sketched the algebraic structure of fusion 2-categorical symmetries, with particular emphasis on  $2\text{Vec}_G$  and  $2\text{Rep}(G)$ , we now turn in this concluding section to their lattice realisations in terms of PEPOs and intertwining MPOs, as well as the dualities associated with these symmetry structures [139, 141, 150, 156]. As one might expect, the two-dimensional framework closely parallels the one-dimensional story told above. Owing to the relative novelty of this framework and technical intricacies of the *Morita theory* of fusion 2-categories, this section remains largely qualitative [157].

For concreteness, let us focus on the case in which the input fusion 2-category is  $2\text{Vec}_G$ . As in one dimension, the input fusion 2-category determines an algebra of local symmetric operators from which a local symmetric Hamiltonian can be constructed. As demonstrated in refs. [141, 150], this algebra can be realised on the lattice via a choice of a *module 2-category*. Such a choice determines in turn a *Morita dual* fusion 2-category describing the symmetry structure of the resulting model. Consequently, the module 2-category naturally has the structure of a *bimodule 2-category*.

One class of indecomposable module 2-categories is in one-to-one correspondence with pairs  $(H \subseteq G, \omega)$  with  $[\omega] \in H^3(H, \text{U}(1))$ . As glossed over above, the Morita dual is defined as the fusion 2-category of *module 2-endofunctors and module 2-natural transformations*. These 2-endofunctors and 2-natural transformations are equipped with coherence data, which in turn determine the components of the PEPO and MPO tensors supported on the corresponding surfaces. In many cases of interest, and in particular for  $2\text{Vec}$  as  $(2\text{Vec}_G, 2\text{Rep}(G))$ -bimodule 2-category, this construction simplifies substantially and admits a more explicit description, as illustrated in the aforementioned references.

Duality transformations amount to a change in the module 2-category,

while keeping the input  $2\text{Vec}_G$  fixed. Indeed, this change can be shown to leave invariant the algebra of local symmetric operators. The corresponding intertwining PEPOs implementing the duality are then naturally described in terms of  $2\text{Vec}_G$ -module 2-functors. As in the one-dimensional setting, the construction simplifies considerably when one of the module 2-categories is chosen to be the regular one.

As an illustrative example, let us finally consider the Kramers-Wannier duality, which is realised by the change of module 2-category  $2\text{Vec}_{\mathbb{Z}_2} \mapsto 2\text{Vec}$  over  $2\text{Vec}_{\mathbb{Z}_2}$ . By specialising the general framework to this case and simplifying the duality operator, one recovers precisely the PEPO defined in eq. (4.25).



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# II

# Applications

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# Les Houches Lecture Notes on Tensor Networks

## 5.1 Synopsis

These lecture notes are based on a series of five lectures taught by Frank Verstraete at the 2025 Les Houches school on *Exact Solvability and Quantum Information*. They cover a variety of topics in the field of tensor networks, with an emphasis on the concept of the MPS manifold, MPO symmetries, integrability and dualities.

## 5.2 Reference

*Les Houches Lecture Notes on Tensor Networks*,  
**B. Vancraeynest-De Cuiper**, W. Wiesiolek, F. Verstraete  
arXiv:2512.24390

## 5.3 Contributions of the author

I wrote the majority of the manuscript, whose content closely follows that of the lectures on which it is based. Apart from some aspects of the exposition, the content is due to the lecturer.

## Les Houches Lecture Notes on Tensor Networks

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### Abstract

Tensor networks provide a powerful new framework for classifying and simulating correlated and topological phases of quantum matter. Their central premise is that strongly correlated matter can best be understood by studying the underlying entanglement structure and its associated (generalised) symmetries. In essence, tensor networks provide a compressed, holographic description of the complicated vacuum fluctuations in strongly correlated systems, and as such they break down the infamous many-body exponential wall. These lecture notes provide a concise overview of the most important conceptual, computational and mathematical aspects of this theory.

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## Introductory remarks

We are currently witnessing a paradigm shift in the way we understand and do quantum many-body physics. The *quantum many-body problem* has been the central problem in theoretical physics and quantum chemistry for the best of the last century, and its associated exponential wall has precluded progress in describing strongly interacting phases of matter. During the last twenty years however, there has been remarkable progress in simulating and classifying those strongly correlated systems using the theory of *entanglement*, as epitomised by the concept of *tensor networks*.

In essence, tensor networks circumvent the exponential quantum many-body wall by providing a *holographic* description of the many-body wave functions. Instead of describing the wave function in an exponentially large Hilbert space, tensor networks model how entanglement is routed between different parts of the system. Underlying the success of tensor networks lies the fact that all the information needed to calculate any observable of a low-energy wave function can be compressed into a local tensor with so-called *entanglement degrees of freedom* making up a *virtual* Hilbert space. The reason that this can be done is related to the existence of *area laws* for the entanglement entropy: tensor networks are concise and precise representations of the manifold of all states satisfying such an area law. By projecting the

Hamiltonian defined in the exponentially large physical Hilbert space on this low-dimensional manifold, it has become possible to construct numerical algorithms to simulate large-scale quantum many-body systems. DMRG (*Density Matrix Renormalisation Group*), MPS (*Matrix Product State*) and PEPS (*Projected Entangled-Pair State*) algorithms are revolutionising the way we simulate state-of-the-art experiments in atomic physics, current-day quantum computers and the low-energy physics of quantum many-body systems.

But there is more: since the pioneering insights of Landau and Anderson, it is known that many-body physics is all about *symmetries* and *symmetry breaking*. It turns out that the global, local and topological symmetries of the full system are reflected in the symmetries of the local tensors of the tensor network, which provide an extremely effective compression of the full many-body wave function. As a consequence, the classification of gapped *phases of matter*, and in particular *topological phases*, is reduced to the problem of classifying the irreducible ways in which tensors transform under a set of relevant symmetries. As those symmetries act on the virtual (entanglement) degrees of freedom, there is no reason why those symmetries should form a group, and more general associative structures emerge that can be described by *fusion categories*. Indeed, tensor networks provide an explicit way of representing those *generalised symmetries* (and the dualities between them) in terms of MPO (*Matrix Product Operator*) algebras and their higher-dimensional analogues, PEPOs (*projected entangled-pair operators*). Several of the most active research areas in mathematical, condensed matter, statistical and high-energy physics are precisely dealing with the consequences of this insight: a holographic description of strongly correlated and topological phases of matter is the only one which reveals its inner structure.

These lecture notes are based on five lectures given by Frank Verstraete at the Les Houches Summer School of Physics in August 2025 on *Exact Solvability and Quantum Information*, covering a selection of elementary computational and theoretical results of tensor networks. They do not aim to provide a comprehensive review of these topics. Rather, they are meant as an invitation for further study and we can refer to a vast literature of existing reviews and lecture notes: [1–9].

## 1 Lecture I: Motivation & the manifold of MPS

### 1.1 The double-exponential wall & the illusion of Hilbert space

Before we dive into these lectures, it is important to recognise the true challenge posed by the quantum many-body problem. As one learns in every course on quantum mechanics, the dimension of the Hilbert space scales exponentially in the number of constituent particles. This is in fact not so different from the classical case where the number of possible strings of classical bits also scales exponentially in the number of bits. However, since quantum mechanics allows for *superpositions* of states and consequently *entanglement*, the natural question to ask is not by how many basis states the Hilbert space is spanned, but rather how many of the states in the Hilbert space can actually be explored by some physical mechanism originating from local (few-body) interactions, and in particular with a quantum computer.

To be more precise and concrete, let us consider the following question raised in ref. [10]. Within the Hilbert space of  $N$  qubits, which portion of the exponentially large many-body Hilbert space can be reached by running a quantum computer starting from a reference state for a time scaling polynomially in system size? To estimate this, we consider so-called  $\varepsilon$ -balls of states around points  $|\psi\rangle$  in the Hilbert space, i.e. we consider states  $|\varphi\rangle$  for which

$$|\langle\psi|\varphi\rangle| \gtrsim 1 - \varepsilon. \quad (1)$$

A simple geometric argument shows that such  $\varepsilon$ -balls occupy a fraction of the total volume of the Hilbert space which scales according to  $\mathcal{O}(\varepsilon^{2^N})$ , i.e. a *double-exponentially* (!) small fraction of the total Hilbert space! Due to this scaling, it would take any physical system (including a quantum computer) a time that is at least exponential in the system size to have overlap with a specific  $\varepsilon$ -ball. We conclude that the vast majority of states is unphysical and is completely irrelevant for the description of physical systems: Hilbert space is a (convenient) illusion, as states with many particles that take exponentially long to create cannot be created by any physically reasonable process.

This dramatic result points to a paradigm shift in our understanding of quantum many-body systems, namely that physical wave functions are those which have low complexity. More specifically, we expect ground states of quantum many-body systems to live on a physically relevant low-complexity *manifold* of states, which for all practical purposes we will want to parametrise with a *small* number of parameters, meaning scaling polynomially in the number of particles. Roughly speaking, a manifold is a space which locally resembles flat Euclidean space. This means that to every point of the manifold one can associate a linear *tangent space* whose dimension is the dimension of the manifold. One of the (many) great insights of Dirac was that one can now actually do physics on such a manifold without ever leaving it by interpreting the manifold as a variational set of states. Let us denote a general variational manifold of states as

$$\mathcal{M} = \{ |\Psi(\mathbf{z})\rangle \mid \mathbf{z} \in \mathbb{C}^n \}. \quad (2)$$

We will write  $T_{\mathbf{z}}\mathcal{M}$  for the linear tangent space at any point  $\mathbf{z}$ . A general state in this tangent space is then of the form

$$\sum_i x^i \frac{\partial}{\partial z^i} |\Psi(\mathbf{z})\rangle, \quad (3)$$

for complex coefficients  $x^i$ . Suppose now that we want to compute the time evolution of a certain state  $|\Psi(\mathbf{z})\rangle$  subject to some many-body Hamiltonian  $H$ :

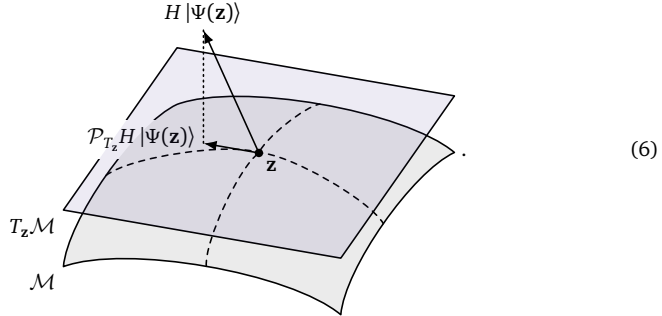
$$i \frac{\partial}{\partial t} |\Psi(\mathbf{z})\rangle = H |\Psi(\mathbf{z})\rangle. \quad (4)$$

Then  $i \frac{\partial}{\partial t} |\Psi(\mathbf{z})\rangle = i \sum_i \dot{z}^i \frac{\partial}{\partial z^i} |\Psi(\mathbf{z})\rangle$  can be thought of as a tangent vector in  $T_{\mathbf{z}}\mathcal{M}$ . However, for a generic Hamiltonian,  $H |\Psi(\mathbf{z})\rangle$  is a vector in the ambient Hilbert space in which  $\mathcal{M}$  is embedded that points *out of* the tangent plane, and thus doesn't belong to the variational class anymore. Dirac's proposal was to define an action on this manifold, but for our purposes it is convenient to think about this as the orthogonal projection of  $H |\Psi(\mathbf{z})\rangle$  on the tangent space of the manifold  $T_{\mathbf{z}}\mathcal{M}$ . Geometrically, one can think of this projection as minimising the distance

$$\min_{\{\dot{z}^i\}_i} \left\| i \sum_i \dot{z}^i \frac{\partial}{\partial z^i} |\Psi(\mathbf{z})\rangle - H |\Psi(\mathbf{z})\rangle \right\|. \quad (5)$$

Denoting the orthogonal projector at a point  $\mathbf{z}$  by  $\mathcal{P}_{T_{\mathbf{z}}}$ , we can depict its action on  $H |\Psi(\mathbf{z})\rangle$

thus as:



The resulting *time-dependent variational principle* (TDVP) equations

$$i \frac{\partial}{\partial t} |\Psi(\mathbf{z})\rangle = \mathcal{P}_{T_z} H |\Psi(\mathbf{z})\rangle, \quad (7)$$

then describe a collection of highly non-linear (due to the appearance of the tangent space projector) differential equations of the generic many-body state on the manifold  $\mathcal{M}$ . Importantly, it can be shown that the manifold  $\mathcal{M}$  is *symplectic*. This means in particular that the expectation value of the energy, generators of symmetries and other constants of motion are exactly conserved under evolution by the TDVP equations.

This manifold approach to the many-body problem has been applied in many fields with exceptional success. In fact, it is merely a small exaggeration to state that all progress in tackling the quantum many-body problem since the discovery of the Schrödinger equation has been in finding such good manifolds of states in which the relevant physics takes place. *Matrix product states* (MPS) and its higher dimensional generalisations such as two-dimensional *projected entangled-pair states* (PEPS) form such classes, and we will introduce them shortly. But before doing so, let us take some time to discuss some other, perhaps to some readers more familiar, variational classes of states from this manifold point of view.

**Free fermions & Hartree-Fock** Probably the most well-known variational manifold of many fermions is that of *Slater determinants* [11]. A Slater determinant parametrises an  $N$ -fermion wave function in terms of  $N$  single-particle orbitals  $\psi_i, i = 1, \dots, N$  as a determinant of the form:

$$\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N) = \begin{vmatrix} \psi_1(\mathbf{x}_1) & \dots & \psi_N(\mathbf{x}_1) \\ \psi_1(\mathbf{x}_2) & \dots & \psi_N(\mathbf{x}_2) \\ \vdots & \ddots & \vdots \\ \psi_1(\mathbf{x}_N) & \dots & \psi_N(\mathbf{x}_N) \end{vmatrix}. \quad (8)$$

By construction, Slater determinants are manifestly anti-symmetric under the exchange of any two fermions, in correspondence with the Pauli exclusion principle. Moreover, note that the number of parameters needed to represent Slater determinants scales *linearly* in the number of electrons and Slater determinants therefore only span an exponentially small submanifold in the Hilbert space. The *Hartree-Fock* method for approximately solving the Schrödinger equation of  $N$  fermions is then a self-consistent variational method over this class of states: approximations of both the time-dependent and time-independent Schrödinger equations are obtained by applying the TDVP principle to this manifold. The success of this method can hardly be overstated. It has allowed us to simulate molecules and materials with great accuracy



which would otherwise be completely intractable with exact methods. In fact, Hartree-Fock allowed us to understand the structure of the periodic table of elements and of band structures from first principles [12, 13].

**Field coherent states of interacting bosons** The bosonic counterpart of Slater determinants also exists in the form of the manifold of *field coherent states*. For the bosonic creation operators<sup>1</sup>  $\hat{\psi}^\dagger(\mathbf{x})$ , one has for any function  $\varphi(\mathbf{x}) : \mathbb{R}^3 \rightarrow \mathbb{C}$  the coherent state

$$|\Psi[\varphi]\rangle = \exp\left(\int d^3\mathbf{x} \varphi(\mathbf{x}) \hat{\psi}^\dagger(\mathbf{x})\right) |\Omega\rangle, \quad (9)$$

where  $|\Omega\rangle$  is the bosonic vacuum characterised by  $\hat{\psi}(\mathbf{x})|\Omega\rangle = 0$ ,  $\forall \mathbf{x}$ . Recall that coherent states are exact eigenstates of the annihilation operators, i.e.  $\hat{\psi}(\mathbf{x})|\Psi[\varphi]\rangle = \varphi(\mathbf{x})|\Psi[\varphi]\rangle$ . Let us now apply the TDVP to the Hamiltonian

$$H = \int d^3\mathbf{x} \hat{\psi}^\dagger(\mathbf{x}) \left( -\frac{1}{2m} \nabla^2 + V(\mathbf{x}) \right) \hat{\psi}(\mathbf{x}) + \frac{U}{2} \hat{\psi}^\dagger(\mathbf{x}) \hat{\psi}^\dagger(\mathbf{x}) \hat{\psi}(\mathbf{x}) \hat{\psi}(\mathbf{x}), \quad (10)$$

describing a dilute boson gas prone to an external potential  $V$  and a contact interaction. One then finds that the TDVP equations give rise to the (time-dependent) *Gross-Pitaevskii* equation:

$$i \frac{\partial}{\partial t} \varphi(\mathbf{x}) = -\frac{\nabla^2}{2m} \varphi(\mathbf{x}) + V(\mathbf{x}) \varphi(\mathbf{x}) + U |\varphi(\mathbf{x})|^2 \varphi(\mathbf{x}). \quad (11)$$

Despite the simplicity of this derivation, the Gross-Pitaevskii has been employed with great success to describe the behaviour of Bose-Einstein condensates [14].

We highlighted here only a few of the most common and successful methods to define and parametrise variational manifolds in the exponentially large Hilbert space. Besides these we have omitted the plethora of post-Hartree-Fock methods such as *coupled cluster methods* [15].

## 1.2 Quantum spin systems: monogamy & frustration

A very useful and insightful idealisation of a generic quantum many-body system or of a quantum field theory is provided by quantum lattice systems. In this framework, the lattice is a discretisation of space that hosts quantum degrees of freedom on its vertices, and in some cases on the edges. From the point of view of quantum field theory, this is the best controlled way of circumventing UV-divergences [16, 17]. From the point of view of condensed matter and atomic physics, lattice systems provide effective (compressed) descriptions of the relevant degrees of freedom.

In these lecture notes, we will be concerned with the low-energy physics of quantum many-body lattice models in one or two spatial dimensions. Restricting to the low-energy sector is often permitted due the observation that for most realistic materials the Fermi temperature is of the order  $10^4 K$ , much larger than the typical temperature scale of an experiment (even room temperature is very cold if you consider systems of electrons). But also in quantum field theory the ground state and its low lying excitations are typically the states of interest when computing for example cross sections of scattering experiments. Typically, the Hilbert spaces under consideration are a tensor product of qudits which we will denote by  $\mathcal{H} = (\mathbb{C}^d)^N \equiv \bigotimes_{i=1}^N \mathbb{C}^d$ , and unless stated otherwise we will consider the thermodynamic limit  $N \rightarrow \infty$ , while neglecting subtleties regarding order of limits. Note however that the dimension of the corresponding

<sup>1</sup>In this paragraph we will exceptionally denote operators with hats to avoid confusion with scalar functions.

Hilbert space is still exponential as a function of the system size, so lattice systems also suffer from the same exponential wall.

Obtaining a good approximation to the ground state of such a model is generically very non-trivial due to *frustration*. In this context, the term frustration refers to the inaptitude of the ground state to simultaneously minimise the energy of all Hamiltonian terms. As an example, consider for instance the paradigmatic spin 1/2 antiferromagnetic Heisenberg chain in  $(1+1)d$ :

$$H = \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j. \quad (12)$$

If we would consider only two spins, the ground state would be the singlet state with density matrix  $\rho = |\phi\rangle\langle\phi|$  where  $|\phi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ . In the case of spin chains with  $N$  spins, the total energy would clearly be minimised if all reduced density matrices  $\rho_{i,i+1}$  of the ground state are equal to this singlet – but this is clearly not possible: the *monogamy of entanglement* dictates that such a state cannot exist [18, 19]. Indeed, there exists no three-partite state  $\rho_{ABC} \geq 0$ , such that  $\text{Tr}_C(\rho_{ABC}) = \text{Tr}_B(\rho_{ABC}) = |\phi\rangle\langle\phi|$ . As such, monogamy implies frustration.

Monogamy of entanglement can also be used to explain why mean field theory becomes increasingly exact when the number of spatial dimensions increases. Consider instead of the one-dimensional chain a lattice with a large coordination number and again a Hamiltonian which is the sum of antiferromagnetic Heisenberg interactions between all neighbouring spins. Intuitively, in that case, for any given spin its entanglement with other spins then needs to be distributed over so many neighbours that the ground state necessarily becomes very close to a separable state  $\otimes_i |\psi_i\rangle$  and mean field theory becomes exact. Though presented in a hand waving fashion here, this argument can be made precise by invoking the *quantum de Finetti theorem* [20]. This theorem characterises multi-partite density matrices that are invariant under permutations of the different parties, and plays an important role in proving the security of quantum cryptographic protocols.

From this discussion we have deduced the fact that the ground state cannot be such that all its  $\rho_{i,i+1}$  are spin singlets. Now we could rather look for a state whose marginals are, in a sense, *as close as possible* to the spin singlet. To this end, one could for example try the *resonating valence bond state* as an ansatz to the ground state:

$$|\Psi\rangle = |\Psi_+\rangle + |\Psi_-\rangle, \quad \text{where} \quad |\Psi_{\pm}\rangle = \prod_i (|\uparrow\downarrow\rangle_{2i,2i\pm 1} - |\downarrow\uparrow\rangle_{2i,2i\pm 1}). \quad (13)$$

Graphically one could depict this state as

$$|\Psi\rangle = \begin{array}{c} \begin{array}{cccc} & 2i-1 & 2i & 2i+1 & 2i+2 \\ & \circ & \circ & \circ & \circ \\ & \text{---} & \text{---} & \text{---} & \text{---} \end{array} \\ + \begin{array}{cccc} & & & & \\ & \circ & & \circ & \\ & \text{---} & & \text{---} & \end{array} \end{array}, \quad (14)$$

where the wiggly lines stand for spin singlets between neighbouring sites. In fact,  $|\Psi_{\pm}\rangle$  span the two-dimensional ground state subspace of the model introduced by Majumdar and Ghosh in ref. [21]. Note that this state has all the symmetries we could expect: it is  $SU(2)$  symmetric and has translational, time-reversal and reflection invariance. This is important as ground states (or more precisely, ground state subspaces) should inherit all the symmetries of the Hamiltonian. More importantly, it has some non-vanishing entanglement, and we would therefore expect the energy of this state to be much closer to the actual ground state energy than the mean field approximation. Nevertheless, the energy of this state turns out to be still a fairly bad approximation of the true ground state energy of the Heisenberg model as it overestimates the energy quite a lot.

At this point it should be clear that writing down entangled states with the correct symmetries that can serve as ansatz for the ground state of our favourite many-body quantum

Hamiltonian, is very hard. However, the valence bond state suggest one particular way of doing this. Inspired by this Majumdar-Ghosh model, Affleck, Kennedy, Tasaki and Lieb (AKLT) proposed in 1987 a state constructed of pairs of entangled spin 1/2 particles which are projected onto a spin 1 particle on each lattice site [22]. Their construction goes as follows. One starts from a chain of spin 1/2 particles where every other pair of neighbouring particles is in a spin singlet. On every site  $i$  the rightmost particle is then taken together with the leftmost particle on the next site  $i + 1$  to form a *physical* site. The two *virtual* spin 1/2 particles that make up the physical site are then projected onto their spin 1 subspace. A graphical depiction of this state is more enlightening:

$$|\Psi_{\text{AKLT}}\rangle = \sim \text{---} \bigcirc \text{---} \bigcirc \text{---} \bigcirc \text{---} \bigcirc \text{---} \sim, \quad (15)$$

where

$$\bigcirc = |+\rangle\langle\uparrow\uparrow| + |0\rangle\frac{\langle\uparrow\downarrow| + \langle\downarrow\uparrow|}{\sqrt{2}} + |-\rangle\langle\downarrow\downarrow|. \quad (16)$$

As shown by the aforementioned authors, this state is the ground state of a Hamiltonian which is very closely related to the spin 1 Heisenberg model, and is believed to belong to the same (symmetry-protected) phase (discussed further in lecture section 3). Since two neighbouring sites exactly involve one spin singlet, the 4 constituent spin 1/2 particles which make up these physical degrees of freedom can not be in a coupled spin 2 state. As such, this state is annihilated by the projector on the spin 2 subspace of two spin 1's. Therefore, this state is the ground state of the AKLT Hamiltonian

$$H = \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1} + \frac{1}{3} (\mathbf{S}_i \cdot \mathbf{S}_{i+1})^2, \quad (17)$$

where the local two-body term can be recognised as the spin 2 projector, up to a shift and a prefactor.

### 1.3 Tensor networks & area laws

A few years after AKLT, in 1992, Fannes, Nachtergaele and Werner [23] introduced the class of one-dimensional *finitely correlated states* which generalises the construction of the AKLT state using ideas of quantum Markov chains. The name *finitely correlated states* stems from the fact that these states provably have exponentially decaying connected correlation functions, as will be demonstrated below in section 2.2. They also proved that all those states are ground states of gapped frustration-free Hamiltonians. The modern point of view, as first advocated in ref. [24], is to construct those states by starting from a chain of maximally entangled *virtual* qudits between neighbouring sites that are projected onto the *physical* spin on each site by a linear map. This linear map is often referred to as *projector* (even though it doesn't have to be an idempotent linear map), which is the origin of the name *projected entangled-pair states*, although this name is now typically used for the higher dimensional generalisations. These projectors contain parameters which play the role of the *variational* degrees of freedom of that state. This class of states coincides with the class of (infinite and translational invariant) uniform *matrix product states* (MPS), and this is the name that is most often used to characterise them. Let us make this more explicit. The construction of MPS thus starts from considering maximally entangled states

$$|I\rangle_i = \sum_{\alpha=1}^{\chi} |\alpha\alpha\rangle_{i,i+1} \quad (18)$$

between each pair of neighbouring sites  $i, i+1, i \in \mathbb{Z}$ . The integer  $\chi$  is called the *bond dimension* and directly controls the number of variational parameters as we will see. We then consider

on every physical site of the chain following linear map

$$A = \sum_{i, \alpha, \beta} A_{\alpha\beta}^i |i\rangle \otimes |\alpha\rangle \otimes \langle\beta| \in \mathbb{C}^{d \times d}, \quad (19)$$

on which we act on the entangled pairs, so that the state corresponding to the tensor  $A$  is given by:

$$|\Psi[A]\rangle = \left(\bigotimes_{i \in \mathbb{Z}} A\right) \cdot \left(\bigotimes_{i \in \mathbb{Z}} |I\rangle_i\right). \quad (20)$$

We will say that  $A$  generates the MPS  $|\Psi[A]\rangle$ . Expanding now the product in this expression yields the more familiar expression of *uniform* matrix product states, directly in the thermodynamic limit:

$$|\Psi[A]\rangle = \sum_{\{i_i\}} (\cdots A^{i_j} A^{i_{j+1}} \cdots) |\dots, i_j, i_{j+1}, \dots\rangle. \quad (21)$$

This expression for the wave function explains the name *matrix product* states, as the wave function amplitudes are computed as the products of the matrices  $\{A^i\}_i$ . In the uniform case the number of variational parameters is  $d\chi^2$ . Notice however that the dimension of the manifold of states spanned by uniform MPSs is strictly smaller than  $d\chi^2$  since for any non-singular matrix  $X$  a transformation of the kind

$$A^i \mapsto XA^iX^{-1}, \quad \forall i = 1, 2, \dots, d, \quad (22)$$

called a *gauge transformation*, leaves the state  $|\Psi[A]\rangle$  invariant. A more precise characterisation of this manifold is relegated to the next section.

Note that we could also have chosen our tensors (projectors) different at every site, leading to a generic non-uniform MPS with bond dimensions that can vary over the different edges of the lattice. This way we can easily deal with finite spin chains with open, periodic or twisted boundary conditions. Remark that, for the uniform case we did not specify the boundary conditions at infinity. For generic matrix product states, which are *injective* (defined below), the choice of these boundary conditions is irrelevant. Throughout these lecture notes we will focus almost exclusively on matrix product states in the thermodynamic limit for which this injectivity property holds. For completeness let us display here generic MPSs on finite spin chains. A generic MPS on an open chain can be written as:

$$|\Psi[A]\rangle = \sum_{\{i_l\}} \mathbf{v}_L^\dagger A^{i_1}[1] A^{i_2}[2] \cdots A^{i_N}[N] \mathbf{v}_R |i_1, i_2, \dots, i_N\rangle. \quad (23)$$

where the vectors  $\mathbf{v}_{L/R}$  encode the boundary conditions, and the integers between square brackets index the physical sites. A translation invariant state on a periodic chain with  $N$  sites on the other hand takes the form

$$|\Psi[A]\rangle = \sum_{\{i_l\}} \text{Tr}(A^{i_1} A^{i_2} \cdots A^{i_N}) |i_1, i_2, \dots, i_N\rangle. \quad (24)$$

As is customary when dealing with tensor networks, we will use a graphical language in which a single MPS tensor is depicted as:

$$\begin{array}{c} i \\ | \\ \alpha \text{---} \textcircled{A} \text{---} \beta \end{array} = A_{\alpha\beta}^i, \quad \begin{array}{l} i=1,2,\dots,d \\ \alpha,\beta=1,2,\dots,\chi \end{array} \quad (25)$$

and where a leg connecting two tensors indicates a contraction over the corresponding indices. With these conventions in mind, we can depict the MPS (21) as following diagram:

$$|\Psi[A]\rangle = \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \dots \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} . \quad (26)$$

Often, we will not explicitly indicate the MPS tensor in the figures when they can be deduced from the context.

Now that we have introduced the variational class of matrix product states, let us argue why these states approximate well the ground states of local gapped Hamiltonians [25]. Physically, this follows from the *area law* of entanglement entropy satisfied by gapped ground states. In generic dimensions, an area law for the entanglement entropy means that the bipartite entanglement entropy between a region  $\mathcal{A}$  and its complement  $\bar{\mathcal{A}}$ , defined as the Von Neumann entropy of the reduced matrix  $\rho_{\mathcal{A}} = \text{Tr}_{\bar{\mathcal{A}}}(\rho)$ , i.e.  $S = -\text{Tr}(\rho_{\mathcal{A}} \log(\rho_{\mathcal{A}}))$ , does not scale with the size, or *volume*, of the region  $\mathcal{A}$  but rather by the size of its boundary  $\partial\mathcal{A}$ , or thus its *area*. In the one-dimensional setting the boundary of a connected subregion of the chain is just a constant, and in this setting the area law was proven in 2007 by Hastings [26]. As we will now demonstrate MPS inherently satisfy an area law.

To this end, we make use of the graphical calculus of MPS. Herein, let us calculate the reduced density matrix of a uniform MPS when tracing out the left half of the system. This can be depicted as

$$\rho_R = \text{Tr}_L(|\Psi[A]\rangle\langle\Psi[A]|) = \dots$$

$$\dots, \quad (27)$$

where the dotted vertical line represents the entanglement cut between the left ( $L$ ) and right ( $R$ ) subsystem. An object which plays a pivotal role in the computation of the reduced density matrix, but also appears in e.g. the computation of correlation functions (see section 2.2), is the *transfer matrix*

$$\mathbb{T}_A = \sum_i A^i \otimes \bar{A}^i, \quad (28)$$

which we have indicated as well in the above figure by the dotted box. Let us write its spectral decomposition<sup>2</sup> as

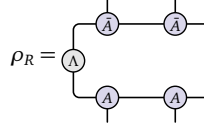
$$\mathbb{T}_A = \sum_{i=1}^{\chi^2} \lambda_i |\rho_i\rangle\langle\lambda_i|, \quad (29)$$

where the eigenvalues are ordered according to decreasing absolute value. The eigenvectors are clearly invariant subspaces of the completely positive linear maps  $S_L(X) = \sum_i A_i^\dagger X A_i$  and  $S_R(X) = \sum_i A_i X A_i^\dagger$ . If those maps are ergodic, the quantum Perron-Frobenius theorem dictates that the largest eigenvalue in magnitude will be unique, real and positive. Furthermore, the corresponding left and right eigenvectors  $|\rho_L\rangle \equiv |\rho\rangle$  and  $|\lambda_L\rangle \equiv |\Lambda\rangle$ , when reshaped as matrices (they have 2 indices), are positive definite. This property of having a unique dominant eigenvector is equivalent to the notion of injectivity mentioned above. Another equivalent characterisation of injectivity is as follows: an MPS is injective if and only if there exists a finite  $L$ , called the *injectivity length*, such that the collection of matrices  $\{A_i^1 A_i^2 \dots A_i^{L_i}\}$  is a basis

<sup>2</sup>More generally, Jordan blocks could be needed, but this is not relevant for the following discussion.

for all  $\chi \times \chi$  matrices. This is also precisely the property needed to guarantee that the MPS is the *unique* ground state of its gapped parent Hamiltonian. For a generic MPS, this will always be the case: the set of non-injective MPSs is of measure zero, as this is only possible when the matrices  $A^i$  have common invariant subspaces.

To ensure that the state is properly normalised to one we will always normalise the MPS tensor  $A$  such that the corresponding eigenvalue  $\lambda_1$  is equal to one (see also section 2.2). From these considerations, it follows that we can express the reduced density matrix as follows:



$$\rho_R = \dots \quad (30)$$

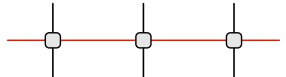
As  $\rho_R$  is the product of a  $d^L \times \chi$  and a  $\chi \times d^L$  matrix with  $L$  the total number of sites on the right hand side, we conclude that the rank of the reduced density matrix  $\rho_R$  is upper bounded by  $\chi$ . The entanglement entropy is hence bounded by  $S \leq \log \chi$ , as this is the maximal entropy for a state of rank  $\chi$ . The theorems in refs. [25, 26] ensure that the converse is also true: whenever a state satisfies an area law for the entanglement entropy, there exists an MPS that will be a faithful approximation of that state. In the next lecture we will study the entanglement properties of general matrix product states in more detail.

#### 1.4 Matrix product operators

Throughout these lecture notes we will also be making use of tensor network representations of linear operators. Uniform *matrix product operators* [27, 28] are written as

$$\sum_{\{i_k, j_k\}} (\dots M^{i_k, j_k} M^{i_{k+1}, j_{k+1}} \dots) |\dots, i_k, i_{k+1}, \dots\rangle \langle \dots, j_k, j_{k+1}, \dots|, \quad (31)$$

and graphically depicted by



$$\quad (32)$$

In analogy with MPS, the generalisation to open boundary conditions, and the non-uniform case is straightforward.

MPOs have since long been used extensively in the field of statistical physics, where MPOs of a particular type are known as transfer matrices (see [29] for a unifying discussion). Similarly to the case of MPS, we call an MPO injective when the matrices  $\{M^{ij}\}_{i,j}$  have no common invariant subspaces and span the entire matrix algebra.

Non-injective MPOs are however also used extensively, for example to encode Hamiltonians of quantum spin chains. Let us e.g. consider the Hamiltonian for the transverse field Ising model with open boundary conditions:

$$H = - \sum_{i=1}^{L-1} \sigma_i^Z \sigma_{i+1}^Z - \lambda \sum_{i=1}^L \sigma_i^X. \quad (33)$$

This Hamiltonian can be represented as an MPO with site-independent tensors<sup>3</sup>  $M$  of bond dimension 3:

$$M_{a\beta} = \begin{pmatrix} \mathbb{1} & -\sigma^Z & -\lambda \sigma^X \\ 0 & 0 & \sigma^Z \\ 0 & 0 & \mathbb{1} \end{pmatrix}_{a\beta}, \quad (34)$$

<sup>3</sup>In this notation, the element  $M_{a\beta}$  is an operator acting on the physical Hilbert space.

and boundary vectors

$$\mathbf{v}_L = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v}_R = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad (35)$$

Note that since we are dealing with a non-injective MPO here (there are invariant subspaces), the boundary vectors must be defined. This MPO is one of the generators of the Onsager algebra. Its exponential,  $\exp(i\tau H)$ , cannot be written anymore as an MPO with a bond dimension independent of the system size – unless  $\lambda = 0$ , then (exercise) it can be written as an injective MPO with  $\chi = 2$ .

### 1.5 The manifold of MPS

Now that we have introduced matrix product states and operators through the lens of the entangled-pair construction, let us investigate in more detail the manifold of uniform matrix product states [4, 30, 31]. To this end we consider all uniform MPSs with a fixed bond dimension  $\chi$ ,  $|\Psi[A]\rangle$ . As a first observation, note that these states don't span a linear subspace of the full Hilbert space since the sum of two MPSs is typically not an MPS of the same bond dimension. Instead, they live on a surface whose local coordinates are given by the variational parameters of the state, i.e.  $A_{\alpha\beta}^i$ . Recall from above that to every point of the manifold we can associate a linear tangent space. For the case at hand, the tangent vectors are of the form

$$|\Psi[B, A]\rangle = \sum_I B^I \frac{\partial}{\partial A^I} |\Psi[A]\rangle, \quad (36)$$

where here and below  $I$  collectively denotes the physical and virtual indices of the MPS tensors. Graphically, such a tangent vector boils down to

$$|\Psi[B, A]\rangle = \sum_n \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \textcircled{B} \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---}, \quad (37)$$

where the sum over  $n$  is a sum over all the positions of the  $B$  tensor in the infinite chain.

At any given point  $|\Psi[A]\rangle$  of this manifold the overlap of two tangent vectors  $|\Psi[B, A]\rangle$  and  $|\Psi[B', A]\rangle$  provides a metric or *Gram matrix*  $G$  via

$$\langle \Psi[\tilde{B}', \tilde{A}] | \Psi[B, A] \rangle = \sum_{I, J} \tilde{B}'^I \underbrace{\left\langle \frac{\partial}{\partial \tilde{A}^I} \Psi[\tilde{A}] \left| \frac{\partial}{\partial A^J} \Psi[A] \right\rangle}_{G(A, \tilde{A})_{I, J}} B^J. \quad (38)$$

Naively one would now assume that the dimension of the manifold is equal to the number of parameters in the tensor  $B$ , viz.  $d\chi^2$ . But note that gauge transformations of the form  $B^i \rightarrow B^i + X A^i - A^i X$  leave the state  $|\Psi[B, A]\rangle$  invariant so that  $d\chi^2$  overcounts the dimension of the MPS manifold. A more careful argument which is made precise in ref. [4] demonstrates that only  $(d-1)\chi^2$  of those parameters can be chosen independently. Provided that  $A_L^i$  is in the so called *left canonical gauge* (see section 2.2), a faithful parametrisation of  $B$  is then  $B^i = V^i X$ , where the  $V^i$  spans the null space of  $(A_L)^i_{\alpha\beta}$  (interpreted as matrix from  $\beta$  to  $\alpha$ ,  $i$ ) and  $X$  encodes the independent degrees of freedom of  $B$ . With this parametrisation of  $B$ , the metric is locally Euclidean, i.e. the Gram matrix is equal to the identity. From the computational point of view this parametrisation is useful as it vastly simplifies the implementation of algorithms and improves their stability.

One such algorithm is the TDVP applied to the class of MPS. Recall from section 1.1 that TDVP aims to approximately solve the time-dependent Schrödinger equation

$$i \frac{\partial}{\partial t} |\Psi[A]\rangle = H |\Psi[A]\rangle \quad (39)$$

by projecting the right-hand side of this equation onto the tangent plane:

$$i \frac{\partial}{\partial t} |\Psi[A(t)]\rangle = \mathcal{P}_{A(t)} H |\Psi[A(t)]\rangle, \quad (40)$$

where  $\mathcal{P}_{A(t)}$  is the projector on the tangent plane. A further manipulation of the TDVP equation eq. (40) reveals that the TDVP is equivalent to a (semi-)classical set of equations of motions governed by a *Poisson bracket* on the MPS manifold. The famous Density Matrix Renormalisation Group (DMRG) algorithm [32] – discussed below in section 2.4.1 – and all its variants can be understood as specific splitting methods for solving this differential equation [33]. Recall that a Poisson bracket is a bilinear, antisymmetric function on phase space which satisfies the Jacobi identity. If we define the Poisson bracket for arbitrary functions  $f$  and  $g$  that depend on  $A$  and  $\bar{A}$  as follows:

$$\{f, g\} = -\frac{\partial f}{\partial A_I} (G^{-1})^{IJ} \frac{\partial g}{\partial \bar{A}_J} + \frac{\partial g}{\partial A_I} (G^{-1})^{IJ} \frac{\partial f}{\partial \bar{A}_J}, \quad (41)$$

and write  $f(A, \bar{A}) = \langle \Psi[A] | F | \Psi[A] \rangle$ , it can be shown that the time evolution of the expectation value  $f$  is governed by

$$\partial_t f = i \{f, h\}. \quad (42)$$

This has two significant consequences. For one, it follows from  $\{h, h\} = 0$  that the energy expectation value is exactly preserved under time evolution. This was anticipated above and implies that the manifold is symplectic. On the other hand, for any operator  $K$  which commutes with the Hamiltonian, and thus constitutes a symmetry generator, as well as with the tangent space projector  $\mathcal{P}_A$ , it can be shown that  $\{K, h\} = 0$  and hence  $\partial_t K = 0$ . In particular, this is the case for symmetry generators which are tensor products of local on-site unitaries.

Besides TDVP it should be mentioned that the manifold picture has also provided the necessary framework for ground state optimisation directly in the thermodynamic limit, the so-called *variational uniform MPS* (VUMPS) algorithm, as well as the construction of *quasiparticle excitations*. These topics are covered in section 2.4 and section 2.5 respectively, but first we now delve deeper into the calculus of MPS.

## 1.6 Synopsis of lecture I

The main message of lecture I is the fact that the full many-body Hilbert space is an illusion: to describe relevant many-body states such as ground states of quantum spin systems, it is sufficient to consider a low-dimensional manifold of states which captures all relevant physics. For the particular case of gapped quantum spin chains, the monogamy of quantum entanglement implies the existence of an area law for the entanglement entropy. This implies that the manifold of MPS is in one-to-one correspondence with all states satisfying an area law. Instead of focusing on the full Hilbert space, we can hence project the Hamiltonian on this MPS manifold using the prescription of the time-dependent variational principle; all tensor network algorithms can be understood within this framework.



## 2 Lecture II: The calculus of MPS & algorithms

In this section we will introduce the calculus of matrix product states. In particular this constitutes the computation of observables, which is simplified greatly by introducing appropriate *normal forms* for the MPS tensors. With these tools at hand we will focus on the renowned *density matrix renormalization group* (DMRG) algorithm for ground state optimisation on finite chains, as well as the *variational uniform matrix product state* (VUMPS) algorithm, which works directly in the thermodynamic limit. We then study quasiparticle excitations on top of the ground state, which are parametrised as the *quasiparticle ansatz*. Let us however kick off with some more examples of matrix product states.

### 2.1 Examples, injectivity & the parent Hamiltonian

**Product states** Any product state  $|\Psi\rangle = |\psi\rangle^{\otimes N}$ ,  $|\psi\rangle = \sum_i \psi_i |i\rangle$  can be written as a bond dimension 1 MPS of the form

$$A_{11}^i = \psi_i. \quad (43)$$

**AKLT** The paradigmatic example of a non-trivial matrix product state encountered above in section 1.3 is the Affleck-Kennedy-Lieb-Tasaki (AKLT) state. This state is generated by the MPS matrices:

$$A^i = \sqrt{\frac{1}{3}} \sigma^i, \quad i = X, Y, Z, \quad (44)$$

where the  $\sigma^i$  are the Pauli matrices, and the prefactor ensures that the state is normalised in the thermodynamic limit.

On periodic boundary conditions, this state is the unique gapped ground state of the spin 1 Hamiltonian (17). As will be explained below in section 3.3, this Hamiltonian has a four-fold ground state degeneracy on open chains, characterised by gapless edge modes. This is a hallmark of its *symmetry-protected topological* (SPT) order.

**GHZ** Another example of a state which can be expressed as a bond dimension 2 MPS is the so called Greenberger–Horne–Zeilinger (GHZ) state [34]:

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle^{\otimes N} + |1\rangle^{\otimes N}). \quad (45)$$

Its MPS representation reads:

$$A^i = \begin{cases} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, & i = 0 \\ \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, & i = 1. \end{cases} \quad (46)$$

Note that the GHZ state may be viewed as exhibiting  $\mathbb{Z}_2$  symmetry breaking. This is reflected in the fact that the MPS matrices are block diagonal, and the blocks are swapped under the action of  $\prod_i \sigma_i^X$ . Clearly, this MPS representation of the GHZ-state is non-injective.

This has to be contrasted with the AKLT state, whose MPS matrices span the entire  $\mathbb{C}^{2 \times 2}$  matrix algebra, therefore cannot be brought in block diagonal form, and thus define an injective MPS. Note that the injectivity length of the AKLT state is 2, which can be checked by an immediate computation.

It holds generically that every uniform MPS can be brought<sup>4</sup> in a *block injective* or a *standard* form whereby all MPS matrices are of the form [35, 36]

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i, \quad (47)$$

and where all  $\{A_k^i\}_k$ ,  $k = 1, \dots, r$  are injective, and sometimes called *injective blocks*. It should be noted that this notion of injectivity should be revisited in the case of *fermionic* MPS, which is the topic of section 3.5.

Interestingly, every MPS can be regarded as a ground state of a corresponding *parent Hamiltonian*, thereby placing the study of MPS and their associated parent Hamiltonians on the same footing. This parent Hamiltonian is gapped, local and frustration-free, meaning that the energy of each term is minimised by the MPS. Its local terms are the projectors on the null space of the local reduced density matrix on the corresponding contiguous region of spins [5, 35]. For the GHZ state for example, the local parent Hamiltonian term is given by  $\frac{1}{2}(\mathbb{1} - \sigma_i^Z \sigma_{i+1}^Z)$ . When an MPS is injective, it can be shown that it is the unique ground state of its parent Hamiltonian. More interestingly, when the parent Hamiltonian commutes with an on-site representation of a symmetry group  $G$ , as in the case of the GHZ state, the action of  $G$  permutes (a minimal irreducible subset of) the different injective blocks. In that case, one finds a maximal subgroup  $H \subseteq G$  whose action leaves a particular reference block invariant [37]. The blocks are then labelled by *cosets* in  $G/H$ .<sup>5</sup>

Below, in lecture 3 and lecture 5, we will re-encounter the parent Hamiltonian as a tool to classify *quantum phases of matter*, which – roughly speaking – are equivalence classes of Hamiltonians which can be adiabatically transmuted into each other.

## 2.2 Normal forms, entanglement & observables

As mentioned in section 1.3, the set of matrices  $\{A^i\}$  is not a unique representation of a given MPS. This is because matrix product states are invariant under reparametrisations of the form

$$A^i \mapsto X A^i X^{-1}, \quad \forall i = 1, 2, \dots, d, \quad (48)$$

for any non-singular matrix  $X$ , since the matrices  $X$  and  $X^{-1}$  cancel each other on the bonds connecting neighbouring MPS matrices. This redundancy is often called *gauge freedom* of the MPS description. It often turns out that this gauge freedom can be heavily exploited to simplify certain computations and improve the numerical stability of MPS algorithms.

By appropriate gauge transformations, the MPS matrices can be brought in the so-called *left or right canonical forms*, i.e. respectively defined via the conditions:

$$\sum_i A^{i\dagger} A^i = \mathbb{1}_\chi, \quad (49)$$

$$\sum_i A^i A^{i\dagger} = \mathbb{1}_\chi. \quad (50)$$

<sup>4</sup>After getting rid of off-diagonal blocks, as reviewed in e.g. ref. [5].

<sup>5</sup>Recall that a (left) coset  $gH$  is the set  $gH = \{gh|h \in H\}$ . The set of all cosets is denoted  $G/H$  and partitions the group  $G$ .

$$\begin{array}{c} \textcircled{A} \\ | \\ \textcircled{A} \end{array} = \left[ \begin{array}{c} \text{---} \\ \text{---} \end{array} \right], \quad (51)$$

5

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this boils down to the contraction

$$\langle O_1[i] O_2[j] \rangle_{\Psi[A]} = \begin{array}{c} \text{---} \text{A} \text{---} \text{A} \text{---} \text{A} \text{---} \\ | \quad | \quad | \\ \text{A} \quad O_1 \quad \text{A} \\ | \quad | \quad | \\ \text{---} \text{A} \text{---} \text{A} \text{---} \text{A} \text{---} \end{array} \dots \begin{array}{c} \text{---} \text{A} \text{---} \text{A} \text{---} \text{A} \text{---} \\ | \quad | \quad | \\ \text{A} \quad O_2 \quad \text{A} \\ | \quad | \quad | \\ \text{---} \text{A} \text{---} \text{A} \text{---} \text{A} \text{---} \end{array} . \quad (58)$$

By assuming injectivity and thus again that the transfer matrix has a unique largest eigenvalue 1 with corresponding left and right eigenvectors  $\Lambda$  and  $\rho$ , we are able to rewrite this contraction as

$$\begin{array}{c} \text{---} \text{A} \text{---} \text{A} \text{---} \\ | \quad | \\ \text{A} \quad O_1 \quad \text{A} \\ | \quad | \\ \text{---} \text{A} \text{---} \text{A} \text{---} \end{array} \dots \begin{array}{c} \text{---} \text{A} \text{---} \text{A} \text{---} \\ | \quad | \\ \text{A} \quad O_2 \quad \text{A} \\ | \quad | \\ \text{---} \text{A} \text{---} \text{A} \text{---} \end{array} \rho , \quad (59)$$

which demonstrates that this correlator can be computed with a cost that scales as  $\mathcal{O}(L\chi^3)$ . Indeed, the contraction of this tensor network proceeds from left to right, and by choosing the contraction order in a judicious way, the main cost is the multiplication of the  $\chi \times \chi$  matrices (see ref. [38] for an algorithm that identifies the optimal contraction order of a generic tensor network). As particular cases of this expression, we have the expectation value of a local operator by taking  $O_1 \equiv O$  and  $O_2 \equiv 1$ :

$$\begin{array}{c} \text{---} \text{A} \text{---} \\ | \\ \text{A} \quad O \quad \text{A} \\ | \\ \text{---} \text{A} \text{---} \end{array} \rho , \quad (60)$$

as well as the norm squared of the MPS

$$\langle \Psi[A] | \Psi[A] \rangle := \begin{array}{c} \text{---} \text{A} \text{---} \\ | \\ \text{A} \quad \text{---} \text{A} \text{---} \\ | \\ \text{---} \text{A} \text{---} \end{array} \rho . \quad (61)$$

If we now define the *connected* two-point correlation function as customary:

$$C(O_1[i], O_2[j]) := \langle O_1[i] O_2[j] \rangle_{\Psi[A]} - \langle O_1[i] \rangle_{\Psi[A]} \langle O_2[j] \rangle_{\Psi[A]}, \quad (62)$$

one can show, by once more making use of the spectral decomposition of the transfer matrix eq. (29) (assuming it exists), that the dominant contribution to this correlation function decays *exponentially* in  $L$ , more precisely:

$$C(O_1[i], O_2[j]) \sim \lambda_2^{L-1} \sim \exp(-L/\xi), \quad (63)$$

if we define the *correlation length*  $\xi$  as  $\xi = -1/\log(|\lambda_2|)$ . This exponential decay of correlation functions is a witness of the fact that MPS describe well the ground state physics of *gapped* Hamiltonians, and can thus not reproduce the algebraic decay of correlations present in *critical* systems. Nevertheless, by making use of the theory of *entanglement scaling*, the topic of the next section, precise approximations to critical correlation functions can be achieved.

### 2.3 Scaling theory

Aside from gapped ground state physics, MPS methods are also a powerful tool to compute properties of *critical* systems. Using uniform methods which work directly in the thermodynamic limit, i.e. on the infinite chain (see Sec. 2.4), it is possible to extract critical exponents and the central charge describing the universality class or CFT of a continuous phase transition. There is a large body of existing research about the topic, originally introduced in refs. [39–42]. Here we will follow the approach developed in the papers [43–45].

Close to a second-order critical point, we expect correlation functions to be of the Ornstein–Zernike form, it is a product of a power law contribution multiplied with an exponential decaying contribution. This can be understood from the Källén–Lehmann representation of correlation functions as a linear combination of a continuum of exponentials [43]. To express the continuum directly using MPS, one would need an infinite bond dimension MPS. Using finite bond dimension MPS is equivalent to a discretisation of this continuum spectrum. We can quantify the ‘discreteness’, or degree of approximation, by the gaps in the transfer matrix spectrum. Let  $\lambda_i$  as before be the eigenvalues of the transfer matrix, with  $\lambda_1 = 1$ , and define  $\epsilon_i = -\ln \lambda_i$ . Take a linear combination of the top  $n$  spectral variables  $\epsilon_i$ , excluding the first one:

$$\delta := \sum_{i=2}^n c_i \epsilon_i \quad \text{with} \quad \sum_{i=2}^n c_i = 0. \quad (64)$$

In refs. [43, 44], it was demonstrated that the  $\epsilon_i$  have a meaning of momenta, and that the difference between consecutive values of  $\epsilon_i$  therefore introduces an inverse length scale  $\delta$  which goes to zero if  $\chi \rightarrow \infty$ , both in the gapped and in the gapless case. In analogy to the finite-size scaling hypothesis of Cardy [46], where the length scale is the system size, it is possible to use this inverse length scale  $\delta$  to formulate a *scaling hypothesis* for any order parameter  $m$  of a second-order phase transition as follows:

$$m(t, \delta) = s^{-\beta/\nu} m(s^{1/\nu} t, s\delta). \quad (65)$$

An analogous scaling hypothesis for the correlation length is:

$$\delta \xi(t, \delta) = \xi(\delta^{-1/\nu} t, 1) = \xi(\delta^{-1/\nu} t). \quad (66)$$

From the forms above, one can also deduce the scaling hypothesis for the entanglement entropy. Using the CFT form for the entanglement entropy  $S$  [47], we know that  $\exp(\frac{6}{c} S)$  scales as length, so the scaling hypothesis for  $S$  is:

$$\exp\left(\frac{6}{c} S(t, \delta)\right) = s \exp\left(\frac{6}{c} S(s^{-1/\nu} t, s\delta)\right), \quad (67)$$

where  $c$  is the central charge. Performing simulations using uniform MPS methods with various bond dimensions  $\chi$  then allows for fitting  $m, \xi, c$  using standard optimisation methods. The whole point of this is the fact that many different simulations for many different parameters of the Hamiltonian close to the critical point and for different bond dimensions can be related to each other, and that the correct values of the critical point and critical exponents will lead to a data collapse of all the ensuing curves. We refer to ref. [45] for details. Working with a finite bond dimension is like opening a gap in the system, and its effect is indistinguishable to adding a relevant perturbation to a critical theory. This way, the finite entanglement scaling theory yields the state-of-the-art way of simulating critical theories – the fact that MPS are always gapped is not a bug, it is a feature!

## 2.4 Variational MPS algorithms

### 2.4.1 Density matrix renormalisation group

The *density matrix renormalisation group* (DMRG) algorithm, originally due to Steven White [32], is a variational method for finding the ground state of *local one-dimensional Hamiltonians*. It has been applied with tremendous success to a plethora of models and is still the current state of the art for ground state optimisation in (1+1)d for strongly-correlated spin chains [2, 8].

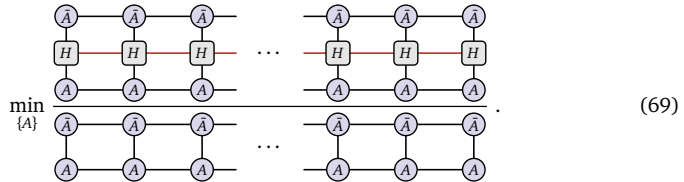
Although DMRG was originally derived and explained in terms of the renormalisation group, its modern interpretation is that of a variational method over the class of matrix product

states, where the optimum is iteratively found by using an alternating least squares method [48]. DMRG minimises the expectation value

$$\min_{\{A\}} \left( \frac{\langle \Psi[A] | H | \Psi[A] \rangle}{\langle \Psi[A] | \Psi[A] \rangle} \right), \quad (68)$$

over all finite MPS of a maximum (bounded) bond dimension  $\chi_{\max}$ . The larger you choose  $\chi_{\max}$ , the better the approximation will be. Even with modest computational resources, one can easily achieve bond dimensions  $\chi \sim \mathcal{O}(10^3)$  by making use of sparse eigensolvers such as a Lanczos method to solve the effective eigenvalue problem (see below), or even  $\mathcal{O}(10^5)$  when symmetries are exploited as well [49–54].

Assuming that the Hamiltonian is written in MPO form (see section 1.4), eq. (68) boils down to:



$$\min_{\{A\}} \left( \frac{\langle \Psi[A] | H | \Psi[A] \rangle}{\langle \Psi[A] | \Psi[A] \rangle} \right). \quad (69)$$

The algorithm is initiated by providing an initial guess for the ground state, typically simply a random MPS, and proceeds with a series of local sequential optimisations of all tensors, sweeping back and forth through the chain, until convergence is reached.

Fixing a site  $i$  and bringing all tensors to its left in left canonical form and vice versa for the tensors to its right, the update of the tensor  $A[i]$  boils down to replacing it by the eigenvector corresponding to the smallest real eigenvalue of the *effective Hamiltonian*



$$\left[ L \right] \left[ H \right] \left[ R \right]. \quad (70)$$

In this figure,  $L$  and  $R$  stand for the (site-dependent) *environments* obtained by contracting all MPS and MPO tensors to the left, respectively right, of site  $i$ .

As such, this version of the algorithm updates each site individually and is referred to as *one-site DMRG*. The downside of this approach is that there is no convenient way of increasing the bond dimension during the sweeps. This is particularly problematic when symmetries are imposed, since in that case the algorithm can not explore distinct symmetry charges on the virtual level. This shortcoming is remedied by a *two-site* version of the algorithm.

This version of the algorithm proceeds by blocking two sites in every step of every sweep, and subsequently updating the blocked tensor  $A'[i, i+1]$  as the minimal eigenvector of the effective two-site Hamiltonian



$$\left[ L \right] \left[ H \right] \left[ H \right] \left[ R \right]. \quad (71)$$

A singular value decomposition is then used to split  $A'$  into  $A[i], A[i+1]$ . It is in this step that the bond dimension can be dynamically controlled by truncating the SVD so as to only keep the largest singular values. The high-level pseudocode of DMRG can be found below.

**Algorithm 1:** Two-site DMRG

---

```

1 Initialize the MPS tensors  $A[i]$  (usually to a random state)
2 for  $k = 1$  to  $\text{num\_sweeps}$  do
    // Left-to-right sweep
3   for  $i = 1$  to  $N - 2$  do
4     Optimise and update pair of sites  $i, i + 1$ 
5     Update left environment at site  $i + 1$ 
    // Right-to-left sweep
6   for  $i = N - 1$  to  $2$  do
7     Optimise and update pair of sites  $i, i + 1$ 
8     Update right environment at site  $i + 1$ 
9 return  $\{A[i]\}$ 

```

---

**2.4.2 Variational uniform matrix product state**

The *variational uniform matrix product state* (VUMPS) algorithm is an infinite-chain equivalent of DMRG, and was introduced in ref. [55] (see ref. [4] for a pedagogical review, and [56] for an alternative method). It works directly in the thermodynamic limit, which makes it particularly useful for the simulation of critical systems and extracting their scaling laws. Of course, in the infinite limit the total ground state energy is unbounded, so that VUMPS rather minimises the *energy density*.

Before introducing the algorithm, let us first fix some notation. Recall the central gauge MPS  $A_C$  and bond tensor  $C$  introduced in section 2.2, alongside the left and right gauges  $A_L, A_R$ . We then define the effective single-site Hamiltonian,  $H_{\text{eff}}$ , as well as the effective environment for the bond tensor  $C$ ,  $H_C$ , as:

$$H_{\text{eff}} := \left[ \begin{array}{c} \text{---} L \text{---} \\ \text{---} H \text{---} \\ \text{---} R \text{---} \end{array} \right], \quad H_C := \left[ \begin{array}{c} \text{---} L \text{---} \\ \text{---} C \text{---} \\ \text{---} R \text{---} \end{array} \right]. \quad (72)$$

Here, the left and right environments  $L, R$  are the fixed points of the transfer matrices  $\mathbb{T}_L^{[H]}, \mathbb{T}_R^{[H]}$  respectively:

$$\mathbb{T}_L^{[H]} := \left[ \begin{array}{c} \hat{A}_L \\ \text{---} H \text{---} \\ \hat{A}_L \end{array} \right], \quad \mathbb{T}_R^{[H]} := \left[ \begin{array}{c} \hat{A}_R \\ \text{---} H \text{---} \\ \hat{A}_R \end{array} \right]. \quad (73)$$

They can be calculated by solving a system of linear equations.

As derived in ref. [55], the variational minimum of the ground state energy density in the manifold of uniform MPS is characterised by a simultaneous solution to

$$\begin{aligned} H_{\text{eff}}(A_C) &\propto A_C, \\ H_C(C) &\propto C, \\ A_C^i &= A_L^i C = C A_R^i. \end{aligned} \quad (74)$$

The last equation is included so as to avoid having to take  $A_L = A_C C^{-1}$ ,  $A_R = C^{-1} A_C$ , which is potentially ill-defined if  $C$  is close to being singular. Given an initial guess for the ground state, the VUMPS algorithm iteratively approaches this solution until convergence is reached. In each step, the current values of  $A_C$  and  $C$  are updated by replacing them by the eigenvectors

with minimal real eigenvalue of the current  $H_{\text{eff}}$  and  $H_C$ . Subsequently, approximate  $\tilde{A}_L$  and  $\tilde{A}_R$  are found from  $\tilde{A}_C$  and  $\tilde{C}$  by invoking the singular value decomposition for  $A_C C$  and  $C A_C$ :

$$\begin{aligned}\tilde{A}_L &= U_L V_L^\dagger, & U_L \Sigma_L V_L^\dagger &= \tilde{A}_C \tilde{C}, \\ \tilde{A}_R &= U_R V_R^\dagger, & \tilde{C}^\dagger \tilde{A}_C &= U_R \Sigma_R V_R^\dagger.\end{aligned}\quad (75)$$

As such, the 2-norms

$$\begin{aligned}\epsilon_L &= \|\tilde{A}_C - \tilde{A}_L \tilde{C}\|_2, \\ \epsilon_R &= \|\tilde{A}_C - \tilde{C} \tilde{A}_R\|_2,\end{aligned}\quad (76)$$

are minimised.

The VUMPS pseudocode for an MPO Hamiltonian can be found below.

---

**Algorithm 2:** VUMPS

---

```

1 Initialise  $\{A_L, A_R, C\}$  (usually to a random state)
2 Initialise current precision  $\epsilon_{\text{cur}}$ 
3 while  $\epsilon_{\text{cur}} > \epsilon_{\text{tol}}$  do
4   Calculate  $\mathbb{T}_L^{[H]}$  from  $A_L, H$ 
5   Calculate  $\mathbb{T}_R^{[H]}$  from  $A_R, H$ 
6   Calculate the left environment  $L$  from  $\mathbb{T}_R^{[H]}$ 
7   Calculate the right environment  $R$  from  $\mathbb{T}_L^{[H]}$ 
8   Construct the effective infinite Hamiltonian  $H_{\text{eff}}$  from  $L, R, H$ 
9   Construct the effective infinite environment  $H_C$  from  $L, R$ 
10  Find  $A_C$ , the leading eigenvector of  $H_{\text{eff}}$ 
11  Find  $C$ , the leading eigenvector of  $E_{\text{eff}}$ 
12  Calculate  $A_L, A_R$  from  $A_C, C$  as in eq. (75)
13  Set  $\epsilon_{\text{cur}} = \max(\epsilon_L, \epsilon_R)$ 
14 return  $\{A_L, A_R, C\}$ 

```

---

## 2.5 Quasiparticle excitations

Up until this point we have mainly focussed on the ground state, and algorithms to variationally approximate it, both in the finite (section 2.4.1) as well as in the infinite (section 2.4.2) setting. A comprehensive understanding of a quantum many-body system should also explain its low-lying excitations. Typically, the spectrum of a gapped quantum many-body Hamiltonian consists of a few isolated branches which have an interpretation in terms of localised *quasiparticle excitations*, as well as a tower of continuum *scattering states* which can be constructed by essentially adding up quasiparticle states. These quasiparticles show up in dynamical correlation functions – which can be probed in e.g. inelastic neutron scattering experiments – but in the case of critical systems also encode universal properties of their effective continuum description<sup>6</sup>.

Matrix product state techniques can be used to capture these quasiparticle excitations and their scattering states by means of the so-called *quasiparticle ansatz*, first introduced in ref. [57] and subsequently generalised in ref. [58] to single-particle excitations in one-dimensional chains and to two-particle states in ref. [59]. Here, we assume a translationally invariant local and gapped Hamiltonian  $H$ . For the time being we moreover assume a unique ground state to exclude domain-wall excitations, which will be discussed below. We denote an MPS

---

<sup>6</sup>Remember that a finite bond dimension gaps out the excitations when approximating critical ground states using MPS.



approximation of the ground state by  $|\Psi[A]\rangle$ . The ansatz for a quasiparticle with momentum  $k$  then takes the form

$$|\Psi[A, B], k\rangle = \sum_n e^{ikn} \text{---} \overset{\text{---}}{\underset{\text{---}}{\text{A}}} \text{---} \overset{\text{---}}{\underset{\text{---}}{\text{A}}} \text{---} \overset{\text{---}}{\underset{\text{---}}{\text{B}}} \text{---} \overset{\text{---}}{\underset{\text{---}}{\text{A}}} \text{---} \overset{\text{---}}{\underset{\text{---}}{\text{A}}} \text{---} . \quad (77)$$

That is, the ansatz replaces one of the ground state tensors by a tensor  $B$  which contains all variational parameters of the ansatz, and we take a momentum superposition over the position of  $B$  in the infinite chain. As such, it can be thought of as a *boosted* version of a tangent vector eq. (37), allowing one to exploit the full arsenal of techniques developed for the tangent space. Also, it is interesting to note that the quasiparticle ansatz can be interpreted as a generalisation of the Feynman-Bijl ansatz, or *single mode approximation* developed to describe collective excitations in liquid Helium [60–62]. Note that due to the finite bond dimension, we should not think about  $|\Psi[A, B], k\rangle$  as describing a localised particle on the location of the tensor  $B$ , but rather as a lump of increased energy density with respect to the uniform vacuum. Given  $A$ , the optimisation of  $B$  boils down to a generalised eigenvalue problem for an effective Hamiltonian, similar in spirit to section 2.4.2. A detailed exposition of the algorithm is beyond the scope of these notes, but let us remark that in it, the use of the efficient parametrisation of  $B$ , very similar to the one described in section 1.5, is crucial.

It is worth emphasising that the ground state encodes information about the excited states. Indeed, for a generic self-adjoint operator, one of its eigenvectors typically does not reveal anything about the others. However, the locality of the Hamiltonian is the distinguishing feature that ensures that eq. (77) provides a *good* ansatz for quasiparticle excitations. Indeed, in an application of the famous Lieb-Robinson bounds [63–65], it was demonstrated in ref. [66] that every (exact) eigenstate  $|\Psi_{k,\alpha}\rangle$  can be approximated arbitrary well via acting with a momentum  $k$  superposition of a localised (in space) operator on the ground state. The ansatz eq. (77) should then be recognised as the extreme case where the MPS ground state is acted upon by said operator whose support is truncated to one site.

Interestingly, in the presence of an internal symmetry the excitations are organised in degenerate multiplets transforming in an (irreducible) representation of the symmetry group. This is not compatible with the quasiparticle approach that we presented so far, as in that case the energy window around an excitation energy  $\Delta$ ,  $[\Delta - \delta, \Delta + \delta]$ , contains more than one excitation for any value of  $\delta$ . To remedy this, one can modify the ansatz by replacing the operator  $\tilde{O}$  by a multiplet of operators  $\tilde{O}_m$ , which in the MPS picture boils down to specifying the charge of the  $B$  tensor [67].

In the case of symmetry breaking and thus a degenerate ground state subspace, the excitations with lowest energy are not of the form eq. (77). Rather, they are *topological domain wall excitations* that form an interface between different ground states. In this context, *topological* refers to the fact that these excitations are created from the ground state by an extensive operator, namely a symmetry operator acting on the half-infinite chain terminated at the location of the domain wall. This can be accounted for by adapting the ansatz so as to encode different ground state tensors on either side of the  $B$  tensor. All computational MPS techniques work equally well in this case, but care has to be taken to contract the right environments with each other.

In fig. 1 we depict the low-energy excitation spectrum of the spin 1 antiferromagnetic Heisenberg model. It was obtained by explicitly enforcing the  $SU(2)$  symmetry, and we labelled the excitations by their spin. The isolated  $S = 1$  branch is clearly visible and reaches its minimum  $\Delta \approx 0.41$  at  $k = \pi$ , famously known as the *Haldane gap* [68, 69]. Astonishingly, this energy gap can be computed using the quasiparticle ansatz with a precision of ca. 10 digits, even with very modest computational resources.

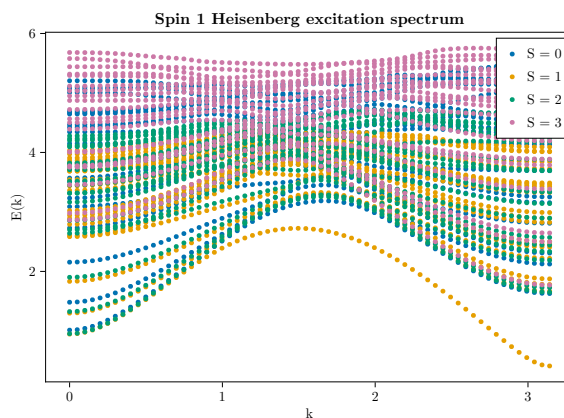


Figure 1: Excitation spectrum of the Heisenberg spin 1 antiferromagnet obtained with the one-site quasiparticle ansatz at bond dimension 48. Excitations are labelled by their  $SU(2)$  spin. Note in particular that the *Haldane* gap is situated at momentum  $\pi$ . Created with MPSKit.jl [70].

## 2.6 Synopsis of lecture II

Lecture II has been dealing with the computational aspects of matrix product states - how to define normal forms, how to minimise the energy on the manifold of MPS using alternating least squares methods (DMRG), how to extend those methods to infinite system sizes (VUMPS), and how to define excitations by projecting the full Hamiltonian on the tangent plane of the manifold of MPS. Furthermore, we touched upon the topic of entanglement scaling - through which critical systems can be studied with unprecedented precision.

The fact that - in all those applications - the computational cost only scales polynomially in both the system size and the bond dimension is a clear witness of the fact that tensor network methods effectively manage to break down the exponential many-body wall - and hence has the potential of resolving central problems in physics and chemistry.

### 3 Lecture III: Phases of matter

#### 3.1 Gapped phases of matter, entanglement & symmetries

A fundamental challenge in quantum many-body physics is to determine the *phase diagram* of a given model. Two models belong to the same phase if there exists a smooth path of gapped, bounded-strength, local Hamiltonians interpolating between them. From a pragmatic point of view, this is equivalent to the existence of a *sublinear-depth quantum circuit* that maps their ground states into each other [65, 71]. Distinct phases are then separated by phase transitions, which in the case of a second-order phase transition – as encountered above in section 2.3 – is typically described by a conformal field theory [72]. For a model with two parameters  $h_{1,2}$  a fictitious phase diagram is sketched below.

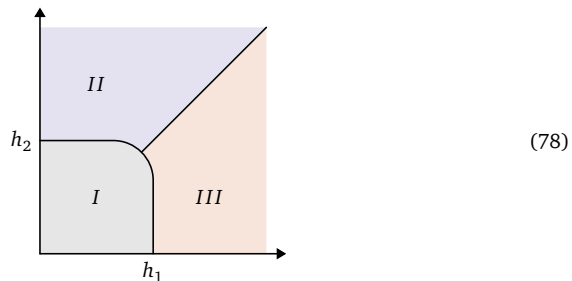
The *Landau paradigm* of phase transitions states that every phase of matter is in one-to-one correspondence with a symmetry breaking pattern [73]. The most famous (classical) example is of course the phase transition of water to ice, in which continuous Euclidean symmetry is broken to one of its 230 subgroups, also known as the *space groups*. Moreover, the paradigm states that this breaking of symmetry can be diagnosed via a local order parameter that is charged under the symmetry group. Vice versa, *symmetric* phases can be identified through *disorder operators*, as argued by Kadanoff and Ceva [74]. The Landau paradigm has since its conception been generalised in a number of ways.

For one, it has since been realised that after breaking its symmetry down to a subgroup, a quantum system can realise its residual symmetry in distinct ways on its entanglement degrees of freedom. These distinguishable entanglement patterns correspond to distinct phases if we require that the adiabatic paths between Hamiltonians commute with the symmetry. Such *symmetry-protected topological* phases are for this reason naturally described and classified in terms of tensor networks [37, 75–77].

The goal of this lecture will be to sketch the classification of one-dimensional quantum phases of matter in the presence of a group symmetry using the framework of MPS. As remarked in section 2.2, symmetry breaking in MPS is manifested by the existence of distinct injective blocks in the ground state MPS. These are labelled by cosets in  $G/H$  for a subgroup  $H \subseteq G$  and are permuted under the group action. The challenge will thus be to classify the *symmetric phases* realised on each of these injective blocks. To this end, we will first derive the *fundamental theorem* of MPS which states that an MPS is invariant under a symmetry if and only if its local tensors transform trivially under that symmetry. This theorem exemplifies the mantra that tensor networks encode *global* properties of states in the *local* tensors. This will bring us in the realm of *projective representations* whose corresponding cohomology classification provides a *topological index* that classifies distinct symmetric phases in the sense that two injective MPSs belong to the same phase if and only if their topological index is the same. Notably, one-dimensional bosonic systems all belong to the same phase if no symmetry is imposed. For fermions the story is more intricate and non-trivial phases can exist due to the always present fermionic parity symmetry; this will be the topic of section 3.5.

Let us also foreshadow our objectives of subsequent lectures. With the discovery of the fractional quantum Hall effect and other two-dimensional *topological phases of matter*, it also became clear that even when a system does not exhibit any symmetry, it can still belong to a non-trivial quantum phase of matter. These phases of matter can be characterised by a ground state degeneracy which depends on the topology of the underlying space, gapped *anyonic excitations* and in some cases chiral edge modes. They are also not detected by any local order parameter. As we will discuss in the next lecture, these phases owe their stability to the entanglement in these systems being *long range*. In particular, in the case of *non-chiral topological order*, the entanglement patterns are also responsible for the ground state degeneracy, as it

has been appreciated that entanglement degrees of freedom exhibit string-like ‘symmetries’ which are described by a *fusion category*. We will encounter a particularly simple class of topological orders below in section 4.6, and a microscopic construction of the generic non-chiral case in section 5.3. The existence of these *generalised symmetries* on the entanglement degrees of freedom implies in particular that the boundary Hamiltonians of these topological orders, describing the dynamics of their edge modes, inherit also these symmetries. As such, we are inclined to consider a Landau paradigm for these generalised symmetries as well, and this will be one of the topics of the last lecture.



### 3.2 Fundamental theorem of MPS

As anticipated above, the *fundamental theorem* of MPS [78] is crucial in the classification of one-dimensional symmetric gapped phases of matter and many other applications of MPS. The incarnation of the theorem that we will apply is as follows. Two uniform *injective* matrix product states represent the same state on any system with periodic boundary conditions, independent of the length, if and only if the MPS tensors are related by a non-singular gauge transformation on the virtual level:

$$|\Psi[A]\rangle \sim |\Psi[B]\rangle \iff A^i = e^{i\chi} X B^i X^{-1}, \quad \forall i. \quad (79)$$

Note that we assumed, without loss of generality, that both MPSs have the same bond dimension. When both MPSs are in canonical form,  $X$  is unitary.

As an application of some of the key concepts introduced earlier in these lectures, we provide a proof here.

Note first that the  $\Leftarrow$  direction is trivial since all appearances of  $X$  and  $X^{-1}$  cancel on the virtual level. To prove the  $\Rightarrow$  direction, let us assume without loss of generality that  $|\Psi[A]\rangle$  is in right canonical form, while  $|\Psi[B]\rangle$  is in left canonical form. Consider now the following mixed transfer matrix:

$$\mathbb{T}_{A,B} = \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \quad (80)$$

Since  $|\langle \Psi[A] | \Psi[B] \rangle| = 1$ , the leading eigenvalue of the mixed transfer has unit norm [79]. We will denote its corresponding right eigenvector by  $X$ , so that  $\mathbb{T}_{A,B} X = e^{i\phi} X$ . Let us now consider the following contraction

$$e^{-i\phi} \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \quad (81)$$

First note that on the one hand this expression is equal to  $\text{Tr}(X^\dagger X)$  via  $\mathbb{T}_{A,B}X = e^{i\phi}X$ . On the other hand, interpreting this expression as the inner product of two vectors with respect to the diagonal cut and applying the Cauchy-Schwarz inequality, we obtain that

$$\text{Tr}(X^\dagger X) = e^{-i\phi} \left( \text{Diagram with } \bar{X}, \bar{B}, A, X \text{ and a diagonal cut} \right) \leq \left| \text{Diagram with } \bar{X}, \bar{A}, A, X \right|^{1/2} \times \left| \text{Diagram with } \bar{B}, \bar{X}, B, X \right|^{1/2} = \text{Tr}(X^\dagger X), \quad (82)$$

where we use that  $A$  and  $B$  are in right and left canonical form. So clearly the equality holds, which implies that  $A^\dagger X = e^{i\chi} X B^\dagger$ , or since  $X$  is full rank thus eq. (79).

### 3.3 SPT classification of one-dimensional bosonic matter

Let us consider a gapped Hamiltonian  $H$  with a unique ground state, and assume it commutes with a symmetry  $G$ :  $[U(g)^{\otimes L}, H] = 0$ , for a representation  $\{U(g), g \in G\}$ . As  $H$  is gapped and has a unique ground state, the ground state must inherit that symmetry. It is guaranteed that this ground state is faithfully represented by an injective MPS. That MPS  $|\Psi[A]\rangle$  must therefore transform trivially under  $G$ , i.e.:

$$U(g)^{\otimes N} |\Psi[A]\rangle \simeq |\Psi[A]\rangle, \quad \forall g \in G. \quad (83)$$

A direct application of the fundamental theorem eq. (79) reveals that the MPS tensor itself has to transform trivially under the symmetry group, namely:

$$\forall g \in G, \exists X_g, \chi_g : \sum_j U(g)_{ij} A^j = e^{i\chi_g} X_g A^i X_g^\dagger \quad (84)$$

where it should be noted that the gauge matrices and phases themselves depend on the group element  $g$ . Let us now consider acting successively with the symmetry operators labelled  $g$  and  $h$  on the state. We can do this in two different ways: either by acting with  $U(gh)$  directly and using the fundamental theorem once or acting sequentially with  $U(g)$  and  $U(h)$  and applying the fundamental theorem twice. Equating these two results in:

$$e^{i\chi_{gh}} X_{gh} A^i X_{gh}^\dagger = e^{i(\chi_g + \chi_h)} X_g X_h A^i X_h^\dagger X_g^\dagger, \quad \forall i, g, h. \quad (85)$$

As  $A^i$  is injective and hence spans the full matrix algebra, this is only possible if both  $\chi_{gh} = \chi_g + \chi_h$  and  $X_g X_h = \psi(g, h) X_{gh}$  for  $U(1)$  phases  $\psi(g, h)$ . The former means that  $\{e^{i\chi_g}, g \in G\}$  constitutes a one-dimensional representation of  $G$ . Note that after blocking a finite number of  $M$  sites, we obtain an enlarged unit cell for which the phase  $e^{iM\chi_g} = 1$  disappears. Therefore we could say that this phase is *unstable against blocking* and we will not take it further into account for phase classification. On the other hand, we have that the set of matrices  $\{X_g, g \in G\}$  obeying

$$X_g X_h = \psi(g, h) X_{gh}, \quad \forall g, h \in G, \quad (86)$$

for some phases  $\psi(g, h) \in U(1)$  forms a so-called  *$\psi$ -projective representation* of  $G$ . Projective representations generalise ordinary *linear* representations by allowing for the projective phase  $\psi(g, h)$  in their multiplication. Due to the associativity of the matrix multiplication the phases  $\psi(g, h)$  have to satisfy

$$\psi(g, h)\psi(g, h, k) = \psi(g, hk)\psi(h, k), \quad \forall g, h, k \in G, \quad (87)$$

a consistency equation commonly known as the *second cocycle equation*. This condition has trivial solutions of the form  $\psi(g, h) = \frac{\phi_{gh}}{\phi_g \phi_h}$  for any function  $\phi : G \rightarrow U(1)$ . In the context of

group cohomology such a trivial cocycle is called a *coboundary*. In order to classify phases we will want to consider solutions of the cocycle equation *modulo* coboundaries. Indeed, from the fundamental theorem it is clear that we can freely redefine  $X_g \mapsto \phi_g X_g$  for any function  $\phi$ . To this end, note that both the cocycles and the coboundaries form abelian groups, which are denoted by  $Z^2(G, U(1))$  and  $B^2(G, U(1))$  respectively. Since clearly  $B^2(G, U(1))$  is a normal subgroup of  $Z^2(G, U(1))$ , we can consider the quotient  $H^2(G, U(1)) := Z^2(G, U(1))/B^2(G, U(1))$ , the *second cohomology group* of  $G$  over  $U(1)$ .

Note that the cocycle equation eq. (87) – after taking the logarithm of both sides, and collecting the unknown phases  $\psi(g, h)$  in a vector  $\psi$  – can be written as a linear system of equations, satisfied modulo  $2\pi$ :

$$\Omega\psi = \mathbf{0}, \quad \text{mod } 2\pi. \quad (88)$$

Computing the second cohomology group and corresponding representative cocycles can then be done by bringing  $\Omega$  in *Smith normal form*, a matrix decomposition of matrices with integer coefficients of the form  $\Omega = PAR$ , where both  $P$  and  $R$  are unimodular integer matrices, and  $\Lambda$  is a diagonal matrix. A detailed explanation is relegated to section A. The main observation is that the second cohomology group is always finite and of the form

$$H^2(G, U(1)) \simeq \mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \cdots \times \mathbb{Z}_{d_d}, \quad (89)$$

where the  $d_i$  are the diagonal entries of  $\Lambda$ . The group structure of  $H^2(G, U(1))$  is physically realised by *stacking* two  $G$ -symmetric chains and considering the diagonal subgroup of  $G \times G$  as the  $G$  symmetry of the whole.

One can prove that if two MPSs transform under a  $G$  symmetry with projective representations corresponding to distinct cohomology classes in  $H^2(G, U(1))$ , these MPSs must represent ground states in different phases [37]. In other words, there is no symmetric adiabatic path one can take to transform one into the other without crossing a phase transition. In that case, the distinct cohomology classes are sometimes said to form a *topological obstruction* to do so, while a representative cocycle is sometimes called a *topological index* characterising that phase. If we were to allow paths that break the symmetry, then all MPSs would again belong to the same trivial phase. In particular, the number of symmetric phases is always finite due to eq. (89). The proof is beyond the scope of these lecture notes, but let us remark that the argument relies heavily on the parent Hamiltonian construction of MPS (see section 2.2).

An important property of any projective representation is that its matrix dimension is necessarily strictly bigger than one, when it corresponds to a non-trivial cohomology class. As an example, consider the smallest group with a non-trivial second cohomology group, namely  $\mathbb{Z}_2 \times \mathbb{Z}_2$ ,  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, U(1)) \simeq \mathbb{Z}_2$ . If we denote its group elements by tuples  $(a, b)$ ,  $a, b = 0, 1$ , a non-trivial projective representation is given by the Pauli matrices as follows:  $X_{(a,b)} = (\sigma^X)^a (\sigma^Z)^b$ , and this representation cannot be further reduced. This fact has an important consequence for the entanglement of an MPS in a non-trivial SPT phase, and also guarantees the existence of gapless symmetry-protected edge modes [41].

Indeed, it turns out that the entanglement spectrum (see section 2.2) is characterised by a degeneracy which is at least the dimension of the smallest irreducible projective representation in that class. This is beautifully illustrated in the spin 1 antiferromagnetic Heisenberg model. In fig. 2 the first 48 Schmidt values are plotted, and the spectrum organises itself visibly in multiplets which are at least twofold degenerate. In this case, the protecting symmetry is  $\text{SO}(3)$  whose second cohomology group indeed has a non-trivial element,  $H^2(\text{SO}(3), U(1)) \simeq \mathbb{Z}_2$ . All irreps of  $\text{SO}(3)$  have an odd dimension, so these degeneracies clearly originate from the half-integer irreps of  $\text{SU}(2)$ , which indeed form projective representations of  $\text{SO}(3)$ . In fact, the phase is also protected by a  $\mathbb{Z}_2 \times \mathbb{Z}_2$  subgroup of  $\text{SO}(3)$ , so that we can even perturb the

Hamiltonian with terms which break part of the  $SO(3)$  symmetry as long as they still commute with  $\mathbb{Z}_2 \times \mathbb{Z}_2$ , without leaving the SPT phase.

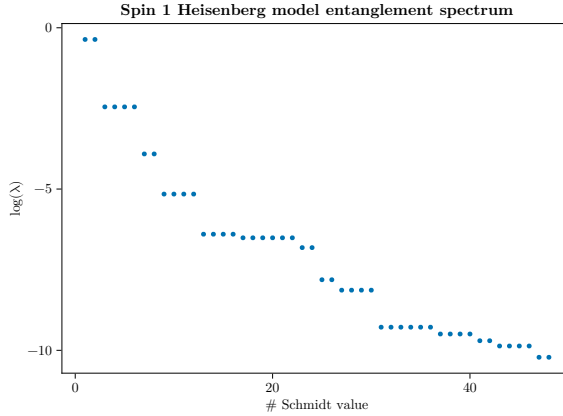


Figure 2: The entanglement spectrum of the spin 1 Heisenberg model ground state in the thermodynamic limit with bond dimension 48. Created with MPSKit.jl [70].

As anticipated in section 2.1, another hallmark of SPT phases is the presence of symmetry-protected gapless *edge modes*, when they are considered on open chains. A feature of matrix product states (and their higher-dimensional generalisations, see section 4.3) is that a tensor product basis for these edge modes is provided by the entanglement degrees of freedom on the boundary. Indeed, suppose we have a uniform MPS  $|\Psi[A]\rangle$  of bond dimension  $\chi$ . We can consider then following  $\chi^2$  states on an open chain of  $N$  sites:

$$|\Psi[A]\rangle = \sum_{\{i_j\}} (A^{i_1} A^{i_2} \cdots A^{i_N})_{a\beta} |i_1, i_2, \dots, i_N\rangle. \quad (90)$$

All states  $|\Psi[A], \alpha, \beta\rangle$ ,  $\forall \alpha, \beta = 1, \dots, \chi$  have the same energy for the parent Hamiltonian, so that seemingly we have  $\chi^2$  degenerate edge modes. However, by perturbing the Hamiltonian, some of these edge modes can be gapped out. Nevertheless, as long as we restrict to symmetry-respecting perturbations, the edge modes will transform under the projective representation on the virtual level according to:

$$U(g)^{\otimes N} |\Psi[A], \alpha, \beta\rangle = \sum_{\alpha', \beta'} (X_g)_{\alpha\alpha'} (\tilde{X}_g)_{\beta\beta'} |\Psi[A], \alpha', \beta'\rangle, \quad (91)$$

and hence, at least a two-fold degeneracy of the edge modes persists.

This concludes the classification of *bosonic*  $G$ -symmetric gapped one-dimensional phases. Provided a symmetry group  $G$ , gapped phases are in one-to-one correspondence with a subgroup  $H \subseteq G$  which represents the *unbroken* symmetry in the ground state subspace, and a cohomology class  $[\psi] \in H^2(H, U(1))$  representing the SPT phase of the surviving  $H$  symmetry.

As an aside we note here that such pairs  $(H, [\psi])$  also characterise the (indecomposable) *module categories* over  $\text{Vec}_G$ , the *fusion category* of  $G$ -graded vector spaces (see also section B). There is thus a one-to-one correspondence between  $\text{Vec}_G$ -module categories and gapped  $G$ -symmetric phases. Of course this observation is completely superfluous for the simple case

of group symmetries, but in the last lecture we will argue that for generic fusion category symmetries this correspondence continues to hold.

### 3.4 Spacetime symmetries

The above discussion was focused on internal symmetries of the Hamiltonian, but applies with some slight modifications to spacetime symmetries as well. When dealing with these symmetries, one has to address two issues which are not present in the internal case. First, note that some spacetime symmetries such as translation symmetry and spatial reflection act in a non-local way. On the other hand, *orientation-reversing* symmetries such as time-reversal or reflection symmetry act in an antiunitary way and thus depend on the chosen basis. The latter has strong consequences for the cohomological classification of SPT phases, as we will see below. Nevertheless, MPS provide a suitable setting in which the classification of symmetric phases protected by spatial symmetries can be derived explicitly [41, 80].

Let us illustrate how a similar reasoning as above leads to the classification of phases protected by time-reversal symmetry. In general, time-reversal symmetry acts as  $UK$  where  $U$  is a unitary operation,  $(\sigma^Y)^{\otimes N}$  in the case of spin  $1/2$ 's as demonstrated by Wigner [81], and  $K$  is complex conjugation. We assume here for simplicity that  $U = \mathbb{1}$ . Acting with complex conjugation on an injective MPS  $|\Psi[A]\rangle$  and applying the fundamental theorem results in

$$\bar{A}^i = e^{i\chi} Q A^i Q^{-1}. \quad (92)$$

Applying time-reversal symmetry then a second time and exploiting injectivity of the MPS shows that:

$$Q\bar{Q} = \pm \mathbb{1}. \quad (93)$$

We thus obtain a non-trivial topological index corresponding to the sign, giving rise to a  $\mathbb{Z}_2$  classification of time-reversal SPT phases. As an illustration consider the AKLT state eq. (44). It is easily checked that in that case  $Q = \sigma^Y$ ,  $Q\bar{Q} = -\mathbb{1}$  so that the AKLT state thus belongs to a non-trivial SPT phase w.r.t. time-reversal symmetry.

Note that the  $Q$  in eq. (93) cannot be interpreted as a non-trivial projective representation of  $\mathbb{Z}_2$  due to the complex conjugation on  $Q$ . Consequently, the phase  $\pm 1$  in eq. (93) also cannot be interpreted as a non-trivial cocycle for  $H^2(\mathbb{Z}_2, U(1))$ , which is in fact trivial. We therefore need a generalised notion of projective representations and cocycles to account for symmetry groups where some group elements act in an antiunitary fashion. It is easy to verify that when a generic symmetry group  $G$  has antiunitary elements, the gauge matrices  $X_g$  multiply according to

$$X_g \beta_g(X_h) = \psi(g, h) X_{gh}, \quad (94)$$

where  $\beta_g(\bullet)$  acts as complex conjugation when  $g$  is an antiunitary element and as the identity otherwise. Clearly, this expression is equivalent to eq. (86) when all group elements act unitarily and to eq. (93) when the symmetry group is  $\mathbb{Z}_2$  time-reversal. It is straightforward to verify as well that the consistency condition on  $\psi(g, h)$  then reads

$$\psi(g, h)\psi(gh, k) = \psi(g, hk)\beta_g(\psi(h, k)), \quad \forall g, h, k \in G. \quad (95)$$

Accordingly, also the notion of coboundaries has to be slightly adapted. Putting things together, this means that in case of antiunitary symmetries, the cohomology classification of SPT phases has to be adapted in a way so as to account for the non-trivial group action  $\beta_g(\bullet)$ . The corresponding cohomology group is then denoted  $H^2(G, U(1)^\beta)$ , and is generically different from  $H^2(G, U(1))$ . Importantly, this group can be computed with the same method as outlined above and detailed in section A. In correspondence with the example of the AKLT state, we have indeed that  $H^2(\mathbb{Z}_2, U(1)^\beta) = \mathbb{Z}_2$ .



### 3.5 Fermionic MPS & symmetric phases

Up until now we have focused solely on *bosonic* matrix product states. The goal of this section is to sketch the extension of the MPS framework to *fermionic systems* and highlight some of the technical complications that arise, thereby closely following ref. [82], see also ref. [83, 84].

Crucial is the notion of a *super vector space*, which is a vector space that admits a direct sum decomposition of the form

$$V = V^0 \oplus V^1, \quad (96)$$

where vectors in  $V^0$ , respectively  $V^1$ , are said to possess a *fermion even*, respectively *odd*, parity. Without loss of generality we assume that basis vectors  $|i\rangle$  have definite parity, which we will denote by  $|i| = 0, 1$ . A super vector space is moreover endowed with a *graded tensor product* behaving as follows under a re-arrangement of the tensor product:

$$|i_1\rangle \otimes_{\mathfrak{g}} |i_2\rangle \mapsto (-1)^{|i_1||i_2|} |i_2\rangle \otimes_{\mathfrak{g}} |i_1\rangle. \quad (97)$$

This property is the mathematical manifestation of fermionic particle exchange statistics. The *supertrace* is then defined as

$$\mathcal{C}(|i\rangle \otimes_{\mathfrak{g}} \langle j|) = (-1)^{|i||j|} \delta_{i,j}. \quad (98)$$

A uniform fermionic MPS (fMPS) is then constructed from an *even-parity* tensor

$$A = \sum_{i,\alpha,\beta} A_{\alpha\beta}^i |\alpha\rangle \otimes_{\mathfrak{g}} |i\rangle \otimes_{\mathfrak{g}} \langle\beta|, \quad (99)$$

i.e.  $|i| + |\alpha| + |\beta| = 0 \pmod{2}$  for all indices. Depending on whether the resulting state has even or odd fermionic parity the construction is slightly different.

**Even-parity states** An even-parity state is constructed from the tensor  $A$  as follows

$$|\Psi\rangle_e = \mathcal{C}_v(A \otimes_{\mathfrak{g}} A \otimes_{\mathfrak{g}} \cdots \otimes_{\mathfrak{g}} A). \quad (100)$$

In this expression,  $\mathcal{C}_v$  stands for the supertrace over the virtual indices. We can now rewrite this fMPS in a more explicit form. Note that the entries of a generic even-parity tensor are of the form

$$A^i = \begin{pmatrix} B^i & 0 \\ 0 & C^i \end{pmatrix}, \quad |i| = 0, \quad A^i = \begin{pmatrix} 0 & D^i \\ F^i & 0 \end{pmatrix}, \quad |i| = 1. \quad (101)$$

Defining the *parity matrix*

$$\mathcal{P} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}, \quad (102)$$

and carefully working out the expression eq. (100) yields then:

$$|\Psi[A]\rangle_e = \sum_{\{i_j\}} \text{Tr}(\mathcal{P} A^{i_1} A^{i_2} \cdots A^{i_N}) |i_1, i_2, \dots, i_N\rangle. \quad (103)$$

**Odd-parity states** The construction of odd-parity states proceeds analogously to the construction of even states, but requires the insertion of an additional matrix in the supertrace:

$$|\Psi\rangle_o = \mathcal{C}_v(Y \otimes_{\mathfrak{g}} A \otimes_{\mathfrak{g}} A \otimes_{\mathfrak{g}} \cdots \otimes_{\mathfrak{g}} A). \quad (104)$$

The requirement to introduce the matrix  $Y$  ultimately stems from the fact that accessing odd-parity states necessitates twisted boundary conditions. We will encounter similar considerations in lecture 5.

Similar manipulations as in the previous case allow us to rewrite the odd-parity fMPS as

$$|\Psi[A]\rangle_o = \sum_{\{i_j\}} \text{Tr}(YA^{i_1}A^{i_2}\dots A^{i_N})|i_1, i_2, \dots, i_N\rangle. \quad (105)$$

Imposing then translation invariance reveals that  $Y$  can be chosen to be of the form

$$Y = \begin{pmatrix} 0 & -\mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}, \quad (106)$$

and in this basis the MPS matrices read:

$$A^i = \begin{pmatrix} B^i & 0 \\ 0 & B^i \end{pmatrix}, |i| = 0, \quad A^i = \begin{pmatrix} 0 & B^i \\ -B^i & 0 \end{pmatrix}, |i| = 1. \quad (107)$$

Contrary to the even case, this fMPS is specified by one block  $B^i$ .

Note that since the odd fMPS matrices commute with  $Y$ , the MPS tensors have a common invariant subspace. Nevertheless, contrary to the bosonic case, the odd-parity states  $|\Psi[A]\rangle_o$  define an irreducible matrix product state when the  $B^i$  span the full  $\mathbb{C}^{\chi/2 \times \chi/2}$  matrix algebra. We refer to appendix D of [85] for a detailed technical discussion and proofs on canonical forms of fermionic graded MPS. Physically, the underlying reason can be traced back to the superselection rule for fermionic parity: there exist no superpositions of odd and even-parity states.

A notable example of an odd fMPS state is the ground state of the Kitaev chain of spinless fermions whose Hamiltonian is a simple sum of pairing terms [86]:

$$H = -i \sum_{i=1}^{N-1} \gamma_{2i} \gamma_{2i+1}, \quad (108)$$

for Majorana modes  $\gamma_i = \gamma_i^\dagger$ ,  $\{\gamma_i, \gamma_j\} = 2\delta_{i,j}$ . The translational invariant fMPS ground state is found to have odd parity and its MPS matrices read

$$A^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A^1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (109)$$

This model is noteworthy for exhibiting Majorana edge modes. Indeed, on open boundary conditions the state takes the form

$$\sum_{\{i_j\}} (A^{i_1}A^{i_2}\dots A^{i_N})_{\alpha\beta} |i_1, i_2, \dots, i_N\rangle. \quad (110)$$

It is then straightforward to check that due to the structure of the tensors, choosing the boundary conditions  $\alpha = \beta = 0$  or  $\alpha = \beta = 1$  yields one and the same ground state. Similarly, both boundary conditions  $\alpha = 0, \beta = 1$  and  $\alpha = 1, \beta = 0$  yield a second ground state, resulting eventually in a two-fold ground state degeneracy, which cannot be lifted by adding an extra local interaction to the Hamiltonian.

Similar to the bosonic case, one can extract from fermionic matrix product states certain discrete topological invariants when global internal or spacetime symmetries are imposed. An illustrious example is the case of interacting time-reversal symmetric fermionic SPT phases. In that case there are three distinct topological indices valued in  $\{0, 1\}$ , one of which is related to whether the fMPS has Majorana edge modes, and two of which are related to the local transformation properties of the fMPS tensors under time-reversal. As carefully demonstrated, in ref. [82] it can be shown that under stacking of the corresponding fMPSs, the  $\mathbb{Z}_8$  classification of Fidkowski and Kitaev is recovered [87].

### 3.6 Synopsis of lecture III

Lecture III explored the manifold of MPS when additional symmetries are imposed on the tensors. The main workhorse of symmetric tensor networks is the *fundamental theorem of MPS*, which states that any *global* symmetry must be reflected in the symmetries of the *local* MPS tensors. Interestingly, those *virtual* symmetries can be represented *projectively*. Any symmetry-preserving adiabatic path between symmetric MPSs which transform according to different projective representations is guaranteed to have a gapless point. As a consequence, all bosonic gapped phases of matter of one-dimensional quantum spin chains with a Hamiltonian that is  $G$ -symmetric are in one to one correspondence with all possible subgroups  $H \subseteq G$  and its possible projective representations labelled by the second group cohomology  $H^2(H, \text{U}(1))$ . The latter classify the so-called *SPT phases*. If no symmetry is considered, then all MPSs can adiabatically converted into each other, and therefore the phases are only protected when the symmetry is imposed. We showed that this classification can be generalised to the case of spacetime symmetry groups. Interestingly, when superselection rules are considered such as the parity superselection rule for fermions, genuine topological phases of matter emerge (i.e. a phase that does not need a symmetry to be protected).

## 4 Lecture IV: MPO algebras: from the algebraic Bethe ansatz to G-injective PEPS

Let us now turn our attention towards integrable models, one of the central topics of this school. It turns out that tensor networks provide a very concise way of rephrasing the *algebraic Bethe ansatz* in terms of matrix product states and operators. In particular, we will see that the *Yang-Baxter equation*, which lies at the heart of its integrability, can be understood as a consequence of the fundamental theorem (section 3.2). Although tensor network techniques are not teaching us anything fundamentally new about the Bethe ansatz, the insights of this lecture will be instrumental in the MPO representation theory of *generalised* or *categorical symmetries* described by *fusion categories*, which is the topic of the final lecture.

After the discussion on the algebraic Bethe ansatz, we will focus in section 4.3 on the two-dimensional generalisation of MPS, namely *projected entangled-pair states*. We will conjecture a generalisation of the fundamental theorem, and discuss the role of MPO symmetries in the tensor network representation of two-dimensional SPT and topological phases, as well as their edge modes.

5

### 4.1 Magnon states

Tensor networks can be used as ansätze for analytic solutions of Hamiltonians, as was exemplified for the AKLT model eq. (44). The example of interest in this lecture however, is a famous family of exactly solvable models, called *integrable systems*. An integrable quantum system is roughly speaking a system with an extensive number of conserved charges – local operators commuting with the Hamiltonian – to which they owe their exact solvability. Integrable systems can be solved with the ansatz wave function called the *Bethe ansatz* (see e.g. [88]). There are two formulations of this approach, called the *coordinate Bethe ansatz* and *algebraic Bethe ansatz*, which produce the same set of solutions. Both formulations can be written down using tensor networks.

For the sake of concreteness, we consider the spin 1/2 XXZ Hamiltonian

$$H_{XXZ} = \sum_{i=1}^L \sigma_i^X \sigma_{i+1}^X + \sigma_i^Y \sigma_{i+1}^Y + \Delta \sigma_i^Z \sigma_{i+1}^Z \quad (111)$$

on periodic boundary conditions, i.e.  $L+1 \equiv 1$ . The *magnon states* used as a basis when solving the Bethe equations can readily be written down as an MPS. For instance, the *periodic* boundary conditions 1-magnon state can be written down as an *open* boundary conditions MPS (see eq. (23)) with left boundary vector  $\mathbf{v}_L^\dagger = \langle 0|$  and right boundary vector  $\mathbf{v}_R = |1\rangle$ , and MPS tensor:

$$A^0(k) = \begin{pmatrix} e^{ik} & 0 \\ 0 & 1 \end{pmatrix}, \quad A^1(k) = \begin{pmatrix} 0 & e^{ik} \\ 0 & 0 \end{pmatrix} \\ \Rightarrow |\Psi_1[A]\rangle = \sum_{n=1}^L e^{ikn} |0_0 \dots 0_{n-1} 1_n 0_{n+1} \dots 0_L\rangle \quad (112)$$

Note that this is precisely how excited states were constructed as boosted tangent vectors on the manifold of MPS in section 2.5. The general n-magnon state can be found in refs. [29, 89, 90], and involves an MPS with a bond dimension that scales exponentially in the number of particles in the system.

#### 4.2 The Yang-Baxter equation as the fundamental theorem of MPS

There is however a completely different way of deriving the exact eigenstates in terms of the algebraic Bethe ansatz (ABA). Before the formulation of both approaches in terms of tensor networks, it was not clear how those two approaches were related to each other; it is now clear that both approaches are related by a non-trivial gauge transformation [91], as has to be because of the fundamental theorem of MPS. In the algebraic Bethe ansatz, all eigenstates of the Hamiltonian can be constructed using the multiplication of MPOs that depend on a continuous *spectral parameter*  $\mu_i$ :

$$(113)$$

Here, all MPO tensors are of the form

$$A_{\alpha\beta}^{ij}(\mu) = \mu \delta_{\alpha\beta} \delta_{ij} + \delta_{\alpha i} \delta_{\beta j} = \mu \begin{array}{c} i \\ \alpha \text{---} \text{---} \beta \\ j \end{array} + \begin{array}{c} i \\ \alpha \text{---} \text{---} \beta \\ j \end{array} . \quad (114)$$

It can easily be checked that every MPO adds one magnon to the system, as the left boundary is  $\langle 0|$  and the right boundary  $|1\rangle$ . The coefficients  $\mu_i$  are functions of the magnons' momenta, and can be determined by solving the so-called *Bethe equations*. A crucial ingredient of the ABA is the fact that all those MPOs commute with each other.

The underlying logic is as follows. We start from a transfer matrix of a two-dimensional statistical mechanical model that can be written in the form of an injective MPO  $T(\mu)$  determined by the same tensor  $A(\mu)$  as in equation 114, but this time with periodic boundary conditions. It turns out that this system will be integrable if and only if all those MPOs commute with each other independent of their spectral parameter:

$$\forall \lambda, \mu : [T(\lambda), T(\mu)] = 0. \quad (115)$$

If this is the case, we can invoke the fundamental theorem of MPS eq. (79) to prove the existence of a gauge transform  $R$  that transforms those products into each other [29]. Indeed, note that we can merge together two legs of an MPO to form a single leg, so that the fundamental theorem of MPS can be used for rank-4 MPOs. Concretely, there is an intertwiner tensor  $R(\lambda, \mu)$  obeying the equation

$$(116)$$

This  $R$ -tensor is a local implementation of the global symmetries given by the commuting MPOs.

Now let us take three copies of the  $R$ -tensor. Due to the associativity of the MPO algebra, these  $R$ -tensors can be fused together in two different ways, giving rise to the following equation:

$$(117)$$

This is precisely the Yang–Baxter equation – hence the fundamental theorem of MPS reproduces this result in a very concise way. Given any integrable transfer matrix, the corresponding intertwiners have therefore to satisfy the Yang-Baxter equations. It can be checked that the tensor in eq. (114) is precisely one of those solutions. A constructive way of finding integrable models is henceforth to turn the tables around: start with finding a solution of the Yang-Baxter equations (which is typically very hard!), leading to explicit representations of the tensors  $R(\lambda, \mu)$ . As a next step, solve for the transfer matrix  $T$  as a solution of eq. (116). What is surprising is the fact that this equation is equivalent to the one of the Yang-Baxter equation by identifying the MPO tensors to be  $A_{\alpha\beta}^{ij}(\mu) := R_{j\beta}^{i\alpha}(\mu, 0)$ . This so-called *fundamental representation* of the Yang-Baxter algebra hence allows to construct a one-parameter family of commuting transfer matrices. Note that this is not the only family of MPOs solving eq. (116). One can construct different representations of the same algebra – later on, we will encounter an analogous situation where we will construct different representations of the same symmetric system. As a last step, it is now possible to construct the magnon MPOs of eq. (113) with the same tensors  $A$ , but with different boundary conditions.

The key principle and advantage of the tensor network formulation is thus as follows: *global* conservation laws and commutation relations can be written down as equations involving *local* degrees of freedom, i.e. MPS and MPO tensors, which are usually easier to solve.

Given the MPO formulation, it is also possible to recover the integrable Heisenberg Hamiltonian. The transfer matrix of the 6-vertex model with all weights equal to each other, the corresponding integrable statistical mechanical model, has exactly the form of eq. (114). By construction, all MPOs  $T(\mu)$  with those tensors  $A(\mu)$  will commute with each other, so the derivative of this MPO is defined univocally. The Heisenberg Hamiltonian is indeed the logarithmic derivative of this transfer matrix; we leave the proof as a (very interesting) exercise (see [90] for an elaborate discussion):

$$H = T^{-1}(0) \frac{dT(\mu)}{d\mu} \Big|_{\mu=0} = \sum_i \cdots \left| \begin{array}{c} \text{diagram of a crossing} \end{array} \right| \cdots \quad (118)$$

In conclusion, both Bethe ansatz formulations (coordinate and algebraic) have a natural representation in terms of tensor networks. Tensor networks and Bethe ansatz clearly share the same DNA, but tensor networks turn out to be more generally applicable – they work equally fine for non-integrable systems and higher dimensions.

### 4.3 Projected entangled-pair states

The natural generalisation of matrix product states to two dimensions are the so-called *projected entangled-pair states* (PEPS) [92]. Similar to the one-dimensional case, a translation invariant PEPS is constructed from a single tensor, which in the case of the square lattice has

one physical and four virtual indices:

$$A = \sum_{\substack{i, \alpha, \beta \\ \gamma, \delta}} A_{\alpha\beta\gamma\delta}^i |i\rangle \otimes |\alpha\rangle \otimes |\beta\rangle \otimes |\gamma\rangle \otimes |\delta\rangle \in \mathbb{C}^{d\chi^4}. \quad (119)$$

Such a tensor is then depicted by

$$\begin{array}{c} \delta \\ | \\ \textcircled{A} \\ | \\ \gamma \end{array} \begin{array}{c} i \\ / \\ \textcircled{A} \\ \backslash \\ \beta \end{array} = A_{\alpha\beta\gamma\delta}^i, \quad \begin{array}{l} i=1,2,\dots,d \\ \alpha,\beta,\gamma,\delta=1,2,\dots,\chi \end{array} \quad (120)$$

so that the full PEPS wave function can be represented by

$$|\Psi[A]\rangle = \begin{array}{c} \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \\ | \quad \quad \quad | \quad \quad \quad | \\ \textcircled{A} \quad \textcircled{A} \quad \textcircled{A} \\ | \quad \quad \quad | \quad \quad \quad | \\ \textcircled{A} \quad \textcircled{A} \quad \textcircled{A} \\ | \quad \quad \quad | \quad \quad \quad | \\ \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \end{array}. \quad (121)$$

Note that PEPS naturally satisfy an area law since the entanglement entropy of a contiguous region of spins scales linearly with the number of bonds on the boundary of the region, and its perimeter. Therefore PEPS are expected to form a good variational class for low-energy states of local gapped Hamiltonians, which have since long been conjectured to satisfy an area law. Conversely, as for MPS, it can be shown that every PEPS is the exact ground state of a local frustration free gapped parent Hamiltonian. It is also obvious how to generalise this construction to open boundary conditions, arbitrary lattices or to higher dimensions. Similarly, one can generalise MPOs to the two-dimensional setting. Such operators are referred to as *projected entangled-pair operators* (PEPOs).

Considerable effort has been made to generalise DMRG and VUMPS algorithms to the case of PEPS. Although it is in principle straightforward to generalise the manifold picture to the case of PEPS, the actual computational scaling with the bond dimension is much worse than in the 1D case. Nevertheless those PEPS algorithms are providing state of the art variational results for strongly correlated lattice systems in 2D such as for the Hubbard model, and it is a very active research field to come up with better algorithms. We refer to refs. [1, 93–96] for state of the art results. In these lectures, we will focus on the theory of PEPS.

Contrary to the case of matrix product states, there exists no completely general fundamental theorem yet that provides necessary and sufficient conditions for two distinct PEPS tensors  $A$  and  $B$  to generate the same state for all system sizes. However, it is clear that a sufficient condition for this to hold is the existence of a matrix product operator that *intertwines* the PEPS tensors according to

$$\begin{array}{c} \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \\ | \quad \quad | \\ \textcircled{A} \quad \textcircled{A} \\ | \quad \quad | \\ \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \end{array} = \begin{array}{c} \text{---} \textcircled{B} \text{---} \textcircled{B} \text{---} \\ | \quad \quad | \\ \textcircled{B} \quad \textcircled{B} \\ | \quad \quad | \\ \text{---} \textcircled{B} \text{---} \textcircled{B} \text{---} \end{array}, \quad (122)$$

and all other ways of pulling the MPO through, such as:

$$\begin{array}{c} \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \\ | \quad \quad | \\ \textcircled{A} \quad \textcircled{A} \\ | \quad \quad | \\ \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \end{array} = \begin{array}{c} \text{---} \textcircled{B} \text{---} \textcircled{B} \text{---} \\ | \quad \quad | \\ \textcircled{B} \quad \textcircled{B} \\ | \quad \quad | \\ \text{---} \textcircled{B} \text{---} \textcircled{B} \text{---} \end{array}. \quad (123)$$

In the absence of a proof, we call this the fundamental *conjecture* of PEPS. Assuming periodic boundary conditions, the symmetry action is then implemented by starting from the PEPS  $|\Psi[A]\rangle$ , nucleating a small bubble of the intertwining MPO, pulling it through the entire lattice and annihilating it with itself so as to end up with the PEPS  $|\Psi[B]\rangle$ . As we will see in the remainder of these lectures, large classes of PEPS implement their symmetries exactly according to this prescription. In some cases, this MPO is just a tensor product (bond dimension 1), so that the virtual symmetry simply acts as a tensor product on the virtual legs.

In the following sections, we will discuss some particularly interesting PEPS wave functions which represent topological phases of matter. The fact that topological wave functions can easily be constructed using the tensor network formalism is clearly one of the main reasons that they are studied so extensively. Additionally, those tensor network representations identify the long-range entanglement with (MPO) symmetries acting on the virtual degrees of freedom. This is the reason that this fundamental conjecture is so relevant: topological order is characterised by non-trivial MPO symmetries.

## 5

#### 4.4 Two-dimensional SPTs: a case study

A paradigmatic example of a PEPS with non-trivial MPO symmetry is the so-called *CZX model* introduced by Chen, Liu and Wen in ref. [76] (see [97–99] for its tensor network representation). Their model is an example of a non-trivial SPT protected by a  $\mathbb{Z}_2$  symmetry which acts in an on-site manner  $U_g^{\otimes N}$  on the physical level of the PEPS. At the virtual level of the PEPS tensors, this symmetry is implemented via a bond dimension 2 MPO as:

$$\text{Diagram (124): } A \text{ with a red line labeled } g \text{ on the top virtual leg} = A \text{ with a red line labeled } U_g \text{ on the top virtual leg and } g \text{ on the bottom virtual leg}.$$

for  $g \in \{0, 1\}$ . The MPOs indexed by  $g$  form a non-local representation of  $\mathbb{Z}_2$ , which graphically can be represented by

$$\text{Diagram (125): } \text{MPO}(g, h) \otimes \text{MPO}(g, h) = \text{MPO}(gh, h).$$

The multiplication eq. (125) can be implemented locally by means of an intertwining tensor satisfying<sup>7</sup> [76, 98, 100]:

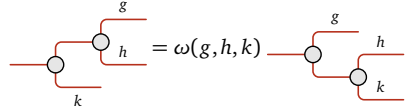
$$\text{Diagram (126): } \text{MPO}(gh, h) \otimes \text{MPO}(g, h) = \text{MPO}(gh, h).$$

Recall that in section 3.3 we argued that in the case of one-dimensional SPT phases, there exists a topological index that takes values in the second cohomology group  $H^2(G, \text{U}(1))$ . This index is fully captured by the multiplication rules of the virtual matrices which form a projective representation of the symmetry. It is now natural to wonder what the corresponding topological index for two-dimensional SPT phases is and how it is encoded in the MPO algebra. As it turns out, if you consider the multiplication of three MPOs representing the  $\mathbb{Z}_2$  symmetry,

<sup>7</sup>The MPO for  $g = 0$  is not injective, but can not be decomposed in injective blocks. This can be explained via the underlying algebraic structure, which is not semisimple [100, 101]. As such, the intertwiners cannot be found directly by invoking the fundamental theorem.



then the fusion tensors defined above can be contracted in two different ways. These two ways have to be compatible, so that there must exist a phase  $\omega$  depending on three group elements satisfying



$$\text{Diagram 1} = \omega(g, h, k) \text{Diagram 2}. \quad (127)$$

Associativity of the matrix multiplication reveals that  $\omega$  has to satisfy

$$\omega(g, h, k)\omega(g, hk, l)\omega(h, k, l) = \omega(gh, k, l)\omega(g, h, kl). \quad (128)$$

This condition is known as the *3-cocycle equation* (see also section A). Note that we can freely multiply the fusion tensors with a phase  $\varphi(g, h)$  and that under this redefinition the 3-cocycle changes according to the prescription

$$\omega(g, h, k) \rightarrow \omega(g, h, k) \frac{\varphi(g, h)\varphi(gh, k)}{\varphi(g, hk)\varphi(h, k)}. \quad (129)$$

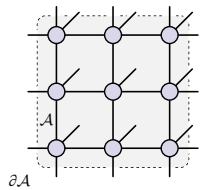
Modding out such redefinitions, we can conclude that the index  $\omega$  takes values in the third cohomology group  $H^3(G, U(1))$ . This group can be computed by the methods laid out in section A, and is in particular also finite. For the case of  $G = \mathbb{Z}_2$  considered above, we have that  $H^3(\mathbb{Z}_2, U(1)) \simeq \mathbb{Z}_2$  so that there exists exactly one non-trivial SPT phase protected by  $\mathbb{Z}_2$  symmetry. A representative 3-cocycle for the corresponding non-trivial class reads

$$\omega(g, h, k) = (-1)^{ghk}, \quad (130)$$

which can be immediately deduced from eq. (126) for the fusion tensors of the CZX model. Just as in the case of quantum spin chains, two PEPS wave functions with different virtual symmetries/3-cocycles cannot be converted into each other via symmetry-preserving adiabatic paths without closing the gap.

#### 4.5 Edge Hamiltonians & anomalies

A noteworthy feature of PEPS emerges when they are defined on an open patch  $\mathcal{A}$ . In that case, the uncontracted bonds define a Hilbert space of dimension  $\chi^{|\partial\mathcal{A}|}$  where  $|\partial\mathcal{A}|$  is the number of links intersecting the boundary. Graphically:



$$\text{Diagram of patch } \mathcal{A} \text{ with boundary } \partial\mathcal{A}. \quad (131)$$

These edge degrees of freedom define a tensor product structure for the edge modes of the corresponding parent Hamiltonian, completely in line with how we identified the SPT edge modes in the one-dimensional case in section 3.3. When the PEPS parent Hamiltonian is subsequently perturbed, this induces an *effective* Hamiltonian on these boundary degrees of freedom, potentially driving them through a phase transition [102]. Interestingly, in the case of PEPS representations of SPTs, these effective boundary Hamiltonians inherit the non-local MPO symmetries of the bulk. We refer to refs. [98, 103] for some examples and their numerical investigation.

Provided an MPO representation of a group raises in turn the question what the phase diagram of a symmetric (edge) Hamiltonian can look like. As we will now demonstrate, the answer to this question depend strongly on whether the 3-cocycle associated to the MPO algebra is trivial or not. As demonstrated in ref. [5, 76] a very elegant and straightforward tensor network argument shows that whenever  $\omega$  is cohomologically non-trivial, the Hamiltonians that commute with it must either break the symmetry spontaneously or be gapless. The cocycle  $\omega$  is in that case said to represent the *anomaly* of the symmetry. This observation closely parallels the celebrated Lieb–Schultz–Mattis theorems [63, 104, 105]. The argument goes as follows.

In the argument we assume an *injective* MPO representation of an arbitrary group  $G$ . Due to the injectivity we can now invoke the fundamental theorem to proof the existence of fusion tensors implementing the group multiplication locally, as in eq. (126). Assume now that the 3-cocycle defined as per eq. (127) is non-trivial and let us assume the existence of a unique trivially gapped symmetric ground state, represented by an injective MPS. Graphically this boils down to:

$$\begin{array}{c} \text{---} \square \text{---} \square \text{---} \square \text{---} \\ | \quad | \quad | \\ \text{---} \square \text{---} \square \text{---} \square \text{---} \end{array} \quad \text{with red line } g \text{ on top} = \begin{array}{c} \text{---} \circ \text{---} \circ \text{---} \circ \text{---} \\ | \quad | \quad | \end{array} \quad (132)$$

An application of the fundamental theorem of MPS then reveals that there must exist a trivalent *action tensor* (in gray) which satisfies:

$$\begin{array}{c} \text{---} \square \text{---} \\ | \quad | \\ \text{---} \square \text{---} \end{array} \quad \text{with red line } g \text{ on top} = \begin{array}{c} \text{---} \circ \text{---} \circ \text{---} \\ | \quad | \end{array} \quad \text{with red line } g \text{ on top} \quad (133)$$

Similar as in the previous section, if we act with two symmetry MPOs on the state, the fusion of the MPO tensors followed by the action on the MPS is up to a phase equal to sequentially acting with the MPOs. Concretely:

$$\begin{array}{c} \text{---} \square \text{---} \square \text{---} \\ | \quad | \\ \text{---} \square \text{---} \square \text{---} \end{array} \quad \text{with red lines } g \text{ and } h = \lambda(g, h) \begin{array}{c} \text{---} \circ \text{---} \circ \text{---} \\ | \quad | \end{array} \quad \text{with red lines } g \text{ and } h \quad (134)$$

If we now consider the action of three such MPOs on our MPS, and consider the two distinct ways in which we can locally recouple these, we can deduce the following condition

$$\frac{\lambda(g, h)\lambda(gh, k)}{\lambda(h, k)\lambda(g, hk)} = \omega(g, h, k), \quad (135)$$

but the left-hand side can be recognized as a coboundary, which would in turn imply that the 3-cocycle is cohomologically trivial. Since this is in contradiction with the assumption that  $\omega$  is non-trivial, we must therefore conclude that there is no symmetric injective MPS, and that every MPO symmetric 1-D Hamiltonian must be symmetry broken or gapless. Note that this argument is true independent of whether the Hamiltonian encodes the edge modes of a 2-D SPT phase or of a genuine quantum spin chain.

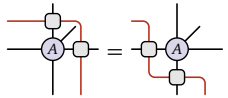
#### 4.6 Topological order: quantum doubles & the tube algebra

Intriguingly, beside SPT phases, there exist in spatial dimensions higher than one robust phases of quantum matter which are not protected by any symmetry. Such phases are said to exhibit *intrinsic topological order* [106–109] which is characterised by a ground state degeneracy that

only depends on the topology of the underlying spatial manifold (sphere, torus,...), gapped excitations with *anyonic* braiding statistics and, depending on its anyon content, a universal negative correction to the area law of entanglement entropy [110–112]. Notably, intimately connected to the absence of symmetry, these phases are not characterised by a local order parameter, and thus fall outside the scope of the traditional Landau paradigm of symmetry breaking [71, 73, 113]. Nevertheless, the fact that there is a ground state degeneracy clearly indicates that some symmetry is broken. It turns out that this symmetry is one that acts purely on the virtual degrees of freedom.

Recall that short-range entangled SPT order gets its stability from the existence of a topological index associated to the transformation properties of the entanglement degrees of freedom under the global symmetry (section 3.3 and 4.4). The robustness of intrinsic topological order on the other hand stems from its long-range entanglement [75]. In a sense, characterising topological orders thus boils down to the classification of equivalence classes of long-range entanglement. Similarly, phase boundaries between these topological phases can be understood from a radical change in the entanglement properties [114, 115]. Having direct access to these entanglement degrees of freedom, as we have in PEPS, suggests that we could further reduce this problem to classifying *local* PEPS tensors that give rise to *global* topological states: the classification of intrinsically topologically ordered systems is equivalent to the construction of MPO-symmetric tensors.

Indeed, in the language of PEPS, the presence of topological order can be traced back to the existence of *virtual* MPO symmetries that commute with the ground state PEPS tensors as:



$$(136)$$

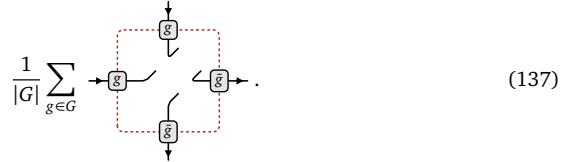
Note that, as opposed to the SPT case, there is no action on the physical degrees of freedom. Consequently, these MPOs are completely deformable – or thus *topological* – on the virtual level. This also implies that the underlying symmetry structure can be much richer than the ones described by groups. As we will see in lecture 5, the symmetry algebra is encoded by a *fusion category*. Since the MPOs are blind to the physical degrees of freedom, such states are resilient to (small) local perturbations on the physical level [115]. However, *anyon condensation* provides a mechanism to drive the system to a trivial phase under sufficiently strong perturbations [116, 117].

It is conjectured that a non-chiral topological order is fully characterised by its anyon content, together with its fusion rules and braiding statistics. Mathematically, this data is encapsulated in a *modular tensor category*, called the *Drinfeld centre*  $\mathcal{Z}(\mathcal{D})$  [118–121]. The full extent of this statement, and the microscopic realisations of generic topological orders will be the topic of section 5.3. In this section we will rather focus on the construction of a class of topological orders called (untwisted) *quantum doubles* [110, 122, 123], as a more gentle introduction to PEPS representations of topological order and some of its properties.

The quantum doubles as introduced by Kitaev in ref. [110] are a family of models constructed from a finite group  $G$  by assigning  $|G|$ -dimensional qudits to the edges of a two-dimensional lattice, subject to a commuting projector Hamiltonian spelled out in the aforementioned reference. They are a lattice Hamiltonian realisation of untwisted Dijkgraaf-Witten topological field theory [122]. For the group  $\mathbb{Z}_2$ , the model reduces to the *toric code*. The ground states of these quantum double models admit notably simple renormalisation fixed point PEPS representations called *G-injective*, or more precisely *G-isometric*, PEPS. The notion of *G-isometry* essentially captures PEPS tensors that are invertible only on a subspace of the virtual degrees

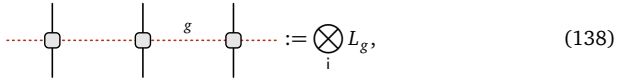
of freedom which is invariant under an action of the group  $G$ , and is a particular incarnation of MPO injectivity.

Up to local physical transformations all  $G$ -isometric PEPS reduce to a universal form that we will now describe. In this section, we assume a square lattice on a two-dimensional torus. We write  $L$  to denote the *left regular representation* of  $G$ , acting as  $L_g |h\rangle = |gh\rangle$ . Define then the projector  $1/|G| \sum_g L_g \otimes L_g \otimes L_g^\dagger \otimes L_g^\dagger$ , which constitutes the ground state PEPS tensor, graphically:



$$\frac{1}{|G|} \sum_{g \in G} \text{[Diagram of PEPS tensor]} \quad (137)$$

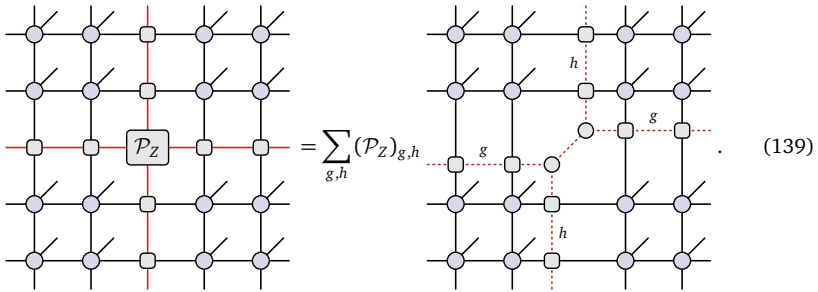
In this figure, the gray boxes represent  $L_g$ , and the bar denotes the inverse group element (recall that  $L_g^\dagger = L_{\bar{g}}$ ). Note that the four innermost legs together form the physical level. As the sum in eq. 137 runs over all group elements, this PEPS tensor satisfies the pulling through conditions eq. (136) for the following MPOs labelled by group elements:



$$\text{[Diagram of MPO]} := \bigotimes_i L_g, \quad (138)$$

where the dotted line is used to emphasise that the bond dimension is 1, i.e. each MPO acts as a tensor product.

The PEPS tensor eq. (137) gives rise to only one of the ground states of the model. As demonstrated in [124], all the other ones can be obtained unitarily from this one by acting with operators that wrap around either or both of the non-contractible cycles of the torus. The extensive nature of these operators are reflective of the local indistinguishability of the ground states. Equivalently, one can obtain all the ground states by wrapping both cycles of the torus with MPOs on the virtual level and joining them on their intersection by a tensor  $\mathcal{P}_Z$  that we need to solve for. In this, we require that the ground states are all translationally invariant and remain normalisable and orthogonal in the thermodynamic limit [125]. Using the fusion tensors for the MPOs eq. (138) we can expand such a state as follows:<sup>8</sup>



$$\text{[Diagram of ground state expansion]} \quad (139)$$

The translation invariance of these states is manifest by the pulling through condition. As demonstrated in refs. [120, 126, 127] a basis of ground states in MPO injective PEPS is in one-to-one correspondence with *minimal central idempotents* or – equivalently – irreducible representations of *Ocneanu's tube algebra* [128, 129]. In fact, the Drinfeld centre  $\mathcal{Z}(\mathcal{D})$  is in

<sup>8</sup>Note that the fusion tensors in this case simply enforce the group multiplication, i.e.  $\sum_{g,h,k} |k\rangle \langle g, h| = \sum_{g,h} |gh\rangle \langle g, h|$ .

practice often computed via the tube algebra. For the particular case of quantum doubles, the relevant tube algebra is spanned by closed MPOs of the form:

$$T_x^g := \text{diagram}, \quad (140)$$

thought of as acting from the outside indices to the inside. Their multiplication is defined by stacking them inside each other. Algebraically this boils down to:

$$T_x^g \circ T_y^h = \delta_{x,hyh^{-1}} T_x^{gh}. \quad (141)$$

This multiplication defines an algebra often called the (*untwisted*) *quantum double*  $\mathcal{D}(G)$  of the group  $G$  [123]. Notably, the tube algebra is endowed with a *star*, which is an antilinear involution that swaps the order of the product. For the case at hand:  $(T_x^g)^* := T_{gxg^{-1}}^{g^{-1}}$ . As such, the tube algebra is endowed with the structure of a  $C^*$ -algebra. Due to a famous result of Artin and Wedderburn, every  $C^*$ -algebra can be decomposed in simple matrix algebras. The aforementioned minimal central idempotents then act as projectors on each of these simple blocks.

Let us compute them explicitly. To this end it is useful to focus first, for fixed  $x \in G$ , on all tubes  $T_x^g$  where  $g \in Z_x = \{g \in G | gx = xg\} \subseteq G$ . These tubes form a representation of the stabiliser group  $Z_x$  since

$$T_x^{g_1} \circ T_x^{g_2} = T_x^{g_1 g_2}, \quad \forall g_1, g_2 \in Z_x, \quad (142)$$

as follows directly from eq. (141). Let  $\mu_x$  denote an irreducible representation of  $Z_x$  of dimension  $d_{\mu_x}$  and with  $\chi^{\mu_x}$  its corresponding character. The operators

$$\mathcal{P}_{(x,\mu)} := \frac{d_{\mu_x}}{|G|} \sum_{k \in Z_x} \chi^{\mu_x}(k) T_x^k, \quad (143)$$

form then hermitian projectors i.e.<sup>9</sup>

$$\mathcal{P}_{(x,\mu_x)} \mathcal{P}_{(y,\nu_y)} = \delta_{x,y} \delta_{\mu_x,\nu_x} \mathcal{P}_{(x,\mu_x)}. \quad (144)$$

These  $\mathcal{P}_{(x,\mu)}$  are *simple* idempotents. The central ones are then obtained by additionally summing over all  $x$  in a conjugacy class  $C_x = \{gxg^{-1} | g \in G\}$ :

$$\mathcal{P}_{(C_x,\mu_x)} \equiv \sum_{y \in C_x} \mathcal{P}_{(y,\mu_x)}. \quad (145)$$

This reproduces the results of refs. [111] that torus ground states of the quantum doubles are labelled by a conjugacy class  $C_x$  and an irreducible representation of the stabiliser subgroup  $Z_x$  of a representative  $x$  of  $C_x$ .

Importantly, also anyonic excitations are labelled by minimal central idempotents of the tube algebra [130, 131]. At the level of the PEPS, one can create pairs of anyonic excitations by replacing two of the ground state tensors by tensors whose support is contained in the image of the corresponding idempotent labelling the excitation, we refer to [132] for details.

<sup>9</sup>Use Schur orthogonality of the characters  $\sum_k \chi^{\mu_x}(k) \chi^{\nu_x}(k^{-1}m) = |Z_x|/d_{\mu} \chi^{\mu}(m)$ .

The (untwisted) quantum doubles considered here are generalised in a number of ways. For one, one could consider a *twisted* version of the model where the MPO symmetries form an anomalous representation of  $G$  with the anomaly taking values in the third cohomology group  $H^3(G, \text{U}(1))$ , as described in section 3.3. For this, we refer to refs. [82, 97, 127].

In section 5.3 we will discuss the PEPS representations of so-called string-net models constructed first by Levin and Wen in ref. [133]. These are a class of models that have a spherical fusion category as input, and realise all non-chiral topological orders  $\mathcal{Z}(\mathcal{D})$ . One way of obtaining the torus ground states and anyonic excitations described by  $\mathcal{Z}(\mathcal{D})$  in these models is exactly via the tube algebra associated to  $\mathcal{D}$ , completely in line with the above construction. For the input  $\mathcal{D} = \text{Vec}_G^\omega$  the corresponding Levin-Wen model coincides with the twisted quantum doubles.

#### 4.7 Synopsis of lecture IV

Lecture IV introduced three new topics: symmetries in the form of MPOs which underlie the integrability of special classes of quantum spin chains; the higher dimensional generalisation of MPS to PEPS, and the corresponding manifold which parametrises low-energy states of gapped higher dimensional lattice models; and MPO symmetries that satisfy a so-called *pulling through symmetry* which characterises the different gapped (topological) phases of such systems. Concerning integrability, we demonstrated that the Yang-Baxter equations are a consequence of the fundamental theorem of MPS. It would be wonderful if this point of view leads to more general classes of integrable models, such as the ones satisfying non-trivial categorical symmetries (see lecture V).

We showed that SPT phases in 2D lattice systems are characterised by non-trivial MPO symmetries whose associators encode 3-cocycles, leading to the classification of 2D SPT phases. However, the pulling through symmetries of PEPS open up completely new ways of realising topological phases of matter in 2D systems if no global symmetries are involved. We demonstrated the existence of intrinsic topologically ordered lattice systems in the form of quantum doubles, and showed that those systems exhibit symmetry breaking with respect to the entanglement degrees of freedom. This was the first hint that the Landau paradigm can be saved if you describe many-body phases of matter in terms of the symmetries of the underlying PEPS representation: different phases realise the symmetries of the tensors in a fundamentally different way.

## 5 Lecture V: Generalised symmetries

### 5.1 Motivation

In lecture 3 we classified phase diagrams of one-dimensional quantum lattice models with a group symmetry. We also saw that in one-dimensional bosonic systems the presence of a symmetry is *necessary* to have a non-trivial phase diagram. Recall from section 3.5 that the story was more subtle for the case of fermions, which can exhibit intrinsic topological order due to a superselection rule. In recent years, substantial research effort has been made to understand systems with symmetries beyond groups [134–144]. As we foreshadowed in the introduction to lecture 3, generalised symmetries described by fusion categories are encountered as the *virtual symmetries* of non-chiral two-dimensional topological orders and therefore also arise as the symmetries of their boundary Hamiltonians or – equivalently – the symmetries of the effective theories describing their edge modes. Studying how those generalised symmetries influence the phase diagrams of systems has led to a *generalised Landau paradigm*, where different phases of matter were found to be in correspondence with different ways to break a generalised symmetry [9, 53, 135, 139, 144]. With non-group symmetries, one cannot simply say that a symmetry is broken to a subgroup, so we should ask: how can a system (partially) break a generalised symmetry? Is there a map between systems exhibiting different ways of generalised symmetry breaking? These questions will all be expanded upon in this section.

Why are tensor networks a good description for such categorical symmetries? A crucial observation is that to encode symmetries beyond on-site, we need operators that act in a correlated way on neighbouring degrees of freedom. Additionally, we want to be able to define these operators on systems of any size, so we need to encode those correlated symmetries locally. MPOs are exactly the right language to use for this purpose. They are parametrised in terms of a single local tensor, can be defined on any number of sites, and share information between neighbouring physical degrees of freedom through their virtual degrees of freedom. Because of these considerations, we will often use the terms *generalised*, *categorical* and *MPO symmetry* interchangeably.

Indeed, the study of general MPO symmetries revealed that, under mild conditions such as injectivity (discussed further in section 5.2), their underlying structure is equivalent to that of fusion categories. In brief, fusion categories are semisimple associative structures of (simple) objects, very much like a group is an associative multiplication rule of its elements (its simple objects). However, unlike the group case, the multiplication of simple objects of a fusion category may lead to a direct sum of simple objects. A canonical example being the fusion category  $\text{Rep}(G)$  whose simple objects are the irreps of a group and whose multiplication rule is dictated by the direct sum decomposition of the tensor product of irreps, e.g.  $1/2 \otimes 1/2 = 0 \oplus 1$ . Fusion category theory is a very well studied topic in mathematics and mathematical physics, and MPOs turn out to be the building blocks of *representations* of those fusion categories on the lattice, much like matrices represent on-site group symmetries in spin systems.

In this final lecture we will first study the MPO representation theory of categorical symmetries and demonstrate how *module categories* label the distinct representations. We will then provide three concrete applications. To start, we will see how these MPO symmetries arise in the Levin-Wen string-net models, which provide microscopic realisations of all non-chiral two-dimensional topological orders. We then provide two approaches to construct *edge theories* of these string-net models. In section 5.4, we will demonstrate how so-called *strange correlators* of these PEPS representations lead to the construction of two-dimensional *classical* partition functions of critical models [145], generalizing a construction of Aasen, Mong and Fendley [146]. The main merit of this strange correlator construction is that it makes the distinction between *topological* and *geometrical* or *dynamical* features of the critical lattice model

manifest: the (2+0)d partition functions inherit the MPO symmetries from the (2+1)d string-nets, and as such provides the lattice analogue of TFT/CFT correspondence. This holographic approach to the construction of edge theories with categorical symmetries can be interpreted as an explicit lattice realisation of the categorical construction of Kong, Wen and Zheng [147] which provides a concrete foundation for what is now widely referred to as *SymTFT* or *topological holography* [9, 140, 148–150]. Finally, we will provide a systematic way to construct local (1+1)d MPO symmetric Hamiltonians and a generalised duality or gauging procedure for generalised symmetries. We conclude by discussing the anticipated generalised Landau paradigm for categorical symmetries, as well as some of its implications.

## 5.2 MPO representation theory

In this section we will follow a logic similar to the one in section 4.2. Assuming a collection of injective MPOs representing a fusion algebra, we deduce the consistency equations satisfied by the fusion tensors. We subsequently show how solutions of these equations can be used to construct representations of these MPOs and fusion tensors. It is demonstrated how distinct representations are constructed from the data encoded in *module categories* over the fusion category describing the symmetry.

Concretely, assume a finite collection of injective MPOs labelled by  $a, b, c, \dots$  with periodic boundary conditions. We are interested in MPO algebras for which the structure constants  $N_{ab}^c$  are independent of the number of sites in the periodic chain:

$$\begin{array}{c} \text{Diagram: A chain of three MPO tensors (squares) with indices } a, b, c \text{ on the horizontal lines and vertical lines on the ends.} \end{array} = \sum_c N_{ab}^c \begin{array}{c} \text{Diagram: A single MPO tensor with index } c \text{ on the horizontal line and vertical lines on the ends.} \end{array} \quad (146)$$

Here and in what follows, the sums are over all distinct MPOs appearing in the algebra. Amongst them, there exists one labelled  $\mathbb{1}$  for which  $N_{a\mathbb{1}}^{a'} = N_{\mathbb{1}a}^{a'} = \delta_{a,a'}$ . The non-negative integer tensor  $N$  defines a *unital associative fusion* structure, meaning in particular that the structure constants satisfy  $\sum_d N_{ab}^d N_{dc}^e = \sum_d N_{bc}^d N_{ad}^e$ . Exploiting the fundamental theorem eq. (79), we find that eq. (146) can be implemented locally by means of pulling the following *fusion tensors* through the MPO tensors:

$$\begin{array}{c} \text{Diagram: A fusion tensor (circle) with index } c \text{ on the left, } a \text{ on the top, and } b \text{ on the bottom.} \end{array} = \begin{array}{c} \text{Diagram: A fusion tensor (circle) with index } c \text{ on the left, } a \text{ on the top, and } b \text{ on the bottom.} \end{array}, \quad (147)$$

where the multiplicity  $i = 1, 2, \dots, N_{ab}^c$  labels the linearly independent fusion tensors. Similarly as in section 4.4, associativity of the MPO multiplication implies the existence of a collection of *unitary* matrices called *F-symbols* or *associators* encoding the basis transformation

$$\begin{array}{c} \text{Diagram: A fusion tensor (circle) with index } d \text{ on the left, } e \text{ on the top, and } b \text{ on the bottom.} \end{array} = \sum_{f,k,l} (F_d^{abc})_{e,ij}^{f,kl} \begin{array}{c} \text{Diagram: A fusion tensor (circle) with index } d \text{ on the left, } a \text{ on the top, and } b \text{ on the bottom.} \end{array} \quad (148)$$

The requirement that the product of four MPOs – which can be recoupled in two distinct ways – is associative, reveals that these *F-symbols* obey a consistency condition colloquially known



as the *pentagon equation*. Schematically, the pentagon equation is a linear multivariate system of equations of the form

$$FF = \sum FFF, \quad (149)$$

or, written out in components:

$$\sum_o (F_e^{fcd})_{g,lm}^{h,no} (F_e^{abh})_{f,ko}^{i,pq} = \sum_{j,rs} (F_g^{abc})_{f,kl}^{j,rs} (F_e^{ajd})_{g,sm}^{i,tq} (F_i^{bcd})_{j,rt}^{h,np}. \quad (150)$$

In essence, the fusion rules specified by the  $N$ -symbol, together with a unitary solution of the pentagon equation constitute the data encoded in a *unitary fusion category*  $\mathcal{C}$ .<sup>10</sup> Importantly, for any valid fusion structure, the number of solutions to the pentagon equation is finite, up to basis transformations on the multiplicity indices.

Given a solution to the pentagon equation, how can we find a corresponding MPO representation in the sense of eq. (146)? Completely in line with the logic of section 4.2, we can find a representation by equating the  $F$ -symbols to (the components of) the fusion tensors defined in eq. (147) as well as to the MPO tensors. In that case, both eq. (148) as well as eq. (147) are equivalent to the pentagon equation (150) [120, 126, 127, 151]. Concretely, the assignment of non-zero components is as follows<sup>11</sup>:

$$\begin{array}{c} (abe,i) \quad (ecd,j) \\ \diagdown \quad \diagup \\ \bigcirc \\ \diagup \quad \diagdown \\ (bck,k) \quad (afd,l) \end{array} = (F_d^{abc})_{e,ij}^{f,kl} = \begin{array}{c} (ecd,j) \\ | \\ \square \\ | \\ (bck,k) \end{array} \quad (eab,i) \quad (daf,l). \quad (151)$$

Note that every index is labelled by a triple of objects  $a, b, c$  and a multiplicity index taking values  $1, 2, \dots, N_{ab}^c$ . In particular, all triples simultaneously have to satisfy the fusion rules in order for a component to be non-zero. Moreover, the structure of these tensors is such that some of the objects are shared between indices. This internal structure of the tensors invites us to depict these tensors as *triple line tensors*:

$$\begin{array}{c} e \\ \diagdown \quad \diagup \\ i \quad b \quad c \quad j \\ \diagup \quad \diagdown \\ a \quad f \quad d \\ | \\ i \end{array} \equiv \begin{array}{c} (abe,i) \quad (ecd,j) \\ \diagdown \quad \diagup \\ \bigcirc \\ \diagup \quad \diagdown \\ (bck,k) \quad (afd,l) \end{array} \equiv \begin{array}{c} (ecd,j) \\ | \\ \square \\ | \\ (bck,k) \end{array} \quad (eab,i) \quad (daf,l) \equiv \begin{array}{c} e \quad j \quad d \\ | \quad | \quad | \\ i \quad a \quad l \\ | \quad | \quad | \\ b \quad c \quad f \end{array}. \quad (152)$$

Contraction of these tensors takes place via identification of the simple objects on strands of concatenated tensors, pairing indices, and summing over them as detailed in e.g. refs. [132, 152].

Due to the fusion rules, the MPOs act on a Hilbert space which is generically not a tensor product space, but rather a Hilbert space with local *constraints* dictated by the fusion rules. In this sense, the MPOs constructed here and generalised below, form a representation and generalisation of the topological symmetries considered in the context of *anyonic chains* [153–158]. The models considered in the original reference [153] for instance are constructed from the Fibonacci fusion category Fib (see section B).

<sup>10</sup>Note the resemblance of the pentagon equation with the 3-cocycle derived in eq. (129) by an analogous reasoning applied to group symmetry. And indeed, a finite group together with a choice of 3-cocycle defines a fusion category called  $\text{Vec}_G^\omega$ , as reviewed in section B, for which the  $F$ -symbol coincides with  $\omega$ .

<sup>11</sup>We omit a conventional prefactor involving quantum dimensions [132, 152].

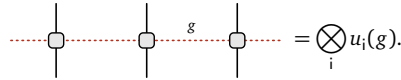
The above provides only one MPO representation of  $\mathcal{C}$ . As demonstrated in ref. [152], any *module category* over the given fusion category  $\mathcal{C}$  provides a distinct way to realise that symmetry on the lattice. Succinctly, a  $\mathcal{C}$ -module category  $\mathcal{R}$  consists of a finite set of simple objects  $A, B, \dots$  together with an *action* of  $\mathcal{C}$  on  $\mathcal{R}$ , which can be expressed as  $a \triangleright A = \sum_B N_{aA}^B B$ . Notably, this  $N$ -symbol is different from the  $N$ -symbol of  $\mathcal{C}$  as it depends on two objects of  $\mathcal{R}$  and one of  $\mathcal{C}$ . A module category also provides a collection of unitary  ${}^\triangleright F$ -symbols or (*left*) *module associators*, which together with the  $F$ -symbols of  $\mathcal{C}$  obey a *mixed pentagon* equation which is of the form:

$${}^\triangleright F {}^\triangleright F = \sum F {}^\triangleright F {}^\triangleright F. \quad (153)$$

Given a  $\mathcal{C}$ -module category and its corresponding module associators, we can construct fusion tensors by equating its components to the  ${}^\triangleright F$ -symbol – completely in line with eq. (151) and as detailed below in eq. (159) – so that the pulling through condition eq. (147) is satisfied by virtue of eq. (153). Remark that every fusion category  $\mathcal{C}$  defines a module category over itself, called the *regular module category*. In that case, we recover the MPO representations of the previous paragraph. Before explaining what the MPO tensors evaluate to, let us establish some intuition for this framework by means of a simple example.

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**Example:**  $\mathcal{C} = \text{Vec}_G$  Formally, a (non-anomalous) finite-group symmetry is described by the fusion category  $\text{Vec}_G$  of  $G$ -graded vector spaces. The simple objects correspond to group elements and the fusion rules are given by the group multiplication, i.e.  $N_{g,h}^k = \delta_{gh,k}$ . Module categories over  $\text{Vec}_G$  are in one-to-one correspondence with pairs  $(H, \psi)$  where  $H \subseteq G$  is a subgroup up to conjugacy and  $\psi$  is a  $U(1)$ -valued 2-cocycle of the subgroup  $H$ . We refer to section B for details. Let us focus on the case where  $H = G$  and  $\psi$  is trivial. The corresponding module category, denoted  $\text{Vec}$ , then results in MPOs



$$= \bigotimes_i u_i(g). \quad (154)$$

The  $u_i$  here are (not necessarily irreducible) unitary representations of  $G$ , and can be chosen independently on each site. The group structure of the MPOs follows from  $u_i(g)u_i(h) = u_i(gh)$ ,  $\forall g, h \in G, \forall i$ . The dotted line is to emphasise that the bond dimension of the MPOs is 1. Notably, the fact that the MPOs act on a tensor product Hilbert space is a direct consequence of the choice of module category  $\text{Vec}$ . For the other allowed module categories over  $\text{Vec}_G$ , the MPOs generically act on constrained Hilbert spaces as mentioned above.

Note that in the case of on-site group symmetry eq. (154), one can bring all MPO tensors  $u_i(g)$  – indexed by the group elements  $g \in G$  – simultaneously in block-diagonal form. The blocks correspond to the irreducible representations contained in the representations  $u_i$ . More generally, we could consider the collection of MPO tensors as operator-valued matrices labelled by the blocks  $a, b, c, \dots$ , and block-decompose the algebra spanned by them in the *vertical* direction. The crucial insight is now that the labels of these injective blocks, and their fusion

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Being a fusion category,  $\mathcal{D}$  comes with its own set of  $F$ -symbols that we shall write as  ${}^{\mathcal{F}}F$ . These satisfy a pentagon equation analogous to eq. (149). The Morita dual  $\mathcal{D}$  is constructed in such a way that  $\mathcal{R}$  is also a module category over  $\mathcal{D}$ , with  $\mathcal{D}$  now acting from the *right* on  $\mathcal{R}$ . As a consequence there are once more corresponding  $F$ -symbols, denoted  ${}^{\mathcal{D}}F$ , encoding the associativity of the  $\mathcal{D}$  action on  $\mathcal{R}$ . These satisfy the fourth pentagon equation we encounter, now involving those of  $\mathcal{D}$  as well:

$$\triangleleft_F \triangleleft_F = \sum \circ_F \triangleleft_F \triangleleft_F. \quad (156)$$

Beyond being a module category for both  $\mathcal{C}$  and  $\mathcal{D}$ ,  $\mathcal{R}$  is an (invertible)  $(\mathcal{C}, \mathcal{D})$ -bimodule category (see section B). Associated to its bimodule structure is a final collection of  $F$ -symbols denoted by  ${}^{\times}F$  obeying

$$\boxtimes F \triangleright F = \sum_i \triangleright F \boxtimes F \boxtimes F, \quad (157)$$

$$\triangleleft F \bowtie F = \sum \bowtie F \bowtie F \triangleleft F. \quad (158)$$

The components of the MPO tensors are equal to the  $\triangleright F$ -symbols, while the  $\triangleleft F$ -symbols are exactly what the fusion tensors in the *vertical* direction evaluate to. Gathering everything:

$$\begin{array}{c}
\begin{array}{c} B \\ \swarrow \quad \searrow \\ l-a \quad i \quad b-k \\ \swarrow \quad \searrow \\ A \quad c \quad C \\ \downarrow \\ j \end{array} \equiv \begin{array}{c} (AaB, l) \quad (BbC, k) \\ \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \\ (AcC, j) \end{array} = (\triangleright F_A^{abc})_{c, ij}^{B, kl}, \\
\\
\begin{array}{c} C \\ \swarrow \quad \searrow \\ i-\alpha \quad k \quad \beta-j \\ \swarrow \quad \searrow \\ A \quad \gamma \quad B \\ \downarrow \\ l \end{array} \equiv \begin{array}{c} (A\alpha C, i) \quad (C\beta B, j) \\ \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \\ (A\gamma B, j) \end{array} = (\triangleleft F_B^{A\alpha\beta})_{C, ij}^{\gamma, kl}, \quad (159)
\end{array}$$

$$\begin{array}{c} C \quad D \\ \begin{array}{c} i \quad a \quad l \\ \alpha \\ k \quad B \end{array} \end{array} \equiv \begin{array}{c} (EaD, j) \\ \begin{array}{c} \square \\ (AaC, i) \quad (D\alpha B, l) \\ (A\alpha B, k) \end{array} \end{array} = \left( {}^{\alpha}F_D^{aA\alpha} \right)_{C, ij}^{B, kl}.$$

Note that in these triple line tensors, objects in the module category always live on the outer loops. One can then verify that the vertical pulling through condition

$$\begin{array}{c} \begin{array}{c} \square \\ \alpha \quad \beta \quad \gamma \end{array} \end{array} = \begin{array}{c} \begin{array}{c} \square \\ \alpha \quad \beta \quad \gamma \end{array} \end{array} \quad (160)$$

is satisfied by virtue of eq. (158), while eq. (147) is equivalent to eq. (157). These fusion tensors in turn are related via a rotated version of the diagram eq. (148) involving these fusion tensors and the  ${}^{\circ}F$ -symbol, equivalent to eq. (156).

In the example  $\mathcal{C} = \text{Vec}_G$ ,  $\mathcal{R} = \text{Vec}$  considered above, the fusion category  $\mathcal{D}$  is  $\text{Rep}(G)$ , whose simple objects are the irreps of the group  $G$ . The corresponding  ${}^{\circ}F$ -symbols are equal to the 6j-symbols of  $G$ , while the  ${}^{\alpha}F$ -symbols are equal to the Clebsch-Gordan coefficients. Those Clebsch-Gordan coefficients decompose the MPO tensors  $u_i(g)$  in its irreducible blocks, and the data in these blocks is encoded in  ${}^{\alpha}F$ , whose values are given by the irreps of  $G$ . In the general (non-group) case, the  ${}^{\alpha}F$ -symbols will below be interpreted as generalised Clebsch-Gordan coefficients for the MPO symmetries constructed here. At the same time, we will interpret eq. (160) as a generalised Wigner-Eckart theorem that states that MPO symmetric tensors are built from these generalised Clebsch-Gordan coefficients.

In conclusion, the following picture has emerged. Given a symmetry specified by a fusion category  $\mathcal{C}$ , an MPO representation is constructed from an invertible  $(\mathcal{C}, \mathcal{D})$ -bimodule category  $\mathcal{R}$ . This structure encodes associator data in the form of five F-symbols ( $F, {}^{\triangleright}F, {}^{\triangleright\triangleright}F, {}^{\triangleleft}F, {}^{\circ}F$ ) that solve six pentagon equations. The bimodule associator  ${}^{\triangleright\triangleright}F$  can be used to construct the injective MPO representation of the fusion category  $\mathcal{C}$ , whose tensors are locally fused horizontally eq. (147) and vertically eq. (160) via the tensors evaluating to  ${}^{\triangleright}F$  and  ${}^{\triangleleft}F$  respectively. Both satisfy a recoupling identity involving  $F$ , respectively  ${}^{\circ}F$ .

### 5.3 Application 1: string-net ground states & intertwiners

*String-net models* were first constructed by Levin and Wen in ref. [133]. They constitute a class of RG fixed point lattice models that takes as an input a (spherical) fusion category, denoted by  $\mathcal{D}$ , and realises the non-chiral topological order given by its Drinfeld centre  $\mathcal{Z}(\mathcal{D})$  [131, 159, 160]. By that, it is meant that the ground states and gapped excitations reproduce the topological sectors and anyonic statistics encoded in  $\mathcal{Z}(\mathcal{D})$ .<sup>13</sup> The quantum doubles encountered in section 4.6 are reproduced for the choice  $\mathcal{D} = \text{Vec}_G$ . The name *string-net* stems from the interpretation of the model as describing a condensed phase of coloured strings, with the allowed string-types and their interactions encoded in  $\mathcal{D}$ . The Hamiltonian is a sum of local commuting projectors on each plaquette, allowing for a straightforward derivation of a PEPS representation of its ground states.

As shown in ref. [152], there exists a ground state PEPS representation for every module category  $\mathcal{R}$  over the input  $\mathcal{D}$ . This observation generalises and unifies earlier references where

<sup>13</sup>Equivalently, they can be thought of as a Hamiltonian realisation of the Turaev–Viro–Barrett–Westbury topological field theory [159, 161, 162].

implicitly the regular module category was assumed [127, 132, 163–165], or in the specific case of the quantum doubles,  $\text{Vec}$  as module category over  $\text{Vec}_G$  [124]. From this point of view, a string-net is thus a PEPS with a particular type of *virtual* symmetry. The MPOs which form representations of the underlying Morita dual fusion category  $\mathcal{C}$  are its symmetries, and they act purely on the virtual degrees of freedom of the tensor network. In accordance with the generalised Wigner-Eckart theorem discussed above, the PEPS tensors evaluate to  $\langle F$  as in eq. (159), with the physical level taking values in  $(\alpha\beta\gamma, k)$ .

Akin to section 4.6, the different torus ground state sectors and anyonic excitations, both labelled by elements in  $\mathcal{Z}(\mathcal{D})$ , can be understood in terms of the *tube algebra*. The tube algebra is constructed from the MPOs in  $\mathcal{C}$ , and their central idempotents are in one-to-one correspondence with elements in  $\mathcal{Z}(\mathcal{C})$  [127, 132]. Recall that  $\mathcal{Z}(\mathcal{D})$  is guaranteed to be equivalent to  $\mathcal{Z}(\mathcal{C})$  due to the invertibility of  $\mathcal{R}$ . Indeed, if we consider a string-net on a torus, the different topological sectors can be obtained by winding the MPO symmetries of the previous section around its uncontractable loops. In their intersection, a tensor is placed whose components are determined by the central idempotents. The insertion of a central idempotent associated to the object  $Z \in \mathcal{Z}(\mathcal{C})$  in a string-net PEPS can be depicted as:

$$\sum_{\substack{a_\alpha, b_\beta, c_\gamma \\ i, j}} (\mathcal{P}_Z)^{i, j}_{a_\alpha, b_\beta, c_\gamma} \quad \text{(161)}$$

The diagram shows a 2D lattice of purple circular tensors connected by blue lines. A central square tensor is highlighted, representing a central idempotent. Red lines, labeled  $a_\alpha$ ,  $b_\beta$ , and  $c_\gamma$ , pass through this central square. The labels  $a_\alpha$ ,  $b_\beta$ , and  $c_\gamma$  are placed near the central square. The equation (161) is to the left of the diagram.

Note that the purple tensors coincide with the fusion tensors of eq. (160), so that the exact location of the central idempotent is undetectable. Here,  $a_\alpha$  stands for  $a_\alpha \equiv (a, \alpha)$  where  $a$  is a simple object in  $\mathcal{C}$  and  $\alpha$  labels the degeneracy of  $a$  in  $Z$ .

Given a PEPS representation based on  $\mathcal{R}$ , one can locally change this to one based on  $\mathcal{R}'$  via a new class of MPO *intertwiners* that implement the change of module category. In particular, when  $\mathcal{R}' = \mathcal{D}$ , these MPO intertwiners are labelled by objects in  $\mathcal{R}$  and are constructed from the module associator<sup>14</sup>. A change in module category hints at an equivalence between edge Hamiltonians for each of these distinct PEPS representations. Later, in section 5.5, we will see that the edge Hamiltonians corresponding to different representations  $\mathcal{R}, \mathcal{R}'$  are related via *duality transformations*, or – equivalently – via gauging (part of) its MPO symmetries. The aforementioned MPO intertwiners play the role of intertwiners between these dual models and can be promoted to isometries, if boundary conditions are adequately accounted for.

Furthermore, the tensor network picture can also be used to explicitly construct *gapped boundaries* of string-net models. This is beyond the scope of these lecture notes and we refer to ref. [152] for details, but let us mention that also in that case module categories show up as labelling different boundary conditions. In conclusion, the complete picture of string-net

<sup>14</sup>In the general case, one requires the notion of  $(\mathcal{D})$ -module functors between module categories, organised in a category denoted by  $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$ . Notably,  $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$  can be endowed with the structure of an invertible  $(\mathcal{D}_{\mathcal{R}}, \mathcal{D}_{\mathcal{R}'})$ -bimodule category.

domain walls and boundaries, as developed by Kitaev and Kong in ref. [166], can be recast in tensor network language, making these structures amenable to simulating them on a computer.

## 5.4 Application 2: classical partition functions as strange correlators

With the tensor network representations of string-net ground states at our disposal, we can now discuss a second application of the general MPO formalism, the so-called *strange correlator* construction. This construction appeared first in ref. [167] as a diagnosis to detect non-trivial SPT phases by considering the matrix element  $C(\mathbf{r}, \mathbf{r}') = \langle \Omega | \phi(\mathbf{r}) \phi(\mathbf{r}') | \Psi \rangle$  where  $|\Psi\rangle$  is the state under scrutiny,  $\phi$  is a local observable, and  $|\Omega\rangle$  denotes a well-chosen short-range entangled trivial state. It was shown for a number of one and two-dimensional states  $|\Psi\rangle$  which exhibit SPT order that the quantity  $C(\mathbf{r}, \mathbf{r}')$  either decays algebraically in the distance  $|\mathbf{r} - \mathbf{r}'|$  or converges to a constant. Hence, this method provides an identification of SPT order in terms of the bulk ground state wave function.

A natural generalisation of this construction amounts to choosing a state  $|\Psi\rangle$  which exhibits intrinsic topological order. Focussing on the case of non-chiral topological order in two dimensions, we can thus construct strange correlators from the overlap of string-net ground states of section 5.3 with a suitable state  $|\Omega\rangle$ . This approach was pioneered in ref. [145]. As we have seen above, the PEPS representations of string-net ground states have exact MPO symmetries acting purely on the virtual level. Consequently, strange correlators constructed from these states inherit these symmetries, regardless of the chosen state  $|\Omega\rangle$ . However, for many interesting examples,  $|\Omega\rangle$  can simply be chosen as a product state. It is then natural to interpret this two-dimensional tensor network as the *partition function of a classical statistical model*<sup>15</sup> with topological defect lines/symmetries. Schematically for the case of  $|\Omega\rangle$  being a (uniform) product state, the partition function thus takes the form

$$Z(\{\beta_i\}) := \langle \Omega | \{\beta_i\} | \Psi \rangle = \text{[Diagram 1]} = \text{[Diagram 2]}, \quad (162)$$

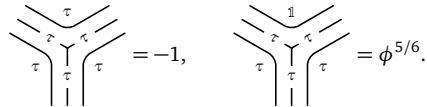
where  $\{\beta_i\}_i$  collectively denotes the parameters defining the state  $|\Omega\rangle$ , and thus indirectly determine the Boltzmann weights of the classical model. For appropriate values of the parameters  $\{\beta_i\}_i$ , the partition function can in fact describe a critical model. In that case, the MPO symmetries can be interpreted as lattice remnants of the continuum topological defects of the CFT. The strange correlator construction thus provides an explicit lattice representation of the correspondence between conformal field theories and topological field theories as envisioned first by Witten [107], Moore and Seiberg [168] and subsequently developed further by Fröhlich, Fuchs, Runkel and Schweigert in refs. [169–173].

**Example: hard hexagon model** Perhaps the simplest example is obtained from the string-net model based on the Fibonacci fusion category on the hexagonal lattice. This fusion category contains two simple objects denoted by  $\{1, \tau\}$ , and the only non-trivial fusion rule reads  $\tau \otimes \tau = 1 \oplus \tau$ . The strange correlator we will consider is a product state that projects all the physical degrees of freedom of the string-net onto the object  $\tau$  – or more precisely, in the

<sup>15</sup>Note however, that depending on the chosen  $|\Omega\rangle$  the classical model may have non-positive Boltzmann weights and requiring these to be positive generically requires fine-tuning the state  $|\Omega\rangle$ .

notation of eq. (159), to  $\phi \cdot (\tau\tau\tau, 1)$ , where  $\phi = \frac{1+\sqrt{5}}{2}$  is the golden ratio. The degrees of freedom on the loops then take on the role of classical binary variables prone to the constraint that no two neighbouring loops can be in the  $\mathbb{1}$  state. It is suggestive to interpret this model as a classical gas of particles on the plaquettes such that a loop labelled by  $\mathbb{1}$  signals the presence of a particle on the corresponding plaquette, and  $\tau$  denotes its absence. This so-called *hard hexagon* model was solved exactly in a tour de force by Baxter in 1980 and demonstrated to exhibit two phases [174]. For a large value of the fugacity the model exhibits order characterised by a full occupancy of either of the three sublattices of the plaquettes of the hexagonal lattice, while for a low fugacity the gas is dilute and homogenous. These phases are separated by a second order phase transition for a critical value  $z_c$  of the fugacity, whose scaling limit is described by the  $c = 4/5$  minimal Potts model [72].

Focussing again on the strange correlator, the tensor from which the partition function is constructed has the following non-zero components:



$$\begin{array}{c} \tau \\ \tau \end{array} \begin{array}{c} \tau \\ \tau \end{array} = -1, \quad \begin{array}{c} \mathbb{1} \\ \tau \end{array} \begin{array}{c} \tau \\ \tau \end{array} = \phi^{5/6}. \quad (163)$$

With the above tensor at hand, it is then straightforward to derive that the partition function can be written as

$$Z = \sum_{n=0}^{N/3} z_c^n g(n, N), \quad (164)$$

where  $N$  denotes the total number of plaquettes and  $g(n, N)$  is a combinatorial factor equal to the number of ways to distribute  $n$  particles over  $N$  plaquettes subject to the aforementioned constraint [175]. Straightforward algebra reveals that  $z_c = \phi^5$ , which is exactly the critical fugacity of the hard hexagon model. This simple strange correlator thus immediately yields the critical fugacity, without having to resort to the vastly more advanced techniques employed by Baxter.

From the transfer matrix one can also extract the finite-size CFT spectra. Above, we have highlighted already the significance of the role of the tube algebra to obtain the topological superselection sectors of quantum doubles and string-net models. As detailed in refs. [136, 145, 176] the transfer matrix of the strange correlator can be supplemented with symmetry-twisted boundary conditions labelled by objects of the symmetry fusion category. These boundary conditions are topological in the sense that they preserve lattice translation symmetry up to a unitary transformation. In their presence, the tube algebra can be used to decompose the spectrum of the transfer matrix in topological sectors. For the case of Fibonacci, there are 4 sectors labelled  $0, \tau, \bar{\tau}, \tau\bar{\tau}$ . Their corresponding spectra are shown in ref. [145], where also the identification with the primaries of the Potts minimal model is explained.

In the previous section we argued that different tensor network representations of string-net ground states correspond to module categories over the input fusion category. Coincidentally, we did not have this freedom for the case study in this section. It turns out that this choice of module category specifies the *modular invariant* of the CFT on the torus as well as its partition function on higher genus surfaces [72, 168, 171]. The so-called *diagonal case* corresponds to choosing the regular module category. Changing this module category is then referred to as *orbifolding* or *gauging*, numerically studied in ref. [177].

There exists a vast literature on tensor network representations of critical partition functions using the strange correlator formalism. We refer to refs. [145, 177–185] for additional case

studies, some analytical results, and spectra on open boundary conditions and unoriented manifolds. Also, in recent work, systematic ways of constructing critical (integrable) strange correlators from purely topological data have been proposed [185, 186]. The appeal is clear: the strange correlator formalism provides a direct way of constructing critical lattice systems with exact categorical defects. The topological part (encoded in the string-net) is completely separated from the geometrical part (encoded in the overlap state  $|\Omega\rangle$ ), thereby making the anomalies in the theory very explicit. One of the central challenges is now to construct critical lattice models corresponding to new CFTs – the Haagerup critical lattice model [157, 181, 185] being the first possible example of such a theory.

### 5.5 Application 3: dualities

In the previous section we constructed two-dimensional classical partition functions with the MPO symmetries classified and constructed in section 5.2. In this final application we will construct (1+1)D Hamiltonians commuting with those symmetries, and study their dualities. In a sense, these models can thus be viewed as *edge Hamiltonians* of the string-nets of section 5.3.

For the case of on-site representations of groups, *Clebsch-Gordan coefficients* are the building blocks of symmetric tensors. Indeed, the *Wigner-Eckart theorem* dictates that symmetric tensors factorise as a product of Clebsch-Gordan coefficients and *reduced matrix elements*. The latter notably only depend on the irrep labels, while the coupling of basis states is fully captured by the Clebsch-Gordan coefficients [81, 187]. This result is not only central in the context of perturbation theory and spectroscopy in which they are often introduced, but also forms the cornerstone of symmetric tensor network algorithms [49–54]. Inspired by this perspective, we can deduce from eq. (160) for the more general case of an MPO symmetry  $\mathcal{C}$ , that generic local operators are constructed from the fusion tensors as follows:

$$h_{i,i+1}^{\mathcal{R}} = h(\alpha, \alpha', \beta, \beta', \gamma, i, i') \quad \begin{array}{c} \alpha' \quad \beta' \\ \diagdown \quad \diagup \\ \text{---} \text{---} \text{---} \\ \diagup \quad \diagdown \\ \alpha \quad \beta \end{array} . \quad (165)$$

Herein, the  $h$  are complex numbers which can be chosen freely and may depend on the indicated labels. Importantly, these coefficients are independent of the internal virtual indices of the tensors, and can thus be interpreted as a generalisation of reduced matrix elements. Commutation of these operators with the MPO symmetries follows from

$$\begin{array}{c} \alpha' \quad \beta' \\ \diagdown \quad \diagup \\ \text{---} \text{---} \text{---} \\ \diagup \quad \diagdown \\ \alpha \quad \beta \end{array} \quad \begin{array}{c} \alpha \\ \text{---} \text{---} \text{---} \\ \alpha' \quad \beta' \end{array} = \begin{array}{c} \alpha' \quad \beta' \\ \diagdown \quad \diagup \\ \text{---} \text{---} \text{---} \\ \diagup \quad \diagdown \\ \alpha \quad \beta \end{array} \quad \begin{array}{c} \alpha \\ \text{---} \text{---} \text{---} \\ \alpha' \quad \beta' \end{array} = \begin{array}{c} \alpha' \quad \beta' \\ \diagdown \quad \diagup \\ \text{---} \text{---} \text{---} \\ \diagup \quad \diagdown \\ \alpha \quad \beta \end{array} , \quad (166)$$

satisfied for all indicated simple objects of  $\mathcal{D}$  and  $\mathcal{C}$ . From eq. (165), a local symmetric Hamiltonian can then be constructed as  $H = \sum_i h_{i,i+1}^{\mathcal{R}}$ . Note that we restrict here to two-body operators, but symmetric operators acting on more sites can be obtained just as easily by taking products of eq. (165). The construction of local operators eq. (165) thus generalises the traditional Wigner-Eckart theorem to MPO symmetries [188].

So far, we have implicitly assumed infinite chains. Completely in line with section 5.4 we can also define these Hamiltonians in the presence of closed periodic and symmetry-twisted boundary conditions [136, 176, 189]. For the latter, an extra degree of freedom is introduced



on the bond between two sites, and the local Hamiltonian term across that bond is modified accordingly, we refer to ref. [189] for details. The importance of including symmetry-twisted boundary conditions will become clear below.

To illustrate the expression eq. (165), consider the spin 1/2 antiferromagnetic Heisenberg model encountered above in eq. (12). By invoking the (traditional) Wigner-Eckart theorem, the local Hamiltonian term  $\mathbf{S}_i \cdot \mathbf{S}_{i+1}$  can be constructed from the SU(2) Clebsch-Gordan coefficients. To rephrase this in the current framework, we need to choose  $\mathcal{D} = \text{Rep}(\text{SU}(2))$ , the category of representations of SU(2), with  $\mathcal{R} = \text{Vec}$ .<sup>16</sup> Spelled out explicitly:

$$\mathbf{S}_i \cdot \mathbf{S}_{i+1} = - \begin{array}{c} \text{2} \quad \text{2} \\ \diagdown \quad \diagup \\ \text{0} \\ \diagup \quad \diagdown \\ \text{2} \quad \text{2} \end{array} = \sum_{\substack{i,j \\ i',j'}} \begin{bmatrix} 2 & 2 & 0 \\ i' & j' & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 & 0 \\ i & j & 1 \end{bmatrix} |i', j'\rangle \langle i, j|, \quad (167)$$

up to an additive constant. Here, **2** and **0** denote the SU(2) doublet and singlet respectively, and  $\begin{bmatrix} 2 & 2 & 0 \\ \bullet & \bullet & 1 \end{bmatrix}$  stands for the (real) Clebsch-Gordan coefficients associated to

$$\begin{bmatrix} 2 & 2 & 0 \\ \bullet & \bullet & 1 \end{bmatrix} : \mathbf{2} \otimes \mathbf{2} \rightarrow \mathbf{0} : |i\rangle \otimes |j\rangle = \begin{bmatrix} 2 & 2 & 0 \\ i & j & 1 \end{bmatrix} |1\rangle, \quad (168)$$

where  $i, j, 1$  denote basis vectors of the respective underlying representation spaces. This Hamiltonian term thus essentially projects two neighbouring spin 1/2's on the singlet representation, and the spin 1 representation appearing in the tensor product  $\frac{1}{2} \otimes \frac{1}{2}$  is given zero weight by the chosen parameters  $h$ .

Let us consider the set of all local symmetric operators (165), as well as their finite products and linear combinations. This constitutes an algebra – sometimes referred to as *bond algebra* [139, 143, 158, 190, 191] – specified by the structure constants

$$h_A^{\mathcal{R}} h_B^{\mathcal{R}} = \sum_C f_{AB}^C({}^{\circ}F) h_C^{\mathcal{R}}, \quad (169)$$

defined for the basis of operators  $\{h_A^{\mathcal{R}}\}$ . Crucially, the *structure constants*  $f_{AB}^C({}^{\circ}F)$  depend on the associators  ${}^{\circ}F$  of the fusion category  $\mathcal{D}$  only, i.e. they are completely independent of the choice of module category  $\mathcal{R}$ . In particular, this observation implies that also the spectrum of the Hamiltonian is – up to degeneracies – independent of  $\mathcal{R}$ . For a fixed  $\mathcal{D}$  and choice of coefficients  $h$  in eq. (165), changing the module category  $\mathcal{R} \rightarrow \mathcal{R}'$  boils down to a *duality transformation*. As such, different module categories over  $\mathcal{D}$ , or equivalently the symmetry category  $\mathcal{C}$ , thus organise an entire web of dualities. Although this framework may appear abstract, most dualities of physical interest, including Kramers–Wannier duality, Kennedy–Tasaki duality, and the Jordan–Wigner transformation, naturally fit within it, as will be illustrated further below. Throughout, we will often refer to the (fictional) model of interest as the *original* model, and compare it to its *dual(s)* model(s). The dual model can look radically different from the original one in the following ways.

As mentioned above, the local symmetric operators eq. (165) and their MPO symmetries act on a Hilbert space which is generically constrained if  $\mathcal{R}$  has more than one simple object. As a consequence, the Hilbert space of one model is typically completely different from that of any of its dual models and their dimensions don't even have to match. This is closely related to

<sup>16</sup>Technically speaking  $\text{Rep}(\text{SU}(2))$  is a tensor category, rather than a fusion category which is required to have a finite number of simple objects. Readers inclined to more mathematical rigour may wish to apply the formalism to a finite subgroup of SU(2) instead.

the fact that dualities of the kind considered here can be realised by gauging (part of) the original MPO symmetry, so that the constrained Hilbert space arises as an *effective* Hilbert space where (part of) the imposed Gauss constraints are resolved [192–195]. Consequently, also the global symmetries of distinct dual models typically differ as they are encoded in  $\mathcal{D}_{\mathcal{R}}^*$  and  $\mathcal{D}_{\mathcal{R}'}^*$  respectively. In other words, given the initial symmetry and the duality transformation, the dual symmetry is fully determined. The existence of a duality between models implies also that every observable in one theory is equivalent to a *dual* observable in the dual theory. However, only the *symmetric*, local observables in one theory get mapped to dual local, symmetric operators in the dual theory, whereas charged operators, such as local order parameters, get mapped to non-local *string order parameters*.

One of the upshots of the tensors network approach to dualities is that we can build explicit *intertwiners* between dual models in terms of (non-invertible) matrix product operators [196]. At the level of the Hamiltonian terms they act according to:

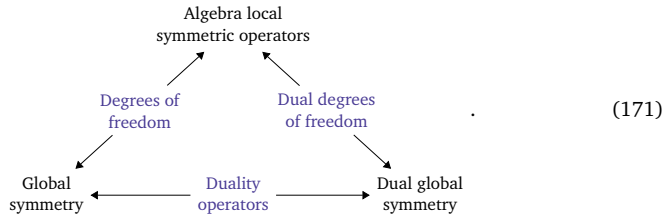
$$h_{i,i+1}^{\mathcal{R}'} \circ D = D \circ h_{i,i+1}^{\mathcal{R}}, \quad (170)$$

from which it follows that acting with  $D$  on an eigenstate of  $H^{\mathcal{R}}$  results in an eigenstate of  $H^{\mathcal{R}'}$  with exactly the same energy.

As demonstrated in ref. [196], such an intertwiner only depends on the categories  $\mathcal{D}$ ,  $\mathcal{R}$  and  $\mathcal{R}'$  and not on the specific model under scrutiny. In fact, these MPO intertwiners are precisely the MPO intertwiners mentioned in section 5.3 that transmute different PEPS representations of a  $\mathcal{D}$  string-net model associated to the module categories  $\mathcal{R}$ ,  $\mathcal{R}'$  into each other.

Another implication of eq. (170) is that given an intertwiner  $D$ , we can multiply  $D$  from the *right* with symmetry operators of the initial symmetry  $\mathcal{C} = \mathcal{D}_{\mathcal{R}}^*$  and from the *left* with the dual symmetry operators in  $\mathcal{D}_{\mathcal{R}'}^*$ . As such, the different intertwiners are themselves organised in an invertible bimodule category – that we will denote by  $\mathcal{M}$  – between the original and dual symmetry categories. The construction is consistent since  $\mathcal{D}_{\mathcal{R}'}^* = \mathcal{C}_{\mathcal{M}}^*$ .<sup>17</sup>

For a pair of dual models, we can collect these structures in a diagram as follows:



In this diagram, every vertex stands for a fusion category. The one on top –  $\mathcal{D}$  – encodes the bond algebra generated by local symmetric operators, whereas the other two –  $\mathcal{D}_{\mathcal{R}}^*$  and  $\mathcal{D}_{\mathcal{R}'}^*$  – encode the global symmetries. The connecting edges represent invertible bimodule categories. The two on the diagonals correspond to  $\mathcal{R}$  and  $\mathcal{R}'$ , encoding the degrees of freedom of either model. The one on the bottom –  $\mathcal{M}$  – encodes the MPO intertwiners.

Invertibility of  $\mathcal{R}$  and  $\mathcal{R}'$  guarantees in particular that the Drinfeld centres of the original symmetry  $\mathcal{D}_{\mathcal{R}}^*$  and the dual symmetry  $\mathcal{D}_{\mathcal{R}'}^*$  are equivalent. In this context, objects in the Drinfeld centre label *topological sectors* of the theory. These sectors correspond to a boundary condition – which breaks part of the symmetry – together with a symmetry charge of the surviving symmetry. By carefully studying the interplay of the duality operators with these sectors, the

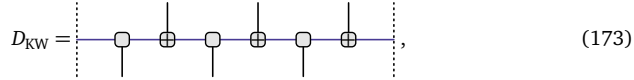
<sup>17</sup>This follows from the observation in footnote 14 that duality operators are labelled by  $\mathcal{D}$ -module functors organised in  $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$ .

non-invertible duality operators can be lifted to unitaries [189]. Under a duality transformation the topological sectors are generically permuted, as exemplified in the Kramers-Wannier duality.

**Example: Kramers-Wannier duality** Let us demonstrate how this formalism reproduces the famous Kramers-Wannier (KW) duality of the transverse field Ising model eq. (33) [197]. To this end, consider the Hamiltonian

$$H_{\text{TFIM}}^{\text{Vec}_{\mathbb{Z}_2}} = - \sum_{i=1}^L \sigma_i^Z \sigma_{i+1}^Z - \lambda \sum_{i=1}^L \sigma_i^X, \quad (172)$$

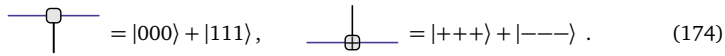
on periodic boundary conditions, i.e.  $L + 1 \equiv 1$ . The Hamiltonian clearly commutes with a  $\mathbb{Z}_2$  symmetry generated by  $\prod_i \sigma_i^X$ . The bond algebra corresponds to the fusion category  $\mathcal{D} = \text{Vec}_{\mathbb{Z}_2}$  and the module category is  $\mathcal{R} = \text{Vec}_{\mathbb{Z}_2}$ . The Kramers-Wannier duality can be thought of as gauging the  $\mathbb{Z}_2$  symmetry, or thus more abstractly, changing the module category  $\mathcal{R} = \text{Vec}_{\mathbb{Z}_2} \rightarrow \mathcal{R}' = \text{Vec}$ , corresponding to  $\mathcal{M} = \text{Vec}$ . The (unique) duality MPO can be depicted as:



$$D_{\text{KW}} = \text{---} \square \text{---} \bigcirc \text{---} \square \text{---} \bigcirc \text{---} \square \text{---} \bigcirc \text{---} \square \text{---} \bigcirc \text{---} \text{---}, \quad (173)$$

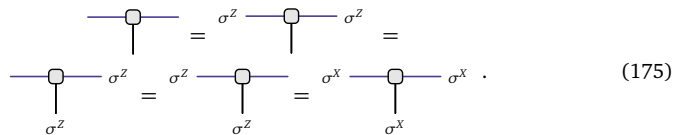
where we have explicitly indicated the periodic boundary conditions with the dotted lines, and here and below depict duality MPOs in blue. Note that the physical legs on either side are shifted by half a site. We emphasise again that this duality MPO is the same for every  $\mathbb{Z}_2$  symmetric model, and as such is not specific to the Ising model.

The MPO tensors are GHZ tensors in the  $\sigma^Z$  and  $\sigma^X$  eigenbasis and explicitly read



$$\text{---} \square \text{---} = |000\rangle + |111\rangle, \quad \text{---} \bigcirc \text{---} = |+++ \rangle + |-- \rangle. \quad (174)$$

All (global) properties of the intertwiner can be deduced from the symmetry properties of these local tensors. In particular, note that the first tensor is invariant under acting on any two legs with a  $\sigma^Z$ , and acting with  $\sigma^X$  on all its legs simultaneously:



$$\text{---} \square \text{---} = \sigma^Z \text{---} \square \text{---} \sigma^Z = \sigma^Z \text{---} \square \text{---} \sigma^Z = \sigma^X \text{---} \square \text{---} \sigma^X. \quad (175)$$

The second tensor has the same symmetries with  $\sigma^Z$  and  $\sigma^X$  interchanged. Hence, it follows that this MPO maps the local Hamiltonian terms according to

$$\sigma_i^Z \sigma_{i+1}^Z \rightarrow \sigma_{i+\frac{1}{2}}^Z, \quad \sigma_i^X \rightarrow \sigma_{i-\frac{1}{2}}^X \sigma_{i+\frac{1}{2}}^X, \quad (176)$$

so that the dual Hamiltonian reads

$$H_{\text{TFIM}}^{\text{Vec}} = - \sum_{i=1}^L \sigma_{i+\frac{1}{2}}^Z - \lambda \sum_{i=1}^L \sigma_{i-\frac{1}{2}}^X \sigma_{i+\frac{1}{2}}^X. \quad (177)$$

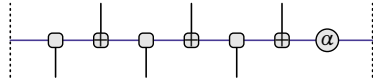
This Hamiltonian has a symmetry generated by  $\prod_i \sigma_{i+1/2}^Z$ , which corresponds to the symmetry category  $\text{Rep}(\mathbb{Z}_2) = (\text{Vec}_{\mathbb{Z}_2})_{\text{Vec}}^*$ , in accordance with the prediction of the framework. The

mapping eq. (176) also shows that the order parameter in the original model gets mapped to the disorder parameter in the dual and vice versa:

$$\sigma_i^Z \sigma_j^Z \rightarrow \prod_{k=i}^{j-1} \sigma_{k+\frac{1}{2}}^Z, \quad \prod_{k=i}^j \sigma_k^X \rightarrow \sigma_{i-\frac{1}{2}}^X \sigma_{j+\frac{1}{2}}^X. \quad (178)$$

The MPO eq. (173) has a null space which contains all  $\mathbb{Z}_2$  odd states:  $\prod_i \sigma_i^X |\Psi\rangle = -|\Psi\rangle$ . Similarly, its image consists of only states which are even under the dual symmetry. In order to access the non-trivial charge sectors in either model, we need to adapt the MPO to account for all symmetry-twisted boundary conditions. For the case at hand, this amounts to either periodic or anti-periodic boundary conditions. At the level of the Hamiltonian, changing the boundary condition boils down to flipping the sign of one of the nearest-neighbour interaction terms, conventionally the one between sites  $L$  and  $1$ .

After having specified the boundary condition and the charge sector of the original model, the topological sector of the dual model is specified by the mapping in table 1 [189, 198–200]. Equivalently, after specifying the boundary conditions in both models, the accessible charge sectors are fully determined. Remark in particular the permutation of topological sectors: in essence the boundary condition and charge sector are interchanged under KW duality. For each of these sectors there is a corresponding MPO intertwiner of the form


(179)

where  $\alpha$  is a placeholder for one of matrices  $\{\mathbb{1}, \sigma^X, \sigma^Y, \sigma^Z\}$  as indicated in table 1 as well.

| Boundary condition | Charge sector | Dual boundary condition | Dual charge sector | Boundary matrix |
|--------------------|---------------|-------------------------|--------------------|-----------------|
| P                  | +1            | P                       | +1                 | $\mathbb{1}$    |
| P                  | -1            | AP                      | +1                 | $\sigma^Z$      |
| AP                 | +1            | P                       | -1                 | $\sigma^X$      |
| AP                 | -1            | AP                      | -1                 | $\sigma^Y$      |

Table 1: Mapping of topological sectors under the Kramers-Wannier duality. Here, (A)P stands for (anti)periodic.

Note that the original model eq. (172) is in the symmetry-broken phase if  $0 \leq \lambda < 1$  and in the symmetric phase if  $\lambda > 1$ ; under Kramers-Wannier duality these phases are interchanged in the dual model. Turning the logic around, for any value of the magnetic field strength  $\lambda$ , there is a model, either eq. (172) or eq. (177), that is (fully) symmetry-broken. We could conjecture that this observation holds generally: every phase can be mapped to a completely symmetry-broken phase – with respect to the dual symmetry – by means of a suitable duality mapping. Further evidence for this speculation is provided by the example of the Kennedy-Tasaki duality.

**Example: Kennedy-Tasaki duality** The *Kennedy-Tasaki* (KT) duality was originally devised to elucidate the ground state physics of the spin 1 antiferromagnetic Heisenberg model, which is now understood to exhibit non-trivial  $\mathbb{Z}_2 \times \mathbb{Z}_2$  SPT order. The duality was introduced as a non-unitary mapping on open chains between the Heisenberg model and a model that spontaneously breaks  $\mathbb{Z}_2 \times \mathbb{Z}_2$  symmetry [201, 202]. Recently, it has been reinterpreted in terms of twisted gauging of the  $\mathbb{Z}_2 \times \mathbb{Z}_2$  symmetry [196, 200, 203–205]. Equivalently, this is described

as the change in module category  $\mathcal{R} = \text{Vec} \rightarrow \mathcal{R}' = \text{Vec}^\psi$  over the input  $\text{Vec}_{\mathbb{Z}_2 \times \mathbb{Z}_2}$ , where  $\psi$  is a non-trivial 2-cocycle of  $\mathbb{Z}_2 \times \mathbb{Z}_2$ . As such, it is more precise to think about the symmetry as being labelled by characters in  $\text{Rep}(\mathbb{Z}_2 \times \mathbb{Z}_2)$ , which is equivalent to  $\text{Vec}_{\mathbb{Z}_2 \times \mathbb{Z}_2}$  as a fusion category. It can be shown that  $\mathcal{M} = \text{Rep}^\psi(\mathbb{Z}_2 \times \mathbb{Z}_2)$  so that given a  $\psi$ -projective representation  $\pi$  of  $\mathbb{Z}_2 \times \mathbb{Z}_2$ , the MPO is constructed from the tensor with components:

$$\begin{array}{c} h \\ | \\ \alpha - \text{---} \square \text{---} \beta \\ | \\ g \end{array} = \delta_{g,h} \pi(g)_{\alpha\beta}. \quad (180)$$

For long it was unclear how to generalise KT from open to closed chains. The MPO framework provides a crisp answer: just take the trace of the MPO to define it on periodic boundary conditions. Also, the original KT transformation on the open chain is obtained by choosing suitable boundary conditions for the MPO eq. (180). Note that the MPO tensors are diagonal in their physical leg, so that KT doesn't change the charge sector, but only permutes the boundary conditions.

Instead of applying it to the Heisenberg model, we focus here on the AKLT state eq. (44), also a non-trivial  $\mathbb{Z}_2 \times \mathbb{Z}_2$  SPT. Recall that the AKLT state can be represented as an MPS with matrices  $A^i = \sigma^i / \sqrt{3}$ . If we take for concreteness the projective representation defined via  $\pi((a, b)) = (\sigma^X)^a (\sigma^Z)^b$ , we obtain that acting with the duality on the AKLT state gives an MPS with matrices  $A^i = \sigma^i \otimes \sigma^i$ . This MPS is clearly reducible as all MPS matrices commute. Indeed, by making use the gauge transformation

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}, \quad (181)$$

this MPS can be reduced to a sum of four product states:

$$\begin{aligned} U(\sigma^X \otimes \sigma^X)U^\dagger &= \text{diag}(1, 1, -1, -1), \\ U(\sigma^Y \otimes \sigma^Y)U^\dagger &= \text{diag}(-1, 1, -1, 1), \\ U(\sigma^Z \otimes \sigma^Z)U^\dagger &= \text{diag}(1, -1, -1, 1). \end{aligned} \quad (182)$$

This illustrates that the duality operator is fully responsible for the twofold degenerate entanglement spectrum of the AKLT state. Turning things around, by acting with the duality on the AKLT state – or any other  $\mathbb{Z}_2 \times \mathbb{Z}_2$  SPT for that matter – we can get rid of the degeneracies of the entanglement spectra. This observation can be used to our advantage as argued below.

In analogy with the example of KW, one can now define four symmetry-twisted boundary conditions, one for each element of  $\mathbb{Z}_2 \times \mathbb{Z}_2$ . We relegate the reader to refs. [200, 205] for a detailed discussion. As shown there, the charge sectors are mapped identically as anticipated, but depending on the charge sector the boundary conditions are permuted into each other.

The observation that there seemingly always exists a unique duality transformation that maps a model to a completely symmetry-broken dual model suggests a one-to-one correspondence between module categories and categorically symmetric gapped phases of matter. And indeed, this is exactly the content of the *generalised Landau paradigm*. Provided the symmetry  $\mathcal{C}$ , the gapped phases of matter are in one-to-one correspondence with  $\mathcal{C}$ -module categories  $\mathcal{P}$  [53, 77, 135]. This module category  $\mathcal{P}$  encodes both the distinct ground states or injective blocks of the

ground state MPS – corresponding to its simple objects – as well as the action of  $\mathcal{C}$  on the ground state subspace – via the module action. By making use of the parent Hamiltonian construction, the generalised Landau paradigm was made explicit in the MPS framework in ref. [77]. In this context, the regular module category  $\mathcal{P} = \mathcal{C}$  represents the completely symmetry-broken phase where the ground states are in bijection with the symmetry operators. As anticipated already in lecture 3, this generalises the case of a group symmetry where both the module categories and phases of matter are characterised by pairs  $(H \subseteq G, \psi)$ ,  $[\psi] \in H^2(H, U(1))$ .

Combining the generalised Landau paradigm and the duality framework has far-reaching implications. For one, note that the symmetry-broken phase is the only gapped phase that is guaranteed to exist.<sup>18</sup> In fact, the existence of the completely symmetric phase is equivalent to the symmetry being anomaly-free and thus fully gaugeable. The corresponding duality operator that interchanges the two then acts by effectively averaging over the symmetry-broken ground states [53, 195]. It is fascinating that, viewed this way, any continuous phase transition can be interpreted as a transition between a fully symmetry-broken phase on one side and a (partially) symmetric phase on the other.

Notably, the symmetry-broken dual model minimises the ground state entanglement entropy. This can be exploited in the DMRG algorithm, and for this reason the symmetry-broken model can be referred to as the *optimal* (dual) model. Indeed, as demonstrated in section 3.3 entanglement spectra can exhibit degeneracies, due to the fact that the entanglement degrees of freedom transform in multiplets of the (projective) virtual symmetry. If this is the case, there is redundancy in the MPS description of the ground state: computational resources are expended to fine-tune some variational parameters to reproduce these spectra. It is therefore beneficial to simulate the optimal dual model that maximally breaks the (dual) symmetry so that no degeneracies remain. In turn, the ground state of the original model is then obtained by acting with the duality operator which reinstates the degeneracies in the entanglement spectrum. A case study was carried out in ref. [53], and the benefit of this approach demonstrated.

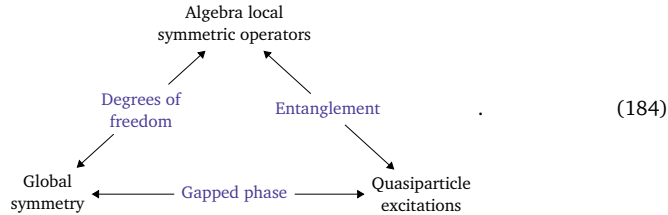
Having identified the optimal dual model additionally provides a unifying perspective on the elementary excitations in any gapped phase  $\mathcal{P}$ . We now reserve the notation  $\mathcal{Q}$  to denote the right module category over  $\mathcal{D}$  corresponding to the optimal model. As argued in section 2.5, the elementary excitations in this optimal and thus symmetry-broken phase  $\mathcal{D}_{\mathcal{Q}}^*$  are topological domain wall excitations. Recall that in the infinite-chain limit such excitations are created by acting on one half of the chain with a symmetry MPO in  $\mathcal{D}_{\mathcal{Q}}^*$ , while its variational degrees of freedom are encoded in a tensor at its endpoint. The quasiparticle excitations in that case are organised in a fusion category,  $\mathcal{E}$ , which coincides with the symmetry category, i.e.  $\mathcal{E} = \mathcal{D}_{\mathcal{Q}}^*$ . Provided this optimal model, its domain wall excitations can in turn be mapped back to the excitations of the original model, which are typically of a completely different nature, such as the magnon excitations in the Heisenberg model (see section 4.1). This situation is depicted as follows:

Herein, the blue MPO represents a duality operator – coincidentally labelled by an object in  $\mathcal{P}$  – between the optimal model corresponding to  $\mathcal{Q}$  and the model of interest  $\mathcal{R}$  as indicated by the coupons. A domain wall excitation is then depicted on the bottom, and parametrised by the tensor  $B$ , similar as in section 2.5. Note that this tensor carries a charge  $e \in \mathcal{E}$  that terminates on the duality operator. This picture also reveals that the labels of  $\mathcal{Q}$  capture the *entanglement degrees of freedom* of the ground state of interest.

<sup>18</sup>For some symmetry categories such as  $\mathcal{C} = \text{Fib}$  it is even the only gapped phase.

In this way, the domain wall excitations of the optimal dual model provide an invariant characterisation for excitations of all of its dual models. And the fundamental excitations in the any model can all be parametrised according to eq. (183).

Combined, these structures related to the quasiparticle excitations and the phase of a model are encapsulated in the following diagram, reproduced from ref. [53]:



## 5.6 Synopsis of lecture V

Lecture V can be seen as a symbiosis of the previous lectures: we encountered the fundamental theorem, MPO algebras, category theory, symmetry-breaking and string-nets models. Viewed from the point of view of tensor networks, topologically ordered systems exhibit symmetry breaking in their entanglement degrees of freedom. Those symmetries are described by MPO algebras which form representations of *fusion categories*.

On the one hand, those symmetries describe the long-range entanglement structure in string-nets, which can themselves readily be expressed in terms of tensor networks. As an application, we introduced the concept of the *strange correlator*. This prescription relates lattice representations of topological field theories to conformal field theories via an explicit holographic construction consisting of an overlap of a string-net with a featureless short-range entangled state. The MPO symmetries of the string-net then become the topological defects in the CFT, thereby providing an explicit lattice realisation of the Fröhlich-Fuchs-Runkel-Schweigert characterisation of topological defects in CFTs.

The same MPO algebras also describe the symmetries of the effective Hamiltonians that govern the physics of the edge modes of those 2+1D systems. In a fascinating twist, the different ways in which those symmetries can be represented – technically described in terms of bimodule categories – lead to a full comprehensive picture for defining dualities between all possible edge Hamiltonians in terms of a set of intertwining MPOs. One of the most fascinating physical consequences of this construction is the fact that every 1D phase of matter is dual to a phase that is completely symmetry broken (with respect to the dual symmetry). Notably, this is irrespective of whether this one-dimensional system is defined on a tensor product structure with on-site symmetries or on a constrained Hilbert space with categorical non-invertible symmetries. Among other things, this means that every possible second order phase transition for such 1D systems is equivalent to one to a completely symmetry broken phase.

In these lectures, we have not touched upon the higher-dimensional versions of MPO symmetries and MPO intertwiners. Interestingly, in higher dimensions, symmetries can act on freely-deformable submanifolds of various dimensionality of the lattice [134, 138, 140, 206–210]. These *higher-form symmetries* are naturally represented in terms of tensor networks as well. In the case of (2+1)D, surface operators are implemented in terms of projected entangled-pair operators (PEPOs), while the 1-form symmetries, living on interfaces between PEPOs, are naturally described by MPOs. As a whole, this symmetry structure and its recoupling theory is encapsulated in a *fusion 2-category* [211, 212]. Similar to the one-dimensional case, distinct ways of representing a fusion 2-categorical symmetry on the lattice are classified by *module 2-categories* [213, 214].

As pioneered in ref. [192], also dualities in 2+1D lattice systems can be implemented on the lattice by PEPOs. Based on this tensor network approach, significant progress in the understanding of higher-categorical symmetries and their dualities has been made [205, 214–219]. Despite these recent advances, plenty of interesting open research directions are still to be explored.

Further research directions include the incorporation of continuous symmetries within this categorical framework. As already touched upon in ref. [53], *Schur-Weyl duality* is completely compatible with the MPO framework for describing dualities. It maps theories with continuous  $SU(N)$  symmetries to theories with discrete  $\text{Rep}(SU(N))$  symmetries. Since the Mermin-Wagner theorem no longer applies for the latter, the duality map can be used to circumvent it. As such, the Schur-Weyl duality maps interchange the symmetric and symmetry-broken phases, making the DMRG simulation of this dual theory much more efficient.

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## A Group cohomology

In this section we explain how cohomology groups  $H^n(G, U(1)^\beta)$  for any finite group  $G$  with a group action  $\beta$  on  $U(1)$  can be computed using simple linear algebra methods developed in [80, 220]. As a side product, we will also obtain representative cocycles for every cohomology class.

Let us first introduce our notation and relevant definitions. We denote the set of all functions  $G^n \rightarrow U(1)$  by  $C^n(G, U(1))$ , these functions are called  $n$ -cochains. The coboundary maps  $\partial^{n+1} : C^n(G, U(1)) \rightarrow C^{n+1}(G, U(1))$  are then defined via:

$$(\partial^{n+1}\omega)(g_1, g_2, \dots, g_{n+1}) := \beta_{g_1}(\omega(g_2, g_3, \dots, g_{n+1})) \times \prod_{i=1}^n [\omega(g_1, g_2, \dots, g_{i-1}, g_i g_{i+1}, \dots, g_{n+1})]^{(-1)^i} \times [\omega(g_1, g_2, \dots, g_n)]^{(-1)^{n+1}} = 0. \quad (\text{A.1})$$

Here,  $\beta$  denotes the group action of  $G$  on (the  $G$ -module)  $U(1)$ . In the case of a symmetry group containing *antiunitary* elements, such as time-reversal and spatial reflection encountered in the main text,  $\beta_g(\bullet)$  acts via complex conjugation when  $g$  is antiunitary and as the identity otherwise. It is easy to check that these maps are nilpotent, i.e.  $\partial^{n+1} \circ \partial^n = 1$ . We then define the abelian groups of  $n$ -cocycles and  $n$ -coboundaries as respectively:

$$Z^n(G, U(1)^\beta) := \ker(\partial^{n+1}), \quad (\text{A.2})$$

$$B^n(G, U(1)^\beta) := \text{im}(\partial^n). \quad (\text{A.3})$$

From the nilpotency of the coboundary maps, it follows that  $B^n(G, U(1)^\beta)$  is a subgroup of  $Z^n(G, U(1)^\beta)$ , so that we can define the  $n$ -th cohomology group of  $G$  as the quotient

$$H^n(G, U(1)^\beta) = Z^n(G, U(1)^\beta) / B^n(G, U(1)^\beta). \quad (\text{A.4})$$

The key insight in computing  $H^n(G, U(1)^\beta)$  and finding representative cocycles is the following. By taking the logarithm of both sides of the cocycle condition  $\partial^{n+1}\omega = 1$ , and collecting the unknown phases  $\omega(g_1, \dots, g_n)$  in a vector  $\omega$ , the condition can be written as a linear system that has to be satisfied modulo  $2\pi$ :

$$\Omega\omega = \mathbf{0}, \quad \text{mod } 2\pi. \quad (\text{A.5})$$

Here, the coefficient matrix  $\Omega$  originates from the coboundary map  $\partial^{n+1}$  and contains only integers. The dimension of  $\omega$  is  $|G|^n$  and  $\Omega$  is  $|G|^{n+1} \times |G|^n$ -dimensional. Since  $\Omega$  only contains integers, we can write  $\Omega$  in its *Smith normal form*:

$$\Omega = P\Lambda R. \quad (\text{A.6})$$

In this decomposition,  $P$  and  $R$  are respectively  $|G|^{n+1} \times |G|^{n+1}$  and  $|G|^n \times |G|^n$  matrices that only contain integers and have determinant one (and thus have integer-valued inverses).  $\Lambda$  also only contains integers, is  $|G|^{n+1} \times |G|^n$ -dimensional and is of the form

$$\Lambda = \left( \begin{array}{c|c} \text{diag}(d_1, d_2, \dots, d_r) & 0_{r, n-r} \\ \hline 0_{n+1-r, r} & 0_{n+1-r, n-r} \end{array} \right), \quad (\text{A.7})$$

in which the non-zero elements  $d_1, \dots, d_r$  along the diagonal, some of which might be one, are in increasing order,  $d_1 \leq d_2 \leq \dots$ , and every element is a divisor of the next,  $d_i | d_{i+1}$ . The Smith

normal form  $\Lambda$  is unique. Inserting this decomposition in the system of equations (A.5) gives rise to the solution

$$\omega = 2\pi R^{-1} \Lambda^+ \nu. \quad (\text{A.8})$$

In this equation  $\Lambda^+$  denotes the (unique) Moore-Penrose pseudoinverse of  $\Lambda$  that satisfies  $\Lambda \Lambda^+ \Lambda = \Lambda$  and which is found to be

$$\Lambda^+ = \left( \begin{array}{c|c} \text{diag}(d_1^{-1}, d_2^{-1}, \dots, d_r^{-1}) & 0_{r, n-r} \\ \hline 0_{n+1-r, r} & 0_{n+1-r, n-r} \end{array} \right). \quad (\text{A.9})$$

The vector  $\nu$  contains arbitrary integers. Writing the solution (A.8) in components yields

$$\omega_i = \sum_{j=1}^r 2\pi (R^{-1})_{ij} \frac{\nu_j}{d_j}. \quad (\text{A.10})$$

Because  $\nu$  can be chosen freely, one can choose subsequently  $\nu_i = \delta_{1,i}, \delta_{2,i}, \dots, \delta_{r,i}$  to obtain a basis of the solution space that can be written as

$$(\omega_j)_i = \frac{2\pi}{d_j} (R^{-1})_{ij}, \quad \forall j \in \{1, \dots, r\}. \quad (\text{A.11})$$

Hence, the  $n$ -cocycles  $\omega$  are found to be the columns of  $R^{-1}$ . Since  $R$  is full rank, all the solutions  $\omega$  are linearly independent. In particular, the non-trivial cocycles (below) can not be related by a coboundary,  $\omega' = \omega + \Omega' \varphi$ , where  $\Omega'$  denotes the coboundary map of degree  $n-1$  and  $\varphi$  is a  $|G|$ -dimensional vector containing arbitrary real numbers. To classify all possible solutions, we now consider the diagonal entries of  $\Lambda$ .

From (A.10) and the fact that  $R^{-1}$  contains only integers, it follows that for every diagonal entry  $d_i = 1$ , a trivial solution  $\omega_i = 0 \pmod{2\pi}$  is obtained. The non-trivial solutions are those that correspond to entries  $d_i > 1$ . From (A.10) and the fact that the solution space is  $\mathbb{Z}$ -linear, it follows that the cocycle  $\omega_j$  corresponding to some  $d_j$  generates a cyclic group. Now note that not all elements of  $\omega_j$  can be divisible by  $d_j$  or any of its prime factors as this would be in contradiction with the fact that  $R^{-1}$  has determinant one. Hence, the cyclic group generated by  $\omega_j$  is  $\mathbb{Z}_{d_j}$ . Finally, the zero entries of  $\Lambda$  can also be discarded in the cohomology as these correspond to trivial solutions of the cocycle equation that can be multiplied by arbitrary phases and thus correspond to coboundaries.

In conclusion following picture arises. Given some group  $G$  one can write down the matrix representation of the coboundary map  $\Omega$  that can be brought in Smith normal form  $P\Lambda R$ . The diagonal entries of  $\Lambda$ ,  $d_1, d_2, \dots, d_r$ , determine the second cohomology group which is then of the form  $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \dots \times \mathbb{Z}_{d_r}$ , with the understanding that  $\mathbb{Z}_1 = \{e\}$  denotes the trivial group and that all zero diagonal entries can be discarded. The non-trivial  $n$ -cocycles in some arbitrary gauge correspond then to the columns of  $R^{-1}$ , weighted by the appropriate factor  $2\pi/d_j$  as per eq. (A.10).

## B Category theory

In this appendix we summarise relevant notions of fusion and (bi)module categories to the extent that they are used in the main text. We refer the reader to the standard references [118, 221, 222] and reviews [111, 121, 223] for a more detailed and technically precise exposition.

### B.1 Fusion categories

Succinctly, a (unitary) *fusion category*  $\mathcal{D}$  consists of

1. A finite set of *simple objects* (topological defects)  $a, b, c, \dots$
2. Fusion rules  $a \otimes b = \bigoplus_c N_{ab}^c c$ , together with a unit for the tensor product denoted  $1$ .
3. Unitary maps called *F-symbols* or *associators*  $F : (\bullet \otimes \bullet) \otimes \bullet \xrightarrow{\sim} \bullet \otimes (\bullet \otimes \bullet)$ .

The F-symbols adhere to a consistency condition, known as the *pentagon equation*. Spelled out, the pentagon equation boils down to linear multivariate system of equations which schematically are of the form

$$FF = \sum FFF, \quad (\text{B.1})$$

and written out in components read

$$\sum_o (F_e^{fcd})_{g,lm}^{h,no} (F_e^{abh})_{f,ko}^{i,pq} = \sum_{j,rst} (F_g^{abc})_{f,kl}^{j,rs} (F_e^{ajd})_{g,sm}^{i,tq} (F_i^{bcd})_{j,rt}^{h,np}. \quad (\text{B.2})$$

Notably, a fusion category is not necessarily endowed with a *braiding*, i.e. unitary maps  $R_{ab} : a \otimes b \xrightarrow{\sim} b \otimes a$ . However, for every fusion category one can construct its *Drinfeld centre*  $\mathcal{Z}(\mathcal{D})$ , which is a *modular* fusion category, meaning in particular that it is endowed with a *non-degenerate* braiding. The centre can be physically realised in the string-net models of section 5.3 as the anyonic excitations of the model. Their exchange statistics are encoded in the braiding of  $\mathcal{Z}(\mathcal{D})$ .

**(Anomalous) Finite group symmetry** For any *finite* group  $G$ , the fusion category  $\text{Vec}_G$  is the category of finite-dimensional  $G$ -graded vector spaces. In essence, the simple objects of  $\text{Vec}_G$  correspond to the group elements of  $G$ , and the fusion rules are dictated by the group multiplication. The *F-symbols* are all trivial. Whenever  $G$  possesses a non-trivial third cohomology group (see section A), any 3-cocycle  $\omega$  can be used to construct  $\text{Vec}_G^\omega$ . This fusion category is constructed analogous to  $\text{Vec}_G$  where now the non-zero components of the *F-symbols* evaluate to  $\omega$ . This cocycle encodes the (*t Hooft*) *anomaly*, i.e. an obstruction to gauge to symmetry, or – equivalently – an obstruction for the symmetry to admit a trivial symmetric ground state as encountered in the main text in section 3.3.

In this sense, a fusion category thus encodes a symmetry structure – symmetry operators, their multiplication rules and junctions – as well as their anomaly – encoded in the *F-symbol*.

**Rep( $G$ ) symmetry** The category  $\text{Rep}(G)$ , for any finite group  $G$ , encodes as objects the representations of  $G$ . The irreducible representations then constitute the *simple* objects. The fusion of those objects is then given by the usual tensor product of group representations. It can then be deduced that the *F-symbols* of  $\text{Rep}(G)$  are given by the 6j-symbols.

**Fibonacci symmetry** The Fibonacci fusion category has two simple objects  $\{1, \tau\}$  whose only non-trivial fusion rule reads  $\tau \otimes \tau = 1 \oplus \tau$ . The F-symbols are spelled out in numerous places such as ref. [223]. The name *Fibonacci* refers to the fact that the multiplicity of  $\tau$  in subsequent tensor powers of  $\tau$  generates the Fibonacci series. It can be shown that also the Fibonacci fusion category admits no local symmetric Hamiltonian with trivial symmetric ground state [77].

## B.2 Module categories

Let  $\mathcal{C}$  be a fusion category. A (unitary left)  $\mathcal{C}$ -module category  $\mathcal{M}$  consists of

1. A finite set of simple objects  $A, B, C, \dots$
2. An action  $\triangleright$  of  $\mathcal{C}$  on  $\mathcal{M}$ :  $a \triangleright A = \bigoplus_B N_{aA}^B B$ , for simple objects  $B$  in  $\mathcal{M}$ .
3. Unitary module  ${}^\triangleright F$ -symbols or (left) module associators  ${}^\triangleright F : (\bullet \otimes \bullet) \triangleright \bullet \xrightarrow{\sim} \bullet \triangleright (\bullet \otimes \bullet)$ .

These  ${}^\triangleright F$ -symbols satisfy in turn a consistency equation involving both the  $F$ -symbols of  $\mathcal{C}$  and the module  ${}^\triangleright F$ -symbols. Schematically:

$${}^\triangleright F {}^\triangleright F = \sum F {}^\triangleright F {}^\triangleright F. \quad (\text{B.3})$$

**Module categories of  $\text{Vec}_G$**  Indecomposable module categories of  $\text{Vec}_G$  are labelled by pairs  $(H, \psi)$ , where  $H \subseteq G$  is a subgroup (up to conjugacy) and  $\psi$  is a 2-cocycle of  $H$ . In this case, the simple objects are labelled by cosets  $r_1 H, r_2 H, \dots \in G/H$ , and the action of  $\text{Vec}_G$  on them is defined as  $g \triangleright rH := (gr)H$ . The module associator  ${}^\triangleright F$  can be constructed from  $\psi$  as detailed in e.g. [224].

**Module categories of  $\text{Rep}(G)$**  Also module categories over  $\text{Rep}(G)$  are classified by tuples  $(H, \psi)$  due to the Morita equivalence of  $\text{Rep}(G)$  and  $\text{Vec}_G$ . The corresponding module category is  $\text{Rep}^\psi(H)$ , the category of  $\psi$ -projective representations of  $H$ . The action of  $\text{Rep}(G)$  on  $\text{Rep}^\psi(H)$  is given via restriction of  $G$ -representations to the subgroup  $H$  and subsequently taking the tensor product or (projective)  $H$ -representations.

## B.3 Bimodule categories

Given a pair of fusion categories  $\mathcal{C}, \mathcal{D}$ , a  $(\mathcal{C}, \mathcal{D})$ -bimodule category  $\mathcal{M}$  is a category which is both a left  $\mathcal{C}$ - and right  $\mathcal{D}$ -module category, together with additional unitary bimodule  ${}^\triangleright F$ -symbols or bimodule associators  ${}^\triangleright F : (\bullet \triangleright \bullet) \triangleleft \bullet \xrightarrow{\sim} \bullet \triangleright (\bullet \triangleleft \bullet)$ .

$\text{Vec} \text{ Vec}$  can be interpreted as a  $(\text{Vec}_G, \text{Rep}(G))$ -bimodule category. In that case, the components of the  ${}^\triangleright F$ -symbols evaluate to the matrix elements of  $G$ -representations.

Throughout this manuscript we exclusively deal with invertible  $(\mathcal{C}, \mathcal{D})$ -bimodule categories. In particular, this guarantees that  $\mathcal{C}$  and  $\mathcal{D}$  are Morita equivalent, or – equivalently – that their Drinfeld centres  $\mathcal{Z}(\mathcal{C})$  and  $\mathcal{Z}(\mathcal{D})$  are equivalent modular fusion categories [225]. In ref. [188], invertibility was shown to be equivalent to the notion of MPO injectivity, and a characterisation in terms of the bimodule associators derived.

Given a fusion category  $\mathcal{C}$  and  $\mathcal{C}$ -module category  $\mathcal{M}$ , there is a unique fusion category, the Morita dual  $\mathcal{C}_{\mathcal{M}}^*$ , that renders  $\mathcal{M}$  an invertible  $(\mathcal{C}, \mathcal{C}_{\mathcal{M}}^*)$ -bimodule category. For example:  $(\text{Vec}_G)^*_{\text{Vec}} = \text{Rep}(G)$ . We refer to the relevant literature for a precise definition of the Morita dual fusion category.

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# One-dimensional symmetric phases protected by frieze symmetries

## 6.1 Synopsis

In section 2.4 we reviewed the classification of one-dimensional SPT phases protected by *internal* symmetries acting in an on-site manner. The present project extends this classification to include quasi-one-dimensional *spatial symmetries*. These are encoded in the so-called frieze groups, of which there are seven. We closely follow an approach analogous to that of section 2.4 and study the action of the symmetry on an injective MPS. We find that these SPT phases are classified by the second cohomology group of the protecting symmetry group over  $U(1)$ , with the additional feature that *orientation-reversing* group elements — encoding spatial reflection, for instance — act on  $U(1)$  via complex conjugation. Additionally, we provide MPS normal forms for each of these phases and explicit renormalisation group fixed-point MPS representations directly in terms of representative cocycles. We also review the SPT classification in the presence of time-reversal symmetry. Finally, we discuss an algorithm that computes cohomology groups of any order and corresponding representative cocycles, including in the presence of non-trivial group action.

## 6.2 Reference

*One-dimensional symmetric phases protected by frieze symmetries*

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## 6.3 Contributions of the author

The project underlying this paper grew out of the Master's thesis of Nicolas Dewolf. Therein the classification of SPT phases protected by frieze symmetries was initiated and the framework established. I completed this classification and their representative cocycles, in full generality, and co-wrote most of the manuscript.

## One-dimensional symmetric phases protected by frieze symmetries

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We make a systematic study of symmetry-protected topological gapped phases of quantum spin chains in the presence of the frieze space groups in one dimension using matrix product states. Here, the spatial symmetries of the one-dimensional lattice are considered together with an additional “vertical reflection,” which we take to be an on-site  $\mathbb{Z}_2$  symmetry. We identify seventeen distinct non-trivial phases, define canonical forms, and compare the topological indices obtained from the MPS analysis with the group cohomological predictions. We furthermore construct explicit renormalization group fixed-point wave functions for symmetry-protected topological phases with global on-site symmetries, possibly combined with time reversal and parity symmetry. En route, we demonstrate how group cohomology can be computed using the Smith normal form.

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### I. INTRODUCTION

Even though there is no intrinsic topological order in gapped one-dimensional quantum spin chains, the phase diagram becomes nontrivial when symmetry constraints are taken into account [1–9]. This gives rise to the well-known paradigm of *symmetry-protected* topological (SPT) order. The lack of topological order in 1D can be understood as follows. Starting from the ground state of a gapped local Hamiltonian, subsequent renormalization group (RG) coarse graining steps do not alter the phase of the system [10]. After sufficiently many steps, the number of which is independent of system size, the state flows towards an RG fixed point exhibiting a valence bond structure [5,9,10]. A tensor product of unitaries on the state then turns this state in a trivial product state, ultimately proving that the state we started from is adiabatically connected to a product state with no topological order. This procedure can be made explicit by writing the state as a matrix product state (MPS) [11,12]. In this formalism, one RG step is equivalent to blocking two sites and acting with an isometry on the blocked site that maximally removes local entanglement inside the block while retaining the entanglement with the rest of the system [10]. When symmetries are taken into account, the RG flow should not break the symmetry. The picture that arises is that the phase diagram, which in the absence of symmetries is simply connected, falls apart in distinct classes that cannot be connected by adiabatic transformations due to topological obstructions.

Chen *et al.* showed that the topological obstructions that prohibit connecting different such SPT phases originate from the fact that physical symmetries can be implemented by *projective* representations of the symmetry group acting on the entanglement degrees of freedom [5–7]. This crucial insight

led Chen *et al.* to a classification of SPT phases in terms of group cohomology [13]. More specifically, the SPT classification corresponds to the second cohomology group  $H^2_\beta(\mathbf{G}, \mathbf{U}_1)$ , where  $\mathbf{G}$  denotes the symmetry group and can contain a global on-site symmetry subgroup, time reversal, parity, or combinations thereof. Here,  $\beta$  denotes the nontrivial action of  $\mathbf{G}$  on the  $\mathbf{U}_1$  module in case of time reversal or parity symmetry.

A folklore example is one where the global on-site symmetry group is  $\mathbb{Z}_2 \times \mathbb{Z}_2$ . Since the second cohomology group of  $\mathbb{Z}_2 \times \mathbb{Z}_2$  is  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbf{U}_1) = \mathbb{Z}_2$ ,  $\mathbb{Z}_2 \times \mathbb{Z}_2$  can protect one nontrivial symmetry-protected phase known as the Haldane phase [2,3,14,15]. In translationally invariant systems, the classification is refined to  $H^1_\alpha(\mathbf{G}, \mathbf{U}_1) \times H^2_\beta(\mathbf{G}, \mathbf{U}_1)$ , where  $H^1$  denotes the first cohomology group [5].

In this paper, we demonstrate that SPT phases can also be protected by quasi-one-dimensional lattice symmetries. The symmetry groups we consider are the seven so-called *frieze groups* [16]. These are defined as being the infinite discrete subgroups of the isometries of a strip,  $\text{Isom}([0, 1] \times \mathbb{R})$ . Apart from translations, the generators of the frieze groups are reflections in the horizontal or vertical direction,  $\pi$  rotations (equivalent to the composition of a horizontal and vertical reflection) and glide reflections. The seven distinct frieze groups these generators give rise to are denoted by  $F_0$  (only translation),  $F_V$  (translation + vertical reflection),  $F_H$  (translation + horizontal reflection),  $F_R$  (translation +  $\pi$ -rotation),  $F_G$  (translation + glide reflection),  $F_{RG}$  (translation +  $\pi$ -rotation + glide reflection), and  $F_{VH}$  (translation + two reflections). In case of a glide reflection, acting with this glide reflection twice is equivalent to the action of the translation generator.

We derive the SPT classification corresponding to these symmetries by imposing the symmetry on a general injective MPS and identifying topologically distinct ways in which this symmetry can be implemented [17]. Here, any on-site  $\mathbb{Z}_2$  symmetry of the system could play the role of the vertical reflection in the frieze groups. However, we preserve the

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geometrical interpretation and represent the vertical reflection as a swap of the two physical degrees of freedom associated with every site (and thus, with every local MPS tensor). The physical models for which this analysis is relevant include ladder systems, systems with two flavors of particles and, notably, the transfer matrix of PEPS. Ultimately, we obtain seventeen nontrivial phases. Furthermore, by imposing the spatial symmetries directly on a generic MPS, the resulting phases are put in one-to-one correspondence with the structure of the local tensors, in particular, their transformation behavior. This structure can in turn be imposed in numerical algorithms. Indeed, by making full use of spatial symmetries and imposing the adequate transformation behavior on the local tensors, for which this paper provides a dictionary, the number of variational degrees of freedom in numerical simulations can be significantly reduced, thus allowing for higher resolution simulations.

### A. Outline

In Sec. II, we begin by providing a review of matrix product states, recapitulating the concepts of MPS injectivity, gauge transformations, the transfer matrix and the fundamental theorem of MPS. After reconsidering the implementation of symmetries in MPS and how this leads to the SPT classification of Chen *et al.* in Sec. III, we present in Sec. III A a method to compute group cohomology and explicit cocycles from the Smith normal form of the coboundary map<sup>1</sup> and in Sec. III B construct explicit correlation length zero MPS tensors transforming according to given cohomology classes characterizing an SPT phase. We explicitly construct our ansatz from the nontrivial 2-cocycle of  $\mathbb{Z}_2 \times \mathbb{Z}_2$  in Sec. III B 1 and find that it reduces to the fixed point cluster state dressed with a trivial dimer state. In Sec. III C, we generalize our ansatz and argue how an MPS transforming on the physical level in some arbitrary representation of a finite symmetry group can be constructed in such a way that the virtual bond dimension is as small as possible. In Sec. IV, we address the main question of this paper, namely we construct the SPT classification for frieze symmetric MPS and construct canonical forms for most of these phases. Finally, to illustrate the generality of our approach, in Sec. V, we reconsider the problem of imposing time reversal symmetry in MPS and combine time reversal with shifts over one site. Some technical details are relegated to Appendix A, and in Appendix B, we demonstrate our algorithm to compute cocycles of  $H_{\beta_P}^2(\mathbb{Z}_2^P, \mathcal{U}(1))$ , the second cohomology of  $\mathbb{Z}_2$  with nontrivial group action originating from parity symmetry.

### B. Summary of results

In Table I, below we give an overview of the symmetry groups we consider and the SPT classification they give rise to.

Translation symmetry in itself does not give rise to nontrivial SPT phases. It can be shown that every translationally

TABLE I. Overview of the SPT classification of frieze group symmetries as well as of time reversal.  $F_1^T$ : independently symmetric under translations over one site and time reversal,  $F_2^T$ : translation invariance over two sites and invariance under combined action of translation over one site and time reversal.

| Symmetry       | SPT classification |
|----------------|--------------------|
| $F_0$          | /                  |
| $F_V$          | $\mathbb{Z}_2$     |
| $F_H$ (parity) | $\mathbb{Z}_2^2$   |
| $F_R$          | $\mathbb{Z}_2^2$   |
| $F_G$          | /                  |
| $F_{RG}$       | $\mathbb{Z}_2^2$   |
| $F_{VH}$       | $\mathbb{Z}_2^4$   |
| $F_1^T$        | $\mathbb{Z}_2$     |
| $F_2^T$        | /                  |

invariant MPS admits a uniform representation, as was shown in Ref. [12]. The reflection in the  $F_V$  symmetry group can be thought of as an on-site  $\mathbb{Z}_2$  symmetry. We rederive that  $F_V$  protects one nontrivial SPT phase, and provide an explicit MPS representation in which the local MPS tensors have definite  $V$ -parity. The reflection in  $F_H$  is equivalent to the parity considered in Ref. [5]. Three nontrivial SPT phases are found, in accordance with [5], and a canonical MPS is constructed in which the tensors have definite  $H$ -parity, possibly at the cost of introducing nontrivial *bond tensors* [19]. The same classification is found for  $F_R$  symmetry, not previously considered elsewhere.  $F_G$  symmetry admits only the trivial phase and we show that every  $F_G$ -symmetric MPS can be brought in a manifestly  $F_G$ -invariant form. In case of the larger symmetry groups  $F_{RG}$  and  $F_{VH}$  there are respectively three and seven nontrivial phases. To our knowledge, also the symmetry groups  $F_G$ ,  $F_{RG}$ , and  $F_{VH}$  have not previously been considered in the literature.

MPS with independent translation symmetry over one site and time reversal symmetry can protect one nontrivial SPT phase as demonstrated in Ref. [5], if time reversal is implemented linearly. We rederive this result and show that in the nontrivial phase the MPS tensors transform according to a quaternionic representation of  $\mathbb{Z}_2^T$ . It is argued that injective MPS cannot be invariant under a projective implementation of time reversal symmetry, an MPS interpretation of the Lieb-Schultz-Mattis theorem. Finally, to demonstrate the generality of our approach we also consider MPS which are symmetric under translations over two lattice sites and the combined action of a one-site translation and time reversal, denoted by the symmetry group  $F_2^T$ . We find that in this case no nontrivial phases are retained.

## II. REVIEW OF MATRIX PRODUCT STATES

In this section, we present a brief review of injective matrix product states. We focus on some key aspects that are used to derive the frieze classification below.

In this paper, we consider bosonic spin systems. The Hilbert space is simply the tensor product of the local  $d$ -dimensional Hilbert spaces of the constituent spins,  $\mathcal{H} \cong (\mathbb{C}^d)^{\otimes N}$ . A matrix product representation of a state in  $\mathcal{H}$  with

<sup>1</sup>After completion of this manuscript we came to understand that this algorithm appeared previously in Ref. [18].



periodic boundary conditions is of the form

$$|\psi\rangle = \sum_{\{i_k\}} \text{Tr}(A_1^{i_1} A_2^{i_2} \dots A_N^{i_N}) |i_1\rangle |i_2\rangle \dots |i_N\rangle. \quad (1)$$

Such a periodic MPS can be pictorially represented as

$$\text{---} \bigcirc_{A_1} \text{---} \bigcirc_{A_2} \text{---} \dots \text{---} \bigcirc_{A_N} \text{---} \quad (2)$$

The variational degrees of freedom are contained in the local tensors  $(A_n^i)_{a\beta} \in \mathbb{C}^D \otimes \mathbb{C}^D \otimes \mathbb{C}^d$ , where  $D$  is called the *bond dimension*. Every state in  $\mathcal{H}$  can be represented with a bond dimension that scales exponentially in the system size, but the power of MPS lies in the fact that ground states of gapped local Hamiltonians can be well approximated by MPS with a bond dimension that scales polynomially in the number of spins [11]. An MPS representation of a state  $|\psi\rangle$  is never unique: a *gauge transformation*  $A'_n \mapsto X_n^{-1} A_n X_{n+1}$  clearly leaves the state invariant because the gauge tensors  $X_i$  cancel on the bonds.

In case of translation invariance, it can be shown that one can always carry out a gauge transformation that brings the translationally invariant MPS in a canonical *uniform* form in which  $A_n \equiv A$ ,  $\forall n$  [12]:

$$|\psi(A)\rangle = \dots \text{---} \bigcirc_A \text{---} \bigcirc_A \text{---} \bigcirc_A \text{---} \dots \quad (3)$$

Gauge transformations can furthermore be used to bring the MPS parametrization in a left- or right-canonical form, characterized respectively by

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \quad \text{---} \bigcirc_A \text{---} \bigcirc_{\bar{A}} \text{---} = \left( \text{---} \right), \quad (4)$$

$$\sum_i A^i (A^i)^\dagger = \mathbb{1}, \quad \text{---} \bigcirc_{\bar{A}} \text{---} \bigcirc_A \text{---} = \left( \text{---} \right). \quad (5)$$

We define the *transfer matrix*  $\mathbb{E}$  as

$$\mathbb{E} = \sum_i A^i \otimes \bar{A}^i = \text{---} \bigcirc_A \text{---} \bigcirc_{\bar{A}} \text{---}. \quad (6)$$

The transfer matrix captures all the relevant information about the entanglement and correlations of the state. Moreover, the transfer matrix determines the MPS uniquely up to a local change in basis. This follows from the observation that the transfer matrix defines a completely positive (CP) map where the local MPS tensors play the role of Kraus operators, combined with the fact that a Kraus decomposition of a CP map is unique up to unitary equivalence [20].

If the matrices  $\{A^i | i = 1, \dots, d\}$  generate (via linear combinations and products) the entire  $D \times D$  matrix algebra, the MPS is said to be *injective*. In that case, the transfer matrix (interpreted as a  $D^2 \times D^2$  matrix from the left pair of indices to the right pair) has a unique eigenvalue of largest magnitude that in an appropriate normalization of the MPS tensors can be

taken to be one. Moreover, when the MPS is in a left or right canonical form, the corresponding eigenvector is  $\mathbb{1}$  as follows from (4) and (5).

By far the most important property of injective MPS is that it satisfies the requirements of the *fundamental theorem* of MPS: two injective uniform MPS defined by local tensors  $A^i$  and  $B^i$  describe the same state  $|\psi(A)\rangle \sim |\psi(B)\rangle$  if and only if there is a gauge transformation  $X$  and a phase  $\theta$  that intertwines the two tensors:  $A^i = e^{i\theta} X^{-1} B^i X$ . If  $A$  and  $B$  are simultaneously in left (or right) canonical form, then  $X$  can be chosen to be unitary. Furthermore,  $e^{i\theta}$  is uniquely defined, whereas  $X$  is only defined up to an overall scaling. Put differently, for  $B^i = A^i$ , the relation  $A^i = e^{i\theta} X^{-1} A^i X$  implies  $\theta = 0 \bmod 2\pi$  and  $X = c\mathbb{1}$  for some  $c \in \mathbb{C}$ , as follows readily from the definition of injectivity.

### III. EXPLICIT SYMMETRIC TENSORS

In this section, we review how symmetries are implemented in the tensor network language and discuss how the SPT classification arises from the projective representation of the underlying symmetry group. We present an algorithm to compute group cohomology and explicit cocycles using the Smith normal form and give an MPS ansatz that realizes the SPT phases classified by a given 1- and 2-cocycle. We work out this ansatz for the nontrivial 2-cocycle of  $\mathbb{Z}_2 \times \mathbb{Z}_2$  and demonstrate that this representative MPS is the tensor product of the correlation length zero cluster state and a trivial dimer state. We then discuss how an MPS can be constructed for every possible physical representation of a given finite symmetry group such that the bond dimension is minimal.

If an injective translationally invariant MPS  $|\psi(A)\rangle$  is invariant under the action of a unitary on-site symmetry transformation, i.e.,  $U_g^{\otimes N} |\psi(A)\rangle \sim |\psi(A)\rangle$  for all  $g \in \mathbf{G}$ , there must exist, for every  $g \in \mathbf{G}$ , a phase  $\varphi(g)$  and a gauge transformation  $X_g$  such that

$$\sum_j (U_g)_{ij} A^j = e^{i\varphi(g)} X_g^{-1} A^i X_g. \quad (7)$$

This equation can only admit solutions if  $U_g$  forms a linear unitary representation of the on-site symmetry group  $\mathbf{G}$ . If the MPS is in either left or right canonical form, the gauge matrices  $X_g$  can be chosen to be unitary. Because of the overall scale freedom in how they are determined, it follows that they only need to constitute a projective representation of  $\mathbf{G}$ , i.e., they form a representation of  $\mathbf{G}$  up to phase:

$$X_g X_h = e^{i\omega(g,h)} X_{gh}. \quad (8)$$

Here, the  $\omega(g, h)$ 's satisfy the well-known 2-cocycle equations

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \omega(h, k) \bmod 2\pi, \quad (9)$$

expressing associativity of the multiplication of the gauge matrices  $X_g$ . Solutions to this constraint are called 2-cocycles. The phase  $\varphi(g)$ , on the other hand, constitutes a one-dimensional linear representation of the symmetry group:

$$\varphi(g) + \varphi(h) = \varphi(gh) \bmod 2\pi. \quad (10)$$

The scale freedom in determining gauge transformations implies that the matrices  $X_g$  can be replaced with an equivalent

choice of the form  $X_g \mapsto e^{i\gamma(g)} X_g$ . Under such a redefinition the cocycles transform according to

$$\omega(g, h) \mapsto \omega(g, h) + \gamma(g) + \gamma(h) - \gamma(gh). \quad (11)$$

Hence, in the classification of projective representations labeled by 2-cocycles, these redefinitions have to be modded out, giving rise to equivalence classes of projective representations. Cocycles  $\omega$  of the form  $\omega(g, h) = \gamma(g) + \gamma(h) - \gamma(gh)$ , which are equivalent to the choice  $\omega(g, h) = 1$ , are called coboundaries. The equivalence classes  $[\omega]$ , defined by  $[\omega] = [\omega'] \iff \omega(g, h) = \omega'(g, h) + \gamma(g) + \gamma(h) - \gamma(gh)$ , are exactly classified by the second cohomology group of  $\mathbf{G}$  with respect to  $\mathbf{U}_1$ ,  $H^2(\mathbf{G}, \mathbf{U}_1)$ . In the context of group cohomology, the one-dimensional linear representation  $\varphi(g)$  is referred to as a 1-cocycle and is correspondingly characterized by  $H^1(\mathbf{G}, \mathbf{U}_1)$ . Chen *et al.* showed that every choice of a 1-cocycle  $\varphi \in H^1(\mathbf{G}, \mathbf{U}_1)$  and cohomology class  $[\omega] \in H^2(\mathbf{G}, \mathbf{U}_1)$  gives rise to a distinct SPT phase [5].

In case of a finite symmetry group, the second cohomology group is always  $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \dots \times \mathbb{Z}_{d_N}$  for topological indices  $d_1, d_2, \dots, d_N$ , giving rise to a finite number of SPT phases. Explicit representative 2-cocycles can be obtained by writing the 2-cocycle condition as a linear system modulo  $2\pi$  and solving it using the Smith normal form, as explained below in Sec. III A.

In case the symmetry group contains time reversal or parity transformations, the above picture has to be modified as follows.

Since the  $\mathbb{Z}_2^T$  time reversal is implemented antiunitarily [21], the time reversal operator can be written as a  $T = UK$ , where  $K$  denotes complex conjugation in the basis with respect to which the MPS tensors  $A^j$  are defined, and  $U$  is a unitary satisfying  $U\bar{U} = \pm \mathbb{1}$  depending on whether time reversal is implemented linearly or projectively (Sec. V A). Hence, on the MPS tensors, a symmetry  $g \in \mathbf{G} = \mathbf{H} \rtimes \mathbb{Z}_2^T$ ,  $\mathbf{H}$  denoting the global on-site symmetry group, acts according to

$$\sum_j (U_g)_{ij} C_g(A^j) = e^{i\varphi(g)} X_g^{-1} A^i X_g. \quad (12)$$

The action of  $C_g$  on  $A^i$  is taking the complex conjugate only if  $g$  contains a time reversal.

From acting with time reversal twice on the MPS tensor, it follows that in this case the matrices  $X_g$  form a *generalized* projective representation of the symmetry group, as their multiplication also contains the action  $C_g$ :

$$X_g C_g(X_h) = e^{i\omega(g, h)} X_{gh}, \quad (13)$$

whereas the phases  $\exp(i\varphi(g))$  obey

$$e^{i\varphi(g)} C_g(e^{i\varphi(h)}) = e^{i\varphi(gh)}. \quad (14)$$

Hence,  $\varphi(g)$  and  $\omega(g, h)$  satisfy the 1- and 2-cocycle constraints with a nontrivial group action that takes the conditioned complex conjugation  $C_g$  into account. The 1-cocycles constraint reads

$$\varphi(g) + \alpha_g^T(\varphi(h)) = \varphi(gh) \mod 2\pi, \quad (15)$$

whereas 2-cocycles satisfy

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \beta_g^T(\omega(h, k)) \mod 2\pi, \quad (16)$$

where the group actions  $\alpha_g^T$  and  $\beta_g^T$  are multiplication by  $-1$  if  $g$  contains time reversal. The SPT classification in this case is given by  $H_{\alpha^T}^1(\mathbf{G}, \mathbf{U}_1) \times H_{\beta^T}^2(\mathbf{G}, \mathbf{U}_1)$ . The equivalence classes of  $H_{\beta^T}^2$  are given by  $[\omega] = [\omega'] \iff \omega(g, h) = \omega'(g, h) + \gamma(g) + \beta_g^T(\gamma(h)) - \gamma(gh)$ , whereas  $H_{\alpha^T}^1$  classifies 1-cocycles up to equivalence of the form  $\varphi(g) \mapsto \varphi(g) + \alpha_g^T(c) - c$  for an arbitrary constant  $c \in [0, 2\pi)$  as coboundaries of the form  $\alpha_g^T(c) - c$  trivially solve the 1-cocycle condition (15).

In case the symmetry group contains global on-site symmetries combined with parity,  $\mathbf{G} = \mathbf{H} \rtimes \mathbb{Z}_2^P$ , we have that

$$\sum_j (U_g)_{ij} T_g(A^j) = e^{i\varphi(g)} X_g^{-1} A^i X_g, \quad (17)$$

where  $T_g$  denotes taking the transpose if  $g$  contains the parity transformation. Similarly as in the case of time reversal, the gauge matrices multiply according to a generalized projective representation:

$$X_g P_g(X_h) = e^{i\omega(g, h)} X_{gh}, \quad (18)$$

where  $P_g$  is taking the inverse transpose if  $g$  contains parity. Note, however, that if  $g$  contains the parity transformation,  $X_g$  can in general not be chosen unitary. As the conditioned transpose does not affect the phases  $\varphi$ , they form a one-dimensional linear representation. The SPT classification for  $\mathbf{H} \rtimes \mathbb{Z}_2^P$  is thus in terms of  $H^1(\mathbf{G}, \mathbf{U}_1) \times H_{\beta^P}^2(\mathbf{G}, \mathbf{U}_1)$ . The 2-cocycles are given by

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \beta_g^P(\omega(h, k)) \mod 2\pi, \quad (19)$$

where, similarly as in the case of time reversal, the group action  $\beta_g^P$  is a multiplication with  $-1$  whenever  $g$  contains a parity transformation. The equivalence classes in the second cohomology group are then  $[\omega] = [\omega'] \iff \omega(g, h) = \omega'(g, h) + \gamma(g) + \beta_g^P(\gamma(h)) - \gamma(gh)$ .

#### A. Computing group cohomology using the Smith normal form

The problem of finding all (generalized) projective representations of a given finite symmetry group  $\mathbf{G}$  or, equivalently, computing its second cohomology group  $H_{\beta}^2(\mathbf{G}, \mathbf{U}_1)$  can be reduced to a problem in linear algebra that can be solved using the Smith normal form [18,22], as we now demonstrate. Our approach works for both trivial and nontrivial group actions  $\beta$  and the same method can be used to compute other cohomology groups.

First note that the 2-cocycle equation (9) or its generalizations with nontrivial group actions (16) and (19) can be written as the linear system

$$\sum_j \Omega_{ij}^{(2, \beta)} \omega_j = 0, \mod 2\pi, \quad (20)$$

which has to be solved modulo  $2\pi$ .  $\Omega^{(2, \beta)}$  is called the 2-boundary map, where  $\beta$  again refers to the group action. Every solution  $\tilde{\omega}$  to this linear system of equations constitutes a valid 2-cocycle.

Since  $\Omega^{(2,\beta)}$  only has entries in  $\mathbb{Z}$ , which forms a principal ideal domain,  $\Omega^{(2,\beta)}$  can be written in Smith normal form as follows [23]:

$$P\Lambda R = \Omega^{(2,\beta)}. \quad (21)$$

In this decomposition,  $P$  and  $R$  are respectively  $|\mathbb{G}|^3 \times |\mathbb{G}|^3$  and  $|\mathbb{G}|^2 \times |\mathbb{G}|^2$  matrices that only contain integers and have determinant one (and thus have integer-valued inverses).  $\Lambda$  also only contains integers, is  $|\mathbb{G}|^3 \times |\mathbb{G}|^2$ -dimensional and is of the form

$$\Lambda = \left( \begin{array}{c|c} \text{diag}(d_1, d_2, \dots, d_r) & 0 \\ \hline 0 & 0 \end{array} \right), \quad (22)$$

in which the nonzero elements  $d_1, \dots, d_r$  along the diagonal, some of which might be one, are in increasing order,  $d_1 \leq d_2 \leq \dots$ , and every element is a divisor of the next,  $d_i | d_{i+1}$ . The Smith normal form  $\Lambda$  is unique. Inserting this decomposition in the system of equations (20) gives rise to the solution

$$\tilde{\omega} = 2\pi R^{-1} \Lambda^+ \tilde{v}. \quad (23)$$

$\Lambda^+$  denotes the (unique) Moore-Penrose pseudoinverse of  $\Lambda$  that satisfies  $\Lambda \Lambda^+ \Lambda = \Lambda$  and which is found to be

$$\Lambda^+ = \left( \begin{array}{c|c} \text{diag}(d_1^{-1}, d_2^{-1}, \dots, d_r^{-1}) & 0 \\ \hline 0 & 0 \end{array} \right). \quad (24)$$

$\tilde{v}$  is an arbitrary vector that only contains integers.

Writing the solution (23) in components yields

$$\omega_i = \sum_{j=1}^r 2\pi (R^{-1})_{ij} \frac{v_j}{d_j}. \quad (25)$$

Because  $\tilde{v}$  can be chosen freely, one can choose subsequently  $v_i = \delta_{1,i}, \delta_{2,i}, \dots, \delta_{r,i}$  to obtain a basis of the solution space that can be written as

$$(\tilde{\omega}_j)_i = \frac{2\pi}{d_j} (R^{-1})_{ij}, \quad \forall j \in \{1, \dots, r\}. \quad (26)$$

Hence, the 2-cocycles  $\tilde{\omega}$  are found to be the columns of  $R^{-1}$ . Since  $R$  is full rank, all the solutions  $\tilde{\omega}$  are linearly independent. In particular, the nontrivial cocycles (below) can not be related by a coboundary,  $\tilde{\omega}' = \tilde{\omega} + \Omega^{(1,\beta)} \tilde{\varphi}$ , where  $\Omega^{(1,\beta)}$  denotes the 1-coboundary map and  $\tilde{\varphi}$  is a  $|\mathbb{G}|$ -dimensional vector containing arbitrary real numbers. To classify all possible solutions, we now consider the diagonal entries of  $\Lambda$ .

From (26) and the fact that  $R^{-1}$  contains only integers, it follows that for every diagonal entry  $d_i = 1$ , a trivial solution  $\omega_i = 0 \pmod{2\pi}$  is obtained. The nontrivial solutions are those that correspond to entries  $d_i > 1$ . From (26) and the fact that the solution space is  $\mathbb{Z}$ -linear, it follows that the cocycle  $\tilde{\omega}_j$  corresponding to some  $d_j$  generates a cyclic group. Now note that not all elements of  $\tilde{\omega}_j$  can be divisible by  $d_j$  (or any of its prime factors) as this would be in contradiction with the fact that  $R^{-1}$  has determinant one. Hence, the cyclic group generated by  $\tilde{\omega}_j$  is  $\mathbb{Z}_{d_j}$ . Finally, the zero entries of  $\Lambda$  can also be discarded in the cohomology as these correspond to trivial solutions of the cocycle equation that can be multiplied by arbitrary phases and thus correspond to coboundaries.

In conclusion following picture arises. Given some group  $\mathbb{G}$  one can write down the coboundary map  $\Omega_j^{(2,\beta)}$  that can

be brought in Smith normal form  $P\Lambda R$ . The diagonal entries of  $\Lambda$ ,  $d_1, d_2, \dots, d_r$ , determine the second cohomology group which is then of the form  $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \dots \times \mathbb{Z}_{d_r}$ , with the understanding that  $\mathbb{Z}_1 = \{e\}$  denotes the trivial group and that all zero diagonal entries can be discarded. The nontrivial 2-cocycles in some arbitrary gauge correspond then to the columns of  $R^{-1}$ , weighted by the appropriate factor  $2\pi/d_j$ .

### B. Zero correlation length SPT ansatz

Given some SPT phase characterized by  $([\varphi], [\omega]) \in H_\alpha^1(\mathbb{G}, \mathbb{U}_1) \times H_\beta^2(\mathbb{G}, \mathbb{U}_1)$ , it is possible to explicitly construct zero-correlation-length MPS tensors that transform according to (7), (12) or (17). Firstly, a projective representation  $X_g$  in the class  $[\omega]$  and with trivial 1-cocycle can be constructed with virtual dimension  $|\mathbb{G}|$ , by taking the  $X_g$  to form the  $\omega$ -projective regular representation of  $\mathbb{G}$  (27). The dimension of the local physical Hilbert space is then  $|\mathbb{G}|^2$ . Concretely, this representative state and the regular representation are given by

$$(X_g)_{g_1, g_2} = \delta_{g_1, g g_2} e^{i\omega(g, g_2)} \quad (27)$$

$$(V^{h_1, h_2})_{g_1, g_2} = \frac{1}{|\mathbb{G}|} \delta_{h_1, g_1} \delta_{h_2, g_2} e^{i\omega(h_2, h_1^{-1} g_1)}. \quad (28)$$

Furthermore, the physical group action  $U_g$  is given by  $L_g \otimes L_g$ , with  $L_g$  the linear left regular representation. Hence,  $(U_g)_{k_1 k_2, h_1 h_2} = \delta_{k_1, g h_1} \delta_{k_2, g h_2}$ . Pictorially:

$$\begin{array}{c} \text{---} \\ | \\ \boxed{V} \\ | \\ \boxed{L_g \otimes L_g} \\ \text{---} \end{array} = \begin{array}{c} \text{---} \boxed{X_g^\dagger} \text{---} \\ | \quad \quad | \\ \boxed{V} \\ | \quad \quad | \\ \text{---} \boxed{X_g} \text{---} \end{array}. \quad (29)$$

Note that the Kronecker deltas in the definition of the MPS tensor indicate its valence bond structure, which is modified only by a unitary diagonal transformation. Hence, it follows readily that the transfer matrix  $\mathbb{E} = \sum_{h_1, h_2} (V^{h_1, h_2})_{g_1, g_2} \otimes (V^{h_1, h_2})_{g'_1, g'_2}$  is idempotent,  $\mathbb{E}^2 = \mathbb{E}$ , implying that the ansatz (28) has zero correlation length and thus defines an RG fixed point [10].

Similarly, in case of a nontrivial 1-cocycle  $\varphi$ , the ansatz is readily modified to also include this cocycle by adding diagonal matrices to the two physical legs:

$$\begin{array}{c} \text{---} \\ | \\ \boxed{V} \\ | \\ \boxed{\sqrt{\varphi} \otimes \sqrt{\varphi}} \\ \text{---} \end{array} = e^{i\varphi(g)} \begin{array}{c} \text{---} \boxed{X_g^\dagger} \text{---} \\ | \quad \quad | \\ \boxed{V} \\ | \quad \quad | \\ \text{---} \boxed{X_g} \text{---} \end{array}, \quad (30)$$

where  $(\sqrt{\varphi})_{g_1, g_2} = \delta_{g_1, g_2} e^{\frac{i}{2}\varphi(g)}$ .

The ansatz can also capture the case of time reversal and parity symmetry. In the first case, we take the physical action to be  $L_g \otimes L_g$  combined with the conditioned complex



We then take this unitary  $U$  as a gauge transformation of our representative MPS ansatz. This gauge transformation then intertwines between the projective regular representation of  $\mathbb{Z}_2 \times \mathbb{Z}_2$  and the block diagonal projective representation given in (45)–(47). Similarly as in the case of linear representation theory, this illustrates how the projective regular representation of  $\mathbb{Z}_2 \times \mathbb{Z}_2$  falls apart in projective irreps, given by the Pauli matrices, where each irrep appears with multiplicity equal to its dimension [28].

The physical basis transformation  $W$  that brings the representative MPS  $V$  in the form of the cluster state fixed point dressed with a trivial dimer state,

$$U^\dagger V W = A, \quad (49)$$

is then immediately found by considering the QR decomposition of our ansatz and the dressed cluster state interpreted as matrix from the physical to virtual level,  $QR = V(\bar{U} \otimes U)$ ,  $\bar{Q}D^\dagger \bar{R} = \mathbb{1} \otimes A \otimes \mathbb{1}$ , where  $D$  is a diagonal matrix containing only phases that fixes the gauge freedom of the QR decomposition in such a way that  $D^\dagger \bar{R} = R$ . The basis transformation then reads  $W = \bar{Q}DQ^\dagger$ . It can then be checked that this  $W$  intertwines the physical representation of the  $\mathbb{Z}_2 \times \mathbb{Z}_2$  symmetry group acting on the dressed cluster state and our ansatz:

$$W^\dagger(\mathbb{1} \otimes \sigma_X \otimes \sigma_X \otimes \mathbb{1})W = L_{(0,1)} \otimes L_{(0,1)} \quad (50)$$

$$W^\dagger(\mathbb{1} \otimes \mathbb{1} \otimes \sigma_X \otimes \mathbb{1})W = L_{(1,0)} \otimes L_{(1,0)} \quad (51)$$

$$W^\dagger(\mathbb{1} \otimes \sigma_X \otimes \mathbb{1} \otimes \mathbb{1})W = L_{(1,1)} \otimes L_{(1,1)}. \quad (52)$$

Notice that in this example we were required to choose the symmetry group of the dimer to be trivial to match the symmetry of our ansatz, even though the dimer has the full  $U_2$  symmetry.

### C. A generalized ansatz

As we have demonstrated, given a 1- and 2-cocycle of some finite symmetry group  $G$ , one can explicitly construct a bond dimension  $|G|$  injective correlation length zero MPS that belongs to the SPT class corresponding to the cohomology classes represented by these cocycles. However, as shown for the explicit example of  $\mathbb{Z}_2 \times \mathbb{Z}_2$ , our ansatz could be written as a cluster state fixed point and a completely disentangled trivial dimer state that doubles the dimension of the virtual Hilbert space. This redundancy is a consequence of the fact that the physical symmetry action is fixed as being the tensor product of two regular representations of  $G$ . Hence, the question rises if an MPS can be constructed for any given representation of the symmetry on the physical level such that the bond dimension is as small as possible. This can in principle be done as follows, possibly at the cost of introducing a correlation length. We restrict to the case of unitary on-site symmetries.

We first note that every projective representation of a finite group  $G$  can be lifted to a linear representation of a larger

finite covering group  $\tilde{G}$  [29]. This covering group fits in following central exact sequence:

$$1 \rightarrow A \rightarrow \tilde{G} \rightarrow G \rightarrow 1, \quad (53)$$

where  $A = H^2(G, U_1) \leq Z(\tilde{G})$  [30]. It should be noted that the covering group is generically not unique. Consider, for example, the case of  $\mathbb{Z}_2 \times \mathbb{Z}_2$ . Two distinct covering groups of  $\mathbb{Z}_2 \times \mathbb{Z}_2$  are the dihedral group  $D_4 = \langle r, s | r^4 = s^2 = (rs)^2 = 1 \rangle$  and the quaternion group  $Q = \langle a, b | a^4 = 1, a^2 = b^2, b^{-1}ab = a^{-1} \rangle$ , both of which are of order 8. Indeed, the two-dimensional projective irrep of  $\mathbb{Z}_2 \times \mathbb{Z}_2$  corresponding to the nontrivial class of  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, U_1) = \mathbb{Z}_2$  given by  $\{\sigma_X, i\sigma_Y, \sigma_Z\}$  is lifted to the faithful two-dimensional irrep of  $D_4$  by taking  $i\sigma_Y = r$  and  $\sigma_Z = s$  as generators, whereas a gauge transformation of this projective representation,  $\{\sigma_X, i\sigma_Y, \sigma_Z\} \mapsto \{i\sigma_X, i\sigma_Y, i\sigma_Z\}$ , yields an equivalent projective representation which is lifted to the faithful two-dimensional irrep of  $Q$  by identifying the generators of the quaternion group as  $a = i\sigma_Y, b = i\sigma_Z$ .

The classification and construction of the projective irreps of a finite group thus reduces in this way to the linear representation theory of its covering group. To construct the aforementioned MPS that transforms according to a given physical representation  $\Pi$  of the symmetry group  $G$ , one chooses the smallest irrep  $\Gamma$  of  $\tilde{G}$ , which projects down to a projective irrep of  $G$  belonging to a certain cohomology class  $[\omega]$ , such that  $\Pi$  is contained in the tensor product  $\bar{\Gamma} \otimes \Gamma$ . The explicit MPS tensor is then chosen as the projector of  $\bar{\Gamma} \otimes \Gamma$  on the  $\Pi$ -sector, and the virtual symmetry action is  $\Gamma$ .

Consider as example again  $\mathbb{Z}_2 \times \mathbb{Z}_2$  and its covering group  $Q$ . The irreps of  $Q$  are the trivial representation  $\mathbb{1}$ , three nontrivial one-dimensional sign representations,  $\Gamma_1, \Gamma_2, \Gamma_3$ , and the two-dimensional faithful representation given by the Pauli matrices,  $\Delta$ . Choosing then a sign representation of  $\mathbb{Z}_2 \times \mathbb{Z}_2$  as physical symmetry, this representation can be lifted to a one-dimensional sign representation  $\Gamma_i$  of  $Q$ . From the fact that  $\Delta \otimes \Delta \cong \mathbb{1} \oplus \Gamma_1 \oplus \Gamma_2 \oplus \Gamma_3$ , it follows that the MPS with the smallest bond dimension which transforms according to a sign representation of  $\mathbb{Z}_2 \times \mathbb{Z}_2$  is exactly one that projects  $\Delta \otimes \Delta$  onto  $\Gamma_i$ . Since the virtual representation  $\Delta$  is exactly the projective irrep of  $\mathbb{Z}_2 \times \mathbb{Z}_2$ , one can immediately conclude that such a state is in the same SPT phase as the cluster state.

This construction also applies to the case of Lie groups. Consider, for example, the symmetry group  $SO_3$ . Choosing the physical symmetry representation to be the  $\mathbb{1}$  of  $SO_3$ , one can choose the virtual representation to be the  $\frac{1}{2}$ . The projector of  $\frac{1}{2} \otimes \frac{1}{2} \cong \mathbb{0} \oplus \mathbb{1}$  on the  $\mathbb{1}$  subspace then exactly results in the MPS description of the AKLT state which belongs to the nontrivial SPT class of  $H^2(SO_3, U_1) = \mathbb{Z}_2$ .

Notice that this constructing generically does not give rise to correlation length zero states due to the projection on the correct physical symmetry sector.

## IV. FRIEZE SYMMETRIC MPS

In this section, we derive the SPT classification of MPS invariant under frieze symmetries. We do so from starting from a general injective MPS and invoking the symmetry,







this way a  $\mathbb{Z}_2^{\times 2} \times \mathbb{Z}_2^{\times 2}$  classification is obtained, exactly in line with the cohomological classification  $H^1(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbf{U}_1) \times H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2^{\times 2} \times \mathbb{Z}_2^{\times 2}$ . The first factors  $\mathbb{Z}_2^{\times 2}$  originating from the first cohomology group correspond to  $\theta_V$  and  $\theta_H$  and the last factors  $\mathbb{Z}_2^{\times 2}$  correspond to the gauge matrix  $Y$  being (anti)symmetric and the generalized (anti)commutation relation among  $X$  and  $Y$ ,  $YX^{-T} = \pm XY$ . The correspondence between the 2-cocycles of  $H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2^{\times 2}$  and the topological indices obtained from the MPS picture can be made a bit more explicit as follows. One can show that there always exists a gauge in which the cocycles of  $H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2^{\times 2}$  are of the form

$$\omega(g, h) = \begin{matrix} & (0, 0) & (1, 0) & (0, 1) & (1, 1) \\ \begin{pmatrix} (0, 0) \\ (1, 0) \\ (0, 1) \\ (1, 1) \end{pmatrix} & \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & \zeta_1 & \zeta_2 & \zeta_1 \zeta_2 \\ 1 & \zeta_1 & \zeta_2 & \zeta_1 \zeta_2 \end{pmatrix} \end{matrix}, \quad (78)$$

where rows are labeled by  $g$  and  $\zeta_1, \zeta_2 = \pm 1$ . These signs  $\zeta_1, \zeta_2$  then exactly correspond to the topological indices obtained from the MPS computation above since it follows from (18) that

$$YX^{-T} = e^{i\omega((0,1),(1,0))} YX = \zeta_1 XY \quad (79)$$

$$YY^{-T} = e^{i\omega((0,1),(0,1))} \mathbb{1} = \zeta_2 \mathbb{1}. \quad (80)$$

Note that the results  $H_{\beta^P}^2(\mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2$  and  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbf{U}_1) = \mathbb{Z}_2$  are well known, corresponding to the existence of one nontrivial SPT phase under either parity or under an on-site  $\mathbb{Z}_2 \times \mathbb{Z}_2$  symmetry. The fact that an a single on-site  $\mathbb{Z}_2$  symmetry (which in itself does not exhibit nontrivial projective representations) combined with the parity  $\mathbb{Z}_2^P$  leads to a richer structure of SPT phases, as expressed by  $H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2^{\times 2}$  is interesting. Using the choice of cocycles in Eq. (78) and the construction from Sec. III B, explicit examples can be constructed for these different phases.

## V. TIME REVERSAL AND LATTICE SYMMETRIES

In this section, we study time reversal symmetry and time reversal combined with translations over one site [21]. The most important feature of time reversal is that it is an antiunitary transformation and can hence be written as  $T = UK$ , where  $U$  is a unitary and  $K$  denotes complex conjugation in a certain basis.  $T$  can be represented linearly or projectively, depending on whether  $U\bar{U} = \pm \mathbb{1}$ .

We revisit the work by Chen *et al.* on the linear implementation of time reversal in MPS and identify the corresponding SPT classification [5]. We demonstrate that injective MPS

can not be invariant under the projective representation of  $T$ , a tensor network manifestation of the Lieb-Schultz-Mattis theorem [32]. Finally we prove that time reversal combined with a shift over one lattice site does not give rise to nontrivial SPT order and construct a canonical form for the trivial phase.

### A. $F_1^T$ : time reversal in TI systems

We consider MPS which are independently symmetric under shifts over one lattice site and  $\mathbb{Z}_2^T$  implemented through  $T = UK$ , where  $U$  is a unitary satisfying  $U\bar{U} = \pm \mathbb{1}$  and  $K$  denotes complex conjugation in the basis of the MPS tensors. Together, these symmetries correspond to the group denoted by  $F_1^T$  mentioned above in the summary of results.

(1)  $T^2 = \mathbb{1}$  and  $U\bar{U} = \mathbb{1}$ . Consider the uniform ansatz (3). Time-reversal symmetry can then be implemented as

$$\sum_j U_{ij} \bar{A}^j = X^{-1} A^i X. \quad (81)$$

Note that without loss of generality we do not need to consider a phase in this transformation because such a phase can be consistently absorbed in the MPS tensor  $A^i$ . For an MPS tensor in left canonical form,  $X$  can furthermore be chosen unitary. Doing a second time reversal results in

$$A^i = \bar{X}^{-1} X^{-1} A^i X \bar{X}. \quad (82)$$

By virtue of the fundamental theorem we have that  $X\bar{X} = e^{i\chi} \mathbb{1}$ , which, combined with unitarity of  $X$ , results in  $e^{i\chi} = \pm 1$  and thus  $X = \pm X^T$ .

If  $X$  is symmetric, writing  $X = VV^T$  (where  $V$  is unitary because  $X$  is) allows us to bring the MPS in a canonical form by means of the gauge transformation  $A^i \mapsto \bar{A}^i = V^\dagger A^i V$ , which now transforms according to

$$\sum_j U_{ij} \bar{A}^j = \bar{A}^i. \quad (83)$$

For a skew-symmetric  $X$ , we write  $X = V\Gamma V^T$ ,  $V$  again being unitary, from which it follows that  $\bar{A}^i = V^\dagger A^i V$  transforms according to a quaternionic representation under  $T$ , up to multiplication by  $U_{ij}$ <sup>3</sup>:

$$\sum_j U_{ij} \bar{A}^j = \Gamma^{-1} \bar{A}^i \Gamma. \quad (84)$$

The cohomological classification corresponds to  $H_{\alpha^T}^1(\mathbb{Z}_2^T, \mathbf{U}_1) \times H_{\beta^T}^2(\mathbb{Z}_2^T, \mathbf{U}_1) = \{e\} \times \mathbb{Z}_2$ , as follows from the Smith normal form (Sec. III A). The  $\mathbb{Z}_2$  is understood as  $X$  being (skew-)symmetric.

We can now also show that the entanglement spectrum in case of the nontrivial SPT phase for which  $X = -X^T$  is at least doubly degenerate [2]. Consider therefore the unique

<sup>3</sup>Consider  $A$  transforming in a quaternionic representation according to  $\bar{A} = X^{-1}AX$ ,  $X$  being skew-symmetric and unitary.  $X$  can be brought in a skew-symmetric tridiagonal form  $W$  by an orthogonal transformation. Unitarity of  $W$  implies that  $X$  can be written as  $X = \bar{Q}\Gamma\bar{Q}^T$  for an orthogonal  $\bar{Q}$ , showing that  $\bar{Q}^T A \bar{Q}$  transforms as  $\bar{Q}^T A \bar{Q} \mapsto \Gamma^{-1} \bar{Q}^T A \bar{Q} \Gamma$ .



leading right eigenvector  $\rho$  of the transfer matrix  $\mathbb{E}$ . In that case,  $\rho$  interpreted as a  $D \times D$  matrix is (Hermitian) positive semidefinite by virtue of the quantum Perron-Frobenius theorem [33,34]. Consider some eigenvector  $\mathbf{x}$  of  $\rho$  with positive eigenvalue  $\lambda$ , then by virtue of  $X\rho X^\dagger = \rho$  (proven in Appendix A),  $\mathbf{x}^\top X^\dagger$  is a left eigenvector of  $\rho$  with the same eigenvalue  $\lambda$ . However, using  $X^\top = -X$  it follows that  $\text{Tr}(X^\dagger(\mathbf{x} \otimes \mathbf{x})) = 0$ , or in other words that  $\mathbf{x}^\top X^\dagger$  and  $\mathbf{x}$  are orthogonal eigenvectors belonging to the same eigenvalue  $\lambda$ .

(2)  $\mathbf{T}^2 = -\mathbb{1}$  and  $\mathbf{U}\bar{\mathbf{U}} = -\mathbb{1}$ . This case is relevant for, e.g.,  $U = \sigma_y$ , which comes into play in the implementation of time reversal symmetry on spin 1/2 particles [21]. Let us not restrict to this particular example and consider a general unitary  $U$  satisfying the aforementioned property  $U\bar{U} = -\mathbb{1}$ . Starting again from the uniform ansatz (3), considering a projective implementation of time reversal and applying it twice leads to

$$A^i \xrightarrow{T} \sum_j U_{ij} \bar{A}^j \xrightarrow{T} -A^i \stackrel{!}{=} A^i, \quad (85)$$

from which we conclude that translationally invariant injective MPS cannot transform projectively under time reversal symmetry. This can be understood as a tensor product interpretation of the celebrated Lieb-Schultz-Mattis theorem [32] that dictates that the ground state of a system of half-integer spins—in which case time reversal acts projectively—should be either symmetry broken (in contradiction with the assumption that the MPS is symmetric under time reversal) or gapless (in which case the matrix product ansatz does not provide a good description).

### B. $F_2^T$ : time reversal combined with a one site shift

We can now break translation invariance over one lattice site to translation invariance over two sites by considering following ansatz:

$$\cdots \text{---} \bigcirc \text{---} \bigcirc \text{---} \bigcirc \text{---} \bigcirc \text{---} \cdots \quad (86)$$

(The diagram shows a sequence of four circles labeled A, B, A, B connected by horizontal lines. Each circle has a vertical line extending downwards from its center.)

The symmetry group we impose,  $F_2^T$ , contains translations over two lattice sites, and the combined action of translations over one site and time reversal, implemented via complex conjugation together with a multiplication with a unitary  $U$  satisfying  $U\bar{U} = \mathbb{1}$ . The transformation of the tensors under the latter then reads

$$\begin{aligned} \sum_j U_{ij} \bar{A}^j &= X^{-1} B^i W, \\ \sum_j U_{ij} \bar{B}^j &= W^{-1} A^i X, \end{aligned} \quad (87)$$

where  $U\bar{U} = \mathbb{1}$ . Note that all phases can be absorbed in the tensors and gauge transformations.

A second transformation results in

$$\begin{aligned} A^i &= \bar{X}^{-1} W^{-1} A^i \bar{X} \bar{W} \\ B^i &= \bar{W}^{-1} X^{-1} B^i W \bar{X}. \end{aligned} \quad (88)$$

From blocking two tensors, we conclude that  $W\bar{X} = e^{i\chi} \mathbb{1}$ . Hence, we can absorb a factor  $e^{i\chi/2}$  in both  $\bar{X}$  and  $W$  such that they become inverses. In conclusion, there is no nontrivial SPT phase and a canonical form is obtained by writing

$$B^i = \sum_j U_{ij} X \bar{A}^j \bar{X} \quad (89)$$

and defining  $C^i = \bar{X} A^i$ :

$$\cdots \text{---} \bigcirc \text{---} \bigcirc \text{---} \bigcirc \text{---} \bigcirc \text{---} \cdots \quad (90)$$

(The diagram shows a sequence of four circles labeled C, C-bar, C, C-bar connected by horizontal lines. Each circle has a vertical line extending downwards from its center. Below the second and fourth circles (labeled C-bar), there is a square box labeled U, connected by a vertical line to the circle's bottom.)

## VI. CONCLUSIONS AND OUTLOOK

In this work, we showed that quasi-one-dimensional spatial symmetries can protect nontrivial SPT phases in quantum spin chains represented by matrix product states, a complete survey of which was lacking in the literature. We identified each of these phases by invoking the symmetries on injective MPS and identifying the topologically distinct ways in which these symmetries can be represented by the local tensors. For most of these phases, we constructed canonical MPS ansätze that are manifestly invariant under the considered symmetries. Finally, we revisited the SPT classification in case of time reversal symmetry and showed that time reversal combined with a translation over one lattice site does not give rise to nontrivial phases.

An important application of the classification is in the simulation of 1 + 1D quantum materials. As we have shown, all frieze SPT phases can be put in one-to-one correspondence with transformation properties of the local MPS tensors, up to the appropriate gauge transformations on the virtual level. This demonstrates that taking into account the spatial symmetries of the spin chains at hand, one can tremendously reduce the number of variational parameters characterizing the state, allowing for simulations with higher bond dimensions.

A natural extension of this work would be to consider the classification of two-dimensional SPT phases protected by space group symmetries. The two-dimensional space groups are known as the wallpaper groups, of which there are seventeen. In this case, the relevant tensor network states are the projected entangled-pair states (PEPS), which form the natural two-dimensional generalization of the MPS considered here. We expect that similarly as in the one-dimensional case, imposing the spatial symmetries directly on the local tensors will also reveal topological obstructions. For each of these phases, canonical ansätze could be constructed that might again prove very useful in numerical simulations of physical systems and materials in which these spatial symmetries are ubiquitous. The framework to investigate these spatial symmetries was laid out in Ref. [35], where it was called the *crystalline equivalence principle*. This principle states that the classification of phases protected by a spatial symmetry group  $G$  is the same as that of the SPT phases with

In particular, it would be interesting to investigate whether some of the symmetry transformations in 2D admit an implementation on the virtual level as string-like matrix product operators (MPOs). The physical application of the symmetry is then “gauged away” by pulling these MPOs through the lattice. Similarly, it might be interesting to demonstrate that also time reversal, which, because of the complex conjugation, contains a priori a very nonlocal symmetry, can be implemented using an MPO of finite bond dimension. In this case, we expect that the consistency equation governing such an SPT phase corresponds to the 3-cocycle equation with a nontrivial group action, which can readily be solved numerically using the algorithm proposed in Sec. III A. Time-reversal in itself gives rise to only trivial 3-cocycles,  $H_{\beta\tau}^3(\mathbb{Z}_2^T, \mathbf{U}_1) = 0$ . The smallest group containing time reversal  $\mathbb{Z}_2^T$  as a direct summand which gives rise to nontrivial 3-cocycles is  $\mathbb{Z}_2 \times \mathbb{Z}_2^T$  since  $H_{\beta\tau}^3(\mathbb{Z}_2 \times \mathbb{Z}_2^T, \mathbf{U}_1) = \mathbb{Z}_2^{\times 2}$ . With the crystalline equivalence principle in mind, another example to consider is  $\mathbb{Z}_4 \times \mathbb{Z}_2^T$ ,  $H_{\beta\tau}^3(\mathbb{Z}_4 \times \mathbb{Z}_2^T, \mathbb{C}^\times) = \mathbb{Z}_2^{\times 2}$ , as the SPT classification of this group is expected to coincide with that of the space group  $\mathbf{D}_4$ . We plan to investigate this in future work.

## 6

## APPENDIX A: PROOF OF $X\bar{\rho}X^\dagger = \rho$

$$\text{Diagram 1} = \text{Diagram 2} \quad (\text{A1})$$

Diagrammatic equation (A3) shows the equivalence between two expressions. On the left, a trace is represented by a circle with a horizontal line through it, containing the operator  $\bar{A}$  above and  $A$  below. This is connected by a vertical line to a square box labeled  $U$ , which is connected by another vertical line to a square box labeled  $U^\dagger$ , which is finally connected by a vertical line to a circle with a horizontal line through it, containing the operator  $A$  above and  $\bar{A}$  below. On the right, the expression is equal to a trace represented by a circle with a horizontal line through it, containing the operator  $\bar{\rho}$  above and  $\rho$  below. The label (A3) is to the right of the equation.

$$\Leftrightarrow \begin{array}{c} \text{---} X^\dagger \\ \text{---} X^\top \end{array} \text{---} \bar{\rho} = \begin{array}{ccccc} & A & & X^\dagger & \\ & | & & & \\ & \bar{A} & & X^\top & \\ & & & & \end{array} \text{---} \bar{\rho} \quad (\text{A5})$$

In this Appendix, we briefly illuminate the algorithm laid out in Sec. III A for computing explicit cocycle representatives. We focus on one of the smallest interesting examples,  $H_{B^P}^2(\mathbb{Z}_2, \mathbf{U}_1)$ , which has a nontrivial group action. The cocycle equations to be solved thus read

where  $\beta_g^P$  is multiplying with  $-1$  when  $g$  is the nontrivial element of  $\mathbb{Z}_2^P$  and the identity otherwise.

The first step of the algorithm consists of filling up the  $\Omega^{(2, \beta^p)}$ -matrix (20), taking into account the nontrivial group action. Denoting the group elements of  $\mathbb{Z}_2^p$  by  $\{0, 1\}$ , we obtain

$$\Omega^{(2,\beta^v)} = \begin{matrix} & \begin{matrix} (0;0) & (0;1) & (1;0) & (1;1) \end{matrix} \\ \begin{matrix} (0;0;0) \\ (0;0;1) \\ (0;1;0) \\ (0;1;1) \\ (1;0;0) \\ (1;0;1) \\ (1;1;0) \\ (1;1;1) \end{matrix} & \begin{pmatrix} 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & -1 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 1 & -2 \end{pmatrix} \end{matrix}. \quad (\text{B2})$$

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these are solved modulo  $2\pi$ . The  $R$  matrix reads

$$R = \begin{pmatrix} (0;0) & (0;1) & (1;0) & (1;1) \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{B3})$$

From the  $\Lambda$  matrix,

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (\text{B4})$$

the group cohomology is immediately found to be  $H_{\beta^e}^2(\mathbb{Z}_2^P, \mathbb{U}_1) = \mathbb{Z}_2$ : the 1 diagonal entries corresponds to trivial cocycles, whereas the entry 2 corresponds to a nontrivial cocycle. We then compute

$$R^{-1} \Lambda + \vec{v} = \begin{pmatrix} (0;0) \\ (0;1) \\ (1;0) \\ (1;1) \end{pmatrix} \begin{pmatrix} v_1 - \frac{v_3}{2} \\ v_2 - \frac{v_3}{2} \\ \frac{v_3}{2} \\ 0 \end{pmatrix}, \quad (\text{B5})$$

where  $\vec{v}$  is an arbitrary integer vector. The nontrivial cocycle valued in  $\mathbb{U}_1$  that generates the  $\mathbb{Z}_2$  cohomology group is then found from (B5) by choosing  $v_3 = 1$  and reads

$$\vec{\omega} = \begin{pmatrix} (0;0) \\ (0;1) \\ (1;0) \\ (1;1) \end{pmatrix} \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \end{pmatrix}. \quad (\text{B6})$$

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# Mapping between Morita equivalent string-net states with a constant depth quantum circuit

## 7.1 Synopsis

As argued in section 1.3, two models belong to the same gapped quantum phase of matter if they are related by an interpolating quantum circuit of sublinear depth. In this project, we explicitly construct such a constant-depth circuit for string-net models whose input UFCs  $\mathcal{D}_1$  and  $\mathcal{D}_2$  are *Morita equivalent*. The circuit is built explicitly from the  $F$ -symbols encoded in an invertible  $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category.

Concretely, we first review in detail the framework of *extended string-net models*, mentioned above in section 3.2.3. These extended string-nets provide a framework in which all anyonic excitations encoded in  $\mathcal{Z}(\mathcal{D})$ , including those violating vertex Hamiltonian terms, can be treated in a unified manner. We then use the aforementioned associator data encoded in an invertible  $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category to define the quantum circuit and compute its action on both ground and excited states. When restricted to the ground state sector, the circuit reduces precisely to the generalised ground-state projector introduced in section 3.4. Its action on excited states on the other hand requires the introduction of ancillary degrees of freedom that account for mismatches in the dimensions of the central idempotents describing the

anyons in the two theories. In that case, the circuit acts on individual plaquettes via *bimodule tubes*, akin to those introduced in section 4.4.

As an illustrative example, we show that, when applied to the *toric code*, our construction reproduces the familiar *electric–magnetic duality*. More generally, for *quantum double models* based on possibly non-abelian input groups, it recovers a quantum circuit previously introduced in refs. [158, 159].

## 7.2 Reference

*Mapping between Morita equivalent string-net states with a constant depth quantum circuit*

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## 7.3 Contributions of the author

The central idea underlying this project, together with its overall technical framework, was developed by Laurens Lootens in collaboration with Norbert Schuch and Frank Verstraete. My primary contribution was the explicit derivation of the quantum circuit in terms of the bimodule data. In addition, I contributed to the writing of the manuscript, including portions of the review of extended string-net models, the presentation of the quantum circuit, and the examples.

## Mapping between Morita-equivalent string-net states with a constant depth quantum circuit

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We construct a constant depth quantum circuit that maps between Morita-equivalent string-net models. Due to its constant depth and unitarity, the circuit cannot alter the topological order, which demonstrates that Morita-equivalent string nets are in the same phase. The circuit is constructed from an invertible bimodule category connecting the two input fusion categories of the relevant string-net models, acting as a generalized Fourier transform for fusion categories. The circuit not only acts on the ground state subspace but also acts unitarily on the full Hilbert space when supplemented with ancillas.

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### I. INTRODUCTION

Ever since their conception, the string-net models as originally proposed by Levin and Wen [1] and their subsequent generalizations [2–7] have provided a rich playground for studying microscopic realizations of nonchiral topologically ordered phases of matter in  $2+1$  dimensions. Taking a unitary fusion category (UFC)  $\mathcal{D}$  [8] as input, these exactly solvable models allow for the explicit realization of several key features of topologically ordered systems, such as ground-state degeneracies that depend on the topology of the space and anyonic quasiparticle excitations that satisfy nontrivial braiding statistics. From the category-theoretical side, the different ground states and excitations are described by the monoidal center  $Z(\mathcal{D})$  of the input UFC  $\mathcal{D}$  [2], which is itself a unitary modular fusion category (UMFC) and describes the topological order. The fact that the UMFC describing the topological order is always the center of some other UFC is the reason why string-net models are only able to describe nonchiral topological order with gappable boundaries; exactly solvable lattice models for chiral topological order have proven more challenging to obtain [9].

An important observation is that the process of going to the center  $Z(\mathcal{D})$  of a UFC  $\mathcal{D}$  is not an injective operation, or, put differently, that distinct UFCs  $\mathcal{D}_1$  and  $\mathcal{D}_2$  can have the same center  $Z(\mathcal{D}_1) \simeq Z(\mathcal{D}_2)$  [2]. This has led to the conjecture that two such string-net models based on different  $\mathcal{D}_1$  and  $\mathcal{D}_2$  belong to the same topological phase, despite the fact that microscopically they can look very different. UFCs with the same center are said to be Morita equivalent [10], and the collection of UFCs Morita equivalent to  $\mathcal{D}$  is called its Morita class. A direct consequence of Morita equivalence is that the topological excitations or anyons of two Morita-equivalent string-net models are in one-to-one correspondence with each other. Kitaev and Kong demonstrated the existence of an invertible domain wall between two Morita-

equivalent string-net models through which anyons can move freely without condensing on the domain wall [2]. Mathematically, the existence of such an invertible domain wall is guaranteed by the fact that, for any two Morita-equivalent UFCs  $\mathcal{D}_1$  and  $\mathcal{D}_2$ , there exists an invertible  $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category  $\mathcal{M}$  [8]. The central goal of this paper is to show that this implies the existence of a constant depth quantum circuit which maps the different string-net states onto each other.

From a quantum information point of view, it has been understood that two states are in the same phase if there exists a constant depth quantum circuit that is able to map between the two [11–13]. In this work, we provide such a constant depth circuit for the case of Morita-equivalent string nets  $\mathcal{D}_1$  and  $\mathcal{D}_2$  that maps the ground states as well as the excitations of one model to those of the other. For the ground state, the quantum circuit can be understood as a generalized version of the Hamiltonian, using the invertible  $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category  $\mathcal{M}$  to intertwine the Hamiltonian of the string-net  $\mathcal{D}_1$  to the Hamiltonian of the string-net  $\mathcal{D}_2$ . Our construction generalizes a previous mapping, which was obtained from Kitaev's quantum double models [14] for a group  $G$  to string-net models  $\text{Rep}(G)$  [15,16], to the general case of two Morita-equivalent string nets.

*Outline.* We begin by providing a brief review of string nets, focusing mainly on a generalization known as the extended string-net models [17]. The reason for this is that these generalizations allow for a more unified treatment of the vertex and plaquette excitations, the two types of excitations in a string-net model. They are understood in the diagrammatic language as idempotents of the tube algebra associated with the string-net UFC  $\mathcal{D}$  [3], which is well known to provide a characterization of the center  $Z(\mathcal{D})$ . Next, we discuss how an invertible  $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category can be used to provide a map between the tube algebras of the UFCs  $\mathcal{D}_1$  and  $\mathcal{D}_2$ , and thereby show how excitations from one model are mapped to excitations of the other. Taking all this together, we are able to explicitly construct a constant depth unitary quantum circuit which we explain in detail.

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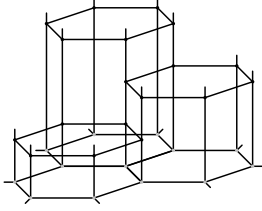


FIG. 1. An illustration of how our circuit acts on the three sublattices of the hexagonal lattice.

For the case of a hexagonal lattice, this circuit is depth three, which follows from the fact that our circuit sequentially acts on three sublattices. This is depicted in Fig. 1 below. The action of the circuit on the three sublattices is worked out in the main text. Each of these local unitaries acts on 18 degrees of freedom associated with a plaquette. We show that the unitary map on the three sublattices is respectively constructed from the different associators of the  $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category  $\mathcal{M}$ , which are the solutions to a set of coupled pentagon equations [8,18]. In this way our construction gives a very concrete quantum information perspective on the categorical notion of Morita equivalence. We also show how the quantum circuit can be generalized to act on the full (extended) string-net Hilbert space and obtain a very similar circuit as in the absence of excitations, the details of which are relegated to the Appendix.

## II. STRING NETS

In this section we give a brief review of Levin and Wen's string-net models on the hexagonal lattice. First, we review the ordinary string-net model, herein closely following the original work of Levin and Wen [1], and then we turn to the extended string nets [17]. These can be regarded as a generalization of the ordinary string nets that provide a more natural and convenient setting in which to study and classify excited states.

### A. Levin and Wen's string nets

A string-net model is defined by the data of some given input unitary fusion category  $\mathcal{D}$ . This input data contains a set of simple objects (string types) that we denote  $a, b, c, \dots$  with corresponding fusion rules  $a \otimes b \simeq \bigoplus_c N_{ab}^c c$ , corresponding quantum dimensions  $d_a, d_b, d_c, \dots$ , and an isomorphism  $F : (a \otimes b) \otimes c \rightarrow a \otimes (b \otimes c)$ . Given some oriented trivalent lattice, such as the hexagonal lattice in this work, the Hilbert space of the string net is defined by configurations of the string types living on the edges of the lattice. Furthermore, for every string type  $a$  there exists a corresponding conjugate string type  $\bar{a}$  such that  $\bar{\bar{a}} \equiv a$  and  $\bar{a}$  can be interpreted as  $a$  living on the same edge with the opposite orientation. The quantum dimensions obey  $d_{\bar{a}} = d_a$ . There is a unique self-conjugate vacuum string  $\mathbf{1}$ ,  $\bar{\mathbf{1}} \equiv \mathbf{1}$ .

The string-net Hamiltonian consists of two terms:

$$H = - \sum_v A_v - \sum_p B_p. \quad (1)$$

Herein the two sums in the Hamiltonian are over the vertices and plaquettes of the lattice, respectively. The sum over  $s$  is a sum over all string types of the input UFC.  $D$  is the total quantum dimension defined as  $D = (\sum_i d_i^2)^{1/2}$ . The vertex operators  $A_v$  act on the vertices according to

$$A_v \left| \begin{array}{c} b \\ \swarrow \quad \searrow \\ i \\ \uparrow \\ a \end{array} \right\rangle = \delta_{a, \bar{b}, \bar{c}} \left| \begin{array}{c} b \\ \swarrow \quad \searrow \\ i \\ \uparrow \\ a \end{array} \right\rangle \quad (2)$$

where  $\delta_{a, \bar{b}, \bar{c}}$  is 1 whenever the fusion of  $a \otimes b$  contains  $\bar{c}$  and is 0 else. The plaquette operators  $B_p$  are defined as

$$B_p = \frac{1}{D} \sum_a d_a B_p^a, \quad (3)$$

where the terms  $B_p^a$  acts on a plaquette by inserting a clockwise-oriented string of type  $a$  in the plaquette and fusing it to the lattice. This is done by making repeated use of the resolution of the identity for UFCs, namely,

$$\begin{array}{c} a \\ \uparrow \\ a \end{array} \begin{array}{c} b \\ \uparrow \\ b \end{array} = \sum_{c, n} \sqrt{\frac{d_c}{d_a d_b}} \begin{array}{c} a \quad b \\ \swarrow \quad \searrow \\ n \\ \uparrow \\ c \end{array}, \quad (4)$$

the  $F$  moves of the given input UFC,

$$\begin{array}{c} b \quad c \\ \swarrow \quad \searrow \\ j \quad k \\ \swarrow \quad \searrow \\ a \quad d \end{array} = \sum_{f, mn} (F_{abc}^{f, mn})_{e, jk} \begin{array}{c} b \quad c \\ \swarrow \quad \searrow \\ m \\ \uparrow \\ f \end{array}, \quad (5)$$

and the “bubble pop” identity

$$\begin{array}{c} c' \\ \uparrow \\ m \\ \downarrow \\ n \\ \uparrow \\ c \end{array} = \sqrt{\frac{d_a d_b}{d_c}} \delta_{m, n} \delta_{c, c'} \begin{array}{c} c \\ \uparrow \\ c \end{array}. \quad (6)$$

The matrix elements of this operator are worked out in detail in Refs. [1,6]. Every operator  $B_p^a$  is a 18-body operator in the sense that its action explicitly depends on the six string types living on the edges of the plaquette  $p$ , the six multiplicities and the six edges adjacent to the plaquette, but acts diagonally on the latter. It can be shown that both  $A_v$  and  $B_p$  are Hermitian projection operators,  $A_v^2 = A_v$  and  $B_p^2 = B_p$ , that moreover all commute. Hence, the string-net condensed renormalization



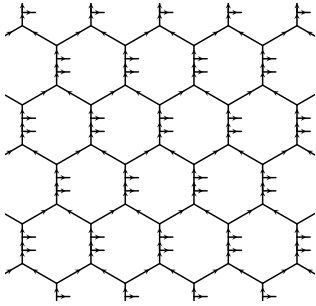


FIG. 2. We adapt the convention that all arrows on the edges point upward and the open legs are attached to the lattice as depicted where every open edge corresponds to the closest vertex of the hexagonal lattice. Multiplicity degrees of freedom are suppressed.

group fixed-point ground states are the simultaneous eigenvectors of all  $A_v$  and  $B_p$ . Quasiparticle excitations on top of these ground states are obtained by violating at least one of the projector constraints. These excitations are gapped and necessarily come in pairs. Vertices—or electric excitations—are obtained by violating the  $A_v$  constraints, whereas plaquettes—or magnetic excitations—violate the  $B_p$  constraints. These excitations are treated in a unified way in the language of extended string nets which we now revisit.

### B. Extended string nets

The extended string-net models [17] are very similar to the ordinary string-net models except for a few key differences which we discuss in this section.

Given a hexagonal lattice we associate with every vertex an additional open edge that is attached to an edge that is connected to the vertex under consideration. These open edges also carry string degrees of freedom. The choice of which open edge belongs to which vertex is to some extent arbitrary and the open edges can even be moved around the plaquette by a series of  $F$  moves but we will adapt the convention denoted in Fig. 2. Furthermore, we demand the fusion rules to be satisfied on every vertex,  $\delta_{a,b,c} = 1$ . Finally, the Hamiltonian of the extended string net is of the same form as the Hamiltonian of the ordinary string-net model:

$$H = - \sum_v A_v - \sum_p B_p, \quad (7)$$

where the  $A_v$  and the  $B_p$  are again mutually commuting Hermitian projectors.  $A_v$  acts on vertices according to

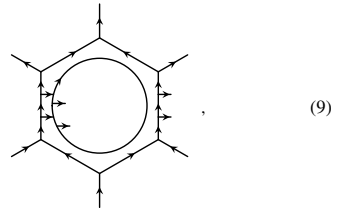
$$A_v \left| \begin{array}{c} b \\ \swarrow \\ i \\ \downarrow a_2 \\ j \\ \downarrow a_1 \end{array} \right\rangle = \delta_{d,1} \left| \begin{array}{c} b \\ \swarrow \\ c \\ \downarrow a_2 \\ j \\ \downarrow a_1 \end{array} \right\rangle. \quad (8)$$

The action of  $B_p$  on the plaquette is explained in detail in Ref. [17] and reduces for  $p_1, p_2 = \mathbf{1}$  to the action of the plaquette operator  $B_p$  of the ordinary model, where  $p_1, p_2$  denote the open edges of the plaquette  $p$ .

The extended string-net model thus deals with vertex excitations in the ordinary string-net model by considering extra degrees of freedom that signal the presence of vertex projector violations.

### C. Tube algebra

In the extended string-net models, vertex and plaquette excitations are both characterized by the *tube algebra* [3,17]. The action of a tube  $T_i$  on a plaquette can graphically be depicted as the insertion of the following diagram in a plaquette:



and fusing it to the boundary; here,  $i$  runs over all possible configurations that give a nonzero tube. The set of tubes is closed under multiplication,

$$T_i T_j = \sum_k f_{ij}^k T_k, \quad (10)$$

defined by stacking:

$$T_j \circ T_i = \sum_k f_{ij}^k T_k, \quad (11)$$

as well as Hermitian conjugation [19,20]:

$$(T_i)^\dagger = \sum_j u_i^j T_j. \quad (12)$$

This turns the tube algebra into a finite  $C^*$  algebra, for which the Artin-Wedderburn theorem dictates that one can find an isomorphism to a direct sum of simple matrix algebras. Explicitly, this implies the existence of simple idempotents  $p_{ii}^a$  and nilpotents  $p_{ij}^a$ ,  $i \neq j$  that satisfy

$$p_{ij}^a = \sum_k x_{ij}^{a,k} T_k, \quad p_{ij}^a p_{kl}^a = \delta_{ab} \delta_{jk} p_{il}^a, \quad (p_{ij}^a)^\dagger = p_{ji}^a. \quad (13)$$

The elementary excitations of the extended string-net model correspond to the simple idempotents  $p_{ii}^a$  of the tube algebra, each of which corresponds to a specific combination of one plaquette excitation and two vertex excitations. These

simple idempotents have to be grouped into central idempotents  $P_a$  satisfying

$$P_a = \sum_{i=0}^{n_a-1} P_{ii}^a, \quad [P_a, T_i] = 0, \quad \sum_a P_a = \mathbb{1}, \quad (14)$$

where  $n_a$  denotes the dimension of the central idempotent  $P_a$ . These central idempotents then correspond to the irreducible representations of the tube algebra and are in one-to-one correspondence with the simple objects of the monoidal center  $Z(\mathcal{D})$ .

### III. BIMODULE CATEGORIES

We consider two string-net models  $\mathcal{D}_1$  and  $\mathcal{D}_2$  that are supposedly in the same phase; this requires that  $\mathcal{D}_1$  and  $\mathcal{D}_2$  are Morita equivalent,  $Z(\mathcal{D}_1) \simeq Z(\mathcal{D}_2)$ . From a categorical perspective, this implies the existence of an invertible  $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category  $\mathcal{M}$ . In this section, we briefly review some properties of a bimodule category, mainly to fix notation,<sup>1</sup> and show how this bimodule category provides a map between the excitations of the two string-net models.

#### A. Bimodule categories

We start with a spherical UFC  $\mathcal{D}_1$ , with objects  $a, b, c, \dots \in \mathcal{D}_1$  and an associator expressed in a basis of simple objects as

$$\begin{array}{c} b \quad c \\ \searrow \quad \swarrow \\ m \\ \swarrow \quad \searrow \\ f \\ \swarrow \quad \searrow \\ a \quad d \end{array} = \sum_{f,m,n} (\mathcal{D}_1 F_d^{abc})_{f,m,n}^{f,m,n} \begin{array}{c} b \quad c \\ \searrow \quad \swarrow \\ j \quad k \\ \swarrow \quad \searrow \\ a \quad d \end{array}, \quad (15)$$

where in the diagrams we color lines labeled by objects in  $\mathcal{D}_1$  in black for the remainder of this work. We also have a second spherical UFC  $\mathcal{D}_2$ , with objects  $\alpha, \beta, \gamma, \dots \in \mathcal{D}_2$  and an associator given by

$$\begin{array}{c} \beta \quad \gamma \\ \searrow \quad \swarrow \\ j \quad k \\ \swarrow \quad \searrow \\ \alpha \quad \delta \end{array} = \sum_{\nu,m,n} (\mathcal{D}_2 F_\delta^{\alpha\beta\gamma})_{\nu,m,n}^{\nu,m,n} \begin{array}{c} \beta \quad \gamma \\ \searrow \quad \swarrow \\ m \\ \swarrow \quad \searrow \\ n \\ \swarrow \quad \searrow \\ \alpha \quad \delta \end{array}, \quad (16)$$

with lines labeled by objects in  $\mathcal{D}_2$  drawn in orange. A left  $\mathcal{D}_1$ -module category  $\mathcal{M}$  is a category  $\mathcal{M}$  with objects  $A, B, C, \dots \in \mathcal{M}$ , a left-action of  $\mathcal{D}_1$  on  $\mathcal{M}$  given by  $a \triangleright A \simeq \bigoplus_B N_{aA}^B B$  and an isomorphism  ${}^\triangleright F : (a \otimes b) \triangleright A \rightarrow$

$a \triangleright (b \triangleright A)$ , which can graphically be expressed as

$$\begin{array}{c} b \quad c \\ \searrow \quad \swarrow \\ j \quad k \\ \swarrow \quad \searrow \\ a \quad \end{array} \begin{array}{c} A \\ \uparrow \\ C \\ \downarrow \\ B \end{array} = \sum_{C,m,n} ({}^\triangleright F_B^{abA})_{C,jk}^{C,mn} \begin{array}{c} b \quad c \\ \searrow \quad \swarrow \\ m \\ \swarrow \quad \searrow \\ a \quad \end{array} \begin{array}{c} A \\ \uparrow \\ C \\ \downarrow \\ B \end{array}, \quad (17)$$

where lines labeled by objects in  $\mathcal{M}$  are blue. Similarly, a right  $\mathcal{D}_2$ -module category  $\mathcal{M}$  has a right-action of  $\mathcal{D}_2$  on  $\mathcal{M}$  given by  $A \triangleleft \alpha \simeq \bigoplus_B N_{A\alpha}^B B$  and an isomorphism  ${}^\triangleleft F : (A \triangleleft \alpha) \triangleleft \beta \rightarrow A \triangleleft (\alpha \otimes \beta)$ , graphically expressed as

$$\begin{array}{c} A \\ \uparrow \\ j \\ \swarrow \quad \searrow \\ C \quad k \\ \swarrow \quad \searrow \\ B \end{array} \begin{array}{c} \alpha \\ \rightarrow \\ m \\ \rightarrow \\ n \\ \rightarrow \\ B \end{array} = \sum_{\gamma,m,n} ({}^\triangleleft F_B^{A\alpha\beta})_{C,jk}^{\gamma,mn} \begin{array}{c} A \\ \uparrow \\ n \\ \swarrow \quad \searrow \\ B \end{array} \begin{array}{c} \alpha \\ \rightarrow \\ m \\ \rightarrow \\ n \\ \rightarrow \\ B \end{array}. \quad (18)$$

A  $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category  $\mathcal{M}$  is then defined as a left- $\mathcal{D}_1$  and right- $\mathcal{D}_2$  module category  $\mathcal{M}$  that additionally is equipped with an isomorphism  ${}^\triangleright\triangleleft F : (a \triangleright A) \triangleleft \alpha \rightarrow a \triangleright (A \triangleleft \alpha)$ , graphically depicted as

$$\begin{array}{c} a \quad \begin{array}{c} A \\ \uparrow \\ j \\ \swarrow \quad \searrow \\ C \quad k \\ \swarrow \quad \searrow \\ B \end{array} \end{array} = \sum_{D,m,n} ({}^\triangleright\triangleleft F_B^{aA\alpha})_{C,jk}^{D,mn} \begin{array}{c} a \quad \begin{array}{c} A \\ \uparrow \\ m \\ \rightarrow \\ n \\ \rightarrow \\ B \end{array} \end{array}. \quad (19)$$

Physically, these bimodule categories arise most evidently in the study of domain walls between different string-net models [2], where the various associativity conditions discussed above amount to the generalizations of the bulk  $F$  moves. For a general domain between Morita-equivalent string-net models, the process of pushing excitations through such a domain wall is not reversible, because several excitations can condense on the domain wall. In the special case where excitations can freely move through the domain wall, we are dealing with an invertible bimodule category. For every two Morita-equivalent UFCs  $\mathcal{D}_1$  and  $\mathcal{D}_2$  one can always write down an invertible  $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category, and we will use this particular bimodule category in the remainder of this work.

The  $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category  $\mathcal{M}$  does not have an intrinsic duality, but one can define a  $(\mathcal{D}_2, \mathcal{D}_1)$ -bimodule category denoted by  $\mathcal{M}^{\text{op}}$  or  $\bar{\mathcal{M}}$  to contain the dual objects of  $\mathcal{M}$ . Crucially, this allows us to impose a generalized notion of sphericity [2,21], which we need to perform certain diagrammatical manipulations. Concretely, invertibility of  $\mathcal{M}$  can be used to generalize the resolution of the identity (4) of UFCs to a resolution of the identity which allows for the

<sup>1</sup>The  $\mathcal{D}_1 F$ ,  ${}^\triangleright F$ ,  ${}^\triangleleft F$ ,  ${}^\triangleright\triangleleft F$ , and  $\mathcal{D}_2 F$  symbols in this work are denoted as  ${}^a F$ ,  $F$ ,  ${}^b F$ ,  ${}^c F$ , and  ${}^d F$  symbols in Ref. [18], respectively.

creation of a  $\mathcal{D}_1$  or  $\mathcal{D}_2$  line from the fusion of an  $\mathcal{M}$  line with an  $\mathcal{M}^{\text{op}}$  line:

$$\begin{array}{c} A \\ \uparrow \\ A \end{array} \quad \begin{array}{c} B \\ \downarrow \\ B \end{array} = \sum_{a,n} \sqrt{\frac{d_a}{d_A d_B}} \begin{array}{c} A \quad B \\ \searrow \quad \swarrow \\ n \\ \swarrow \quad \searrow \\ A \quad B \end{array} \quad (20)$$

$$\begin{array}{c} A \\ \downarrow \\ A \end{array} \quad \begin{array}{c} B \\ \uparrow \\ B \end{array} = \sum_{\alpha,n} \sqrt{\frac{d_\alpha}{d_A d_B}} \begin{array}{c} A \quad B \\ \searrow \quad \swarrow \\ n \\ \swarrow \quad \searrow \\ A \quad B \end{array} \quad (21)$$

### B. Tube algebra bimodules

Using the bimodule categories  $\mathcal{M}$  and  $\overline{\mathcal{M}}$  we can write down bimodule tubes for the tube algebras  $\mathcal{T}^{\mathcal{D}_1}$  and  $\mathcal{T}^{\mathcal{D}_2}$  of  $\mathcal{D}_1$  and  $\mathcal{D}_2$ , respectively. These bimodule tube algebras, denoted by  $\mathcal{T}^{\mathcal{M}}$  and  $\mathcal{T}^{\overline{\mathcal{M}}}$ , are generated by

$$\mathcal{T}_i^{\mathcal{M}} = \begin{array}{c} \text{---} \rightarrow \text{---} \\ \text{---} \rightarrow \text{---} \end{array} \quad \mathcal{T}_i^{\overline{\mathcal{M}}} = \begin{array}{c} \text{---} \leftarrow \text{---} \\ \text{---} \leftarrow \text{---} \end{array} \quad (22)$$

They satisfy

$$\mathcal{T}_i^{\mathcal{M}} \mathcal{T}_j^{\mathcal{D}_1} = \sum_k d_{ij}^k \mathcal{T}_k^{\mathcal{M}}, \quad \mathcal{T}_i^{\mathcal{D}_2} \mathcal{T}_j^{\mathcal{M}} = \sum_k b_{ij}^k \mathcal{T}_k^{\mathcal{M}}, \quad (23)$$

$$\mathcal{T}_i^{\mathcal{D}_1} \mathcal{T}_j^{\overline{\mathcal{M}}} = \sum_k c_{ij}^k \mathcal{T}_k^{\overline{\mathcal{M}}}, \quad \mathcal{T}_i^{\overline{\mathcal{M}}} \mathcal{T}_j^{\mathcal{D}_1} = \sum_k d_{ij}^k \mathcal{T}_k^{\overline{\mathcal{M}}}, \quad (24)$$

$$\mathcal{T}_i^{\overline{\mathcal{M}}} \mathcal{T}_j^{\mathcal{M}} = \sum_k e_{ij}^k \mathcal{T}_k^{\mathcal{D}_1}, \quad \mathcal{T}_i^{\mathcal{M}} \mathcal{T}_j^{\overline{\mathcal{M}}} = \sum_k g_{ij}^k \mathcal{T}_k^{\mathcal{D}_2}, \quad (25)$$

where the last line requires the invertibility of the bimodule category, manifested in the identities (20) and (21). These bimodule tubes are mapped into one another under Hermitian conjugation:

$$\left( \begin{array}{c} A_4 \uparrow \\ c_1 \rightarrow i_4 \\ A_3 \uparrow \\ i_3 \rightarrow \gamma_1 \\ A_2 \uparrow \\ c_2 \rightarrow i_2 \\ A_1 \uparrow \\ i_1 \rightarrow \gamma_2 \\ A \uparrow \end{array} \right)^\dagger = \sqrt{\frac{d_{c_1} d_{c_2}}{d_{\gamma_1} d_{\gamma_2}}} \begin{array}{c} A_4 \uparrow \\ i_4 \rightarrow c_1 \\ A_3 \uparrow \\ i_3 \rightarrow \gamma_1 \\ A_2 \uparrow \\ i_2 \rightarrow c_2 \\ A_1 \uparrow \\ i_1 \rightarrow \gamma_2 \\ A \uparrow \end{array} \quad (26)$$

This Hermitian conjugation is defined such that the operators obtained from inserting  $\mathcal{T}_i^{\mathcal{M}}$  and  $(\mathcal{T}_i^{\mathcal{M}})^\dagger$  in a plaquette expressed as matrices acting on the relevant degrees of freedom are related via the usual Hermitian conjugation. Using

these properties, we can define a big  $C^*$  algebra generated by  $\{\mathcal{T}_i^{\mathcal{D}_1}, \mathcal{T}_j^{\mathcal{M}}, \mathcal{T}_k^{\overline{\mathcal{M}}}, \mathcal{T}_l^{\mathcal{D}_2}\}$ , where we take undefined tube multiplications to be zero. This big  $C^*$  algebra contains the  $C^*$  algebras  $\mathcal{T}^{\mathcal{D}_1}$  and  $\mathcal{T}^{\mathcal{D}_2}$  as subalgebras. Using the Artin-Wedderburn theorem for this big (finite)  $C^*$  algebra, there exists an isomorphism to a direct sum of simple matrix algebras, the number of which equals the number of central idempotents of  $\mathcal{T}^{\mathcal{D}_1}$  and  $\mathcal{T}^{\mathcal{D}_2}$ . In addition to the simple idempotents and nilpotents of  $\mathcal{T}^{\mathcal{D}_1}$  and  $\mathcal{T}^{\mathcal{D}_2}$ , we can define simple bimodules:

$$p_{ij}^{a,\mathcal{M}} = \sum_k y_{ij}^{a,k} \mathcal{T}_k^{\mathcal{M}}, \quad p_{ji}^{a,\overline{\mathcal{M}}} = \sum_k z_{ji}^{a,k} \mathcal{T}_k^{\overline{\mathcal{M}}}, \quad (27)$$

$$(p_{ij}^{a,\mathcal{M}})^\dagger = p_{ji}^{a,\overline{\mathcal{M}}}, \quad (28)$$

with  $0 \leq i < n_a^{\mathcal{D}_1}$  and  $0 \leq j < n_a^{\mathcal{D}_2}$ , such that they satisfy

$$\begin{aligned} p_{ij}^{a,\mathcal{D}_1} p_{kl}^{b,\mathcal{D}_1} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{D}_1}, & p_{ij}^{a,\mathcal{D}_2} p_{kl}^{b,\mathcal{D}_2} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{D}_2}, \\ p_{ij}^{a,\mathcal{M}} p_{kl}^{b,\mathcal{D}_1} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{M}}, & p_{ij}^{a,\mathcal{D}_2} p_{kl}^{b,\mathcal{M}} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{M}}, \\ p_{ij}^{a,\mathcal{D}_1} p_{kl}^{b,\overline{\mathcal{M}}} &= \delta_{ab} \delta_{jk} p_{il}^{a,\overline{\mathcal{M}}}, & p_{ij}^{a,\overline{\mathcal{M}}} p_{kl}^{b,\mathcal{D}_2} &= \delta_{ab} \delta_{jk} p_{il}^{a,\overline{\mathcal{M}}}, \\ p_{ij}^{a,\overline{\mathcal{M}}} p_{kl}^{b,\mathcal{M}} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{D}_1}, & p_{ij}^{a,\mathcal{M}} p_{kl}^{b,\overline{\mathcal{M}}} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{D}_2}, \end{aligned} \quad (29)$$

where from now on the range of the indices  $i, j, k, l$  should be inferred from the context. The simple bimodules will be the essential building blocks of the quantum circuit we aim to construct, providing a map between the ground states and excitations of the string-net model  $\mathcal{D}_1$  to the ground states and excitations of the string-net model  $\mathcal{D}_2$ .

### IV. QUANTUM CIRCUIT

Before we construct our quantum circuit, we have to address the point that the total dimension of the Hilbert space need not match between the string-net model  $\mathcal{D}_1$  and  $\mathcal{D}_2$ ; the fact that they are Morita equivalent only ensures that number of ground states on some closed manifold is the same. The number of distinct excitations which form a complete basis for the Hilbert space will in general differ between the two models, which is reflected in their tube algebras; since we have  $Z(\mathcal{D}_1) \simeq Z(\mathcal{D}_2)$ , the number of central idempotents  $P_a^{\mathcal{D}_1}$  and  $P_a^{\mathcal{D}_2}$  is the same, but their dimensions  $n_a^{\mathcal{D}_1}$  and  $n_a^{\mathcal{D}_2}$  can differ. This discrepancy in the dimensions of the Hilbert spaces poses an obstacle if we want to construct a unitary quantum circuit that maps the full Hilbert space of one model to the other.

To solve this problem, we will place an ancillary qudit  $|i\rangle$  of dimension  $n_a^{\mathcal{D}_2} = \max_a n_a^{\mathcal{D}_2}$  in each plaquette of the string-net model  $\mathcal{D}_1$ . The original Hilbert space of the string-net model is obtained by fixing these ancillas to  $|0\rangle$ ; we denote states in this Hilbert space as  $|\psi^{\mathcal{D}_1}, 0\rangle$ . We can then define the following operator:

$$V_{\mathcal{D}_1}^{\mathcal{D}_2} := \left( \sum_{a,i,j} p_{ij}^{a,\mathcal{M}} \otimes |j\rangle \langle i| \right)^{\otimes N}, \quad (30)$$

with  $N$  being the number of plaquettes in the system, where  $p_{ij}^{a,\mathcal{M}}$  acts on a plaquette as defined before, and  $|j\rangle \langle i|$  acts on the corresponding ancilla. The Hermitian conjugate of this

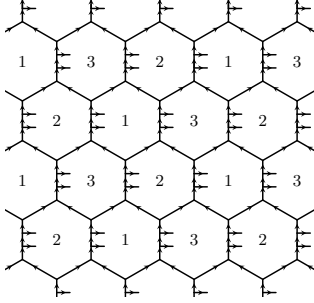


FIG. 3. Division of the hexagonal lattice into three sublattices. We build the quantum circuit by acting on the plaquettes in the order denoted in the figure.

operator is given by

$$(V_{\mathcal{D}_1}^{\mathcal{D}_2})^\dagger = \left( \sum_{a,ij} p_{ij}^{a,\overline{\mathcal{M}}} \otimes |j\rangle\langle i| \right)^{\otimes N}, \quad (31)$$

and the product of  $V_{\mathcal{D}_1}^{\mathcal{D}_2}$  with its Hermitian conjugate is

$$\begin{aligned} (V_{\mathcal{D}_1}^{\mathcal{D}_2})^\dagger V_{\mathcal{D}_1}^{\mathcal{D}_2} &= \left( \sum_{ab,ii'jj'} p_{ji'}^{a,\overline{\mathcal{M}}} p_{ij}^{a,\mathcal{M}} |i'\rangle\langle j'| |i\rangle\langle i| \right)^{\otimes N} \\ &= \left( \sum_{a,ij} p_{jj}^{a,\mathcal{D}_1} \otimes |i\rangle\langle i| \right)^{\otimes N} \\ &= \left( \sum_a p_a^{\mathcal{D}_1} \otimes \mathbb{1}_{n_a^{\mathcal{D}_2}} \right)^{\otimes N}. \end{aligned} \quad (32)$$

In general this is a projector implying  $V_{\mathcal{D}_1}^{\mathcal{D}_2}$  is an isometry, but restricting to the original Hilbert space where all ancillas are in the  $|0\rangle$  state we get

$$\begin{aligned} (V_{\mathcal{D}_1}^{\mathcal{D}_2})^\dagger V_{\mathcal{D}_1}^{\mathcal{D}_2} |\psi^{\mathcal{D}_1}, 0\rangle &= \left( \sum_a p_a^{\mathcal{D}_1} \otimes \mathbb{1}_{n_a^{\mathcal{D}_2}} \right)^{\otimes N} |\psi^{\mathcal{D}_1}, 0\rangle \\ &= |\psi^{\mathcal{D}_1}, 0\rangle, \end{aligned} \quad (33)$$

meaning that the isometry  $V_{\mathcal{D}_1}^{\mathcal{D}_2}$  is in fact a unitary on the original Hilbert space of the string-net model.

A constant depth quantum circuit is obtained by applying the isometry  $V_{\mathcal{D}_1}^{\mathcal{D}_2}$  in such a way that it is a constant depth operator. For the hexagonal lattice, this can be achieved by dividing the lattice into three sublattices as shown in Fig. 3 and acting sequentially on the first, second, and third sublattice. This implementation of the isometry  $V_{\mathcal{D}_1}^{\mathcal{D}_2}$  turns it into a depth-three quantum circuit. We now discuss this quantum circuit in more detail, first for the ground states, and then for a generic excited state in the original string-net Hilbert space spanned by  $|\psi^{\mathcal{D}_1}, 0\rangle$ .

### A. Ground state

The ground states of the string-net model  $\mathcal{D}_1$ , which we will denote as  $|\psi_{\text{GS}}^{\mathcal{D}_1}, 0\rangle$ , are defined as the eigenvectors of the ground-state projector  $P_{\text{GS}}^{\mathcal{D}_1}$ :

$$\begin{aligned} P_{\text{GS}}^{\mathcal{D}_1} |\psi_{\text{GS}}^{\mathcal{D}_1}, 0\rangle &= (p_{00}^{0,\mathcal{D}_1} \otimes |0\rangle\langle 0|)^{\otimes N} |\psi_{\text{GS}}^{\mathcal{D}_1}, 0\rangle \\ &= |\psi_{\text{GS}}^{\mathcal{D}_1}, 0\rangle, \end{aligned} \quad (34)$$

projecting each plaquette onto the simple idempotent  $p_{00}^{0,\mathcal{D}_1}$  which corresponds to the vacuum; it is equal to the plaquette term of the Hamiltonian,

$$p_{00}^{0,\mathcal{D}_1} = \frac{1}{D} \sum_a d_a \mathcal{B}_p^{a,\mathcal{D}_1}, \quad (35)$$

which also projects onto the ground state of the vertex term by fixing the extended string-net legs to the trivial string. At first glance, one would expect the simple idempotent  $p_{00}^{0,\mathcal{D}_1}$  corresponding to the vacuum to be central; the reason this is not the case is that in the extended string-net model, the simple idempotents keep track of one plaquette and two vertex excitations. If the two vertices are excited with a particle-antiparticle pair then together they are in the vacuum, and the corresponding simple idempotent  $p_{ij}^{0,\mathcal{D}_1}$ ,  $i \neq 0$  will also be contained in the central idempotent  $P_{\frac{1}{2}}^{\mathcal{D}_1}$  corresponding to the vacuum in  $Z(\mathcal{D}_1)$ .

Acting with our quantum circuit on the ground state, we obtain

$$V_{\mathcal{D}_1}^{\mathcal{D}_2} |\psi_{\text{GS}}^{\mathcal{D}_1}, 0\rangle = (p_{00}^{0,\mathcal{M}} \otimes |0\rangle\langle 0|)^{\otimes N} |\psi_{\text{GS}}^{\mathcal{D}_1}, 0\rangle. \quad (36)$$

One can readily verify that

$$\begin{aligned} P_{\text{GS}}^{\mathcal{D}_2} V_{\mathcal{D}_1}^{\mathcal{D}_2} |\psi_{\text{GS}}^{\mathcal{D}_1}, 0\rangle &= (p_{00}^{0,\mathcal{M}} \otimes |0\rangle\langle 0|)^{\otimes N} |\psi_{\text{GS}}^{\mathcal{D}_1}, 0\rangle \\ &= |\psi_{\text{GS}}^{\mathcal{D}_2}, 0\rangle, \end{aligned} \quad (37)$$

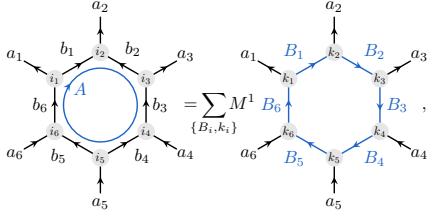
showing that  $V_{\mathcal{D}_1}^{\mathcal{D}_2}$  does indeed provide a unitary map between the ground states of the string-net model  $\mathcal{D}_1$  and  $\mathcal{D}_2$ . The operator  $p_{00}^{0,\mathcal{M}}$  can be interpreted as a generalized plaquette term, in the sense that

$$p_{00}^{0,\mathcal{M}} = \frac{1}{D} \sum_A d_A \mathcal{B}_p^{A,\mathcal{M}}, \quad (38)$$

i.e., it is the result of fusing a weighted sum of all loops labeled by  $A \in \mathcal{M}$  with the plaquette. When applied to the string-net configuration where all edges are fixed to the trivial label (a product state),  $(p_{00}^{0,\mathcal{M}})^{\otimes N}$  projects onto the ground state of the string-net model  $\mathcal{D}_2$ . This operator can be used to derive a projected entangled pair state (PEPS) representation of the string-net ground state, generalizing the constructions in Refs. [22,23] to the tensor network representations obtained in Ref. [18].

More explicitly, the application of  $\mathcal{B}_p^{A,\mathcal{M}}$ ,  $\mathcal{B}_p^{D,\mathcal{M}}$ , and  $\mathcal{B}_p^{G,\mathcal{M}}$  to the three sublattices in the order depicted in Fig. 3 can be worked out by a series of recouplings involving the bimodule

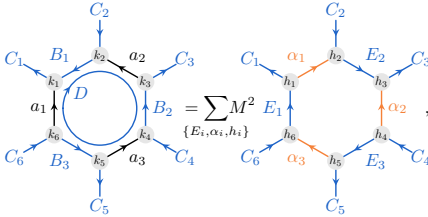
$F$  symbols. For a plaquette of the first kind, we get



where the matrix  $M^1$  depends on  $\{a_i, A\}$ , which we suppress for convenience, and its components read

$$(M^1)_{\{b_i, i_i\}}^{\{B_i, k_i\}} = \sum_{\{j_i\}} \sqrt{\frac{d_{b_1} d_{b_2} d_{b_4} d_{b_5} d_{B_6}}{d_{a_2} d_{a_5} d_{b_3} d_{b_6} d_A^2}} (\triangleright F_{B_6}^{a_1 b_1 A})_{b_6, i_1 j_6}^{B_1, j_1 k_1} \times (\triangleright F_{B_1}^{b_1 b_2 B_2})_{a_2, i_2 k_2}^{A, j_1 j_2} (\triangleright F_{A}^{b_2 a_3 B_3})_{b_3, i_3 j_3}^{B_2, k_3 j_2} (\triangleright F_{A}^{b_4 a_4 B_5})_{b_5, i_4 j_5}^{B_4, k_4 j_4} \times (\triangleright F_{B_5}^{b_5 b_6 B_6})_{a_5, i_5 k_5}^{A, j_4 j_5} (\triangleright F_{B_6}^{a_6 b_6 A})_{b_6, i_6 j_6}^{B_5, j_5 k_6}. \quad (39)$$

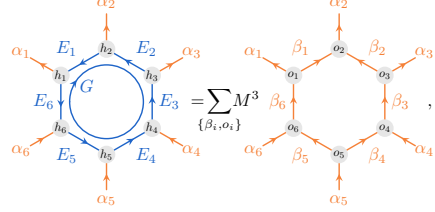
For a plaquette of the second kind, we get



where the matrix  $M^2$  depends on  $\{C_i, D\}$  and its components read

$$(M^2)_{\{a_i, B_i, k_i\}}^{\{E_i, \alpha_i, h_i\}} = \sum_{\{l_i\}} \sqrt{\frac{d_{a_1} d_{a_2} d_{a_3} d_{B_2} d_{E_1} d_{a_4} d_{a_5} d_{a_6}}{d_{C_1} d_{C_3} d_{C_4} d_{C_6} d_D^4}} (\triangleright F_{E_1}^{a_1 B_1 \alpha_1})_{C_1, k_1 l_1}^{D, l_2 l_1} \times (\triangleright \triangleleft F_D^{a_2 C_2 \alpha_2})_{B_1, k_2 l_2}^{E_2, h_2 l_2} (\triangleright \triangleleft F_{B_2}^{a_3 E_2 \alpha_3})_{D, l_3 l_4}^{C_3, h_3 k_3} (\triangleright \triangleleft F_{B_2}^{a_4 E_3 \alpha_4})_{D, l_5 l_4}^{C_4, h_4 k_4} \times (\triangleright \triangleleft F_D^{a_5 C_5 \alpha_5})_{B_3, k_5 l_5}^{E_3, h_5 l_5} (\triangleright \triangleleft F_{E_1}^{a_6 B_3 \alpha_6})_{C_6, k_6 h_6}^{D, l_6 l_1}. \quad (40)$$

Finally, for the third kind of plaquette we get



where the matrix  $M^3$  depends on  $\{\alpha_i, G\}$  and its components read

$$(M^3)_{\{E_i, h_i\}}^{\{\beta_i, \alpha_i\}} \equiv \sum_{\{n_i\}} \sqrt{\frac{d_{\beta_1} d_{\beta_2} d_{\beta_4} d_{\beta_5} d_{E_3} d_{E_6}}{d_{\alpha_2} d_{\alpha_5} d_{\beta_3} d_{\beta_6} d_G^2}} (\triangleleft F_G^{E_6 \alpha_1 \beta_1})_{E_1, h_1 n_1}^{\beta_6, \alpha_1 n_6} \times (\triangleleft F_{E_2}^{E_1 \beta_1 \beta_2})_{G, n_1 n_2}^{\alpha_2, \alpha_2 h_2} (\triangleleft F_{E_3}^{G \beta_2 \alpha_3})_{E_2, n_2 h_3}^{\beta_3, \alpha_3 n_3} (\triangleleft F_{E_3}^{G \beta_4 \alpha_4})_{E_4, n_4 h_4}^{\beta_5, \alpha_4 n_5} \times (\triangleleft F_{E_4}^{E_3 \beta_5 \beta_6})_{G, n_5 n_4}^{\alpha_5, \alpha_5 h_5} (\triangleleft F_G^{E_4 \alpha_6 \beta_5})_{E_5, h_5 n_5}^{\beta_6, \alpha_6 n_6}. \quad (41)$$

The explicit expression for the operator  $(p_{00}^{0, \mathcal{M}})^{\otimes N}$  is then obtained by taking the appropriate linear combinations of the matrices  $M^i$  and acting on their respective sublattices in the order described above. While the local Hilbert space of the two string-net models looks different, they have the same number of ground states, and therefore  $(p_{00}^{0, \mathcal{M}})^{\otimes N}$  is represented as a square matrix. To show that it is also unitary, we note that

$$(p_{00}^{0, \mathcal{M}})^{\dagger} p_{00}^{0, \mathcal{M}} = p_{00}^{0, \mathcal{D}_1}, \quad (42)$$

implying that the Hermitian conjugate of the matrix representation of  $p_{00}^{0, \mathcal{M}}$  should correspond to the matrix representation of  $(p_{00}^{0, \mathcal{M}})^{\dagger}$ . By virtue of Eq. (26),

$$(p_{00}^{0, \mathcal{M}})^{\dagger} = \frac{1}{D} \sum_A d_A \mathcal{B}_p^{A, \overline{\mathcal{M}}}, \quad (43)$$

which implies that the matrix representation of  $(p_{00}^{0, \mathcal{M}})^{\dagger}$  is obtained as a linear combination of matrices  $\overline{M}^i$  obtained in similar fashion as the matrices  $M^i$ . Using the fact that the different  $F$  symbols themselves are unitary matrices since we are dealing with unitary bimodule categories, one can indeed show that  $(M^i)^{\dagger} = \overline{M}^i$  through a cumbersome but straightforward computation. Taking everything together, this implies that  $(p_{00}^{0, \mathcal{M}})^{\otimes N}$  is represented as a unitary matrix on the ground state.

## B. Excited states

When applied to a state with excitations, the mapping provided by our circuit is less obvious since, as stressed before, the dimension of the Hilbert spaces does not match between the string-net model  $\mathcal{D}_1$  and  $\mathcal{D}_2$ . We compensated for this by introducing an ancillary qudit in each plaquette, that we initialize to be in the  $|0\rangle$  state. Looking at the action of  $V_{\mathcal{D}_1}^{\mathcal{D}_2}$

on the space spanned by  $|\psi^{D_1}, 0\rangle$ , we find

$$V_{D_1}^{D_2} |\psi^{D_1}, 0\rangle = \left( \sum_{a,j} p_{0j}^{a,\mathcal{M}} \otimes |j\rangle \langle 0| \right)^{\otimes N} |\psi^{D_1}, 0\rangle. \quad (44)$$

Acting on any plaquette of the resulting state with a simple idempotent  $p_{ii}^{a,D_2}$  only gives a nonzero result if  $i = 0$ , meaning that for every central idempotent  $P_a$  we only get the excitation corresponding to  $p_{00}^a$ . Additionally, this state is no longer part of the original Hilbert space spanned by  $|\psi^{D_2}\rangle$ , as the ancilla  $|j\rangle$  is now entangled with the other degrees of freedom of the string-net model  $\mathcal{D}_2$ .

There are a number of ways to disentangle these ancillas, but due to the mismatch in Hilbert space it is impossible to do so in a unique or unitary way. One option is to act with the operator  $V_{D_2}^{D_2}$ , which gives

$$\begin{aligned} V_{D_2}^{D_2} V_{D_1}^{D_2} |\psi^{D_1}, 0\rangle &= \left( \sum_a \sum_{i=0}^{m_a-1} p_{ii}^{a,\mathcal{M}} \otimes |0\rangle \langle 0| \right)^{\otimes N} |\psi^{D_1}, 0\rangle \\ &= |\psi^{D_2}, 0\rangle, \end{aligned} \quad (45)$$

where  $m_a = \min(n_a^{D_1}, n_a^{D_2})$ . This state is again part of the original Hilbert space, and the ancilla can be discarded. If  $m_a = n_a^{D_1}$ , not all excitations of type  $a \in Z(D_2)$  in the string-net model  $\mathcal{D}_2$  are in the image of this map, while if  $m_a = n_a^{D_2}$  some excitations of the string-net model  $\mathcal{D}_1$  are projected out, all depending on the choice of enumeration of the simple idempotents. A different approach might be to make sure the energy of the state is conserved when mapping back to the original Hilbert space; we leave the exploration of the different possibilities to future work.

An explicit expression generalizing the matrices  $M^i$  to the case of excitations is given in the Appendix. Unitarity of the full circuit can again be shown from unitarity of the  $F$  symbols and the fact that the Hermitian conjugate of the bimodule tubes defined in Eq. (26) coincides with the Hermitian conjugate of the matrix representation of the corresponding operator, by definition.

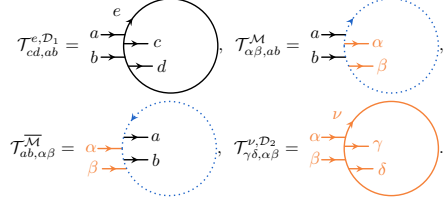
## V. EXAMPLES

To illustrate the simple idempotents and simple bimodules that build up our circuit, we write them down for the case of a constant depth circuit that maps the toric code to itself. While this example is almost trivial, it provides enough insight to understand the more general case. We also present the required categorical data for the generic case of a quantum double.

### A. Toric code to toric code

The toric code [14] is built from the fusion category  $\mathcal{D}_1 = \text{Vec}_{\mathbb{Z}_2}$ , the category of  $\mathbb{Z}_2$  graded vector spaces with two simple objects  $\{0, 1\}$  that fuse according to  $\mathbb{Z}_2$ . It only has one nontrivial module category  $\mathcal{M} = \text{Vec}$ , the category of finite-dimensional vector spaces with a single simple object  $\mathbb{C}$  from which one can construct an invertible bimodule category to  $\mathcal{D}_2 = \text{Rep}(\mathbb{Z}_2) \simeq \text{Vec}_{\mathbb{Z}_2}$ . We can use this bimodule category to construct a constant depth quantum circuit that maps the

toric code to itself, but is not completely trivial. The relevant tubes of this bimodule category can be written as



The central idempotents with their corresponding simple idempotents and nilpotents of the tube algebra of  $\mathcal{D}_1$  are labeled by the four excitations in the toric code. These are the vacuum  $\mathbb{1}$ :

$$\begin{aligned} p_{00}^{\mathbb{1},D_1} &= \frac{1}{2}(\mathcal{T}_{00,00}^{0,D_1} + \mathcal{T}_{00,00}^{1,D_1}), & p_{01}^{\mathbb{1},D_1} &= \frac{1}{2}(\mathcal{T}_{00,11}^{0,D_1} + \mathcal{T}_{00,11}^{1,D_1}), \\ p_{10}^{\mathbb{1},D_1} &= \frac{1}{2}(\mathcal{T}_{11,00}^{0,D_1} + \mathcal{T}_{11,00}^{1,D_1}), & p_{11}^{\mathbb{1},D_1} &= \frac{1}{2}(\mathcal{T}_{11,11}^{0,D_1} + \mathcal{T}_{11,11}^{1,D_1}), \end{aligned}$$

the electric excitation  $e$ :

$$\begin{aligned} p_{00}^{e,D_1} &= \frac{1}{2}(\mathcal{T}_{01,01}^{0,D_1} + \mathcal{T}_{01,01}^{1,D_1}), & p_{01}^{e,D_1} &= \frac{1}{2}(\mathcal{T}_{01,10}^{0,D_1} + \mathcal{T}_{01,10}^{1,D_1}), \\ p_{10}^{e,D_1} &= \frac{1}{2}(\mathcal{T}_{10,01}^{0,D_1} + \mathcal{T}_{10,01}^{1,D_1}), & p_{11}^{e,D_1} &= \frac{1}{2}(\mathcal{T}_{10,10}^{0,D_1} + \mathcal{T}_{10,10}^{1,D_1}), \end{aligned}$$

the magnetic excitation  $m$ :

$$\begin{aligned} p_{00}^{m,D_1} &= \frac{1}{2}(\mathcal{T}_{00,00}^{0,D_1} - \mathcal{T}_{00,00}^{1,D_1}), & p_{01}^{m,D_1} &= \frac{1}{2}(\mathcal{T}_{00,11}^{0,D_1} - \mathcal{T}_{00,11}^{1,D_1}), \\ p_{10}^{m,D_1} &= \frac{1}{2}(\mathcal{T}_{11,00}^{0,D_1} - \mathcal{T}_{11,00}^{1,D_1}), & p_{11}^{m,D_1} &= \frac{1}{2}(\mathcal{T}_{11,11}^{0,D_1} - \mathcal{T}_{11,11}^{1,D_1}), \end{aligned}$$

and the fermion  $f$ :

$$\begin{aligned} p_{00}^{f,D_1} &= \frac{1}{2}(\mathcal{T}_{01,01}^{0,D_1} - \mathcal{T}_{01,01}^{1,D_1}), & p_{01}^{f,D_1} &= \frac{1}{2}(\mathcal{T}_{01,10}^{0,D_1} - \mathcal{T}_{01,10}^{1,D_1}), \\ p_{10}^{f,D_1} &= \frac{1}{2}(\mathcal{T}_{10,01}^{0,D_1} - \mathcal{T}_{10,01}^{1,D_1}), & p_{11}^{f,D_1} &= \frac{1}{2}(\mathcal{T}_{10,10}^{0,D_1} - \mathcal{T}_{10,10}^{1,D_1}). \end{aligned}$$

These central idempotents are all built from two simple idempotents, reflecting the fact that these tube algebras describe the excitations of two vertices and one plaquette. For instance, the simple idempotent  $p_{00}^{\mathbb{1},D_1}$  corresponds to the true vacuum, where neither the vertices nor the plaquette is excited. The other simple idempotent  $p_{11}^{\mathbb{1},D_1}$  represents the case where both vertices are excited; together, they are in the vacuum, and therefore they also contribute to the vacuum central idempotent. The other central idempotents follow a similar pattern. The simple idempotents and nilpotents of the tube algebra of  $\mathcal{D}_2$  are related to those of the tube algebra of  $\mathcal{D}_1$  as follows:

$$\begin{aligned} p_{ij}^{\mathbb{1},D_2} &= p_{ij}^{\mathbb{1},D_1}, & p_{ij}^{e,D_2} &= p_{ij}^{m,D_1}, \\ p_{ij}^{m,D_2} &= p_{ij}^{e,D_1}, & p_{ij}^{f,D_2} &= p_{ij}^{f,D_1}. \end{aligned}$$

We note that the electric and magnetic excitations are swapped, which is a manifestation of the  $e$ - $m$  duality in the toric code. From these idempotents and nilpotents, one can compute the simple bimodules of  $\mathcal{M}$ . We find

$$\begin{aligned} p_{00}^{\mathbb{1},\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{00,00}^{\mathcal{M}}, & p_{01}^{\mathbb{1},\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{00,11}^{\mathcal{M}}, \\ p_{10}^{\mathbb{1},\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{11,00}^{\mathcal{M}}, & p_{11}^{\mathbb{1},\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{11,11}^{\mathcal{M}}, \end{aligned}$$

for the vacuum,

$$\begin{aligned} p_{00}^{e,\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{00,01}^{\mathcal{M}}, & p_{01}^{e,\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{00,10}^{\mathcal{M}}, \\ p_{10}^{e,\mathcal{M}} &= \frac{-1}{\sqrt{2}} \mathcal{T}_{11,01}^{\mathcal{M}}, & p_{11}^{e,\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{11,10}^{\mathcal{M}}, \end{aligned}$$

for the electric excitation,

$$\begin{aligned} p_{00}^{m,\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{01,00}^{\mathcal{M}}, & p_{01}^{m,\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{01,11}^{\mathcal{M}}, \\ p_{10}^{m,\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{10,00}^{\mathcal{M}}, & p_{11}^{m,\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{10,11}^{\mathcal{M}}, \end{aligned}$$

for the magnetic excitation and

$$\begin{aligned} p_{00}^{f,\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{01,01}^{\mathcal{M}}, & p_{01}^{f,\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{01,10}^{\mathcal{M}}, \\ p_{10}^{f,\mathcal{M}} &= \frac{-1}{\sqrt{2}} \mathcal{T}_{10,01}^{\mathcal{M}}, & p_{11}^{f,\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{10,10}^{\mathcal{M}}, \end{aligned}$$

for the fermion. The simple bimodules of  $\overline{\mathcal{M}}$  are obtained by Hermitian conjugation of the simple bimodules of  $\mathcal{M}$ :

$$\begin{aligned} p_{ji}^{1,\overline{\mathcal{M}}} &= (p_{ij}^{1,\mathcal{M}})^\dagger, & p_{ji}^{e,\overline{\mathcal{M}}} &= (p_{ij}^{e,\mathcal{M}})^\dagger, \\ p_{ji}^{m,\overline{\mathcal{M}}} &= (p_{ij}^{m,\mathcal{M}})^\dagger, & p_{ji}^{f,\overline{\mathcal{M}}} &= (p_{ij}^{f,\mathcal{M}})^\dagger. \end{aligned}$$

### B. Ground states of Kitaev's quantum doubles to string nets

The previous example of the toric code can now be generalized to a circuit mapping between quantum doubles and string-net ground states. In those examples the group  $\mathbb{Z}_2$  is replaced by an arbitrary finite group  $G$ . This then amounts to choosing the UFC  $\mathcal{D}_1$  to be  $\mathcal{D}_1 = \text{Vec}_G$ , the category of all finite-dimensional  $G$ -graded vector spaces which is the input category of Kitaev's quantum doubles [14]. The simple objects of  $\text{Vec}_G$  are in one-to-one correspondence with the group elements of  $G$  such that the fusion of elements in  $\text{Vec}_G$  is then equivalent to the group multiplication in  $G$ . The ground states of these quantum doubles can then be mapped to string nets with input category  $\mathcal{D}_2 = \text{Rep}(G)$ , the category of all representations of  $G$  in which the simple objects are the irreducible representations. The invertible bimodule category connecting  $\mathcal{D}_1$  and  $\mathcal{D}_2$  is  $\mathcal{M} = \text{Vec}$ , as in the case of the toric code. A crucial difference with the previous example is that for a generic group, there is no monoidal equivalence between  $\text{Vec}_G$  and  $\text{Rep}(G)$ . The data of this bimodule category is as follows [18]:

(i)  $(\mathcal{D}_1 F_{g_1, g_2, g_3}^{g_1, g_2, g_3})_{g_{12}, 11}^{g_{23}, 11} = \omega(g_1, g_2, g_3)$  where  $\omega$  is a three-cocycle belonging to the trivial class. One can choose a gauge of the  $\mathcal{D}_1 F$  symbol in which this cocycle is identically one,  $\omega(g_1, g_2, g_3) \equiv 1, \forall g_1, g_2, g_3 \in G$ .

(ii) Writing  $(\mathcal{D}_1 F_{C, 11}^{g_1, g_2, C})_{g_{12}, 11} = \psi(g_1, g_2)$ , it can be shown that by virtue of the pentagon equation between  $\mathcal{D}_1 F$  and  $\mathcal{D}_2 F$ ,  $\psi$  is a two-cocycle classified by  $H^2(G, U(1))$ .

(iii)  $(\mathcal{D}_2 F_{C, 11}^{g, j, 1})_{C, 11} = D^g(g_i^j)$  where  $D^g(g)$  is the matrix representation of the group element  $g \in G$  in the irreducible representation  $\alpha$  and where  $i, j$  are  $d_\alpha$ -dimensional indices in that representation.

(iv)  $(\mathcal{F}_C^{C\alpha_1\alpha_2})_{C, i_1 i_2}^{\alpha_3, k i_3} = c_{i_1 i_2 i_3, k}^{\alpha_1 \alpha_2 \alpha_3}$  are the Clebsch-Gordan coefficients of the group. In this notation  $k$  denotes the possible number of ways in which  $\alpha_1$  and  $\alpha_2$  can fuse to  $\alpha_3$ .

(v) The  $\mathcal{D}_2 F$ -symbols can then be recognized as the Racah  $W$  coefficients which are equal to the  $6j$  symbols of  $G$ . These  $6j$  symbols serve as the input of the  $\text{Rep}(G)$  string net.

Substituting these  $F$  symbols in the expressions of the quantum circuit then results in a circuit that allows one to map the ground states of all of Kitaev's quantum double models to corresponding string nets. A similar circuit was found in Refs. [15,16]. In their map the  $\text{Rep}(G)$  degrees of freedom where introduced by means of a Fourier transform on  $G$ . In our circuit these irreps appear through the  $\mathcal{D}_2 F$  symbol of the bimodel category. We also recover the  $6j$  symbols that appeared in their construction in the form of the  $\mathcal{D}_2 F$  symbols. We note that the quantum doubles as defined in Refs. [16,22] are defined on a triangular lattice and are shown to be equivalent to the string-net model on the hexagonal lattice; our construction does not change the lattice, and a direct comparison is therefore somewhat cumbersome.

## VI. CONCLUSION AND OUTLOOK

In this work, we have constructed a constant depth quantum circuit that is able to map between Morita-equivalent ground states. Crucially, the circuit is unitary on the full Hilbert space, meaning that it can be implemented in a truly local way without performing a measurement as would be required to construct the string-net state from a product state. We have shown that the mapping is exact for the ground states, which implies that they are in the same phase as they can be adiabatically connected without closing a gap.

One immediate application of our circuit exploits the recent insight that string-net models can be used to construct partition functions of classical statistical mechanics models [24,25]. From the tensor network point of view, this is understood as the strange correlator [26], which boils down to taking the overlap of the string-net ground-state PEPS with some unentangled product state to obtain a tensor network representation of a classical partition function. The existence of a quantum circuit between Morita-equivalent string nets allows one to construct different lattice models with the same partition functions. We plan to explore this further in future work.

A natural generalization would be to no longer restrict to closed manifolds but also include the case of boundaries to the vacuum or domain walls between different string-net models. In general this requires the use of a four-object bicategory containing four fusion categories and six invertible bimodule categories [18], the recoupling theory of which has not been written down explicitly in the literature.

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## APPENDIX: THE QUANTUM CIRCUIT FOR THE EXTENDED STRING NET

$$\begin{aligned}
 & \left[ M^1_{\{a_i, d_i, T_A\}} \right]_{\{B_i, k_i\}} \equiv \sum_{\{j_i\}} \left( \frac{d_{b_1} d_{b_2} d_{b_3} d_{b_4} d_{c_1} d_{c_2} d_{A_1} d_{A_2} d_{B_3} d_{B_4}}{d_A^3 d_{a_2} d_{a_3} d_{b_5} d_{b_6} d_{A_2}} \right)^{1/2} \left( \varphi_{F_{B_{10}}^{b_{10} c_1 A_3}} \right)^{A_1, i_{14} j_{12}} \left( \varphi_{F_{B_9}^{b_9 A_3 \gamma_1}} \right)^{A_2, i_{13} j_{10}} \\
 & \times \left( \varphi_{F_{B_8}^{b_8 c_2 A_1}} \right)^{A_2, i_{12} j_{10}} \left( \varphi_{F_{B_8}^{b_8 A_1 \gamma_2}} \right)^{A_3, i_{11} j_8} \left( \varphi_{F_{B_{10}}^{a_1 b_1 A}} \right)^{B_1, j_1 k_1} \left( \varphi_{F_{B_1}^{b_1 b_2 B_2}} \right)^{A_1, j_1 j_2} \left( \varphi_{F_A^{b_2 a_3 B_3}} \right)^{B_2, k_3 j_2} \\
 & \times \left( \varphi_{F_A^{b_4 d_2 B_5}} \right)^{B_4, k_5 j_4} \left( \varphi_{F_A^{b_3 d_1 B_4}} \right)^{B_3, k_4 j_3} \left( \varphi_{F_A^{b_6 a_4 B_5}} \right)^{B_6, k_6 j_6} \left( \varphi_{F_{B_7}^{b_7 b_6 B_6}} \right)^{A_1, j_6 j_7} \left( \varphi_{F_{B_8}^{a_6 b_7 A}} \right)^{B_7, j_7 k_8} \\
 & \times \left( \varphi_{F_{B_8}^{b_8 a_5 B_5}} \right)^{B_8, i_8 j_8}
 \end{aligned}$$

7

$$\begin{aligned}
 & \left[ M^2_{\{C_i, \gamma_i, T_D\}} \right]_{\{E_i, \alpha_i, m_i\}} \equiv \sum_{\{l_i\}} \left( \frac{d_{a_1} d_{a_2} d_{a_3} d_{c_1} d_{c_2} d_{D_1} d_{D_2} d_{B_3} d_{E_3} d_{E_4} d_{\alpha_1} d_{\alpha_2} d_{\alpha_3}}{d_D^2 d_{C_1} d_{C_3} d_{C_4} d_{C_6} d_{D_2} d_{E_4}} \right)^{1/2} \left( \varphi_{F_{E_5}^{a_5 c_1 D_3}} \right)^{D_1, k_{14} l_{12}} \left( \varphi_{F_{E_4}^{a_4 D_3 \beta_1}} \right)^{D_2, k_{13} l_{10}} \\
 & \times \left( \varphi_{F_{E_4}^{a_4 c_2 D_1}} \right)^{D_2, k_{12} l_{10}} \left( \varphi_{F_{E_3}^{a_3 D_1 \beta_2}} \right)^{D_1, k_{11} l_8} \left( \varphi_{F_{E_5}^{a_5 B_1 \alpha_1}} \right)^{D_1, l_1 l_{12}} \left( \varphi_{F_D^{a_1 C_2 \alpha_1}} \right)^{E_1, m_2 l_2} \left( \varphi_{F_{B_2}^{a_1 E_1 \alpha_2}} \right)^{C_3, m_3 k_3} \\
 & \times \left( \varphi_{F_{B_3}^{D \alpha_3 \gamma_1}} \right)^{a_3, m_4 l_4} \left( \varphi_{F_{B_4}^{D \alpha_3 \gamma_2}} \right)^{a_4, m_5 l_5} \left( \varphi_{F_{B_4}^{a_2 E_2 \alpha_4}} \right)^{C_4, m_6 k_6} \left( \varphi_{F_D^{a_2 C_5 \alpha_5}} \right)^{E_2, m_7 l_6} \left( \varphi_{F_{E_3}^{a_3 B_5 \alpha_5}} \right)^{D_1, l_7 l_8} \\
 & \times \left( \varphi_{F_{E_3}^{a_3 B_5 \alpha_5}} \right)^{C_6, k_8 m_8}
 \end{aligned}$$



$$\begin{aligned}
& \sum_{\{\delta_i, o_i\}} \left[ M_{\{\alpha_i, \gamma_i, G\}}^3 \right]_{\{E_i, m_i\}}^{\{\delta_i, o_i\}} \\
& \equiv \sum_{\{n_i\}} \left( \frac{d d_1 d_2 d_{E_1} d_{E_2} d_{E_3} d_{E_5} d_{E_6} d_{E_7} d_{E_8}^2 d_{E_{10}} d_{G_2} d_{\delta_1} d_{\delta_2} d_{\delta_6} d_{\delta_7}}{d_G^2 d_{G_1} d_{G_3} d_{\alpha_2} d_{\alpha_5} d_{\delta_8}} \right)^{1/2} \left( \bowtie_{F_G^{d_1 E_9 \delta_{10}}}^{G_3, n_9 m_{14}} \right)_{E_{10}, m_{10} n_{10}} \\
& \times \left( \bowtie_{G_2^{E_9 \delta_{10} \delta_1}}^{\delta_9, o_{10} n_8} \right)_{G_3, n_9 m_{13}} \left( \bowtie_{G_2^{d_2 E_8 \delta_9}}^{G_1, n_{11} m_{12}} \right)_{E_9, m_9 n_{12}} \left( \bowtie_{G_1^{E_8 \delta_9 \delta_{12}}}^{\delta_8, o_9 n_{10}} \right)_{G_1, n_{11} m_{11}} \left( \bowtie_{E_1^{E_{10} \alpha_1 \delta_1}}^{\delta_{10}, o_1 n_{12}} \right)_{E_1, m_1 n_1} \left( \bowtie_{G_2^{E_1 \delta_1 \delta_2}}^{\alpha_2, o_2 m_2} \right)_{G, n_1 n_2} \\
& \times \left( \bowtie_{E_3^{G \delta_2 \alpha_3}}^{\delta_3, o_3 n_3} \right)_{E_2, n_2 m_3} \left( \bowtie_{E_3^{G_4 \delta_1}}^{\delta_4, o_4 n_4} \right)_{E_3, n_3 m_4} \left( \bowtie_{E_5^{G \delta_4 \gamma_2}}^{\delta_5, o_5 n_5} \right)_{E_4, n_4 m_5} \left( \bowtie_{E_6^{G \delta_6 \alpha_4}}^{\delta_6, o_6 n_6} \right)_{E_6, n_6 m_6} \left( \bowtie_{G_7^{E_7 \delta_7 \delta_6}}^{\alpha_5, o_7 m_7} \right)_{G, n_7 n_6} \left( \bowtie_{E_7^{m_8 n_7}}^{G_8, o_8 n_8} \right)_{E_7, m_8 n_7}
\end{aligned}$$

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# From gauging to duality in one-dimensional quantum lattice models

## 8.1 Synopsis

It is well known that gaugings of  $\mathcal{C}$ -symmetric two-dimensional quantum field theories are encoded by (Frobenius) *algebra objects*  $A$  in  $\mathcal{C}$  [132, 154]. These algebra objects specify both a gaugeable collection of topological lines and a generalised notion of *discrete torsion* through their multiplication map. In this project, we extend the gauging map of Haegeman et al. [139], initially defined for non-anomalous group symmetries, to incorporate genuine categorical symmetries and show that a choice of algebra object precisely captures the data required for this generalisation. Our construction recovers the results of ref. [139] for the symmetry category  $\text{Vec}_G$ , and agrees with a parallel work that discusses gauging algebra objects on the lattice, where they overlap [131]. We further demonstrate that, up to constant-depth unitary quantum circuits, the resulting gauging maps are equivalent to the duality intertwiners introduced in refs. [97, 116] and discussed in section 4.2. In addition, we consider the inclusion of symmetry-twisted boundary conditions, present several non-trivial examples, and prove the existence of a symmetric and injective matrix product state when  $\mathcal{C}$  describes an anomaly-free symmetry and is thus fully gaugeable.

## 8.2 Reference

*From gauging to duality in one-dimensional quantum lattice models*

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## 8.3 Contributions of the author

This project was conceived by José Garre-Rubio and Laurens Lootens. My main contributions consisted of working out the technical details, completing and systematising the derivations, and incorporating the examples. I wrote the majority of the manuscript, with the exception of section 5 which is due to Laurens Lootens.

## From gauging to duality in one-dimensional quantum lattice models

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### Abstract

Gauging and duality transformations, two of the most useful tools in many-body physics, are shown to be equivalent up to constant depth quantum circuits in the case of one-dimensional quantum lattice models. This is demonstrated by making use of matrix product operators, which provide the lattice representation theory for global (categorical) symmetries as well as a classification of duality transformations. Our construction makes the symmetries of the gauged theory manifest and clarifies how to deal with static background fields when gauging generalised symmetries.



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## 1 Introduction

Recent years have witnessed an increased interest in generalised notions of finite global symmetry in quantum theories [23, 29]. These generalisations encompass in particular *non-invertible* symmetries in which the algebraic structure governing the symmetry and its 't Hooft anomaly is no longer a group, but instead a (*higher*) *fusion category* [25, 62, 67, 68, 73]. Such symmetries occur ubiquitously in both the continuum [25, 29, 47, 73] as well as in lattice models [3, 35, 50, 56, 58, 69, 70, 74, 78]. The focus of this manuscript is on fusion categorical symmetries in one-dimensional lattice models.

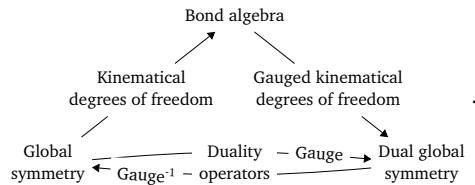
Based on tensor network representations of string-net ground state models [4, 9, 18, 30, 54, 80] – which are a particular case of the Turaev-Viro-Barrett-Westbury state-sum construction with symmetry category  $\mathcal{C}$  as input – explicit *matrix product operator* (MPO) representations of fusion categorical symmetries have been classified and realised [9, 12, 69, 70, 74, 82]. Succinctly, for a categorical symmetry  $\mathcal{C}$  each choice of *module category*  $\mathcal{R}$  over it provides an explicit MPO representation of  $\mathcal{C}$  [21, 62]. Subsequently, for the chosen module category, local operators that commute with the categorical symmetry can be built from *generalised Clebsch-Gordan coefficients* [8], thereby providing a systematic machinery to construct local lattice Hamiltonians with a certain categorical symmetry. In particular, the *algebra of local symmetric operators* [15, 42, 44] is encapsulated in the *Morita dual* fusion category  $\mathcal{C}_{\mathcal{R}}^*$ , and is thus fixed by the choice of symmetry and its representation. Generically these MPOs and local Hamiltonians are represented on Hilbert spaces without local tensor product structure, but which rather obey local constraints [1, 7, 28].

Given such a  $\mathcal{C}$ -symmetric model, it was demonstrated in a recent series of papers [50, 51], how different choices of module categories  $\mathcal{M}$  over  $\mathcal{C}$  lead to distinct *dual* theories. Notable examples of such exact lattice dualities include Kramers-Wannier, Jordan-Wigner and Kennedy-Tasaki duality [45, 48, 49, 61]. Under this operation the symmetry of the dual model is given by the Morita dual of  $\mathcal{C}$  with respect to  $\mathcal{M}$ . Physically, these categorical dualities are characterized by the fact that *local symmetric* operators are mapped to *dual local* operators which commute with the dual symmetry and *charged local* operators are mapped to *string operators* [15, 59]. From the skeletal data of  $\mathcal{M}$ , MPOs can be constructed which intertwine the dual models.

On the other hand, it is known that gaugings of  $\mathcal{C}$ -symmetric models in two spacetime dimensions are given by *special haploid symmetric Frobenius algebra objects*  $A$  in  $\mathcal{C}$  [25, 62, 66]. Such an algebra object encodes both a collection of symmetry lines that are gauged as well as a choice of *discrete torsion*. On the level of the partition function the gauged model is obtained

from the ungauged partition function by inserting a network of lines labelled by  $A$  on a dual triangulation of the underlying manifold [11, 16, 19, 72]. The defining properties of  $A$  then guarantee invariance of the gauged partition functions under recouplings of the  $A$  defects. In this setting it is understood that the topological defects of the gauged model correspond to *bimodules* over  $A$ , encoded in the fusion category  $\text{Bimod}_{\mathcal{C}}(A)$  [11, 25]. Notably, the gauging is not determined by  $A$  itself, but rather by its *Morita class*. Indeed, given  $A$ , the category  $\text{Mod}_{\mathcal{C}}(A)$  of  $A$ -modules can be constructed, which constitutes a module category over  $\mathcal{C}$ . As such, different algebra objects can produce the same category of modules, which ultimately dictates the gauging.

In this work, we make this connection between dualities and gauging of categorical symmetries via algebra objects explicit. Given a symmetry  $\mathcal{C}$  and MPO representation specified by  $\mathcal{R}$ , we construct a *gauging map* following the approach taken in refs. [39, 71]. More precisely, from a chosen  $A$  we construct a local *Gauss law*, from which in turn a global projector onto the gauge-invariant subspace is made. This point-of-view provides an interpretation of the defects contained in  $A$  as being the *gauge degrees of freedom* for the symmetry, in accordance with ref. [24, 71]. We then proceed by showing that the gauging map is equivalent to an MPO duality operator of ref. [50, 51] by means of a constant-depth unitary quantum circuit. This circuit maps the gauging map into an *effective Hilbert space* characterized by local constraints encoded in the module category  $\mathcal{R}'$  over  $\mathcal{C}_{\mathcal{R}}^*$ . As detailed in the main text,  $\mathcal{R}'$  is fully specified by the choice of  $\mathcal{M}$  and  $\mathcal{R}$ . As such, the dual global symmetry is manifestly given by  $\mathcal{C}_{\mathcal{M}}^*$ , which is equivalent to  $\text{Bimod}_{\mathcal{C}}(A)$  [11, 53]. The procedure can be summarized in a triangle diagram:



This diagram commutes in the sense that fixing the module category  $\mathcal{R}$  representing the global symmetry  $\mathcal{C}$  on the lattice – corresponding to the left side of the triangle – and a duality  $\mathcal{M}$  – the bottom line – the dual symmetry and its lattice representation is fixed. Also note that the bond algebra of local symmetric operators, encoded in  $\mathcal{C}_{\mathcal{R}}^*$ , is shared by both the ungauged and the gauged model.

The dualities of ref. [50, 51] are isometries only when the Hilbert space is supplemented with an extra degree of freedom representing the *symmetry-twisted boundary condition*. In the presence of such twists, the Hilbert space decomposes in *topological sectors* which are permuted under the duality mappings [6, 55, 64, 65]. We also provide a version of our gauging map in the presence of such a twist and demonstrate how it maps to a collection of duality *tubes*.

Our results can be extended in a number of ways. For one, the algebra objects considered in this manuscript admit higher-dimensional generalisations. In particular for the case of two spatial dimensions, it is expected that higher-dimensional generalisations of the gauging map [39, 58, 77] can be related to the previously constructed duality maps of ref. [20] via a circuit generalising the one presented here. On the other hand, throughout the manuscript we restrict to internal symmetries. A natural question is how spacetime symmetries such as reflection or time-reversal symmetry fit in the framework. This would require addressing the notion of orientation-reversing domain walls and their Gauss law. We also expect that in this case the tensor network formalism will provide insight, given existing partial progress in this direction [17, 43, 75].

**Organisation of the manuscript:** We begin sec. 2 by reviewing generic properties of the MPO symmetries considered in this manuscript, before proceeding to the construction of the gauging map from a choice of algebra object. To this end we define a generalised commuting Gauss law from which a global projector onto the gauge-invariant subspace is defined. In sec. 3 we review the construction of duality MPOs and show the unitary equivalence with the gauging map in sec. 4. In section 5 we comment on the case of full gaugeability and how this implies the existence of a short-range entangled symmetric state, and the absence of an anomaly. We then provide a gauging map acting on symmetry-twisted boundary conditions in sec. 6. Algebra objects and their corresponding module categories for the category of representations of the symmetric group  $S_3$  and the Haagerup fusion categories are presented in sec. 7. A review of the relevant category theory used throughout the manuscript is relegated to the appendix. A guide to our notation is provided in the following table.

Table 1: Overview of the key physical concepts, corresponding categorical structures, and their notation used throughout the manuscript.

| Physical notion                       | Categorical notion                             | Notation                                                           |
|---------------------------------------|------------------------------------------------|--------------------------------------------------------------------|
| Global symmetry ungauged model        | Fusion category                                | $\mathcal{C}$                                                      |
| Kinematical degrees of freedom        | $\mathcal{C}$ -module category                 | $\mathcal{R}$                                                      |
| Algebra symmetric operators           | Morita dual fusion category                    | $\mathcal{C}_{\mathcal{R}}^*$                                      |
| Gaugeable subsystem                   | Frobenius algebra in $\mathcal{C}$             | $A$                                                                |
| Duality operators                     | $\mathcal{C}$ -module category                 | $\mathcal{M} \simeq \text{Mod}_{\mathcal{C}}(A)$                   |
| Gauged kinematical degrees of freedom | $\mathcal{C}_{\mathcal{M}}^*$ -module category | $\mathcal{R}'$                                                     |
| Global symmetry gauged model          | Morita dual fusion category                    | $\mathcal{C}_{\mathcal{M}}^* \simeq \text{Bimod}_{\mathcal{C}}(A)$ |

## 2 Gauging procedure for algebra objects

We construct a gauging map for any finite symmetry described by a unitary fusion category  $\mathcal{C}$  from a choice of haploid symmetric separable Frobenius algebra object in  $\mathcal{C}$ . Such an algebra object fully characterises an anomaly-free subset of symmetries in  $\mathcal{C}$  together with a choice of generalised discrete torsion. This gauging map directly generalises the one from ref. [39], which was formulated for an anomaly-free invertible symmetry.

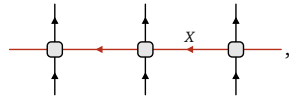
In this manuscript we are concerned with generalised finite symmetries described by *unitary fusion categories* (UFCs). It was shown in ref. [54] that any UFC admits a faithful and internal matrix product operator (MPO) representation of finite bond dimension provided an extra collection of data encoded in a choice of (left) *module category* over the symmetry category. Since the proposed generalised gauging procedure is oblivious to the concrete lattice realisation of the symmetry, we first review the main properties of the symmetry MPOs before turning to the construction of the gauging map.

Throughout this manuscript we rely heavily on the graphical calculus for tensor networks in which a generic MPO tensor is depicted as

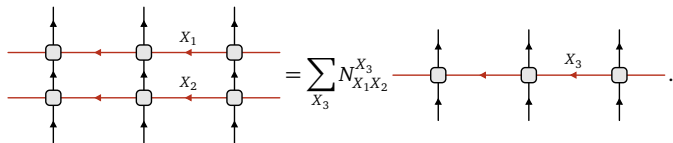
$$\begin{array}{c} \updownarrow \\ \text{---} \square \text{---} \\ \updownarrow \end{array} \stackrel{X}{=} \sum_{\substack{i,j \\ \alpha,\beta}} \alpha \begin{array}{c} \updownarrow \\ \text{---} \square \text{---} \\ \updownarrow \end{array} \beta \quad |\alpha\rangle\langle\beta| \otimes |i\rangle\langle j|. \quad (1)$$



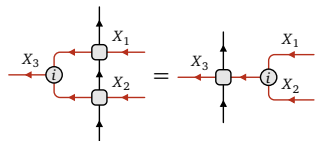
We refer to  $i, j$  and  $\alpha, \beta$  in the above equation as *physical* and *virtual indices* respectively, borrowing standard tensor network language. All vector spaces we consider are finite-dimensional. In this depiction, and below, the label  $X$  is used to both denote the MPO tensor and the virtual space. A uniform and periodic MPO can then be constructed by concatenating individual MPO tensors and contracting basis vectors on the virtual level. In the presence of symmetry twists the symmetry operators have to be modified in a way that is explained below in sec. 6. Graphically, the periodic matrix product operators are


(2)

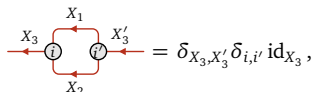
with the convention that the left and rightmost virtual indices are identified and summed over so as to implement the periodic boundary condition. Note that the (closed) MPOs are invariant under gauge transformations on the virtual level. From now on we assume without loss of generality that, potentially after carrying out such a gauge transformation, the symmetry MPOs are block-injective, meaning that the MPO tensors are block diagonal and each block generates an injective symmetry MPO [38]. At the level of the virtual spaces, this boils down to a block decomposition of the form  $X \simeq \bigoplus_i N_{X_i}^X X_i$ , where some of the blocks may appear multiple times given by the integer  $N_{X_i}^X$ , and the total number of distinct blocks is finite. Furthermore we assume that the MPOs are closed under composition. Concretely, this means that for any triple of blocks  $X_1, X_2, X_3$  there exists an integer denoted by  $N_{X_1 X_2}^{X_3}$ , referred to as *N-symbol*, such that independent of system size one has:


(3)

Herein, the sum is interpreted so as to run over all injective blocks with the convention that  $N_{X_1 X_2}^{X_3} = 0$  whenever  $X_3$  does not appear in the block decomposition of  $X_1 \otimes X_2$ . Among the injective blocks there is a distinguished one labelled by  $\mathbb{1}$  such that  $N_{X_1 \mathbb{1}}^{X_2} = N_{\mathbb{1} X_1}^{X_2} = \delta_{X_1, X_2}$  for all  $X_1, X_2$ . As such,  $N$  defines a *fusion structure* on the collection of blocks  $\{X_i\}_i$ . Further requiring that (3) holds for all boundary conditions, the fundamental theorem of MPOs implies there exists an intertwining *fusion tensor* that implements the above decomposition on the level of the individual MPO tensors [9]. Graphically, this intertwining property can be depicted as


(4)

Herein,  $i = 1, \dots, N_{X_1 X_2}^{X_3}$  labels the  $N_{X_1 X_2}^{X_3}$  linearly independent fusion tensors and corresponds to the multiplicity of the block  $X_3$  in the decomposition of  $X_1$  and  $X_2$ . We are also guaranteed the existence of a corresponding right inverse, or *splitting tensor*, satisfying the orthogonality condition


(5)

so that the multiplicity indices  $i$  and  $i'$  label dual orthogonal bases for the *fusion* and *splitting* space situated on the corresponding tensors, as well as a completeness condition spelled out in [9]. As demonstrated therein, associativity of the multiplication of three injective MPOs requires the existence of a set of invertible matrices, henceforth called *F-symbols*, that encode the two distinct ways in which three injective MPOs can be recoupled. Graphically this statement boils down to the recoupling identity

$$\text{Diagram (6)} = \sum_{X_6, k, l} (F_{X_4}^{X_1 X_2 X_3})_{X_5, i, j}^{X_6, k, l} \text{Diagram (6)} \quad (6)$$

which has to be satisfied for every choice of the external labels  $X_1, X_2, X_3$  and  $X_5, i, j$ . Consistency of the recoupling of four MPOs reveals that the F-symbols need to satisfy a coherence condition known as the *pentagon equation*, which in turn guarantees that every allowed recoupling of fusion tensors is consistent. Crucially, for the symmetries considered in this manuscript, there exists an adequate basis for the junction spaces in eq. (6) in which all F-symbols are unitary as matrices in the sense that

$$\sum_{X_6, k, l} (F_{X_4}^{X_1 X_2 X_3})_{X_5, i, j}^{X_6, k, l} (\overline{F}_{X_4}^{X_1 X_2 X_3})_{X_5', i', j'}^{X_6, k, l} = \delta_{X_5, X_5'} \delta_{i, i'} \delta_{j, j'}, \quad (7)$$

where the overline denotes complex conjugation. Up to basis transformations on the Hom spaces, it turns out that the number of solutions to the pentagon equation is finite for any valid fusion structure as specified by an N-symbol.

The set of labels  $\{X_i\}_i$ , assuming the existence of a unique trivial label  $\mathbb{1}$  and the notion of duals, endowed with a fusion structure and a unitary solution to the pentagon equation exactly constitutes the defining data of a *unitary fusion category* – conventionally denoted by  $\mathcal{C}$  – as demonstrated in refs. [9, 54, 82]. Some relevant notions from the theory of fusion categories and the notation we use are summarized in appendix A. In this sense, the MPOs considered in this paper form explicit finite-dimensional lattice representations of the (abstract) graphical calculus of fusion category theory, cfr. eq. (6). In particular this means that we can identify the collection of injective blocks with the set of (representatives of isomorphism classes of) simple objects of  $\mathcal{C}$ , denoted by  $\mathcal{I}_{\mathcal{C}}$ , and the F-symbols appearing in the recoupling identity eq. (6) coincide with the matrix elements of the components of the monoidal associator of  $\mathcal{C}$ , as defined per eq. (A.1). We now proceed to discuss gauging (part of) the  $\mathcal{C}$  symmetry.

Following the paradigm of ref. [39], gauging happens by introducing new gauge degrees of freedom between physical sites and enforcing a projector onto the gauge-invariant subspace of this enlarged Hilbert space. This projector is constructed as the product of local commuting projectors that represent a Gauss law. In the invertible symmetry case covered in ref. [39] introducing the gauge degrees of freedom boils down to tensoring the initial – ‘matter’ – Hilbert space with copies of the group algebra  $\mathbb{C}[G]$  localized on the links of the lattice. Subsequently, the gauge degrees of freedom are initialized in a configuration which can then be absorbed in the global projection, so as to end up with a gauging map acting on the initial Hilbert space.

As anticipated in the introduction, a choice of *special haploid symmetric Frobenius algebra object* in  $\mathcal{C}$  – henceforth often referred to as simply (Frobenius) algebra object – exactly encodes the data needed to generalise the gauging procedure of ref. [39] to fusion-categorical symmetries. Recall that loosely speaking a Frobenius algebra object in  $\mathcal{C}$  is a (generically non-simple) object  $A \simeq \bigoplus_{X \in \mathcal{I}_{\mathcal{C}}} N_X^A X$  together with multiplication and comultiplication maps, denoted by  $\mu : A \otimes A \rightarrow A$  and  $\Delta : A \rightarrow A \otimes A$  respectively, which are (co)associative on the nose and satisfy a number of further criteria spelled out in appendix A. In particular, a finite symmetry  $\mathcal{C}$  is called

(fully) gaugeable when the quantum dimensions of all simple objects are integer and the object  $\bigoplus_{X \in \mathcal{I}_C} d_X X$  can be endowed with a Frobenius structure. Physically, as an object in  $\mathcal{C}$ ,  $A$  exactly encodes the symmetries that are being gauged while the Frobenius structure on  $A$  encodes a particular way of gauging them, thereby providing a more general notion of *discrete torsion*. An obstacle arising in the general setting is that the symmetries  $\mathcal{C}$  are typically represented on Hilbert spaces that are not naturally equipped with a tensor product structure. Hence, it is not immediately apparent what the appropriate notion of a gauge degree of freedom is in this case. This issue is remedied in the following section where we demonstrate how symmetry twists labelled by  $A$  can be inserted in the non-tensor product Hilbert space and how these defects play the role of gauge field in this setting.

In the remainder of this section we will absorb the trivial gauge field directly into the gauging map to obtain the gauging map that acts on the original Hilbert space. Given a Frobenius algebra  $A$  in  $\mathcal{C}$ , we now proceed by proposing a generalised notion of Gauss law [31]. Drawing upon our graphical calculus, we consider the following tensor network operator:

$$\mathcal{P} := \begin{array}{c} A \quad A \\ \downarrow \quad \downarrow \\ \mu \quad \square \quad \Delta \\ \uparrow \quad \uparrow \\ A \quad A \end{array}. \quad (8)$$

In this we have defined

$$\begin{array}{c} \downarrow \\ \square \\ \uparrow \end{array} \begin{array}{c} A \\ \leftarrow \end{array} := \sum_{X \in \mathcal{I}_C} N_X^A \begin{array}{c} \downarrow \\ \square \\ \uparrow \end{array} \begin{array}{c} X \\ \rightarrow \end{array}, \quad (9)$$

and the trivalent tensors are constructed from the fusion and splitting tensors introduced above, with the components of  $\mu$  and  $\Delta$  defined per. eq. (A.9) as follows:

$$\begin{array}{c} A \quad A \\ \downarrow \quad \downarrow \\ \mu \\ \uparrow \\ A \end{array} := \sum_{\substack{X_1, X_2, X_3 \\ i_1, i_2, i_3, l}} \mu_{(X_3, i_3), l}^{(X_1, i_1), (X_2, i_2)} \begin{array}{c} X_1 \quad X_2 \\ \downarrow \quad \downarrow \\ \downarrow \\ X_3 \end{array}, \quad \begin{array}{c} A \\ \downarrow \\ \Delta \\ \uparrow \quad \uparrow \\ A \quad A \end{array} := \sum_{\substack{X_1, X_2, X_3 \\ i_1, i_2, i_3, l}} \Delta_{(X_1, i_1), (X_2, i_2)}^{(X_3, i_3), l} \begin{array}{c} X_3 \\ \downarrow \\ \downarrow \\ X_1 \quad X_2 \end{array}. \quad (10)$$

In this definition  $i_1, i_2, i_3$  stand for the multiplicity of  $X_1, X_2, X_3$  in  $A$  respectively, and take values in  $i_j = 1, 2, \dots, N_{X_j}^A$ ,  $j = 1, 2, 3$ . The operator (8) should be interpreted as acting on a physical site and two neighboring gauge fields taking values in the object  $A$ . As such, it can be thought of as a localized version of the symmetry MPO (2) restricted to the subset of symmetries encoded in  $A$ .

We now show that the  $\mathcal{P}$ 's are (local) mutually commuting projection operators. Whenever two of these operators overlap in a common gauge degree of freedom, their commutation is graphically depicted as

$$\begin{array}{c} \downarrow \quad \downarrow \quad \downarrow \\ \mu \quad \square \quad \Delta \\ \uparrow \quad \uparrow \quad \uparrow \\ A \quad A \quad A \end{array} \stackrel{(A.8)}{=} \begin{array}{c} \downarrow \quad \downarrow \quad \downarrow \\ \square \quad \mu \quad \Delta \\ \uparrow \quad \uparrow \quad \uparrow \\ A \quad A \quad A \end{array}. \quad (11)$$

In this figure and those that follow we suppressed  $A$  labels when they can be inferred from the context. This commutation property can be directly deduced from the first Frobenius condition (A.8) which is satisfied by the multiplication and comultiplication maps. The fact that  $\mathcal{P}$  is a

projector follows in turn from the (co)associativity of  $\mu$  and  $\Delta$  together with the intertwining property (4) and the normalization of the Frobenius algebra (A.7):

$$\begin{aligned}
 \mathcal{P}^2 &= \text{Diagram 1} \stackrel{(A.4)}{=} \text{Diagram 2} \stackrel{(4)}{=} \text{Diagram 3} \stackrel{(A.7)}{=} \mathcal{P}.
 \end{aligned} \tag{12}$$

The fact that these projectors mutually commute allows us to unambiguously define the following (global) projection operator on periodic boundary conditions:

$$\text{Diagram 4} \tag{13}$$

In order to define a gauging map acting on the original Hilbert space as anticipated we choose a trivial initial gauge configuration which is acted upon by the gauging map. Such a trivial gauge configuration is provided by the unit structure of the Frobenius algebra. Indeed, the unit provides a vector, denoted by  $|\mathbb{1}_A\rangle \in \mathcal{C}(\mathbb{1}, A)$ , such that

$$\mathcal{G}_A := \text{Diagram 5} = \text{Diagram 6}, \tag{14}$$

is exactly the sought-after gauging map. The action of the gauging map  $\mathcal{G}_A$  on a local operator  $O$  that commutes with the symmetry MPOs eq. (2) is then given via the prescription

$$O_g \cdot \mathcal{G}_A = \mathcal{G}_A \cdot O, \tag{15}$$

with  $O_g$  the local gauged operator.

Despite being explicit, the gauging map (14) suffers from a number of shortcomings. Most notably, it is not obvious from its presentation in eq. (14) that the global symmetries of the gauged model are encoded in the fusion category  $\text{Bimod}_{\mathcal{C}}(A)$  of  $A$ - $A$ -bimodules in  $\mathcal{C}$ , as argued in ref. [11]. This is particularly troublesome when trying to consider symmetry-twisted boundary conditions in the gauged model, which is necessary to obtain an invertible version of the gauging map. Moreover, gauging maps constructed from *Morita equivalent* Frobenius algebras should be considered physically equivalent even though their MPO representations (14) are distinct. These weaknesses are remedied by considering a basis in which the above gauging map boils down to the duality matrix product operators constructed in ref. [50, 51] based on the MPO representations of bimodule categories of ref. [54]. In the following section we review the duality matrix product operators pioneered in ref. [50, 51] before demonstrating the unitary equivalence with the gauging maps constructed above.

### 3 Duality matrix product operators

Given a finite symmetry  $\mathcal{C}$  we review the construction of ref. [54] of tensor network representations of the  $\mathcal{C}$  symmetry operators, labelled by indecomposable module categories over  $\mathcal{C}$ , and the local operators invariant under them. We review the category theoretic approach to dualities proposed in ref. [50, 51] in which different lattice realisations of this local operator algebra amount to performing a duality. In preparation for the following section we revisit the matrix product operators intertwining the dual local operator algebras.

From the results of ref. [8] one can infer a generalised Wigner-Eckart theorem which states that local operators that commute with a finite symmetry  $\mathcal{C}$  can be decomposed into *generalised Clebsch-Gordan coefficients*. Distinct collections of such coefficients are in one-to-one correspondence with distinct lattice realisations of the symmetry  $\mathcal{C}$  and are classified by (finite indecomposable) (left) *module categories*  $\mathcal{R}$  over  $\mathcal{C}$ , whose definition is revisited in appendix A.2. More precisely, these coefficients boil down to (the components of) the module associator of  $\mathcal{R}$  as a (right) module category over the *Morita dual* fusion category  $\mathcal{C}_{\mathcal{R}}^*$ . Recall that  $\mathcal{C}_{\mathcal{R}}^*$  is defined as  $\text{Func}(\mathcal{R}, \mathcal{R})$ , i.e. the category of  $\mathcal{C}$ -module endofunctors of  $\mathcal{R}$ . As such,  $\mathcal{R}$  is endowed with the structure of an *invertible*  $(\mathcal{C}, \mathcal{C}_{\mathcal{R}}^*)$ -bimodule category. Below, we denote simple representative objects of  $\mathcal{R}$  and  $\mathcal{C}_{\mathcal{R}}^*$  by  $\{R_m\}_m$  and  $\{Y_m\}_m$  respectively. The general framework of tensor network representations of bimodule categories, pioneered in ref. [54], suggests to depict these coefficients in terms of *triple line tensors*. These tensors generalise those introduced in refs. [4, 30] in the context of PEPS representations of string-net ground states. Concretely, their non-vanishing components evaluate to:

$$\begin{aligned}
 \text{Diagram 1} &\equiv \sum_{\{R_n\}_n} \sum_{i,j,l} \text{Diagram 2} |R_1 Y_1 R_3, i\rangle \otimes |R_3 Y_2 R_2, j\rangle \otimes \langle R_1 Y_3 R_2, l| \\
 &\equiv \sum_{\{R_n\}_n} \sum_{i,j,l} (F_{R_2}^{R_1 Y_1 Y_2})_{R_3, ij}^{Y_3, kl} |R_1 Y_1 R_3, i\rangle \otimes |R_3 Y_2 R_2, j\rangle \otimes \langle R_1 Y_3 R_2, l|, \\
 \text{Diagram 3} &\equiv \sum_{\{R_n\}_n} \sum_{i,j,l} \text{Diagram 4} |R_1 Y_3 R_2, l\rangle \otimes \langle R_1 Y_1 R_3, i| \otimes \langle R_3 Y_2 R_2, j| \\
 &\equiv \sum_{\{R_n\}_n} \sum_{i,j,l} (\bar{F}_{R_2}^{R_1 Y_1 Y_2})_{R_3, ij}^{Y_3, kl} |R_1 Y_3 R_2, l\rangle \otimes \langle R_1 Y_1 R_3, i| \otimes \langle R_3 Y_2 R_2, j|.
 \end{aligned} \tag{16}$$

Note that every index of these tensors is labelled by a triple of objects  $(R, Y, R')$  and a multiplicity label taking values  $1, 2, \dots, N_{RY}^{R'}$ , and  $k = 1, 2, \dots, N_{Y_1 Y_2}^{Y_3}$ . More precisely these label basis vectors in the Hom spaces:

$$\begin{aligned}
 |R_1 Y_1 R_3, i\rangle &\in \mathcal{R}(R_1 \triangleleft Y_1, R_3), \\
 |R_3 Y_2 R_2, j\rangle &\in \mathcal{R}(R_3 \triangleleft Y_2, R_2), \\
 |R_1 Y_3 R_2, l\rangle &\in \mathcal{R}(R_1 \triangleleft Y_3, R_2).
 \end{aligned} \tag{17}$$

Throughout, we assume that these bases are so that the F-symbols are unitary when interpreted as linear maps  $F_{R_2}^{R_1 Y_1 Y_2} : \bigoplus_{Y_3} \mathcal{C}_{\mathcal{R}}^*(Y_1 \otimes Y_2, Y_3) \otimes \mathcal{R}(R_1 \triangleleft Y_3, R_2) \xrightarrow{\sim} \bigoplus_{R_3} \mathcal{R}(R_1 \triangleleft Y_1, R_3) \otimes \mathcal{R}(R_3 \triangleleft Y_2, R_2)$ .

Note that all triples of objects  $(R, Y, R')$  as well as  $(Y_1, Y_2, Y_3)$  simultaneously have to obey the fusion rules in order for the corresponding tensor component to be non-vanishing. Moreover, the objects  $R_{1,2,3}$  are shared between tensor legs as is reflected in the graphical depiction of these tensors.

In order to connect to the previous section, we remark that by combining the triples and corresponding multiplicity index into a single index, one can depict these tensors in a more standard tensor network notation:

$$\begin{array}{c} \text{Diagram 1} \end{array} \equiv \begin{array}{c} \text{Diagram 2} \end{array} \quad (18)$$

Note that in the right figure the choice of module category is implicit, which would obfuscate the manipulations of the next sections. Therefore we will stick to triple line tensors in what follows.

In eq. (16), and in similar expressions below, we adapt the graphical convention that unlabelled blue strands are summed over, and unlabelled gray patches stand for a sum over basis vectors, i.e.

$$\begin{array}{c} \text{Diagram 1} \end{array} \equiv \sum_{\substack{R_1, R_2 \\ \in \mathcal{L}_{\mathcal{R}}}} \begin{array}{c} \text{Diagram 2} \end{array} \equiv \sum_{\substack{R_1, R_2 \\ \in \mathcal{L}_{\mathcal{R}}}} \sum_i \begin{array}{c} \text{Diagram 3} \end{array} |R_1 Y R_2, i\rangle, \quad (19)$$

$$\begin{array}{c} \text{Diagram 4} \end{array} \equiv \sum_{\substack{R_1, R_2 \\ \in \mathcal{L}_{\mathcal{R}}}} \begin{array}{c} \text{Diagram 5} \end{array} \equiv \sum_{\substack{R_1, R_2 \\ \in \mathcal{L}_{\mathcal{R}}}} \sum_i \begin{array}{c} \text{Diagram 6} \end{array} \langle R_1 Y R_2, i|. \quad (19)$$

Contraction of these triple line tensors takes place via identification of the simple objects on module strands of concatenated tensors, pairing basis vectors, and summing over them as detailed in ref. [50].

Under the above conventions, a local  $\mathcal{C}$ -symmetric two-site operator can be depicted as

$$\approx_{i,n}^{\mathcal{R}} := \sum_{\{Y_m\}_m} \sum_{i,j} \lambda_{i,n}(\{Y_m\}_m, i, j) \begin{array}{c} \text{Diagram 1} \end{array}, \quad (20)$$

where  $\{\lambda_{i,n}\}$  is a collection of complex numbers which may depend on all internal labels of the tensor on the right. The underlying Hilbert space can be written as

$$\begin{array}{c} \text{Diagram 1} \end{array} \left| \{Y_{i+\frac{1}{2}}\}_i, \{R_i\}_i, \{i_{i+\frac{1}{2}}\}_i \right\} \quad (21)$$

$$\equiv \text{Span}_{\mathcal{C}} \left\{ \bigotimes_i |R_i Y_{i+\frac{1}{2}} R_{i+1}, i_{i+\frac{1}{2}}\rangle \left| \{Y_{i+\frac{1}{2}}\}_i, \{R_i\}_i, \{i_{i+\frac{1}{2}}\}_i \right. \right\},$$

where periodic boundary conditions are implemented by identifying  $R_1 \equiv R_{L+1}$  for some chain length  $L$ . Note that this Hilbert space is generically constrained since the Hom space at  $i$  depends on those at  $i-1$  and  $i+1$  via the module objects  $R_i$  and  $R_{i+1}$  they share. In particular,

the direct sum over  $\{R\}$  is absent whenever  $\mathcal{R}$  contains a single simple object, and  $\mathcal{R}$  is thus equivalent to the category of (finite-dimensional) vector spaces. Hence, in that case, we obtain a tensor product Hilbert space.

A generic symmetric local Hamiltonian acting on this space is then constructed as  $\mathbb{H}^{\mathcal{R}} = \sum_i \sum_n \tilde{\kappa}_{i,n}^{\mathcal{R}}$ . Note that one can always choose  $\mathcal{R} = \mathcal{C}$ , in which case the above Clebsch-Gordan coefficients boil down to the F-symbols of  $\mathcal{C}$ .<sup>1</sup> As such, the lattice models considered in refs. [50,51] and this manuscript generalise the *anyonic spin chains* first considered in ref. [28].

A *duality* in the sense of ref. [50] then amounts to changing the module category  $\mathcal{R}$  while keeping the category  $\mathcal{C}_{\mathcal{R}}^*$  and the coefficients  $\{\lambda_{i,n}\}$  fixed. Since  $\mathcal{C}_{\mathcal{R}}^*$  is kept fixed throughout, we move to a lightweight notation in the expressions below by writing  $\mathcal{D}$  for  $\mathcal{C}_{\mathcal{R}}^*$ , thereby suppressing the dependency of  $\mathcal{D}$  on the initial symmetry and choice of module category. Under such a duality transformation the local symmetric operators (20) are mapped to dual local symmetric operators, charged local operators are mapped to disorder operators and the spectrum is preserved up to degeneracies if closed boundary conditions are accounted for as discussed below in sec. 6. The latter can heuristically be understood from the fact that the algebra of local operators, spanned by operators of the form (20), in particular its structure constants, is fully specified by the fixed fusion category  $\mathcal{D}$ .

Within the tensor network paradigm this change of module category can be implemented explicitly as a matrix product *intertwiner* that transmutes the dual local operators into each other. Indeed, for a generic duality between module categories  $\mathcal{R}$  and  $\mathcal{R}'$  the intertwiners are, by the result of ref. [50], labelled by  $\mathcal{D}$ -module functors between  $\mathcal{R}$  and  $\mathcal{R}'$  organized in  $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$ , which itself is a *right* module category  $\mathcal{M}$  over  $\mathcal{C}$ . To a given simple object  $X \in \mathcal{M}$  there corresponds for every  $R_1 \in \mathcal{I}_{\mathcal{R}}$ ,  $Y \in \mathcal{I}_{\mathcal{D}}$  and  $R'_2 \in \mathcal{I}_{\mathcal{R}'}$  a matrix  ${}^X\omega_{R'_2}^{R_1 Y}$  defined in eq. (A.13), from which we construct an MPO tensor whose components are

$$\begin{array}{c} \begin{array}{ccccc} & & j & & \\ & & \uparrow & & \\ R'_1 & & X & & R'_2 \\ & & \downarrow & & \\ i & & Y & & l \\ & & \downarrow & & \\ R_1 & & k & & R_2 \end{array} \end{array} := ({}^X\omega_{R'_2}^{R_1 Y})_{R'_1, ij}^{R_2, kl}. \quad (22)$$

Without loss of generality  ${}^X\omega_{R'_2}^{R_1 Y}$  is assumed to be a unitary map  ${}^X\omega_{R'_2}^{R_1 Y} : \bigoplus_{R_2} \mathcal{R}(R_1 \triangleleft Y, R_2) \otimes \mathcal{R}'(X \triangleright R_2, R'_2) \xrightarrow{\sim} \bigoplus_{R'_1} \mathcal{R}'(X \triangleright R_1, R'_1) \otimes \mathcal{R}(R'_1 \triangleleft Y, R'_2)$ . Notably,  ${}^X\omega$  satisfies a consistency equation involving the F-symbols appearing in eq. (16) for the module categories  $\mathcal{R}$  and  $\mathcal{R}'$  which can be graphically depicted as the following tensor equality:

$$\begin{array}{c} \begin{array}{ccccc} & & k & & l \\ & & \uparrow & & \uparrow \\ R'_1 & & R'_3 & & R'_2 \\ & & \downarrow & & \downarrow \\ i & & X & & j \\ & & \downarrow & & \downarrow \\ R_1 & & Y_1 & & Y_2 & & R_2 \\ & & \downarrow & & \downarrow & & \\ & & m & & \\ & & \downarrow & & \\ & & Y_3 & & \\ & & n & & \end{array} \end{array} = \begin{array}{c} \begin{array}{ccccc} & & k & & l \\ & & \uparrow & & \uparrow \\ R'_3 & & Y_1 & & Y_2 \\ & & \downarrow & & \downarrow \\ & & m & & \\ & & \downarrow & & \\ R'_1 & & Y_3 & & R'_2 \\ & & \downarrow & & \downarrow \\ i & & X & & j \\ & & \downarrow & & \downarrow \\ R_1 & & n & & R_2 \end{array} \end{array}, \quad (23)$$

<sup>1</sup>The term generalised Clebsch-Gordan coefficients stems from the fact that for the case of an invertible symmetry as described by the fusion category  $\mathcal{C} = \text{Vec}_G$ , the category of  $G$ -graded vector spaces, the F-symbols appearing in (16) coincide with the usual Clebsch-Gordan coefficients when the choice  $\mathcal{R} = \text{Vec}$  is made. Here,  $\text{Vec}$  stands for the category of complex finite-dimensional vector spaces. In that case the Morita dual is found to be  $(\text{Vec}_G)_{\text{Vec}}^* \simeq \text{Rep}(G)$ , the fusion category of complex finite-dimensional representations of  $G$  [21,54].

for all valid labellings of the objects and multiplicity indices. A similar property is satisfied by the second tensor in eq. (16). This equality demonstrates that the MPO constructed from the tensor (22) labelled by the object  $X \in \mathcal{M}$  transmutes local operators of one model corresponding to the choice of module category  $\mathcal{R}$  to the one of the model based on  $\mathcal{R}'$ , thereby implementing the duality transformation as foreseen. Importantly, choosing both module categories in eq. (22) equal to  $\mathcal{R}$  exactly realises the lattice representation of the symmetry operators encoded in  $\mathcal{C}$ , here thought of as  $\mathcal{C} \equiv \text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}) = \mathcal{D}_{\mathcal{R}}^*$ . In that instance it is more precise to refer to (22) as the *bimodule associator* of  $\mathcal{R}$  as  $(\mathcal{C}, \mathcal{D})$ -bimodule category. Choosing both module categories equal to  $\mathcal{R}'$  on the other hand reveals that the global symmetries of the dual model are encoded in the fusion category  $\mathcal{D}_{\mathcal{R}'}^*$ , equivalently  $\mathcal{C}_{\mathcal{M}}^*$ . The global symmetries of the gauged model are thus encoded in the Morita dual of the symmetry category of the ungauged model with respect to the category of duality operators. Note that all duality operators labelled by module functors in  $\mathcal{M}$  implement physically the same duality operation. Being itself an invertible  $(\mathcal{C}_{\mathcal{M}}^*, \mathcal{C})$ -bimodule category,  $\mathcal{M}$  can be acted upon from the *right* by  $\mathcal{C}$  and from the *left* by  $\mathcal{C}_{\mathcal{M}}^*$ . As such, distinct duality operators can be converted into each other by pre- or postcomposing with symmetry operators in  $\mathcal{C}$ , respectively  $\mathcal{C}_{\mathcal{M}}^*$ .

Note that in eq. (22) – similar to eq. (18) – we can again conflate the indices so as to obtain a conventional tensor of the form

$$\begin{array}{c} \begin{array}{c} R'_1 \quad j \\ \downarrow \\ i \leftarrow X \rightarrow l \\ \uparrow \\ R_1 \quad k \end{array} \quad R'_2 \\ \equiv (R_1 X R'_1, i) \leftarrow X \rightarrow (R_2 X R'_2, l), \end{array} \quad (24)$$

where on the right the choice of module categories is again implicit. Consequently, by specifying to  $\mathcal{R} = \mathcal{R}'$ , we recover the symmetry MPO tensors of eq. (1).

Given such  $\mathcal{C}$  symmetry MPOs, what are the fusion and splitting tensors employed in the construction of the gauging map proposed in the previous section? Akin to the definition (16) we can consider tensors evaluating to the components of the *left* module associators of  $\mathcal{R}$  over  $\mathcal{C}$ :

$$\begin{array}{c} \begin{array}{c} l \quad R_2 \quad k \\ \downarrow \\ X_1 \leftarrow i \rightarrow X_2 \\ \uparrow \\ R_3 \quad X_3 \quad R_1 \end{array} \\ := (F_{R_3}^{X_1 X_2 R_1})_{X_3, ij}^{R_2, kl} \end{array} \quad \text{and} \quad \begin{array}{c} \begin{array}{c} R_3 \quad j \\ \downarrow \\ X_1 \leftarrow i \rightarrow X_2 \\ \uparrow \\ R_2 \quad k \end{array} \\ := (\bar{F}_{R_3}^{X_1 X_2 R_1})_{X_3, ij}^{R_2, kl} \end{array} \quad (25)$$

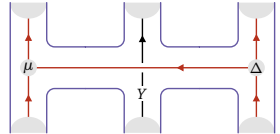
Again by virtue of two mixed pentagon equations these fusion tensors can be shown to satisfy the intertwining property (4) for the symmetry MPO tensors (22) and the recoupling identity (6) involving the  $\mathcal{C}$  F-symbol, as showcased in ref. [54]. As such, given an algebra  $A$  in  $\mathcal{C}$ , explicit lattice representations of the multiplication and comultiplication maps read:

$$\begin{array}{c} \begin{array}{c} \begin{array}{c} \downarrow \\ A \end{array} \\ \downarrow \\ \begin{array}{c} \downarrow \\ A \end{array} \end{array} := \sum_{\substack{X_1, X_2, X_3 \\ i_1, i_2, i_3, l}} \mu_{(X_3, i_3), l}^{(X_1, i_1), (X_2, i_2)} \begin{array}{c} \begin{array}{c} \downarrow \\ X_1 \end{array} \end{array}, \end{array} \quad (26)$$

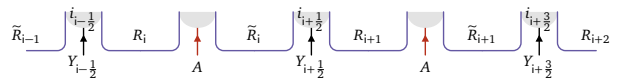
$$\begin{array}{c} \begin{array}{c} \begin{array}{c} \downarrow \\ A \end{array} \\ \downarrow \\ \begin{array}{c} \downarrow \\ A \end{array} \end{array} := \sum_{\substack{X_1, X_2, X_3 \\ i_1, i_2, i_3, l}} \Delta_{(X_1, i_1), (X_2, i_2)}^{(X_3, i_3), l} \begin{array}{c} \begin{array}{c} \downarrow \\ X_3 \end{array} \end{array},$$



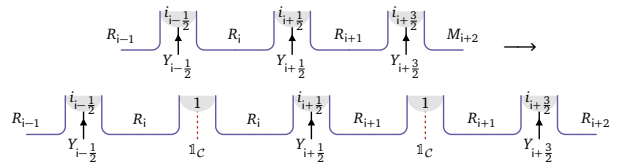
where the sums should be interpreted as explained surrounding eq. (10). Collecting everything, for any choice of module category over the symmetry category  $\mathcal{C}$  one finds a corresponding MPO representation of that symmetry so that for any algebra  $A$  in  $\mathcal{C}$  the generalised Gauss operator is given by:


(27)

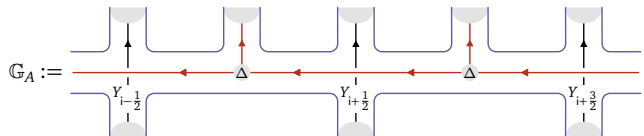
This Gauss operator acts on the Hilbert space (21) supplemented with symmetry twists labelled by the object  $A$ . Following ref. [50] states in this enlarged Hilbert space can be depicted as


(28)

where the span is now taken over additional module labels  $\tilde{R}_i$ . Note that since we have assumed  $A$  to be haploid, meaning that the decomposition of  $A$  contains a single copy of the monoidal unit  $\mathbb{1}_{\mathcal{C}}$ , there is a canonical way in which a basis state in the initial Hilbert space (21) can be embedded in this enlarged Hilbert space, namely via


(29)

Equivalently, in line with the previous section, we can incorporate this choice of gauge field directly into the gauging map so as to obtain a gauging map acting on the original Hilbert space (21). The resulting operator can be depicted as:


(30)

As such, this procedure provides an interpretation of the  $A$  symmetry twists in the extended Hilbert space (28) as the ‘gauge field’ for the subsymmetry  $\mathcal{C}$  which is being gauged, in accordance with ref. [71]. Notably, the insertion of the defects in (28) is independent of the Frobenius structure on  $A$ , which only appears in the definition of the Gauss constraint. In that sense, the Frobenius structure can be thought of as generalising the notion of discrete torsion to the categorical setting, as anticipated above.

This can be compared to ref. [39] whose results are obtained as a specific instance of this formalism. Indeed, the results of ref. [39] are recovered by choosing  $\mathcal{R} = \text{Vec}$ , the category of (complex finite-dimensional) vector spaces, as module category over the fusion category  $\mathcal{C} = \text{Vec}_G$  of  $G$ -graded vector spaces which describes an invertible  $G$  symmetry. In that case the objects  $\{Y_i\}_i$  label representations of  $G$ . Gauging the full  $G$  symmetry then happens via choosing  $A \simeq \mathbb{C}[G]$ , here thought of as the regular representation of  $G$ . The Hilbert space (28) then effectively boils down to the initial ‘matter’ Hilbert space tensored with copies of the group algebra  $\mathbb{C}[G]$  in between the physical sites. Furthermore, one could twist the gauging map by

a choice of 2-cocycle, i.e.  $[\psi] \in H^2(G, U(1))$ , by choosing  $A$  to be the  $\psi$ -twisted group algebra  $\mathbb{C}^\psi[G]$  which as a vector space is spanned by states  $\{|g\rangle, g \in G\}$  with algebra multiplication given by  $|g\rangle \cdot |h\rangle = \psi(g, h)|gh\rangle$  [50, 56, 76].

In preparation for the following section, let us note that the trivalent tensors evaluating to the left module associators (25) are a special instance of a more general kind of tensors which encode the composition of module functors. Indeed, given  $\mathcal{M}, \mathcal{N}, \mathcal{O}$ , all right module categories over  $\mathcal{D}$ , the unitary F-symbols introduced in (A.14) and their conjugates can be organised in triple line tensors of the form

$$\begin{array}{c} \text{Diagram 1: A trivalent tensor with three legs labeled } l, k, j \text{ and three regions labeled } N, O, M. \text{ The central node is labeled } X_1, X_2, X_3. \\ \text{Diagram 2: A trivalent tensor with three legs labeled } l, k, j \text{ and three regions labeled } O, N, M. \text{ The central node is labeled } X_1, X_2, X_3. \end{array} := (F_O^{X_1, X_2, M})_{X_3, ij}^{N, kl} \quad \text{and} \quad \begin{array}{c} \text{Diagram 3: A trivalent tensor with three legs labeled } l, k, j \text{ and three regions labeled } O, N, M. \text{ The central node is labeled } X_1, X_2, X_3. \\ \text{Diagram 4: A trivalent tensor with three legs labeled } l, k, j \text{ and three regions labeled } O, N, M. \text{ The central node is labeled } X_1, X_2, X_3. \end{array} := (\bar{F}_O^{X_1, X_2, M})_{X_3, ij}^{N, kl}, \quad (31)$$

where  $M \in \mathcal{I}_{\mathcal{M}}, N \in \mathcal{I}_{\mathcal{N}}, O \in \mathcal{I}_{\mathcal{O}}$  and  $X_1 \in \mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{N}, \mathcal{O})}, X_2 \in \mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{N})}, X_3 \in \mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{O})}$ . From  $\mathcal{D}_{\mathcal{R}}^* \equiv \mathcal{C}$  it follows that for the case where all three module categories are chosen to be equal to  $\mathcal{R}$  these tensors indeed reduce to the matrix components of the left module associators of  $\mathcal{R}$  over  $\mathcal{C}$  as per eq. (25).

## 4 Equivalence gauging map and duality operator

In this section, we show how the gauging map (30) can be converted to a duality MPO from the previous section by means of a finite sequence of local unitary transformations, i.e. a constant depth quantum circuit.

8

Given a finite symmetry  $\mathcal{C}$  represented on the lattice via a choice of module category  $\mathcal{R}$ , how can the gauging map constructed from a Frobenius algebra object  $A$  in  $\mathcal{C}$  be related to a duality operator of the kind described in the previous section? By the results of refs. [62, 66] there exists a right  $\mathcal{C}$ -module category such that  $A$  corresponds to the *internal hom* of a particular simple module object. More precisely, this module category is exactly equivalent to the category  $\mathcal{M} = \text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$  of duality operators, for a particular  $\mathcal{R}'$ . Indeed,  $\mathcal{R}'$  is determined by the fact that  $\mathcal{M}$  is equivalent to the category  $\text{Mod}_{\mathcal{C}}(A)$  of left  $A$ -modules [25, 62]. The internal hom reconstruction of  $A$  is reviewed in appendix A.3. In what follows  $X_A$  denotes the simple object such that  $A \equiv \underline{\text{Hom}}(X_A, X_A)$ . The simple module object  $X_A \in \mathcal{M}$  is precisely the one labelling the duality MPO to which the gauging map  $\mathbb{G}_A$  (30) is unitarily equivalent by means of a unitary constant-depth quantum circuit, as we now demonstrate.

The internal hom reconstruction allows us to ‘inflate’ the comultiplication tensors (26) appearing in the gauging map (30) as follows [26, 66]:

$$\begin{array}{c} \text{Diagram 1: A trivalent tensor with three legs labeled } i_1, i_2, i_3 \text{ and three regions labeled } X_1, X_2, X_3. \\ \text{Diagram 2: A trivalent tensor with three legs labeled } i_1, i_2, i_3 \text{ and three regions labeled } X_1, X_2, X_3. \end{array} = \sum_{\substack{X_1, X_2, X_3 \\ i_1, i_2, i_3}} \frac{\sqrt{d_{X_1} d_{X_2} d_{X_3}}}{d_{X_A}^2} \begin{array}{c} \text{Diagram 3: A trivalent tensor with three legs labeled } i_1, i_2, i_3 \text{ and three regions labeled } X_1, X_2, X_3. \\ \text{Diagram 4: A trivalent tensor with three legs labeled } i_1, i_2, i_3 \text{ and three regions labeled } X_1, X_2, X_3. \end{array}, \quad (32)$$

where, as in eq. (25), the sum is over the simple objects appearing in the decomposition of  $A$ , and where in virtue of the defining property of the internal hom (A.15),

$N_{X_j}^A = \dim_{\mathbb{C}} \mathcal{C}(X_j, A) = \dim_{\mathbb{C}} \mathcal{M}(X_A \triangleleft X_j, X_A)$  [66]. Also remark that the innermost closed module loop on the right-hand side is valued in  $\mathcal{R}'$  over whose simple objects we sum. Furthermore, the normalisation is chosen to match the one in eq. (A.16). In appendix C we demonstrate, based on our graphical calculus for triple line tensors, that the inflated representation of the comultiplication (32) satisfies the defining coalgebra property.

Before applying the aforementioned circuit, we first rewrite the gauging map by making use of eq. (32). We begin by inserting eq. (32) in the gauging map eq. (30), followed by using the orientation-reversing ‘flags’ introduced in appendix B to reverse some of the arrows in eq. (32). We then exploit the unitarity of the trivalent tensors, which in our graphical notation reads

$$\sum_{X,i} \begin{array}{c} \text{Diagram: A trivalent tensor with inputs } k, l \text{ and output } i, \text{ and another with inputs } i, k' \text{ and output } l'. \end{array} = \delta_{R'_1, R'_2} \delta_{k, k'} \delta_{l, l'} \begin{array}{c} \text{Diagram: Two separate vertical strands with inputs } k, l \text{ and outputs } k, l. \end{array}, \quad (33)$$

so as to rewrite the gauging map as follows:

$$\mathbb{G}_A = \sum_{\substack{\{X_i\}_i \\ \{j_i\}_i}} d_{X_A}^{-\frac{1}{2}} \begin{array}{c} \text{Diagram: A complex circuit with multiple layers of module strands (R, R') and dual strands (X_A, \bar{X}_A). It includes summations over indices i, j, and k.} \end{array}, \quad (34)$$

where the letters  $\mathcal{R}, \mathcal{R}'$  denote the module categories in which surrounding module strands are valued. Here and below,  $\bar{X}_A \in \mathcal{I}_{\mathcal{M}^{\text{op}}}$ , where  $\mathcal{M}^{\text{op}} = \text{Fun}_{\mathcal{D}}(\mathcal{R}', \mathcal{R})$ , labels the unique ‘opposite’ duality strand specified by the fact that the composition of  $\bar{X}_A$  and  $X_A$  exactly yields the algebra object  $A$  in  $\mathcal{C}$ .

Let us stress again that eq. (34) is an exact rewriting of the gauging map. In order to bring it into the form of a duality operator, we now invoke the following unitary gate defined in ref. [53] constructed from the fusion tensors eq. (31):

$$\begin{array}{c} \text{Diagram: A unitary gate with inputs } \bar{X}_A, X_A \text{ and outputs } \bar{X}_A, X_A. \end{array} \equiv \sum_X \begin{array}{c} \text{Diagram: A trivalent tensor with inputs } X, R_1 \text{ and output } R_2. \end{array} \equiv \sum_{X, R_1, R_2, R'} \begin{array}{c} \text{Diagram: A trivalent tensor with inputs } R_1, R_2 \text{ and output } X. \end{array}, \quad (35)$$

which should be interpreted as unitary between the Hilbert spaces  $\bigoplus_{X, R_1, R_2} \mathcal{R}(X \triangleright R_2, R_1) \otimes \mathcal{C}(X, \bar{X}_A \otimes X_A) \xrightarrow{\sim} \bigoplus_{R_1, R_2, R'} \mathcal{M}(X_A \triangleright R_1, R') \otimes \mathcal{M}^{\text{op}}(\bar{X}_A \triangleright R', R_2)$ . Graphically, unitarity of this gate can be expressed by a rotated version of eq. (33).

Acting then with a layer of this gate on the strands labelled by objects  $\{X_i\}_i$  in eq. (34), eq. (34) can be brought in the following form:

$$\mathbb{G}_A \cong d_{X_A}^{-1/2} \cdot \quad (36)$$

At the bottom of eq. (36) we recognize the duality MPO implementing the duality  $\mathcal{R} \rightarrow \mathcal{R}'$  labelled by  $X_A$ . The 'cups' on top can be removed by means of a local unitary transformation. To this end we also define the following unitary gate whose entries are expressed in terms of a rotated version of the  $X_A \omega$  tensors introduced in above eq. (22):

$$\quad \equiv \sum_Y \quad \equiv \sum_{\substack{R_1, R_2 \\ R'_1, R'_2}} R_1 \quad R'_2 \quad (37)$$

This gate can be used to thread the strands labelled by  $X_A$  to the left through the physical degrees of freedom  $\{Y_{i+1/2}\}_i$ . Subsequently, by making use of the conjugate of the gate eq. (35) the  $\bar{X}_A$  and  $X_A$  strands can be fused together unitarily, in the process picking up an extra overall factor  $d_{X_A}^{L/2}$ . Due to the presence of the cup in eq. (36) the only allowed fusion outcome is  $\mathbb{1}_{\mathcal{C}_{\mathcal{M}}^*}$  which appears with multiplicity one in  $X_A \otimes \bar{X}_A$ . As such, one recovers eventually the duality MPO labelled by  $X_A$  as anticipated:

$$\mathbb{G}_A \cong \mathbb{D}_{X_A} := \quad (38)$$

We commented earlier that the global symmetries of the dual model are encoded in the fusion category  $\mathcal{C}_{\mathcal{M}}^*$ . In virtue of the fact that the category  $\mathcal{M}$  labelling the duality operators is equivalent to the category  $\text{Mod}_{\mathcal{C}}(A)$  of  $A$ -modules,  $\mathcal{C}_{\mathcal{M}}^*$  is indeed equivalent to  $\text{Bimod}_{\mathcal{C}}(A)$ , in agreement with the results of refs. [11, 25]. This can be verified more explicitly by conjugating the symmetry MPOs labelled by objects in  $\mathcal{C}_{\mathcal{M}}^*$  with the circuit constructed above. Making use of the inflation trick eq. (32) it is then easy to verify that the resulting operator, labelled by some object in  $\mathcal{C}$ , indeed is endowed with the structure of an  $A$ - $A$  bimodule in  $\mathcal{C}$ . More precisely, starting from the (non-simple) symmetry operator in the dual model labelled by  $X_A \otimes X \otimes \bar{X}_A \in \mathcal{C}_{\mathcal{M}}^*$ , where  $X \in \mathcal{C}$ , we can conjugate it with the circuit to obtain the MPO corresponding to the object  $A \otimes X \otimes A \in \mathcal{C}$ . Interpreting this object as the *induced bimodule*  $\text{Ind}_{A|A}(X)$  defined in [27, Def. 5.1], it then follows from [27, Lemma 5.2], that every simple  $A$ - $A$  bimodule is obtained in the decomposition of this MPO for a certain  $X$ .

We conclude this section with a few additional remarks. First, notice that the depth of the quantum circuit being constant and small is a direct consequence of the fact that the construction of the gauging map involves the introduction of an extensive number of defects labelled

by  $A$ . As such it should be compared to the quantum circuit representation of the duality operator showcased in ref. [53], whose depth scales linearly in the system size in the generic case, and is constant in the nilpotent case when supplemented with measurements and feedforward operations.

A further natural question is whether every (simple) duality operator labelled by  $X \in \mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{N})}$  is equivalent to a gauging map for a suitable choice of algebra object. The answer is affirmative and the algebra object is provided by the internal hom  $\underline{\text{Hom}}(X, X)$ . In that case, the circuit that transmutes the gauging map  $\mathbb{G}_{\underline{\text{Hom}}(X, X)}$  to the duality MPO  $\mathbb{D}_X$  exactly mimics the one above with  $\underline{\text{Hom}}(X, X)$  substituted for  $A$  and  $X$  for  $X_A$ .

As mentioned above,  $\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{N})$  carries the structure of an invertible  $(\mathcal{D}_{\mathcal{N}}^*, \mathcal{C})$ -bimodule category. In particular this implies that  $\underline{\text{Hom}}(X_A, X_A)$  can be interpreted as an internal hom object in the category  $\mathcal{D}_{\mathcal{N}}^*$ , namely via  $\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{N})(X \triangleright X_A, X_A) \simeq \mathcal{D}_{\mathcal{N}}^*(X, \underline{\text{Hom}}(X_A, X_A))$ . As such,  $\underline{\text{Hom}}(X_A, X_A)$  is also endowed with the structure of a Frobenius algebra in  $\mathcal{D}_{\mathcal{N}}^*$  which can be expressed in terms of the left  $\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{N})$  module associators, akin to eq. (A.16). This algebra object thus also provides a gauging map which in this case turns out to be equivalent to the duality MPO  $\mathbb{D}_{X_A}$  implementing the duality  $\mathcal{N} \rightarrow \mathcal{M}$ . Note however that  $\mathbb{G}_{X_A} \circ \mathbb{G}_{X_A}$  is not equal to the identity operator on the periodic Hilbert space eq. (21). In fact,  $\mathbb{G}_{X_A} \circ \mathbb{G}_{X_A}$  yields the symmetry MPO labelled by the algebra object  $A$ , which is typically a non-invertible symmetry operator. It turns out that when the original and dual Hilbert spaces are supplemented with symmetry-twists, the duality MPOs can be implemented unitarily. This is the topic of section 6.

## 5 Anomalies and short-range symmetric states

*We comment on the specific case where the symmetry is fully gaugeable and show that in that case we can think of gauging as averaging over a local version of the symmetry. We further argue that full gaugeability implies the existence of a short-range entangled  $\mathcal{C}$ -symmetric state, and therefore the absence of an anomaly.*

For invertible symmetries described by  $\text{Vec}_G^\omega$ , the 3-cocycle  $[\omega] \in H^3(G, U(1))$  poses an obstruction to gauging. Indeed, the possible algebra objects in  $\text{Vec}_G^\omega$  are  $\psi$ -twisted group algebras  $\mathbb{C}^\psi[K]$ , where  $\omega|_K$  is trivial,  $[\psi] \in H^2(K, U(1))$ . Therefore,  $\omega$  characterizes the anomaly of such a symmetry, and gauging with an algebra object  $\mathbb{C}^\psi[K]$  amounts to  $\psi$ -twisted gauging of the non-anomalous subgroup  $K$ . As argued in [14], the anomaly  $\omega$  also provides an obstruction to the existence of a short-range entangled state that is symmetric, which is the generally accepted definition for an anomaly in (1+1)d.

For general non-invertible symmetries, the interpretation of the algebra object  $A$  as a subset of symmetries that can be gauged is less clear. The reason for this is that in this case, one generally cannot think of the projectors in eq. (8) as an averaging over localised versions of the symmetry operators as in the invertible case. The exception to this is when the symmetry  $\mathcal{C}$  is anomaly-free; in this case, it can be gauged completely, even if it is non-invertible. To see this, we rely again on the equivalence between algebra objects in  $\mathcal{C}$  and internal Hom objects. Since gapped phases of a  $\mathcal{C}$  symmetric theory are classified by module categories  $\mathcal{P}$  over  $\mathcal{C}$  [32, 73], being anomaly-free implies the existence of  $\text{Vec}$  as a  $\mathcal{C}$ -module category, also referred to as a *fiber functor*. This requires  $\mathcal{C}$  to be the representation category of a Hopf algebra  $\mathcal{A}$ , and we denote  $\mathcal{C} = \text{Rep}(\mathcal{A})$ . The internal Hom construction provides an algebra object  $A \equiv \underline{\text{Hom}}(\mathbb{C}, \mathbb{C})$ , with  $\mathbb{C}$  the unique simple object in  $\text{Vec}$ . As an object of  $\text{Rep}(\mathcal{A})$ ,  $A$  is the regular representation,  $A \simeq \bigoplus_{X \in \mathcal{I}_{\mathcal{C}}} d_X \cdot X$ . It is this choice of algebra object that amounts to the complete gauging of the symmetry  $\mathcal{C}$ , as we show below.

Using the expression of the local Gauss constraint from section 2, we can write

$$\mathcal{P} = \begin{array}{c} \text{Diagram: A square with four legs labeled } A, \text{ containing a circle with a dot.} \end{array} = \sum_{a,i} \begin{array}{c} \text{Diagram: A square with four legs labeled } A, \text{ containing a circle with a dot and a small triangle labeled } a. \end{array} =: \sum_a \frac{d_a}{\text{FPdim } \mathcal{M}} \mathcal{P}^a, \quad (39)$$

where the triangular tensor is the projector of  $A$  on the block labeled by  $a$  and  $i$  is the multiplicity of the simple object  $a$  in  $A$ , cf. eq. (A.9).  $\text{FPdim } \mathcal{M} = \text{FPdim } \mathcal{C} = \sum_{X \in \mathcal{I}_{\mathcal{C}}} d_X^2$  denotes the total quantum dimension of  $\mathcal{M}$ , which in our conventions is equal to that of  $\mathcal{C}$ . We can show using the graphical calculus and the explicit expressions of the multiplication and comultiplication maps that

$$\mathcal{P}^a \mathcal{P}^b = \sum_c N_{ab}^c \mathcal{P}^c, \quad (40)$$

where the  $N$  symbol is that of the symmetry category  $\mathcal{C}$ . Indeed:

$$\begin{aligned} \mathcal{P}^a \mathcal{P}^b &\stackrel{(A.4)}{=} \frac{(\text{FPdim } \mathcal{M})^2}{d_a d_b} \sum_{i,j} \begin{array}{c} \text{Diagram: Two squares with legs labeled } A, \text{ connected by lines labeled } a, b. \end{array} \\ &\stackrel{(A.16)}{=} \frac{(\text{FPdim } \mathcal{M})^2}{d_a d_b} \sum_{\substack{c,k, \\ k',l}} \underbrace{\left( \sum_{i,j} \mu_{(c,k),l}^{(a,i),(b,j)} \Delta_{(a,i),(b,j)}^{(c,k'),l} \right)}_{\frac{d_a d_b}{d_c} \frac{1}{\text{FPdim } \mathcal{M}} \delta_{k,k'}} \begin{array}{c} \text{Diagram: A square with legs labeled } A, \text{ containing a circle with a dot and a small triangle labeled } c. \end{array} \\ &\stackrel{(7)}{=} \sum_{c,k} N_{ab}^c \frac{\text{FPdim } \mathcal{M}}{d_c} \begin{array}{c} \text{Diagram: A square with legs labeled } A, \text{ containing a circle with a dot and a small triangle labeled } c. \end{array} = \sum_c N_{ab}^c \mathcal{P}^c. \end{aligned} \quad (41)$$

This demonstrates that the operators  $\mathcal{P}^a$  are local versions of the (non-invertible) symmetry  $\mathcal{C}$ , and we can interpret the gauging by  $A$  as gauging the full symmetry.

The equivalence between these gauging maps and duality operators from the previous section can now also be used to show that if a symmetry can be completely gauged, there must exist a symmetric short-range entangled state. Indeed, it suffices to consider the gauging map  $\mathcal{G}_A$ , which as we have shown corresponds to the duality operator labelled by  $X_A$  up to local unitary transformations. Applied to the above setting, the gauging map for the algebra object  $A$  in  $\text{Rep}(\mathcal{A})$  is equivalent to the duality operator labelled by  $\mathbb{C} \in \text{Vec}$ . Acting with  $\mathcal{G}_A^\dagger$  on a generic product state therefore produces an injective  $\mathcal{C}$ -symmetric MPS, since  $\mathcal{G}_A$  is an injective MPO. This way of producing  $\mathcal{C}$ -symmetric MPS by ungauging the symmetry  $\mathcal{C}$  lies at the heart of symmetric tensor network methods, as shown in [52]. Note that the same procedure can also be applied to the setting where the symmetry category  $\mathcal{C}$  contains an anomaly-free subcategory  $\mathcal{C}' \subset \mathcal{C}$ , despite  $\mathcal{C}$  itself being anomalous. In that case one can, by choosing the appropriate algebra object, still construct a short-range entangled  $\mathcal{C}'$ -symmetric state. One notable example is the  $\text{Vec}_{\mathbb{Z}_2}$  subcategory of Ising [70]; another is the  $\text{Vec}_{\mathbb{Z}_3}$  subcategory of the Haagerup category  $\mathcal{H}_3$ , which is discussed in more detail below.

## 6 Symmetry-twisted boundary conditions

We propose a generalisation of the gauging map eq. (30) to act on a Hilbert space twisted by a symmetry defect. We demonstrate the equivalence with the symmetry-twisted duality operators introduced in ref. [51].

In the construction of the anyonic spin chains eq. (21), we identified the module labels  $R_1 \equiv R_{L+1}$  so as to implement *periodic* boundary conditions. More generally, one can insert a *symmetry defect*, labelled by an object  $B \in \mathcal{I}_C$ , before closing the chain. This results in a *symmetry-twisted* Hilbert space, given by

$$\text{Span}_C \left\{ \begin{array}{c} \text{Diagram: A chain of boxes labeled } R_L, R_{L+1}, R_1, R_2 \text{ connected by vertical lines. The first line has a label } Y_{L+\frac{1}{2}} \text{ and the last line has a label } Y_{\frac{1}{2}}. A red arrow labeled } B \text{ points from the } R_{L+1} \text{ box to the } R_1 \text{ box.} \\ \{Y_{i+\frac{1}{2}}\}_i, \{R_i\}_i, \{i_{i+\frac{1}{2}}\}_i \end{array} \right\}. \quad (42)$$

The symmetry twist is *topological* in the sense that the Hilbert space obtained by moving the twist by one site is related to the original one by a unitary transformation given by the bimodule associator of  $\mathcal{R}$ . Note that for the choice  $B = \mathbb{1}$  the periodic chain eq. (21) is retrieved.

Local operators acting on the twisted Hilbert space away from the twist are still given by ladder diagrams of the form eq. (20), whereas local operators that act across the defect are given by a modified version of eq. (20) in which a rotated version of the tensor eq. (22) with  $\mathcal{R} = \mathcal{R}'$  is inserted, as detailed in ref. [51]. As before, a duality amounts to changing the choice of module category  $\mathcal{R} \rightarrow \mathcal{R}'$ . The duality MPOs defined above in eq. (38) now need to be promoted to *tubes*, given by matrix product operators of the form

$$\mathfrak{T}_{\mathcal{R}'|\mathcal{R}}^{B, B', X, X', k, k'} := \begin{array}{c} \text{Diagram: A tube diagram with two horizontal strands. The top strand has boxes labeled } X \text{ and } X'. \text{ The bottom strand has boxes labeled } Y_{L+\frac{1}{2}} \text{ and } Y_{\frac{1}{2}}. A red arrow labeled } B' \text{ points from the } X \text{ box to the } X' \text{ box. A red arrow labeled } B \text{ points from the } Y_{L+\frac{1}{2}} \text{ box to the } Y_{\frac{1}{2}} \text{ box.} \end{array}, \quad (43)$$

where  $X, X' \in \mathcal{I}_{\mathcal{M}}$ ,  $B' \in \mathcal{I}_{\mathcal{C}_{\mathcal{M}}^*}$  and  $X'$  is contained in both  $B \otimes X$  and  $X \otimes B'$ . The trivalent tensors used to fuse the twists with the duality strand are those defined in eq. (31) for the appropriate choice of module categories. Similar to the untwisted case, tubes of the type  $\mathfrak{T}_{\mathcal{R}|\mathcal{R}}$  and  $\mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}$  commute with the aforementioned local operators so that they constitute the symmetry operators of the original and the dual model. Together, the collection of tubes  $\{\mathfrak{T}_{\mathcal{R}|\mathcal{R}}, \mathfrak{T}_{\mathcal{R}|\mathcal{R}'}, \mathfrak{T}_{\mathcal{R}'|\mathcal{R}}, \mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}\}$  is endowed with a multiplication defined via stacking of tubes as well as a dagger. As such they span a finite  $C^*$ -algebra [51, 57, 60]. This algebra admits an Artin-Wedderburn decomposition in terms of simple matrix algebras which are identified with the *topological sectors* of the model. In particular, this algebra consists of two subalgebras spanned by tubes of the type  $\mathfrak{T}_{\mathcal{R}|\mathcal{R}}$  and  $\mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}$  respectively, whose blocks are in one-to-one correspondence with simple objects of the *Drinfel'd center*  $\mathcal{Z}(\mathcal{C})$  labelling the superselection sectors of the original and dual model. Such a (simple) sector  $Z \in \mathcal{I}_{\mathcal{Z}(\mathcal{C})}$  provides a certain (typically non-simple) boundary condition as well as a charge leaving that boundary condition invariant. In virtue of the Morita equivalence between  $\mathcal{C}$  and  $\mathcal{C}_{\mathcal{M}}^*$ ,  $\mathcal{Z}(\mathcal{C})$  is equivalent to  $\mathcal{Z}(\mathcal{C}_{\mathcal{M}}^*)$  as a braided fusion category, and projectors on the topological sectors in both models are given by the minimal central idempotents of the  $\mathfrak{T}_{\mathcal{R}|\mathcal{R}}$  and  $\mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}$  tube algebras. The duality  $\mathcal{R} \rightarrow \mathcal{R}'$  then induces a permutation of the topological sectors, i.e.  $\mathcal{Z}(\mathcal{C}) \xrightarrow{\sim} \mathcal{Z}(\mathcal{C}_{\mathcal{M}}^*)$ , in accordance with the fact that not all boundary conditions in the dual model are compatible with a certain boundary condition in the original model, as is apparent from eq (43).

Let us now proceed by proposing a gauging map acting on the twisted Hilbert space eq. (42) which is unitarily equivalent to a (linear combination) of intertwining tubes eq. (43) [81]. Requiring that the gauging map is constructed from a product of local commuting projectors acting on an initial trivial gauge field and that it acts diagonally on the boundary condition suggests following definition:

$$\mathbb{G}_A^B := \sqrt{d_A} \left( \text{Diagram 1} + \text{Diagram 2} \right), \quad (44)$$

where, as above, every red strand is labelled by the algebra object  $A$ . The unitary circuit that converts  $\mathbb{G}_A^B$  in a linear combination of tubes  $\mathfrak{T}_{\mathcal{R}'|\mathcal{R}}$  parallels the one from the previous section. We begin by inflating all occurrences of the comultiplication tensor in  $\mathbb{G}_A^B$  using eq. (32). Subsequently we apply the gates defined in eq. (35). This results again in a collection of ‘cups’ which in turn can be removed by a depth 2 circuit, as discussed around eq. (37). As such, one ends up with

$$\mathbb{G}_A^B \cong \sqrt{d_A} \left( \text{Diagram 3} + \text{Diagram 4} \right). \quad (45)$$

Finally, the boundary condition  $B$  can be fused to the duality strands  $X_A$  and  $\bar{X}_A$  by sequentially applying a variation of the gate eq. (35). The following linear combination of tubes is obtained:

$$\mathbb{G}_A^B \cong \sqrt{d_{X_A}} \sum_{\substack{X', B' \\ k, k'}} \left( \text{Diagram 5} \right). \quad (46)$$

The orientation reversing flag can then be absorbed in the fusion tensor to its right, at the cost of introducing extra quantum dimensions as per eq. (B.5). In the end, we obtain that

$$\mathbb{G}_A^B \cong \sum_{\substack{X', B' \\ k, k'}} \sqrt{\frac{d_{X_A} d_{X'}}{d_{B'}}} \times \mathfrak{T}_{\mathcal{R}'|\mathcal{R}}^{B, B', X, X', k, k'}. \quad (47)$$

## 7 Examples

We illustrate our formalism by means of two examples corresponding to symmetries encoded in the fusion category of representations of  $S_3$  and the Haagerup categories.

### 7.1 $\text{Rep}(S_3)$

As a first example, consider the fusion category  $\text{Rep}(S_3)$  of (finite-dimensional complex) representations of the permutation group  $S_3$ . An example Hamiltonian exhibiting  $S_3$  symmetry is the spin-1/2 XXZ model. This model and all its dualities were considered in ref. [51].



The simple objects of  $\text{Rep}(S_3)$  are denoted by  $\mathcal{I}_{\text{Rep}(S_3)} = \{\underline{1}, \underline{1}^*, \underline{2}\}$ , and correspond to the trivial, sign and 2-dimensional irreducible representation respectively. Its non-trivial fusion rules are given by

$$\begin{aligned}\underline{1}^* \otimes \underline{1}^* &\simeq \underline{1}, \\ \underline{1}^* \otimes \underline{2} &\simeq \underline{2} \oplus \underline{1}^* \simeq \underline{2}, \\ \underline{2} \otimes \underline{2} &\simeq \underline{1} \oplus \underline{1}^* \oplus \underline{2}.\end{aligned}\quad (48)$$

Recall that the inequivalent indecomposable module categories of  $\text{Rep}(S_3)$  are in one-to-one correspondence with subgroups of  $S_3$  up to conjugacy, of which there are four:  $S_3$ ,  $\mathbb{Z}_3$ ,  $\mathbb{Z}_2$  and  $\{1\}$ . Given such a subgroup  $H \subseteq S_3$  the corresponding module category is  $\text{Rep}(H)$  which is acted upon by  $\text{Rep}(S_3)$  via  $\lambda \triangleleft \rho \simeq \lambda \otimes \text{Res}_H^{S_3}(\rho)$ , where  $\text{Res}_H^{S_3}$  denotes the restriction of  $S_3$ -representations to the subgroup  $H$  and  $\otimes$  stands for the usual tensor product of  $H$ -representations. For each of these module categories their Morita duals are computed to be

$$\begin{aligned}\text{Rep}(S_3)_{\text{Rep}(S_3)}^* &\simeq \text{Rep}(S_3), & \text{Rep}(S_3)_{\text{Rep}(\mathbb{Z}_3)}^* &\simeq \text{Vec}_{S_3}, \\ \text{Rep}(S_3)_{\text{Rep}(\mathbb{Z}_2)}^* &\simeq \text{Rep}(S_3), & \text{Rep}(S_3)_{\text{Vec}}^* &\simeq \text{Vec}_{S_3}.\end{aligned}\quad (49)$$

We now proceed to discuss the algebra objects in  $\text{Rep}(S_3)$  to which each of these module categories correspond.

Let us denote the irreps of  $\mathbb{Z}_2$  by  $\mathcal{I}_{\text{Rep}(\mathbb{Z}_2)} = \{\underline{1}_{\mathbb{Z}_2}, \underline{1}_{\mathbb{Z}_2}^*\}$  where  $\underline{1}_{\mathbb{Z}_2}^*$  stands for the sign representation of  $\mathbb{Z}_2$ . The restriction functor for this subgroup applied to the  $S_3$ -irreps yields  $\text{Res}_{\mathbb{Z}_2}^{S_3}(\underline{1}) \simeq \underline{1}_{\mathbb{Z}_2}$ ,  $\text{Res}_{\mathbb{Z}_2}^{S_3}(\underline{1}^*) \simeq \underline{1}_{\mathbb{Z}_2}^*$  and  $\text{Res}_{\mathbb{Z}_2}^{S_3}(\underline{2}) \simeq \underline{1}_{\mathbb{Z}_2} \oplus \underline{1}_{\mathbb{Z}_2}^*$ . Making use of def. (A.15) and the  $\text{Rep}(\mathbb{Z}_2)$  fusion rules it is then straightforward to find that at the level of the objects, the internal Homs correspond to:

$$\begin{aligned}\underline{\text{Hom}}(\underline{1}_{\mathbb{Z}_2}, \underline{1}_{\mathbb{Z}_2}) &\simeq \underline{1} \oplus \underline{2}, \\ \underline{\text{Hom}}(\underline{1}_{\mathbb{Z}_2}^*, \underline{1}_{\mathbb{Z}_2}^*) &\simeq \underline{1} \oplus \underline{2}.\end{aligned}\quad (50)$$

All components of their respective multiplication morphisms evaluate to one except those given in terms of the F-symbols

$$\begin{aligned}(F_{\underline{1}_{\mathbb{Z}_2}}^{\underline{1}_{\mathbb{Z}_2} \underline{2} \underline{2}})^{\underline{1}, 11} &= (F_{\underline{1}_{\mathbb{Z}_2}}^{\underline{1}_{\mathbb{Z}_2} \underline{2} \underline{2}})^{\underline{1}, 11} = \frac{1}{\sqrt{2}}, \\ (F_{\underline{1}_{\mathbb{Z}_2}}^{\underline{1}_{\mathbb{Z}_2} \underline{2} \underline{2}})^{\underline{2}, 11} &= \frac{1}{\sqrt{2}}, \quad (F_{\underline{1}_{\mathbb{Z}_2}^*}^{\underline{1}_{\mathbb{Z}_2}^* \underline{2} \underline{2}})^{\underline{2}, 11} = -\frac{1}{\sqrt{2}}.\end{aligned}\quad (51)$$

These algebra structures are readily checked to be isomorphic. Equivalently, both inner Homs can be converted into each other by an automorphism of the module category given by the module functor  $- \triangleleft \underline{1}^*$ .

We write  $\mathcal{I}_{\text{Rep}(\mathbb{Z}_3)} = \{\underline{1}_{\mathbb{Z}_3}, \underline{\omega}_{\mathbb{Z}_3}, \underline{\omega}_{\mathbb{Z}_3}^2\}$  for the three one-dimensional representations of  $\mathbb{Z}_3$ . Restrictions of the  $S_3$ -irreps to  $\mathbb{Z}_3$  are computed to be  $\text{Res}_{\mathbb{Z}_3}^{S_3}(\underline{1}) \simeq \underline{1}_{\mathbb{Z}_3}$ ,  $\text{Res}_{\mathbb{Z}_3}^{S_3}(\underline{1}^*) \simeq \underline{1}_{\mathbb{Z}_3}$  and  $\text{Res}_{\mathbb{Z}_3}^{S_3}(\underline{2}) \simeq \underline{\omega}_{\mathbb{Z}_3} \oplus \underline{\omega}_{\mathbb{Z}_3}^*$ . The internal Homs are given by  $\underline{\text{Hom}}(\lambda, \lambda) \simeq \underline{1} \oplus \underline{1}^*$  for all  $\lambda \in \mathcal{I}_{\text{Rep}(\mathbb{Z}_3)}$ . The components of the multiplication map all evaluate to  $\mu_{\lambda_1 \otimes \lambda_2}^{\lambda_1, \lambda_2} = \frac{1}{\sqrt{2}}$  for all  $\lambda_1, \lambda_2 \in \{\underline{1}, \underline{1}^*\}$ .

Finally, considering  $\text{Vec}$  as module category, one finds as anticipated the regular representation of  $S_3$  as corresponding algebra object:

$$\underline{\text{Hom}}(\mathbb{C}, \mathbb{C}) \simeq \underline{1} \oplus \underline{1}^* \oplus 2 \cdot \underline{2}.\quad (52)$$

Its algebra morphism is then specified by the F-symbol  $F_{\mathbb{C}}^{C\lambda_1\lambda_2}$  whose components evaluate to the (unitary) Clebsch-Gordan coefficients according to the prescription:

$$(F_{\mathbb{C}}^{C\lambda_1\lambda_2})_{\mathbb{C}, ij}^{\lambda_3, k1} = \begin{bmatrix} \lambda_1 & \lambda_2 & \lambda_3 \\ i & j & k \end{bmatrix},\quad (53)$$

where  $\lambda_3 \in \lambda_1 \otimes \lambda_2$ .

## 7.2 Haagerup categories

The Haagerup Morita class consists of three distinct unitary fusion categories denoted by  $\mathcal{H}_1$ ,  $\mathcal{H}_2$  and  $\mathcal{H}_3$ . The first two arise from the *Haagerup subfactor* proposed and constructed in refs. [2, 34], while in ref. [33] the existence of  $\mathcal{H}_3$  was proven by providing an appropriate algebra object in  $\mathcal{H}_1$  whose corresponding category of bimodules was hitherto unknown. The interest in the Haagerup subfactor stems from Jones' conjectured correspondence between subfactors and conformal field theories [41]. Since the conjecture does not provide an explicit construction of the CFT, this has led to a plethora of recent studies in which both one-dimensional quantum models [10, 13, 37, 40, 50] as well as classical models [79] with  $\mathcal{H}_3$  symmetry have been constructed and investigated using analytical and numerical methods, so as to shed light on the alleged Haagerup CFT.

The unitary fusion categories  $\mathcal{H}_2$  and  $\mathcal{H}_3$  are both rank six and have the same simple objects, denoted by  $\mathcal{I}_{\mathcal{H}_2} = \mathcal{I}_{\mathcal{H}_3} = \{\mathbb{1}, \alpha, \alpha^2, \rho, \alpha\rho, \alpha^2\rho\}$ . They satisfy the same fusion rules, the non-trivial ones being given by

$$\begin{aligned} \alpha \otimes \alpha &\simeq \alpha^2, & \alpha^i \otimes \rho &\simeq \alpha^i \rho \simeq \rho \otimes \alpha^i, \\ \alpha^2 \otimes \alpha &\simeq \mathbb{1}, & \rho \otimes \rho &\simeq \mathbb{1} \oplus \rho \oplus \alpha\rho \oplus \alpha^2\rho. \end{aligned} \quad (54)$$

The two categories differ in their unitary F-symbols, which can not be transmuted into each other via basis transformations on their multiplicity spaces. Both  $\mathcal{H}_2$  and  $\mathcal{H}_3$  possess a  $\text{Vec}_{\mathbb{Z}_3}$  subcategory consisting of the simple objects  $\{\mathbb{1}, \alpha, \alpha^2\}$ .

The fusion category  $\mathcal{H}_1$  on the other hand has four simple objects denoted by  $\mathcal{I}_{\mathcal{H}_1} = \{\mathbb{1}, \mu, \eta, \nu\}$ . Its fusion rules are symmetric, and the non-trivial ones read

$$\begin{aligned} \mu \otimes \mu &\simeq \mathbb{1} \oplus \nu, & \nu \otimes \nu &\simeq \mathbb{1} \oplus 2\nu \oplus 2\eta \oplus \mu, \\ \mu \otimes \nu &\simeq \mu \oplus \nu \oplus \eta, & \nu \otimes \eta &\simeq 2\nu \oplus \eta \oplus \mu, \\ \mu \otimes \eta &\simeq \nu \oplus \eta, & \eta \otimes \eta &\simeq \mathbb{1} \oplus \nu \oplus \eta \oplus \mu. \end{aligned} \quad (55)$$

The F-symbols of  $\mathcal{H}_1$ ,  $\mathcal{H}_2$  and  $\mathcal{H}_3$  were first calculated in refs. [5, 36, 63] respectively.

Since these three fusion categories are Morita equivalent, there exist invertible bimodule categories between them [33]. Let us denote these bimodule categories by  $\mathcal{H}_{i,j}$ ,  $i, j = 1, 2, 3$ , such that  $(\mathcal{H}_i)_{\mathcal{H}_{i,j}}^* \simeq \mathcal{H}_j$ , with the understanding that  $\mathcal{H}_{j,i} \equiv \mathcal{H}_{i,j}^{\text{op}}$  and  $\mathcal{H}_{i,i} \equiv \mathcal{H}_i$ .

The aforementioned quantum models with  $\mathcal{H}_3$  symmetry can be obtained by choosing  $\mathcal{M} = \mathcal{D} = \mathcal{H}_3$  as input to the models defined in eq. (20), and appropriate choices for the coefficients  $\lambda_{i,n}$ . Dual models with symmetries encoded in  $\mathcal{H}_1$ ,  $\mathcal{H}_2$  are then obtained by choosing the module categories  $\mathcal{H}_{1,3}$ , respectively  $\mathcal{H}_{2,3}$ , over the input category whose module associators are attached to ref. [5]. Coincidentally, in virtue of the fact that  $\text{Fun}_{\mathcal{D}}(\mathcal{D}, \mathcal{M}) \simeq \mathcal{M}$ , the category of duality operators is in those respective cases also given by  $\mathcal{H}_{1,3}, \mathcal{H}_{2,3}$ .

Let us first consider  $\mathcal{H}_{1,3}$ , whose simple objects we write as  $\{\Gamma, \Gamma\alpha, \Gamma\alpha^2, \Lambda\}$ . It follows from  $N_{\Gamma, \alpha\rho}^{\Gamma} = N_{\Gamma, \alpha^2\rho}^{\Gamma} = 1$  that  $\underline{\text{Hom}}(\Gamma, \Gamma) \simeq \mathbb{1} \oplus \alpha\rho \oplus \alpha^2\rho$ , as an object. The subset of F-symbols appearing in the algebra structure on  $\underline{\text{Hom}}(\Gamma, \Gamma)$  are given in ref. [5]. In similar fashion one constructs the algebra objects  $\underline{\text{Hom}}(\Gamma\alpha, \Gamma\alpha)$ ,  $\underline{\text{Hom}}(\Gamma\alpha^2, \Gamma\alpha^2)$ , which are related to  $\underline{\text{Hom}}(\Gamma, \Gamma)$  by conjugation with the invertible objects  $\alpha^2$  and  $\alpha$  of  $\mathcal{H}_3$  respectively. Finally one constructs also the algebra object  $\underline{\text{Hom}}(\Lambda, \Lambda)$  which turns out to be equivalent to  $\underline{\text{Hom}}(\Lambda, \Lambda) \simeq \mathbb{1} \oplus \alpha\rho \oplus \alpha^2\rho$ .

$\mathcal{H}_{2,3}$  has two simple objects  $\mathcal{I}_{\mathcal{H}_{2,3}} \simeq \{\Omega, \Omega\rho\}$ . Its fusion rules can be inferred from  $\Omega \triangleleft \alpha^i \simeq \Omega$  and  $\Omega \triangleleft \rho \simeq \Omega\rho$ . From the first set of fusion rules it follows that the duality  $\mathcal{H}_{2,3}$  amounts to gauging the  $\text{Vec}_{\mathbb{Z}_3}$  symmetry generated by  $\alpha$ , as indeed  $\underline{\text{Hom}}(\Omega, \Omega) \simeq \mathbb{1} \oplus \alpha \oplus \alpha^2$ . The other module object gives rise to the Morita equivalent algebra object  $\underline{\text{Hom}}(\Omega\rho, \Omega\rho) \simeq \mathbb{1} \oplus \alpha \oplus \alpha^2 \oplus 3 \cdot (\rho \oplus \alpha\rho \oplus \alpha^2\rho)$ . Likewise, one shows that  $\mathcal{H}_3$  is obtained from gauging the  $\text{Vec}_{\mathbb{Z}_3}$  subsymmetry inside  $\mathcal{H}_2$ .

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## A Categorical prerequisites

In this section we introduce the necessary category theoretic tools, notations and conventions used throughout the manuscript. We refer to the standard references [21, 25] for a more elaborate exposition.

### A.1 Fusion categories and algebra objects

Succinctly, a *unitary fusion category* (UFC)  $\mathcal{C}$  consists of a collection of objects labeling topological defects whose set of (representatives of isomorphism classes of) simple objects is denoted  $\mathcal{I}_{\mathcal{C}}$ . The collection of defects is endowed with a monoidal product  $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$  encoding their fusion. It is useful to introduce an integer-valued rank 3 tensor  $N$  encoding the fusion rules as  $X_1 \otimes X_2 \simeq \bigoplus_{X_3 \in \mathcal{I}_{\mathcal{C}}} N_{X_1 X_2}^{X_3} X_3$ . Among the simple objects there exists a unique trivial defect  $\mathbb{1}$  satisfying  $\mathbb{1} \otimes X \simeq X \otimes \mathbb{1} \simeq X$ ,  $\forall X \in \mathcal{I}_{\mathcal{C}}$ . The fusion rules are associative up to an isomorphism which satisfies a consistency axiom known as the *pentagon equation*. Given an appropriate choice of basis for the junction spaces  $\mathcal{C}(X_1 \otimes X_2, X_3)$ , the components of this isomorphism can be expressed as unitary *F-symbols* defined graphically via:

$$\begin{array}{c} X_1 \quad X_2 \quad X_3 \\ \diagdown \quad \diagup \quad \diagdown \\ \text{---} i \text{---} j \text{---} \\ \diagup \quad \diagdown \quad \diagup \\ X_5 \quad X_4 \end{array} = \sum_{X_6, k, l} (F_{X_4}^{X_1 X_2 X_3})_{X_5, ij}^{X_6, kl} \begin{array}{c} X_1 \quad X_2 \quad X_3 \\ \diagdown \quad \diagup \quad \diagdown \\ \text{---} k \text{---} l \text{---} \\ \diagup \quad \diagdown \quad \diagup \\ X_5 \quad X_6 \end{array}. \quad (\text{A.1})$$

In the unitary gauge we additionally assume that

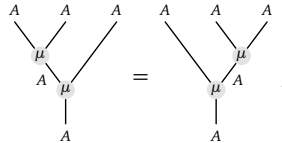
$$(F_X^{X \bar{X} X})_{\mathbb{1}, \mathbb{1} \mathbb{1}}^{\mathbb{1}, \mathbb{1}} = \frac{x_X}{d_X}, \quad (\text{A.2})$$

where  $x_X = \pm 1$  denotes the *Frobenius-Schur indicator* of  $X$ , which satisfies  $x_{\bar{X}} = x_X^*$ , and the *quantum dimension*  $d_X$  of the defect  $X$  is the unique solution to

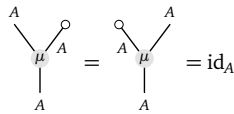
$$\begin{aligned} d_X &> 0, \\ d_{X_1} d_{X_2} &= \sum_{X_3 \in \mathcal{I}_{\mathcal{C}}} N_{X_1 X_2}^{X_3} d_{X_3}, \end{aligned} \quad (\text{A.3})$$

for all  $X, X_1, X_2 \in \mathcal{I}_C$ .

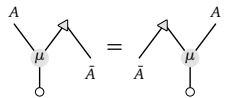
An *algebra object*, henceforth often called simply *algebra*, in a UFC  $\mathcal{C}$  is a triple  $(A, \mu, \eta)$  consisting of an object  $A \simeq \bigoplus_{X \in \mathcal{I}_C} N_X^A X$ , with  $N_X^A = \dim_{\mathcal{C}} \mathcal{C}(X, A)$ , together with a *multiplication morphism*  $\mu \in \mathcal{C}(A \otimes A, A)$  that satisfies following associativity condition on the nose


(A.4)

and *unit map*  $\eta \in \mathcal{C}(\mathbb{1}, A)$ , denoted in our graphical calculus by a white node, satisfying

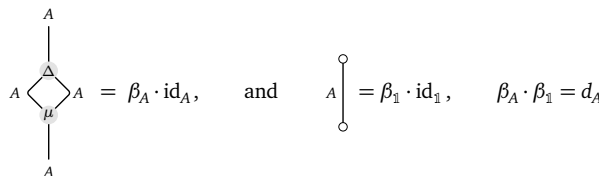

(A.5)

It is a useful fact that  $N_X^A \leq [d_X]$  for the algebra objects considered in this manuscript. We also assume that all algebra objects are *haploid*, meaning that  $A$  contains the monoidal unit  $\mathbb{1}$  of  $\mathcal{C}$  with multiplicity 1, or thus  $N_{\mathbb{1}}^A = 1$ . Indeed, by virtue of [62, Thm. 3.1] every Morita class of (semisimple) indecomposable algebra objects contains at least one haploid representative. An algebra object  $(A, \mu, \eta)$  is called *symmetric* if there exists a morphism  $\varepsilon \in \mathcal{C}(A, \mathbb{1})$  so that


(A.6)

where the flags depict respectively the right and left coevaluation morphisms of  $A$ . It can be shown [25, Cor. 3.10] that Haploid algebras are symmetric for all  $\varepsilon \in \mathcal{C}(A, \mathbb{1})$  in virtue of their defining property  $\dim_{\mathcal{C}} \mathcal{C}(\mathbb{1}, A) = 1$ .

One can then define an analogous *coalgebra object* as a triple  $(A, \Delta, \varepsilon)$  where  $A$  is an object in  $\mathcal{C}$ ,  $\Delta \in \mathcal{C}(A, A \otimes A)$  is a *comultiplication map* satisfying a coassociativity condition akin to eq. (A.4) and  $\varepsilon \in \mathcal{C}(A, \mathbb{1})$  is a *counit map* which together with  $\Delta$  is required to obey a mirrored version of eq. (A.5). A *special algebra object* is an object that is both an algebra  $(A, \mu, \eta)$  and coalgebra  $(A, \Delta, \varepsilon)$  for which


(A.7)

with the quantum dimension of the algebra object  $A$  given by  $d_A = \sum_{X \in \mathcal{I}_C} N_X^A d_X$ . To reduce the occurrence of quantum dimensions in the main text, we adopt without loss of generality the convention that  $\beta_A = 1$ ,  $\beta_{\mathbb{1}} = d_A$ .

Proceeding as such, a *Frobenius algebra object* is a quintuple  $(A, \mu, \eta, \Delta, \varepsilon)$  where  $(A, \mu, \eta)$  constitutes an algebra and  $(A, \Delta, \varepsilon)$  a coalgebra, such that the multiplication and comultipli-

cation map jointly obey the *Frobenius relations*

$$\text{Diagram (A.8): } \mu \circ \delta = \delta \circ \mu \text{ (in diagrammatic notation)} \quad (\text{A.8})$$

Considering an algebra  $(A, \Delta, u)$  in  $\mathcal{C}$ , the multiplication morphism can be written in components by means of

$$\text{Diagram (A.9): } \mu_{(X_1, X_2), X_3} = \sum_l \mu_{(X_1, X_2), l}^{(X_3, k, l)} \delta_{(X_1, i), (X_2, j)} \quad (\text{A.9})$$

where  $i, j, k$  and  $l$  label orthonormal basis vectors in the vector spaces  $\mathcal{C}(X_1, A)$ ,  $\mathcal{C}(X_2, A)$ ,  $\mathcal{C}(A, X_3)$  and  $\mathcal{C}(X_1 \otimes X_2, X_3)$  respectively. Analogously one defines the tensor  $\Delta_{(X_1, i), (X_2, j)}^{(X_3, k, l)}$  for the comultiplication map.

Two algebra objects are *isomorphic* when they are related by unitary basis transformations on the multiplicity spaces  $\mathcal{C}(X_i, A)$  and  $\mathcal{C}(A, X_i)$ .

## A.2 Module categories and module functors

Given a UFC  $\mathcal{D}$ , a (unitary finite indecomposable semisimple) *right module category*  $\mathcal{R}$  over  $\mathcal{D}$  is a semisimple category whose set of (representative objects of isomorphism classes of) simple objects is written as  $\mathcal{I}_{\mathcal{R}}$  and which is equipped with a right action bifunctor  $\triangleleft: \mathcal{R} \times \mathcal{D} \rightarrow \mathcal{R}$ . Up to an isomorphism the action is associative. In an appropriate basis of the junction spaces, this isomorphism can once more be expressed as following unitary F-symbols:

$$\text{Diagram (A.10): } F_{R_2}^{R_1 Y_1 Y_2}_{R_3, i j} = \sum_{Y_3, i, j} (F_{R_2}^{R_1 Y_1 Y_2})_{R_3, i j}^{Y_3, k l} \quad (\text{A.10})$$

These F-symbols satisfy a mixed pentagon equation which also involves the F-symbols of  $\mathcal{D}$ . We choose a basis for the Hom spaces such that

$$(F_{R_2}^{R_1 1 X_2})_{R_1, 1 i}^{X_2, 1 j} = (F_{R_2}^{R_1 X_1 1})_{R_2, i 1}^{X_1, 1 j} = \delta_{i, j}. \quad (\text{A.11})$$

Also for the module objects  $R \in \mathcal{I}_{\mathcal{R}}$  we define quantum dimensions, which are uniquely specified by [22]:

$$\begin{aligned} d_R &> 0, \\ d_{R_1} d_X &= \sum_{R_2 \in \mathcal{I}_{\mathcal{R}}} N_{R_1 X}^{R_2} d_{R_2}, \\ \sum_{R \in \mathcal{I}_{\mathcal{R}}} d_R^2 &= \sum_{X \in \mathcal{I}_{\mathcal{C}}} d_X^2, \end{aligned} \quad (\text{A.12})$$

for all  $R, R_1 \in \mathcal{I}_{\mathcal{R}}, X \in \mathcal{I}_{\mathcal{C}}$ . Analogously, one defines a *left* module category over  $\mathcal{D}$  by mirroring the above definition.

Crucial to the construction of the duality intertwiners and symmetry MPOs is the notion of  $\mathcal{D}$ -module functors between (right)  $\mathcal{D}$ -module categories  $\mathcal{R}$  and  $\mathcal{R}'$ . Concretely, a module functor between  $\mathcal{R}$  and  $\mathcal{R}'$  is a functor  $\mathfrak{F} : \mathcal{R} \rightarrow \mathcal{R}'$ , which respects the module structures of  $\mathcal{R}$  and  $\mathcal{R}'$  in the sense that there exists an isomorphism  $\omega$  whose components are given by  $\omega^{RY} : \mathfrak{F}(R \triangleleft Y) \xrightarrow{\sim} \mathfrak{F}(R) \triangleleft Y$  for all  $R, Y$  in  $\mathcal{I}_{\mathcal{R}}$  and  $\mathcal{I}_{\mathcal{D}}$  respectively and where  $\triangleleft$  and  $\triangleleft$  denote the action functors of  $\mathcal{R}$  and  $\mathcal{R}'$  respectively. Such module functors are organized in a (semisimple) category  $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$  [21]. Given a simple object  $X$  in  $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$ , let us write  ${}^X\omega$  for its corresponding isomorphism. Its components can be expressed via the diagram

$$= \sum_{R_2, k, l} ({}^X\omega^{R_1 Y})_{R'_2, ij}^{R_2, kl} . \quad (\text{A.13})$$

Consider now three right module categories  $\mathcal{M}, \mathcal{N}, \mathcal{O}$  over  $\mathcal{D}$ . Module functors in  $\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{N})$  and  $\text{Fun}_{\mathcal{D}}(\mathcal{N}, \mathcal{O})$  can be composed, resulting in a module functor in  $\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{O})$ . Being semisimple, we can consider simple module functors  $X_1, X_2$  in  $\mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{N}, \mathcal{O})}$  and  $\mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{N})}$  respectively. Decomposing the composition of  $X_1$  and  $X_2$  in simples in  $\mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{O})}$  then happens via unitary complex matrices whose entries are defined in terms of following string diagrams:

$$= \sum_{N, kl} (F_O^{X_1 X_2 M})_{X_3, ij}^{N, kl} , \quad (\text{A.14})$$

in which  $i, j, k, l$  are indices in the appropriate junction spaces.

### A.3 Internal Hom objects and algebras

Given a right module category  $\mathcal{M}$  over  $\mathcal{C}$ , an *internal Hom object* [62] for any two objects  $M_1, M_2$  is an object  $\underline{\text{Hom}}(M_1, M_2)$  in  $\mathcal{C}$  together with a collection of isomorphisms

$$\mathcal{M}(M_1 \triangleleft X, M_2) \simeq \mathcal{C}(X, \underline{\text{Hom}}(M_1, M_2)), \quad (\text{A.15})$$

satisfying a number of naturality conditions specified in ref. [66]. Crucially, for any simple object  $M \in \mathcal{I}_{\mathcal{M}}$ ,  $\underline{\text{Hom}}(M, M)$  is guaranteed to be a special haploid symmetric Frobenius algebra object, as demonstrated in ref. [66]. Let us spell this out explicitly. Notating  $A \equiv \underline{\text{Hom}}(M, M)$ , it readily follows from (A.15) that the decomposition of  $A$  in simple objects  $\mathcal{C}$  is given by  $A \simeq \bigoplus_{X \in \mathcal{I}_{\mathcal{C}}} N_{MX}^M X$ , where  $N_{MX}^M \equiv \dim_{\mathcal{C}} \mathcal{M}(M \triangleleft X, M)$ . The components of the multiplication and comultiplication maps can be related to the right module associator (A.10) via

$$\mu_{(X_3, k), l}^{(X_1, i), (X_2, j)} = \frac{1}{d_M} \sqrt{\frac{d_{X_1} d_{X_2}}{d_{X_3}}} (F_M^{MX_1 X_2})_{M, ij}^{X_3, lk}, \quad (\text{A.16})$$

$$\Delta_{(X_1, i), (X_2, j)}^{(X_3, k), l} = \frac{1}{d_M} \sqrt{\frac{d_{X_1} d_{X_2}}{d_{X_3}}} (\bar{F}_M^{MX_1 X_2})_{M, ij}^{X_3, lk},$$

in which once more the isomorphisms (A.15) were leveraged. In this normalisation of the multiplication and comultiplication map and the F-symbol eq. (A.11), the unique component of the unit and counit labelled by  $\eta_1$  and  $\varepsilon_1$  respectively is given by

$$\eta_1 = \varepsilon_1 = d_M, \quad (\text{A.17})$$

which is verified to be compatible with the normalization in eq. (A.7).

## B Orientation-reversing flags

Let  $X$  be a simple object in  $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$  in what follows. Orientation-reversing ‘flags’ for  $X$  are defined in terms of the fusion (splitting) tensors by terminating them in the vacuum label  $\mathbb{1}$ :

$$\begin{aligned} \begin{array}{c} R' \\ \text{---} X \text{---} \bar{X} \text{---} \\ R \end{array} &\equiv \sum_{i,j} \begin{array}{c} R' \\ i \text{---} X \text{---} \bar{X} \text{---} j \\ R \end{array} |RXR', i\rangle \otimes |R'\bar{X}R, j\rangle \\ &\equiv \sum_{i,j} \sqrt{d_X} \times \begin{array}{c} \mathbb{1} \\ \downarrow \\ R' \quad \downarrow \quad R' \\ \text{---} X \text{---} \mathbb{1} \text{---} \bar{X} \text{---} \\ R \end{array} |RXR', i\rangle \otimes |R'\bar{X}R, j\rangle, \end{aligned} \quad (\text{B.1})$$

and

$$\begin{aligned} \begin{array}{c} R \\ \text{---} X \text{---} \bar{X} \text{---} \\ R' \end{array} &\equiv \sum_{i,j} \begin{array}{c} R \\ i \text{---} X \text{---} \bar{X} \text{---} j \\ R' \end{array} \langle RXR', i| \otimes \langle R'\bar{X}R, j| \\ &\equiv \sum_{i,j} \sqrt{d_X} \times \begin{array}{c} i \quad j \\ \downarrow \quad \downarrow \\ R' \quad \downarrow \quad R' \\ \text{---} X \text{---} \mathbb{1} \text{---} \bar{X} \text{---} \\ R \end{array} \langle RXR', i| \otimes \langle R'\bar{X}R, j|. \end{aligned} \quad (\text{B.2})$$

In these definitions,  $\mathbb{1}$  denotes the monoidal unit in the fusion category  $\mathcal{C}_{\mathcal{R}'}$ , and  $1$  stands for the sole basis vector in  $\mathcal{R}'(\mathbb{1} \triangleright R, R) \simeq \mathbb{C}$ .

The Frobenius-Schur indicator, which for module objects is defined in a way analogous to eq. (A.2), can be used to reverse the orientation of the flags, concretely:

$$\begin{array}{c} \text{---} X \text{---} \bar{X} \text{---} \end{array} = \varkappa_X^* \begin{array}{c} \text{---} X \text{---} \bar{X} \text{---} \end{array}, \quad (\text{B.3})$$

$$\begin{array}{c} \text{---} X \text{---} \bar{X} \text{---} \end{array} = \varkappa_X \begin{array}{c} \text{---} X \text{---} \bar{X} \text{---} \end{array}.$$

The flags satisfy a number of properties generalising those spelled out in refs. [9, 54]. Most notably, they satisfy a collection of ‘snake axioms’ stating that oppositely paired flags cancel in the sense that eg.

$$\begin{array}{c} \text{---} X \text{---} \bar{X} \text{---} \end{array} = \begin{array}{c} \text{---} X \text{---} \end{array}. \quad (\text{B.4})$$

As such, these flags implement the evaluation and coevaluation maps of  $X$ ,  $\text{ev}_X : \bar{X} \otimes X \rightarrow \mathbb{1}$ , respectively  $\text{coev}_X : \mathbb{1} \rightarrow X \otimes \bar{X}$ , in terms of triple line tensors.

Moreover, these flags can be used to bend lines on the fusion and splitting tensors as follows:

$$\begin{aligned}
 & \text{Diagram 1} = \sum_j \frac{\sqrt{d_{X_1}} (F_{X_2}^{\bar{X}_1 X_1 X_2})_{\mathbb{1}, 11}^{X_3, ij}}{\sqrt{\frac{d_{X_3}}{d_{X_2}} (A_{X_3}^{X_1 X_2})_{ij}}} \text{Diagram 2}, \\
 & \text{Diagram 3} = \sum_j \frac{\sqrt{d_{X_2}} (F_{X_1}^{X_1 X_2 \bar{X}_2})_{X_3, ij}^{\mathbb{1}, 11}}{\sqrt{\frac{d_{X_3}}{d_{X_1}} (B_{X_3}^{X_1 X_2})_{ij}}} \text{Diagram 4}.
 \end{aligned} \tag{B.5}$$

The unitary matrices  $A$  and  $B$  are defined as [46]

$$\begin{aligned}
 (A_{X_3}^{X_1 X_2})_{ij} &= \sqrt{\frac{d_{X_1} d_{X_2}}{d_{X_3}}} (F_{X_2}^{\bar{X}_1 X_1 X_2})_{\mathbb{1}, 11}^{X_3, ij}, \\
 (B_{X_3}^{X_1 X_2})_{ij} &= \sqrt{\frac{d_{X_1} d_{X_2}}{d_{X_3}}} (F_{X_1}^{X_1 X_2 \bar{X}_2})_{X_3, ij}^{\mathbb{1}, 11}.
 \end{aligned} \tag{B.6}$$

## C Proof (co)algebra multiplication

Here, we showcase that the inflated comultiplication tensor (32) satisfies the comultiplication axiom by making use of our graphical calculus [26, 66]. Invoking the orientation-reversing flags from appendix B one finds:

$$\sum_{\substack{\{X_j\}_j, \\ \{i_j\}_j}} \frac{\sqrt{d_{X_1} d_{X_2} d_{X_3} d_{X_4} d_{X_5}}}{d_{X_A}^4} \text{Diagram} \tag{C.1}$$



$$\begin{aligned}
&= \sum_{\{X_j\}_j, \{i_j\}_j} \frac{\sqrt{d_{X_1} d_{X_2} d_{X_3} d_{X_4}}}{d_{X_A}^3} \\
&= \sum_{\{X_j\}_j, \{i_j\}_j} \frac{\sqrt{d_{X_1} d_{X_2} d_{X_3} d_{X_4} d_{X_6}}}{d_{X_A}^4}
\end{aligned}$$

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# Systematic construction of stabilizer codes via gauging abelian boundary symmetries

## 9.1 Synopsis

In this project, we generalise a construction of the second author, published in ref. [160], to arbitrary spatial dimensions. Starting from a generic  $d$ -dimensional quantum state with an *abelian* symmetry, we iteratively gauge the symmetry and its dual to obtain a state in one dimension higher. The admissible initial symmetries include global 0-form, higher-form, *subsystem* and *fractal symmetry*.

The resulting  $(d + 1)$ -dimensional state naturally inherits this original symmetry at its boundary and can be shown to be the ground state of a local commuting projector Hamiltonian. The construction relies on an extension, developed in this work, of Williamson's gauging procedure of *submanifold symmetries* [161]. We formulate the framework in terms of tensor networks, where gauging maps are implemented as tensor network operators akin to those of ref. [139]. The composition of these then provides a tensor network representation of the one-higher-dimensional state.

We illustrate the approach through several examples, including three-dimensional *Clifford-deformed surface codes* and *type-I fracton phases* obtained by gauging linear subsystem and *Sierpinski* fractal symmetries.

## 9.2 Reference

*Systematic construction of stabilizer codes via gauging abelian boundary symmetries*

**B. Vancraeynest-De Cuiper**, J. Garre-Rubio

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## 9.3 Contributions of the author

The current project is a generalisation of the construction developed in ref. [160] by José Garre-Rubio to arbitrary dimensions. My main contribution was in generalising the gauging procedure of Williamson, which in its original form was formulated for the symmetry group  $\mathbb{Z}_2$ , to arbitrary abelian groups. This extension constitutes the main technical ingredient of the work. The remaining constructions follow by applying the procedure to the relevant symmetry structures.

I co-wrote the majority of the manuscript and worked out the example of the three-dimensional Clifford-deformed surface codes. The other examples are fully due to José Garre-Rubio.

# Systematic construction of stabilizer codes via gauging abelian boundary symmetries

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We propose a systematic framework to construct a  $(d+1)$ -dimensional stabilizer model from an initial generic  $d$ -dimensional abelian symmetry. Our approach builds upon the *iterative gauging procedure*, developed by one of the authors in [J. Garre-Rubio, *Nature Commun.* **15**, 7986 (2024)], in which an initial symmetric state is repeatedly gauged to obtain an emergent model in one dimension higher that supports the initial symmetry at its boundary. This method not only enables the construction of emergent states and corresponding commuting stabilizer Hamiltonians of which they are ground states, but it also provides a way to construct gapped boundary conditions for these models that amount to spontaneously breaking part of the boundary symmetry.

In a detailed introductory example, we showcase our paradigm by constructing three-dimensional Clifford-deformed surface codes from iteratively gauging a global 0-form symmetry that lives in two dimensions. We then provide a proof of our main result, hereby drawing upon a slight extension of the gauging procedure of Williamson. We additionally provide two more examples in  $d=2$  in which different type-I fracton orders emerge from gauging initial linear subsystem and Sierpinski fractal symmetries. En passant, we provide explicit tensor network representations of all of the involved gauging maps and the emergent states.

## Introduction

Different incarnations of *bulk-boundary correspondence* are ubiquitous in condensed matter physics. Generically, a bulk-boundary correspondence establishes a relation between the non-trivial physics in the bulk of a certain physical system and the physics constrained to its boundary. One example of such a relation is the conjecture that the bulk of certain *anomaly-free* topological orders is fully determined by

their gapped boundary theory by means of an appropriate notion of the categorical *center* [1–3]. A bulk-boundary correspondence has also proven to be the key for finding cohomological invariants that differentiate certain symmetry-protected topological phases (SPTs) protected by an invertible symmetry, whereby these invariants pertain to the transformation properties of the edge modes under the symmetry group. For the particular case of  $(1+1)d$ , this approach has led to a complete classification of gapped one-dimensional phases with group-like symmetry [4, 5]. More recently, it has also been argued how the boundaries of particular fracton phases host *emergent* subsystem symmetries whose explicit realization can be traced back to the characteristics of the bulk excitation spectrum [6]. Contrary to the case of intrinsic topological orders, it was demonstrated that the *bulk reconstruction* of the fracton orders under scrutiny depends in a non-trivial way on the relative orientation of the boundary with respect to the bulk. In a certain sense all of the above examples of bulk-boundary correspondence can thus be brought together under the umbrella term of a *holographic principle*.

In another recent series of developments an opposite avenue has been pursued in which a given quantum field theory (QFT) with a (higher) *categorical symmetry* is regarded as an interval compactification of a topological field theory in one dimension higher, coined ‘SymTFT’, with a *topological* gapped boundary condition determining the symmetry of the QFT and a not necessarily gapped *physical* boundary condition that dictates the dynamics of the QFT [7, 8]. In this picture the focus is very much on the understanding of certain features of the symmetric QFT, such as (dis)order parameters, gapped phases and anomalies, from the higher-dimensional SymTFT. Aspects of this paradigm have previously appeared under the name of ‘strange correlator’, which admits an explicit lattice realization in terms of tensor network representations of the higher-dimensional topological order [9–11].

In this manuscript we realize an explicit bulk-boundary correspondence for generic abelian *boundary symmetries* in any dimension by making use of the *iterative gauging procedure* proposed by one of the authors in [12] for *one-dimensional abelian bound-*

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ary symmetries. That work builds on the observation that gauging an *initial* abelian (anomaly-free global 0-form) symmetry gives rise to *dual* on-site global symmetry generators labeled by the one-dimensional representations of the underlying symmetry group. Since this emergent symmetry is on-site, a tensor product of local unitaries, it is free of any 't Hooft anomaly, and so it can be gauged, thereby recovering the initial  $G$  symmetry. The crux of the paradigm put forward in [12] is that these gauging procedures can be repeated indefinitely and that in this way a state emerges which naturally lives in one dimension higher than the initial symmetry. Such an *emergent state* was shown in [12] to be stabilized by certain local operators whose origin can be traced back to the Gauss constraints of the involved gauging procedures. In this picture, the initial symmetry which is being gauged can be thought of as a *boundary symmetry* for the emergent state. This idea is made very tangible by making use of the explicit *gauging map* for quantum states proposed in [13] and generalized later in [14]. One of the main advantages of using this approach is that a tensor network description of the gauging maps is readily available, which can in turn be used to write down an explicit tensor network representation of the emergent state, facilitating the analytical study of its properties and its numerical simulation.

In the current manuscript, Ref. [12] is extended to arbitrary dimensions and generic abelian symmetries. Specifically, we propose a slight generalization of the gauging procedure of [14] to arbitrary abelian groups, and demonstrate that the iterative gauging procedure of [12] in *any dimension* gives rise to emergent states with corresponding *local stabilizer* Hamiltonians. The abelian symmetries which are being gauged in the framework of [14] are completely specified by a *complete collection* of 'checks' which commute with all symmetry generators. As such, that framework includes the gauging of *fractal* and *subsystem* symmetries as specific cases. In our general framework we are also able to relate inequivalent choices of *gapped boundary stabilizers* to different spontaneous symmetry breaking patterns of the boundary symmetry.

To showcase our framework we provide three detailed worked examples where the initial boundary symmetry is two-dimensional. Gauging a global on-site 0-form symmetry in  $(2+1)d$  gives rise to emergent 1-form – or *Wilson loop*– symmetries. In this case the iterative gauging framework gives rise to three-dimensional *Clifford-deformed surface codes* [15], which are up to a local basis change equivalent to the three-dimensional abelian quantum doubles [16]. Two-dimensional linear subsystem symmetries map to dual linear subsystem symmetries on the dual lattice. Applying the framework in this case gives rise to foliated anisotropic type-I fracton orders [17]. Finally, we leverage the full potential of our framework and the gauging procedure of [14] to

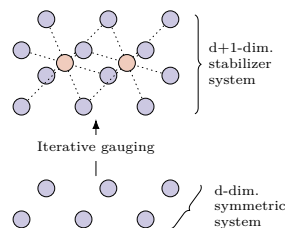


Figure 1: Schematic overview of the iterative gauging procedure. Given a  $d$ -dimensional symmetric state, iterative gauging of its symmetry and the emerging dual symmetries give rise to a  $(d+1)$ -dimensional stabilizer model with the initial state as boundary.

gauge fractal boundary symmetry, specifically of Sierpinski type, and obtain the fracton model proposed by Caseltnovo and Chamon in [18] for the study of quantum glassiness. For each of these examples we also provide a tensor network description of the involved gauging maps, and hence the emergent state, which is based on the tensor network description of the gauging map for global on-site 0-form symmetries of [13].

Our manuscript is organized as follows. In section 1 we provide an introductory example to explicitly demonstrate the iterative gauging procedure applied to an initial two-dimensional global on-site symmetry. All the details of our framework including the study of symmetry-breaking boundary conditions and the tensor network representation of the emergent Clifford-deformed CSS codes are discussed in detail. This section also serves as an introduction to many of the conventions and notations used throughout this manuscript. We then proceed by putting these observations on systematic footing in section 2. First, we slightly generalize the gauging procedure of [14] to generic abelian groups and then prove that the iterative gauging procedure in any dimension gives rise to emergent states with a corresponding commuting stabilizer Hamiltonian of which it is a ground state. We then provide a way to add boundary stabilizers to the emergent Hamiltonian which commute with all terms in the bulk and among each other, thereby preserving the gap. We argue that different gapped boundary stabilizers correspond to certain symmetry breaking patterns of the boundary symmetry. We then conclude with two more examples in section 3 and section 4 that pertain to the derivation of different type-I fracton models from the iterative gauging of linear subsystem boundary symmetry and Sierpinski fractal symmetry respectively.

## 1 Introductory example: Clifford-deformed surface codes from abelian higher-form symmetries

In this section we review the gauging procedure of abelian on-site global 0-form symmetries proposed in [13] for two-dimensional symmetric states. We argue that the gauged state is in the even sector of a dual global 1-form  $\hat{G}$  symmetry acting non-trivially solely on the introduced gauge degrees of freedom. We use an explicit gauging procedure for this 1-form symmetry, following refs. [19, 14], that yields in turn again the original  $G$  symmetry. For both gauging maps we construct a tensor network representation of bond dimension  $|G|$ . We show that the concatenation of these 2D gauging maps as in [12] produces 3D Clifford-deformed surface codes [15], which are unitarily equivalent to abelian  $G$  quantum doubles [16].

Let us consider the two-dimensional torus endowed with a lattice  $\Lambda$ , whose sets of vertices, oriented edges and plaquettes will be denoted by  $V(\Lambda)$ ,  $E(\Lambda)$  and  $P(\Lambda)$  respectively. The starting point of the gauging procedure laid out in [13] is then a two-dimensional state  $|\psi\rangle$  living in a tensor product Hilbert space of the so-called *matter* degrees of freedom supported on the vertices of  $\Lambda$ ,  $|\psi\rangle \in \bigotimes_{v \in V(\Lambda)} \mathcal{H}^v$ . The local Hilbert space  $\mathcal{H}^v$  forms a unitary representation of  $G$ , which is assumed to be abelian throughout the manuscript, and whose matrix representation is written as  $\{u_g | g \in G\}$ . We adapt the notation  $\bar{g}$  to denote the inverse group element of  $g \in G$  and 1 to denote the identity element. The state  $|\psi\rangle$  is then required to transform trivially under the tensor product representation  $\{\bigotimes_{v \in V(\Lambda)} u_g^v | g \in G\}^1$ , i.e.:

$$\left( \bigotimes_{v \in V(\Lambda)} u_g^v \right) |\psi\rangle = |\psi\rangle, \quad \forall g \in G. \quad (1.1)$$

Notice that the symmetry operators considered here act on all matter degrees of freedom  $\mathcal{H}^v$ ,  $v \in V(\Lambda)$ , simultaneously. Such a global  $G$  symmetry is in modern jargon referred to as an *invertible global 0-form* symmetry [20].

Here and below, for  $q = 0, 1$ , a  $q$ -form operator is a *topological* operator that acts non-trivially only on a codimension  $q$  submanifold of the spatial manifold, i.e. the two-dimensional torus. The case of the  $\bigotimes_v u_g^v$  symmetry operators then naturally corresponds to  $q = 0$ .

For the sake of being concrete we will henceforth make an explicit choice of local Hilbert space and its rep-

<sup>1</sup>Indeed, if  $|\psi\rangle$  transforms non-trivially under  $G$ , the gauging map defined below annihilates  $|\psi\rangle$ . However, a different gauging map can be defined for each symmetry sector.

resentation.<sup>2</sup> To this end we consider the character group  $\hat{G} := \text{Hom}(G, \text{U}(1))$  of  $G$ , which is isomorphic to  $G$  here since  $G$  is abelian. We then choose the local Hilbert space as  $\mathcal{H}^v = \mathbb{C}[\hat{G}]$  where  $\mathbb{C}[\hat{G}] := \text{Span}_{\mathbb{C}}\{|\chi\rangle | \chi \in \hat{G}\}$  is endowed with the inner product  $\langle \chi_1 | \chi_2 \rangle = \delta_{\chi_1, \chi_2}$ . This local Hilbert space is then assumed to carry the  $G$ -representation  $u_g = Z_g$  defined as  $Z_g := \sum_{\chi \in \hat{G}} \chi(g) |\chi\rangle\langle\chi|$ ,  $\forall g \in G$ . Let us also introduce the shorthand  $\mathcal{U}_g := \bigotimes_{v \in V(\Lambda)} Z_g^v$ .

### 1.1 Review: gauging two-dimensional abelian 0-form symmetries

Gauging a global on-site 0-form  $G$  symmetry à la [13]<sup>3</sup> begins with supplementing the Hilbert space with gauge degrees of freedom supported on the edges  $E(\Lambda)$  and valued in  $\mathbb{C}[G] := \text{Span}_{\mathbb{C}}\{|g\rangle | g \in G\}$ ,  $\langle g_1 | g_2 \rangle = \delta_{g_1, g_2}$ . The group algebra  $\mathbb{C}[G]$  is naturally endowed with the left regular representation given by:

$$X_g := \sum_{h \in G} |gh\rangle\langle h|, \quad \forall g \in G. \quad (1.2)$$

On this enlarged Hilbert space

$$\left( \bigotimes_{v \in V(\Lambda)} \mathbb{C}[\hat{G}] \right) \otimes \left( \bigotimes_{e \in E(\Lambda)} \mathbb{C}[G] \right) \quad (1.3)$$

we then define a *gauge transformation* for every vertex  $v \in V(\Lambda)$  and every  $g \in G$ :

$$\mathcal{G}_g^v := Z_g^v \bigotimes_{e \rightarrow v} X_{\bar{g}}^e \bigotimes_{e \leftarrow v} X_g^e. \quad (1.4)$$

Here,  $e(\rightarrow v)$ , respectively  $e(\leftarrow v)$ , stands for the set of edges of  $\Lambda$  that point into, respectively point away from, the fixed vertex  $v$ .

Notice that the operators  $\mathcal{G}_g^v$  in eq. (1.4) form a *local* representation of  $G$  in the sense that they are supported on a small neighborhood around  $v$  and obey  $\mathcal{G}_{g_1}^v \cdot \mathcal{G}_{g_2}^v = \mathcal{G}_{g_1 g_2}^v$  for every  $v \in V(\Lambda)$  and for every  $g_1, g_2 \in G$ . Moreover, notice that the global symmetry operators are recovered by taking the product of the local symmetry operators on all vertices, namely

$$\prod_{v \in V(\Lambda)} \mathcal{G}_g^v = \mathcal{U}_g, \quad \forall g \in G. \quad (1.5)$$

<sup>2</sup>It should be noted that the gauging procedure which follows can be carried out for any on-site realization of the symmetry [13]. However, in the construction of the emergent state below, the choice of the representation of the  $G$  symmetry on the boundary does not influence the bulk stabilizers of the emergent state.

<sup>3</sup>Notice that the framework of [13] applies more generally to non-abelian groups as well. The gauged theory in that case enjoys Wilson loop symmetries labeled by representations of the symmetry group. Wilson loops corresponding to higher-dimensional representations of the group can be realized on the lattice by non-trivial *matrix product operators* [21–23].

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Next, a *local* projector is obtained by averaging  $\mathcal{G}_g^\nu$  over  $G$ :

$$\mathcal{P}^\nu := \frac{1}{|G|} \sum_{g \in G} \mathcal{G}_g^\nu. \quad (1.6)$$

Indeed it is straightforward to verify that  $\mathcal{P}^\nu \cdot \mathcal{P}^\nu = \mathcal{P}^\nu$ ,  $\forall \nu \in V(\Lambda)$  by a relabeling of summation variables.

The commutation of the local projectors  $[\mathcal{P}^\nu, \mathcal{P}^{\nu'}] = 0$  allows us to unambiguously define the *global* projector onto the gauge invariant subspace of the enlarged Hilbert space:

$$\mathcal{P} := \prod_{\nu \in V(\Lambda)} \mathcal{P}^\nu. \quad (1.7)$$

The state  $|\psi\rangle$  is then gauged by applying the projector  $\mathcal{P}$  to the initial state  $|\psi\rangle$  coupled to a trivial flat gauge field  $\bigotimes_{e \in E(\Lambda)} |1\rangle^e$ . Hence, we define the *gauging map*  $\mathcal{G}^0$  for the global 0-form  $G$  symmetry as:

$$\mathcal{G}^0 := \mathcal{P} \left( \bigotimes_{e \in E(\Lambda)} |1\rangle^e \right) : \quad (1.8)$$

$$\left( \bigotimes_{\nu \in V(\Lambda)} \mathbb{C}[\widehat{G}] \right) \rightarrow \left( \bigotimes_{\nu \in V(\Lambda)} \mathbb{C}[\widehat{G}] \right) \otimes \left( \bigotimes_{e \in E(\Lambda)} \mathbb{C}[G] \right),$$

and the *gauged state* is by definition  $\mathcal{G}^0 |\psi\rangle$ .

In order to elucidate the above procedure and facilitate the construction of the emergent state below, let us present a tensor network representation of the gauging map  $\mathcal{G}^0$ . To this end we introduce two types of tensors. For every vertex of the lattice with  $m$  incoming edges and  $n$  outgoing edges, we introduce the rank- $(m+n)$  *vertex tensor*, or *Z-type tensor*, defined as follows:

$$\begin{aligned} & \text{Diagram of a vertex tensor } Z_g \text{ with } m \text{ incoming and } n \text{ outgoing legs.} \\ & := \sum_{\substack{\chi_1, \chi_2 \\ \{g_k\}_k}} \frac{1}{|G|} \sum_{g_{m+1}, \dots, g_{m+n}} |g_{m+1}, \dots, g_{m+n}\rangle \langle g_1, \dots, g_m| \\ & := \sum_{g \in G} \frac{1}{|G|} Z_g \otimes |g, \dots, g\rangle \langle g, \dots, g|. \end{aligned} \quad (1.9)$$

In this and subsequent figures, the dots are a placeholder for tensor legs which are omitted for clarity.

In order to implement the flatness of the gauge field supported on the edges of the lattice we consider the rank-3 *+type* tensors given by:

$$\begin{aligned} & \text{Diagram of a +type tensor } \oplus \text{ with } m \text{ incoming and } n \text{ outgoing legs.} \\ & := \sum_{g, h, k \in G} g \otimes |g, h\rangle \langle k| \\ & := \sum_{g, h \in G} |g, h\rangle \langle gh|, \end{aligned} \quad (1.10)$$

and

$$\begin{aligned} & \text{Diagram of a +type tensor } \oplus \text{ with } m \text{ incoming and } n \text{ outgoing legs.} \\ & := \sum_{g, h, k \in G} k \otimes |g, h\rangle \langle k| \\ & := \sum_{g, h \in G} |gh\rangle \langle g, h|. \end{aligned} \quad (1.11)$$

Using these tensors, the local projector  $\mathcal{P}^\nu$  introduced in eq. (1.6) admits the simple tensor network representation

$$\mathcal{P}^\nu = \text{Diagram of a local projector } \mathcal{P}^\nu \text{ as a tensor network.} \quad (1.12)$$

where the orientation of the horizontal red legs is assumed to be in correspondence with the orientation of the edges of the lattice. Note that the commutation of two local projectors  $\mathcal{P}^\nu, \mathcal{P}^{\nu'}$  on different vertices  $\nu \neq \nu'$  follows immediately from the definition of the  $+$ -type tensors. Combining this observation with the fact that projecting one of the incoming legs of the  $+$ -type tensor eq. (1.11) onto  $|1\rangle$  effectively removes that leg, it follows that the gauging map  $\mathcal{G}^0$  admits a representation as projected entangled-pair operator (PEPO) of bond dimension  $|G|$ , whose unit cell can be depicted as:

$$\text{Diagram of the PEPO unit cell.} \quad (1.13)$$

In this construction it should be understood that the full PEPO on the torus is obtained by placing copies of this unit cell on all vertices of the lattice and contracting along the virtual red legs according to the connectivity of the lattice.

Let us proceed by dissecting the symmetry properties of the gauging map  $\mathcal{G}^0$ . One of the upshots of the PEPO representation is that these symmetry properties automatically follow from the symmetry properties of the local tensors from which the PEPO is made up. It follows readily from its definition that the vertex tensor satisfies

$$\text{Diagram of a vertex tensor } Z_g \text{ with } m \text{ incoming and } n \text{ outgoing legs.} = X_g^\dagger \otimes \text{Diagram of a vertex tensor } Z_g \text{ with } m \text{ incoming and } n \text{ outgoing legs.} = X_g \otimes \text{Diagram of a vertex tensor } Z_g \text{ with } m \text{ incoming and } n \text{ outgoing legs.}, \quad (1.14)$$

whereas the edge tensor obeys the relations

$$\text{Diagram of an edge tensor } X_g \text{ with } m \text{ incoming and } n \text{ outgoing legs.} = X_g \otimes \text{Diagram of an edge tensor } X_g \text{ with } m \text{ incoming and } n \text{ outgoing legs.} \quad (1.15)$$

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Consider first gauge invariance of the gauging map, that is  $\mathcal{G}_g^v \cdot \mathcal{G}^0 = \mathcal{G}^0$  for all  $g \in G$  and  $v \in V(\square)$ . Note that it is exactly in this sense that this gauging procedure *localizes* the global 0-form  $G$  symmetry. From the PEPO representation this property can be deduced by combining the second relation in eq. (1.14) with both identities in eq. (1.15) and noticing that all  $X_g^{(t)}$ 's on the virtual level exactly cancel with each other. Note that algebraically this follows directly from a renaming of summation variable in the definition of  $\mathcal{P}^v$  eq. (1.6). One also has that  $\mathcal{G}^0 \cdot \mathcal{U}_g = \mathcal{G}^0$  for all  $g \in G$ . Physically this means that the gauging map projects out states which transform non-trivially under the global symmetry  $G$ . As such,  $\mathcal{G}^0$  has a non-trivial kernel spanned by states in the original matter Hilbert space  $\bigotimes_{v \in V(\Lambda)} \mathbb{C}[\widehat{G}]$  which are charged under the  $G$  symmetry. This can be seen from application of the first identity in eq. (1.14) on every vertex together with the invariance of the edge tensors, namely

$$x_g \leftarrow \bigoplus \leftarrow x_g^\dagger = \bigoplus \leftarrow x_g, \quad (1.16)$$

which is deduced from eq. (1.15).

As mentioned in the introduction, the gauged states exhibit at least a global 1-form  $\widehat{G}$  symmetry as a consequence of the imposed Gauss laws. This 1-form symmetry is the topic of the next section.

To conclude this paragraph, a comment which is of importance for what follows is in place. As argued in [13] it is possible to disentangle the initial matter degrees of freedom from the newly introduced gauge degrees of freedom by means of a finite depth unitary quantum circuit, rendering the Gauss constraints  $\mathcal{G}_g^v \stackrel{!}{=} 1$  trivial and effectively freezing the matter degrees of freedom to the  $+1$  eigenstate of  $Z_g$ . As showed below, we will construct a three-dimensional state by concatenating gauging operators, wherein the ‘matter’ degrees of freedom in every layer will play the role of physical degrees of freedom of the three-dimensional state. We therefore refrain from applying this unitary circuit.

## 1.2 The dual 1-form symmetry

Let us consider  $\ell \in Z_1(\Lambda, \mathbb{Z})$  to be an oriented closed path supported on the edges of the lattice  $\Lambda$ , and introduce

$$Z_\chi := \sum_{g \in G} \chi(g) |g\rangle\langle g|, \quad \forall \chi \in \widehat{G}, \quad (1.17)$$

which together with  $X_g$  obeys the following commutation relation:

$$Z_\chi X_g = \chi(g) X_g Z_\chi. \quad (1.18)$$

We define then following *Wilson loop* operators acting exclusively on the gauge degrees of freedom:

$$\mathcal{U}_\chi[\ell] := \bigotimes_{e \in \ell} Z_{o_\ell^e}^e(\chi). \quad (1.19)$$

Here,  $o_\ell^e(\chi)$  is defined so that  $o_\ell^e(\chi) = \chi$  if the orientation of  $e$  and  $\ell$  coincide and  $\bar{\chi}$  otherwise. As we will momentarily show, it holds that

$$\mathcal{U}_\chi[\ell] \cdot \mathcal{G}^0 = \mathcal{G}^0, \quad (1.20)$$

$\forall \chi \in \widehat{G}, \ell \in Z_1(\Lambda, \mathbb{Z})$ . This implies that the gauged state  $\mathcal{G}^0 |\psi\rangle$  transforms trivially under  $\mathcal{U}_\chi[\ell]$  for every valid choice of  $\ell$  and every character  $\chi \in \widehat{G}$ . Making use of the fact that

$$\chi_1(g)\chi_2(g) = (\chi_1 \otimes \chi_2)(g), \quad (1.21)$$

$\forall \chi_1, \chi_2 \in \widehat{G}, g \in G$ , it follows that the operators  $\mathcal{U}_\chi[\ell]$ , with same  $\ell$ , multiply according to the product of characters in  $\widehat{G}$ , i.e.:

$$\mathcal{U}_{\chi_1}[\ell] \cdot \mathcal{U}_{\chi_2}[\ell] = \mathcal{U}_{\chi_1 \otimes \chi_2}[\ell]. \quad (1.22)$$

Also note that the operators  $\mathcal{U}_\chi[\ell]$  are invariant under continuous deformations of  $\ell$ , rendering them *topological*. These observations demonstrate that after gauging a global 0-form  $G$  symmetry, a *global 1-form  $\widehat{G}$  symmetry* emerges.<sup>4</sup> Alternatively, one could think of (1.20) as a *kinematical* constraint satisfied by all allowed states in the gauge invariant Hilbert space. In order to verify eq. (1.20) we can again resort to the PEPO representation eq. (1.13). The  $+$ -type tensor eq. (1.11) satisfies following invariance property

$$z_x \leftarrow \bigoplus \leftarrow z_x^\dagger = \bigoplus \leftarrow z_x, \quad (1.23)$$

whereas the  $Z$ -type tensor eq. (1.9) obeys

$$\bigoplus \leftarrow z_x \leftarrow \bigoplus = \bigoplus \leftarrow z_x \leftarrow \bigoplus. \quad (1.24)$$

Both these invariance properties once more follow directly from the definitions of these respective tensors. From them, the claim eq. (1.20) follows readily as again all occurrences of  $Z_\chi^{(t)}$  that appear on the virtual level after application of  $\mathcal{U}_\chi[\ell]$  cancel. For the sake of completeness, let us also verify this more explicitly algebraically. To this end, one checks that for every such loop  $\ell$  this line operator commutes with the local gauge transformations:

$$\mathcal{U}_\chi[\ell] \cdot \mathcal{G}_g^v = \mathcal{G}_g^v \cdot \mathcal{U}_\chi[\ell], \quad \forall g \in G, v \in V(\Lambda). \quad (1.25)$$

To prove this, first notice that if  $\mathcal{U}_\chi[\ell]$  and  $\mathcal{G}_g^v$  overlap, they overlap in an even number of edges. Let us

<sup>4</sup>In fact, one should think of the emergent 1-form  $\widehat{G}$  symmetry, or more precisely  $\text{Rep}(G)$  symmetry, as being the invertible component of the full symmetry structure of the gauge theory. The complete symmetry structure was shown to constitute the 1-form symmetry together with all its 0-form *condensation descendants* and is organized in the *fusion 2-category*  $2\text{Rep}(G)$  [24–27, 23, 28].

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consider just the case of two overlapping edges that we call  $e_1, e_2$ . Then for the given vertex  $v$ , if both  $e_1, e_2$  have the same orientation as  $\ell$  or both have the opposite orientation as  $\ell$ , commuting  $\mathbb{U}_\chi[\ell]$  through  $\mathcal{G}_g^v$  yields two cancelling phases  $\chi(g)$  and  $\bar{\chi}(g)$  by virtue of the definition of  $\mathcal{G}_g^v$  and eq. (1.18). If on the other hand  $\ell$  has the same orientation as  $e_1$  but has the opposite orientation as  $e_2$ , then we also obtain, due to the relative difference in orientation, phases  $\chi(g)$  and  $\bar{\chi}(g)$  that cancel. From eq. (1.25) together with the fact that

$$\mathbb{U}_\chi[\ell] \left( \bigotimes_{e \in E(\Lambda)} |1\rangle^e \right) = \bigotimes_{e \in E(\Lambda)} |1\rangle^e \quad (1.26)$$

we can conclude eq. (1.20).

### 1.3 Gauging the dual 1-form symmetry

Since the emergent 1-form symmetry is realized on-site, it is anomaly-free in the sense of [29] and can hence be gauged. To this end we use a similar procedure as in [13, 19, 14]. After gauging this 1-form symmetry one expects the original 0-form  $G$  symmetry to re-emerge [20, 19].

Following [19, 14, 30], gauging of the 1-form symmetry happens via introduction of gauge degrees of freedom on the vertices  $V(\Lambda)$  of the lattice valued in the group algebra  $\mathbb{C}[\hat{G}]$ .<sup>5</sup> A local 1-form gauge transformation for every edge  $e \in E(\Lambda)$  is then constructed as follows:

$$\mathcal{G}_\chi^e := X_\chi^{v(\leftarrow e)} \otimes Z_\chi^e \otimes X_\chi^{v(\rightarrow e)}, \quad (1.27)$$

where we used an analogue notation to the one in eq. (1.4) denoting by  $v(\leftarrow e)$  the vertex which is the target of  $e$  and by  $v(\rightarrow e)$  the source vertex of the edge  $e$ . Naturally,  $X_\chi$  is defined as

$$X_\chi := \sum_{\eta \in \hat{G}} |\chi\eta\rangle\langle\eta|. \quad (1.28)$$

Akin to the case of the gauging of a global 0-form symmetry, these operators form a local representation of  $\hat{G}$  supported near  $e$ ,  $\mathcal{G}_{\chi_1}^e \cdot \mathcal{G}_{\chi_2}^e = \mathcal{G}_{\chi_1\chi_2}^e$ ,  $\forall \chi_1, \chi_2 \in \hat{G}$ . Notice that the global 1-form symmetries can be recovered by taking the (oriented) product of the 1-form gauge transformations along the loop  $\ell$ , i.e.:

$$\mathbb{U}_\chi[\ell] = \prod_{e \in \ell} \mathcal{G}_{\sigma_\ell^e(\chi)}^e, \quad (1.29)$$

$\forall \chi \in \hat{G}$ ,  $\ell \in Z_1(\Lambda, \mathbb{Z})$ , analogous to eq. (1.5).

Group-averaging over  $\hat{G}$  allows us to construct the local projectors

$$\mathcal{P}^e := \frac{1}{|\hat{G}|} \sum_{\chi \in \hat{G}} \mathcal{G}_\chi^e, \quad (1.30)$$

<sup>5</sup>By invoking a form of Poincaré duality, one can identify the vertices of the primal lattice  $\Lambda$  with the plaquettes of the dual lattice  $\Lambda^\vee$ , i.e.  $V(\Lambda) \equiv P(\Lambda^\vee)$ . In that sense the gauge field can indeed be thought of as a 2-form gauge field for the global 1-form symmetry [30].

where we made use of  $|G| = |\hat{G}|$ . From these we can again unambiguously construct the global projector onto the gauge invariant subspace as:

$$\mathcal{P} := \prod_{e \in E(\Lambda)} \mathcal{P}^e. \quad (1.31)$$

Gauging the 1-form symmetry then ultimately happens in a similar fashion as in the 0-form case. We construct a gauging map by initializing all newly introduced gauge degrees of freedom in the trivial representation  $\mathbb{1} \in \hat{G}$  followed by acting with the projector onto the gauge invariant subspace:

$$\mathcal{G}^1 := \mathcal{P} \left( \bigotimes_{v \in V(\Lambda)} |\mathbb{1}\rangle^v \right) : \quad (1.32)$$

$$\left( \bigotimes_{e \in E(\Lambda)} \mathbb{C}[G] \right) \rightarrow \left( \bigotimes_{e \in E(\Lambda)} \mathbb{C}[G] \right) \otimes \left( \bigotimes_{v \in V(\Lambda)} \mathbb{C}[\hat{G}] \right).$$

Let us draw attention to the fact that without loss of generality the gauging map  $\mathcal{G}^1$  has as its domain only the edge degrees of freedom. This can be traced back to the fact that the emergent 1-form symmetry generators  $\mathbb{U}_\chi[\ell]$  only act (non-trivially) on these degrees of freedom.

Let us now proceed by showcasing the tensor network representation of the gauging map  $\mathcal{G}^1$ , and its symmetry properties. In a similar vein as for the gauging map of the global 0-form  $G$  symmetry, we introduce tensors that will be placed on the edges of the lattice that read:

$$\begin{aligned} & \text{Diagram: A vertex with four edges. The top edge has a red arrow pointing up, labeled  $g_2$ . The bottom edge has a red arrow pointing down, labeled  $g_1$ . The left edge has a blue arrow pointing left, labeled  $\chi_2$ . The right edge has a blue arrow pointing right, labeled  $\chi_1$ .} \\ & := \sum_{\substack{g_1, g_2 \\ \chi_1, \chi_2}} \frac{1}{|G|} |\chi_2\rangle\langle g_2| \otimes |g_1\rangle\langle \chi_1| \\ & := \sum_{\chi \in \hat{G}} \frac{1}{|\hat{G}|} Z_\chi \otimes |\chi\rangle\langle\chi|. \end{aligned} \quad (1.33)$$

Tensors incorporating the flatness of the (2-form) gauge field placed on the vertices read then

$$\begin{aligned} & \text{Diagram: A vertex with four edges. The top edge has a blue arrow pointing up, labeled  $\chi_{m+n}$ . The bottom edge has a blue arrow pointing down, labeled  $\chi_m$ . The left edge has a blue arrow pointing left, labeled  $\chi_{m+1}$ . The right edge has a blue arrow pointing right, labeled  $\chi_{m+n+1}$ .} \\ & := \sum_{\{\chi_k\}_k} \chi_{m+2} \cdots \chi_{m+n} |\chi_{m+1}, \dots, \chi_{m+n}\rangle \otimes \langle \chi_1, \dots, \chi_m| \\ & := \sum_{\{\chi_k\}_k} \delta_{\prod_{k=1}^m \chi_k \cdot \prod_{k=m+1}^{m+n} \chi_k} |\chi_{m+1}, \dots, \chi_{m+n}\rangle \otimes \langle \chi_1, \dots, \chi_m|. \end{aligned} \quad (1.34)$$

Their relevant symmetry properties are:

$$\text{Diagram: A vertex with four edges. The top edge has a red arrow pointing up, labeled  $Z_\chi$ . The bottom edge has a red arrow pointing down, labeled  $Z_\chi$ . The left edge has a blue arrow pointing left, labeled  $X_\chi^\dagger$ . The right edge has a blue arrow pointing right, labeled  $X_\chi$ .} = X_\chi^\dagger \otimes X_\chi = Z_\chi \otimes Z_\chi, \quad (1.35)$$



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and

$$(1.36)$$

Making use of these tensors then results in following PEPO representation of the Gauss law  $\mathcal{P}^e$ :

$$(1.37)$$

From this, the PEPO representation of the full gauging map  $\mathcal{G}^1$  then readily follows. One unit cell is given by

$$(1.38)$$

where, as alluded to above, the blue  $+$ -type tensors are placed in alignment with the vertices of the primal lattice.

Gauge invariance which in this case reads  $\mathcal{G}_\chi^e \cdot \mathcal{G}^1 = \mathcal{G}^1$  for all  $\chi \in \hat{G}$  and  $e \in E(\Lambda)$  is a direct consequence of the second equality in eq. (1.35) combined with the identities of the vertex tensor in eq. (1.36). Also the property  $\mathcal{G}^1 \cdot \mathbb{U}_\chi[\ell] = \mathcal{G}^1$  holds for all closed paths  $\ell$  and  $\chi \in \hat{G}$ . In other words, the gauging map  $\mathcal{G}^1$  annihilates states which are not in the even sector of the global 1-form  $\hat{G}$  symmetry. This follows from the first equation in eq. (1.35) and the invariance properties of the vertex tensor, one of which reads:

$$(1.39)$$

This is not an independent property as it can be derived immediately from eq. (1.36).

Finally and crucially, after gauging the 1-form symmetry the initial global 0-form  $G$  symmetry reappears as anticipated. Indeed, it is found that

$$\mathbb{U}_g \cdot \mathcal{G}^1 = \mathcal{G}^1, \quad (1.40)$$

for all  $g \in G$ .

For future reference we demonstrate this here algebraically. To this end, first note that for two vertices  $v, v'$  which share an edge  $e$ ,  $Z_g^v \otimes Z_g^{v'}$  commutes with the gauge transformation  $\mathcal{G}_\chi^e$ ,  $\forall g \in G$  by virtue of the identity

$$Z_g X_\chi = \chi(g) X_\chi Z_g. \quad (1.41)$$

Combined with the observation that

$$\mathbb{U}_g \left( \bigotimes_{v \in V(\Lambda)} |1\rangle^v \right) = \bigotimes_{v \in V(\Lambda)} |1\rangle^v, \quad (1.42)$$

we recover indeed eq. (1.40).

In order to verify this using the PEPO representation eq. (1.38) on the other hand, the relevant invariance properties this time read:

$$(1.43)$$

and

$$(1.44)$$

From these properties, eq. (1.40) follows readily since after application of  $\mathbb{U}_g$  on the gauging map, all occurrences of  $Z_g^{(\dagger)}$  again cancel on the virtual bonds.

## 1.4 3D local symmetries from iterative gauging

Constructing a three-dimensional state from the above gauging maps now happens as follows. We start by choosing a two-dimensional state  $|\psi\rangle$  that will serve as the boundary condition of the three-dimensional state and fixing a choice of abelian group  $G$ . We assume, as anticipated above, that this state lives in  $\bigotimes_{v \in V(\Lambda)} \mathbb{C}[\hat{G}]$  and transforms trivially under the abelian 0-form  $G$  symmetry  $\mathbb{U}_g$ , i.e.  $\mathbb{U}_g |\psi\rangle = |\psi\rangle$ ,  $\forall g \in G$ . Throughout the above discussions  $\Lambda$  was kept completely general, and as such the construction that follows also can be applied to this most general case.

On  $|\psi\rangle$  we can thus first act with the gauging operator  $\mathcal{G}^0$  defined explicitly in eq. (1.8), thereby introducing degrees of freedom valued in  $\mathbb{C}[G]$  and supported on the edges of  $\Lambda$ . Indeed, recall that  $\mathcal{G}^0$  acts according to  $\mathcal{G}^0 : \mathbb{C}[\hat{G}]^{\otimes V} \rightarrow \mathbb{C}[\hat{G}]^{\otimes V} \otimes \mathbb{C}[G]^{\otimes E}$ , in which  $\otimes E$  and  $\otimes V$  stand for tensor products over the vertices and edges of the lattice. The gauged state  $\mathcal{G}^0 |\psi\rangle$  has then as argued in section 1.2 an emergent dual 1-form  $\hat{G}$  symmetry which is completely supported on the edges of  $\Lambda$  and that in turn can be gauged using the procedure outlined in section 1.3. Notice that the gauging map for this 1-form symmetry,  $\mathcal{G}^1$  defined in (1.32), acts only on the  $\mathbb{C}[G]$ -valued degrees of freedom supported on the edges, leaving the original ‘matter’ degrees of freedom on the vertices  $v \in V(\Lambda)$  valued in  $\mathbb{C}[\hat{G}]$  untouched. With slight abuse of notation:  $\mathcal{G}^1 : \mathbb{C}[G]^{\otimes E} \rightarrow \mathbb{C}[G]^{\otimes E} \otimes \mathbb{C}[\hat{G}]^{\otimes V}$ . In turn we can then again gauge the emergent 0-form  $G$  symmetry that arises after the application of  $\mathcal{G}^1$  using the gauging map  $\mathcal{G}^0$ . In this step  $\mathcal{G}^0$  does not act on the degrees of freedom in  $\mathbb{C}[\hat{G}]$  living on  $E(\Lambda)$ . By repeatedly concatenating the gauging maps  $\mathcal{G}^0$  and  $\mathcal{G}^1$  we thus in every step are left with degrees of freedom in  $\mathbb{C}[\hat{G}]$  and  $\mathbb{C}[G]$  which are not acted upon by the next gauging map. These are the degrees of freedom which, following the paradigm of Ref. [12], are

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interpreted as *physical* degrees of freedom of a *foliated* three-dimensional state  $|\Psi[\psi]\rangle$  whose 2D layers are made up of the ‘matter’ degrees of freedom in every application of the gauging maps. Formally  $|\Psi[\psi]\rangle$  reads:

$$|\Psi[\psi]\rangle := \cdots \circ \mathcal{G}_2^1 \circ \mathcal{G}_{\frac{3}{2}}^0 \circ \mathcal{G}_1^1 \circ \mathcal{G}_{\frac{1}{2}}^0 |\psi\rangle. \quad (1.45)$$

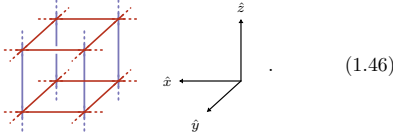
The exact meaning of the subscripts will be explained momentarily.

For concreteness we specialize to the square lattice  $\square = \Lambda$  henceforth. It can then be argued that the physical degrees of freedom of  $|\Psi[\psi]\rangle$  naturally organize themselves on the edges of a three-dimensional cubic lattice that we will denote by  $\mathcal{B}$ , and where the initial two-dimensional state serves as a physical boundary of the lattice.

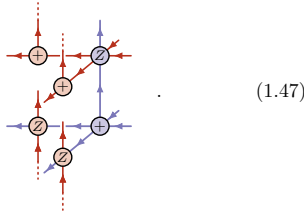
We therefore consider the cubic lattice as being made up of alternating layers of vertical edges followed by a layer of horizontal edges, that we label by  $\frac{1}{2}, \frac{3}{2}, \dots$  and  $1, 2, \dots$  respectively. The cubic lattice hence has a physical boundary made up of vertical edges.

For every layer of vertical edges we associate these edges with the vertices of a copy of  $\square$  and the corresponding  $\mathbb{C}[\widehat{G}]$ -valued ‘matter’ degrees of freedom that are not acted upon by the gauging maps  $\mathcal{G}^1$ . Analogously, we associate with every horizontal edge the  $\mathbb{C}[G]$ -valued degrees of freedom that are untouched by the  $\mathcal{G}^0$ ’s.

Using a blue color for edges that carry a copy of the local Hilbert space  $\mathbb{C}[\widehat{G}]$  and red for those with a copy of  $\mathbb{C}[G]$ , one can depict the cubic lattice away from the boundary as follows:



One unit cell of the state  $|\Psi[\psi]\rangle$  away from the boundary then can be depicted by combining eq. (1.13) and eq. (1.38) as follows:



In this figure it is understood that all horizontal legs are contracted with neighboring unit cells and the dots represent a contraction with unit cells in the vertical direction. The physical degrees of freedom

of the state then correspond to the uncontracted vertical legs.

By virtue of the local symmetries of all the gauging maps appearing in eq. (1.45), the obtained state  $|\Psi[\psi]\rangle$  also inherits those local symmetries in a slightly modified way. More precisely, the emergent state  $|\Psi[\psi]\rangle$  is stabilized by plaquette operators for every plaquette of the cubic lattice, coming from the localization of the 1-form symmetry, and vertex operators for every vertex, from gauging the global 0-form symmetry. We denote the plaquette stabilizers which are labeled by  $\chi \in \widehat{G}$  by  $\mathbb{B}^\chi$ :

$$\mathbb{B}_\chi^{p_{zz}} |\Psi[\psi]\rangle = |\Psi[\psi]\rangle, \quad (1.48)$$

$$\mathbb{B}_\chi^{p_{yz}} |\Psi[\psi]\rangle = |\Psi[\psi]\rangle, \quad (1.49)$$

$$\mathbb{B}_\chi^{p_{xy}} |\Psi[\psi]\rangle = |\Psi[\psi]\rangle, \quad (1.50)$$

and the vertex stabilizers  $\phi^v$  carry a group label  $g \in G$ :

$$\phi_g^v |\Psi[\psi]\rangle = |\Psi[\psi]\rangle. \quad (1.51)$$

Explicitly, the plaquette and vertex terms are given by

$$\mathbb{B}_\chi^{p_{zz}} := \begin{array}{c} \text{---} Z_x^\dagger \text{---} \\ | \\ X_x \text{---} \text{---} X_x^\dagger \\ | \\ \text{---} Z_x \text{---} \end{array}, \quad (1.52)$$

$$\mathbb{B}_\chi^{p_{yz}} := \begin{array}{c} \text{---} Z_x^\dagger \text{---} \\ | \\ X_x \text{---} \text{---} X_x^\dagger \\ | \\ \text{---} Z_x \text{---} \end{array}, \quad (1.53)$$

$$\mathbb{B}_\chi^{p_{xy}} := \begin{array}{c} \text{---} Z_x \text{---} Z_x^\dagger \text{---} \\ | \\ X_x \text{---} \text{---} X_x^\dagger \end{array}, \quad (1.54)$$

$$\phi_g^v := \begin{array}{c} \text{---} Z_g^\dagger \text{---} \\ | \\ X_g \text{---} \text{---} X_g^\dagger \\ | \\ \text{---} Z_g \text{---} \end{array}. \quad (1.55)$$

Let us first show algebraically that  $\mathbb{B}_\chi^{p_{zz}}$  and  $\mathbb{B}_\chi^{p_{yz}}$  stabilize  $|\Psi[\psi]\rangle$  before residing to our graphical PEPO representation of the state. To this end we single out one of those plaquette term. From the commutation relation eq. (1.41) together with the definition of  $\mathcal{G}^0$  in terms of the local gauge transformations (1.4), it follows that

$$\begin{aligned} & (X_\chi^{v(\leftarrow e)} \otimes Z_\chi^e \otimes X_\chi^{v(\rightarrow e)}) \cdot \mathcal{G}^0 \\ &= \mathcal{G}^0 \cdot (X_\chi^{v(\leftarrow e)} \otimes X_\chi^{v(\rightarrow e)}). \end{aligned} \quad (1.56)$$

Recognizing then  $X_\chi^{v(\leftarrow e)} \otimes Z_\chi \otimes X_\chi^{v(\rightarrow e)}$  as the local 1-form gauge transformation (1.27), which leaves  $\mathcal{G}^1$  invariant, together with the fact that  $Z_\chi \cdot \mathcal{G}^1 = \mathcal{G}^1 \cdot Z_\chi$  proves that  $\mathbb{B}_\chi^{p_{zz}}$  and  $\mathbb{B}_\chi^{p_{yz}}$  are stabilizers of the state. Let us now give an interpretation of these stabilizers in terms of our PEPO representation of the state

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eq. (1.47). Note first that from the commutation relation eq. (1.41) it follows that

$$\text{Diagram 1} = \text{Diagram 2} = \text{Diagram 3}, \quad (1.57)$$

from which, together with eq. (1.35), one finds that in the  $xz$ -plane eq. (1.56) can be depicted as:

$$\text{Diagram 1} = \text{Diagram 2}, \quad (1.58)$$

and similar in the  $yz$ -plane. Considering now a slice of the tensor network representation of our state eq. (1.47) along the  $xz$ -plane, we compute that:

$$\mathbb{B}_\chi^{\text{P}_{xz}} |\Psi[\psi]\rangle = \text{Diagram 1} = \text{Diagram 2} = |\Psi[\psi]\rangle, \quad (1.59)$$

where in the second equality we made use of the fact that the operator  $Z_x^\dagger$  on top can be freely *pulled through* the  $Z$ -type tensor on which it acts, together with eq. (1.58). In the final equality we then recognized the gauge invariance of  $\mathcal{G}^1$ . The  $yz$ -plaquette term can be derived in similar fashion.

The final  $xy$ -plaquette terms can be recognized as the (minimal) global 1-form symmetries of the  $\mathcal{G}^0$  gauging map. In a similar vein to eq. (1.56), one can check that

$\mathfrak{o}_g^y$  applied to  $\mathcal{G}^1$  maps to  $\mathcal{G}_g^y$  which is exactly the local symmetry of  $\mathcal{G}^0$ , from which it follows that  $\mathfrak{o}_g^y$  is also a stabilizer.

These observations allow us to consider the state  $|\Psi[\psi]\rangle$  a ground state of the commuting projector bulk Hamiltonian (a stabilizer code):

$$\mathbb{H}^{\text{blk}} := - \sum_{v \in V(\mathcal{C})} \mathbb{P}^v - \sum_{p_{xz} \in P(\mathcal{C})} \mathbb{P}^{p_{xz}} - \sum_{p_{yz} \in P(\mathcal{C})} \mathbb{P}^{p_{yz}} - \sum_{p_{xy} \in P(\mathcal{C})} \mathbb{P}^{p_{xy}}. \quad (1.60)$$

In this expression the sum over  $V(\mathcal{C})$  is over all vertices of the cubic lattice, and the sums over  $P(\mathcal{C})$  are over plaquettes in the  $xz$ ,  $yz$  and  $xy$  planes respectively. The projectors are explicitly given by

$$\mathbb{P}^v := \frac{1}{|G|} \sum_{g \in G} \mathfrak{o}_g^v, \quad (1.61)$$

and

$$\mathbb{P}^{p_{xz}} := \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \mathbb{B}_\chi^{\text{P}_{xz}}, \quad (1.62)$$

$$\mathbb{P}^{p_{yz}} := \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \mathbb{B}_\chi^{\text{P}_{yz}}, \quad (1.63)$$

$$\mathbb{P}^{p_{xy}} := \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \mathbb{B}_\chi^{\text{P}_{xy}}. \quad (1.64)$$

It turns out that the Hamiltonian in eq. (1.60) is unitarily equivalent to the  $G$  abelian quantum double [16]. In its displayed form (1.60) it appeared previously in [15] under the name of *Clifford-deformed surface code*. It can be brought in its more familiar form [16] defined on  $\bigotimes_{e \in E(\mathcal{C})} \mathbb{C}[G]$  by applying the generalized Hadamard transformation

$$\mathcal{U} := \sum_{g \in G, \chi \in \widehat{G}} \frac{\chi(g)}{\sqrt{|G|}} |g\rangle\langle\chi| \quad (1.65)$$

to every vertical edge carrying a copy of  $\mathbb{C}[\widehat{G}]$ , as it can be checked that

$$\mathcal{U} Z_g \mathcal{U}^\dagger = X_g^\dagger, \quad (1.66)$$

$$\mathcal{U} X_\chi \mathcal{U}^\dagger = Z_\chi. \quad (1.67)$$

### 1.5 Gapped boundary conditions

The local stabilizers analyzed so far act on the bulk of the three-dimensional state  $|\Psi[\psi]\rangle$ . In this section we study the boundary stabilizers. These naturally are given by a subset of truncated bulk stabilizers commuting with the bulk stabilizers  $\mathbb{b}$  and  $\mathfrak{o}$  defined above, where the subset depends on the choice of the boundary state  $|\psi\rangle$ , in particular the way in which  $|\psi\rangle$  is invariant under the  $G$  0-form symmetry. Indeed, for

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every choice of subgroup  $H \leq G$  one can construct explicitly a  $G$ -symmetric boundary state  $|\psi\rangle_H$  for which  $|\Psi[\psi_H]\rangle$  is a ground state of the bulk Hamiltonian  $\mathbb{H}^{\text{blk}}$  (1.60) plus additional boundary terms which are projectors and which all commute with the terms in  $\mathbb{H}^{\text{blk}}$  (1.60) and among themselves.

Let us proceed by constructing these boundary states and the corresponding boundary Hamiltonian terms explicitly. To this end we first define the following operators near the boundary  $\partial\mathcal{D}$  (effectively truncating  $\mathbb{b}$ ) for any  $\chi \in \widehat{G}$ :

$$\mathbb{d}_\chi^{\text{p}xz} := \begin{array}{c} \text{---} Z_\chi^\dagger \text{---} \\ | \\ X_\chi^\dagger \quad X_\chi \end{array}, \quad (1.68)$$

$$\mathbb{d}_\chi^{\text{p}yz} := \begin{array}{c} \text{---} Z_\chi^\dagger \text{---} \\ | \\ X_\chi \quad X_\chi^\dagger \end{array}. \quad (1.69)$$

These terms commute both with the bulk vertex operators  $\mathbb{e}_g^\vee$  and plaquette operators  $\mathbb{b}_\chi^\vee$ . Naturally, they also commute all among each other. From the explicit form of the gauging maps  $\mathcal{G}_\frac{0}{2}$  and  $\mathcal{G}_\frac{1}{2}$  it follows that given an irrep  $\chi \in \widehat{G}$  the operators (1.68-1.69) are stabilizers of  $|\Psi[\psi]\rangle$  if and only if  $(X_\chi^\vee \otimes X_\chi^\vee)|\psi\rangle = |\psi\rangle$ , equivalently

$$\begin{aligned} \mathbb{d}_\chi |\Psi[\psi]\rangle &= |\Psi[\psi]\rangle \iff \\ (\psi | (X_\chi^\vee \otimes X_\chi^\vee) | \psi) &= 1 \end{aligned} \quad (1.70)$$

for neighboring vertices  $\mathbf{v}, \mathbf{v}' \in \mathbb{V}(\square)$ . Notice that  $X_\chi \otimes X_\chi$  is an order parameter for the (partial) breaking of the global 0-form  $G$  symmetry. Its expectation value is a two-point symmetric correlation function.

In what follows we show that given a choice of subgroup  $H \leq G$ ,  $\mathbb{d}_\chi$  is a stabilizer of  $|\Psi[\psi_H]\rangle$  for an appropriate boundary state  $|\psi\rangle_H$  depending on that subgroup, provided that  $\chi$  is trivial when restricted to  $H$ . It is a fact that characters in  $\widehat{G}$  that are trivial when restricted to  $H$  form a subgroup of  $\widehat{G}$  isomorphic to  $\widehat{G/H}$ . From now on we will abuse notation and denote this subgroup simply by  $\widehat{G/H} \leq \widehat{G}$ . For the chosen subgroup  $H \leq G$  we then propose following boundary state  $|\psi\rangle_H$ . First, we define the normalized states

$$|+\rangle_H := \frac{1}{\sqrt{|\widehat{G/H}|}} \sum_{[\chi] \in \widehat{G/H}} |\chi\rangle. \quad (1.71)$$

Notice that  $|+\rangle_H$  is stabilized by  $H$  in the sense that  $Z_h |+\rangle_H = |+\rangle_H$  for all  $h \in H$ . Given  $|+\rangle_H$  we define

$$|\psi\rangle_H := \frac{1}{\sqrt{|\widehat{G/H}|}} \sum_{[g] \in \widehat{G/H}} \left( \bigotimes_{\mathbf{v} \in \mathbb{V}(\square)} Z_g^\vee |+\rangle_H \right). \quad (1.72)$$

By construction  $|\psi\rangle_H$  is symmetric under the 0-form  $G$  symmetry and, as anticipated, is symmetric under

$\{X_\chi^\vee \otimes X_\chi^\vee, \chi \in \widehat{G/H}\}$  since  $X_\chi^\vee$  leaves invariant  $|+\rangle_H$  and  $X_\chi^\vee \otimes X_\chi^\vee$  commutes with  $Z_g^\vee \otimes Z_g^\vee$ . Notice one can think of  $|\psi\rangle_H$  as an equal weight superposition of the  $|\widehat{G/H}|$  ground states of a  $G$  symmetric Hamiltonian whose symmetry is spontaneously broken down to  $H$ . It is therefore meaningful to refer to these boundaries as *symmetry breaking* boundaries. We also define the following operators acting on the boundary:

$$\mathbb{e}_g^{\text{e}z} := \begin{array}{c} | \\ Z_g \end{array}, \quad (1.73)$$

which are symmetries of  $|\Psi[\psi_H]\rangle$ , and which commute with  $\mathbb{d}_\chi$  only if  $g \in H$ . Defining then

$$\mathbb{Q}_H^{\text{p}xz} := \frac{1}{|\widehat{G/H}|} \sum_{\chi \in \widehat{G/H}} \mathbb{d}_\chi^{\text{p}xz}, \quad (1.74)$$

$$\mathbb{Q}_H^{\text{p}yz} := \frac{1}{|\widehat{G/H}|} \sum_{\chi \in \widehat{G/H}} \mathbb{d}_\chi^{\text{p}yz}, \quad (1.75)$$

$$\mathbb{Q}_H^{\text{e}z} := \frac{1}{|H|} \sum_{h \in H} \mathbb{e}_h^{\text{e}z}, \quad (1.76)$$

the Hamiltonian in the presence of such boundary terms reads

$$\begin{aligned} \mathbb{H}_H &= \mathbb{H}^{\text{blk}} + \sum_{\mathbf{p}xz \in \mathbb{P}(\partial\mathcal{D})} \mathbb{Q}_H^{\text{p}xz} \\ &+ \sum_{\mathbf{p}yz \in \mathbb{P}(\partial\mathcal{D})} \mathbb{Q}_H^{\text{p}yz} + \sum_{\mathbf{e}z \in \mathbb{E}(\partial\mathcal{D})} \mathbb{Q}_H^{\text{e}z}, \end{aligned} \quad (1.77)$$

where the dependence on the choice of subgroup  $H \leq G$  is explicit. As an example, one can consider the extreme cases of  $H = G$  and  $H = \{1\}$  that can be identified with *Dirichlet* and *Neumann* boundary conditions respectively. In the context of lattice gauge theory as of here it is also customary to refer to these cases as realizing the *smooth* and *rough* boundary conditions.<sup>6</sup>

As a final comment we consider an analogous construction of a three-dimensional state with a corresponding stabilizer Hamiltonian, which is constructed by starting from a state  $|\hat{\psi}\rangle$  invariant under a 1-form  $\widehat{G}$  symmetry generated by the Wilson loop operators (1.19):

$$|\hat{\Psi}[\hat{\psi}]\rangle := \cdots \circ \mathcal{G}_2^1 \circ \mathcal{G}_\frac{0}{2} \circ \mathcal{G}_1^1 |\hat{\psi}\rangle. \quad (1.78)$$

<sup>6</sup>Notice that we don't recover *twisted* boundary conditions that amount to pasting an SPT phase on top of the symmetry broken boundary phase [31–35]. It is expected that all such boundary conditions, which are in one-to-one correspondence with pairs  $(H \leq G, [\psi] \in H^3(H, \mathbb{U}(1)))$ , are recovered by appropriately twisting the boundary terms akin as in the lower-dimensional setting [12, 32]. This is most easily facilitated for by the triangular lattice.

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The boundary terms in that case are given by truncating  $c_g^v$  and  $b_\chi^g$  as follows:

$$e_g^v := \begin{array}{c} \text{---} X_g \text{---} \\ \text{---} X_g \text{---} \\ \text{---} X_g \text{---} \end{array} \begin{array}{c} Z_g^\dagger \\ X_g^\dagger \\ X_g^\dagger \end{array} \quad (1.79)$$

$$f_\chi^e := \text{---} Z_\chi \text{---} \quad (1.80)$$

$$f_\chi^e := \text{---} Z_\chi \text{---} \quad (1.81)$$

Given a  $g \in G$ ,  $e_g^v$  is a symmetry of  $|\hat{\Psi}[\hat{\psi}]\rangle$  if and only if

$$\text{---} X_g \text{---} \begin{array}{c} X_g^\dagger \\ X_g^\dagger \end{array} \text{---} |\hat{\psi}\rangle = |\hat{\psi}\rangle. \quad (1.82)$$

Similarly as in the previous case one could define  $|\hat{\psi}\rangle_{\hat{H}}$  hosting a (partial) symmetry breaking phase of the 1-form symmetry with unbroken symmetry group  $\hat{H} \leq \hat{G}$ . Commuting boundary stabilizers for such a state are given by

$$\mathbb{R}_H^v := \frac{1}{|H|} \sum_{h \in H} e_h^v, \quad (1.83)$$

$$\mathbb{R}_H^e := \frac{1}{|G/H|} \sum_{\chi \in G/H} f_\chi^e. \quad (1.84)$$

Notice thus in particular that  $|\hat{\psi}\rangle$  can host non-trivial 2D topological order. It is understood that besides these boundaries and the  $G$  symmetry breaking ones considered above, it is also possible to define additional gapped boundaries which amount to (trivially) stacking a 2D topological order or stacking a 2D topological order while condensing composites of boundary anyons of the emergent state and (bulk) anyons from the stacked topological order [34].

## 2 General framework

What are the main takeaways from this introductory example? From iteratively gauging a global abelian 0-form symmetry and its dual 1-form symmetry we obtain stabilizer codes (local commuting projector Hamiltonians) in one dimension higher. Moreover, the obtained codes can be readily extended to include gapped boundary terms, and we have written down the gauging maps and states explicitly in terms of tensor networks.

The goal of this section is to provide a general framework for iteratively gauging a state  $|\psi\rangle$  which is symmetric under a collection of abelian symmetries defined via certain *check operators* which commute with them. Hereby we extend the formalism of [14] to arbitrary abelian groups. We argue that in this framework the concatenated gauging maps generically exhibit *local symmetries* originating from the Gauss constraints

of the symmetries under scrutiny. These local symmetries can in turn be organized in an *emergent* local commuting projector Hamiltonian that naturally lives in one dimension higher than the initial state  $|\psi\rangle$  which is being gauged. Towards the end of this section we comment on the inclusion of the boundary terms in the emergent Hamiltonian which can be identified with certain spontaneous symmetry breaking patterns of the boundary symmetry realized by  $|\psi\rangle$ . Since these gauging maps can be defined in any dimension, we recover the results from [12] as a specific case.

### 2.1 Gauging abelian symmetries

To set the scene, we assume a lattice  $\Lambda$  in which each vertex  $v \in V(\Lambda)$  supports a copy of  $\mathbb{C}[\hat{G}]$ . The tensor product Hilbert space  $\bigotimes_{v \in V(\Lambda)} \mathbb{C}[\hat{G}]$  will be written as  $\mathcal{H} := \bigotimes_{v \in V(\Lambda)} \mathbb{C}[\hat{G}]$  henceforth. Notice that we don't make any assumption about the underlying topology or dimension of the spatial manifold on which  $\Lambda$  is defined.

Throughout this section we deal with symmetry operators that are fully specified by a *complete collection of check operators*, labeled  $c \in \mathcal{C}$ , that commute with all of them. The formal gauging procedure of Williamson [14] then happens via the introduction of a gauge degree of freedom for every such check. Formally, consider for any  $g \in G$  the operators

$$\mathbb{U}_g[\Omega] := \bigotimes_{v \in \Omega} Z_g^v, \quad g \in G, \quad (2.1)$$

where  $\Omega$  is a subset of the vertices, i.e.  $\Omega \subseteq V(\Omega)$ . Borrowing the notation of [14], we will denote by  $\eta$  a function

$$\begin{aligned} \eta : \mathcal{C} &\rightarrow P(V(\Lambda)) \\ c &\mapsto \eta(c) \subseteq V(\Lambda) \end{aligned} \quad (2.2)$$

from  $\mathcal{C}$  to the power set  $P(V(\Lambda))$ . Physically,  $\eta$  maps a given constraint  $c$  from the collection  $\mathcal{C}$ , that will below be identified with the gauge degrees of freedom, to a corresponding set of vertices  $\eta(c) \subseteq V(\Lambda)$ . Furthermore we introduce the involution  $\alpha_c^v(\chi)$  which evaluates to  $\chi$  or  $\bar{\chi}$  depending on the *relative orientation* of the constraint  $c$  and the vertex  $v$ . An operator  $\mathbb{U}_g[\Omega]$  is then a *valid symmetry generator* if it commutes with the checks

$$X_\chi^c := \bigotimes_{v \in \eta(c)} X_{\alpha_c^v(\chi)}^v \quad (2.3)$$

for all  $c \in \mathcal{C}$  and all  $\chi \in \hat{G}$ . Below, we will sometimes refer to the symmetry structure generated by these valid symmetry generators as the *initial* symmetry. To connect with the previous section, let us remark that the global 0-form  $G$  symmetry corresponds to the case  $\Omega = V(\Lambda)$ , and thus  $\mathbb{U}_g[V(\Lambda)] \equiv \mathbb{U}_g$ . For

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these symmetry generators the checks can be chosen so that there is exactly one check per edge of the lattice, i.e.  $\mathcal{C} \equiv \mathcal{E}(\Lambda)$ . Explicitly, the checks then read  $X_{\chi}^e = X_{\chi}^{v(\rightarrow e)} \otimes X_{\bar{\chi}}^{v(\leftarrow e)}$ , where we refer to the previous section for a detailed explanation of the notation.

An important observation is that the local constraints can be interpreted as local Hamiltonian terms that commute with the symmetry [14]. As such, they can be used to construct *generalized Ising models* and when perturbed, for example with single-vertex  $Z_g$  terms, they host symmetry breaking phases of such symmetries.

We now proceed by gauging the symmetry generated by the operators  $\mathcal{U}_g[\Omega]$ , essentially mimicking the procedure used in the previous section based on [13, 14] adapted to this more general setting. To this end we first enlarge the ‘matter’ Hilbert space  $\mathcal{H}$  with a  $\mathbb{C}[G]$ -valued *gauge degree of freedom* for every check in  $\mathcal{C}$ . We will denote the physical support of the gauge degrees of freedom by  $\mathfrak{c} \in \mathcal{C}(\Lambda)$  where it has to be understood that there is bijection between  $\mathcal{C}(\Lambda)$  and  $\mathcal{C}$ . Typically,  $\mathcal{C}(\Lambda)$  can be identified with a (subset) of  $k$ -cells of the lattice  $\Lambda$  rendering the aforementioned bijection canonical.

Having again the example from the previous section in mind,  $\mathcal{C}(\Lambda)$  can thus be identified with  $\mathcal{E}(\Lambda)$  as anticipated above.

Writing  $\hat{\mathcal{H}} := \bigotimes_{\mathfrak{c} \in \mathcal{C}} \mathbb{C}[G]$ , the first step of the gauging procedure thus entails enlarging  $\mathcal{H}$  according to  $\mathcal{H} \rightarrow \mathcal{H} \otimes \hat{\mathcal{H}}$ . Given the map  $\eta$  defined above, we define  $\eta^{-1}(\mathfrak{v})$  as the preimage of  $\mathfrak{v}$  under  $\eta$ . In other words,  $\eta^{-1}$  maps the matter degree of freedom localized on  $\mathfrak{v}$  to its associated gauge degrees of freedom. In light of the above identification of gauge degrees of freedom and constraints, we will also interpret the set of constraints  $\eta^{-1}(\mathfrak{v})$  as its corresponding subset of  $\mathcal{C}(\Lambda)$ . For every  $g \in G$  we then define the *gauge transformation*  $\mathcal{G}_g^v$  on the enlarged Hilbert space  $\mathcal{H} \otimes \hat{\mathcal{H}}$  as:

$$\mathcal{G}_g^v := Z_g^v \bigotimes_{\mathfrak{c} \in \eta^{-1}(\mathfrak{v})} X_{o_{\mathfrak{c}}^v(g)}^c. \quad (2.4)$$

Out of these gauge transformations, a *local projector* is constructed by taking the  $G$ -average:

$$\mathcal{P}^v := \frac{1}{|G|} \sum_{g \in G} \mathcal{G}_g^v, \quad (2.5)$$

$\forall \mathfrak{v} \in \mathcal{V}(\Lambda)$ . Indeed remark that from a redefinition of summation variables it follows that  $\mathcal{P}^v \cdot \mathcal{P}^v = \mathcal{P}^v$  for all  $\mathfrak{v} \in \mathcal{V}(\Lambda)$ .

The *gauging map* is then finally defined as the operator obtained by acting with the local projectors on the trivial gauge field  $\bigotimes_{\mathfrak{c} \in \mathcal{C}(\Lambda)} |1\rangle^{\mathfrak{c}}$ :

$$\mathcal{G} := \left( \prod_{\mathfrak{v} \in \mathcal{V}(\Lambda)} \mathcal{P}^v \right) \left( \bigotimes_{\mathfrak{c} \in \mathcal{C}(\Lambda)} |1\rangle^{\mathfrak{c}} \right) : \mathcal{H} \rightarrow \mathcal{H} \otimes \hat{\mathcal{H}}. \quad (2.6)$$

Let us summarize the most important properties of  $\mathcal{G}$  before proceeding. We refer to [14] for a more detailed digression. For a given  $\mathfrak{v} \in \mathcal{V}(\Lambda)$ , the gauge transformations  $\mathcal{G}_g^v$  form by construction a  $G$ -representation supported on  $\eta^{-1}(\mathfrak{v})$ . It then follows from the definitions that  $\mathcal{G}_g^v \cdot \mathcal{G} = \mathcal{G}$ . By a relabeling of summation variable in the definition of  $\mathcal{P}^v$  it also follows immediately that  $\mathcal{G} \cdot \mathcal{U}_g[\Omega] = \mathcal{G}$  for all allowed symmetry operators. As illustrated in section 1 this property implies that only states  $|\psi\rangle$  transforming trivially under the symmetry, i.e.  $\mathcal{U}_g[\Omega]|\psi\rangle = |\psi\rangle$ ,  $\forall g \in G$  and for all allowed  $\Omega$ , are *gaugeable*. Given such a state, we coin  $\mathcal{G}|\psi\rangle$  the *gauged state*. From  $\mathcal{G}_g^v \cdot \mathcal{G} = \mathcal{G}$  it readily follows that the gauged state is gauge invariant as expected. Even though we only deal with gauging *states* throughout this manuscript, it should be noted that also operators can be gauged using this prescription. In particular, *local* and *symmetric* operators in the initial matter theory are mapped to *dual* local gauge invariant operators under the gauging procedure in such a way that the matrix elements of these symmetric operators w.r.t. a symmetric state are preserved.

As an example, consider once more the case of a 0-form symmetry. Notice that with the choices of relative orientations  $o_{\mathfrak{c}}^e$  implicitly defined above, we recover the Gauss constraints defined above in eq. (1.4), from which eventually the gauging map given in eq. (1.8) of [13] is recovered.

Crucially, the gauge theory obtained via the aforementioned proposal exhibits global *emergent* symmetries that, by construction, are anomaly-free and hence gaugeable [29]. Let us continue by identifying these emergent symmetries. First notice that even though the constraints  $X_{\chi}^c$  do not commute with the gauging map, the *dressed constraints*  $Z_{\chi}^c \otimes X_{\chi}^c$  do commute with all Gauss operators  $\mathcal{G}_g^v$  and, a fortiori, with the gauging map  $\mathcal{G}$ . Indeed, this follows from a direct computation. For fixed  $\mathfrak{v} \in \mathcal{V}(\Lambda)$  and  $\mathfrak{c} \in \mathcal{C}$  let us assume without loss of generality that  $\mathfrak{v} \in \eta(\mathfrak{c})$ , which necessarily implies that  $\mathfrak{c} \in \eta^{-1}(\mathfrak{v})$ :

$$\begin{aligned} & \left( Z_{\chi}^c \otimes X_{\chi}^c \right) \cdot \mathcal{G}_g^v \\ &= \left( Z_{\chi}^c \bigotimes_{\mathfrak{w} \in \eta(\mathfrak{c})} X_{o_{\mathfrak{w}}^c(\chi)}^w \right) \cdot \left( Z_g^v \bigotimes_{\mathfrak{d} \in \eta^{-1}(\mathfrak{v})} X_{o_{\mathfrak{d}}^d(g)}^d \right) \\ &= o_{\mathfrak{c}}^c(\bar{\chi})(g) \cdot \chi(o_{\mathfrak{c}}^c(g)) \cdot \mathcal{G}_g^v \cdot \left( Z_{\chi}^c \otimes X_{\chi}^c \right) \\ &= \mathcal{G}_g^v \cdot \left( Z_{\chi}^c \otimes X_{\chi}^c \right). \end{aligned} \quad (2.7)$$

In the second line the definition of  $X_{\chi}^c$  and  $\mathcal{G}_g^v$  was employed. In the third line we used the commutation relations eq. (1.18) and eq. (1.41). Combined with the fact that  $Z_{\chi}^c$  evaluates to 1 on the trivial gauge field  $\bigotimes_{\mathfrak{c} \in \mathcal{C}(\Lambda)} |1\rangle^{\mathfrak{c}}$ , this implies that the constraint  $X_{\chi}^c$  for every  $\mathfrak{c}$  and  $\chi \in \hat{G}$  is the preimage of  $Z_{\chi}^c \otimes X_{\chi}^c$  under

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the gauging map  $\mathcal{G}$ . In particular, it follows that for a choice of constraints  $\Gamma \subseteq \mathcal{C}$  and choice of involution  $\delta^c(\chi)$ ,  $c \in \Gamma$  such that

$$\prod_{c \in \Gamma} X_{\delta^c(\chi)}^c = \text{id}, \quad (2.8)$$

there exists a corresponding operator in the gauge theory

$$\mathbb{U}[\Gamma, \delta^c] = \prod_{c \in \Gamma} Z_{\delta^c(\chi)}^c \quad (2.9)$$

which is an emergent symmetry of  $\mathcal{G}$  in the sense that

$$\mathbb{U}[\Gamma, \delta^c] \cdot \mathcal{G} = \mathcal{G}. \quad (2.10)$$

Applying the above to the example of  $\mathcal{G}^0$  (1.8) considered in section 1, we note that eq. (2.8) is exactly satisfied for any choice  $\Gamma$  which coincides with a closed loop  $\ell$  supported on the edges  $\mathbf{E}(\Lambda)$  of the lattice. With the above choice of  $\delta_v^c$ ,  $\delta^c$  exactly takes the relative orientation of  $\ell$  with respect to the orientation of the edges into account.

In order to gauge the emergent symmetry operators we adapt a similar approach as for the initial symmetry (2.1). Notice that the symmetry generators defined in eq. (2.9) commute with the local checks

$$X_g^v := \bigotimes_{c \in \eta^{-1}(v)} X_{\delta_v^c(g)}^c \quad (2.11)$$

which are naturally localized around the vertices  $\mathbf{V}(\Lambda)$ . Therefore we will identify the set of these checks with the set of vertices  $\mathbf{V}(\Lambda)$ . To prove that these checks indeed commute with the emergent symmetry generators, consider some vertex  $v \in \eta(\Gamma)$ . The identity (2.8) guarantees the existence of constraints  $c, d \in \eta^{-1}(v)$  such that  $\delta^c(\delta_v^d(\chi)) = \delta^d(\delta_v^c(\chi))$ . It follows immediately that  $Z_{\delta^c(\chi)}^c \otimes Z_{\delta^d(\chi)}^d$  commutes with  $X_{\delta_v^c(g)}^c \otimes X_{\delta_v^d(g)}^d$ , from which the claim follows. In order to define the corresponding dual gauging map  $\widehat{\mathcal{G}}$ , we once more enlarge the Hilbert space according to  $\widehat{\mathcal{H}} \rightarrow \widehat{\mathcal{H}} \otimes \mathcal{H}$ , on which we define the Gauss operators

$$\mathcal{G}_\chi^c := Z_\chi^c \bigotimes_{v \in \eta(c)} X_{\delta_v^c(\chi)}^v. \quad (2.12)$$

Notice the complex conjugate appearing in  $\delta_v^c$ . Corresponding local projectors localized around  $c$  follow from averaging over  $\widehat{\mathcal{G}}$ :

$$\mathcal{P}^c := \frac{1}{|\widehat{\mathcal{G}}|} \sum_{\chi \in \widehat{\mathcal{G}}} \mathcal{G}_\chi^c \quad (2.13)$$

Ultimately the dual gauging map reads

$$\widehat{\mathcal{G}} := \left( \prod_{c \in \mathbf{C}(\Lambda)} \mathcal{P}^c \right) \left( \bigotimes_{v \in \mathbf{V}(\Lambda)} |\mathbb{1}\rangle^v \right). \quad (2.14)$$

It follows that after gauging these emergent symmetries using the gauging map  $\widehat{\mathcal{G}}$  the original  $G$  symmetry is recovered, as expected. This is quickly demonstrated by noticing that all symmetry generators  $\mathbb{U}_g[\Omega]$  commute by definition with the checks  $X_\chi^c$ , and by extension with the Gauss constraints  $\mathcal{G}_\chi^c$ , so that  $\mathbb{U}_g[\Omega] \cdot \widehat{\mathcal{G}} = \widehat{\mathcal{G}}$ .

Finalizing the example of the previous section, one verifies immediately that the 1-form Gauss constraints eq. (1.27) coincide precisely with the definition provided in eq. (2.12) applied to the case at hand.

## 2.2 Concatenation of gauging maps

Let us now consider the composition of the gauging operators defined above. With slight abuse of notation we consider in this section thus the composed operators

$$\widehat{\mathcal{G}} \circ \mathcal{G} : \mathbb{C}[\widehat{\mathcal{G}}]^{\otimes |\mathbf{V}|} \rightarrow \mathbb{C}[\widehat{\mathcal{G}}]^{\otimes |\mathbf{V}|} \otimes \mathbb{C}[\mathcal{G}]^{\otimes |\mathbf{C}|} \otimes \mathbb{C}[\widehat{\mathcal{G}}]^{\otimes |\mathbf{V}|},$$

and

$$\mathcal{G} \circ \widehat{\mathcal{G}} : \mathbb{C}[\mathcal{G}]^{\otimes |\mathbf{C}|} \rightarrow \mathbb{C}[\mathcal{G}]^{\otimes |\mathbf{C}|} \otimes \mathbb{C}[\widehat{\mathcal{G}}]^{\otimes |\mathbf{V}|} \otimes \mathbb{C}[\mathcal{G}]^{\otimes |\mathbf{C}|}.$$

Herein,  $|\mathbf{V}|$  stands for the number of vertices of the lattice  $\Lambda$  and by definition  $|\mathbf{C}|$  is equal to the number of constraints defining the initial symmetry (2.1), i.e.  $|\mathbf{C}| = |\mathcal{C}|$ . As anticipated above, these concatenated gauging maps exhibit local symmetries that can be traced back to the Gauss constraints enforced by  $\mathcal{G}$  and  $\widehat{\mathcal{G}}$ . Crucial in the derivation of the local symmetries are following identities which can be derived in a similar fashion as eq. (2.7):

$$\left( Z_g^v \bigotimes_{c \in \eta^{-1}(v)} X_{\delta_v^c(g)}^c \right) \cdot \widehat{\mathcal{G}} \quad (2.15)$$

$$= \widehat{\mathcal{G}} \cdot \left( \bigotimes_{c \in \eta^{-1}(v)} X_{\delta_v^c(g)}^c \right),$$

$$\left( Z_\chi^c \bigotimes_{v \in \eta(c)} X_{\delta_v^c(\chi)}^v \right) \cdot \mathcal{G} \quad (2.16)$$

$$= \mathcal{G} \cdot \left( \bigotimes_{v \in \eta(c)} X_{\delta_v^c(\chi)}^v \right).$$

Given these identities, it is now straightforward to argue that the mutually commuting operators

$$\mathfrak{a}_g^v := Z_g^v \otimes \left( \bigotimes_{c \in \eta^{-1}(v)} X_{\delta_v^c(g)}^c \right) \otimes Z_g^v, \quad (2.17)$$

$$\in \text{End}(\mathbb{C}[\widehat{\mathcal{G}}]^{\otimes |\mathbf{V}|} \otimes \mathbb{C}[\mathcal{G}]^{\otimes |\mathbf{C}|} \otimes \mathbb{C}[\widehat{\mathcal{G}}]^{\otimes |\mathbf{V}|})$$

and

$$\mathfrak{b}_\chi^c := Z_\chi^c \otimes \left( \bigotimes_{v \in \eta(c)} X_{\delta_v^c(\chi)}^v \right) \otimes Z_\chi^c \quad (2.18)$$

$$\in \text{End}(\mathbb{C}[\mathcal{G}]^{\otimes |\mathbf{C}|} \otimes \mathbb{C}[\widehat{\mathcal{G}}]^{\otimes |\mathbf{V}|} \otimes \mathbb{C}[\mathcal{G}]^{\otimes |\mathbf{C}|})$$



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are symmetries of the concatenated gauging maps  $\widehat{G} \circ \mathcal{G}$  and  $\mathcal{G} \circ \widehat{G}$  respectively. Indeed:

$$\mathfrak{o}_g^\vee \cdot (\widehat{G} \circ \mathcal{G}) = \widehat{G} \circ \mathcal{G}, \quad (2.19)$$

$$\mathfrak{b}_\chi^c \cdot (\mathcal{G} \circ \widehat{G}) = \mathcal{G} \circ \widehat{G}, \quad (2.20)$$

for all  $\mathbf{v} \in \mathbf{V}(\Lambda)$ ,  $\mathbf{c} \in \mathbf{C}(\Lambda)$ ,  $g \in G$ ,  $\chi \in \widehat{G}$ . The iterated gauging procedure put forward in [12] now dictates that given any state  $|\psi\rangle \in \mathcal{H}$  stabilized by the initial  $G$  symmetry, that is

$$\mathbb{U}_g[\Omega]|\psi\rangle = |\psi\rangle \quad (2.21)$$

for every *valid* symmetry operator  $\mathbb{U}_g[\Omega]$ , one can define the *emergent* state  $|\Psi[\psi]\rangle$  as

$$\begin{aligned} |\Psi[\psi]\rangle &:= \cdots \mathcal{G}_{\frac{1}{2}} \circ \widehat{G}_1 \circ \mathcal{G}_{\frac{1}{2}} |\psi\rangle \\ &\in \bigotimes_{\frac{1}{2}} \mathbb{C}[\widehat{G}]^{\otimes |\mathbf{V}|} \bigotimes_i \mathbb{C}[G]^{\otimes |\mathbf{C}|}, \end{aligned} \quad (2.22)$$

where the big tensor products run over all appearances of the gauging maps in the definition.

This state is then by construction stabilized by the operators  $\mathfrak{o}_g^\vee$  and  $\mathfrak{b}_\chi^c$  defined above. Given these stabilizers one can subsequently construct a stabilizer Hamiltonian for which  $|\Psi[\psi]\rangle$  is a ground state. Remark that additional stabilizers are given by the emergent symmetries of every gauging map. For the examples that follow we will refrain from including these stabilizers as they are in those cases non-local operators. However, note that this is by no means always the case. Recall that in the example of section 1, the emergent 1-form symmetries around individual plaquettes gave rise to the plaquette terms defined in eq. (1.54). In fact, we rather would like to think of these terms as kinematical constraints on the Hilbert space on which the emergent state is defined and as such we are allowed to include them in the Hamiltonian.

Given the foliated way in which the emergent state (2.22) is constructed it is now a sensible thing to consider this foliation as an additional dimension so that  $|\Psi[\psi]\rangle$  is a state that lives in exactly one dimension higher than the initial state  $|\psi\rangle$ . In this interpretation  $|\psi\rangle$  behaves as a boundary condition for the emergent state. The lattice on which  $|\Psi[\psi]\rangle$  is supported can then also be thought of as *emerging* from  $\Lambda$  on which  $|\psi\rangle$  is defined. More precisely, the lattice is made up of layers which each carry a copy of  $\Lambda$ . In every other layer  $\mathbb{C}[\widehat{G}]$ -valued degrees of freedom are placed on  $\mathbf{V}(\Lambda)$  and  $\mathbb{C}[G]$ -valued degrees of freedom supported on  $\mathbf{C}(\Lambda)$ . This is most easily demonstrated by concrete examples.

### 2.3 Gapped boundary conditions

In the interpretation of  $|\Psi[\psi]\rangle$  as a state living in one dimension higher than  $|\psi\rangle$ , the stabilizers  $\mathfrak{o}_g^\vee$  and  $\mathfrak{b}_\chi^c$  are interpreted as acting on the *bulk* of the state

$|\Psi[\psi]\rangle$ . The question of how to include boundary terms that stabilize  $|\Psi[\psi]\rangle$  poses itself. As it turns out, the answer to this question is intimately interwoven with the exact way in which the initial  $G$  symmetry is realized on the boundary state  $|\psi\rangle$ .

Let us begin by considering the simplest local candidate stabilizer. It can be verified readily that for a given  $g \in G$  and  $\mathbf{v} \in \mathbf{V}(\Lambda)$

$$\begin{aligned} &Z_g^\vee \otimes \text{id} \\ &\in \text{End}(\mathbb{C}[\widehat{G}]^{\otimes |\mathbf{V}|} \otimes \mathbb{C}[G]^{\otimes |\mathbf{C}|}) \end{aligned} \quad (2.23)$$

commutes with the gauging map  $\mathcal{G}$  since it obviously commutes with the local gauge transformations  $\mathcal{G}_g^\vee$ . Hence, it is a stabilizer of the emergent state exactly when it is one of the boundary condition, i.e.  $Z_g^\vee |\psi\rangle = |\psi\rangle$ . Note that one could think of this operator as a ‘truncated’ vertex term  $\mathfrak{o}_g^\vee$ .

Consider likewise the truncated  $\mathfrak{b}_\chi^c$  term

$$\begin{aligned} &\text{id} \otimes Z_\chi^c \bigotimes_{\mathbf{v} \in \eta(\mathbf{c})} X_{\sigma_\mathbf{c}^\vee(\chi)}^\vee \\ &\in \text{End}(\mathbb{C}[\widehat{G}]^{\otimes |\mathbf{V}|} \otimes \mathbb{C}[G]^{\otimes |\mathbf{C}|} \otimes \mathbb{C}[\widehat{G}]^{\otimes |\mathbf{V}|}) \end{aligned} \quad (2.24)$$

Recall that the preimage of this operator under  $\widehat{G} \circ \mathcal{G}$  was shown to be

$$\bigotimes_{\mathbf{v} \in \eta(\mathbf{c})} X_{\sigma_\mathbf{c}^\vee(\chi)}^\vee, \quad (2.25)$$

in eq. (2.16). This observation thus implies that the truncated term (2.24) is a stabilizer of the emergent state, whenever  $|\psi\rangle$  is stabilized by the check  $X_\chi^c$ . Since these local constraints  $X_\chi^c$  can be understood as local symmetric order parameters for the global  $G$  symmetry (2.1), inequivalent boundary conditions realizing different symmetry breaking patterns of this symmetry can be devised. More formally, consider for any subgroup  $H \leq G$  the state  $|+\rangle_H$  which we recall was defined in section 1.5 as  $|+\rangle_H := \frac{1}{\sqrt{|G/H|}} \sum_{[\chi] \in \widehat{G/H}} |\chi\rangle$ . Constructing then the tensor product state

$$\bigotimes_{\mathbf{v} \in \mathbf{V}(\Lambda)} |+\rangle_H^\vee, \quad (2.26)$$

it is clear that this reference state is stabilized by  $Z_h^\vee$  and  $X_\chi^c$  provided that  $h \in H$ , and  $\chi \in \widehat{G/H}$ . Remember indeed from section 1.5 that  $\widehat{G/H}$  stands for the subgroup of characters in  $\widehat{G}$  which are trivial when restricted to  $H$ . From this it also follows that the corresponding boundary stabilizers of the emergent state  $|\Psi[\psi_H]\rangle$  commute. More generally, we could equally well consider boundary states of the form

$$\left( \sum_{[g] \in G/H} \mathbb{U}_g[\Omega] \right) \left( \bigotimes_{\mathbf{v} \in \mathbf{V}(\Lambda)} |+\rangle_H^\vee \right) \quad (2.27)$$

for any allowed  $\Omega$ , which enjoy the same stabilizers.



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Also remark that, depending on the specifics of the  $G$  symmetry generators, even more generic boundary conditions can be defined, for example by also (partially) breaking the translational invariance of the boundary state. An exhaustive study of gapped boundary conditions for the models under scrutiny will be the topic of future work.

### 3 Foliated type-I fracton order from 2D linear subsystem symmetries

*In this section we showcase the gauging of abelian linear subsystem symmetries and their dual emergent symmetries, which are themselves linear subsystem symmetries on the dual lattice, following the prescription of section 2. Concatenation of these gauging maps results in foliated type-I fracton codes, generalizing the anisotropic models studied in [17] defined for cyclic groups to arbitrary finite abelian groups.*

#### 3.1 Gauging 2D abelian linear subsystem symmetries and their emergent symmetries

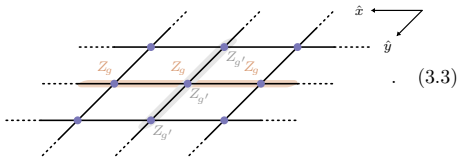
Throughout this section we consider again the two-dimensional torus, now endowed with a square lattice denoted by  $\square$ . The initial  $\mathbb{C}[\hat{G}]$ -valued ‘matter’ degrees of freedom are located on the vertices  $\mathbf{v} \in \mathbf{V}(\square)$  as customary. It will be useful to express the vertices in terms of their Cartesian coordinates as  $\mathbf{v} \equiv (i, j) \in \mathbb{Z}_{L_x} \times \mathbb{Z}_{L_y}$ . Naturally,  $L_x$  and  $L_y$  stand for the linear system size in the  $x$  and  $y$  direction respectively. The linear subsystem symmetry (LSS) generators are then represented on the matter Hilbert space  $\mathcal{H} := \bigotimes_{\mathbf{v} \in \mathbf{V}(\square)} \mathbb{C}[\hat{G}]$  as

$$Z_g^{\hat{x},j} := \bigotimes_{i \in \mathbb{Z}_{L_x}} Z_g^{(i,j)}, \quad (3.1)$$

$$Z_g^{\hat{y},i} := \bigotimes_{j \in \mathbb{Z}_{L_y}} Z_g^{(i,j)}. \quad (3.2)$$

There thus exists an extensive number of symmetry generators, one for every  $j$  in the  $\hat{x}$  direction of the square lattice,  $Z_g^{\hat{x},j}$ , and one for every  $i$  in the  $\hat{y}$  direction,  $Z_g^{\hat{y},i}$ . Notice however that not all of the symmetries generators are independent since clearly  $\prod_{j \in \mathbb{Z}_{L_y}} Z_g^{\hat{x},j} = \prod_{i \in \mathbb{Z}_{L_x}} Z_g^{\hat{y},i}$ ,  $\forall g \in G$ .

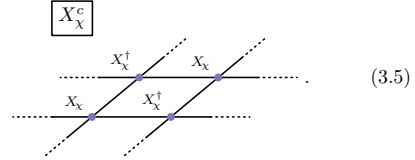
Graphically, an example of symmetry operators  $Z_g^{\hat{x},j}$  and  $Z_g^{\hat{y},i}$  on a portion of the square lattice looks like:



In order to employ the recipe of the general framework laid out in section 2 we need to consider the checks that fully specify these symmetry generators. An adequate choice is given by

$$X_\chi^c = X_\chi^{(i,j)} \otimes X_\chi^{(i+1,j)} \otimes X_\chi^{(i,j+1)} \otimes X_\chi^{(i+1,j+1)}, \quad (3.4)$$

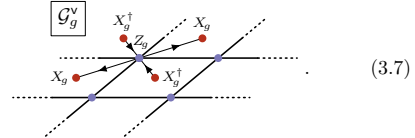
$\forall \chi \in \hat{G}$ ,  $\forall i, j$ . These operators are naturally associated with the plaquettes  $\mathbf{P}(\square)$ , and we will therefore identify the constraints with the plaquettes, i.e.  $\mathcal{C} \ni c \equiv \mathbf{p} \in \mathbf{P}(\square)$ . Indeed, graphically, one such operator can be depicted as



Let us continue by gauging this symmetry. To this end we introduce thus  $\mathbb{C}[G]$ -valued gauge fields on the plaquettes  $\mathbf{p} \in \mathbf{P}(\square)$  and thereby extend the Hilbert space according to  $\mathcal{H} \rightarrow \mathcal{H} \otimes \hat{\mathcal{H}}$ , where  $\hat{\mathcal{H}} := \bigotimes_{\mathbf{p} \in \mathbf{P}(\square)} \mathbb{C}[\hat{G}]$ . The local gauge transformations in the neighborhood of the vertex  $\mathbf{v} \in \mathbf{V}(\square)$  read then

$$\mathcal{G}_g^{\mathbf{v}} := Z_g^{\mathbf{v}} \bigotimes_{\mathbf{p} \rightarrow \mathbf{v}} X_g^{\mathbf{p}} \bigotimes_{\mathbf{p} \leftarrow \mathbf{v}} X_g^{\mathbf{p}}, \quad (3.6)$$

for every  $g \in G$ . The tensor products appearing in this definition are over the sets of plaquettes  $\mathbf{p} \rightarrow \mathbf{v}$ , respectively  $\mathbf{p} \leftarrow \mathbf{v}$ , which point into, respectively point away from, the vertex  $\mathbf{v}$  according to the following orientations:



As such, the definition of  $\mathcal{G}_g^{\mathbf{v}}$  is in accordance with eq. (2.4). Notice that here the same blue and red color code of the previous sections for depicting degrees of freedom in respectively  $\mathbb{C}[\hat{G}]$  and  $\mathbb{C}[G]$  is recycled.

As prescribed by our general gauging procedure, we define the following local projector:

$$\mathcal{P}^{\mathbf{v}} := \frac{1}{|G|} \sum_{g \in G} \mathcal{G}_g^{\mathbf{v}}, \quad (3.8)$$

for every vertex  $\mathbf{v} \in \mathbf{V}(\Lambda)$ , out of which we construct the projector onto the gauge invariant subspace as:

$$\mathcal{P} := \prod_{\mathbf{v} \in \mathbf{V}(\square)} \mathcal{P}^{\mathbf{v}}. \quad (3.9)$$

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Gauging of the LSS then happens by initializing the gauge fields in the trivial tensor product configuration  $\bigotimes_{\mathbf{p} \in \mathbf{P}(\square)} |1\rangle^{\mathbf{p}}$  and applying the above projector  $\mathcal{P}$  so as to obtain

$$\mathcal{G}^{(0,1)} := \mathcal{P} \left( \bigotimes_{\mathbf{p} \in \mathbf{P}(\square)} |1\rangle^{\mathbf{p}} \right). \quad (3.10)$$

The notation  $\mathcal{G}^{(0,1)}$ , following [19], is motivated by the fact that there is a sense in which linear subsystem symmetries can be thought of as ‘(0,1)-form’ symmetries: 0-form symmetries acting on a codimension 1 foliation of the spatial manifold. From the above definition of the gauging map it follows that it satisfies:

$$\mathcal{G}^{(0,1)} \cdot Z_g^{\hat{x},j} = \mathcal{G}^{(0,1)}, \quad (3.11)$$

$$\mathcal{G}^{(0,1)} \cdot Z_g^{\hat{y},i} = \mathcal{G}^{(0,1)}, \quad (3.12)$$

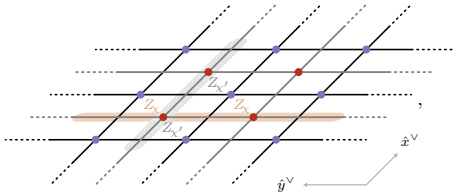
$\forall g \in G, i \in \mathbb{Z}_{L_x}, j \in \mathbb{Z}_{L_y}$ . Hence, in line with the expectation from the general framework, states transforming non-trivially under one or more of the global linear subsystem symmetry generators are annihilated by the gauging map  $\mathcal{G}^{(0,1)}$ .

Let us now present the emergent symmetries. These happen to be themselves linear subsystem symmetries supported on the dual square lattice  $\square^\vee$  and are labeled by characters  $\chi \in \hat{G}$ . Similarly as in the case of the initial LSS it proves useful to define the vertices of the dual square lattice,  $\mathbf{v}^\vee \in \mathbf{V}(\square^\vee)$ , in terms of their Cartesian coordinates, i.e.  $\mathbf{v}^\vee \equiv (i^\vee, j^\vee) \in \mathbb{Z}_{L_y} \times \mathbb{Z}_{L_x}$ . Also notice that the dual vertices can be identified under Poincaré duality with the plaquettes of the primal square lattice, formally:  $\mathbf{V}(\square^\vee) \ni \mathbf{v}^\vee \equiv \mathbf{p} \in \mathbf{P}(\square)$ . We then define for any character  $\chi \in \hat{G}$  the dual LSS generators:

$$Z_\chi^{\hat{x},j^\vee} := \bigotimes_{i^\vee \in \mathbb{Z}_{L_y}} Z_\chi^{(i^\vee, j^\vee)}, \quad (3.13)$$

$$Z_\chi^{\hat{y},i^\vee} := \bigotimes_{j^\vee \in \mathbb{Z}_{L_x}} Z_\chi^{(i^\vee, j^\vee)}. \quad (3.14)$$

Examples of  $Z_\chi^{\hat{x},j^\vee}$  and  $Z_\chi^{\hat{y},i^\vee}$  then look like



wherein the dual lattice  $\square^\vee$  is drawn in gray on top of the primal lattice  $\square$  drawn in black.

The emergent symmetries are apparent from

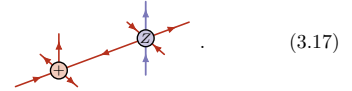
$$Z_\chi^{\hat{x},j^\vee} \cdot \mathcal{G}^{(0,1)} = \mathcal{G}^{(0,1)}, \quad (3.15)$$

$$Z_\chi^{\hat{y},i^\vee} \cdot \mathcal{G}^{(0,1)} = \mathcal{G}^{(0,1)}, \quad (3.16)$$

for all  $\chi \in \hat{G}, i^\vee \in \mathbb{Z}_{L_y}, j^\vee \in \mathbb{Z}_{L_x}$ .

Despite these identities being a mere application of eq. (2.10) to the case at hand, let us nevertheless derive these emergent symmetries explicitly for completeness. To this end, note that the operators  $Z_\chi^{\hat{x},j^\vee}, Z_\chi^{\hat{y},i^\vee}$  commute with the projector in eq. (3.9) and leave invariant the initial trivial gauge field on the plaquettes  $\bigotimes_{\mathbf{p} \in \mathbf{P}(\square)} |1\rangle^{\mathbf{p}}$ . The former is satisfied because along dual horizontal and vertical directions  $\hat{x}^\vee/\hat{y}^\vee$ , the operators  $Z_\chi^{\hat{x},j^\vee}, Z_\chi^{\hat{y},i^\vee}$  overlap in two faces with the projectors  $\mathcal{P}_g^{\mathbf{v}}$  appearing in the definition of  $\mathcal{G}^{(0,1)}$  that act as  $X_g \otimes X_g^\dagger$  on them.

To conclude the construction of the  $\mathcal{G}^{(0,1)}$  gauging map, let us provide its PEPO representation. To this end we use tensors which slightly generalize those introduced in section 1. Their symmetry properties are therefore also completely analogous and are omitted here. Again restricting to one unit cell of the PEPO, which in this case is made up of one Z-type tensor on the vertex  $\mathbf{v} \in \mathbf{V}(\square)$  of the unit cell and one +type tensor on an adjacent plaquette  $\mathbf{p} \in \mathbf{P}(\square)$ , we claim that the PEPO is constructed from:



Indeed, let us verify its relevant properties. In the first place it follows from eq. (1.14) combined with eq. (1.16) that the gauging map absorbs the initial LSSs supported on the vertices of the primal lattice, i.e. (3.11-3.12). Analogously, the gauge invariance of the gauging map, i.e.  $\mathcal{G}_g^{\mathbf{v}} \cdot \mathcal{G}^{(0,1)}, \forall \mathbf{v} \in \mathbf{V}(\square), g \in G$ , is proven by making use of eq. (1.14) and (1.15). Finally, the existence of the emergent symmetries is a direct consequence of the flatness of the gauge field enforced by the +-tensors, combined with the observation that the Z-type tensors are diagonal in its virtual indices.

### 3.2 Iterative gauging of linear subsystem symmetries

Due to the structural similarity between the dual LSS generators (3.13, 3.14) and the initial ones (3.1, 3.2), the gauging procedure of the latter can be readily adopted to gauge the emergent LSS symmetries. To this end, we introduce  $\mathbb{C}[\hat{G}]$ -valued gauge fields on the plaquettes  $\mathbf{p}^\vee$  of the dual lattice  $\square^\vee$ , which, as we mentioned, can equivalently be thought of as the vertices  $\mathbf{v} \in \mathbf{V}(\square)$ . As is custom by now, we initiate these gauge fields as  $\bigotimes_{\mathbf{v} \in \mathbf{V}(\square)} |1\rangle^{\mathbf{v}}$ . On this enlarged Hilbert space we define the local gauge transformation

$$\mathcal{G}_\chi^{\mathbf{p}} := Z_\chi^{\mathbf{p}} \bigotimes_{\mathbf{v} \rightarrow \mathbf{p}} X_\chi^{\mathbf{v}} \bigotimes_{\mathbf{v} \leftarrow \mathbf{p}} X_\chi^{\mathbf{v}}. \quad (3.18)$$

The orientations determining the sets of ‘incoming’ and ‘outgoing’ plaquettes on the dual square lattice, respectively  $\mathbf{v} \rightarrow \mathbf{p}$  and  $\mathbf{v} \leftarrow \mathbf{p}$ , are dual to those on

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the primal lattice as per the convention in (3.7). From the perspective of the primal lattice this gauge transformation can hence be depicted as

$$(3.19)$$

Using the local projectors

$$\mathcal{P}^p := \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \mathcal{G}_\chi^p, \quad (3.20)$$

the following gauging map is then constructed:

$$\widehat{\mathcal{G}}^{(0,1)} := \left( \prod_{\mathbf{p} \in \mathbf{P}(\square)} \mathcal{P}^p \right) \left( \bigotimes_{\mathbf{v} \in \mathbf{V}(\square)} |\mathbb{1}\rangle^{\mathbf{v}} \right). \quad (3.21)$$

After the application of  $\widehat{\mathcal{G}}^{(0,1)}$  LSSs on the primal lattice represented by (3.1-3.2) are reattained.

The PEPO representation of  $\widehat{\mathcal{G}}^{(0,1)}$  in turn is constructed from the unit cell

$$(3.22)$$

In this depiction it has to be understood that the dual Z-type tensor is placed on a plaquette  $\mathbf{p} \in \mathbf{P}(\square)$  of the lattice whereas the dual + -type tensor is located on a vertex  $\mathbf{v} \in \mathbf{V}(\square)$ . The symmetries of this PEPO are immediately found from dualizing those of the above PEPO defined in eq. (3.17); from them the properties of  $\widehat{\mathcal{G}}^{(0,1)}$  are recovered immediately.

An initial state  $|\psi\rangle$  that is in the even sector of the LSSs (3.1-3.2) can now be iteratively gauged by using the usual prescription as follows:

$$|\Psi[\psi]\rangle := \dots \circ \mathcal{G}_{\frac{1}{2}}^{(0,1)} \circ \widehat{\mathcal{G}}_1^{(0,1)} \circ \mathcal{G}_{\frac{1}{2}}^{(0,1)} |\psi\rangle. \quad (3.23)$$

Following our general paradigm we then interpret  $|\Psi[\psi]\rangle$  as a three-dimensional state living on a cubic lattice  $\mathfrak{C}$  endowed with a copy of  $\mathbb{C}[\widehat{G}]$  to every edge in the  $\hat{z}$  direction (blue) and a local Hilbert space  $\mathbb{C}[G]$  (red) associated to every plaquette in the  $xy$ -plane. Putting the PEPO representations eqs. (3.17) and (3.22) together, it then follows that the emergent state away from the boundary can be represented by the PEPO unit cell

$$(3.24)$$

From the general procedure outlined in section 2, it then follows that the bulk stabilizers of the state  $|\Psi[\psi]\rangle$  can be pictorially represented as

$$\mathfrak{o}_g^{\mathbf{v}} = \quad (3.25)$$

$$\mathfrak{b}_\chi^{\mathbf{c}} = \quad (3.26)$$

for every vertex  $\mathbf{v} \in \mathbf{V}(\mathfrak{C})$  and every cube  $\mathbf{c} \in \mathbf{C}(\mathfrak{C})$  respectively. The state (3.23) is thus a ground state of the commuting projector Hamiltonian

$$\mathbb{H}^{\text{blk}} := - \sum_{\mathbf{v} \in \mathbf{V}(\mathfrak{C})} \mathbb{P}^{\mathbf{v}} - \sum_{\mathbf{c} \in \mathbf{C}(\mathfrak{C})} \mathbb{P}^{\mathbf{c}}, \quad (3.27)$$

in which

$$\mathbb{P}^{\mathbf{v}} := \frac{1}{|G|} \sum_{g \in G} \mathfrak{o}_g^{\mathbf{v}}, \quad (3.28)$$

$$\mathbb{P}^{\mathbf{c}} := \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \mathfrak{b}_\chi^{\mathbf{c}}. \quad (3.29)$$

This generalizes the anisotropic model introduced in [17], which exhibits foliated type-I fracton order, to arbitrary finite groups. Our representation can be brought in the form showcased in [17] by applying the local basis transformation defined in eq. (1.65) to the red plaquettes. It is expected that our approach will facilitate the classification and construction of these fracton orders.

### 3.3 Gapped boundary conditions

Let us now also extend the bulk Hamiltonian to the boundary of the three-dimensional cubic lattice. To this end we consider following operators near the boundary:

$$\mathfrak{d}_\chi^{\mathbf{c}} := \quad (3.30)$$

$$\mathfrak{o}_g^{e_z} := \quad (3.31)$$

for all  $\chi \in \widehat{G}$  and  $g \in G$ .

Being truncated versions of the bulk stabilizers  $\mathfrak{b}_\chi^{\mathbf{c}}$  and  $\mathfrak{o}_g^{\mathbf{v}}$ , they naturally commute with all bulk terms appearing in the bulk Hamiltonian  $\mathbb{H}^{\text{blk}}$ . Note that,

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similar as to the boundary terms defined for the introductory example in section 1.5,  $\mathfrak{d}_\chi^c$  and  $\mathfrak{e}_g^{e_s}$  only commute for particular choices of  $\chi$  and  $g$  such that  $\chi(g) = 1$ . This statement will be refined below. From the general framework it follows that  $\mathfrak{d}_\chi^c$  stabilizes the three-dimensional state  $|\Psi[\psi]\rangle$  if the boundary condition  $|\psi\rangle$  is symmetric under the operator

$$\boxed{X_\chi^c} \quad \begin{array}{c} \text{---} X_\chi \text{---} \\ \text{---} X_\chi^\dagger \text{---} \\ \text{---} X_\chi \text{---} \\ \text{---} X_\chi^\dagger \text{---} \end{array} \quad (3.32)$$

for the boundary plaquette  $\mathfrak{p} \in \mathcal{P}(\square)$  corresponding to  $c$ . One recognizes these operators as the checks  $X_\chi^c$  defined above. For illustrative purposes, let us construct one particular reference boundary before proceeding to the more general case.

A suitable choice for  $|\psi\rangle$  is any of the extensive number of ground states of the two-dimensional classical  $\widehat{G}$  Ising plaquette model. Indeed, recall that the Hamiltonian of this model is exactly given by  $\mathbb{H}^{\text{Ising}} := -\sum_c \sum_{\{\chi_c\}_c} X_\chi^c$ . One such ground state is for example constructed by starting from the product state

$$\bigotimes_{v \in V(\square)} |+\rangle_{\{1\}}^v, \quad (3.33)$$

where the notation introduced in eq. (1.71) was used, and acting on this reference state with all LSS generators according to

$$|\psi\rangle_{\{1\}} := \left( \sum_{j=0}^{L_y-1} \sum_{\{g_j\}_j \in G} Z_{g_j}^{x,j} \right) \cdot \left( \sum_{i=0}^{L_x-1} \sum_{\{g_i\}_i \in G} Z_{g_i}^{y,i} \right) \cdot \left( \bigotimes_{v \in V(\square)} |+\rangle_{\{1\}}^v \right). \quad (3.34)$$

We stress that in this definition every summation over group elements is over the entire group  $G$ . From the observation that  $X_\chi^c (\bigotimes_{v \in V(\square)} |+\rangle_{\{1\}}^v) = \bigotimes_{v \in V(\square)} |+\rangle_{\{1\}}^v$  for every  $\chi \in \widehat{G}$ , together with the fact that the checks commute by definition with all LSS generators, it indeed follows that  $|\psi\rangle_{\{1\}}$  is a good boundary condition for the emergent state  $|\Psi[\psi_{\{1\}}]\rangle$  for which additional boundary stabilizers are given by

$$\mathbb{Q}_{\{1\}}^c := \sum_{\chi \in \widehat{G}} \mathfrak{d}_\chi^c, \quad \forall c \in C(\partial\mathfrak{D}). \quad (3.35)$$

Similar as to the introductory example in section 1, and as per the general paradigm in section 2, we can also define more general boundary conditions for different symmetry breaking patterns of the LSS on the boundary. Indeed, given again the state  $|+\rangle_H$  defined

in eq. (1.71), it is natural to define

$$|\psi\rangle_H := \left( \sum_{j=0}^{L_y-1} \sum_{\{g_j\}_j \in G/H} Z_{g_j}^{x,j} \right) \cdot \left( \sum_{i=0}^{L_x-1} \sum_{\{g_i\}_i \in G/H} Z_{g_i}^{y,i} \right) \cdot \left( \bigotimes_{v \in V(\square)} |+\rangle_H^v \right). \quad (3.36)$$

Given this choice of boundary condition, appropriate commuting boundary stabilizers are given by:

$$\mathbb{Q}_H^c := \frac{1}{|G/H|} \sum_{\chi \in G/H} \mathfrak{d}_\chi^c, \quad (3.37)$$

$$\mathbb{Q}_H^{e_s} := \frac{1}{|H|} \sum_{h \in H} \mathfrak{e}_h^{e_s}. \quad (3.38)$$

Indeed given that  $h \in H$ , and thus  $\chi(h) = 1$  for any  $\chi \in G/H$ , the boundary operators  $\mathfrak{d}_\chi^c$  and  $\mathfrak{e}_h^{e_s}$  defined in (3.30,3.31) commute. These boundary terms are then added to the bulk Hamiltonian (3.27) to obtain the Hamiltonian

$$\mathbb{H}_H := \mathbb{H}^{\text{blk}} + \sum_{c \in C(\partial\mathfrak{D})} \mathbb{Q}_H^c + \sum_{e_s \in E(\partial\mathfrak{D})} \mathbb{Q}_H^{e_s}. \quad (3.39)$$

Before proceeding we note that the boundary conditions presented here are by no means exhaustive. In particular it should be pointed out that the boundary states defined in eq. (3.36) are manifestly translationally invariant, which is a condition which can be relaxed [6]. A complete study of all (gapped) boundary conditions for these models will be presented elsewhere.

## 4 Type-I fracton order from 2D Sierpinski fractal symmetry

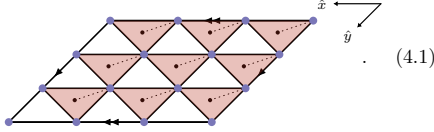
In this section we consider  $\mathbb{Z}_2$  Sierpinski fractal symmetry as initial symmetry to our iterative gauging procedure. It is shown that the original Castelnovo-Chamon fractal model proposed in [18] emerges.

### 4.1 Gauging Sierpinski fractal symmetry and its emergent dual fractal symmetry

Let us begin by defining a set of generators for the fractal Sierpinski symmetry. Therefore we consider the two-dimensional torus endowed with a triangular lattice that we will denote by  $\triangle$ . Given a vertex  $v \in V(\triangle)$ , we will denote its rectangular coordinates in the basis  $\{\hat{x}, \hat{y}\}$ , indicated in the figure below, by  $v \equiv (i, j)$ ,  $i, j \in \mathbb{Z}_L$ ,  $L$  being the linear system size. Every vertex  $v$  carries a copy of the local Hilbert space

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$\mathbb{C}[\mathbb{Z}_2] \simeq \mathbb{C}^2 = \text{Span}_{\mathbb{C}}\{|0\rangle, |1\rangle\}$  on which the Pauli operators  $X$  and  $Z$  are defined.<sup>7</sup> We will consider a subset  $P(\Delta)^+ \subset P(\Delta)$  of the plaquettes whose member plaquettes are shaded in red in the figure below, foregrounding that we will put gauge degrees of freedom on these plaquettes when gauging the fractal symmetry. Note that the number of such plaquettes equals the number of vertices, i.e.  $|P(\Delta)^+| = |V(\Delta)|$ .<sup>8</sup> This allows us to assign to every  $\mathbf{p}^+ \in P(\Delta)^+$  a *base point*  $\text{bp}(\mathbf{p}^+)$  which we conventionally choose as dictated by the dotted lines in the figure:



As pointed out in [36] the number of symmetry generators depends in a non-trivial way on the (linear) system size  $L$ . In this section we will for demonstrative purposes restrict to the case of  $L = 2^m - 1$  for an integer  $m$ , in which case the number of symmetry generators is exactly equal to  $L - 1$ . The Sierpinski symmetry generators are then of the form

$$\mathbb{U}[S_q] := \prod_{i,j} (Z^{(i,j)})^{S_q(i,j)}, \quad (4.2)$$

where  $S_q$  is a function  $S_q : \mathbb{Z}_L \times \mathbb{Z}_L \rightarrow \mathbb{F}_2$  defined such that  $\mathbb{U}[S_q]$  commutes with the checks  $X^{\mathbf{p}^+}$  which are given by

$$X^{\mathbf{p}^+} := X^{(i,j)} \otimes X^{(i+1,j)} \otimes X^{(i,j+1)}, \quad (i,j) = \text{bp}(\mathbf{p}^+), \quad (4.3)$$

for all  $\mathbf{p}^+ \in P(\Delta)^+$ . A check thus consists of three Pauli  $X$  operators acting on the three vertices surrounding a plaquette  $\mathbf{p}^+ \in P(\Delta)^+$ .

As follows from a direct computation, imposing commutation of the generators (4.2) with all these checks requires  $S_q$  to satisfy the recursive identity

$$S_q(i, j+1) := S_q(i+1, j) + S_q(i, j), \quad (4.4)$$

$\forall i, j \in \mathbb{Z}_L$ . A *valid symmetry generator* is hence fully specified by a *valid* choice of ‘initial condition’ at  $j = 0$ :

$$S_q(i, 0) := q(i) \in \mathbb{F}_2^L. \quad (4.5)$$

The function  $S_q(i, j)$ , for  $j \neq 0$  follows then from applying the recursion relation to this choice of  $q$ , and as

<sup>7</sup>In fact, in the remainder of this section we will identify  $\mathbb{C}[\mathbb{Z}_2]$  and  $\mathbb{C}[\widehat{\mathbb{Z}}_2]$  by making use of the obvious isomorphism  $\mathbb{Z}_2 \simeq \widehat{\mathbb{Z}}_2$ .

<sup>8</sup>Indeed, this follows from the fact that the Euler characteristic of the torus is zero, which can be expressed as  $|V(\Delta)| - |E(\Delta)| + |P(\Delta)| = 0$ , combined with the facts that  $|P(\Delta)| = 2|P(\Delta)^+|$  and  $|E(\Delta)| = 3|V(\Delta)|$ .

such the corresponding generator  $\mathbb{U}[S_q]$  commutes by construction with all checks. The  $L - 1$  independent choices for  $q$  which give rise to the  $L - 1$  independent Sierpinski symmetry generators that we expect can be chosen as

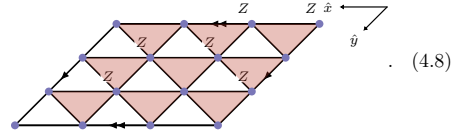
$$q^{(\alpha)}(i) = \delta_{i,\alpha} + \delta_{i,\alpha+1}, \quad \alpha = 1, 2, \dots, L - 1. \quad (4.6)$$

Let us mention as a side remark that it turns out that one could think of  $S_q$  as a *valid history* of a *Sierpinski cellular automaton* in the following sense. For fixed  $j$ ,  $S(i, j)$  corresponds to a state valued in  $\mathbb{F}_2^L$ . Given this state, the state at  $j + 1$ , i.e.  $S_q(i, j + 1)$  for fixed  $j + 1$ , is fully specified by the above recursion relation from  $S_q(i, j)$ . Therefore  $j$  can be thought of as labeling discrete ‘time’ steps, where this notion of time thus flows in the direction of increasing  $j$ . In this picture  $q(i)$  serves as an initial condition for the automaton, a posteriori justifying our jargon introduced above.

As a concrete example take  $L = 3$ , and  $q^{(1)}(i) = [1 \ 1 \ 0]$ . The corresponding history  $S_q$  is then computed to be

$$S_q(i, j) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}_{j,i}, \quad (4.7)$$

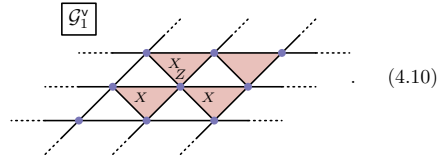
whose matching lattice operator  $\mathbb{U}[S_q]$  can be depicted as:



Let us now gauge the Sierpinski symmetry using the general paradigm of section 2. To this end we introduce qubit gauge degrees of freedom on the plaquettes in  $P(\Delta)^+$ . A local gauge transformation for every  $\mathbf{v} \in V(\Delta)$  reads then

$$\mathcal{G}_1^{\mathbf{v}} := Z^{\mathbf{v}} \bigotimes_{\mathbf{p}^+ \in P(\Delta)^+ | \mathbf{v}} X^{\mathbf{p}^+}, \quad (4.9)$$

wherein the tensor product is understood to be over plaquettes  $\mathbf{p}^+ \in P(\Delta)^+ | \mathbf{v}$  that contain  $\mathbf{v}$ . Graphically:



From the local projector

$$\mathcal{P}^{\mathbf{v}} := \frac{1}{2} (\text{id} + \mathcal{G}_1^{\mathbf{v}}), \quad (4.11)$$

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one constructs, as is by now routine, the global projector  $\mathcal{P} = \prod_{\mathbf{v} \in V(\Delta)} \mathcal{P}^{\mathbf{v}}$  from which one obtains the gauging map by trivially initializing the gauge field:

$$\mathcal{G} := \mathcal{P} \left( \bigotimes_{\mathbf{p} \in P(\Delta)^+} |0\rangle^{\mathbf{p}} \right). \quad (4.12)$$

Evidently, one has  $\mathcal{G}_1^{\mathbf{v}} \cdot \mathcal{G} = \mathcal{G}$ . Let us also verify explicitly that this gauging map indeed projects out states transforming non-trivially under the Sierpinski symmetry, i.e.  $\mathcal{G} \cdot \mathcal{U}[\mathcal{S}_q] = \mathcal{G}$ , for all *valid* symmetry generators defined in eq. (4.2). To this end, we could expand the local projectors appearing in the definition of the gauging map  $\mathcal{G}$ :

$$\prod_{\mathbf{v} \in V(\Delta)} \prod_{\{g_{\mathbf{v}}\}_{\mathbf{v}}} (\mathcal{G}_1^{\mathbf{v}})^{g_{\mathbf{v}}}. \quad (4.13)$$

From the definition of  $\mathcal{G}_1^{\mathbf{v}}$  in eq. (4.9) it then follows that three  $\mathcal{G}_1^{\mathbf{v}}$ 's in the expansion (4.13) on vertices  $(i+1, j)$ ,  $(i, j)$  and  $(i, j+1)$  overlap in exactly one plaquette  $\mathbf{p}^+ \in P(\Delta)^+$ , namely the one such that  $(i, j) = \text{bp}(\mathbf{p}^+)$ . Locally, this means that the property  $\mathcal{G} \cdot \mathcal{U}[\mathcal{S}_q] = \mathcal{G}$  is equivalent to  $\mathcal{S}_q$  satisfying (4.4), which it has to in order to define a genuine Sierpinski symmetry generator.

We now turn to the emergent symmetries. As a matter of fact, after gauging we obtain a *dual* Sierpinski fractal symmetry supported on the gauge degrees of freedom. Given a plaquette  $\mathbf{p}^+ \in P(\Delta)^+$ , we define its *rectangular coordinates* as  $(i^{\mathbf{v}}, j^{\mathbf{v}}) := \text{bp}(\mathbf{p}^+)$ . Mimicking the definition of  $\mathcal{S}_q$ , let us define  $\mathcal{S}_q^{\mathbf{v}} : \mathbb{Z}_L \times \mathbb{Z}_L \rightarrow \mathbb{F}_2$  by means of the recursive identity

$$\begin{aligned} \mathcal{S}_q^{\mathbf{v}}(i^{\mathbf{v}}, j^{\mathbf{v}}) &:= \mathcal{S}_q^{\mathbf{v}}(i^{\mathbf{v}}, j^{\mathbf{v}} + 1) \\ &\quad + \mathcal{S}_q^{\mathbf{v}}(i^{\mathbf{v}} - 1, j^{\mathbf{v}} + 1), \end{aligned} \quad (4.14)$$

$$\mathcal{S}_q^{\mathbf{v}}(i^{\mathbf{v}}, 0) := q(i^{\mathbf{v}}), \quad (4.15)$$

for all  $i^{\mathbf{v}}, j^{\mathbf{v}} \in \mathbb{Z}_L$ . By comparison with the definition of  $\mathcal{S}_q$  in (4.4) we can infer that  $\mathcal{S}_q^{\mathbf{v}}$  defines a Sierpinski fractal symmetry in which the 'time steps' labeled by  $j$  flow in the opposite direction compared to the one in (4.4) and in which the 'spatial direction'  $i$  is reflected as well. It can then be shown that  $\mathcal{S}_q^{\mathbf{v}}$  exactly specifies the emergent fractal symmetries after gauging. Namely, defining

$$\mathcal{U}^{\mathbf{v}}[\mathcal{S}_q^{\mathbf{v}}] := \prod_{i^{\mathbf{v}}, j^{\mathbf{v}}} (Z^{(i^{\mathbf{v}}, j^{\mathbf{v}})})^{\mathcal{S}_q^{\mathbf{v}}(i^{\mathbf{v}}, j^{\mathbf{v}})}, \quad (4.16)$$

it follows that

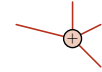
$$\mathcal{U}^{\mathbf{v}}[\mathcal{S}_q^{\mathbf{v}}] \cdot \mathcal{G} = \mathcal{G}. \quad (4.17)$$

Indeed, this follows from the fact that exactly for this choice of  $\mathcal{S}_q^{\mathbf{v}}$  the operator  $\mathcal{U}^{\mathbf{v}}[\mathcal{S}_q^{\mathbf{v}}]$  commutes with every  $\mathcal{G}_1^{\mathbf{v}}$ ,  $\mathcal{U}^{\mathbf{v}}[\mathcal{S}_q^{\mathbf{v}}] \cdot \mathcal{G}_1^{\mathbf{v}} = \mathcal{G}_1^{\mathbf{v}} \cdot \mathcal{U}^{\mathbf{v}}[\mathcal{S}_q^{\mathbf{v}}]$ , together with the trivial observation that the initial flat gauge configuration  $\bigotimes_{\mathbf{p}^+ \in P(\Delta)^+} |0\rangle^{\mathbf{p}^+}$  is a  $+1$  eigenstate of  $\mathcal{U}^{\mathbf{v}}[\mathcal{S}_q^{\mathbf{v}}]$ .

Since we are dealing with the symmetry group  $\mathbb{Z}_2$  here, we can in fact significantly simplify the tensor network calculus introduced in section 1 by removing the arrows and using a **red** color for all tensor legs since we have identified  $\mathbb{C}[\mathbb{Z}_2]$  and  $\mathbb{C}[\mathbb{Z}_2]$  above. By virtue of these simplifications, we will also only need two type of tensors for both gauging maps which are given by:

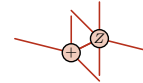

(4.18)

which in the construction of the  $\mathcal{G}$ -PEPO will be placed on every vertex  $\mathbf{v} \in V(\Delta)$  and in the case of the  $\hat{\mathcal{G}}$ -PEPO on every plaquette in  $P(\Delta)^+$ , and


(4.19)

placed on the plaquettes  $P(\Delta)^+$  when it appears in the PEPO representation of  $\mathcal{G}$  and on the vertices  $\mathbf{v} \in V(\Delta)$  when it appears in the  $\hat{\mathcal{G}}$ -PEPO.

For the PEPO representations of both gauging maps we make a choice of unit cell such that a plaquette tensor on a given  $\mathbf{p}^+ \in P(\Delta)^+$  is combined with a vertex tensor placed on the corresponding base point  $\text{bp}(\mathbf{p}^+)$ . For the gauging map  $\mathcal{G}$ , the PEPO unit cell is then explicitly given by:


(4.20)

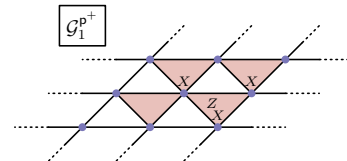
where, as mentioned above, the  $+$ -type tensor is placed on a certain plaquette  $\mathbf{p}^+ \in P(\Delta)^+$  and the  $Z$ -type tensor on the corresponding base point  $\text{bp}(\mathbf{p}^+)$ .

## 4.2 Iterative gauging of Sierpinski fractal symmetry

Gauging of the dual Sierpinski symmetry happens via the introduction of qubits on the vertices  $V(\Delta)$ . In the neighborhood of a plaquette  $\mathbf{p}^+ \in P(\Delta)^+$ , following gauge transformation is defined:

$$\mathcal{G}_1^{\mathbf{p}^+} := Z^{\mathbf{p}^+} \bigotimes_{\mathbf{v} \in V(\Delta)|_{\mathbf{p}^+}} X^{\mathbf{v}}, \quad (4.21)$$

which forms a local  $\mathbb{Z}_2$ -representation. In this,  $V(\Delta)|_{\mathbf{p}^+}$  stands for the set of three vertices surrounding  $\mathbf{p}^+$ . Concretely:


(4.22)

Based on the local projector, defined as

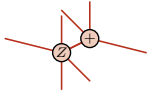
$$\mathcal{P}^{\mathbf{p}^+} := \frac{1}{2} (\text{id} + \mathcal{G}_1^{\mathbf{p}^+}), \quad (4.23)$$

the corresponding gauging map reads:

$$\hat{\mathcal{G}} := \mathcal{P} \left( \bigotimes_{\mathbf{v} \in \mathbf{V}(\Delta)} |0\rangle^{\mathbf{v}} \right), \quad (4.24)$$

in which  $\mathcal{P} := \prod_{\mathbf{p}^+ \in \mathbf{P}(\Delta)^+} \mathcal{P}^{\mathbf{p}^+}$ . The expected properties  $\mathbb{U}[\mathcal{S}_q] \cdot \hat{\mathcal{G}} = \hat{\mathcal{G}}$  and  $\hat{\mathcal{G}} \cdot \mathbb{U}^{\mathbf{V}}[\mathcal{S}_q^{\mathbf{V}}] = \hat{\mathcal{G}}$ , for all valid symmetry generators  $\mathbb{U}[\mathcal{S}_q]$  and  $\mathbb{U}^{\mathbf{V}}[\mathcal{S}_q^{\mathbf{V}}]$ , hold by construction.

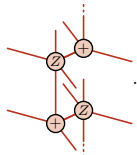
Under the aforementioned conventions, the unit cell of the PEPO representation of  $\hat{\mathcal{G}}$  can be depicted as



$$, \quad (4.25)$$

where now the Z-type tensor is placed on a plaquette  $\mathbf{p}^+ \in \mathbf{P}(\Delta)^+$  and the second tensor on the corresponding base point.

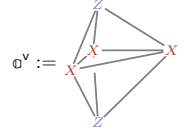
What is the emergent state obtained from these gauging procedures? It turns out that the emergent state can be interpreted as living on the vertices of a *hexagonal close-packed* (HCP) lattice, which consists of layers of triangular lattices whose vertices are slightly displaced with respect to each other [37]. A slightly different realization of this lattice which is closer to our setup is the following. We consider also layers of two-dimensional triangular lattices  $\Delta$ , but in such a way that the vertices of all layers align along the direction perpendicular to the stack of layers. In every other layer we then place qubits on the vertices of the triangular lattice, i.e. on  $\mathbf{V}(\Delta)$ , and on the plaquettes  $\mathbf{P}(\Delta)^+$ . The PEPO representation of the emergent state for one unit cell is then given by:



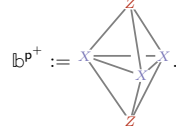
$$. \quad (4.26)$$

Using a blue color for the degrees of freedom on  $\mathbf{V}(\Delta)$  and a red color for those on the plaquettes  $\mathbf{P}(\Delta)^+$ , the bulk terms of the emergent Hamiltonian are then

the following:



$$, \quad (4.27)$$



$$. \quad (4.28)$$

With slight abuse of notation,  $\mathbf{v}$  refers to the vertex of the emergent lattice on which the bottom blue qubit appearing in the  $\mathbf{e}$  operator is supported. On the other hand,  $\mathbf{p}^+$  refers to the support of the red qubit on the bottom of the depiction of the  $\mathbf{b}^{\mathbf{p}^+}$  operator. It is clear that these operators all mutually commute. We can construct projectors out of these operators as

$$\mathbb{P}^{\mathbf{v}} := \frac{1}{2} (\text{id} + \mathbf{e}^{\mathbf{v}}), \quad (4.29)$$

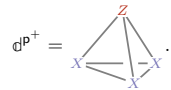
$$\mathbb{P}^{\mathbf{p}^+} := \frac{1}{2} (\text{id} + \mathbf{b}^{\mathbf{p}^+}). \quad (4.30)$$

The emergent bulk Hamiltonian then reads

$$\mathbb{H}^{\text{blk}} := - \sum_{\mathbf{v} \in \mathbf{V}} \mathbb{P}^{\mathbf{v}} - \sum_{\mathbf{p}^+ \in \mathbf{P}^+} \mathbb{P}^{\mathbf{p}^+}, \quad (4.31)$$

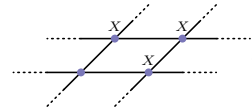
where the sums are over all (bulk) vertices and  $\mathbf{p}^+$ -plaquettes of the HCP lattice. This model was proposed by Caselnovo and Chamon in [18], is covered in the family of models by Yoshida [38] and was recently studied in detail in [37].

If we start the iterative gauging by  $\mathcal{G}$ , the possible boundary terms correspond to the truncation of (4.27) near the boundary or individual Z terms. Explicitly the truncated (4.27) terms read:



$$. \quad (4.32)$$

which also have been identified and studied in [37]. The term  $\mathbf{e}^{\mathbf{p}^+}$  acts on the boundary state  $|\psi\rangle$ , which is symmetric under the fractal Sierpinski symmetry, as:



$$, \quad (4.33)$$

which can be recognized as the local checks  $X^{\mathbf{p}^+}$  and serve as a local order parameter for the initial fractal symmetry. As such, there are two possible boundaries depending on whether the fractal symmetry is spontaneously broken by  $|\psi\rangle$  or not. In the first case, the

boundary stabilizers are given by

$$\mathbb{Q}^{p^+} := \frac{1}{2} \left( \text{id} + X^{p^+} \right), \quad (4.34)$$

for all  $p^+ \in P(\Delta)^+$  on the boundary  $\Delta$ , whereas in the symmetry breaking case the stabilizers are given by

$$\mathbb{Q}^{p^+} := \frac{1}{2} \left( \text{id} + Z^v \right). \quad (4.35)$$

## 5 Conclusion and outlook

In this work we have put the iterative gauging paradigm of [12] for abelian symmetries on a systematic footing and extended it to arbitrary dimensions and generic types of abelian symmetries. We provided explicit examples where the emergent states are three-dimensional. Concretely we have shown the emergence of surface codes, foliated type-I fracton codes and non-foliated type-I fracton codes from the iterative gauging of 0-form, 1-form, linear subsystem and Sierpinski fractal symmetry respectively.

The method proposed in this work resembles the *sandwich* construction of SymTFT. In particular, we constructed states of the following form:

$$\langle \hat{\psi} | \mathcal{G} \circ \hat{\mathcal{G}} \circ \cdots \circ \mathcal{G} \circ \hat{\mathcal{G}} \circ \mathcal{G} | \psi \rangle,$$

made up of the gauging operator  $\mathcal{G}$  of a  $G$  symmetry, its dual  $\hat{\mathcal{G}}$  and states  $|\psi\rangle, |\hat{\psi}\rangle$  symmetric under the initial abelian symmetry and its dual. For the case where the boundary supports an abelian 1-form symmetry, our method exactly recovers the quantum double in correspondence with the SymTFT proposal. The symmetries of this sandwich state are considered as stabilizer local Hamiltonian terms giving rise to the Hamiltonian:

$$\mathbb{H}^{\text{blk}} + \mathbb{H}^{\text{bdry}},$$

where the bulk terms only depend on the symmetry  $G$  and the boundary terms depend on the quantum phase of  $|\psi\rangle$ .

We predict a number of ways in which this framework can be further refined and propose the following open problems. From the case of the emergent surface codes, it is clear that further refinement is required to accommodate boundary conditions hosting SPT phases on top of the partial symmetry breaking. It is plausible that akin to the lower-dimensional case [12] this requires modification of the boundary stabilizers. A similar question arises regarding the boundary conditions of the emergent anisotropic models studied in section 3. In particular, it would be revealing to study boundaries that (potentially partially) break translation symmetry. It is plausible to expect that by including boundary states which realize different symmetry breaking patterns in distinct regions, we could realize domain walls in the emergent orders such as the electromagnetic domain wall

in the three-dimensional toric code [22]. One could also wonder to what extent our framework is exhaustive in producing classes stabilizer codes.

The main assumption of this work is the symmetry group being abelian, which results in abelian higher-dimensional codes. As such we leave the study of non-abelian, and more general categorical, symmetries in the iterative gauging framework for future work. As a first step, we will generalize the one-dimensional gauging map of [13] to generalized symmetries described by unitary fusion categories in forthcoming work [39]. An obvious family of models not treated in this work are type-II fracton models, which differ from type-I fracton models in the mobility of their excitations. Adapting our approach to accommodate these fracton orders could prove to be a splendid challenge. We further foresee that the explicit tensor network constructions of the states provided here will enhance their numerical study. Having direct access to the boundary states in this framework could facilitate the study of boundary phase transitions of these emergent models.

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# Twisted gauging and topological sectors in (2+1)d abelian lattice gauge theories

## 10.1 Synopsis

An open problem in the study of three-dimensional quantum field theories is the construction of genuinely *self-dual* models — that is, models that are mapped to themselves under some duality transformation. Necessarily, such a self-duality should preserve the symmetry structure of either model. A central observation of ref. [162] is that a specific class of gauging and twisted gauging transformations appears to be the only mechanism capable of producing such self-dual theories in this dimension.

The duality transformations we study in this project intertwine (2+1)-dimensional *twisted Abelian lattice gauge theories* whose symmetry structure is encoded by the fusion 2-category  $2\text{Rep}(G)$ , which simultaneously captures Wilson line operators as 1-form symmetry operators, their condensation defects as 0-form symmetry operators, as well as topological interfaces between those membrane operators. The duality transformation amounts to the change in module 2-category  $2\text{Vec} \mapsto 2\text{Vec}^\alpha$  over the input  $2\text{Vec}_G$ , with  $[\alpha] \in H^3(G, \text{U}(1))$  encoding the aforementioned twist or discrete torsion. Physically, this map amounts to the gauging of the 1-form symmetry of the theory, followed by a twisted gauging of the resulting 0-form  $2\text{Vec}_G$  symmetry. We provide lattice representations of these symmetry operators and duality intertwiners in the form of topological matrix product operators and projected entangled-pair operators. We characterise the 1-form symmetry-

twisted boundary conditions and charge sectors, and construct the unitary duality transformations that relate the dual theories to one another.

We first illustrate the mechanism in one spatial dimension, where the construction for the case of  $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ , with the non-trivial twist  $[\psi]$  valued in  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1)) \simeq \mathbb{Z}_2$ , reduces to the *Kennedy-Tasaki duality* [123] and which serves as an introduction to the higher-dimensional case. In two spatial dimensions, we work out an explicit realisation of the duality between the *toric code* and the *double semion* model corresponding to the string-net model with  $\text{Vec}_{\mathbb{Z}_2}$ , respectively  $\text{Vec}_{\mathbb{Z}_2}^\alpha$ , as input,  $[\alpha]$  being the non-trivial class in  $H^3(\mathbb{Z}_2, \text{U}(1)) \simeq \mathbb{Z}_2$  [66, 71, 163]. Finally, we show that the sum of these two models admits an interpretation as a *higher gauge theory*, in which the duality transformation is promoted to a genuine internal symmetry.

## 10.2 Reference

*Twisted gauging and topological sectors in (2+1)d abelian lattice gauge theories*  
**B. Vancraeynest-De Cuiper**, C. Delcamp  
SciPost Phys. **19**, 054 (2025), arXiv:2501.16301

## 10.3 Contributions of the author

This project was mainly conceived by Clement Delcamp, who developed the overarching conceptual framework based on earlier work [141, 150] and wrote the majority of the manuscript. My contributions focused primarily on the derivations related to the one-dimensional dualities, with additional contributions to parts of the two-dimensional framework. I also co-wrote parts of the paper.

# Twisted gauging and topological sectors in (2+1)d Abelian lattice gauge theories

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## Abstract

Given a two-dimensional quantum lattice model with an Abelian gauge theory interpretation, we investigate a duality operation that amounts to gauging its invertible 1-form symmetry, followed by gauging the resulting 0-form symmetry in a twisted way via a choice of discrete torsion. Using tensor networks, we introduce explicit lattice realisations of the so-called condensation defects, which are obtained by gauging the 1-form symmetry along submanifolds of spacetime, and employ the same calculus to realise the duality operators. By leveraging these tensor network operators, we compute the non-trivial interplay between symmetry-twisted boundary conditions and charge sectors under the duality operation, enabling us to construct isometries relating the dual Hamiltonians. Whenever a lattice gauge theory is left invariant under the duality operation, we explore the possibility of promoting the self-duality to an internal symmetry. We argue that this results in a symmetry structure that encodes the 2-representations of a 2-group.



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## 1 Introduction

Given a quantum theory, a modern viewpoint identifies the existence of a collection of *topological defects* as the definition of a global *symmetry* [40]. Aside from producing notions of symmetry that go beyond the ordinary one—and its axiomatisation in terms of abstract *groups*—this viewpoint has led to a calculus of symmetries that takes full advantage of well-established methods from topological quantum field theory [35]. In particular, this calculus greatly facilitates the *gauging* of finite symmetries and the study of the consequences of such an operation, both in the continuum [1, 14, 18, 23, 39, 44] and in the discrete [6, 27, 48, 59].

In this manuscript, we are interested in two-dimensional quantum lattice models that result from gauging an ordinary finite Abelian 0-form symmetry. These are defined on a constrained Hilbert space of states satisfying magnetic *Gauß constraints*. As such, these can be interpreted as *Abelian lattice gauge theories*. It is well established that theories obtained in this way can be coupled to topological lines labelled by the *Pontrjagin dual* of the initial Abelian group, which define a 1-form symmetry. But, it was recently understood that in addition to these topological lines, such theories also host topological surfaces referred to as *condensation defects* [56, 65], which have the peculiarity of being typically non-invertible as linear maps on the constrained Hilbert space [16, 19, 27]. The resulting complete symmetry structure is axiomatised in terms of a higher mathematical structure encoding the so-called *2-representations* of the group [4, 5, 13, 22, 27].

Given a (2+1)d Abelian lattice gauge theory, we consider an operation that amounts to ungauging its original 0-form symmetry, before gauging it back in a twisted way via a choice of *discrete torsion*, which is classified by the third cohomology of the group. The (untwisted) gauging operation alone is intimately related to the celebrated *Kramers–Wannier* transformation [47, 75], which is known to be a *non-invertible* operation on a fixed Hilbert space. Indeed, assuming closed periodic boundary conditions, gauging projects both theories onto the *singlet* sectors of their respective symmetries. Therefore, turning the Kramers–Wannier transformation into a duality requires addressing the non-trivial interplay between closed boundary conditions and charge sectors. An isometry acting on the resulting total Hilbert space that relate the spectra of the dual models can then be found by promoting the choice of boundary condition to a dynamical degree of freedom [49, 57, 59, 70]. Calculating the mapping of topological sectors under the combination of untwisted and twisted gaugings described above is the main objective of this work.

One merit of the dualities investigated in this manuscript is the ability to define them in any spacetime dimension. More precisely, one can apply the same strategy to any quantum theory that results from gauging the finite invertible 0-form symmetry of another theory. For

instance, it was demonstrated in ref. [48] that in  $(1+1)d$  the so-called *Kennedy–Tasaki* transformation was precisely of this type [45, 63]. Another attractive trait of these dualities is that they preserve the symmetry structure. This is especially relevant in  $(2+1)d$  where most gauging procedures are found to modify the symmetry structure. As a matter of fact, it was conjectured in ref. [27] that most non-trivial dualities in  $(2+1)d$  sharing this feature are of the type investigated in this manuscript, later corroborated in ref. [28]. Since the symmetry structure is preserved under the duality, one can ask for models that are left invariant under the transformation. Whenever this is the case, one can further ask to promote the resulting self-duality to an internal symmetry, which mathematically amounts to performing a *group extension* of the corresponding symmetry structure [30, 33, 73]. In  $(2+1)d$ , we argue that this results into the emergence of lattice *higher gauge theories* [8, 9, 11, 25, 26, 46, 77] whose symmetry operators are organised into higher mathematical structures encoding the 2-representations of 2-groups.

Concretely, given a finite symmetry acting on a one-dimensional quantum lattice model, a framework was developed in ref. [48] in order to systematically gauge any of its subsymmetries, effectively realising the approach of ref. [14, 34] in the discrete. This framework relies on a generalisation of the *anyonic chain* construction [10, 36, 37], which, in addition to clearly distinguishing an abstract symmetry from its explicit realisation in a physical system, makes *tube algebra* techniques [61, 62] amenable to the computation of *topological sectors* [2, 49, 53]. In this formulation, the action of a finite symmetry, which is encoded into a so-called *fusion category* [33], requires in particular a choice of *module category* [31]. Two physical systems that only differ in such a choice of module category are related by some gauging procedure. Then, the type of duality considered in the present manuscript relates two physical systems that differ in a choice of module category that is equivalent to the category of complex vector spaces. A generalisation of this framework to  $(2+1)d$  was initiated in ref. [27]. In this higher-dimensional context, the action of a finite symmetry, which is now encoded into a so-called *fusion 2-category* [24, 38], requires in particular a choice of *module 2-category* [21, 22, 28, 29]. The type of duality we are interested in then relates two Abelian lattice gauge theories that only differ in a choice of module 2-category that is equivalent to the 2-category of 2-vector spaces. A general strategy to realise on the lattice the various symmetry operators evoked above was presented in ref. [27, 42]. Following ref. [49], the same strategy can be employed to compute the lattice operators that transmute the dual Hamiltonians onto one another. Here, we realise these various operators very explicitly in the form of *tensor network operators* using the calculus outlined in ref. [22], partially extending to higher dimensions the results of ref. [3, 20, 51, 66, 74]. Using these operators, we are then able to compute the mapping of topological sectors under the dualities. In Table 1, we have summarised the symmetry structures, dualities and permutation of the topological sectors under scrutiny in this manuscript, as a guide to the reader.

**Organisation of the manuscript:** We begin by reviewing in sec. 2 the one-dimensional scenario. After introducing a tensor network operator implementing the duality that amounts to the gauging of the Abelian invertible symmetry followed by the twisted gauging of the dual symmetry, we derive the interplay between symmetry-twisted boundary conditions and charge sectors. This section is concluded by exploring concrete examples. In sec. 3, we begin by introducing families of two-dimensional quantum lattice models with a lattice gauge theory interpretation. We then unpack the symmetry encoded into the fusion 2-category of 2-representations of the gauge group, realising in particular the condensation defects in the form of tensor networks. We then employ the same calculus to realise the duality operator, which we leverage to compute the permutation of topological sectors under the duality. Finally, we discuss lifting the self-duality to a genuine internal symmetry through the lens of a simple theory built from the so-called toric code and the double semion model.

Table 1: Summary of the main results. We consider a duality operation in (1+1)d and (2+1)d that amounts to gauging a 0-form and 1-form  $\text{Rep}(G)$  symmetry, respectively, followed by a twisted gauging of the resulting dual invertible 0-form symmetry. Both in (1+1)d and (2+1)d, the symmetry structures are the same on both sides of the duality.

|                     | Sec. 2 — (1+1)d                                  | Sec. 3 — (2+1)d                                                    |
|---------------------|--------------------------------------------------|--------------------------------------------------------------------|
| Discrete torsion    | $[\psi] \in H^2(G, \text{U}(1))$                 | $[\lambda] \in H^3(G, \text{U}(1))$                                |
| Symmetry structure  | $\text{Rep}(G) \simeq \text{Mod}(\mathbb{C}[G])$ | $2\text{Rep}(G) \simeq \text{Mod}(\text{Vec}_G)$                   |
| Duality operators   | $\pi \in \text{Rep}^\psi(G)$                     | $\mathcal{M}(A, \phi) \in 2\text{Rep}^\lambda(G)$                  |
| Topological sectors | $(g, \eta) \in G \times \widehat{G}$             | $(g_1, g_2, \eta) \in G \times G \times \widehat{G}$               |
| Permutation sectors | $(g, \eta) \mapsto (g, t_g(\psi)\eta)$           | $(g_1, g_2, \eta) \mapsto (g_1, g_2, t_{g_1, g_2}^2(\lambda)\eta)$ |

## 2 Twisted gauging of (1+1)d spin chains

Given an arbitrary one-dimensional quantum lattice model with an Abelian invertible symmetry, we review the non-trivial interplay between closed symmetry-twisted boundary conditions and charge sectors under dualities that amount to the (untwisted) gauging of the symmetry, followed by the twisted gauging of the dual symmetry. Given pairs of compatible boundary conditions and charge sectors, the unitary operators relating the spectra of dual Hamiltonians are explicitly constructed in the form of tensor network operators.

### 2.1 Symmetric Hamiltonians and closed symmetry-twisted boundary conditions

Let  $G$  be a finite Abelian group  $G$  and  $\psi$  a normalised representative 2-cocycle of a cohomology class  $[\psi]$  in  $H^2(G, \text{U}(1))$ . We begin by constructing a family of Hamiltonians on a closed one-dimensional lattice that commutes with symmetry operators labelled by characters in the Pontrjagin dual  $\widehat{G} = \text{Hom}(G, \text{U}(1))$  of  $G$ . Given a closed one-dimensional lattice, we denote the sets of vertices and oriented edges by  $V$  and  $E$ , respectively. Notice that  $|V| = |E|$ . Choosing a total ordering of the vertices induces an orientation of the edges. For any edge  $e \in E$  we write its source and target vertex as  $\partial_- e$  and  $\partial_+ e$ , respectively, so that we identify  $e \equiv (\partial_- e, \partial_+ e)$ . Throughout this section, we assume without loss of generality that all the edges share the same orientation. To every edge, we assign a copy of  $\mathbb{C}[G] = \text{Span}_{\mathbb{C}}\{|g\rangle \mid g \in G\}$ , with  $\langle g_1 | g_2 \rangle = \delta_{g_1, g_2}$ , so that the microscopic Hilbert space of the model is given by  $\bigotimes_{e \in E} \mathbb{C}[G]$ . Given a function  $\mathfrak{g} : E \rightarrow G$ , we notate via  $|g\rangle = \bigotimes_{e \in E} |g(e)\rangle$  the function  $\mathfrak{g}$  regarded as a basis element of the microscopic Hilbert space. Moreover, for every vertex  $v \in V$ , we introduce a function  $\mathfrak{r}_v : V \rightarrow G$  such that  $\mathfrak{r}_v(v_1) = \mathbb{1}$ , whenever  $v_1 \neq v$ . The set of such functions is naturally isomorphic to  $G$ .<sup>1</sup> Finally, to every  $v \in V$ , we assign the local operators

$$\mathbb{h}_{v,n}^\psi := \sum_{\mathfrak{g}, \mathfrak{r}_v} h_n(\mathfrak{g}, \mathfrak{r}_v) \psi(\mathfrak{g}, \mathfrak{r}_v) |\mathfrak{g} \mathfrak{d}\mathfrak{r}_v\rangle \langle \mathfrak{g}|, \quad (1)$$

wherein  $n$  is some label,  $h_n(\mathfrak{g}, \mathfrak{r}_v)$  are complex coefficients,  $\mathfrak{g} \mathfrak{d}\mathfrak{r}_v : E \rightarrow G$  is the function such that

$$(\mathfrak{g} \mathfrak{d}\mathfrak{r}_v)(e) := \begin{cases} \mathfrak{r}_v(v) \mathfrak{g}(e), & \text{if } e \equiv (v, v_2), \\ \mathfrak{g}(e) \mathfrak{r}_v(v)^{-1}, & \text{if } e \equiv (v_1, v), \\ \mathfrak{g}(e), & \text{otherwise,} \end{cases} \quad (2)$$

<sup>1</sup>Note that by summing over  $\mathfrak{r}_v$ , we are summing over all possible values in  $G$  such a function can assign to  $v$ . Although this notation may seem excessive at this point, we introduce it here in preparation for the higher-dimensional study.



and  $\psi(\mathfrak{g}, \mathfrak{r}_v)$  is the phase factor

$$\psi(\mathfrak{g}, \mathfrak{r}_v) := \frac{\psi(\mathfrak{g}(v_1 v) \mathfrak{r}_v(v)^{-1}, \mathfrak{r}_v(v))}{\psi(\mathfrak{r}_v(v), \mathfrak{g}(v v_2))}, \quad (3)$$

where  $v_1 < v < v_2$ , so that  $(v_1 v)$  and  $(v v_2)$  are the edges whose target and source vertices are  $v$ , respectively.

For any possible choice of complex coefficients  $h_n(\mathfrak{g}, \mathfrak{r}_v)$ , for every  $\mathfrak{g} : E \rightarrow G$  and  $\mathfrak{r}_v : V \rightarrow G$ , and cohomology class  $[\psi]$ , it follows from the defining property of group characters that local operators (1) commute with (0-form) symmetry operators of the form

$$\mathbb{U}^\chi := \sum_{\mathfrak{g}} \left( \prod_{e \in E} \chi(\mathfrak{g}(e)) \right) |\mathfrak{g}\rangle \langle \mathfrak{g}|, \quad (4)$$

for any  $\chi \in \widehat{G}$ . From  $(\chi_1 \otimes \chi_2)(\mathfrak{g}) = \chi_1(\mathfrak{g})\chi_2(\mathfrak{g})$  we readily infer that the symmetry operators compose according to  $\mathbb{U}^{\chi_1} \circ \mathbb{U}^{\chi_2} = \mathbb{U}^{\chi_1 \otimes \chi_2}$ . Although it is somewhat superfluous for an Abelian group, we can state that symmetry operators are organised into the fusion (1-)category  $\text{Rep}(G)$  of (finite-dimensional complex) representations of  $G$ . Finally, we construct arbitrary  $\widehat{G}$ -symmetric models as  $\mathbb{H}^\psi = \sum_{v \in V} \sum_n \mathbb{H}_{v,n}^\psi$ . Notice that we restricted to local operators acting on nearest neighbours for simplicity but nothing prevents us from defining local operators acting on additional sites.

Up until this point we have implicitly assumed periodic boundary conditions. Symmetry-twisted closed boundary conditions are then obtained by inserting symmetry twists, which—in contrast to symmetry operators that extend over the whole space—are localised at one spatial point and extend in the time direction. The resulting symmetry-twisted boundary conditions preserve the translation invariance of the system, in the sense that there is an isomorphism between Hilbert spaces obtained by moving the defect by one site [2, 49, 53, 67]. More specifically, we promote symmetry-twisted boundary conditions to genuine physical degrees of freedom by tensoring the microscopic Hilbert space  $\text{Span}_{\mathbb{C}}\{|\mathfrak{g}\rangle : \mathfrak{g} : E \rightarrow G\}$  with a copy of  $\mathbb{C}[\widehat{G}]$  associated with a vertex conventionally denoted by  $v_0$ . Fixing the boundary condition to be that labelled by  $\eta \in \widehat{G}$  then amounts to setting the  $\mathbb{C}[\widehat{G}]$  degree of freedom to  $|\eta\rangle$ , and we denote the resulting kinematical Hilbert space by  $\mathcal{H}^\eta$ . Local operators across the symmetry defect labelled by  $\eta \in \widehat{G}$  are acted upon by the symmetry action as follows:

$$\mathbb{H}_{v_0,n}^{\psi,\eta} := \sum_{\mathfrak{g}, \mathfrak{r}_{v_0}} h_n(\mathfrak{g}, \mathfrak{r}_{v_0}) \psi(\mathfrak{g}, \mathfrak{r}_{v_0}) \eta(\mathfrak{r}_{v_0}) |\mathfrak{g} \mathfrak{d} \mathfrak{r}_{v_0}\rangle \langle \mathfrak{g}|, \quad (5)$$

with  $\eta(\mathfrak{r}_{v_0}) \equiv \eta(\mathfrak{r}_{v_0}(v_0))$ . The Hamiltonian in the presence of this symmetry twist hence reads

$$\mathbb{H}^{\psi,\eta} := \sum_{\substack{v \in V \\ v \neq v_0}} \sum_n \mathbb{H}_{v,n}^\psi + \sum_n \mathbb{H}_{v_0,n}^{\psi,\eta}. \quad (6)$$

In order to accommodate the presence of a symmetry twist labelled by  $\eta \in \widehat{G}$ , the symmetry operators (4) need to be modified as follows:

$$\mathbb{U}_\eta^\chi := \sum_{\mathfrak{g}} \left( \prod_{e \in E} \chi(\mathfrak{g}(e)) \right) |\mathfrak{g}\rangle \langle \mathfrak{g}| \otimes |\eta\rangle \langle \eta|. \quad (7)$$

Again, it readily follows from the defining property of group characters that  $[\mathbb{H}_{v_0,n}^{\psi,\eta}, \mathbb{U}_\eta^\chi] = 0$ . Given a choice of symmetry-twisted closed boundary condition  $\eta \in \widehat{G}$ , it follows from the  $\widehat{G}$ -symmetry that a Hamiltonian  $\mathbb{H}^{\psi,\eta}$  decomposes into charge sectors labelled by group elements

$g \in G$  corresponding to the holonomy of  $\mathfrak{g}$  around the spatial manifold. Pairs  $(g, \eta) \in G \times \widehat{G}$  label the (simple) *topological sectors* of the theory that decompose the action of the algebra of operators (7) onto the microscopic Hilbert space. The projector onto the topological sector  $(g, \eta)$  is then given by  $\sum_{\chi \in \widehat{G}} \chi^\vee(g) \mathbb{U}_\eta^\chi$ , where  $\chi^\vee$  denotes the complex conjugate character of  $\chi$  and where we are using the fact

$$\sum_{\chi \in \widehat{G}} \chi^\vee(g) \prod_{e \in E} \chi(\mathfrak{g}(e)) = \sum_{\chi \in \widehat{G}} \chi^\vee(g) \chi(\text{hol}(\mathfrak{g})) = \delta_{g, \text{hol}(\mathfrak{g})}, \quad (8)$$

with  $\text{hol}(\mathfrak{g}) := \prod_{e \in E} \mathfrak{g}(e)$ .

## 2.2 Duality operators

Given a finite Abelian group  $G$ , we constructed in sec. 2.1 families of one-dimensional quantum lattice models with  $\text{Rep}(G)$  symmetry, which are parameterised by normalised representatives  $\psi$  of a cohomology class  $[\psi]$  in  $H^2(G, \text{U}(1))$  as well as sets of complex coefficients  $\{h_n\}_n$ . We claim that models constructed in this way that only differ in the choice of cohomology class  $[\psi]$  are dual to one another. One way to confirm this statement would be to verify that (von Neumann) algebras of local operators generated by the Hamiltonian terms are isomorphic, which follows from the 2-cocycle condition of  $\psi$ . Another possibility is to explicitly compute the lattice operators transmuting Hamiltonians associated with distinct choices of  $[\psi]$  onto one another. Without loss of generality, let us focus on the relation between Hamiltonian models  $\mathbb{H}^\psi$  and  $\mathbb{H}$  with and without non-trivial 2-cocycle. We know from the results of ref. [48] that operators transmuting local symmetric operators of two Hamiltonians that only differ in such a choice of 2-cocycle  $\psi$  onto one another are labelled by projective representations of  $G$  with Schur's multiplier  $\psi$ . Formally, this is the statement that these operators are encoded into the category  $\text{Rep}^\psi(G)$  of  $\psi$ -projective representations of  $G$ .

Let us verify this statement explicitly. To this end, let  $\pi$  denote the representative of an isomorphism class of simple objects in  $\text{Rep}^\psi(G)$ . The operator transmuting the Hamiltonian  $\mathbb{H}$  into  $\mathbb{H}^\psi$  with periodic boundary conditions explicitly reads

$$\mathbb{D}^\pi := \sum_{\mathfrak{g}} \text{tr} \left( \prod_{e \in E} \pi(\mathfrak{g}(e)) \right) |\mathfrak{g}\rangle \langle \mathfrak{g}|, \quad (9)$$

where the product over edges  $e \in E$  is ordered. The operator  $\mathbb{D}^\pi$  acts diagonally in the computational basis. We claim that  $\mathbb{D}^\pi$  transmutes  $\mathbb{H}$  into  $\mathbb{H}^\psi$ , i.e.,  $\mathbb{D}^\pi \circ \mathbb{H} = \mathbb{H}^\psi \circ \mathbb{D}^\pi$ . Indeed letting  $v \in V$  and  $\mathfrak{r}_v : V \rightarrow G$  such that  $\mathfrak{r}_v(v_1) = \mathbb{1}$ , whenever  $v_1 \neq v$ , it follows from the definition of a projective representation that

$$\begin{aligned} \pi((\mathfrak{g} \, d\mathfrak{r}_v)(v_1 v)) &= \pi(\mathfrak{g}(v_1 v) \mathfrak{r}_v(v)^{-1}) = \frac{1}{\psi(\mathfrak{g}(v_1 v), \mathfrak{r}_v(v)^{-1})} \pi(\mathfrak{g}(v_1 v)) \pi(\mathfrak{r}_v(v)^{-1}), \\ \pi((\mathfrak{g} \, d\mathfrak{r}_v)(v v_2)) &= \pi(\mathfrak{r}_v(v) \mathfrak{g}(v v_2)) = \frac{1}{\psi(\mathfrak{r}_v(v), \mathfrak{g}(v v_2))} \pi(\mathfrak{r}_v(v)) \pi(\mathfrak{g}(v v_2)). \end{aligned} \quad (10)$$

Bringing everything together, one obtains

$$\prod_{(v_1 v_2)} \pi((\mathfrak{g} \, d\mathfrak{r}_v)(v_1 v_2)) = \psi(\mathfrak{g}, \mathfrak{r}_v) \cdot \prod_{(v_1 v_2)} \pi(\mathfrak{g}(v_1 v_2)), \quad (11)$$

where  $\psi(\mathfrak{g}, \mathfrak{r}_v)$  was defined in eq. (3). This guarantees that  $\mathbb{D}^\pi \circ \mathbb{H} = \mathbb{H}^\psi \circ \mathbb{D}^\pi$ , as desired. Notice how this operation only depends on  $\psi$  and not the specific choice of simple object in  $\text{Rep}^\psi(G)$ .

Before proceeding further, let us introduce a tensor network representation of the duality operator (9). Given the tensor

$$\begin{array}{c} \uparrow \\ \boxed{\pi} \\ \downarrow \end{array} \begin{array}{c} \leftarrow \\ \boxed{\pi} \\ \rightarrow \end{array} \equiv \sum_{g_1, g_2 \in G} \sum_{i_1, i_2=1}^{\dim_{\mathbb{C}} \pi} i_1 \begin{array}{c} \uparrow \\ \boxed{\pi} \\ \downarrow \end{array} \begin{array}{c} \leftarrow \\ \boxed{\pi} \\ \rightarrow \end{array} i_2 \begin{array}{c} \uparrow \\ g_2 \\ \downarrow \end{array} \begin{array}{c} \leftarrow \\ g_1 \\ \rightarrow \end{array} \equiv \sum_{g \in G} \sum_{i_1, i_2} \pi(g)_{i_1 i_2} |g\rangle \langle g| \otimes |i_1\rangle \langle i_2|, \quad (12)$$

one can realise the duality operator  $\mathbb{D}^\pi$  with periodic boundary conditions as a tensor network of the form

$$\mathbb{D}^\pi \equiv \begin{array}{c} \uparrow \\ \boxed{\pi} \\ \downarrow \end{array} \begin{array}{c} \leftarrow \\ \boxed{\pi} \\ \rightarrow \end{array} \begin{array}{c} \uparrow \\ \boxed{\pi} \\ \downarrow \end{array} \begin{array}{c} \leftarrow \\ \boxed{\pi} \\ \rightarrow \end{array} \begin{array}{c} \uparrow \\ \boxed{\pi} \\ \downarrow \end{array}, \quad (13)$$

where opposite indices in the horizontal direction are contracted so as to implement periodic boundary conditions. Restricting to the charge sector labelled by  $g \in G$ , it follows from the defining property of the projective representation  $\pi$  that the duality operator explicitly depends on the complex number  $\text{tr } \pi(g) = \text{tr } \pi(\text{hol}(g))$ . Indeed, repeatedly using the defining property of  $\psi$ -projective representation  $\pi$  one finds

$$\mathbb{D}^\pi = \sum_{\mathfrak{g}} \psi(\mathfrak{g}(v_1 v_2), \mathfrak{g}(v_2 v_3)) \psi(\mathfrak{g}(v_1 v_2) \mathfrak{g}(v_2 v_3), \mathfrak{g}(v_3 v_4)) \cdots \psi(\mathfrak{g}(v_1 v_2) \mathfrak{g}(v_2 v_3) \cdots \mathfrak{g}(v_{|V|-1} v_{|V|}), \mathfrak{g}(v_{|V|} v_1)) \text{tr } \pi(\text{hol}(g)) |g\rangle \langle g|, \quad (14)$$

where  $v_1 < v_2 < \cdots < v_{|V|}$ . Graphically, this follows from

$$\begin{array}{c} \uparrow \\ \boxed{\pi} \\ \downarrow \end{array} \begin{array}{c} \leftarrow \\ \boxed{\pi} \\ \rightarrow \end{array} \begin{array}{c} \uparrow \\ \boxed{\pi} \\ \downarrow \end{array} = \sum_{g_1, g_2 \in G} \begin{array}{c} \begin{array}{c} \uparrow \\ \psi \\ \downarrow \end{array} \begin{array}{c} \leftarrow \\ \boxed{\pi} \\ \rightarrow \end{array} \begin{array}{c} \uparrow \\ \boxed{1} \\ \downarrow \end{array} \end{array}, \quad (15)$$

where we introduced the tensors

$$\begin{array}{c} \begin{array}{c} \uparrow \\ \psi \\ \downarrow \end{array} \begin{array}{c} \leftarrow \\ \psi \\ \rightarrow \end{array} \end{array} \equiv \sum_{g_1, g_2, g_3 \in G} \begin{array}{c} \begin{array}{c} \uparrow \\ \psi \\ \downarrow \end{array} \begin{array}{c} \leftarrow \\ \psi \\ \rightarrow \end{array} \end{array} \begin{array}{c} \uparrow \\ g_3 \\ \downarrow \end{array} |g_1, g_2\rangle \langle g_3| \equiv \sum_{g_1, g_2 \in G} \psi(g_1, g_2) |g_1, g_2\rangle \langle g_1 g_2|, \quad (16)$$

$$\begin{array}{c} \begin{array}{c} \uparrow \\ \psi \\ \downarrow \end{array} \begin{array}{c} \leftarrow \\ \psi \\ \rightarrow \end{array} \end{array} \equiv \sum_{g_1, g_2, g_3 \in G} \begin{array}{c} \begin{array}{c} \uparrow \\ \psi \\ \downarrow \end{array} \begin{array}{c} \leftarrow \\ \psi \\ \rightarrow \end{array} \end{array} \begin{array}{c} \uparrow \\ g_3 \\ \downarrow \end{array} |g_3\rangle \langle g_1, g_2| \equiv \sum_{g_1, g_2 \in G} \frac{1}{\psi(g_1, g_2)} |g_1 g_2\rangle \langle g_1, g_2|,$$

for any normalised representative  $\psi$  of  $[\psi] \in H^2(G, \mathbb{U}(1))$ . Besides, we can intuitively infer from this graphical identity that duality operators transmuting Hamiltonians  $\mathbb{H}^{\psi'}$  into  $\mathbb{H}^\psi$  would be labelled by simple objects in  $\text{Rep}^{\psi/\psi'}(G)$ .

The duality operator acting diagonally in the computational basis, any charge sector  $g \in G$  is preserved as we transmute  $\mathbb{H}$  into  $\mathbb{H}^\psi$ . But, notice that

$$\text{tr } \pi(g) = \text{tr } \pi(g x x^{-1}) = \frac{\psi(g, x)}{\psi(x, g)} \text{tr } \pi(g), \quad (17)$$

for all  $x \in G$ . This implies that the duality operator forces both  $\mathbb{H}$  and  $\mathbb{H}^\psi$  onto charge sectors  $g \in G$  for which  $\frac{\psi(g, -)}{\psi(-, g)} = 1$ . This follows from the fact that we are implicitly assuming periodic

boundary conditions for both models. Accessing other charge sectors requires modifying the duality operator, which in turn forces one of the models to have a different symmetry-twisted boundary condition. This is the statement that topological sectors are not mapped identically under the duality. We compute the corresponding permutation of topological sectors in the following.

### 2.3 Permutation of topological sectors and non-invertibility

In order to lift the duality operator  $\mathbb{D}^\pi$  to a unitary transformation, we need to characterise the interplay between symmetry-twisted closed boundary conditions and charge sectors. First, we need to modify the duality operator so as to accommodate the presence of a symmetry twist. Let  $\pi$  be a (unitary) irreducible  $\psi$ -projective representation and  $\chi \in \widehat{G}$ . Naturally, the tensor product  $\chi \otimes \pi$  is isomorphic to a  $\psi$ -projective representation  $\tilde{\pi}$  of the same dimension as  $\pi$  via an intertwining unitary map

$$\begin{bmatrix} \chi & \pi \\ 1 & - \end{bmatrix} \in \text{Hom}_{\text{Rep}^\psi(G)}(\chi \otimes \pi, \tilde{\pi}), \quad (18)$$

satisfying

$$(\chi \otimes \pi)(g)_{i_1 i_2} = \sum_{i_3, i_4} \begin{bmatrix} \chi & \pi \\ 1 & i_1 \end{bmatrix}^*_{i_3} \tilde{\pi}(g)_{i_3 i_4} \begin{bmatrix} \chi & \pi \\ 1 & i_2 \end{bmatrix}_{i_4}. \quad (19)$$

A similar definition holds for  $\pi \otimes \chi$ . From these intertwining maps, we construct the following tensor

$$\begin{aligned} \text{Diagram} &\equiv \sum_{i_1, i_2} i_1 \text{Diagram} i_2 |\eta_2\rangle\langle\eta_1| \otimes |i_1\rangle\langle i_2| \\ &\equiv \sum_{i_1, i_2, i_3} \begin{bmatrix} \pi & \eta_2 \\ i_1 & 1 \end{bmatrix}^*_{i_3} \begin{bmatrix} \eta_1 & \pi \\ 1 & i_2 \end{bmatrix}_{i_3} |\eta_2\rangle\langle\eta_1| \otimes |i_1\rangle\langle i_2|, \end{aligned} \quad (20)$$

for any  $\eta_1, \eta_2 \in \widehat{G}$ , out of which we construct the duality operator  $\mathbb{D}^\pi_{\eta_1 \rightarrow \eta_2} : \mathcal{H}^{\eta_1} \rightarrow \mathcal{H}^{\eta_2}$  depicted as

$$\mathbb{D}^\pi_{\eta_1 \rightarrow \eta_2} \equiv \text{Diagram}. \quad (21)$$

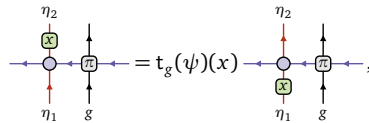
By definition of the map (18) implementing an isomorphism of projective representations, the rank 4 tensor (20) satisfies the following invariance property:

$$\text{Diagram} = \text{Diagram}, \quad (22)$$

for all  $x \in G$ , where we introduced the matrices

$$\text{Diagram} \equiv \sum_{\eta \in \widehat{G}} \text{Diagram} |\eta\rangle\langle\eta| \equiv \sum_{\eta \in \widehat{G}} \eta(x) |\eta\rangle\langle\eta|, \quad \text{Diagram} \equiv \sum_{i_1, i_2=1}^{\dim_{\mathbb{C}} \pi} \text{Diagram} |i_1\rangle\langle i_2| \equiv \sum_{i_1, i_2=1}^{\dim_{\mathbb{C}} \pi} \pi(x)_{i_1 i_2} |i_1\rangle\langle i_2|. \quad (23)$$

It follows from eq. (10) in combination with the invariance property (22) that  $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^\pi \circ \mathbb{H}^{1, \eta_1} = \mathbb{H}^{\psi, \eta_2} \circ \mathbb{D}_{\eta_1 \rightarrow \eta_2}^\pi$ . As suggested above, such a duality operator mapping boundary condition  $\eta_1 \in \widehat{G}$  onto  $\eta_2 \in \widehat{G}$  puts some constraints on the charge sectors allowed, which must be the same on both side of the duality. Conversely, a choice of boundary condition  $\eta_1 \in \widehat{G}$  and charge sector  $g \in G$ , fully specifies the boundary condition  $\eta_2 \in \widehat{G}$ . Concretely, invoking eq. (14), one can consider the duality operator effectively acting on a single site whose degree of freedom is labelled by the charge sector  $g$ , which combined with the invariance property (22) and  $\pi(x)\pi(g) = \frac{\psi(x, g)}{\psi(g, x)}\pi(g)\pi(x)$  results for every  $x \in G$  in



$$\text{Diagram (24)} \quad (24)$$

where in the spirit of eq. (15), the diagrams should here be interpreted as tensor networks whose opposite indices in the horizontal direction are contracted so as to impose periodic boundary conditions. Here, we have made use of  $t_g : Z^2(G, U(1)) \rightarrow Z^1(G, U(1))$  defined according to

$$t_g(\psi)(x) := \frac{\psi(x, g)}{\psi(g, x)}, \quad (25)$$

for all  $x, g \in G$  [76]. Since the previous derivation holds for any  $x \in G$ , we find that the boundary condition  $\eta_2$  is related to  $\eta_1$  via

$$\eta_2 \stackrel{!}{=} t_g(\psi) \eta_1. \quad (26)$$

Putting everything together, upon our duality operation, the group of topological sectors undergo the following automorphism in  $\text{Aut}(G \times \widehat{G})$  [60]:

$$(g, \eta) \mapsto (g, \eta_g^\psi), \quad \text{with} \quad \eta_g^\psi := t_g(\psi) \eta. \quad (27)$$

In particular, it follows from the normalisation of  $\psi$  that in the singlet sector  $g = \mathbb{1}$  the boundary condition is always preserved.

Before considering some examples, let us comment on the *non-invertibility* of the duality operations studied in this section. We already noticed that fixing periodic boundary conditions on both sides of the duality, the operator  $\mathbb{D}_{1 \rightarrow 1}^\pi$  projects onto certain charge sectors, and as such it is non-invertible. Nevertheless, restricting to such a sector, acting with the corresponding projector on  $\mathbb{D}^\pi \equiv \mathbb{D}_{1 \rightarrow 1}^\pi$  produces the unitary map relating the spectra of the dual Hamiltonians. More generally, given compatible sectors as per eq. (27), projecting the duality operator performing the mapping  $\eta \mapsto \eta_g^\psi$  onto the charge sector  $g \in G$  produces a unitary map. More specifically about the non-invertibility, instead of transmuting  $\mathbb{H}$  into  $\mathbb{H}^\psi$ , we can transmute  $\mathbb{H}^\psi$  into  $\mathbb{H}$  via a duality operator labelled by a projective representation in  $\text{Rep}^{1/\psi}(G)$ . Given a projective representation  $\pi$  with Schur's multiplier  $\psi$ , its contragredient representation  $\pi^\vee$  is a projective representation with Schur's multiplier  $1/\psi$ . Indeed,  $\pi^\vee$  is defined as  $\pi^\vee(g) := ((\pi(g)^{-1})^\vee : V^\vee \rightarrow V^\vee)$ , where  $\pi(g)$  is the transpose of  $\pi(g)$  and  $V^\vee := \text{Hom}(V, \mathbb{C})$ . It follows from  $\pi(g)^{-1} = \frac{1}{\psi(g, g^{-1})}\pi(g^{-1})$  that

$$\pi^\vee(g_1)\pi^\vee(g_2) = \frac{1}{\psi(g_1, g_2)}\pi^\vee(g_1g_2), \quad (28)$$

for all  $g_1, g_2 \in G$ , as expected. In particular, we have that  $\pi^\vee \otimes \pi$  is a linear representation of  $G$ , where  $\otimes$  is the usual tensor product, and thus labels a symmetry operator of  $\mathbb{H}$ .

Defining duality operators  $\mathbb{D}_{\eta_2 \rightarrow \eta_1}^{\pi^\vee}$  in the same vein as  $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^\pi$ , it is clear for instance that  $\mathbb{D}_{\eta_2 \rightarrow \eta_1}^{\pi^\vee} \circ \mathbb{D}_{\eta_1 \rightarrow \eta_2}^\pi = \mathbb{D}_{\eta_1 \rightarrow \eta_1}^{\pi^\vee \otimes \pi}$  does not act as the trivial symmetry operator in general. Nevertheless, it is possible to find a combination of duality operators that boils down to the trivial symmetry operator. For every  $\eta \in \widehat{G}$  such that  $\pi \otimes \eta \cong \pi$ , consider for instance duality operators  $\mathbb{D}_{1 \rightarrow \eta}^\pi$ . But,

$$\dim_{\mathbb{C}} \text{Hom}_{\text{Rep}^\psi(G)}(\pi \otimes \eta, \pi) = \dim_{\mathbb{C}} \text{Hom}_{\text{Rep}(G)}(\eta, \pi^\vee \otimes \pi) \leq 1, \quad (29)$$

for every  $\eta \in \widehat{G}$ . In particular, characters in  $\widehat{G}^\pi := \{\eta \in \widehat{G} \mid \dim_{\mathbb{C}} \text{Hom}_{\text{Rep}(G)}(\eta, \pi^\vee \otimes \pi) = 1\}$  form a subgroup of  $\widehat{G}$  such that  $|\widehat{G}^\pi| = (\dim_{\mathbb{C}} \pi)^2$ . It follows from the unitarity of the intertwining map

$$\left[ \begin{array}{cc|c} \pi^\vee & \pi & - \\ - & - & 1 \end{array} \right] \in \text{Hom}_{\text{Rep}(G)} \left( \pi^\vee \otimes \pi, \bigoplus_{\eta \in \widehat{G}^\pi} \eta \right), \quad (30)$$

that the operator  $\sum_{\eta \in \widehat{G}^\pi} \mathbb{D}_{\eta \rightarrow 1}^{\pi^\vee} \circ \mathbb{D}_{1 \rightarrow \eta}^\pi$  does act like the trivial symmetry operator on the microscopic Hilbert space associated with periodic boundary conditions.

## 2.4 Examples

Let us illustrate our formalism by specialising to the group  $G = \mathbb{Z}_p \times \mathbb{Z}_p$ , with  $p$  prime. We begin by introducing some notations: Throughout this part, we denote group elements of  $\mathbb{Z}_p \times \mathbb{Z}_p$  by  $g \equiv (g^{(1)}, g^{(2)})$  where  $g^{(1)}, g^{(2)} \in \{0, 1, \dots, p-1\}$  and we define the multiplication rule of  $\mathbb{Z}_p$  as addition modulo  $p$ . Introducing the primitive  $p^{\text{th}}$  root of unity  $\omega^{(p)} := \exp(\frac{2\pi i}{p})$ , a normalised representative 2-cocycle  $\psi$  generating the second cohomology group  $H^2(\mathbb{Z}_p \times \mathbb{Z}_p, \text{U}(1)) \cong \mathbb{Z}_p$  is given by<sup>2</sup>

$$\psi(g_1, g_2) := (\omega^{(p)})^{g_1^{(2)} g_2^{(1)}}. \quad (31)$$

It turns out that for any cohomology class in  $H^2(\mathbb{Z}_p \times \mathbb{Z}_p, \text{U}(1))$ , there is exactly one irreducible projective representation, up to equivalence, which is of dimension  $p$  [17]. In other words, the category  $\text{Rep}^\psi(\mathbb{Z}_p \times \mathbb{Z}_p)$  admits one isomorphism class of simple objects. Given the generalised Pauli matrices

$$X^{(p)} := \sum_{g=0}^{p-1} |g+1\rangle \langle g|, \quad \text{and} \quad Z^{(p)} := \sum_{g=0}^{p-1} (\omega^{(p)})^g |g\rangle \langle g|, \quad (32)$$

which satisfy the commutation relation  $Z^{(p)} X^{(p)} = \omega^{(p)} X^{(p)} Z^{(p)}$ , the projective representation  $\pi_k$  associated with the cocycle  $\psi^k(g_1, g_2)$  with  $k = 0, \dots, p-1$  explicitly reads

$$\pi_k(g) := (X^{(p)})^{g^{(1)}} (Z^{(p)})^{k g^{(2)}}, \quad (33)$$

for all  $g \equiv (g^{(1)}, g^{(2)}) \in \mathbb{Z}_p \times \mathbb{Z}_p$ . Indeed, one can readily check that

$$\pi_k(g_1) \pi_k(g_2) = (\omega^{(p)})^{k g_1^{(2)} g_2^{(1)}} \pi_k(g_1 + g_2).$$

Elements of the character group of  $\mathbb{Z}_p \times \mathbb{Z}_p$  are denoted by  $\chi_l \equiv \chi_{(l^{(1)}, l^{(2)})}$  with  $l^{(1)}, l^{(2)} \in \{0, 1, \dots, p-1\}$  and are defined via

$$\chi_l : g \mapsto (\omega^{(p)})^{g^{(1)} l^{(1)} + g^{(2)} l^{(2)}}. \quad (34)$$

<sup>2</sup>In defining  $\psi(g_1, g_2)$ , we are implicitly promoting  $\mathbb{Z}_p$  to the corresponding Galois field  $\mathbb{F}_p$ .

The tensor product of two characters simply reads

$$\chi_{l_1} \otimes \chi_{l_2} \cong \chi_{(l_1^{(1)}+l_2^{(1)}, l_1^{(2)}+l_2^{(2)})}, \quad (35)$$

where the sums are again modulo  $p$ . Finally, bases of intertwiners realising the isomorphisms  $\chi_l \otimes \pi_k \xrightarrow{\sim} \pi_k$  and  $\pi_k \otimes \chi_l \xrightarrow{\sim} \pi_k$  are provided by coefficients

$$\left[ \begin{array}{cc|c} \pi_k & \chi_l & \pi_k \\ i_1 & 1 & i_2 \end{array} \right] = \left[ \begin{array}{cc|c} \chi_l & \pi_k & \pi_k \\ 1 & i_1 & i_2 \end{array} \right] = \left[ (X^{(p)})^m (Z^{(p)})^{-l^{(1)}} \right]_{i_2 i_1}, \quad (36)$$

for all  $k, l^{(1)}, l^{(2)} \in \{0, \dots, p-1\}$  and  $i_1, i_2 \in \{1, \dots, p\}$ , where  $m$  is a solution of  $mk = l^{(2)} \pmod{p}$ , which is guaranteed to exist since  $p$  is prime. Let us confirm the mapping (27) of topological sectors using the explicit formula (36) for the intertwiners. For concreteness, let us assume periodic boundary conditions for the Hamiltonian  $\mathbb{H}$ . Given the charge sector  $g \in \mathbb{Z}_p \times \mathbb{Z}_p$ , let us compute the dual symmetry-twisted boundary condition for the Hamiltonian  $\mathbb{H}^{\psi^k}$ . We find that the duality operator explicitly depends on the complex number

$$\text{tr} \left[ (Z^{(p)})^{l^{(1)}} (X^{(p)})^{-m} (X^{(p)})^{g^{(1)}} (Z^{(p)})^{kg^{(2)}} \right] = \text{tr} \left[ (X^{(p)})^{g^{(1)}-m} (Z^{(p)})^{kg^{(2)}+l^{(1)}} \right], \quad (37)$$

where  $mk = l^{(2)} \pmod{p}$ . But, since  $p$  is prime, this factor is non-vanishing if and only if  $m = g^{(1)}$  and  $l^{(1)} = -kg^{(2)} \pmod{p}$ . This indicates that the boundary condition of the dual theory must be provided by the character

$$g \mapsto (\omega^{(p)})^{-g^{(1)}kg^{(2)}+g^{(2)}g^{(1)}k}, \quad (38)$$

which indeed coincides with  $\psi^k(-, g)/\psi^k(g, -)$ .

It is instructive to further specialise to the case  $p = 2$ . Let us consider a representative of the gapped phase where the  $\widehat{\mathbb{Z}}_2 \times \widehat{\mathbb{Z}}_2$  is spontaneously broken in the ground state subspace for periodic boundary conditions. To make contact with the literature, we think of the edges of the lattice as the sites of a chain of length  $|E|$ , which we enumerate  $i = 1, \dots, |E|$ . The microscopic Hilbert space is then given by  $\bigotimes_{i=1}^{|E|} \mathbb{C}[\mathbb{Z}_2 \times \mathbb{Z}_2]$  and the Hamiltonian encodes a pure ferromagnet:

$$\mathbb{H} = - \sum_{i=1}^{|E|} [(X\mathbb{I})_i (X\mathbb{I})_{i+1} + (\mathbb{I}X)_i (\mathbb{I}X)_{i+1}], \quad (39)$$

where  $X \equiv X^{(2)}$  is the usual Pauli matrix. In the notation of eq. (1), this is the Hamiltonian  $\mathbb{H} = \sum_v \mathbb{h}_{v,1}$  obtained by choosing non-vanishing coefficients  $h_1(g, \mathbf{r}_v) = -1$  for any  $g : E \rightarrow G$  and  $\mathbf{r}_v$  such that  $\mathbf{r}_v(v) \equiv (1, 0)$  or  $(0, 1)$ . Replacing the implicit trivial 2-cycle in  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$  by the normalised representative  $\psi : (g_1, g_2) \mapsto (-1)^{g_1^{(2)}g_2^{(1)}}$  in the unique non-trivial class yields the dual model

$$\mathbb{H}^\psi = - \sum_{i=1}^{|E|} [(XZ)_i (X\mathbb{I})_{i+1} + (\mathbb{I}X)_i (ZX)_{i+1}], \quad (40)$$

which we recognise as the (1+1)d *cluster state* model [12]. This model is the stable renormalisation group fixed point of the non-trivial  $\widehat{\mathbb{Z}}_2 \times \widehat{\mathbb{Z}}_2$  symmetry protected topological (SPT) phase. Within this context, the duality operator  $\mathbb{D}^\pi$  implements the celebrated *Kennedy–Tasaki* duality transformation, which was first introduced in [45] as a means to elucidate the Haldane phase of the antiferromagnetic spin-1 Heisenberg XXZ model.<sup>3</sup> We count four charge sectors

<sup>3</sup>The fact that the duality as formulated in this manuscript coincides with the Kennedy–Tasaki duality in the context of the spin-1 Heisenberg XXZ model was demonstrated in ref. [48].

and four boundary conditions that amount to imposing periodic/antiperiodic boundary conditions with respect to each copy of  $\mathbb{Z}_2$ . The mapping of the 16 topological sectors given by eq. 27 coincides with that obtained in ref. [54].<sup>4</sup>

As another application of our formalism, let us determine the fate of the  $\widehat{\mathbb{Z}_2} \times \widehat{\mathbb{Z}_2}$  ferromagnetic order parameters under the duality. The symmetric ferromagnetic order parameter acting on the first  $\mathbb{Z}_2$  component on sites  $i < j$  away from the boundary condition reads  $(X\mathbb{I})_i(X\mathbb{I})_j$ . The duality operator  $\mathbb{D}^\pi$  transmutes this local order parameter into the string order parameter  $(XZ)_i(\prod_{k=i+1}^{j-1}(\mathbb{I}Z)_k)(X\mathbb{I})_j$ , where the product is over all sites between  $i$  and  $j$ . One proceeds similarly for the second  $\mathbb{Z}_2$  component. Graphically, the mapping simply follows from

$$\begin{array}{c} \uparrow \\ \text{---} \text{X} \text{---} \text{I} \text{---} \text{X} \text{---} \\ \downarrow \\ g \end{array} = \frac{\psi(g, x)}{\psi(x, g)} \begin{array}{c} \uparrow \\ \text{---} \text{I} \text{---} \\ \downarrow \\ g \end{array}, \quad (41)$$

as well as the symmetry

$$\begin{array}{c} \uparrow \\ \text{---} \text{I} \text{---} \\ \downarrow \\ g \end{array} = \psi(g, x) \begin{array}{c} \uparrow \\ \text{---} \text{I} \text{---} \text{X} \text{---} \text{I} \text{---} \\ \downarrow \\ g \end{array} = \psi(x, g) \begin{array}{c} \uparrow \\ \text{---} \text{X} \text{---} \text{I} \text{---} \text{X} \text{---} \\ \downarrow \\ g \end{array}, \quad \text{with} \quad \begin{array}{c} \uparrow \\ \text{---} \text{X} \text{---} \\ \downarrow \\ g \end{array} \equiv \sum_{g \in \mathbb{Z}_2} |g\rangle \langle g+x|, \quad (42)$$

for any  $x \in \mathbb{Z}_2 \times \mathbb{Z}_2$ .

## 2.5 Category theoretic underpinnings

In preparation for the following section, let us delve a little bit into the mathematical formalism underlying the results presented above, following the approach of ref. [48–50]. Consider a one-dimensional quantum lattice model with a non-anomalous Abelian symmetry group  $G$ . In modern parlance, the model is said to admit topological lines valued in  $G$  [40]. The requirement that we should be able to construct junctions (in spacetime) of such topological lines invites us to promote the group  $G$  to a higher mathematical structure [14], namely the *fusion category*  $\text{Vec}_G$  of  $G$ -graded vector spaces. In particular, simple objects in  $\text{Vec}_G$  are one-dimensional vector spaces  $\mathbb{C}_g$ , for every  $g \in G$ , such that  $(\mathbb{C}_{g_1})_{g_2} = \delta_{g_1, g_2} \mathbb{C}$ ,  $\text{Hom}_{\text{Vec}_G}(\mathbb{C}_{g_1}, \mathbb{C}_{g_2}) \cong \delta_{g_1, g_2} \mathbb{C}$  and  $\mathbb{C}_{g_1} \otimes \mathbb{C}_{g_2} = \mathbb{C}_{g_1 g_2}$ . Moreover, the *monoidal associator* evaluates to the identity so that  $1_{\mathbb{C}_{g_1 g_2 g_3}} : (\mathbb{C}_{g_1} \otimes \mathbb{C}_{g_2}) \otimes \mathbb{C}_{g_3} \xrightarrow{\sim} \mathbb{C}_{g_1} \otimes (\mathbb{C}_{g_2} \otimes \mathbb{C}_{g_3})$ . Given a non-anomalous symmetry  $G$ —or rather, a symmetry  $\text{Vec}_G$ —it is always possible to gauge a subsymmetry  $A$ , where  $A$  is a subgroup of  $G$ . There are several ways to perform such a gauging, classified by the second cohomology group  $H^2(A, \text{U}(1))$ . Whenever the whole symmetry is gauged, the resulting theory possesses a  $\text{Rep}(G)$ -symmetry, i.e., a symmetry whose operators are labelled by representations of the group  $G$  and compose according to the tensor product of representations. Since the group  $G$  is assumed to be Abelian, we further have the equivalence  $\text{Rep}(G) \simeq \text{Vec}_{\widehat{G}}$ .

Mathematically, a pair  $(A, \psi)$  consisting of a subgroup  $A \subseteq G$  and a normalised representative  $\psi$  of a cohomology  $[\psi] \in H^2(A, \text{U}(1))$  specifies a so-called (finite semisimple indecomposable) *module category* over  $\text{Vec}_G$  [64]. Let  $\mathcal{M}(A, \psi)$  be the category whose simple objects are

<sup>4</sup>Note that we can add to the Hamiltonian  $\mathbb{H}$  symmetric terms of the form  $-\sum_i (Z\mathbb{I})_i$ . The resulting model would describe a symmetry-breaking ferromagnetic interaction supported on the second copy of  $\mathbb{Z}_2$  and a decoupled critical transverse-field Ising chain supported on the first copy of  $\mathbb{Z}_2$ . The corresponding dual Hamiltonian would then be given by  $\mathbb{H}^\psi - \sum_i (Z\mathbb{I})_i$ . An effective model describing the low-energy physics of this Hamiltonian was first examined in ref. [69] and was termed a *gapless SPT phase*. In ref. [55], this and other gapless SPT phases were constructed by making use of the Kennedy–Tasaki duality. It is plausible that our approach would facilitate the construction of other examples of gapless SPT phases as well as the study of their properties.



one-dimensional vector spaces  $\mathbb{C}_{rA}$ , for every  $rA \in G/A$ , equipped with the (left)  $\text{Vec}_G$ -module structure provided by the  $G$ -action of left cosets in  $G/A$ , i.e.  $\mathbb{C}_g \triangleright \mathbb{C}_{rA} := \mathbb{C}_{g \triangleright rA} = \mathbb{C}_{(gr)A}$ , and module associator  $\alpha_\psi^\triangleright$  specified by isomorphisms

$$\alpha_\psi^\triangleright(g_1, g_2)(rA) \cdot 1_{\mathbb{C}_{(g_1 g_2) \triangleright rA}} : (\mathbb{C}_{g_1} \otimes \mathbb{C}_{g_2}) \triangleright \mathbb{C}_{rA} \xrightarrow{\sim} \mathbb{C}_{g_1} \triangleright (\mathbb{C}_{g_2} \triangleright \mathbb{C}_{rA}), \quad (43)$$

for every  $g_1, g_2 \in G$  and  $rA \in G/A$ . The 2-cochain  $\alpha_\psi^\triangleright \in C^2(G, \text{Fun}(G/A, \text{U}(1)))$  is defined in terms of the 2-cocycle  $\psi$  via

$$\alpha_\psi^\triangleright(g_1, g_2)(rA) := \psi(a_{g_1, g_2 \triangleright rA}, a_{g_2, rA}), \quad (44)$$

where given  $g \in G$  and  $rA, sA \in G/A$  such that  $g \triangleright rA = sA$ ,  $a_{g, rA}$  is the unique group element in  $A$  satisfying  $g = s a_{g, rA} r^{-1}$ . It follows from the 2-cocycle condition satisfied by  $\psi$  and the property  $a_{g_1, g_2 \triangleright rA} a_{g_2, rA} = a_{g_1 g_2, rA}$ , for every  $g_1, g_2 \in G$  and  $rA \in G/A$ , that

$$\frac{\alpha_\psi^\triangleright(g_2, g_3)(rA) \alpha_\psi^\triangleright(g_1, g_2 g_3)(rA)}{\alpha_\psi^\triangleright(g_1 g_2, g_3)(rA) \alpha_\psi^\triangleright(g_1, g_2)(g_3 \triangleright rA)} = 1, \quad (45)$$

for every  $g_1, g_2, g_3 \in G$  and  $rA \in G/A$ . Identity (45) guarantees the so-called *pentagon axiom* of  $\mathcal{M}(A, \psi)$ .

Performing the  $\psi$ -twisted gauging of the symmetry  $G$  amounts to picking the  $\text{Vec}_G$ -module category  $\mathcal{M}(G, \psi) \equiv \text{Vec}^\psi$ , which is equivalent to  $\text{Vec}$  as a category but whose  $\text{Vec}_G$ -module structure depends on  $\psi$ . The symmetry structure of the theory resulting from the  $\psi$ -twisted gauging of the subsymmetry  $A$  is provided by the fusion category  $(\text{Vec}_G)_{\mathcal{M}(A, \psi)}^\vee := \text{Fun}_{\text{Vec}_G}(\mathcal{M}(A, \psi), \mathcal{M}(A, \psi))$  defined as the category of  $\text{Vec}_G$ -module functors from  $\mathcal{M}(A, \psi)$  to itself, with fusion structure provided by the composition of  $\text{Vec}_G$ -module functors [31]. In particular, we have  $(\text{Vec}_G)_{\text{Vec}^\psi}^\vee \simeq \text{Rep}(G)$ , for any  $[\psi] \in H^2(G, \text{U}(1))$ , as expected.

We can think of the  $\widehat{G}$ -symmetric models defined in sec. 2.1 as resulting from the  $\psi$ -twisted gauging of  $G$ -symmetric models. Therefore, two models with Hamiltonians  $\mathbb{H}$  and  $\mathbb{H}^\psi$ —only differing in a choice of 2-cocycle  $\psi$ —are associated with the  $\text{Vec}_G$ -module categories  $\text{Vec}$  and  $\text{Vec}^\psi$ , respectively. This can be made very explicit employing the formalism of ref. [48, 49], from which it also follows that a duality operator transmuting  $\mathbb{H}$  into  $\mathbb{H}^\psi$  is associated with a simple object in  $\text{Fun}_{\text{Vec}_G}(\text{Vec}, \text{Vec}^\psi) \simeq \text{Rep}^\psi(G)$ , as expected. For future reference, it is useful to observe that the category  $\text{Rep}^\psi(G)$  of  $\psi$ -projective representations of  $G$  is equivalent to the category  $\text{Mod}(\mathbb{C}[G]^\psi)$  of modules over the  $\psi$ -twisted group algebra  $\mathbb{C}[G]^\psi$ . Explicitly,  $\mathbb{C}[G]^\psi$  is the algebra with underlying vector space  $\text{Span}_{\mathbb{C}}\{|g\rangle \mid g \in G\}$  and multiplication  $|g_1\rangle \cdot |g_2\rangle = \psi(g_1, g_2) |g_1 g_2\rangle$ , for every  $g_1, g_2 \in G$ . In this context, topological sectors of a  $\widehat{G}$ -symmetric model correspond to simple objects in the so-called *Drinfel'd center*  $\mathcal{Z}(\text{Vec}_{\widehat{G}})$  of the symmetry fusion category  $\text{Vec}_{\widehat{G}}$  [49, 53], which are indeed labelled by group elements in  $G \times \widehat{G}$ .

## 2.6 Self-duality

As announced in the introduction, an appealing feature of the dualities we consider is that they strictly preserve the symmetry structure. Therefore, it is sensible to ask for (1+1)d models that would be left invariant under such duality transformations. Supposing we have such a model, one can further ask to promote the self-duality to a genuine internal symmetry, which amounts to performing a group extension of the symmetry fusion category.

Let us illustrate this procedure for the case  $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ . The Hamiltonian  $\mathbb{H} + \mathbb{H}^\psi$  constructed from the Hamiltonians (39) and (40), respectively, is left invariant under the action

of the duality operator (9). Out of the possible extensions of  $\text{Rep}(\mathbb{Z}_2 \times \mathbb{Z}_2)$  by  $\mathbb{Z}_2$  computed in ref. [73], promoting the self-duality operator to a symmetry operator results in the fusion category  $\text{Rep}(\mathbb{D}_8)$  of representations of the dihedral group  $\mathbb{D}_8$  of order eight [72], which, as a category, is equivalent to  $\text{Rep}(\mathbb{Z}_2 \times \mathbb{Z}_2) \boxplus \text{Rep}^\psi(\mathbb{Z}_2 \times \mathbb{Z}_2)$ . One could verify this fact by computing tensors implementing junctions between the symmetry operators in  $\text{Rep}(\mathbb{Z}_2 \times \mathbb{Z}_2)$  and the self-duality operator, and show that these correspond to morphisms in  $\text{Rep}(\mathbb{D}_8)$ . Instead, we will show that the Hamiltonian  $\mathbb{H} + \mathbb{H}^\psi$  can be constructed from local operators that are manifestly  $\text{Rep}(\mathbb{D}_8)$  symmetric.

We begin by noticing that the dihedral group  $\mathbb{D}_8$  with presentation  $\langle r, s \mid r^4 = s^2 = (rs)^2 = 1 \rangle$  can be realised as the *central extension*

$$1 \rightarrow \mathbb{Z}_2 \rightarrow \mathbb{D}_8 \cong \mathbb{Z}_2 \times_\phi (\mathbb{Z}_2 \times \mathbb{Z}_2) \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow 1, \quad (46)$$

specified by the cohomology class in  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{Z}_2) \cong \mathbb{Z}_2^3$  with normalised representative  $\phi$  such that  $\phi(g_1, g_2) = g_1^{(2)} + g_2^{(1)} \in \mathbb{Z}_2$  for every  $g_1, g_2 \in \mathbb{Z}_2 \times \mathbb{Z}_2$ . In particular, we notice the expected close similarity between the definitions of the 2-cocycles  $\psi$  and  $\phi$ . Identifying  $1 \equiv (0, (0, 0))$ ,  $s \equiv (0, (0, 1))$  and  $r \equiv (0, (1, 1))$ , one finds that the set  $\mathbb{Z}_2 \times (\mathbb{Z}_2 \times \mathbb{Z}_2)$  equipped with the multiplication rule

$$(n_1, q_1) \cdot (n_2, q_2) := (n_1 + n_2 + \phi(q_1, q_2), q_1 + q_2), \quad (47)$$

for every  $q_1, q_2 \in \mathbb{Z}_2 \times \mathbb{Z}_2$  and  $n_1, n_2 \in \mathbb{Z}_2$ , is indeed isomorphic to  $\mathbb{D}_8$ . Families of one-dimensional quantum lattice models that possess a  $\text{Rep}(\mathbb{D}_8)$  symmetry can be easily constructed mimicking the definitions of sec. 2.1. Consider the microscopic Hilbert space  $\bigotimes_{e \in E} \mathbb{C}[\mathbb{D}_8]$  spanned by states  $|g\rangle$  where  $g : E \rightarrow \mathbb{D}_8$ . Defining functions  $r_v : V \rightarrow \mathbb{D}_8$  as in sec. 2.1, for every  $v \in V$ , consider the following local operators acting on the the kinematical Hilbert space:

$$\mathbb{h}_{v,n} := \sum_{g, r_v} h_n(g, r_v) |g \, d r_v\rangle \langle g|, \quad (48)$$

wherein  $n$  is some label,  $h_n(g, r_v)$  are complex coefficients, and  $g \, d r_v$  was defined in eq. (2). For any possible choice of complex coefficients  $h_n(g, r_v)$ , for every  $g : E \rightarrow \mathbb{D}_8$  and  $r_v$ , it follows from the defining property of group representations that local operators (48) commute with symmetry operators of the form

$$\mathbb{U}^\rho := \sum_g \text{tr} \left( \prod_{e \in E} \rho(g(e)) \right) |g\rangle \langle g|, \quad (49)$$

where the product over edges  $e \in E$  is ordered, for any simple object  $\rho$  in  $\text{Rep}(\mathbb{D}_8)$ . By virtue of  $(\rho_1 \otimes \rho_2)(g) = \rho_1(g) \otimes \rho_2(g)$ , for every  $g \in \mathbb{D}_8$ , the symmetry operators compose according to  $\mathbb{U}^{\rho_1} \circ \mathbb{U}^{\rho_2} = \sum_{\rho_3 \in \rho_1 \otimes \rho_2} \mathbb{U}^{\rho_3}$ . In particular, symmetry operators associated with higher-dimensional representations are non-invertible. Finally, we construct arbitrary  $\text{Rep}(\mathbb{D}_8)$ -symmetric models as  $\sum_{v \in V} \sum_n \mathbb{h}_{v,n}$ .

Let us now reveal how the Hamiltonian  $\mathbb{H} + \mathbb{H}^\psi$  is effectively constructed from local operators of the form (48). By definition, the Hamiltonian  $\mathbb{H} + \mathbb{H}^\psi$ , which acts on the kinematical Hilbert space  $\bigotimes_i \mathbb{C}[\mathbb{Z}_2 \times \mathbb{Z}_2]$ , is made of local operators that act as  $-(\mathbb{L}X)_i(\mathbb{L}X)_{i+1} - (\mathbb{L}X)_i(ZX)_{i+1}$  or  $-(X\mathbb{L})_i(X\mathbb{L})_{i+1} - (XZ)_i(X\mathbb{L})_{i+1}$ . We argue that there is a choice of complex coefficients  $h_1(g, r_v)$  such that local operators of the form (48) effectively act in the same way. The crux is to restrict the microscopic Hilbert space to states  $|g\rangle \equiv |(n, q)\rangle$ , with  $q : E \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2$  and  $n : E \rightarrow \mathbb{Z}_2$ , such that  $n$  assigns the identity group element to every edge/site, effectively reducing the microscopic Hilbert space to  $\bigotimes_i \mathbb{C}[\mathbb{Z}_2 \times \mathbb{Z}_2]$ . This can be enforced via a choice of complex coefficients  $h_1(g, r_v) \equiv h_1((n, q), r_v)$ . One can further choose these coefficients such that the only

non-vanishing ones are such that  $\mathfrak{r}_v(v) = (0, (0, 1)) \equiv s$  or  $\mathfrak{r}_v(v) = (0, (1, 0)) \equiv rs$  and evaluate to  $-2$ . Proceeding as such, it follows from  $\phi((0, 1), (1, 0)) = 1$  that the local operator acting at site  $i$  is a sum of two operators, effectively acting as  $-2(\mathbb{I}X)_i(\mathbb{I}X)_{i+1}$  and  $-2(X\mathbb{I})_i(X\mathbb{I})_{i+1}$ , respectively, while projecting out states  $|g\rangle$  such that  $g(i+1) \notin \{1, s\}$  and  $g(i) \notin \{1, rs\}$ , respectively. On the effective Hilbert space, this recovers the local operators constitutive of  $\mathbb{H} + \mathbb{H}^\psi$ . One can now confirm that the two-dimensional representation  $\rho$  of  $\mathbb{D}_8$ , which is such that  $\rho(r) = XZ$  and  $\rho(s) = Z$  reproduces the action of the self-duality operator.

### 3 Twisted gauging of (2+1)d lattice gauge theories

*Given an arbitrary two-dimensional lattice gauge theory with an Abelian invertible 1-form symmetry, we study the non-trivial interplay between symmetry-twisted closed boundary conditions and charge sectors under the duality that amounts to the (untwisted) gauging of the 1-form symmetry, followed by the twisted gauging of the dual 0-form symmetry. Given pairs of compatible boundary conditions and charge sectors, the unitary operators relating the spectra of dual Hamiltonians are explicitly constructed in the form of tensor network operators.*

#### 3.1 Local Hamiltonians

Let us begin by constructing a family of two-dimensional lattice gauge theories on the two-torus  $\mathbb{T}^2$ . Let  $\mathbb{T}_\Delta^2$  be a triangulation of  $\mathbb{T}^2$ , whose sets of vertices, edges and plaquettes are denoted by  $V(\mathbb{T}_\Delta^2)$ ,  $E(\mathbb{T}_\Delta^2)$  and  $P(\mathbb{T}_\Delta^2)$ , respectively. Given an edge  $e \in E(\mathbb{T}_\Delta^2)$ ,  $-e$  denotes the edge with opposite orientation. For simplicity, we shall often assume that  $\mathbb{T}_\Delta^2$  is the triangular lattice, but our results typically hold more generally. We equip  $\mathbb{T}_\Delta^2$  with a total ordering of its vertices, which induces a relative orientation for its edges and plaquettes. Given an oriented edge  $e \in E(\mathbb{T}_\Delta^2)$ ,  $\partial_-e$  and  $\partial_+e$  still denote its source and target vertices, respectively, and we identify  $e \equiv (\partial_-e \partial_+e)$ . Given a finite Abelian group  $G$ ,<sup>5</sup> we consider a differential on the spaces of simplicial  $G$ -valued forms:

$$C^0(\mathbb{T}_\Delta^2, G) \xrightarrow{d^{(0)}} C^1(\mathbb{T}_\Delta^2, G) \xrightarrow{d^{(1)}} C^2(\mathbb{T}_\Delta^2, G), \quad (50)$$

such that for any  $e \in E(\mathbb{T}_\Delta^2)$  and  $p \in P(\mathbb{T}_\Delta^2)$

$$(d^{(0)}\mathfrak{r})(e) := \mathfrak{r}(\partial_-e)\mathfrak{r}(\partial_+e)^{-1}, \quad \text{and} \quad (d^{(1)}\mathfrak{g})(p) := \prod_{e \in \partial p} \mathfrak{g}(e), \quad (51)$$

where the product  $\prod_{e \in \partial p}$  is over edges in the boundary of  $p$ , whose orientations are assumed to be induced by that of  $p$ . The differential is clearly nilpotent, i.e.,  $d \circ d = 0$ , and we define the groups  $Z^n(\mathbb{T}_\Delta^2, G)$  of closed  $G$ -valued forms and  $B^n(\mathbb{T}_\Delta^2, G)$  of exact  $G$ -valued forms as  $\ker d^{(n)}$  and  $\text{im } d^{(n-1)}$ , respectively.

To every edge  $e \in E(\mathbb{T}_\Delta^2)$ , we assign a microscopic degree of freedom valued in  $\mathbb{C}[G]$ . Given  $\mathfrak{g} \in C^1(\mathbb{T}_\Delta^2, G)$  such that  $\mathfrak{g}(-e) = \mathfrak{g}(e)^{-1}$ , for every  $e \in E(\mathbb{T}_\Delta^2)$ , we notate via  $|g\rangle$ , the function  $\mathfrak{g}$  regarded as an element of the tensor product Hilbert space  $\bigotimes_e \mathbb{C}[G]$ . The kinematical Hilbert space of the two-dimensional lattice gauge theories on the two-torus is chosen to be the subspace of the microscopic Hilbert space  $\bigotimes_e \mathbb{C}[G]$  that is spanned by vectors  $|g\rangle$ ,

<sup>5</sup>Although we focus on finite Abelian groups throughout this manuscript, and occasionally use formalism specific to Abelian groups, most of our results readily generalise to the non-Abelian case, sometimes requiring little to no modification in the derivations.

where  $g \in Z^1(\mathbb{T}_\Delta^2, G)$ . In other words, at every plaquette  $p \in P(\mathbb{T}_\Delta^2)$ , we impose the kinematical constraint that  $\prod_{e \in \partial p} g(e) = \mathbb{1}$ , promoting  $g$  to a (1-form) flat gauge field. Since the group  $G$  is Abelian, we can view these kinematical constraints as Gauß constraints for a ‘magnetic’ gauge field defined on the Poincaré dual lattice, so that the resulting theory does have the interpretation of a lattice gauge theory.

Let  $\lambda$  be a normalised representative 3-cocycle of a cohomology class  $[\lambda] \in H^3(G, U(1))$ . Given a vertex  $v \in V(\mathbb{T}_\Delta^2)$ , we denote by  $r_v \in C^0(\mathbb{T}_\Delta^2, G)$  a function  $V(\mathbb{T}_\Delta^2) \rightarrow G$  such that  $r_v(v_1) = \mathbb{1}$ , whenever  $v_1 \neq v$ . We then consider the following local operators acting on the kinematical Hilbert space:

$$\mathbb{H}_{v,n}^\lambda := \sum_{g \in Z^1(\mathbb{T}_\Delta^2, G)} \sum_{r_v} h_n(g, r_v) \lambda(g, r_v) |g d^{(0)} r_v\rangle \langle g|, \quad (52)$$

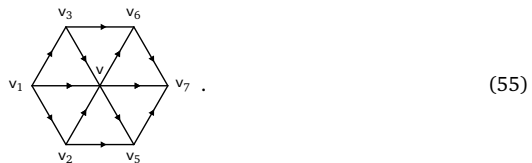
where  $n$  is some label,  $h_n(g, r_v)$  are arbitrary complex coefficients,  $g d^{(0)} r_v$  is by definition the function  $E[\mathbb{T}_\Delta^2] \rightarrow G$  such that

$$(g d^{(0)} r_v)(e) = \begin{cases} r_v(v)g(e), & \text{if } e \equiv (v v_1), \\ g(e)r_v(v)^{-1}, & \text{if } e \equiv (v_1 v), \\ g(e), & \text{otherwise,} \end{cases} \quad (53)$$

and  $\lambda(g, r_v)$  is the phase factor

$$\begin{aligned} \lambda(g, r_v) &:= \prod_{(v_1 v_2 v)} \lambda(g(v_1 v_2), g(v_2 v) r_v(v)^{-1}, r_v(v))^{\epsilon(v_1 v_2 v)} \\ &\cdot \prod_{(v_1 v v_2)} \lambda(g(v_1 v) r_v(v)^{-1}, r_v(v), g(v v_2))^{-\epsilon(v_1 v v_2)} \\ &\cdot \prod_{(v v_1 v_2)} \lambda(r_v(v), g(v v_1), g(v_1 v_2))^{\epsilon(v v_1 v_2)}. \end{aligned} \quad (54)$$

In the definition (54) of the phase factor  $\lambda(g, r_v)$ , the products are over plaquettes adjacent to  $v$  such that  $v_1 < v_2 < v$ ,  $v_1 < v < v_2$  and  $v < v_1 < v_2$ , respectively, while  $\epsilon(p) = \pm 1$  depends on the relative orientation of the plaquette  $p$ . Notice that  $\mathbb{H}_{v,n}^\lambda$  only acts non-trivially within a neighbourhood of the vertex  $v$ . Concretely, suppose  $\mathbb{T}_\Delta^2$  is the triangular lattice, and choose a total ordering of the vertices such that edges of the hexagonal subcomplex centered around  $v$  are oriented as follows:



Orientations above are compatible for instance with a total ordering such that  $v_1 < v_2 < v_3 < v < v_5 < v_6 < v_7$ . Under these assumptions, the phase factor (54) explicitly reads

$$\lambda(g, r_v) = \frac{\lambda(g(v_1 v_2), g(v_2 v) r_v(v)^{-1}, r_v(v)) \lambda(g(v_2 v) r_v(v)^{-1}, r_v(v), g(v v_5)) \lambda(r_v(v), g(v v_5), g(v_5 v_7))}{\lambda(g(v_1 v_3), g(v_3 v) r_v(v)^{-1}, r_v(v)) \lambda(g(v_3 v) r_v(v)^{-1}, r_v(v), g(v v_6)) \lambda(r_v(v), g(v v_6), g(v_6 v_7))}. \quad (56)$$

Finally, arbitrary combinations of local operators (52) can be organised into translation invariant local Hamiltonians  $\mathbb{H}^\lambda = \sum_{v \in V(\mathbb{T}_\Delta^2)} \mathbb{H}_v^\lambda \equiv \sum_{v \in V(\mathbb{T}_\Delta^2)} \sum_n \mathbb{H}_{v,n}^\lambda$ . By construction, the Hamiltonian  $\mathbb{H}^\lambda$  assumes periodic boundary conditions.

### 3.2 Symmetry operators

Given a choice of normalised representative 3-cocycle in a cohomology class  $[\lambda] \in H^3(G, \mathbb{U}(1))$ , consider a local Hamiltonian  $\mathbb{H}^\lambda$  defined in terms of local operators (52). Regardless of the choices of complex coefficients  $h_n(\mathfrak{g}, \mathfrak{x}_v)$ , for every  $\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)$  and  $\mathfrak{x}_v$ , and cohomology class  $[\lambda]$ , Hamiltonians  $\mathbb{H}^\lambda$  share the same symmetry structure. First of all, the model with Hamiltonian  $\mathbb{H}^\lambda$  possesses a 1-form  $\widehat{G}$  symmetry—or rather,  $\text{Rep}(G)$  symmetry—generated by the *Wilson loop* operators

$$\chi(\ell) \equiv \bigotimes_{e \in \ell} \chi(e) := \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)} \left( \prod_{e \in \ell} \chi(\mathfrak{g}(e)) \right) |\mathfrak{g}\rangle \langle \mathfrak{g}|, \quad (57)$$

where  $\chi \in \widehat{G}$ ,  $\ell \in Z_1(\mathbb{T}_\Delta^2, \mathbb{Z})$  is any oriented (simplicial) 1-cycle of  $\mathbb{T}_\Delta^2$  and the product  $\prod_{e \in \ell}$  is over edges along  $\ell$  whose orientations are assumed to be induced by that of  $\ell$ . In particular, whenever the orientation of  $e \subset \ell$  does not agree with that of  $\ell$ , it contributes a factor  $\chi(\mathfrak{g}(-e)) = \chi(\mathfrak{g}(e)^{-1}) = \chi^\vee(\mathfrak{g}(e))$  to the product. Invoking that  $\chi_1(g)\chi_2(g) = (\chi_1 \otimes \chi_2)(g)$ , for every  $g \in G$  and  $\chi_1, \chi_2 \in \widehat{G}$ , the composition of operators  $\chi_1(\ell) \circ \chi_2(\ell)$  equals  $(\chi_1 \otimes \chi_2)(\ell)$ . Clearly, it follows from the definition of the kinematical constraints that  $\chi(\ell)$  is invariant under continuous deformations of  $\ell$ , making it a topological line operator. In particular, it implies that  $\chi(\ell) = 1$ , whenever  $\ell$  is a contractible 1-cycle, i.e.,  $\ell$  is homologically trivial. Whenever  $\ell$  is a non-contractible 1-cycle, the commutation relation  $[\mathbb{H}^\lambda, \chi(\ell)] = 0$  follows from the fact that

$$\prod_{e \in \ell} \chi(\mathfrak{g} d^{(0)} \mathfrak{x}_v(e)) = \prod_{e \in \ell} \chi(\mathfrak{g}(e)), \quad (58)$$

for every vertex  $v \in V(\mathbb{T}_\Delta^2)$  and function  $\mathfrak{x}_v \in C^0(\mathbb{T}_\Delta^2, G)$  as defined above. This is essentially the ‘gauge invariance’ of Wilson loop operators. It follows from the 1-form  $\text{Rep}(G)$  symmetry that the Hamiltonian  $\mathbb{H}^\lambda$  decomposes into charge sectors labelled by  $(g_1, g_2) \in G \times G$  corresponding to holonomies of the flat gauge field  $\mathfrak{g}$  along both non-contractible 1-cycles of the two-torus.

Importantly, we can also construct topological junctions of topological Wilson lines.<sup>6</sup> Consider any number of oriented Wilson line operators meeting at any given vertex  $v \in V(\mathbb{T}_\Delta^2)$ . As long as the Wilson lines are chosen so that the tensor product of their corresponding representations, or their duals depending on the orientations, is isomorphic to the trivial representation, then it is a valid symmetry operator. As a matter of fact, since  $G$  is Abelian, it simply boils down to a composition of Wilson loop operators. Therefore, any network of topological Wilson lines thus constructed is a valid symmetry operator. More formally, consider the non-degenerate pairing

$$\begin{aligned} \langle -, - \rangle : \widehat{G} \times G &\longrightarrow \mathbb{U}(1), \\ (\chi, g) &\longmapsto \langle \chi, g \rangle := \chi(g). \end{aligned} \quad (59)$$

It enables us to define the following *cup product* [7]:

$$\begin{aligned} \langle -, - \rangle : C^1(\mathbb{T}_\Delta^2, \widehat{G}) \times C^1(\mathbb{T}_\Delta^2, G) &\longrightarrow C^2(\mathbb{T}_\Delta^2, \mathbb{U}(1)), \\ (\chi, \mathfrak{g}) &\longmapsto (\langle \chi, \mathfrak{g} \rangle : (v_1 v_2 v_3) \longmapsto \langle \chi(v_1 v_2), \mathfrak{g}(v_2 v_3) \rangle). \end{aligned} \quad (60)$$

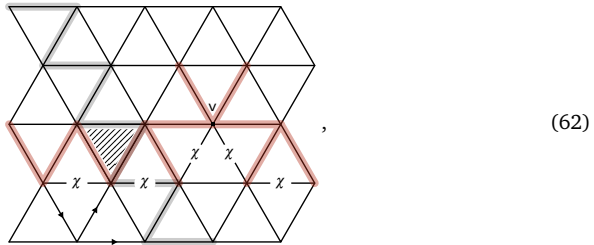
Given the differential on the spaces of simplicial  $\widehat{G}$ -valued forms defined as in eq. (50), we can

<sup>6</sup>This ability to construct junctions of topological lines is precisely why it is more accurate to talk about a 1-form  $\text{Rep}(G)$  symmetry rather than a 1-form  $\widehat{G}$  symmetry, as a higher mathematical structure is required to encompass both lines and their junctions.

consider groups  $Z^n(\mathbb{T}_\Delta^2, \widehat{G})$  and  $B^n(\mathbb{T}_\Delta^2, \widehat{G})$ . Let  $\mathfrak{X} \in Z^1(\mathbb{T}_\Delta^2, \widehat{G})$  and consider the operator

$$\mathfrak{X}(\mathbb{T}_\Delta^2) := \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)} \int_{\mathbb{T}_\Delta^2} \langle \mathfrak{X}, \smile \mathfrak{g} \rangle |g\rangle \langle g| \equiv \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)} \left( \prod_{p \in P(\mathbb{T}_\Delta^2)} \langle \mathfrak{X}, \smile \mathfrak{g} \rangle(p) \right) |g\rangle \langle g|. \quad (61)$$

It follows from the definition of the cup product that  $\mathfrak{X}(\mathbb{T}_\Delta^2)$  precisely amounts to inserting a certain combination of Wilson loops as defined in eq. (57). Let us consider a concrete example in the case where  $\mathbb{T}_\Delta^2$  is the triangular lattice. Let  $\chi(\ell)$  be a Wilson loop operator with  $\chi \in \widehat{G}$  and  $\ell \in Z_1(\mathbb{T}_\Delta^2, \mathbb{Z})$ . One can always associate to it a 1-cocycle  $\mathfrak{X} \in Z^1(\mathbb{T}_\Delta^2, \widehat{G})$  such that the action of  $\mathfrak{X}(\mathbb{T}_\Delta^2)$  coincides with that of  $\chi(\ell)$ . We depict such a configuration below:<sup>7</sup>



where the bold red lines represent the only edges to which  $\mathfrak{X}$  assigns non-trivial elements in  $\widehat{G}$ , to wit  $\chi$  or  $\chi^\vee$  depending on the orientations. We also depicted in bold gray lines a choice of gauge field  $\mathfrak{g}$ . Finally, the hashed plaquette is the only one that contributes non-trivially to  $\langle \mathfrak{X}, \smile \mathfrak{g} \rangle$ . It follows from the properties of the cup product that  $\langle \mathfrak{X}, \smile \mathfrak{g} \rangle$  yields the same result regardless of the choices of representatives in  $[\mathfrak{g}]$  and  $[\mathfrak{X}]$ , respectively. In particular, picking a different representative in  $[\mathfrak{X}]$  simply amounts to performing a continuous deformation of  $\ell$ .

It was recently pointed out that the 1-form  $\text{Rep}(G)$  symmetry is only a component of the total symmetry structure [4, 13, 22]. In particular, topological surface operators commuting with the Hamiltonian can also be defined. These are often referred to as *condensation operators* in the literature [56, 65]. A lattice implementation of condensation operators was first presented in ref. [27], employing the tensor network calculus developed in ref. [22]. We shall provide here particularly explicit expressions for these operators in the form of tensor networks. First of all, we define the topological surface operator  $\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2)$ , which acts on the whole kinematical Hilbert space, by summing over all possible insertions of Wilson loops. Another way to describe this operator, is that we insert a network of Wilson line operators associated with the representation  $\bigoplus_{\chi \in \widehat{G}} \chi$ , in such a way that every junction implements the tensor product of representations—a process that amounts to gauging the  $\text{Rep}(G)$  symmetry along the space-time submanifold  $\mathbb{T}^2$  [65]. The notation  $\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2)$  then stems from the fact that the induced representation  $\text{Ind}_{\{1\}}^G(\underline{0}_{\{1\}})$  of the trivial representation  $\underline{0}_{\{1\}}$  of  $\{1\} \subset G$  in  $G$  is isomorphic to the regular representation  $\mathbb{C}^G \cong \bigoplus_{\chi \in \widehat{G}} \chi$ . Invoking eq. (61), we can equivalently express this operator as follows:

$$\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2) := \frac{|Z^0(\mathbb{T}_\Delta^2, \widehat{G})|}{|Z^1(\mathbb{T}_\Delta^2, \widehat{G})|} \sum_{\mathfrak{X} \in Z^1(\mathbb{T}_\Delta^2, \widehat{G})} \mathfrak{X}(\mathbb{T}_\Delta^2). \quad (63)$$

<sup>7</sup>Notice how deforming the Wilson loop so as to ‘straighten’ it amounts to multiplying  $\mathfrak{X}$  by a 1-coboundary. Specifically, the coboundary is given by  $b_v$  wherein  $v$  stands for the vertex highlighted in eq. (62) and which is defined as  $b_v(v_1) = \underline{0}$  if  $v_1 \neq v$ ,  $\underline{0}$  being the trivial representation of  $G$ , and  $b_v(v) = \chi$ .

But, whenever  $\mathfrak{X}$  is cohomologically trivial,  $\mathfrak{X}(\mathbb{T}_\Delta^2) = 1$ . Decomposing the sum over  $Z^1(\mathbb{T}_\Delta^2, \widehat{G})$  into a sum over  $H^1(\mathbb{T}_\Delta^2, \widehat{G}) = Z^1(\mathbb{T}_\Delta^2, \widehat{G})/B^1(\mathbb{T}_\Delta^2, \widehat{G}) \cong \widehat{G} \times \widehat{G}$  and a sum over  $B^1(\mathbb{T}_\Delta^2, \widehat{G})$  yields the alternative formula

$$\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2) = \frac{1}{|G|} \sum_{[\mathfrak{x}] \in H^1(\mathbb{T}_\Delta^2, \widehat{G})} \mathfrak{X}(\mathbb{T}_\Delta^2), \quad (64)$$

where we used the fact that  $|\widehat{G}| = |G|$ , together with  $|Z^0(\mathbb{T}_\Delta^2, \widehat{G})| = |\widehat{G}|$ . What is the effect of such a condensation operator? It acts as a projector onto the singlet sector of the 1-form  $\text{Rep}(G)$ -symmetry. Indeed, denoting by  $(\gamma_1, \gamma_2)$  a generator of  $H_1(\mathbb{T}_\Delta^2, \mathbb{Z}) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ , i.e. two inequivalent non-contractible 1-cycles along the one-skeleton of  $\mathbb{T}_\Delta^2$ , it follows from  $\sum_{\chi \in \widehat{G}} \chi(g) = |G| \delta_{g,1}$ , for every  $g \in G$ , that

$$\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2) = \frac{1}{|G|} \left( \sum_{\chi_1 \in \widehat{G}} \chi_1(\gamma_1) \right) \circ \left( \sum_{\chi_2 \in \widehat{G}} \chi_2(\gamma_2) \right), \quad (65)$$

enforces holonomies  $\text{hol}_{\gamma_1}(\mathfrak{g})$  and  $\text{hol}_{\gamma_2}(\mathfrak{g})$  of the gauge field  $\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)$  along  $\gamma_1$  and  $\gamma_2$ , respectively, to evaluate to the identity element in  $G$ . In other words, it projects onto the subspace of the kinematical Hilbert space spanned by vectors  $|d^{(0)}m\rangle$ , where  $m \in C^0(\mathbb{T}_\Delta^2, G)$ . Accordingly, we have  $\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2) \circ \chi(\ell) = \chi(\ell) \circ \mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2) = \mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2)$ , for every  $\chi \in \widehat{G}$  and  $\ell \in Z_1(\mathbb{T}_\Delta^2, \mathbb{Z})$ . Moreover, it is immediate from eq. (65) that acting twice with  $\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2)$  amounts to acting with  $|G| \cdot \mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2)$ .

In a similar vein, for every subgroup  $A \subseteq G$ , one defines a condensation operator  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$  obtained by introducing a network of Wilson line operators associated with the representation  $\text{Ind}_A^G(\mathbb{Q}_A) \cong \mathbb{C}^{G/A}$ . But, by Frobenius reciprocity, the representations that appear in  $\text{Ind}_A^G(\mathbb{Q}_A)$  are precisely the representations in  $G$  that are trivial when restricted to  $A$ , so that the set of such representations is isomorphic to  $\widehat{G/A}$ .<sup>8</sup> Bringing everything together, one defines

$$\mathbb{U}^A(\mathbb{T}_\Delta^2) := \frac{1}{|G/A|} \left( \sum_{\chi_1 \in \text{Ind}_A^G(\mathbb{Q}_A)} \chi_1(\gamma_1) \right) \circ \left( \sum_{\chi_2 \in \text{Ind}_A^G(\mathbb{Q}_A)} \chi_2(\gamma_2) \right). \quad (66)$$

It follows from  $\sum_{\chi \in \text{Ind}_A^G(\mathbb{Q}_A)} \chi(g)$  being equal to  $|G/A|$  whenever  $g \in A$ , and zero otherwise,<sup>9</sup> that  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$  has the effect of projecting onto charge sectors where holonomies along both  $\gamma_1$  and  $\gamma_2$  evaluate to group elements in  $A$ . Moreover, we have

$$\mathbb{U}^A(\mathbb{T}_\Delta^2) \circ \mathbb{U}^A(\mathbb{T}_\Delta^2) = |G/A| \cdot \mathbb{U}^A(\mathbb{T}_\Delta^2). \quad (67)$$

Naturally,  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$  boils down to  $\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2)$  when specialising to  $A = \{1\}$ . By construction, Wilson lines labelled by representations in  $\text{Ind}_A^G(\mathbb{Q}_A)$  condense on this condensation operator in the sense that  $\mathbb{U}^A(\mathbb{T}_\Delta^2) \circ \chi(\ell) = \chi(\ell) \circ \mathbb{U}^A(\mathbb{T}_\Delta^2) = \mathbb{U}^A(\mathbb{T}_\Delta^2)$ , for every  $\chi \in \text{Ind}_A^G(\mathbb{Q}_A)$  and  $\ell \in Z_1(\mathbb{T}_\Delta^2, \mathbb{Z})$ , whereas Wilson lines generating the  $\text{Rep}(A)$  1-form subsymmetry survive on it. In other words, even while acting with  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ , the model still possesses a faithful  $\text{Rep}(A)$  1-form symmetry. Importantly, these are not the only topological line operators that can equip

<sup>8</sup>Since  $G$  is Abelian, every subgroup is normal and thus the quotient  $G/A$  is always a group. As a matter of fact, since  $G$  is finite Abelian,  $G$  admits a subgroup that is isomorphic to  $G/A$ , for any  $A \subseteq G$ .

<sup>9</sup>Given any subgroup  $R \subseteq G$ , there exists  $A \subseteq G$  and a group isomorphism  $\varphi : G/A \xrightarrow{\sim} R$ . Let  $g \in G$  such that  $g \notin A$ ; it follows from  $G = \bigsqcup_{r \in G/A} rA$  that  $g = ra$  with  $a \in A$  and  $r \neq 1$ . Therefore,  $\sum_{\chi \in \text{Ind}_A^G(\mathbb{Q}_A)} \chi(g) = \sum_{\chi \in \text{Ind}_A^G(\mathbb{Q}_A)} \chi(r) = \sum_{\chi \in \widehat{R}} \chi(\varphi(rA)) = 0$ , as desired.

the topological surface operator  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ . Indeed, one can also decorate  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$  with networks of 't Hooft lines labelled by left cosets in  $G/A$  so that the resulting surface operator projects onto charge sectors where holonomies along both  $\gamma_1$  and  $\gamma_2$  evaluate to any group element in  $G$ . Topological interfaces between condensation operators  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$  associated with different choices of  $A \subseteq G$  can be constructed in a similar fashion [22, 27].

Before introducing the remaining surface operators, it is very useful to provide a microscopic realisation of  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ , for every  $A \subseteq G$ , in terms of tensor networks, applying the general framework outlined in ref. [22]. First of all, notice that the condensation operator  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$  can be equivalently expressed as follows:

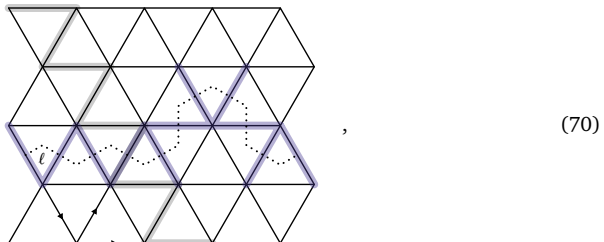
$$\mathbb{U}^A(\mathbb{T}_\Delta^2) = \sum_{\substack{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G) \\ \mathfrak{m} \in C^0(\mathbb{T}_\Delta^2, G/A)}} \left( \prod_{(v_1 v_2)} \delta_{\mathfrak{g}(v_1 v_2) \triangleright \mathfrak{m}(v_2), \mathfrak{m}(v_1)} \right) |\mathfrak{g}\rangle \langle \mathfrak{g}|. \quad (68)$$

Indeed, since for every  $e \equiv (v_1 v_2) \in E(\mathbb{T}_\Delta^2)$ , there is a Kronecker delta imposing  $\mathfrak{g}(v_1 v_2) \triangleright \mathfrak{m}(v_2) = \mathfrak{m}(v_1)$ , it follows from the comments in sec. 2.5 that there exists a unique group element  $a_{\mathfrak{g}(v_1 v_2), \mathfrak{m}(v_2)} \in A$  such that  $\mathfrak{g}(v_1 v_2) = \text{rep}(\mathfrak{m}(v_1)) \cdot a_{\mathfrak{g}(v_1 v_2), \mathfrak{m}(v_2)} \cdot \text{rep}(\mathfrak{m}(v_2))^{-1}$ , where  $\text{rep} : G/A \rightarrow G$  assigns to every left coset in  $G/A$  its representative in  $G$ . Moreover, by definition, we have  $a_{g_1, g_2 \triangleright rA} \cdot a_{g_2, rA} = a_{g_1, g_2, rA}$ , for every  $g_1, g_2 \in G$  and  $rA \in G/A$ . Consequently, defining  $\alpha_{\mathfrak{g}, \mathfrak{m}} \in C^1(\mathbb{T}_\Delta^2, A)$  via  $\alpha_{\mathfrak{g}, \mathfrak{m}}(v_1 v_2) := a_{\mathfrak{g}(v_1 v_2), \mathfrak{m}(v_2)}$ , for every  $(v_1 v_2) \in E(\mathbb{T}_\Delta^2)$  such that  $\mathfrak{g}(v_1 v_2) \triangleright \mathfrak{m}(v_2) = \mathfrak{m}(v_1)$ , one finds  $\alpha_{\mathfrak{g}, \mathfrak{m}}(v_1 v_2) \alpha_{\mathfrak{g}, \mathfrak{m}}(v_2 v_3) = \alpha_{\mathfrak{g}, \mathfrak{m}}(v_1 v_3)$ , for every  $(v_1 v_2 v_3) \in P(\mathbb{T}_\Delta^2)$ , and thus  $\alpha_{\mathfrak{g}, \mathfrak{m}} \in Z^1(\mathbb{T}_\Delta^2, A)$ . This confirms that  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$  acts as a projector onto the subspace spanned by states  $|a \, d^{(0)} \text{rep}(\mathfrak{m})\rangle$ , where  $a \in Z^1(\mathbb{T}_\Delta^2, A)$  and  $\mathfrak{m} \in C^0(\mathbb{T}_\Delta^2, G/A)$ . Furthermore, we recover the fact that  $\mathbb{U}^A(\mathbb{T}_\Delta^2) \circ \mathbb{U}^A(\mathbb{T}_\Delta^2) = |G/A| \cdot \mathbb{U}^A(\mathbb{T}_\Delta^2)$ , as following from the fact that the Cartesian product  $G/A \times G/A$  decomposes into a disjoint union of  $|G/A|$  copies of  $G/A$  under the diagonal action of  $G$ .

Invoking (68), one can explain how to construct the surface operator  $\mathbb{U}^A(\mathbb{T}_\Delta^2)[\mathfrak{f}]$  that is decorated by a network  $\mathfrak{f}$  of 't Hooft lines. Let us consider a concrete example. Let  $\ell$  be an oriented 1-cycle along the Poincaré dual  $\mathbb{T}_\Delta^{2V}$  of  $\mathbb{T}_\Delta^2$ . Consider the 't Hooft loop labelled by  $fA \in G/A$  with support  $\ell$ . One defines the 1-cocycle  $\mathfrak{f} \in Z^1(\mathbb{T}_\Delta^2, G/A)$  associated with this 't Hooft loop as follows:

$$\mathfrak{f}(e) = \begin{cases} fA, & \text{if } \iota(e) \subset \ell, \\ f^{-1}A, & \text{if } \iota(-e) \subset \ell, \\ \mathbb{1}, & \text{otherwise,} \end{cases} \quad (69)$$

for every oriented edge  $e \in E(\mathbb{T}_\Delta^2)$ , where  $\iota : E(\mathbb{T}_\Delta^2) \xrightarrow{\sim} E(\mathbb{T}_\Delta^{2V})$ . When defining the corresponding surface operator  $\mathbb{U}^A(\mathbb{T}_\Delta^2)[\mathfrak{f}]$ , we replace the Kronecker deltas entering the definition (68) by  $\delta_{\mathfrak{g}(v_1 v_2) \triangleright \mathfrak{m}(v_2), \mathfrak{f}(v_1 v_2) \mathfrak{m}(v_1)}$  for every  $(v_1 v_2) \in E(\mathbb{T}_\Delta^2)$ . Supposing that  $\ell \equiv \gamma_1$  is one of the generators of  $H_1(\mathbb{T}_\Delta^{2V}, \mathbb{Z})$ , we depict such a configuration on the triangular lattice below





where the bold purple lines represent the only edges to which  $f$  assigns non-trivial elements in  $G/A$ . There is a choice of  $m \in C^0(\mathbb{T}_\Delta^2, G/A)$  such that a gauge field satisfying  $g(v_1 v_2) \triangleright m(v_2) = f(v_1 v_2) m(v_1)$ , at every edge  $(v_1 v_2)$ , is non-vanishing along the bold gray lines only. It follows from the various definitions that  $\text{hol}_{\gamma_2}(g) \in fA$ , for any continuous deformation of the 't Hooft loop.

Still specialising to the case where  $\mathbb{T}_\Delta^2$  is the triangular lattice, let us invoke eq. (68) to express  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$  as a tensor network. One begins by introducing a rank- $(p+q)$  tensor in  $\mathbb{C}[G/A]^{\otimes p} \rightarrow \mathbb{C}[G/A]^{\otimes q}$  of the form  $\sum_{\{r_i A\}_i} \left( \prod_{i=1}^{p+q-1} \delta_{r_i A, r_{i+1} A} \right) \otimes_{i=p+1}^{p+q} |r_i A\rangle \otimes_{i=1}^p \langle r_i A|$ , graphically depicted as

$$\begin{aligned}
 \text{Diagram} &\equiv \sum_{\{r_i A\}_{i=1}^{p+q}} \text{Diagram} \equiv \sum_{\{r_i A\}_{i=1}^{p+q}} \left( \prod_{i=1}^{p+q-1} \delta_{r_i A, r_{i+1} A} \right) \otimes_{i=p+1}^{p+q} |r_i A\rangle \otimes_{i=1}^p \langle r_i A|, \\
 &\equiv \sum_{\{r_i A\}_{i=1}^{p+q}} \left( \prod_{i=1}^{p+q-1} \delta_{r_i A, r_{i+1} A} \right) \otimes_{i=p+1}^{p+q} |r_i A\rangle \otimes_{i=1}^p \langle r_i A|,
 \end{aligned} \quad (71)$$

where the sums are over left cosets in  $G/A$ . Henceforth, we refer to such a tensor as a ‘Kronecker delta tensor’. We also require a tensor implementing the  $G$ -action on  $G/A$ :

$$\begin{aligned}
 \text{Diagram} &\equiv \sum_{\substack{g \in G \\ r_1 A, r_2 A \in G/A}} r_1 A \text{Diagram} r_2 A \langle g | \otimes | g \rangle \langle g | \otimes | r_1 A \rangle \langle r_2 A | \\
 &\equiv \sum_{\substack{g \in G \\ r A \in G/A}} |g\rangle \langle g| \otimes |g\rangle \langle g| \otimes |g\rangle \langle g| \otimes |r A\rangle \langle r A|,
 \end{aligned} \quad (72)$$

as well as a tensor implementing the flatness of the gauge field:

$$\begin{aligned}
 \text{Diagram} &\equiv \sum_{g_1, g_2, g_3 \in G} \text{Diagram} |g_1, g_2\rangle \langle g_3| \equiv \sum_{g_1, g_2 \in G} |g_1, g_2\rangle \langle g_1 g_2|, \\
 \text{Diagram} &\equiv \sum_{g_1, g_2, g_3 \in G} \text{Diagram} |g_3\rangle \langle g_1, g_2| \equiv \sum_{g_1, g_2 \in G} |g_1 g_2\rangle \langle g_1, g_2|.
 \end{aligned} \quad (73)$$

Bringing everything together, we can realise  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$  as a tensor network whose unit cell is given by

$$\text{Diagram} \quad (74)$$

Let us dissect this tensor network. First of all, to every vertex of the triangular lattice, one assigns such a unit cell, in such a way that the Kronecker delta tensor coincides with the vertex,

and the purple lines with the edges of the lattice. Comparing with eq. (68), the Kronecker delta tensors perform the summation over  $C^0(\mathbb{T}_\Delta^2, G/A)$ , tensors (72) impose the constraints encoded into the Kronecker delta tensors, whereas the remaining tensors enforce the flatness of the gauge field. Notice that when realising  $\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2)$ , it follows from  $d^{(1)}(d^{(0)}m) = 0$ , for every  $m \in C^0(\mathbb{T}_\Delta^2, G)$ , that the tensors imposing the flatness of the gauge field are superfluous and can thus be omitted.

It turns out that the model with Hamiltonian  $\mathbb{H}^\lambda$  hosts as many topological surface operators as there are (indecomposable) module categories over the category  $\text{Vec}_G$ . We reviewed in sec. 2.5 that indecomposable  $\text{Vec}_G$ -module categories are labelled by pairs  $(A, \psi)$  consisting of a subgroup  $A \subseteq G$  and a normalised representative  $\psi$  of a cohomology class  $[\psi] \in H^2(A, U(1))$ . Whenever the 2-cocycle  $\psi$  is trivial, the topological surface operator associated with the  $\text{Vec}_G$ -module category  $\mathcal{M}(A, 1)$  is the condensation operator  $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ , as defined above [22]. A non-trivial 2-cocycle  $\psi$  modifies it in the following way:

$$\mathbb{U}^{A,\psi}(\mathbb{T}_\Delta^2) := \sum_{\substack{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G) \\ m \in C^0(\mathbb{T}_\Delta^2, G/A)}} \left( \prod_{\mathfrak{g}(v_1 v_2) \triangleright m(v_2), m(v_1)} \delta_{\mathfrak{g}(v_1 v_2) \triangleright m(v_2), m(v_1)} \prod_{(v_1 v_2 v_3)} \psi(a_{\mathfrak{g},m}(v_1 v_2), a_{\mathfrak{g},m}(v_2 v_3))^{\epsilon(v_1 v_2 v_3)} \right) |\mathfrak{g}\rangle \langle \mathfrak{g}|, \quad (75)$$

where we recognise the 2-cochain  $\alpha_\psi^\triangleright \in C^2(G, \text{Fun}(G/A, U(1)))$  defined in eq. (44) the module associator of  $\mathcal{M}(A, \psi)$  evaluates to. The commutation relation  $[\mathbb{H}^\lambda, \mathbb{U}^{A,\psi}(\mathbb{T}_\Delta^2)] = 0$  follows from the 2-cocycle condition satisfied by  $\psi$ . Indeed, letting  $v \in V(\mathbb{T}_\Delta^2)$  and  $\mathfrak{r}_v \in C^0(\mathbb{T}_\Delta^2, G)$  such that  $\mathfrak{r}_v(v_1) = 1$ , whenever  $v_1 \neq v$ , the 2-cocycle condition ensures that

$$\prod_{(v_1 v_2 v_3)} \psi(a_{\mathfrak{g},d^{(0)}_{\mathfrak{r}_v, \mathfrak{r}_v} \triangleright m}(v_1 v_2), a_{\mathfrak{g},d^{(0)}_{\mathfrak{r}_v, \mathfrak{r}_v} \triangleright m}(v_2 v_3))^{\epsilon(v_1 v_2 v_3)} = \prod_{(v_1 v_2 v_3)} \psi(a_{\mathfrak{g},m}(v_1 v_2), a_{\mathfrak{g},m}(v_2 v_3))^{\epsilon(v_1 v_2 v_3)}, \quad (76)$$

for every  $\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)$  and  $m \in C^0(\mathbb{T}_\Delta^2, G/A)$  such that  $\mathfrak{g}(v_1 v_2) \triangleright m(v_2) = m(v_1)$  at every edge  $(v_1 v_2) \in E(\mathbb{T}_\Delta^2)$ , and where  $(\mathfrak{r}_v \triangleright m)(v_1) := \mathfrak{r}_v(v_1) \triangleright m(v_1)$  for every  $v_1 \in V(\mathbb{T}_\Delta^2)$ . Introducing the tensors

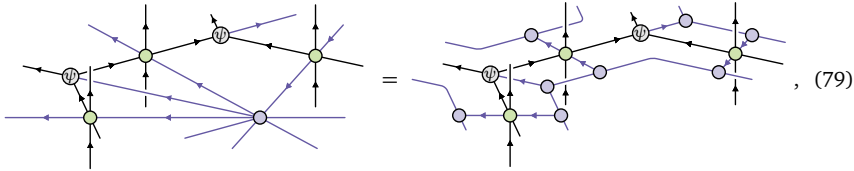
$$\begin{aligned} \text{Diagram} &\equiv \sum_{\substack{g_1, g_2, g_3 \in G \\ rA \in G/A}} \text{Diagram} |g_1, g_2\rangle \langle g_3| \otimes |rA\rangle \\ &\equiv \sum_{\substack{g_1, g_2 \in G \\ rA \in G/A}} \psi(a_{g_1, g_2 \triangleright rA}, a_{g_2, rA}) |g_1, g_2\rangle \langle g_1 g_2| \otimes |rA|, \end{aligned} \quad (77)$$

and

$$\begin{aligned} \text{Diagram} &\equiv \sum_{\substack{g_1, g_2, g_3 \in G \\ rA \in G/A}} \text{Diagram} |g_3\rangle \langle g_1, g_2| \otimes |rA\rangle \\ &\equiv \sum_{\substack{g_1, g_2 \in G \\ rA \in G/A}} \psi(a_{g_1, g_2 \triangleright rA}, a_{g_2, rA})^{-1} |g_1 g_2\rangle \langle g_1, g_2| \otimes |rA\rangle, \end{aligned} \quad (78)$$

we can realise  $\mathbb{U}^{A,\psi}(\mathbb{T}_\Delta^2)$  on the triangular lattice as a tensor network whose unit cell is given

by<sup>10</sup>



where, in order to obtain the r.h.s., we used the fact that contracting Kronecker delta tensors results in another Kronecker delta tensor. As a special case, the topological surface operator associated with  $\mathcal{M}(G, \psi)$  reads

$$\begin{aligned} \mathbb{U}^{G, \psi}(\mathbb{T}_{\Delta}^2) &= \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_{\Delta}^2, G)} \left( \prod_{(v_1, v_2, v_3)} \psi(\mathfrak{g}(v_1 v_2), \mathfrak{g}(v_2 v_3))^{\epsilon(v_1 v_2 v_3)} \right) |\mathfrak{g}\rangle \langle \mathfrak{g}| \\ &= \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_{\Delta}^2, G)} \frac{\psi(\text{hol}_{\gamma_1}(\mathfrak{g}), \text{hol}_{\gamma_2}(\mathfrak{g}))}{\psi(\text{hol}_{\gamma_2}(\mathfrak{g}), \text{hol}_{\gamma_1}(\mathfrak{g}))} |\mathfrak{g}\rangle \langle \mathfrak{g}|, \end{aligned} \quad (80)$$

where we invoked the 2-cocycle condition of  $\psi$  to go from the first line to the second line.

We commented earlier that the condensation operator (68) could be obtained by summing over networks of Wilson lines valued in a subgroup of  $\hat{G}$  that is isomorphic to  $G/A$ , which should be thought as topological lines decorating the identity surface operator  $\mathbb{U}^G(\mathbb{T}_{\Delta}^2)$ . Conversely, the same operator  $\mathbb{U}^A(\mathbb{T}_{\Delta}^2)$  can be obtained, starting from  $\mathbb{U}^{\{1\}}(\mathbb{T}_{\Delta}^2)$ , by summing over all possible networks of topological lines labelled by group elements in  $A \subseteq G$ , in such a way that the group multiplication in  $A$ —or rather the multiplication rule of the group algebra  $\mathbb{C}[A]$ —is implemented at every junction. In this formulation, the operator  $\mathbb{U}^{G, \psi}(\mathbb{T}_{\Delta}^2)$  is obtained starting from  $\mathbb{U}^{\{1\}}(\mathbb{T}_{\Delta}^2)$  by summing over all possible networks of topological lines in  $G$  such that the multiplication rule of the twisted group algebra  $\mathbb{C}[G]^{\psi}$  is implemented at every junction.

We have established that for every  $\text{Vec}_G$ -module category, one can construct a topological surface operator. Moreover, these topological surfaces can be decorated with networks of topological lines. In particular, the surface operator  $\mathbb{U}^{A, \psi}(\mathbb{T}_{\Delta}^2)$  associated with the  $\text{Vec}_G$ -module category  $\mathcal{M}(A, \psi)$  can be decorated with lines labelled by simple objects in the category  $(\text{Vec}_G)_{\mathcal{M}(A, \psi)}^{\vee}$  defined in sec. 2.5, which happens to be equivalent to  $\text{Rep}(A)^{\boxplus |G/A|}$  since the group  $G$  is Abelian, as expected. Bringing everything together, this indicates that the symmetry structure of Hamiltonians  $\mathbb{H}^{\lambda}$  is provided by the fusion 2-category  $\text{Mod}(\text{Vec}_G)$  of  $\text{Vec}_G$ -module categories and  $\text{Vec}_G$ -module functors. By analogy with  $\text{Rep}(G) \simeq \text{Mod}(\mathbb{C}[G])$ , it is enlightening to write  $2\text{Rep}(G) \equiv \text{Mod}(\text{Vec}_G)$ , where  $\text{Vec}_G$  is thought as a categorification of  $\mathbb{C}[G]$  obtained by promoting the ring  $\mathbb{C}$  to the fusion (1-)category  $\text{Vec}$ , and refer to objects in  $2\text{Rep}(G)$  as 2-representations of  $G$ .

### 3.3 Symmetry twists

As mentioned earlier, the  $2\text{Rep}(G)$ -symmetric Hamiltonians  $\mathbb{H}^{\lambda} = \sum_v \sum_n \mathbb{H}_{v,n}^{\lambda}$  defined in terms of local operators (52) assume periodic boundary conditions. Let us now consider symmetry-twisted closed boundary conditions obtained by inserting symmetry twists at various loci

<sup>10</sup>Notice that we can unambiguously omit certain arrows for visual convenience.

of the spatial manifold. The same way we distinguish two types of symmetry operators, namely topological surfaces and topological lines, we distinguish two types of symmetry twists. These are localised along specific spatial 1-cycles and 0-cycles, respectively, while extending in the time direction. Henceforth, we refer to the resulting boundary conditions as being 0-form symmetry-twisted or 1-form symmetry-twisted, respectively. When implementing such symmetry-twisted boundary conditions, the kinematical Hilbert space typically requires to be supplemented with appropriate degrees of freedom, but not always. In any case, the resulting symmetry-twisted boundary conditions preserve the translation invariance of the system, in the sense that there is a unitary isomorphism between Hilbert spaces associated with symmetry twists that are related by continuous deformations. Generally, the presence of 0-form and 1-form symmetry-twisted boundary conditions can be witnessed by the corresponding point-like and line-like charges by moving them along 1- and 2-cycles, respectively. Throughout this section, the focus will be on 1-form symmetry-twisted boundary conditions, since the 0-form ones can be deduced from them, the same way the topological surface operators descend from the topological line operators.

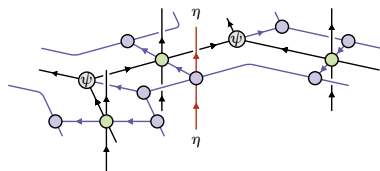
Starting from the kinematical Hilbert space  $\text{Span}_{\mathbb{C}}\{|g\rangle \mid g \in Z^1(\mathbb{T}_{\Delta}^2, G)\}$ , consider inserting at a vertex  $v_0 \in V(\mathbb{T}_{\Delta}^2)$  the topological line defect labelled by an irreducible representation  $\eta : G \rightarrow \text{U}(1)$ . Concretely, we perform this operation by considering the tensor product between the kinematical Hilbert space and  $\mathbb{C}[\widehat{G}]$  before enforcing the degree of freedom in  $\mathbb{C}[\widehat{G}]$  to be  $|\eta\rangle$ . We denote the resulting kinematical space by  $\mathcal{H}^{\eta}$ , which is still isomorphic to  $\text{Span}_{\mathbb{C}}\{|g\rangle \mid g \in Z^1(\mathbb{T}_{\Delta}^2, G)\}$ . Local symmetric operators acting at  $v_0$  are modified in the following way:

$$\mathbb{H}_{v_0, n}^{\lambda, \eta} := \sum_{g \in Z^1(\mathbb{T}_{\Delta}^2, G)} \sum_{\mathbb{F}_{v_0}} h_n(g, \mathbb{F}_{v_0}) \eta(\mathbb{F}_{v_0}) \lambda(g, \mathbb{F}_{v_0}) |g\rangle d^{(0)}_{\mathbb{F}_{v_0}} |g\rangle, \quad (81)$$

where  $\eta(\mathbb{F}_{v_0}) := \eta(\mathbb{F}_{v_0}(v_0))$ . All the other local operators are left intact so that the resulting local Hamiltonians read

$$\mathbb{H}^{\lambda, \eta} = \sum_{\substack{v \in V(\mathbb{T}_{\Delta}^2) \\ v \neq v_0}} \sum_n \mathbb{H}_{v, n}^{\lambda} + \sum_n \mathbb{H}_{v_0, n}^{\lambda, \eta}. \quad (82)$$

Since the group  $G$  is Abelian, the presence of the symmetry twist does not alter the symmetry structure of the Hamiltonian  $\mathbb{H}^{\lambda, \eta}$ , in the sense that symmetry operators are still organised in  $2\text{Rep}(G)$ . Nonetheless, they need to be adapted so as to accommodate the presence of the symmetry twist. This is very simple in the case of topological surface operators  $\mathbb{U}_{\eta}^{G, \psi}(\mathbb{T}_{\Delta}^2) : \mathcal{H}^{\eta} \rightarrow \mathcal{H}^{\eta}$  as it simply requires taking the tensor product of the vanilla operator (80) with  $|\eta\rangle\langle\eta|$ . However, specialising to the case where  $\mathbb{T}_{\Delta}^2$  is the triangular lattice, the topological surface operator  $\mathbb{U}_{\eta}^{\{1\}}(\mathbb{T}_{\Delta}^2) : \mathcal{H}^{\eta} \rightarrow \mathcal{H}^{\eta}$  is obtained by replacing the unit cell of its tensor network parametrisation located at  $v_0$  by the following one:


(83)

where we introduced the tensor

$$\begin{aligned}
 & \equiv \sum_{\{g_i\}_{i=1}^{p+q}} \text{[Diagram with legs } g_1, g_2, \dots, g_{p+q}] \\
 & \equiv \sum_{\{g_i\}_{i=1}^{p+q}} \left( \prod_{i=1}^{p+q-1} \delta_{g_i, g_{i+1}} \right) \eta_1(g_1) \eta_2^\vee(g_1) \bigotimes_{i=p+1}^{p+q} |g_i\rangle \bigotimes_{i=1}^p \langle g_i|,
 \end{aligned} \tag{84}$$

for every  $\eta_1, \eta_2 \in \widehat{G}$ . More generally, for an arbitrary triangulation of  $\mathbb{T}_\Delta^2$ , it suffices to replace the tensor of the form (71) located at  $v_0$  by the corresponding one of the form (84). Importantly, tensors (84) satisfy the following invariance property:

$$\begin{aligned}
 & \text{[Diagram with legs } \eta_1, \eta_2 \text{ and } x] = \text{[Diagram with legs } \eta_1, \eta_2 \text{ and } x] \\
 & \text{with} \quad | \eta \rangle \equiv \sum_{\eta \in \widehat{G}} | \eta \rangle \langle \eta | \equiv \sum_{\eta \in \widehat{G}} \eta(x) | \eta \rangle \langle \eta |, \\
 & \quad \quad \quad | g \rangle \equiv \sum_{g \in G} | g \rangle \langle g |,
 \end{aligned} \tag{85}$$

for every  $x \in G$ . Together with (76), this symmetry condition guarantees that the symmetry operator  $\mathbb{U}^{\{\mathbb{1}\}}(\mathbb{T}_\Delta^2)$  commutes with local operators  $\mathbb{h}_{v_0, n}^{\lambda, \eta}$  acting at the locus of the 1-form symmetry twist. The remaining symmetry operators can be treated in a similar fashion but the tensors (84) take a more complicated form. For any choice of 1-form symmetry-twisted boundary condition  $\eta \in \widehat{G}$ , it follows from the  $2\text{Rep}(G)$  symmetry—more specifically, the 1-form  $\text{Rep}(G)$  symmetry—that Hamiltonians  $\mathbb{H}^{\lambda, \eta}$  still decompose into charge sectors labelled by  $(g_1, g_2) \in G \times G$ . Triples  $(g_1, g_2, \eta) \in G \times G \times \widehat{G}$  label the *topological sectors* of the theory.

We mentioned above that we could also consider 0-form symmetry-twisted boundary conditions. For instance, consider inserting along a non-contractible cycle  $\gamma_1 \in Z_1(\mathbb{T}_\Delta^2, \mathbb{Z})$  the topological surface defect labelled by simple 2-representations of the form  $\mathcal{M}(G, \psi) \simeq \text{Vec}^\psi \in 2\text{Rep}(G)$ , whose corresponding symmetry operator  $\mathbb{U}^{G_\psi}(\mathbb{T}_\Delta^2)$  was defined in eq. (80). Similarly to the 1-form symmetry-twisted boundary conditions, this specific type of topological surface defect results in boundary conditions that do not alter the kinematical Hilbert space of the system. Local symmetric operators acting at vertices  $v \subset \gamma_1$  are modified in the following way:

$$\mathbb{h}_{v, n}^{\lambda, \psi} := \sum_{g \in Z^1(\mathbb{T}_\Delta^2, G)} \sum_{\mathbf{r}_v} h_n(g, \mathbf{r}_v) \psi(g, \mathbf{r}_v) \lambda(g, \mathbf{r}_v) |g d^{(0)} \mathbf{r}_v\rangle \langle g|, \tag{86}$$

where

$$\psi(g, \mathbf{r}_v) := \prod_{(v_1, v) \subset \gamma_1} \psi(g(v_1, v) \mathbf{r}_v(v)^{-1}, \mathbf{r}_v(v)) \prod_{(v, v_2) \subset \gamma_1} \psi(\mathbf{r}_v(v), g(v, v_2))^{-1}. \tag{87}$$

All the other local operators are left intact so that the resulting local Hamiltonians read

$$\mathbb{H}^{\lambda, \psi} = \sum_{\substack{v \in V(\mathbb{T}_\Delta^2) \\ v \notin \gamma_1}} \sum_n \mathbb{h}_{v, n}^\lambda + \sum_{\substack{v \in V(\mathbb{T}_\Delta^2) \\ v \subset \gamma_1}} \sum_n \mathbb{h}_{v, n}^{\lambda, \psi}. \tag{88}$$

The remaining symmetry twists can be implemented in a similar fashion.

### 3.4 Duality operators

Given a finite Abelian group  $G$ , we constructed in sec. 3.1 families of two-dimensional quantum lattice models with  $2\text{Rep}(G)$  symmetry, which are parametrised by normalised representatives  $\lambda$  of a cohomology class  $[\lambda]$  in  $H^3(G, \text{U}(1))$  as well as sets of complex coefficients  $\{h_n\}_n$ . We claim that models constructed in this way that only differ in the choice of cohomology class  $[\lambda]$  are dual to one another. For now, let us assume periodic boundary conditions and construct the operator transmuting Hamiltonians  $\mathbb{H}$  and  $\mathbb{H}^\lambda$  into each other. Postponing more formal justifications to sec. 3.7, let us first motivate the algebraic structure encoding such duality operators by analogy with the one-dimensional case. In (1+1)d, we showed that Hamiltonians of  $\text{Rep}(G)$ -symmetric models only differing in a choice of 2-cocycle  $\psi$  could be transmuted into each other by any duality operator encoded into the (1-)category  $\text{Rep}^\psi(G) \simeq \text{Mod}(\mathbb{C}[G]^\psi)$  of  $\psi$ -projective representations of  $G$ . Thus we naively expect Hamiltonians of  $2\text{Rep}(G)$ -symmetric models only differing in a choice of 3-cocycle  $\lambda$  to be transmutable into each other by any duality operator encoded into the 2-category  $2\text{Rep}^\lambda(G)$  of ‘ $\lambda$ -projective 2-representations of  $G$ ’. Consider the fusion category  $\text{Vec}_G$  of  $G$ -graded vector spaces, whose associator isomorphism, we recall, is given by the identity. Now, considering instead the associator

$$\lambda(g_1, g_2, g_3) \cdot 1_{\mathbb{C}_{g_1 g_2 g_3}} : (\mathbb{C}_{g_1} \otimes \mathbb{C}_{g_2}) \otimes \mathbb{C}_{g_3} \xrightarrow{\sim} \mathbb{C}_{g_1} \otimes (\mathbb{C}_{g_2} \otimes \mathbb{C}_{g_3}), \quad (89)$$

the resulting fusion category is denoted by  $\text{Vec}_G^\lambda$ . We then define the fusion 2-category  $2\text{Rep}^\lambda(G)$  as the 2-category  $\text{Mod}(\text{Vec}_G^\lambda)$  of  $\text{Vec}_G^\lambda$ -module categories. Indecomposable module categories over  $\text{Vec}_G^\lambda$  are defined similarly to module categories over  $\text{Vec}_G$  with the following difference: Only subgroups  $A$  such that  $\lambda|_{A \times A \times A}$  is cohomologically trivial are admissible, and the module associator (43) now evaluates to a 2-cochain  $\alpha_\phi^\triangleright \in C^2(G, \text{Fun}(G/A, \text{U}(1)))$ , associated with a choice of 2-cochain  $\phi \in C^2(A, \text{U}(1))$  such that  $d^{(2)}\phi = \lambda^{-1}|_{A \times A \times A}$ ,<sup>11</sup> that satisfy

$$d^{(2)}\alpha_\phi^\triangleright(g_1, g_2, g_3)(rA) := \frac{\alpha_\phi^\triangleright(g_2, g_3)(rA) \alpha_\phi^\triangleright(g_1, g_2 g_3)(rA)}{\alpha_\phi^\triangleright(g_1 g_2, g_3)(rA) \alpha_\phi^\triangleright(g_1, g_2)(g_3 \triangleright rA)} = \lambda(g_1, g_2, g_3)^{-1}, \quad (90)$$

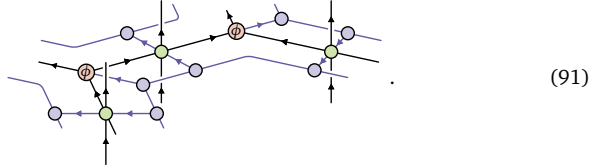
for every  $g_1, g_2, g_3 \in G$  and  $rA \in G/A$ . The condition (90) guarantees the pentagon axiom, which now involves the monoidal associator (89). We denote the resulting  $\text{Vec}_G^\lambda$ -module category by  $\mathcal{M}(A, \phi)$ .

Given the data of a  $\text{Vec}_G^\lambda$ -module category  $\mathcal{M}(A, \phi)$ , consider the tensors

$$\begin{aligned} \text{Diagram 1} &\equiv \sum_{\substack{g_1, g_2, g_3 \in G \\ rA \in G/A}} |g_1, g_2\rangle \langle g_3| \otimes \langle rA| \equiv \sum_{\substack{g_1, g_2 \in G \\ rA \in G/A}} \alpha_\phi^\triangleright(g_1, g_2)(rA) |g_1, g_2\rangle \langle g_1 g_2| \otimes \langle rA|, \\ \text{Diagram 2} &\equiv \sum_{\substack{g_1, g_2, g_3 \in G \\ rA \in G/A}} |g_3\rangle \langle g_1, g_2| \otimes \langle rA| \equiv \sum_{\substack{g_1, g_2 \in G \\ rA \in G/A}} \alpha_\phi^\triangleright(g_1, g_2)(rA)^{-1} |g_1 g_2\rangle \langle g_1, g_2| \otimes \langle rA|, \end{aligned}$$

<sup>11</sup>Since we do not require it, we shall omit to write any explicit relation between the 2-cochains  $\phi \in C^2(A, \text{U}(1))$  and  $\alpha_\phi^\triangleright \in C^2(G, \text{Fun}(G/A, \text{U}(1)))$ . Nonetheless, it is important to keep in mind that even if  $\phi$  is trivial, the corresponding 2-cochain  $\alpha_\phi^\triangleright$  may not be. For instance, whenever  $A = \{1\}$ , there is no choice of  $\phi$ , and yet we have  $\alpha_1^\triangleright(-, -)(-) = \lambda(-, -, -)$ .

and denote by  $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$  the tensor network operator whose unit cell on the triangular lattice with periodic boundary conditions is given by



We claim that the operator  $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$  transmutes  $\mathbb{H}$  into  $\mathbb{H}^\lambda$ , i.e.,  $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2) \circ \mathbb{H} = \mathbb{H}^\lambda \circ \mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$ . This computation is best carried out invoking the explicit expression

$$\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2) := \sum_{\substack{g \in Z^1(\mathbb{T}_\Delta^2, G) \\ m \in C^0(\mathbb{T}_\Delta^2, G/A)}} \left( \prod_{(v_1, v_2)} \delta_{g(v_1 v_2) \triangleright m(v_2), m(v_1)} \prod_{(v_1, v_2, v_3)} \alpha_\phi^\triangleright(g(v_1 v_2), g(v_2 v_3))(m(v_3))^{e(v_1 v_2 v_3)} \right) |g\rangle \langle g|. \quad (92)$$

Indeed, letting  $v \in V(\mathbb{T}_\Delta^2)$  and  $r_v \in C^0(\mathbb{T}_\Delta^2, G)$  such that  $r_v(v_1) = 1$ , whenever  $v_1 \neq v$ , it immediately follows from condition (90) that

$$\begin{aligned} & \alpha_\phi^\triangleright(g d^{(0)} r_v)(v_1 v_2), (g d^{(0)} r_v)(v_2 v_3) \chi(r_v \triangleright m)(v) \\ &= \frac{\alpha_\phi^\triangleright(g(v_2 v) r_v(v)^{-1}, r_v(v)) \chi(m(v)) \alpha_\phi^\triangleright(g(v_1 v_2), g(v_2 v_3)) \chi(m(v)) \lambda(g(v_1 v_2), g(v_2 v) r_v(v)^{-1}, r_v(v))}{\alpha_\phi^\triangleright(g(v_1 v) r_v(v)^{-1}, r_v(v)) \chi(m(v))} \\ & \alpha_\phi^\triangleright(g d^{(0)} r_v)(v_1 v), (g d^{(0)} r_v)(v v_2) \chi(r_v \triangleright m)(v_2) \\ &= \frac{\alpha_\phi^\triangleright(g(v_1 v), g(v v_2)) \chi(m(v_2)) \alpha_\phi^\triangleright(g(v_1 v) r_v(v)^{-1}, r_v(v)) \chi(g(v v_2) \triangleright m(v_2))}{\alpha_\phi^\triangleright(r_v(v), g(v v_2)) \chi(m(v_2)) \lambda(g(v_1 v) r_v(v)^{-1}, r_v(v), g(v v_2))}, \quad (93) \\ & \alpha_\phi^\triangleright(g d^{(0)} r_v)(v v_1), (g d^{(0)} r_v)(v_1 v_2) \chi(r_v \triangleright m)(v_2) \\ &= \frac{\alpha_\phi^\triangleright(g(v v_1), g(v_1 v_2)) \chi(m(v_2)) \alpha_\phi^\triangleright(r_v(v), g(v v_2)) \chi(m(v_2)) \lambda(r_v(v), g(v v_1), g(v_1 v_2))}{\alpha_\phi^\triangleright(r_v(v), g(v v_1)) \chi(g(v_1 v_2) \triangleright m(v_2))}. \end{aligned}$$

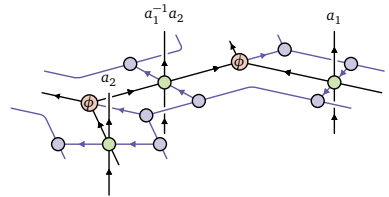
Bringing everything together, one obtains

$$\begin{aligned} & \prod_{(v_1, v_2, v_3)} \alpha_\phi^\triangleright(g d^{(0)} r_v)(v_1 v_2), (g d^{(0)} r_v)(v_2 v_3) \chi(r_v \triangleright m)(v_3))^{e(v_1 v_2 v_3)} \\ &= \lambda(g, r_v) \cdot \prod_{(v_1, v_2, v_3)} \alpha_\phi^\triangleright(g(v_1 v_2), g(v_2 v_3)) \chi(m(v_3))^{e(v_1 v_2 v_3)}, \quad (94) \end{aligned}$$

where  $\lambda(g, r_v)$  was defined in eq. (54). This guarantees that  $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2) \circ \mathbb{H} = \mathbb{H}^\lambda \circ \mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$ , as desired. Notice that the above derivation only relies on (90), which is verified for any  $\text{Vec}_G^\lambda$ -module category, and thus does not depend on a specific choice  $\mathcal{M}(A, \phi)$ . From this case, it would be easy to infer the duality operators transmuted Hamiltonians  $\mathbb{H}^\lambda$  and  $\mathbb{H}^{\lambda'}$  into each other.

Suppose  $\lambda$  is a non-trivial 3-cocycle. It follows from (90) that it is not possible to choose the whole group  $G$  to define a  $\text{Vec}_G^\lambda$ -module category, as otherwise it would require  $\phi$  to be a 2-cochain in  $C^2(G, \text{U}(1))$  trivialising  $\lambda$ , which would be a contradiction. Therefore, a duality operator  $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$  is necessarily associated with a  $\text{Vec}_G^\lambda$ -module category  $\mathcal{M}(A, \phi)$  with  $A$  a proper subgroup of  $G$ . This implies in particular that any duality operator  $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$  necessarily

projects the Hamiltonians  $\mathbb{H}$  and  $\mathbb{H}^\lambda$  onto specific charge sectors. Consider for instance the duality operator  $\mathbb{D}^{\{1\}}(\mathbb{T}_\Delta^2)$  associated with the  $\text{Vec}_G^\lambda$ -module category  $\mathcal{M}(\{1\}, 1) \simeq \text{Vec}_G^\lambda$ , whose module associator evaluates to  $\lambda$  so that (90) merely follows from  $d^{(3)}\lambda = 1$ . Similarly to the 0-form symmetry operator  $\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2)$ , the duality operator  $\mathbb{D}^{\{1\}}(\mathbb{T}_\Delta^2)$  enforces holonomies along both non-contractible cycles to be trivial, and thus acts as a projector onto the singlet sector of the 1-form  $\text{Rep}(G)$  symmetry, making this duality operator non-invertible, and a fortiori non-unitary. Nonetheless, upon restricting to the singlet sector, the operator  $\mathbb{D}^{\{1\}}(\mathbb{T}_\Delta^2)$  does become invertible and relates the spectra of the dual Hamiltonians. In order to access additional charge sectors, one option is to decorate the topological surface with topological lines labelled by elements in  $G$ , in the same vein as for the symmetry operator  $\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2)$  (see sec. 3.5). However, this is insufficient as the duality operator typically imposes additional constraints on the charge sectors that depend on the choice of  $\lambda$ . In the same vein, employing a different duality operator associated with a distinct  $\mathcal{M}(A, \phi)$  could a priori permit to access additional charge sectors, as it would not necessarily project onto the singlet sector, but there would still be additional constraints that depend on  $\lambda$ . Indeed, suppose we are restricting to charge sectors  $(a_1, a_2) \in A \times A$ . It follows from the cocycle condition (90) that the duality explicitly depends on the complex number<sup>12</sup>



$$= \sum_{rA \in G/A} \frac{\alpha_\phi^\lambda(a_1, a_1^{-1}a_2)(rA)}{\alpha_\phi^\lambda(a_1^{-1}a_2, a_1)(rA)}, \quad (95)$$

where the l.h.s. should be interpreted as a single unit cell of the tensor network whose opposite indices are contracted so as to implement periodic boundary conditions. Consider now shifting the summation variable as follows:  $rA \mapsto x \triangleright rA$ , for any  $x \in G$ . But, repeatedly using the cocycle conditions (90) and  $d^{(3)}\lambda = 1$ , as well as the fact that  $a \triangleright rA = rA$  for every  $a \in A$  and  $rA \in G/A$ , one finds that

$$\frac{\alpha_\phi^\lambda(a_1, a_1^{-1}a_2)(x \triangleright rA)}{\alpha_\phi^\lambda(a_1^{-1}a_2, a_1)(x \triangleright rA)} = \frac{t_x(\lambda)(a_1, a_1^{-1}a_2)}{t_x(\lambda)(a_1^{-1}a_2, a_1)} \cdot \frac{\alpha_\phi^\lambda(a_1, a_1^{-1}a_2)(rA)}{\alpha_\phi^\lambda(a_1^{-1}a_2, a_1)(rA)}, \quad (96)$$

where  $t_x : Z^3(G, U(1)) \rightarrow Z^2(G, U(1))$  is defined via [76]

$$t_x(\lambda)(g_1, g_2) := \frac{\lambda(x, g_1, g_2)\lambda(g_1, g_2, x)}{\lambda(g_1, x, g_2)}, \quad (97)$$

for every  $g_1, g_2, x \in G$ . Additional manipulations reveal that

$$\frac{t_x(\lambda)(a_1, a_1^{-1}a_2)}{t_x(\lambda)(a_1^{-1}a_2, a_1)} = t_{a_1, a_1^{-1}a_2}^2(\lambda)(x) = t_{a_1, a_2}^2(\lambda)(x), \quad (98)$$

where  $t_{x,y}^2 : Z^3(G, U(1)) \rightarrow Z^1(G, U(1))$  is defined via

$$t_{x_1, x_2}^2(\lambda)(g) := \frac{t_{x_1}(\lambda)(x_2, g)}{t_{x_1}(\lambda)(g, x_2)}, \quad (99)$$

<sup>12</sup>To establish this fact, we are employing a higher dimensional version of the trick presented in eq. (15).



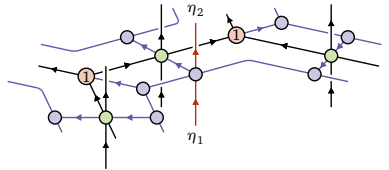
for every  $g, x_1, x_2 \in G$ . Bringing everything together, we showed that within the charge sector  $(a_1, a_2) \in A \times A$ ,  $\mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2) = t_{a_1, a_2}^2(\lambda)(x) \cdot \mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2)$ , for every  $x \in G$ . Since it is true for every  $x \in G$  and  $t_{a_1, a_2}^2(\lambda) \in \hat{G}$ , we further have

$$\mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2) = \left( \frac{1}{|G|} \sum_{x \in G} t_{a_1, a_2}^2(\lambda)(x) \right) \mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2) = \begin{cases} \mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2), & \text{if } t_{a_1, a_2}^2(\lambda) = 1, \\ 0, & \text{otherwise,} \end{cases} \quad (100)$$

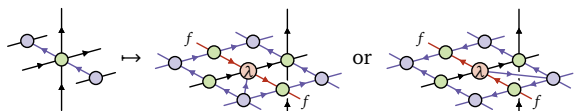
which means that charge sectors  $(a_1, a_2) \in A \times A$ , for which  $t_{a_1, a_2}^2(\lambda)$  is not the trivial representation are projected out. In order to access all the charge sectors—regardless of the choice of duality operator—it is required to allow for the duality operator to modify the closed boundary conditions. We compute in the following the relevant permutation of topological sectors.

### 3.5 Permutation of topological sectors

As alluded to above, in order to lift a duality operator  $\mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2)$  to a unitary transformation, it is required to characterise the interplay between boundary conditions and charge sectors. First, we need to modify the duality operators so as to accommodate the presence of symmetry twists, the same way we modified the symmetry operators. Since for every group  $G$  and  $\lambda$ , the only duality operator whose existence we can guarantee is  $\mathbb{D}^{\{1\}}(\mathbb{T}_\Delta^2)$ , we shall focus on this case throughout the remainder of this section. Suppose a 1-form symmetry twist is inserted at the vertex  $v_0$ . Specialising to the triangular lattice, we define the duality operator  $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{1\}}(\mathbb{T}_\Delta^2) : \mathcal{H}^{\eta_1} \rightarrow \mathcal{H}^{\eta_2}$  by replacing the unit cell (91) of its tensor network parametrisation located at  $v_0$  by the following one:


(101)

where we are employing the tensors defined in eq. (84). It is clear from the previous derivations that  $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{1\}}(\mathbb{T}_\Delta^2) \circ \mathbb{H}^{1, \eta_1} = \mathbb{H}^{\lambda, \eta_2} \circ \mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{1\}}(\mathbb{T}_\Delta^2)$ . Now, recall that the duality operator  $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{1\}}(\mathbb{T}_\Delta^2)$  projects out states  $|g\rangle$  such that  $\text{hol}_{\gamma_1}(g)$  and  $\text{hol}_{\gamma_2}(g)$  are not the identity element. We can lift this limitation by decorating  $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{1\}}(\mathbb{T}_\Delta^2)$  by appropriate topological lines in  $G$ , thereby allowing to consider any charge sector  $(g_1, g_2) \in G \times G$ . More formally, the duality operator  $\mathbb{D}^{\{1\}}(\mathbb{T}_\Delta^2)$  being associated with the  $\text{Vec}_G^\lambda$ -module category  $\mathcal{M}(\{1\}, 1) \simeq \text{Vec}_G^\lambda$ , it can be decorated with 't Hooft lines labelled by simple objects in the category  $(\text{Vec}_G^\lambda)_{\mathcal{M}(\{1\}, 1)}^\vee \simeq \text{Vec}_G^\lambda$ . Therefore, although the 't Hooft lines are effectively labelled by group elements in  $G$ , their action is more subtle than in (70) and depends in particular on  $\lambda$ . Correspondingly, it is not enough to implement the group multiplication at junctions in order to ensure topological invariance of network of lines. Since any network of 't Hooft lines boils down to a collection of single 't Hooft loops, by virtue of the group  $G$  being Abelian, let us focus for now on a single 't Hooft loop labelled by  $f \in G$ . Consider an oriented closed loop in  $Z_1(\mathbb{T}_\Delta^2, \mathbb{Z})$ . For every edge in this closed loop, we modify the corresponding local patch of the tensor network as follows


(102)

depending on the orientation of the edge relative to that of the loop, where we are using tensors adapted from eq. (72) in an obvious way and

$$\begin{aligned} \text{Diagram 1} &\equiv \sum_{f_1, f_2, g_1, g_2, x \in G} f_1 \rightarrow f_2 |f_2\rangle\langle f_1| \otimes |g_2\rangle\langle g_1| \otimes |x\rangle \equiv \sum_{f, g, x \in G} \lambda(g, x, f) |f\rangle\langle f| \otimes |g\rangle\langle g| \otimes |x\rangle, \\ \text{Diagram 2} &\equiv \sum_{f_1, f_2, g_1, g_2, x \in G} f_1 \rightarrow f_2 |f_1\rangle\langle f_2| \otimes |g_2\rangle\langle g_1| \otimes |x\rangle \equiv \sum_{f, g, x \in G} \lambda(g, x, f)^{-1} |f\rangle\langle f| \otimes |g\rangle\langle g| \otimes |x\rangle. \end{aligned}$$

One can check that the cocycle condition  $d^{(3)}\lambda = 1$  ensures that the resulting duality operator is left invariant under continuous deformations of the closed loop. Although the definition of these operators may seem somewhat ad hoc, it is actually a reinterpretation of another tensor network calculus that is closely related to that of fusion 2-categories [22]. For instance, given the data of a  $\text{Vec}_G^\lambda$ -module endofunctor of  $\mathcal{M}(A, \phi)$ , there is a systematic way of constructing a tensor network operator, which boils down to that defined in eq. (102) whenever  $\mathcal{M}(A, \phi) \simeq \text{Vec}_G^\lambda$ . In order to construct the operator  $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^\perp(\mathbb{T}_\Delta^2)[g_1, g_2]$  projecting onto a charge sector labelled by  $[f] \in H^1(G, \mathbb{T}_\Delta^2)$  with holonomies  $\text{hol}_{\gamma_1}(f) = g_1$  and  $\text{hol}_{\gamma_2}(f) = g_2$ , it suffices to insert two loop operators as defined above along both non-contractible cycle of the torus. However, as mentioned above, the junction of these two loops operators requires some care so as to be topologically invariant. Immediately specialising to our configuration of interest, one finds that the duality operator  $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{1\}}(\mathbb{T}_\Delta^2)[g_1, g_2] : \mathcal{H}^{\eta_1} \rightarrow \mathcal{H}^{\eta_2}$  explicitly depends on the complex number

$$\begin{aligned} &\text{Diagram (103)} \\ &= \sum_{x \in G} \frac{\lambda(g_1, g_1^{-1}g_2, g_1x) \lambda(g_2, x, g_1) \lambda(xg_1, g_2g_1^{-1}, g_1)}{\lambda(g_1^{-1}g_2, g_1, x) \lambda(g_1, x, g_2) \lambda(x, g_1, g_1^{-1}g_2)} \eta_1^\vee(g_1x) \eta_2(g_1x) \\ &= \sum_{x \in G} t_{g_1, g_2}^2(\lambda)(x) \eta_1(g_1x) \eta_2^\vee(g_1x), \end{aligned} \tag{103}$$

where, as in eq. (95), the l.h.s. is interpreted as a single unit cell of the tensor network whose opposite indices are contracted so as to implement periodic boundary conditions. In addition to the unit cell (101) and tensors of the form (102), one introduced the following tensors that implement the fusion of topological lines in  $\text{Vec}_G^\lambda$ :

$$\text{Diagram 3} \equiv \sum_{f_1, f_2, f_3, x \in G} f_1 \rightarrow f_2 |f_1, f_2\rangle\langle f_3| \otimes |x\rangle \equiv \sum_{f_1, f_2, x \in G} \lambda(x, f_2, f_1) |f_1, f_2\rangle\langle f_1f_2| \otimes |x\rangle,$$

$$\text{Diagram} \equiv \sum_{\substack{f_1, f_2 \\ f_3, x \in G}} \text{Diagram} |f_3\rangle\langle f_1, f_2| \otimes |x\rangle \equiv \sum_{f_1, f_2, x \in G} \lambda(x, f_2, f_1)^{-1} |f_1 f_2\rangle\langle f_1, f_2| \otimes |x\rangle.$$

It simply follows from the orthogonality of representations in  $G$  that, in order for the operator  $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{\mathbb{1}\}}(\mathbb{T}_{\Delta}^2)[g_1, g_2]$  not to vanish, we must have  $\eta_2(-) = t_{g_1, g_2}^2(\lambda)(-)\eta_1(-)$ . Bringing everything together, one obtains the following mapping of topological sectors:

$$(g_1, g_2, \eta) \mapsto (g_1, g_2, \eta_{g_1, g_2}^{\lambda}) \quad \text{with} \quad \eta_{g_1, g_2}^{\lambda}(-) := t_{g_1, g_2}^2(\lambda)(-)\eta(-). \quad (104)$$

In particular, it follows from the normalisation of  $\lambda$  that the topological sectors of the form  $(\eta, \mathbb{1}, \mathbb{1})$  are mapped to themselves under  $\mathbb{D}^{\mathcal{A}_{\psi}}(\mathbb{T}_{\Delta}^2)$  for any  $\eta \in \widehat{G}$ . Finally, promoting the 1-form symmetry twist to a dynamical degree of freedom, the invertible duality operators between compatible topological sectors constructed above organise into a unitary operator onto the resulting total Hilbert space.

### 3.6 Examples

Let illustrate our formalism with a couple of examples. Let  $G = \mathbb{Z}_2$ . As in sec. 2.4, we write the multiplication rule of  $\mathbb{Z}_2 = \{0, 1\}$  as addition modulo 2. A normalised representative of the cohomology class  $[\lambda]$  generating  $H^3(\mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$  is provided by  $\lambda(g_1, g_2, g_3) = (-1)^{g_1 g_2 g_3}$ , for every  $g_1, g_2, g_3 \in \mathbb{Z}_2$ . Given a two-dimensional quantum lattice model with a 0-form  $\mathbb{Z}_2$  symmetry, we have two gauging procedures at our disposal, namely the untwisted one and the one twisted by the representative  $\lambda$  defined above. The two Hamiltonians resulting from these two gauging procedures possess a  $2\text{Rep}(\mathbb{Z}_2)$  symmetry. Moreover, since they only differ in the choice of 3-cocycle  $\lambda$ , the corresponding duality operator is encoded into the unique simple object in the 2-category  $2\text{Rep}^{\lambda}(\mathbb{Z}_2)$ , which is  $\text{Vec}_{\mathbb{Z}_2}^{\lambda}$  as a module category over itself. Let us consider the special Hamiltonian whose local operators (52) are obtained by choosing the non-vanishing complex coefficients  $h(\mathfrak{g}, \mathfrak{r}_v)$  to be equal to  $-1$ , for every  $\mathfrak{g} \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$  and  $\mathfrak{r}_v \in C^0(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$  such that  $\mathfrak{r}_v(v) = 1$ , i.e.,

$$\mathbb{H} = - \sum_v \prod_{e \supset v} X_e. \quad (105)$$

Recall that the kinematical Hilbert space is spanned by states  $|g\rangle$ , where  $g \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$ . Making the kinematical constraint  $\ker d^{(1)}g = 0$  dynamical would result in the celebrated (2+1)d toric code model [43]. Let us briefly comment on the  $2\text{Rep}(\mathbb{Z}_2)$  symmetry structure. By definition, there are two topological surface operators, namely the identity operator and the condensation operator<sup>13</sup>

$$\mathbb{U}^{\{\mathbb{1}\}}(\mathbb{T}_{\Delta}^2) = \sum_{\substack{g \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2) \\ m \in C^0(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)}} \delta_{g, d^{(0)}m} |g\rangle\langle g| = \frac{1}{2^{P(\mathbb{T}_{\Delta}^2)}} \left( \bigotimes_v \langle + | \right) \prod_e (\mathbb{1} + Z_{\partial_- e} Z_e Z_{\partial_+ e}) \left( \bigotimes_v | + \rangle \right), \quad (106)$$

where  $|+\rangle := \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ . Decorating the trivial surface operator  $\mathbb{U}^{\mathbb{Z}_2}(\mathbb{T}_{\Delta}^2)$  with the network of Wilson lines encoded into the 1-cocycle  $\mathfrak{X} \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$  results into the surface operator<sup>14</sup>

$$\mathbb{U}^{\mathbb{Z}_2}(\mathbb{T}_{\Delta}^2)[\mathfrak{X}] \equiv \mathfrak{X}(\mathbb{T}_{\Delta}^2) = \sum_{g \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)} \exp \left( i\pi \int_{\mathbb{T}_{\Delta}^2} \mathfrak{X} \smile g \right) |g\rangle\langle g|. \quad (107)$$

<sup>13</sup>Recall that  $P(\mathbb{T}_{\Delta}^2) = E(\mathbb{T}_{\Delta}^2) - V(\mathbb{T}_{\Delta}^2)$  since the Euler characteristic of  $\mathbb{T}^2$  equals 0.

<sup>14</sup>Here, we are exploiting the isomorphism  $\mathbb{Z}_2 \cong \widehat{\mathbb{Z}_2}$  in order to simplify the definition of the cup product (60).

Decorating  $\mathbb{U}^{\{1\}}(\mathbb{T}_{\Delta}^2)$  with the network of 't Hooft lines encoded into the 1-cocycle  $f \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$  results into the surface operator

$$\mathbb{U}^{\{1\}}(\mathbb{T}_{\Delta}^2)[f] = \sum_{\substack{g \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2) \\ m \in C^0(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)}} \delta_{g, d^{(0)}m + f} |g\rangle \langle g| = \frac{1}{2^{\mathcal{P}(\mathbb{T}_{\Delta}^2)}} \left( \bigotimes_v \langle + | \right) \prod_e \left( \mathbb{I} + (-1)^{f(e)} Z_{\partial_- e} Z_e Z_{\partial_+ e} \right) \left( \bigotimes_v | + \rangle \right).$$

Replacing in the definition of the local operators the trivial 3-cocycle by  $\lambda$ , while keeping everything else the same, yields the Hamiltonian  $\mathbb{H}^{\lambda}$ . Note that  $\mathbb{H}^{\lambda}$  does not admit a particularly compact expression. However, specialising to the scenario where  $\mathbb{T}_{\Delta}^2$  is the triangular lattice, one can perform the unitary transformation [68]

$$U := \sum_g \int_{\mathbb{T}_{\Delta}^2} i^g |g\rangle \langle g|, \quad (108)$$

so as to obtain

$$U \mathbb{H}^{\lambda} U^{\dagger} = \sum_v \prod_{e \supset v} X_e \prod_{\substack{(v_1 v_2) \\ (v_1 v v_2) \\ (v_1 v_2 v)}} S_{(v_1 v_2)}, \quad (109)$$

where  $S = |0\rangle \langle 0| + i|1\rangle \langle 1|$ . At this point, making the kinematical constraint  $\ker d^{(1)}g = 0$  dynamical would result in the Hamiltonian of the so-called *double semion* model [52, 58], which is the Hamiltonian realisation of the Turaev–Viro–Barrett–Westbury state-sum invariant with input fusion category  $\text{Vec}_{\mathbb{Z}_2}^{\lambda}$  [15, 71]. Assuming periodic boundary conditions for both  $\mathbb{H}$  and  $\mathbb{H}^{\lambda}$ , the duality operator  $\mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2)$  transmuting the Hamiltonians  $\mathbb{H}$  and  $\mathbb{H}^{\lambda}$  into each other takes the following form:

$$\begin{aligned} \mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2) &= \sum_{\substack{g \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2) \\ m \in C^0(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)}} \delta_{g, d^{(0)}m} \left( \prod_{(v_1 v_2 v_3)} \lambda(g(v_1 v_2), g(v_2 v_3), m(v_3))^{\epsilon(v_1 v_2 v_3)} \right) |g\rangle \langle g| \\ &= \sum_{m \in C^0(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)} \left( \prod_{(v_1 v_2 v_3)} (-1)^{m(v_1)m(v_3)(m(v_2)+1)} \right) |d^{(0)}m\rangle \langle d^{(0)}m|. \end{aligned} \quad (110)$$

One can now verify that  $\mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2) \circ \mathbb{H} = \mathbb{H}^{\lambda} \circ \mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2)$ . What about the mapping of topological sectors? Since  $\lambda(g_1, g_2, g_3) = \lambda(g_2, g_3, g_1) = \lambda(g_3, g_1, g_2)$ , for every  $g_1, g_2, g_3 \in \mathbb{Z}_2$ , it turns out that  $t_{g_1, g_2}^2(x) = 1$ , for every  $x \in \mathbb{Z}_2$ . Therefore, any 1-form symmetry twisted boundary condition is exceptionally preserved under the duality transmuting  $\mathbb{H}$  into  $\mathbb{H}^{\lambda}$ . Naturally,  $\mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2)$  is not a unitary operator on the total kinematical Hilbert space, since it projects both  $\mathbb{H}$  and  $\mathbb{H}^{\lambda}$  onto the singlet sector of the 1-form  $\text{Rep}(\mathbb{Z}_2)$ -symmetry. In order to obtain the unitary operator relating the full spectra of  $\mathbb{H}$  and  $\mathbb{H}^{\lambda}$ , it is required to decorate  $\mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2)$  with networks of 't Hooft lines, following the procedure described in sec. 3.5.

Since the previous example is not accompanied with a non-trivial permutation of topological sectors, let us briefly consider a slightly more complicated example. Let  $G = \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \ni (g^{(1)}, g^{(2)}, g^{(3)}) \equiv g$ . There is a normalised representative of a cohomology class in  $H^3(\mathbb{Z}_2^3, \text{U}(1))$  defined by

$$\lambda(g_1, g_2, g_3) = \exp(i\pi g_1^{(1)} g_2^{(2)} g_3^{(3)}), \quad (111)$$

for every  $g_1, g_2, g_3 \in \mathbb{Z}_2^3$ . It follows from the general derivation that upon restricting to the charge sector  $(g_1, g_2) \in \mathbb{Z}_2^3 \times \mathbb{Z}_2^3$ , the duality operator  $\mathbb{D}_{1 \rightarrow \eta}^{\{a\}}(\mathbb{T}_\Delta^2)[g_1, g_2]$  would be non-vanishing if and only if  $\eta(-) = \mathbf{t}_{g_1, g_2}^2(-)$  with

$$\mathbf{t}_{g_1, g_2}^2(x) = \exp \left( i\pi \begin{vmatrix} g_1^{(1)} & g_2^{(1)} & x^{(1)} \\ g_1^{(2)} & g_2^{(2)} & x^{(2)} \\ g_1^{(3)} & g_2^{(3)} & x^{(3)} \end{vmatrix} \right), \quad (112)$$

for every  $x \in \mathbb{Z}_2^3$ . Therefore, for a generic charge sector, the 1-form symmetry twisted boundary condition of the Hamiltonian  $\mathbb{H}^\lambda$  would need to be non-trivial.

### 3.7 Higher category theoretic underpinnings

Transposing sec. 2.5, let us sketch the mathematical formalism underlying the results presented above, following the approach of ref. [27]. Consider a two-dimensional quantum lattice model with a non-anomalous Abelian symmetry group  $G$ , which admits topological surfaces valued in  $G$ . The requirement that we should be able to construct topological interfaces between surfaces and topological junctions of interfaces invites us to rather consider a higher category theoretic structure, namely the *fusion 2-category*  $2\text{Vec}_G$  of  $G$ -graded 2-vector spaces, where a 2-vector space is defined to be a finite semisimple  $\mathbb{C}$ -linear (1-)category. In particular, simple objects in  $2\text{Vec}_G$  are 2-vector spaces  $\text{Vec}_g$ , for every  $g \in G$ , such that  $(\text{Vec}_{g_1})_{g_2} = \delta_{g_1, g_2} \text{Vec}$  with hom-categories  $\text{Hom}_{2\text{Vec}_G}(\text{Vec}_{g_1}, \text{Vec}_{g_2}) \cong \delta_{g_1, g_2} \text{Vec}$ , while the monoidal structure reads  $\text{Vec}_{g_1} \odot \text{Vec}_{g_2} = \text{Vec}_{g_1 g_2}$ . Moreover, the monoidal associator evaluates to the identity 1-morphism, and it satisfies the pentagon axiom up to an invertible modification, known as the *monoidal pentagonator*, which evaluates to the identity 2-morphism. Given a non-anomalous symmetry  $G$  in (2+1)d—or rather, a symmetry  $2\text{Vec}_G$ —it is always possible to gauge a sub-symmetry  $A$ , where  $A$  is a subgroup of  $G$ . As in (1+1)d, we identify several ways to perform such a gauging, which are now labelled by cohomology classes in  $H^3(A, \text{U}(1))$ . Whenever the whole symmetry is gauged, the resulting theory possesses a  $2\text{Rep}(G)$ -symmetry, i.e., a symmetry structure encompassing both 0-form symmetry operators labelled by 2-representations of the group  $G$ , which compose according to the fusion of  $\text{Vec}_G$ -module categories [41], as well as 1-form symmetry operators labelled by  $\text{Vec}_G$ -module functors. Crucially, even though the group is Abelian,  $2\text{Vec}_G$  and  $2\text{Rep}(G)$  are clearly not equivalent, even as 2-categories.

Mathematically, a pair  $(A, \lambda)$  consisting of a subgroup  $A \subseteq G$  and a normalised representative  $\lambda$  of a cohomology class  $[\lambda] \in H^3(A, \text{U}(1))$  specifies a so-called *module 2-category* over  $2\text{Vec}_G$  [21, 22]. Let  $\mathcal{M}(A, \lambda)$  be the (finite semisimple) 2-category whose simple objects are 2-vector spaces  $\text{Vec}_{rA}$ , for every  $rA \in G/A$ . It is equipped with the (left)  $2\text{Vec}_G$ -module structure provided by the  $G$ -action of left cosets in  $G/A$ , i.e.  $\text{Vec}_g \triangleright \text{Vec}_{rA} := \text{Vec}_{g \triangleright rA} \simeq \text{Vec}_{(gr)A}$ , such that the module associator evaluates to the identity 1-morphisms, and it satisfies the pentagon axiom up to an invertible modification  $\pi_\lambda^\triangleright$  referred as the *module pentagonator*. Components of the module pentagonator  $\pi_\lambda^\triangleright$  evaluate to the 3-cochain  $\pi_\lambda^\triangleright \in C^3(G, \text{Fun}(G/A, \text{U}(1)))$  defined in terms of the 3-cocycle  $\lambda$  via

$$\pi_\lambda^\triangleright(g_1, g_2, g_3)(rA) := \lambda(a_{g_1, g_2 g_3 \triangleright rA}, a_{g_2, g_3 \triangleright rA}, a_{g_3, rA}), \quad (113)$$

for every  $g_1, g_2, g_3 \in G$  and  $rA \in G/A$ . It follows from the 3-cocycle condition satisfied by  $\lambda$  that

$$\frac{\pi_\lambda^\triangleright(g_2, g_3, g_4)(rA) \pi_\lambda^\triangleright(g_1, g_2 g_3, g_4)(rA) \pi_\lambda^\triangleright(g_1, g_2, g_3)(g_4 \triangleright rA)}{\pi_\lambda^\triangleright(g_1 g_2, g_3, g_4)(rA) \pi_\lambda^\triangleright(g_1, g_2, g_3 g_4)(rA)} = 1, \quad (114)$$

for every  $g_1, g_2, g_3, g_4 \in G$  and  $rA \in G/A$ . Identity (114) guarantees the so-called *associativity axiom* of  $\mathcal{M}(A, \lambda)$  [21, 22].

Performing the  $\lambda$ -twisted gauging of a symmetry  $2\text{Vec}_G$  amounts to picking the  $2\text{Vec}_G$ -module 2-category  $\mathcal{M}(G, \lambda) \equiv 2\text{Vec}^\lambda$ , which is equivalent to  $2\text{Vec}$  as a 2-category but whose  $2\text{Vec}_G$ -module structure depends on  $\lambda$ . The same data encodes a choice of fiber 2-functor  $2\text{Vec}_G \rightarrow 2\text{Vec}$ . The symmetry structure of the theory resulting from the  $\lambda$ -twisted gauging of the subsymmetry  $A$  is provided by the fusion 2-category  $(2\text{Vec}_G)_{\mathcal{M}(A, \lambda)}^\vee := 2\text{Fun}_{2\text{Vec}_G}(\mathcal{M}(A, \lambda), \mathcal{M}(A, \lambda))$  defined as the category of  $2\text{Vec}_G$ -module 2-functors from  $\mathcal{M}(A, \lambda)$  to itself, with fusion structure provided by the composition of  $2\text{Vec}_G$ -module 2-functors [4, 5, 22, 27]. In particular, we have  $(2\text{Vec}_G)_{2\text{Vec}^\lambda}^\vee \simeq 2\text{Rep}(G)$ , for any  $[\lambda] \in H^3(G, \text{U}(1))$ , as expected.

We can think of the  $2\text{Rep}(G)$ -symmetric models defined in sec. 3.1 as resulting from the  $\lambda$ -twisted gauging of  $G$ -symmetric models. Therefore, two models with Hamiltonians  $\mathbb{H}$  and  $\mathbb{H}^\lambda$ —only differing in a choice of 3-cocycle  $\lambda$ —are associated with the  $2\text{Vec}_G$ -module categories  $2\text{Vec}$  and  $2\text{Vec}^\lambda$ , respectively. This can be made very explicit employing the formalism of ref. [27], from which it also follows that a duality operator transmuting  $\mathbb{H}$  into  $\mathbb{H}^\lambda$  corresponds to a choice of simple object in  $2\text{Fun}_{2\text{Vec}_G}(2\text{Vec}, 2\text{Vec}^\lambda)$ , which can be shown to be equivalent to  $2\text{Rep}^\lambda(G)$ , as expected.

### 3.8 Self-duality and lattice higher gauge theories

As in (1+1)d, a particularly attractive feature of the dualities considered in this section is that they preserve the symmetry structure, namely  $2\text{Rep}(G)$  for some finite Abelian group  $G$ . As a matter of fact, it seems that all non-trivial dualities of bosonic two-dimensional quantum lattice models preserving symmetry structures are of this flavour, i.e., dualities between models only differing in a choice of fiber 2-functor over some input fusion 2-category [28]. This property suggests that it may be possible to define a model that is self-dual with respect to such a duality. Given such a self-dual model, one could then consider promoting the self-duality to a genuine internal symmetry, which mathematically amounts to considering an extension of the symmetry fusion 2-category.

Let us illustrate this principle with the first duality studied in sec. 3.6. There, we constructed two manifestly  $2\text{Rep}(\mathbb{Z}_2)$ -symmetric Hamiltonians  $\mathbb{H}$  and  $\mathbb{H}^\lambda$ , which are associated with fiber 2-functors  $2\text{Vec}_{\mathbb{Z}_2} \rightarrow 2\text{Vec}$  and  $2\text{Vec}_{\mathbb{Z}_2} \rightarrow 2\text{Vec}^\lambda$ , respectively. Clearly, the Hamiltonian  $\mathbb{H} + \mathbb{H}^\lambda$  is left invariant under the action of the duality operator (110), making  $\mathbb{H} + \mathbb{H}^\lambda$  self-dual. We claim that promoting this self-duality to a genuine internal symmetry results in the fusion 2-category  $2\text{Rep}(\mathbb{Z}_2^{[0] \bullet_\mu \mathbb{Z}_2^{[1]}})$  of 2-representations of a 2-group  $\mathbb{Z}_2^{[0] \bullet_\mu \mathbb{Z}_2^{[1]}}$  with homotopy group  $\mathbb{Z}_2$  both in degree one and two. Generally, a 2-group is defined as a monoidal category such that every object admits a weak inverse and every morphism is invertible. In particular, given two finite Abelian groups  $G$  and  $H$ , and a normalised representative 3-cocycle  $\mu$  of a cohomology class in  $H^3(G, H)$ , the 2-group  $G^{[0] \bullet_\mu H^{[1]}}$  is the monoidal category such that the objects form the group  $G$ , automorphisms of the monoidal unit forms the group  $H$ , while the monoidal associator is provided by  $\mu$ .<sup>15</sup> The fusion 2-category  $2\text{Rep}(G^{[0] \bullet_\mu H^{[1]}})$  is then defined as the fusion 2-category of pseudofunctors from the so-called delooping of the 2-group to  $2\text{Vec}$ .<sup>16</sup> The underlying 2-category was studied in detail in ref. [32]:

$$2\text{Rep}(G^{[0] \bullet_\mu H^{[1]}}) \simeq \bigoplus_{\chi \in \widehat{H}} 2\text{Rep}^{\chi \circ \mu}(G). \quad (115)$$

We are now ready to show that the self-dual Hamiltonian  $\mathbb{H} + \mathbb{H}^\lambda$  has a  $2\text{Rep}(\mathbb{Z}_2^{[0] \bullet_\mu \mathbb{Z}_2^{[1]}})$  symmetry structure, where  $\mu(g_1, g_2, g_3) = 1 \in \mathbb{Z}_2$  if  $g_1 = g_2 = g_3 = 1$ , and 0 otherwise.

<sup>15</sup>More generally, the group  $H$  can be endowed with the structure of a  $G$ -module, but we choose the module structure to be trivial.

<sup>16</sup>Similarly, the fusion 2-category of 2-representations of a finite group  $G$  can be defined as the fusion 2-category of pseudofunctors from the delooping of the group to  $2\text{Vec}$ .

Our strategy is to demonstrate that  $\mathbb{H} + \mathbb{H}^\lambda$  has a lattice *higher gauge theory* interpretation [9, 9, 11, 25, 26, 46, 77] and results from gauging a two-dimensional quantum lattice model with a  $2\text{Vec}_{\mathbb{Z}_2^{[0]}, \mu \mathbb{Z}_2^{[1]}}$  symmetry [27].

We begin by constructing a family of two-dimensional quantum lattice models with a lattice higher gauge theory interpretation that possess a  $2\text{Rep}(G^{[0]}, \mu H^{[1]})$  symmetry, before specialising to  $2\text{Rep}(\mathbb{Z}_2^{[0]}, \mu \mathbb{Z}_2^{[1]})$ . The kinematical Hilbert space is spanned by vectors  $|g, h\rangle$ , where  $g \in Z^1(\mathbb{T}_\Delta^2, G)$  and  $h \in C^2(\mathbb{T}_\Delta^2, H)$ . Let  $\mu$  be a normalised representative 3-cocycle of a cohomology class  $[\mu] \in H^3(G, H)$ . Given a vertex  $v \in V(\mathbb{T}_\Delta^2)$ , we denote by  $r_v \in C^0(\mathbb{T}_\Delta^2, G)$  a function  $V(\mathbb{T}_\Delta^2) \rightarrow G$  such that  $r_v(v_1) = 1$  whenever  $v_1 \neq v$ . Moreover, we denote by  $\mathfrak{z}_v \in C^1(\mathbb{T}_\Delta^2, H)$  a function  $E(\mathbb{T}_\Delta^2) \rightarrow H$  such that  $\mathfrak{z}_v(v_1 v_2) = 1$  whenever  $v \notin \partial(v_1 v_2)$ . We consider the following local operators acting on the kinematical Hilbert space:

$$\mathbb{h}_{v,n} := \sum_{\substack{g \in Z^1(\mathbb{T}_\Delta^2, G) \\ h \in C^2(\mathbb{T}_\Delta^2, H)}} \sum_{r_v, \mathfrak{z}_v} h_n(g, h, r_v, \mathfrak{z}_v) |g d^{(0)} r_v, h d^{(1)} \mathfrak{z}_v \mu(g, r_v)^{-1}\rangle \langle g, h|, \quad (116)$$

where  $n$  is some label,  $h_n(g, h, r_v, \mathfrak{z}_v)$  are arbitrary complex coefficients,  $h d^{(1)} \mathfrak{z}_v \mu(g, r_v)^{-1}$  is by definition the function  $P[\mathbb{T}_\Delta^2] \rightarrow G$  such that

$$(h d^{(1)} \mathfrak{z}_v \mu(g, r_v)^{-1})(p) = \begin{cases} h(p) \mathfrak{z}_v(v_2 v_3) \mathfrak{z}_v(v_1 v_2)^{-1} \mu(g(v_1 v_2), g(v_2 v_3) r_v(v)^{-1}, r_v(v))^{-1}, & \text{if } p \equiv (v_1 v_2 v_3), \\ h(p) \mathfrak{z}_v(v_2 v_3) \mathfrak{z}_v(v_1 v_2) \mu(g(v_1 v_2) r_v(v)^{-1}, r_v(v), g(v_2 v_3)), & \text{if } p \equiv (v_1 v v_2), \\ h(p) \mathfrak{z}_v(v v_1) \mathfrak{z}_v(v v_2)^{-1} \mu(r_v(v), g(v v_1), g(v_1 v_2))^{-1}, & \text{if } p \equiv (v v_1 v_2), \\ h(p), & \text{otherwise,} \end{cases} \quad (117)$$

where we defined the 2-cochain  $\mu(g, r_v)$  analogously to the phase factor (54).

We claim that any Hamiltonian built from local operators (116) possesses a  $2\text{Rep}(G^{[0]}, \mu H^{[1]})$  symmetry. Let us focus on the 0-form symmetry operators. It follows from eq. (115) that simple objects  $2\text{Rep}(G^{[0]}, \mu H^{[1]})$  are labelled by pairs  $(\chi \in \widehat{H}, \mathcal{M}(A, \phi) \in 2\text{Rep}(\chi^\circ, \mu(G)))$ . In particular, whenever  $\chi$  is the trivial representation of  $H$ , the topological surface operator is labelled by a 2-representation of  $G$ , acts as in eq. (75), and commutes with local operators  $\mathbb{h}_{v,n}$  for the same reasons as before. More generally, let us denote by  $\mathbb{U}^{A, \phi, \chi}(\mathbb{T}_\Delta^2)$  the topological surface operator associated with the pair  $(\chi, \mathcal{M}(A, \phi))$ :

$$\mathbb{U}^{A, \phi, \chi}(\mathbb{T}_\Delta^2) := \sum_{\substack{g \in Z^1(\mathbb{T}_\Delta^2, G) \\ h \in C^2(\mathbb{T}_\Delta^2, H) \\ m \in C^0(\mathbb{T}_\Delta^2, G/A)}} \left( \prod_{(v_1 v_2)} \delta_{g(v_1 v_2) \triangleright m(v_2), m(v_1)} \cdot \prod_{(v_1 v_2 v_3)} [\chi(h) \alpha_\phi^>(g(v_1 v_2), g(v_2 v_3))(m(v_3))]^{e(v_1 v_2 v_3)} \right) |g, h\rangle \langle g, h|. \quad (118)$$

Letting  $r_v \in C^0(\mathbb{T}_\Delta^2, G)$  and  $\mathfrak{z}_v \in C^1(\mathbb{T}_\Delta^2, H)$  defined as above, it follows from eq. (94) as well as  $\prod_{(v_1 v_2 v_3)} \chi(d^{(1)} \mathfrak{z}_v) = 1$  that

$$\begin{aligned} & \prod_{(v_1 v_2 v_3)} [\chi(h d^{(1)} \mathfrak{z}_v \mu(g, r_v)^{-1}) \alpha_\phi^>((g d^{(0)} r_v)(v_1 v_2), (g d^{(0)} r_v)(v_2 v_3))(r_v \triangleright m)(v_3))]^{e(v_1 v_2 v_3)} \\ &= \prod_{(v_1 v_2 v_3)} [\chi(h) \alpha_\phi^>(g(v_1 v_2), g(v_2 v_3))(m(v_3))]^{e(v_1 v_2 v_3)}, \end{aligned} \quad (119)$$

for every  $g \in Z^1(\mathbb{T}_\Delta^2, G)$  and  $h \in C^2(\mathbb{T}_\Delta^2, H)$ , thereby confirming that  $\mathbb{U}^{A, \phi, \chi}(\mathbb{T}_\Delta^2)$  commutes with local operators  $\mathbb{h}_{v,n}$ .

Let us now specialise to the 2-group  $\mathbb{Z}_2^{[0]} \bullet_\mu \mathbb{Z}_2^{[1]}$ . By definition, the Hamiltonian  $\mathbb{H} + \mathbb{H}^\lambda$  constructed from eq. (105) and eq. (109) is made of local operators that acts as  $-\prod_{e \ni v} X_e$ , for every  $v \in V(\mathbb{T}_\Delta^2)$ , while projecting out states  $|g\rangle$  satisfying  $\lambda(g, r_v) = -1$ . We argue that there is a choice of complex coefficients  $h_1(g, h, r_v, \mathfrak{z}_v)$  such that local operators of the form (116) effectively act in the same way. Similarly to the example of sec. 2.6, the crux is to restrict the kinematical Hilbert space to states  $|g, h\rangle$ , for which the 2-cochain  $h \in C^2(\mathbb{T}_\Delta^2, \mathbb{Z}_2)$  assigns the identity group element to every plaquette. This can be enforced via a choice of coefficients  $h_1(g, h, r_v, \mathfrak{z}_v)$ . One can further choose these coefficients such that the only non-vanishing ones are equal to  $-1$ , for every  $g \in Z^1(\mathbb{T}_\Delta^2, \mathbb{Z}_2)$ ,  $\mathfrak{z}_v \in C^1(\mathbb{T}_\Delta^2, \mathbb{Z}_2)$ , and  $r_v(v) = 1$ . Proceeding as such,  $(h d^{(1)}_{\mathfrak{z}_v} \mu(g, r_v)^{-1})(p) = 0$  enforces that  $(d^{(1)}_{\mathfrak{z}_v} \mu(g, r_v)^{-1})(p) = 0$ , for every  $p \in P(\mathbb{T}_\Delta^2)$ . For any  $\mathfrak{z}_v \in C^1(\mathbb{T}_\Delta^2, \mathbb{Z}_2)$ , the previous condition enforces that  $\mu(g, r_v)(p)$  is distinct from the identity element in  $\mathbb{Z}_2$  for an even number of plaquettes, precisely constraining  $g$  to be such that  $\lambda(g, r_v) = 1$ , as desired. Bringing everything together, this confirms that promoting the self-duality of  $\mathbb{H} + \mathbb{H}^\lambda$  to an internal symmetry does result in a  $2\text{Rep}(\mathbb{Z}_2^{[0]} \bullet_\mu \mathbb{Z}_2^{[1]})$  symmetry.

Naturally, one can construct much more general lattice higher gauge theories with a  $2\text{Rep}(\mathbb{Z}_2^{[0]} \bullet_\mu \mathbb{Z}_2^{[1]})$  symmetry than the one discussed above. Given such a theory, one can then ask for a duality relation of the same type as those considered so far, i.e. combining twisted and untwisted gauging operations. This is possible since the fusion 2-category  $2\text{Vec}_{\mathbb{Z}_2^{[0]} \bullet_\mu \mathbb{Z}_2^{[1]}}$  admits two inequivalent fiber 2-functors, which differ in a choice of 2-group 3-cocycle. Indeed, representatives of cohomology classes in  $H^3(\mathbb{Z}_2^{[0]} \bullet_\mu \mathbb{Z}_2^{[1]}, U(1))$  are labelled by pairs  $(\epsilon, \gamma)$  consisting of a 3-cochain  $\gamma \in C^3(\mathbb{Z}_2, U(1))$  and a homomorphism  $\epsilon : \mathbb{Z}_2 \rightarrow \widehat{\mathbb{Z}_2}$  such that  $d\gamma = \langle \epsilon, \smile \mu \rangle$  [46]. The unique non-trivial solution is provided by choosing  $\epsilon$  to be the non-trivial character of  $\mathbb{Z}_2$  and  $\gamma(g_1, g_2, g_3) = i^{g_1 g_2 g_3}$ . Given a model that would happen to be self-dual under the resulting duality, lifting the self-duality to an internal symmetry would result in one of the Tambara–Yamagami fusion 2-categories considered in ref. [28].

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## Conclusions and outlook

The central theme of this thesis has been the concrete tensor network realisation and study of various (generalised) symmetry structures. In conclusion, tensor networks provide a natural and effective framework in which these generalised symmetries can be explicitly represented on the lattice and manipulated. Not only do tensor networks provide a graphical calculus for dealing with these symmetries analytically, they also make it possible to put models with such symmetries on a computer and obtain their low-lying spectrum using variational methods. We reviewed this formalism from the ground up and illustrated it through several applications and research results.

Concretely, we saw that by phrasing topological order in the language of tensor networks, one can reveal hidden symmetry structures acting on the virtual degrees of freedom encoding the entanglement of the ground states. These symmetry structures are naturally described by fusion categories and realised as matrix product operators. Their relevance for the description of topological phases of matter can hardly be overstated. We demonstrated that they can be used naturally to construct distinct ground states as well as excited states, and argued that these symmetries persist beyond renormalisation fixed points.

In essence, these fusion categories describe simple symmetry operators, their fusion rules, and their anomalies. They therefore naturally generalise finite group and representation theory. Constructing all distinct representations of these categorical symmetries however requires bimodule categories. Although these may appear abstract at first sight, we have shown their importance in the construction of gapped boundaries of string-net models, and we have used them in the construction of a concrete quantum circuit that

maps between Morita equivalent string-net models belonging to the same topological phase.

We then argued that these categorical symmetries also arise as internal symmetries of general Hamiltonians or classical models in one lower dimension. This holographic perspective allowed us to construct symmetric Hamiltonians, understand duality transformations as changes of module category, and relate the resulting dual models via bimodule tubes.

Finally, we showed that the same holographic tensor network perspective extends naturally to one higher dimension. We have shown that the symmetries of the  $(2+1)$ -dimensional models considered here are encoded in fusion 2-categories. This provides a higher-dimensional analogue of the categorical symmetry and duality framework developed earlier in the thesis.

Within this framework, a number of open questions persist. Some of which, that spark my own interest, are the following.

One low-hanging fruit is to incorporate open boundary conditions into the symmetry and duality framework. In essence, this amounts to classifying the different ways in which symmetry operators can terminate at a physical boundary, and at the same time constructing the corresponding symmetric boundary Hamiltonian terms. This is particularly relevant for critical lattice models, where the infinite correlation length makes the bulk physics highly sensitive to the choice of boundary conditions [54, 97, 154, 164].

From a technical point of view, module categories are expected to classify such boundary conditions in one dimension, and module 2-categories in two dimensions. However, it remains to understand how this classification interacts with the bulk degrees of freedom, and how to explicitly construct the corresponding boundary tensors for both the MPO and PEPO symmetries and the Hamiltonians.

Once such boundary conditions are incorporated, the next question is how to classify and explicitly construct the corresponding topological sectors, and how these sectors are mapped under duality transformations. This is particularly relevant for the numerical study of these open lattice models. Although the categorical structures expected to classify these sectors are formally known, turning this classification into an explicit tensor network construction remains an important open problem [97, 165, 166].

Also, it remains largely unexplored how to construct integrable lattice models with categorical symmetries within the tensor network formalism. In the main text, we only briefly touched upon the more natural setting for this question, namely two-dimensional classical lattice models. Such models can be constructed using the strange correlator construction which provides a



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systematic way of generating MPO symmetric transfer matrices depending on only a small number of parameters. The natural question is then which models within this low-dimensional family are (Yang-Baxter) integrable, and hence exactly solvable.

One promising approach is that of discrete holomorphicity, which imposes consistency equations on the Boltzmann weights guaranteeing the existence of a conserved current [111]. It was shown how this naturally leads to known families of solutions to the Yang-Baxter equation. An interesting extension of the existing framework would be to allow currents described not by the input category itself, but by its Drinfeld center, thereby also covering input categories that are not braided. This is reminiscent of recent progress on the construction of critical lattice models with exotic symmetries [167, 168].

By employing the tensor network formalism, it becomes possible to study these models numerically and potentially reconstruct the full conformal field theory spectrum from the obtained lattice models.

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